

BILL WOODWARD

FIBER OPTICS INSTALLER (FOI) CERTIFICATION EXAM GUIDE



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Fiber Optics Installer (FOI) Certification Exam Guide



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Best regards,

A handwritten signature in black ink, appearing to read "Chris Webb", written in a cursive style.

Chris Webb
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*To Dick Glass and Charlie Husson, thanks for being great mentors
and friends.*

—BW

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About the Author

Bill Woodward is the director of CSISR Engineering Products with UrsaNav, Inc., an engineering services company. Bill has been teaching fiber optics and other technical courses since 1992. He has more than 25 years of experience in the design, operation, maintenance, troubleshooting, and repair of electronic and electrical systems.

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Introduction

The term “broadband” commonly refers to high-speed Internet access that is always on and faster than the traditional dial-up access. Without fiber optics, broadband as we know it today would not exist. Fiber optics is the backbone of the global telecommunications system. No other transmission medium can move the high rates of data over the long distances required to support the global telecommunications system. This technology works so well that the typical user may not be aware that it even exists.

This book focuses on building a solid foundation in fiber-optic theory and application. It describes in great detail fiber-optic cable technology, connectorization, and splicing. It examines the electronic technology built into fiber-optic receivers, transmitters, and test equipment that makes incredible broadband download and upload speeds possible. In addition, many current industry standards pertaining to optical fiber, connector, splice, and network performance are discussed in detail.

Who This Book Is For

If you are standing in your neighborhood bookstore browsing through this book, you may be asking yourself whether you should buy it. The procedures in this book are illustrated and written in English rather than “technospeak.” That’s because this book was designed specifically to help unlock the mysteries of fiber optics. Fiber optics can be a confusing topic; it has its own language, acronyms, and standards. This book was developed with the following types of people in mind:

- Information technology (IT) professionals who can use this book to gain a better understanding and appreciation of a structured fiber-optic cabling system
- IT managers who are preparing to install a new computer system
- Do-it-yourselfers who need to install a few new cabling runs in their facility and want to get it right the first time
- New cable installers who want to learn more than just what it takes to pull a cable through the ceiling and terminate it to the patch panel
- Students taking introductory courses in LANs and cabling
- Students preparing for the ETA fiber-optics installer (FOI) certification

This book is an excellent reference for anyone currently working in fiber optics as well as those who are just starting to learn about fiber optics. The book covers in detail all of the competencies of the Electronics Technicians Association International (ETA) (FOI) certification.

ETA's FOI Certification Program

The ETA's FOI certification program is the most comprehensive in the industry. It requires students to attend an ETA-approved training school. Each student must achieve a score of 75 percent or greater on the written exam and satisfactorily complete all the hands-on requirements. Those who are interested in obtaining ETA FOI certification can visit the ETA's website at www.eta-i.org and get the most up-to-date information on the program and a list of approved training schools.

The ETA FOI certification requires no prerequisite and is designed for anyone who is interested in learning how to become a fiber-optic installer. The FOI certification is recommended as a prerequisite for the FOT certification, for those who want to learn how to test a fiber-optic link to the current industry standards and how to troubleshoot. Fiber-optic certification demonstrates to your employer that you have the knowledge and hands-on skills required to install, test, and troubleshoot fiber-optic links and systems. With the push to bring fiber optics to every home, these skills are highly sought after.

What This Book Covers

This book's topics run the gamut of fiber-optics technology and application; they include the following:

- The history of fiber optics and broadband access
- The principles of fiber-optic transmission
- The basic principles of light
- Optical fiber construction and theory
- Optical fiber characteristics
- Safety
- Fiber-optic cables
- Fusion and mechanical splicing
- Connectors
- Fiber-optic light sources and transmitters
- Fiber-optic detectors and receivers
- Cable installation and hardware
- Fiber-optic system advantages
- Test equipment and link/cable testing

This book provides you with a solid foundation in fiber-optic technologies and practices. The book is loaded with valuable information, including the following elements:

Assessment test Directly following this introduction is an assessment test that you should take. It is designed to help you determine how much you already know. Each question is tied to a topic discussed in the book. Using the results of the assessment test, you can figure out the areas where you need to focus your study. Of course, we do recommend that you read the entire book.

Competency-by-competency coverage of the topics you need to know Each chapter lists the ETA FOI exam knowledge competencies covered in that chapter, followed by a detailed discussion of each competency.

Chapter exercises You'll find exercises designed to give you the important hands-on experience that is critical for your exam preparation. The exercises support the topics of the chapter, and they walk you through the steps necessary to perform a particular function.

Real World Scenarios Because reading a book isn't enough for you to learn how to apply these topics in your everyday duties, we have provided Real World Scenarios in special sidebars. These explain when and why a particular solution would make sense in a working environment that you'd actually encounter.

Exam Essentials To highlight what you learn, you'll find a list of Exam Essentials at the end of each chapter. The Exam Essentials section briefly highlights the topics that need your particular attention as you prepare for the FOI exam.

Review questions, complete with detailed explanations Each chapter is followed by a set of review questions that test what you learned in the chapter. The questions are written with the exam in mind, meaning that they are designed to have the same look and feel as what you'll see on the exam. Answers to the review questions with detailed explanations can be found in the appendix.

Glossary Throughout each chapter, you will be introduced to important terms and concepts that you will need to know for the FOI exam. These terms appear in italics within the chapters. At the end of the book, a detailed glossary gives definitions for these terms, as well as other general terms you should know.

How to Use This Book

To understand the way this book is put together, you must learn about a few of the special conventions that were used. Here are some of the items you will commonly see.

Italicized words indicate new terms. After each italicized term, you will find a definition.



Tips will be formatted like this. A tip is a special bit of information that can make your work easier or make an installation go more smoothly.



Notes are formatted like this. When you see a note, it usually indicates some special circumstance to make note of. Notes often include out-of-the-ordinary information about working with a telecommunications infrastructure.



Warnings are found within the text whenever a technical situation arises that may cause damage to a component or cause a system failure of some kind. Additionally, warnings are placed in the text to call particular attention to a potentially dangerous situation.



Key terms are used to introduce a new word or term that you should be aware of. Just as in the worlds of networking, software, and programming, the world of cabling and telecommunications has its own language.

Sidebars

This special formatting indicates a sidebar. *Sidebars* are entire paragraphs of information that, although related to the topic being discussed, fit better into a standalone discussion. They are just what their name suggests: a sidebar discussion.

This book provides a solid foundation for the serious effort of preparing for the ETA FOI certification exam. To best benefit from this book, you might want to use the following study method:

1. Take the assessment test to identify your weak areas.
2. Study each chapter carefully. Do your best to fully understand the information.
3. Read over the Real World Scenarios to improve your understanding of how to use what you learn in the book.
4. Study the Exam Essentials to make sure that you are familiar with the areas you need to focus on.
5. Answer the review questions at the end of each chapter.
6. Work through the Chapter Exercises at the end of the chapter.
7. Take note of the questions you did not understand, and study the corresponding sections of the book again.
8. Go back over the Exam Essentials.

To learn all the material required to pass the exam, you will need to study regularly and with discipline before and while attending an ETA-approved training course. Try to set aside the same time every day to study, and select a comfortable and quiet place in which to do it. Do not wait until the break before the exam to start studying. Remember: if you have any questions about the material you are learning, ask your instructor.

Enjoy!

Have fun reading this book—it has been fun writing it. I hope that it will be a valuable resource to you and will answer at least some of your questions on fiber optics. As always, I love to hear from readers, you can reach me at wrwoodward2013@gmail.com.

Assessment Test

1. The decade that saw the first full-scale commercial application of fiber-optic communication systems was
 - A. The 1880s
 - B. The 1920s
 - C. The 1960s
 - D. The 1970s
2. Which of the following groups describes the components in a fiber-optic link?
 - A. Transmitter, receiver, optical fiber, decoder
 - B. Transmitter, receiver, optical fiber, connectors
 - C. Transmitter, receiver, optical fiber
 - D. Transmitter, receiver, optical fiber, cabling
3. In a fiber-optic link, which component has an input of electrical energy and an output of light energy?
 - A. Connector
 - B. Optical fiber
 - C. Receiver
 - D. Transmitter
4. A 50 percent reduction in signal strength is a loss of _____.
 - A. 3dB
 - B. 3dBm
 - C. 7dB
 - D. 7dBm
5. Refraction is caused by _____.
 - A. Light changing color as it passes from one medium into another
 - B. Light changing speed as it passes from one medium into another
 - C. Light changing frequency as it passes from one medium into another
 - D. Light changing direction within a single medium
6. A Fresnel reflection will occur because light changes _____ as it moves from a material with a given n into a material with a different n .
 - A. Direction
 - B. Speed
 - C. The critical angle
 - D. The index of refraction

7. Listed from innermost to outermost, the components of an optical fiber are
 - A. Cladding, core, coating
 - B. Core, cladding, coating
 - C. Cladding, coating, core
 - D. Core, coating, cladding
8. Modal dispersion is caused when
 - A. Light escapes into the cladding from the core.
 - B. Light rays follow different paths along the fiber core.
 - C. A fiber is split into several parts.
 - D. Light encounters a break in the fiber.
9. For a fiber of a given bandwidth, as the length of the fiber increases, the bandwidth
 - A. Increases
 - B. Decreases
 - C. Does not change
 - D. Is independent of the length
10. Because infrared lasers cannot be seen, they _____.
 - A. Cannot hurt your eyes
 - B. Are not useful in fiber optics
 - C. Are especially dangerous to your eyes
 - D. Require special instruments
11. Which of the following describes the four components of a fiber optic cable?
 - A. Core, cladding, coating, buffer
 - B. Fiber, buffer, strength member, jacket
 - C. Core, buffer, jacket, coating
 - D. Fiber, buffer, strength member, coating
12. Article _____ of the National Electric Code covers optical fiber cables and raceways.
 - A. 250
 - B. 660
 - C. 770
 - D. 810
13. Which part of the connector holds the fiber in place?
 - A. The ferrule
 - B. The cap
 - C. The boot
 - D. The body

14. Which geometry of the connector endface ensures a physical contact (PC) finish?
- A. Flat
 - B. Rough
 - C. Curved
 - D. Lensed
15. The puck is used to
- A. Keep a firm grip on the connector for polishing.
 - B. Hold down the abrasive material during polishing.
 - C. Keep the abrasive surface clean.
 - D. Keep the ferrule perpendicular to the polishing surface.
16. Per ANSI/TIA-568-C.3 the strain relief for a single-mode connector should be which color?
- A. Aqua
 - B. Black
 - C. Blue
 - D. Beige
17. Lasers emit photons that are in phase through a process called _____ emission.
- A. Coherent
 - B. Incoherent
 - C. Stimulated
 - D. Spontaneous
18. Fiber-optic receivers use a(n) _____ to convert the light energy from the optical fiber into electrical energy.
- A. Photoresistor
 - B. Photodiode
 - C. Phototransistor
 - D. LED
19. A cable's minimum bend radius specification is greater
- A. During installation
 - B. After installation
 - C. When the temperature is higher
 - D. When it is in a cable tray

20. A pulling eye is attached to
- A. The cable's outer jacket
 - B. The cable's strength member
 - C. The fiber
 - D. The armor
21. What is the jacket color of laser-optimized simplex cordage?
- A. Orange
 - B. Slate
 - C. Blue
 - D. Aqua
22. Hybrid cable as applied to fiber optics combines
- A. Loose and tight buffers
 - B. Single-mode and multimode optical fibers
 - C. Optical fibers and current-carrying electrical conductors
 - D. Plastic and glass optical fibers
23. A mechanical splice typically uses _____ to reduce or eliminate Fresnel reflections.
- A. Epoxy
 - B. A high-voltage electric arc
 - C. Index matching gel
 - D. Isopropyl alcohol
24. Only a _____ laser is considered eye safe.
- A. Class I
 - B. Class II
 - C. Class III
 - D. Class IV
25. The _____ is the laser primarily used in multimode transmitters.
- A. DFB
 - B. VCSEL
 - C. Fabry-Pérot
 - D. LED

26. Which types of optical fibers should be used for the physical layer when multiple transmitters are operating in parallel?
- A. OS1, OS2
 - B. OS3, OS4
 - C. OM1, OM2
 - D. OM3, OM4
27. _____ is required to ensure the proper operation of bidirectional fiber-optic communication systems that use separate transmit and receive optical fibers.
- A. Common equipment
 - B. A demarcation point
 - C. Polarity
 - D. A first-level backbone
28. When is a splice enclosure used?
- A. Whenever a fiber has been spliced
 - B. When a splice must be placed underground
 - C. When a splice must be placed underwater
 - D. Splice enclosures are optional.
29. What is the purpose of a patch panel?
- A. It takes the place of a splice enclosure.
 - B. It fills a hole where fiber has been installed.
 - C. It is used to route signals between cables.
 - D. It provides a permanent link between two pieces of hardware.
30. _____ is the administration standard for telecommunications infrastructures.
- A. ANSI/TIA-568-C.3
 - B. ANSI/TIA-606-B
 - C. ANSI/TIA-568-C.0
 - D. Article 770
31. Optical fiber offers _____ bandwidth and _____ attenuation than twisted-pair or coaxial cable.
- A. Less, more
 - B. Equal, more
 - C. Greater, more
 - D. Greater, less

32. Encircled flux launch conditions were developed to improve the accuracy of _____ insertion loss measurements.
- A. Multimode
 - B. Single-mode
 - C. Multimode and single-mode
 - D. Multifiber
33. The _____ is a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back.
- A. Continuity tester
 - B. OLTS
 - C. VFL
 - D. OTDR
34. Which broadband technology moves the highest rates of data over the greatest distances?
- A. Fiber optics
 - B. Digital Subscriber Line (DSL)
 - C. Broadband over Power Lines (BPL)
 - D. Wi-Fi
35. In fiber optics, _____ is typically used to describe the output of a light source.
- A. Frequency
 - B. Wavelength
 - C. Hertz
 - D. Kilometers
36. Light traveling across a small air gap between two optical fibers will produce _____.
- A. Total internal reflection
 - B. Electromagnetic radiation
 - C. A Fresnel reflection
 - D. The Index of Refraction
37. The optical trench surrounds the _____ of multimode or single-mode bend-insensitive optical fibers.
- A. Core
 - B. Cladding
 - C. Coating
 - D. Buffer

- 38.** Which optical fiber designation offers a minimum effective modal bandwidth-length product of 4700 MHz · km?
- A.** OM1
 - B.** OM2
 - C.** OM3
 - D.** OM4
- 39.** What type of bending loss may not be visible when looking at the fiber-optic cable?
- A.** Macrobend
 - B.** Microbend
 - C.** Intrinsic
 - D.** Attenuation
- 40.** The maximum attenuation allowable for OM1 fiber at 850nm is _____ .
- A.** 0.5 dB/km
 - B.** 1.0 dB/km
 - C.** 1.5 dB/km
 - D.** 3.5 dB/km

Answers to Assessment Test

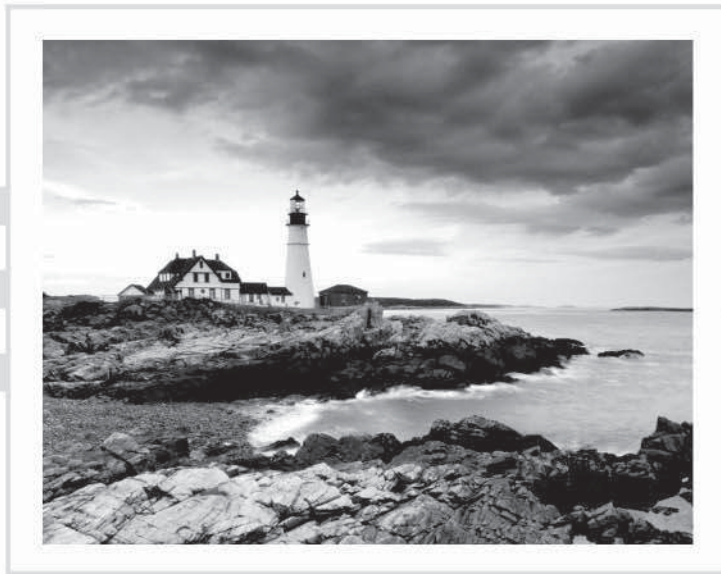
1. D. The first full-scale commercial application of fiber-optic communication systems occurred in 1977, when both AT&T and GTE began using fiber-optic telephone systems for commercial customers.
2. B. The fiber-optic link requires a transmitter to send the signal, a receiver to capture the signal, optical fiber to carry the signal, and connectors to provide an interface for the optical fiber and the equipment.
3. D. In order for a signal to be sent through optical fiber, the transmitter must first convert it from electrical energy into light.
4. A. Reducing signal strength by one-half is a 3dB loss.
5. B. Refraction, or the change in the direction of light as it changes speed passing from one material into another, is a key component in fiber-optic transmissions.
6. B. When light changes speed as it moves from a material with a given index of refraction into a material with a different index of refraction, a Fresnel reflection occurs.
7. B. The core is at the center of the optical fiber; the cladding surrounds the core, and the coating surrounds the cladding.
8. B. Modal dispersion is caused when light rays follow different paths in the fiber.
9. B. Because dispersion increases with the length of the fiber, the usable bandwidth of the fiber decreases as the signal pulses must be kept farther apart to avoid overlapping.
10. C. Because infrared lasers are invisible, you cannot know when they are on, even if you are looking directly into a fiber. You should always make sure the fiber is disconnected from the light source before looking into it with your bare eyes or optical instruments.
11. B. The core, cladding, and coating of the fiber itself are considered one component of the cable, with the rest being the buffer, strength member, and jacket.
12. C. Article 770 of the National Electrical Code defines requirements for optical fiber cables and raceways.
13. A. Beginning at the working end of the connector, the ferrule holds the fiber in place. The ferrule must hold the fiber exactly centered in its endface for the best possible connection, so its construction is critical.
14. C. To ensure physical contact (PC) between optical fiber ends, the best endface geometry is a convex curve. This curved, or PC finish, ensures that the highest feature or apex on the endface is the center of the optical fiber end.
15. D. The polishing fixture, or puck as it is typically referred to, is used to ensure that the ferrule stays perpendicular to the polishing film during the polishing process.

16. C. Per ANSI/TIA-568-C.3 the strain relief for a single-mode connector should be blue in color.
17. C. Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This process of generating light energy is called stimulated emission.
18. B. In a fiber-optic receiver, it's the job of the photodiode to convert the light energy received from the optical fiber into electrical energy.
19. A. The minimum installation bend radius is the short-term bend radius, and the minimum operational bend radius is the long-term bend radius. The minimum short-term bend radius is actually the larger of the two because of the tensile stresses that may be placed on the cable during installation.
20. B. The pulling eye is specially designed to attach to the cable's strength member at one end and a pulling line at the other.
21. D. TIA-598-C defines premises cable jacket colors; aqua is the jacket color for laser-optimized multimode optical fiber used for non-military applications.
22. B. Hybrid cable, as applied to fiber optics, combines multimode and single-mode optical fibers in one cable. Hybrid cable should not be confused with composite cable, although the terms have been used interchangeably in the past.
23. C. A mechanical splice aligns the cleaved optical fibers and holds them in place. Index matching gel inside the mechanical splice reduces or eliminates Fresnel reflections.
24. A. Under normal operating conditions, Class I lasers do not producing damaging radiation levels.
25. B. Currently, VCSEL multimode transmitters support only 850nm operation with 50/125µm or 62.5/125µm optical fiber. The VCSEL can support data rates as high as 10Gbps.
26. D. As defined in IEEE Standard 802.3-2012, Section 6, when multiple transmitters are operating in parallel, multimode OM3 or OM4 optical fiber should be used for the physical layer.
27. C. Polarity is required to ensure the proper operation of bidirectional fiber-optic communication systems that use separate transmit and receive optical fibers. It ensures that there is an end-to-end transmission path between the transmitter and the receiver of a channel.
28. A. A splice enclosure is used whenever two fibers have been spliced to protect the splice from exposure and strain.
29. C. A patch panel is an interconnection point for fiber-optic cables. They allow signals to be routed from one cable to another with a patch cord or jumper.
30. B. ANSI/TIA-606-B is the administration standard for telecommunications infrastructures. This document contains detailed information on conventions for labeling optical

fiber connections within buildings, datacenters, and between buildings that share a telecommunication network.

- 31. D. Optical fiber offers the highest bandwidth and the least attenuation of any medium currently available.
- 32. A. Encircled flux launch conditions for all multimode insertion loss measurements has been added to IEC 61280-4-1 Edition 2, ANSI/TIA-568-C.0, and ANSI/TIA-526-14-B. The addition of this requirement arose from the need to improve the accuracy of multimode insertion loss measurements.
- 33. D. The OTDR is nothing more than a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back. It provides a graphical representation of what is happening in the fiber-optic link or cable under test.
- 34. A. Without fiber optics, broadband as we know it today would not exist. No other transmission medium can move the high rates of data over the long distances required to support the global telecommunications system.
- 35. B. Wavelength or frequency can be used to describe electromagnetic energy; however, wavelength is typically used to describe the output of a fiber-optic light source.
- 36. C. A Fresnel reflection occurs when light changes speed as it moves from the optical fiber to the air and from the air to the optical fiber.
- 37. A. The optical trench surrounds the core of the optical fiber. It has a lower refractive index than the core or the cladding.
- 38. D. The minimum effective bandwidth-length product for a 50/125 μ m laser optimized OM4 multimode optical fiber is 4700MHz · km.
- 39. B. Microbends are small distortions of the boundary layer between the core and cladding caused by crushing or pressure. Microbends are very small and may not be visible when looking at the fiber-optic cable.
- 40. D. ISO/IEC 11801 defines the performance of OM1; the maximum attenuation at 850nm is 3.5dB for a length of one kilometer.

Fiber Optics Installer (FOI) Certification Exam Guide



Chapter 1

History of Fiber Optics and Broadband Access

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Trace the evolution of light in communication.
- ✓ Summarize the evolution of optical fiber manufacturing technology.
- ✓ Track the evolution of optical fiber integration and application.
- ✓ Describe the role of fiber optics in high-speed Internet access.





Like many technological achievements, fiber-optic communications grew out of a succession of quests, some of them apparently unrelated. It is important to study the history of

fiber optics to understand that the technology as it exists today is relatively new and still evolving.

This chapter discusses the major accomplishments that led to the creation of high-quality optical fibers and their use in high-speed communications and data transfer, as well as their integration into existing communications networks.

In this chapter, you will learn to

- Recognize the refraction of light
- Identify total internal reflection
- Detect crosstalk between multiple optical fibers
- Recognize attenuation in an optical fiber

Evolution of Light in Communication

Hundreds of millions of years ago, the first bioluminescent creatures began attracting mates and luring food by starting and stopping chemical reactions in specialized cells. Over time, these animals began to develop distinctive binary, or on-off, patterns to distinguish one another and communicate intentions quickly and accurately. Some of them have evolved complex systems of flashing lights and colors to carry as much information as possible in a single glance. These creatures were the first to communicate with light, a feat instinctive to them but tantalizing and elusive to modern civilization until recently.

Early Forms of Light Communication

Some of the first human efforts to communicate with light consisted of signal fires lit on hilltops or towers to warn of advancing armies, and lighthouses that marked dangerous coasts for ancient ships and gave them reference points in their journeys. To the creators of these signals, light's tremendous speed (approximately 300,000 kilometers per second) made its travel over great distances seem instantaneous.

An early advance in these primitive signals was the introduction of relay systems to extend their range. In some cases, towers were spread out over hundreds of kilometers, each one in the line of sight of the next. With this system, a beacon could be relayed in the time it took each tower guard to light a fire—a matter of minutes—while the fastest transportation might have taken days. Because each tower only needed in its line of sight the sending and

receiving towers, the light, which normally travels in a straight line, could be guided around obstacles such as mountains as well as over the horizon. As early as the fourth century A.D., Empress Helena, the mother of Constantine, was believed to have sent a signal from Jerusalem to Constantinople in a single day using a relay system.



The principle behind signal relay towers is still used today in the form of repeaters, which amplify signals attenuated by travel over long distances through optical fibers.

Early signal towers and lighthouses, for all their usefulness, were still able to convey only very simple messages. Generally, no light meant one state, whereas a light signaled a change in that state. The next advance needed was the ability to send more detailed information with the light. A simple but notable example is the signal that prompted Paul Revere's ride at the start of the American Revolution. By prearranged code, one light hung in the tower of Boston's Old North Church signaled a British attack by land; two lights meant an invasion by sea. The two lamps that shone in the tower not only conveyed a change in state, but also provided a critical detail about that change.

The Quest for Data Transmission

Until the 1800s, light had proven to be a speedy way to transmit simple information across great distances, but until new technologies were available, its uses were limited. It took a series of seemingly unrelated discoveries and inventions to harness the properties of light through optical fibers.

The first of these discoveries was made by Willebrord van Roijen Snell, a Dutch mathematician who in 1621 wrote the formula for the principle of *refraction*, or the bending of light as it passes from one material into another. The phenomenon is easily observed by placing a stick into a glass of water. When viewed from above, the stick appears to bend because light travels more slowly through the water than through the air. Snell's formula, which was published 70 years after his death, stated that every transparent substance had a particular *index of refraction*, and that the amount that the light would bend was based on the relative refractive indices of the two materials through which the light was passing. Air has an approximate refractive index of 1 and water has a refractive index of 1.33.

The next breakthrough came from Jean-Daniel Colladon, a Swiss physicist, and Jacques Babinet, a French physicist. In 1840, Colladon and Babinet demonstrated that bright light could be guided through jets of water through the principle of *total internal reflection*. In their demonstration, light from an arc lamp was used to illuminate a container of water. Near the bottom of the container was a hole through which the water could escape. As the water poured out of the hole, the light shining into the container followed the stream of water. Their use of this discovery, however, was limited to illuminating decorative fountains and special effects in operas. It took John Tyndall, a natural philosopher and physicist from Ireland, to bring the phenomenon to greater attention. In 1854, Tyndall performed the demonstration before the British Royal Society and made it part of his published works in 1871, casting a shadow over the contribution of Colladon and Babinet. Tyndall is now widely credited with discovering total internal reflection, although Colladon and Babinet had demonstrated it 14 years previously.

Total internal reflection takes place when light passing through a material with a higher index of refraction (the water in the experiment) hits a boundary layer with a material that has a lower index of refraction (the air). When this takes place, the boundary layer becomes reflective, and the light bounces off the boundary layer, remaining contained within the material with the higher index of refraction.

Shortly after Tyndall, Colladon, and Babinet laid the groundwork for routing light through a curved material, another experiment took place that showed how light could be used to carry higher volumes of data.

In 1880, Alexander Graham Bell demonstrated his photophone, one of the first true attempts to carry complex signals with light. It was also the first device to transmit signals wirelessly. The photophone gathered sunlight onto a mirror attached to a mouthpiece that vibrated when a user spoke into it. The vibrating mirror reflected the light onto a receiver coated with selenium, which produced a *modulated* electrical signal that varied with the light coming from the sending device. The electrical signal went to headphones where the original voice input was reproduced.

Bell's invention suffered from the fact that outside influences such as dust or stray light confused the signals, and clouds or other obstructions to light rendered the device inoperable. Although Bell had succeeded in transmitting a modulated light signal nearly 200 meters, the photophone's limitations had already fated it to be eclipsed by Bell's earlier invention, the telephone. Until the light could be modulated and guided as well as electricity could, inventions such as the photophone would continue to enjoy only novelty status.

Evolution of Optical Fiber Manufacturing Technology

John Tyndall's experiment in total internal reflection had led to attempts to guide light with more control than could be achieved in a stream of water. One such effort by William Wheeler in 1880, the same year that Bell's photophone made its debut, used pipes with a reflective coating inside that guided light from a central arc lamp throughout a house. As with other efforts of the time, there was no attempt to send meaningful information through these conduits—merely to guide light for novelty or decorative purposes. The first determined efforts to use guided light to carry information came out of the medical industry.

Controlling the Course of Light

Doctors and researchers had long tried to create a device that would allow them to see inside the body with minimal intrusion. They had begun experimenting with bent glass and quartz rods, bringing them tantalizingly close to their goal. These tools could transmit light into the body, but they were extremely uncomfortable and sometimes dangerous for the patient, and there was no way yet to carry an image from the inside of the body out to doctors. What they needed was a flexible substance or *medium* that could carry whole images for about half a meter.

One such material was in fact pioneered for quite a different purpose. Charles Vernon Boys was a British physics teacher who needed extremely sensitive instruments for his continuing research in heat and gravity. In 1887, to provide the materials he needed, he began drawing fine fibers out of molten silica. Using an improvised miniature crossbow, he shot a needle that dragged the molten material out of a heat source at high speed. The resulting fiber—more like quartz in its crystalline structure than glass—was finer than any that had been made to date, and was also remarkably even in its thickness. Even though glass fibers had already been available for decades before this, Boys' ultra-fine fibers were the first to be designed for scientific purposes and were also the strongest and smallest that had been made to date. He did not, however, pursue research into the optical qualities of his fibers.

Over the next four decades, attempts to use total internal reflection in the medical industry yielded some novel products, including glass rods designed by Viennese researchers Roth and Reuss to illuminate internal organs in 1888, and an illuminated dental probe patented in 1898 by David Smith. A truly flexible system for illuminating or conveying images of the inside of the body remained elusive, however.

The next step forward in the optical use of fibers occurred in 1926. In that year, Clarence Weston Hansell, an electrical engineer doing research related to the development of television at RCA, filed a patent for a device that would use parallel quartz fibers to transmit a lighted image over a short distance. The device remained in the conceptual stage, however, until a German medical student, Heinrich Lamm, developed the idea independently in an attempt to form a flexible gastroscope. In 1930, Lamm bundled commercially produced fibers and managed to transmit a rough image through a short stretch of the first fiber-optic cable. The process had several problems, however, including the fact that the fiber ends were not arranged exactly, and they were not properly cut and polished. Another issue was to prove more daunting. The image quality suffered from the fact that the quartz fibers were bundled against each other. This meant that the individual fibers were no longer surrounded by a medium with a lower index of refraction. Much of the light from the image was lost to *crosstalk*. Crosstalk or optical coupling is the result of light leaking out of one fiber into another fiber.

The poor focus and resolution of Lamm's experimental image meant that a great deal more work would be needed, but Lamm was confident enough to write a paper on the experiment. The rise of the Nazis, however, forced Lamm, a Jew, to leave Germany and abandon his research. The dream of Hansell and Lamm languished until a way could be found to solve the problems that came with the materials available at the time.

Also in 1930, the chemical company DuPont invented a clear plastic material that it branded Lucite. This new material quickly replaced glass as the medium of choice for lighted medical probes. The ease of shaping Lucite pushed aside experiments with bundles of glass fiber, along with the efforts to solve the problems inherent in Lamm's probe.

The problems surfaced again 20 years later, when the Dutch government began looking for better periscopes for its submarines. They turned to Abraham van Heel, who was at the time the president of the International Commission of Optics and a professor of physics at the Technical University of Delft, the Netherlands. Van Heel and his assistant, William Brouwer, revived the idea of using fiber bundles as an image-transmission medium. Fiber bundles, Brouwer pointed out, had the added advantage of being able to scramble and then unscramble an image—an attractive feature to Dutch security officials.

When van Heel attempted to build his image carrier, however, he rediscovered the problem that Lamm had faced. The refractive index of adjacent fibers reduced a fiber's ability to achieve total internal reflection, and the system lost a great deal of light over a short distance. At one point, van Heel even tried coating the fibers with silver to improve their reflectivity, but the effort provided little benefit.

At his government's suggestion, van Heel approached Brian O'Brien, president of the Optical Society of America, in 1951. O'Brien suggested a procedure that is still the basis for fiber optics today: surrounding, or "cladding," the fiber with a layer of material with a lower refractive index.

Following O'Brien's suggestion, van Heel ran the fibers through a liquid plastic that coated them, and in April 1952, he succeeded in transmitting an image through a 400-fiber bundle over a distance of half a meter.

Van Heel's innovation—along with research performed by Narinder Singh Kapany, who also coined the term *fiber optics*, and Harold Hopkins—helped make the 1950s the pivotal decade in the development of modern fiber optics.

Working in England, Kapany and Hopkins developed a method for ensuring that the fibers at each end of a cable were in precise alignment. They wound a single fine strand several thousand times in a figure-eight pattern and sealed a section in clear epoxy to bind the fibers together throughout the bundle. They then sawed the sealed portion in half, leaving the fiber ends bonded in exact alignment. The image transmitted with this arrangement was clearly an improvement, but the brightness degraded quickly since the fibers were unclad.

Extending Fiber's Reach

In January 1954, the British journal *Nature* chanced to publish papers on the findings of van Heel as well as Kapany and Hopkins in the same issue. Although their placement in the journal was apparently coincidental, the two advancements were precisely the right combination of ideas for Professor Basil Hirschowitz, a gastrosurgeon from South Africa who was working on a fellowship at the University of Michigan. Hirschowitz assembled a team to study the uses of these new findings as a way to finally build a flexible endoscope for peering inside the body. Assisting Hirschowitz were physicist C. Wilbur Peters and a young graduate student named Lawrence Curtiss.

Curtiss studied the work of Kapany and Hopkins and used their winding method to create a workable fiber bundle, but his first attempt at cladding used van Heel's suggestion of cladding glass fibers with plastic. The results were disappointing.

In 1956, Curtiss began working with a new type of glass from Corning, one with a lower refractive index than the glass he was using in his fibers. He placed a tube made of the new glass around a core made from the higher refractive index glass and melted the two together. The cladded glass fiber that he drew from this combination was a success. On December 8, 1956, Curtiss made a fiber with light-carrying ability far superior to that of any fiber before it. Even when he was 12 meters away from the glass furnace, he could see the glow of the fire inside the fiber that was being drawn from it. By early 1957, Hirschowitz and Curtiss had created a working endoscope, complete with lighting

and optics. This event marked the first practical use of optical fibers to transmit complex information.

Curtiss' fibers were well suited for medical applications, but their ability to carry light was limited. Suffering a signal loss of one *decibel* per meter, the fibers were still not useful for long-distance communications. Many thought that glass was inherently unusable for communications, and research in this area remained at a minimal level for nearly a decade.

In the meantime, the electronic communications industry had been experimenting with methods of improving bandwidth for the higher volumes of traffic they expected to carry. The obvious choice for increasing the amount of information a signal could carry was to increase the frequency, and throughout the 1950s, researchers had pushed frequencies into the tens of *gigahertz*, which produced *wavelengths* of only a few millimeters. Frequencies in this range—just below the lowest infrared frequencies—required hollow pipes to be used as *waveguides*, because the signals were easily disturbed by atmospheric conditions such as fog or dust.

With the invention of the laser in 1960, the potential for increasing communication bandwidths literally increased exponentially. Wavelengths had been slashed from the millimeter range to the micrometer range, and true optical communications seemed within reach. The problems of atmospheric transmission remained, however, and waveguides used for lower frequencies were proving inadequate for optical wavelengths unless they were perfectly straight. Optical fibers, too, were all but ruled out as a transmission medium because of the loss of light or *attenuation*. The loss of 1,000 decibels per kilometer was still too great.

One researcher did not give up on fiber, however; Charles K. Kao, working at Standard Telecommunications Laboratories, began studying the problems encountered in optical fibers. His conclusions revived interest in the medium after he announced in 1966 that signal losses in glass fibers were not caused by inherent deficiencies of the material, but by flaws in the manufacturing process. Kao proposed that improved manufacturing processes could lower attenuation to levels of 20 decibels per kilometer or better, while providing the ability to carry up to 200,000 telephone channels in a single fiber.

Kao's pronouncement sparked a race to find the lower limit of signal loss in optical fibers. In 1970, Corning used pure silica to create a fiber with a loss that achieved Kao's target of only 20 decibels per kilometer. That was just the beginning. Six years later, the threshold had dropped to just half a decibel per kilometer, and in 1979 the new low was 0.2 decibel per kilometer. Optical fiber had passed well into the realm of practicality for communications and could begin showing its promise as a superior medium to copper.

Evolution of Optical Fiber Integration and Application

Once signal losses in fiber dropped below Kao's projected figure of 20 decibels per kilometer, communications companies began looking seriously at fiber optics as a new transmission medium. The technology required for this fledgling medium was still expensive,

however, and fiber-optic communications systems remained in closed-circuit, experimental stages until 1973. In that year, the U.S. Navy installed a fiber-optic telephone link aboard the *USS Little Rock*. Fiber optics had left the lab and started working in the field. Further military tests showed fiber's advantages over copper in weight and information-carrying capacity.

The first full-scale commercial application of fiber-optic communication systems occurred in 1977, when both AT&T and GTE began using fiber-optic telephone systems for commercial customers. During this period, the U.S. government breakup of the Bell Telephone system monopoly ushered in a boom time for smaller companies seeking to market long-distance service. A number of companies had positioned themselves to build microwave towers throughout the country to create high-speed long-distance networks. With the rise of fiber-optic technology, however, the towers were obsolete before they had even been built. Plans for the towers were scrapped in the early 1980s in favor of fiber-optic links between major cities. These links were then connected to local telephone companies that leased their capacity from the operators. The result was a bandwidth feeding frenzy. The fiber-optic links had such high capacities that extra bandwidth was leased to other local and long-distance carriers, which often undercut the owners of the lines, driving some out of business.

Following the success of fiber optics in the telecommunications industry, other sectors began taking advantage of this medium. During the 1990s, fiber-optic networks began to dominate in the fields of industrial controls, computers, and information systems. Improvements in lasers and fiber manufacturing continued to drive data rates higher and bring down operating costs.

Today, fiber optics have become commonplace in many areas as the technology continues to improve. Until recently, the transition to fiber optics was cost effective only for business and industry; equipment upgrades made it too expensive for telephone and cable companies to run fiber to every home. Manufacturing improvements have reduced costs, however, so that running fiber to the home is now an affordable alternative for telephone and cable companies.

Broadband since the Turn of the Century

A search on the Internet for the definition of broadband will yield many different results. Which result is correct? For this chapter, the definition published by the Federal Communications Commission (FCC) is correct. The FCC states that the term broadband commonly refers to high-speed Internet access that is always on and faster than the traditional dial-up access.

Broadband can be accessed using different high-speed transmission technologies over different mediums. Typical broadband connections include:

- Fiber optics
- Wireless

- Cable modem
- Satellite
- Digital Subscriber Line (DSL)
- Broadband over Power Lines (BPL)

In June of 2013, the United States Office of Science and Technology Policy and The National Economic Council published a report entitled *Four Years of Broadband Growth*. Many of the facts and definitions presented in this section of the chapter were obtained from that report.

The Role of Fiber Optics in Broadband

Today broadband can be accessed using different transmission technologies over different mediums. On the road, you may access the Internet using your cell phone and *fourth-generation* (4G) technology. At your local coffee shop, that same phone may access the Internet using the coffee shop's *Wi-Fi* connection. When you arrive home, your phone connects to your wireless router that provides Internet access over a *cable modem*. Later in the day, you place the phone on the charger and turn on a high-definition television with Internet capabilities that connects to your router with a cable. While your favorite show is playing in the background, you turn on your laptop and check email over your wireless connection.

The state of broadband technology today makes all this connectivity relatively easy, and to most users it is completely transparent. In other words, you do not need to understand anything about the infrastructure that supports the global telecommunications system to communicate and share information with nearly anyone in the world.

Without fiber optics, broadband as we know it today would not exist. Fiber optics is the backbone of the global telecommunications system. No other transmission medium can move the high rates of data over the long distances required to support the global telecommunications system. This technology works so well that the typical user may not be aware that it even exists.

Cell phone towers like the one shown in Figure 1.1 are everywhere. From a distance, you can see the antennas at the top of the tower. As you get closer to the tower, you can see the copper cables running up the tower to the antennas. However, what you do not see are the optical fibers typically buried underground moving the data to and from the cell tower.

Broadband Speed and Access at the Turn of the Century and Today

As stated in the Four Years of Broadband Growth Report, at the turn of the century broadband speed was considered 200,000 bits per second or 200 kilobits per second (kbps), while dial-up Internet connections were typically 28.8kbps or 56kbps. Only 4.4 percent of the households in America had a broadband connection to their home. However, 41.5 percent had a dial-up Internet connection.

FIGURE 1.1 Cell phone tower

In 2013, the basic broadband speed was defined as 3,000,000 bits per second or 3 megabits per second (Mbps) *downstream* and 768kbps *upstream*. Downstream describes the number of bits that travel from the Internet service provider (ISP) to the person accessing broadband. This is often referred to as *download speed*. Upstream describes the number of bits being sent to the ISP. This is often referred to as *upload speed*.

While the basic broadband speed was defined with a 3Mbps download speed, more than 94 percent of the homes in America exceed 10Mbps. More than 75 percent have download speeds greater than 50Mbps, 47 percent have download speeds greater than 100Mbps, and more than 3 percent enjoy download speeds greater than 1 billion bits, or a gigabit, per second (Gbps).

Summary

This chapter discussed the history of fiber optics in communications, beginning with the first use of light to carry messages. It covered early experiments in the control of light for carrying sound, along with the problems faced by early experimenters.

This chapter also covered the discovery of fundamental principles that are essential to fiber optics and the ways in which they were adapted to control the path of light. It described the invention of processes used to improve the ability of glass fibers to carry light, as well as the refinements that made efficient, long-distance light transmission possible.

This chapter described the growth of fiber optics as an experimental data transmission medium, as a new technology for the telecommunications industry, and in its role in broadband since the turn of the century.

Exam Essentials

Understand the evolution of light in communication. Make sure that you understand the qualities of light that make it a desirable form of communication, the limitations of early communication with light, and some of the ways in which those limitations were overcome. Also, be sure you understand the principles behind early experiments with materials used to modulate and guide light.

Understand the evolution of optical fiber manufacturing technology. Make sure that you understand the problems encountered by researchers looking for more efficient ways to guide light over long distances and the breakthroughs that made modern optical fibers practical.

Understand the evolution of optical fiber integration and application. Make sure that you understand how the drop in signal losses helped optical fiber transition from experiments in guided light to a widespread communication medium. Pay attention to events that first used fiber optics in “real world” applications and proved that the technology could be used practically.

Be able to describe the role played by fiber optics in high-speed Internet access. Be able to explain that without fiber optics, broadband as we know it today would not exist. Know that only fiber optics can move the high rates of data over the long distances required to support the global telecommunications system.

Review Questions

1. What was one of the earliest advantages that light held over other forms of communication?
 - A. Ability to communicate complex ideas
 - B. Ability to carry messages quickly
 - C. Ability to carry messages privately
 - D. Ability to carry coded messages

Hint: Light travels at approximately 300,000 kilometers per second in air.

2. The demonstrations in which light could be seen in water draining from a vessel proved that:
 - A. Light could travel through water.
 - B. Light was refracted in some materials.
 - C. Light could be guided through the principle of total internal reflection.
 - D. Light could travel in a straight line.

Hint: Light can be guided through water because the air around it has a lower index of refraction.

3. Some of the earliest attempts to carry light through glass fibers took place because doctors and researchers were trying to:
 - A. View images from inside the human body.
 - B. Send messages from one building to another.
 - C. Compete with the telephone.
 - D. Collect numerous light sources in one area.

Hint: Doctors and researchers experimented with bent glass and quartz.

4. One of the first goals in creating a fiber that could be used for communications was to:
 - A. Make the fibers smaller.
 - B. Make the fibers longer.
 - C. Reduce signal loss to 40 decibels per kilometer.
 - D. Reduce signal loss to 20 decibels per kilometer.

Hint: In 1966, Charles K. Kao announced that signal losses in glass fibers could be reduced through improved manufacturing processes.

5. The decade that saw the first full-scale commercial application of fiber-optic communication systems was:
 - A. The 1880s
 - B. The 1920s

- C. The 1960s
- D. The 1970s

Hint: AT&T and GTE began using fiber-optic telephone systems for commercial customers.

6. Which broadband technology moves the highest rates of data over the greatest distances?
- A. Fiber optics
 - B. Digital Subscriber Line (DSL)
 - C. Broadband over Power Lines (BPL)
 - D. Wi-Fi

Hint: No other transmission medium can move such high rates of data over the long distances.

Chapter Exercises

Recognize the Refraction of Light

Refraction is the bending of light as it passes from one material into another.

You are cleaning your pool with a small net at the end of a pole when you notice a large bug that appears to be 2' below the surface. You place the net where you believe the bug to be and move it through the water. When you lift the net from the pool, the bug is not in the net. Why did the net miss the bug?

Remember refraction is easily observed by placing a stick into a glass of water. When viewed from above, the stick appears to bend because light travels more slowly through the water than through the air. The same is true for the bug. The bug was viewed from above the water and refraction bent the light so the bug appeared to be in a different location than it actually was.

Identify Total Internal Reflection

In 1840, Colladon and Babinet demonstrated that bright light could be guided through jets of water through the principle of total internal reflection.

You just removed your fish from a dirty 10-gallon aquarium you are preparing to clean when a friend shows up with a laser pointer. Your friend energizes the laser pointer and directs the light into the side of the tank. The laser light illuminates the small dirty particles in the tank and you and your friend observe the light entering one end of the tank and exiting the other. As you friend aims the laser pointer at an angle toward the surface of the water, the light does not exit; instead it bounces off the surface of the water at an angle. Why did this happen?

Remember that Colladon and Babinet demonstrated that bright light could be guided through jets of water through the principle of total internal reflection. In their

demonstration, light from an arc lamp was used to illuminate a container of water. You essentially re-created their experiment. Instead of an arc lamp and a small barrel of water, you had a laser pointer and a 10-gallon aquarium full of dirty water. The light reflected off the surface of the tank because of total internal reflection. You were able to see this happen, because the water was cloudy. Had you attempted this after the tank was cleaned the laser would be very difficult to view because there would be no particles to scatter the light. Scattering will be covered in detail in Chapter 5, “Optical Fiber Characteristics.”

Detect Crosstalk between Multiple Optical Fibers

Crosstalk or optical coupling is the result of light leaking out of one fiber into another fiber.

You have bundled six flexible clear plastic strands together in an effort to make a fiber-optic scope that will allow you to look into the defroster vent in your car in hope of locating your missing Bluetooth headset. You insert your fiber-optic scope into the defroster vent and are disappointed with the image you see. What is one possible cause for the poor performance of your fiber-optic scope?

Remember that in 1930, Lamm bundled commercially produced fibers and managed to transmit a rough image through a short stretch of the first fiber-optic cable. The process had several problems, including the fact that the fiber ends were not arranged exactly, and they were not properly cut and polished. Another issue was to prove more daunting. The image quality suffered from the fact that the quartz fibers were bundled against each other. This meant that the individual fibers were no longer surrounded by a medium with a lower index of refraction. Much of the light from the image was lost to crosstalk. Crosstalk or optical coupling is the result of light leaking out of one fiber into another fiber.

Recognize Attenuation in an Optical Fiber

In 1960 after the invention of the laser, optical fibers were all but ruled out as a transmission medium because of the loss of light or attenuation.

You are troubleshooting a clog in a drainpipe and because of the size of your flashlight and the location of the drain, you cannot illuminate the drain adequately to see the clog. You decide to modify a flashlight to illuminate the inside of drainpipe in hope of identifying the source of the clog. You purchase a small diameter flexible clear plastic rod approximately 12" in length and secure one end over the bright LEDs in the center of the flashlight. After powering up the flashlight, you are disappointed that the light exiting the optical fiber is much dimmer than you expected; however, you still attempt to identify the clog. Why did the fiber-optic flashlight fail to illuminate the inside of the drainpipe?

Remember that in 1960 optical fibers were all but ruled out as a transmission medium because of their attenuation. However, in 1966 Charles K. Kao announced that flaws in the manufacturing processes caused signal losses in glass fibers. The plastic rod purchased for the flashlight was not engineered to transmit light and experienced the high attenuation that limited the use of optical fibers prior to 1970 when Corning developed an optical fiber that achieved Kao's target of only 20 decibels per kilometer. Decibels are covered in Chapter 2, “Principles of Fiber-Optic Transmission.”

Chapter 2

Principles of Fiber-Optic Transmission

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe the basic parts of a fiber-optic link.
- ✓ Describe the basic operation of a fiber-optic transmitter.
- ✓ Describe the basic operation of a fiber-optic receiver.
- ✓ Explain how to express gain or loss using the decibel (dB).
- ✓ Explain how to express optical power in dBm.





Like Bell's photophone, the purpose of fiber optics is to convert a signal to light, move the light over a distance, and then reconstruct the original signal from the light. The equipment

used to do this job has to overcome all of the same problems that Bell encountered, while carrying more data over a much greater distance.

In this chapter, you will learn about the basic components that transmit, receive, and carry the optical signal. You will also learn to

- Calculate the decibel value of a gain or loss in power
- Calculate the gain or loss in power from a known decibel value
- Calculate the gain or loss in power using the dB rules of thumb
- Convert dBm to a power measurement
- Convert a power measurement to dBm

The Fiber-Optic Link

A *link* is a transmission pathway between two points using some kind of generic cable. The pathway includes a means to couple the signal to the cable and a way to receive it at the other end in a useful way.

Any time we send a signal from one point to another over a wire, we are using a link. A simple intercom, for example, consists of the sending station (which converts voice into electrical signals), the wire over which the signals are transmitted, and the receiving station (which converts the electrical signal back into voice).

Links are often described in terms of their ability to send and receive signals as part of a communication system. When described in these terms, they are broken down into *simplex* and *duplex*. Simplex means that the link can only send at one end and receive at the other end. In other words, the signal goes only one way. An example is the signal from a radio station. Duplex means that the link has a transmitter and a receiver at each end. A *half-duplex* system allows signals to go only one way at a time—an example is an intercom system. A *full-duplex* system allows users to send and receive at the same time. A telephone is a common example of a full-duplex system.

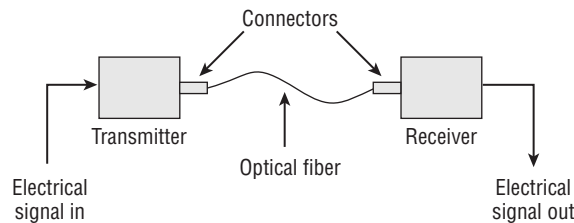


The definition of the term *link* is not universal; it varies from industry standard to standard. Several definitions of link are listed here.

- ARINC Report 805: A fiber optic cable with connectors attached to a transmitter (source) and receiver (detector).
- ANSI/TIA-568-C: A transmission path between two points, not including equipment and cords.
- IEEE Std 802.3-2012: The transmission path between any two interfaces of generic cabling.

A fiber-optic link, shown in Figure 2.1, is like any other link, except that it uses optical fiber instead of wire. A fiber-optic link consists of four basic components:

FIGURE 2.1 The fiber-optic link



- The transmitter that converts a signal into light and sends the light
- The receiver that captures the light and converts it back to a signal
- The optical fiber that carries the light
- The connectors that couple the optical fiber to the transmitter and receiver

Now let's look at each component in a little more detail.

Transmitter

The transmitter, shown in Figure 2.2, converts an electrical signal into light energy to be carried through the optical fiber. The signal can be generated by many sources, such as a computer, a voice over a telephone, or data from an industrial sensor.

FIGURE 2.2 The fiber-optic transmitter



Receiver

The receiver is an electronic device that collects light energy from the optical fiber and converts it into electrical energy, which can then be converted into its original form, as shown

in Figure 2.3. The receiver typically consists of a *photodiode* to convert the received light into electricity and circuitry to amplify and process the signal.

FIGURE 2.3 The fiber-optic receiver



Optical Fibers

Optical fibers carry light energy from the transmitter to the receiver. An optical fiber may be made of glass or plastic, depending on the requirements of the job that it will perform. The advantage of light transmission through optical fiber as compared to the transmission of light through air is that the fiber can carry light around corners and over great distances.

Many fibers used in a fiber-optic link have a *core* between 8 and 62.5 microns (millionths of a meter) in diameter. For comparison, a typical human hair is about 100 microns in diameter. The *cladding* that surrounds the core is typically 125 microns in diameter.

The optical fiber's coating protects the cladding from abrasion. The thickness of the coating is typically half the diameter of the cladding, which increases the overall size of the optical fiber to 250 microns. Even with the additional thickness of the coating, optical fiber cabling is much smaller and lighter than copper cabling, as shown in Figure 2.4, and can carry many times the information.

FIGURE 2.4 Comparison of copper cable (top) and fiber cable (bottom)



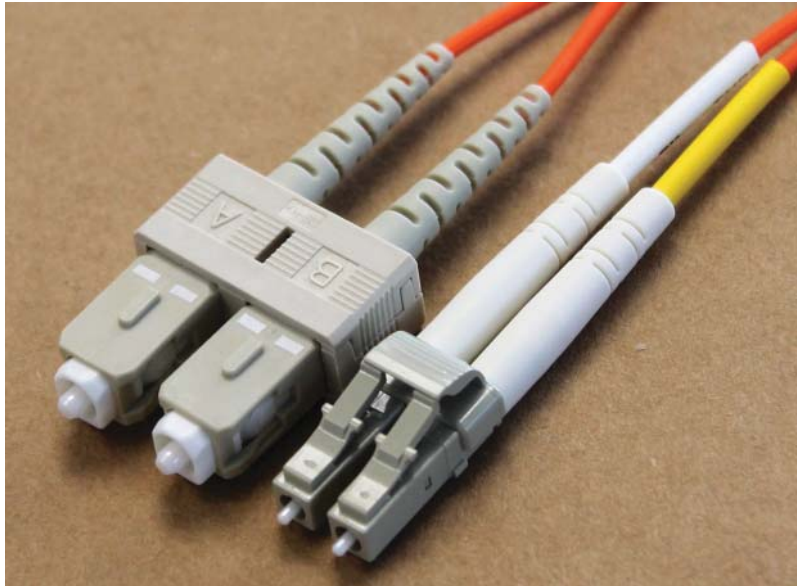
Connectors

The connector is attached to the optical fiber and the fiber-optic cable. The connector allows the optical fiber to be mated to the transmitter or receiver. Transmitters or

receivers typically have a *receptacle* that securely holds the connector and provides solid contact between the optical fiber and the optical subassembly of the device. The connector must align the fiber end precisely with the light source or photodiode to minimize signal loss.

The connectors, shown in Figure 2.5, could be considered the elements that make it possible for us to use fiber optics because they allow large hands to handle the small, fragile fibers. They are also the only place in the link where the optical fiber is exposed.

FIGURE 2.5 Fiber-optic connectors



Decibels (dB)

As light travels away from its source through the link, it loses energy. Energy loss can be caused by several factors, such as the absorption or scattering of light by impurities in the fiber, or by light passing through the core and cladding and being absorbed in the coating.

It is important to be able to measure the amount of light energy lost in a fiber-optic link. Knowing the loss allows us to predict the strength of the light energy at the receiving equipment. The receiving equipment needs a minimum amount of light energy to accurately convert the light energy to the original signal. In addition, an understanding of how and where light is lost in a link can be helpful when troubleshooting the link.

One of the more common terms used when discussing the quality of a signal in fiber optics is the *decibel (dB)*. The decibel was originally used to measure the strength of sounds as perceived by the human ear. Its name means “one-tenth of a bel.”



A *bel* is a sound measurement named for telephone inventor Alexander Graham Bell.

Calculating dB Power Loss and Power Gain

The decibel can be used to express power gain or power loss relative to a known value. In fiber optics, the decibel is most commonly used to describe optical power, which is also known as signal power. Optical power is typically measured with an *optical power meter* (OPM). Measuring optical power with an OPM is covered in detail in Chapter 14, “Test Equipment and Link/Cable Testing.”

Recall that in the 1960s, a signal loss of 20 decibels per kilometer was considered the goal for making fiber optics practical for communications. A 20-decibel loss means that of the original power put into the signal, only 1 percent remains.

Modern optical fiber has very low signal power loss and many fiber-optic links do not require the signal to be amplified. The decibel is used to express the signal power loss in all fiber-optic links. In links that extend many kilometers, it is used to express the signal power gain provided by the amplifier.

To calculate the decibel value of a gain or loss in optical power, use the following equation:

$$\text{dB} = 10\text{Log}_{10} (P_{\text{out}} \div P_{\text{in}})$$

Let's apply this equation to the loss associated with a length of optical fiber. The transmitter couples 10 microwatts (μW) into the optical fiber and the power at the receiver is $3\mu\text{W}$. Since both values are in microwatts, we do not have to use microwatts in the equation. The equation would be written as follows:

$$\text{dB} = 10\text{Log}_{10} (3 \div 10)$$

$$\text{dB} = 10\text{Log}_{10} 0.3$$

$$\text{dB} = -5.23$$

$$\text{Loss} = 5.23\text{dB}$$

Because the calculated value is negative, it is a loss. When a loss is stated, the negative sign is dropped. The loss is 5.23dB.

Most fiber-optic applications do not require amplification. However, amplification may be required for long transmission distances. If the input power and output power of the amplifier are known, the gain in dB can be calculated. We can solve for the gain of an

amplifier where the input power is $1\mu\text{W}$ and the output power is $23\mu\text{W}$ by solving the equation as shown here:

$$\text{dB} = 10\text{Log}_{10}(23 \div 1)$$

$$\text{dB} = 10\text{Log}_{10} 23$$

$$\text{dB} = 13.6$$

$$\text{Gain} = 13.6\text{dB}$$

If you know the decibel value and want to calculate the gain or loss, you will have to rearrange the equation as shown here:

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(\text{dB} \div 10)$$

Let's apply this equation to the loss associated with a length of optical fiber. To solve for the gain or loss we do not need to know the input power or output power; we just need the gain or loss in dB. Remember the loss in dB is a ratio of power out divided by power in. This length of optical fiber has a loss of 3.5dB. The equation would be written as follows:

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(\text{dB} \div 10)$$

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(-3.5 \div 10)$$

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog} -0.35$$

$$(P_{\text{out}} \div P_{\text{in}}) = 0.447$$

If one power is known, the other power can be calculated. In this example, the input power is 13 milliwatt. To calculate the output power the equation would be written as follows:

$$(P_{\text{out}} \div 13\text{mW}) = 0.447$$

$$P_{\text{out}} = 0.447 \times 13\text{mW}$$

$$P_{\text{out}} = 5.8\text{mW}$$

Signals may be decreased, or attenuated, just about anywhere in the link. In addition to attenuation in the optical fiber itself, connectors, splices, and bends in the fiber-optic cable can also cause loss in the signal—sometimes considerable loss.

Expressing dB in Percentages

When measuring signal power gain or loss, decibels are calculated relative to the original power, rather than as an absolute number. For example, a loss of 0.1 decibel means that the signal has 97.7 percent of its power remaining, and a loss of 3 decibels means that only 50 percent of the original power remains. This relationship is logarithmic, rather than linear, meaning that with each 10 decibels of loss, the power is 10 percent of what it was (as shown in Table 2.1).

With each 10 decibels of gain, the power is 10 times what it was (as shown in Table 2.2).

TABLE 2.1 Decibel losses expressed in percentages

Loss in dB	% power lost	% power remaining
0.1	2.3%	97.7%
0.2	4.5%	95.5%
0.3	6.7%	93.3%
0.4	8.8%	91.2%
0.5	10.9%	89.1%
0.6	12.9%	87.1%
0.75	15.9%	84.1%
0.8	16.8%	83.2%
0.9	18.7%	81.3%
1	21%	79%
3	50%	50%
6	75%	25%
7	80%	20%
9	87%	13%
10	90%	10%
13	95.0%	5%
16	97%	3%
17	98.0%	2.0%
19	98.7%	1.3%
20	99.00%	1.00%
23	99.50%	0.50%

Loss in dB	% power lost	% power remaining
30	99.9%	0.1%
33	99.95%	0.05%
40	99.99%	0.01%
50	99.999%	0.001%
60	99.9999%	0.0001%
70	99.99999%	0.00001%

TABLE 2.2 Decibel gains expressed in percentages

Gain in dB	% power increase	% total power
0.1	2.3%	102.3%
0.2	4.7%	104.7%
0.3	7.2%	107.2%
0.4	9.6%	109.6%
0.5	12.2%	112.2%
0.6	14.8%	114.8%
0.75	18.9%	118.9%
0.8	20.2%	120.2%
0.9	23.0%	123.0%
1	26%	126%
3	100%	200%
6	298%	398%
7	401%	501%

TABLE 2.2 Decibel gains expressed in percentages *(continued)*

Gain in dB	% power increase	% total power
9	694%	794%
10	900%	1,000%
13	1,895%	1,995%
16	3,881%	3,981%
17	4,912%	5,012%
19	7,843%	7,943%
20	9,900%	10,000%
23	19,853%	19,953%
30	99,900%	100,000%
33	199,426%	199,526%
40	999,900%	1,000,000%
50	9,999,900%	10,000,000%
60	99,999,900%	100,000,000%
70	999,999,900%	1,000,000,000%

One of the advantages of using decibels when calculating gain and loss is their relative ease of use compared to percentages. When measuring loss through different components, you can algebraically add the decibel loss from each component and arrive at a total signal loss for the system. If you were to use percentages, you would have to calculate the remaining power after the signal passed through each component, then take another percentage off for the next component, and so on, until the loss through the entire system had been calculated.

The Rules of Thumb

When calculating gain or loss in a system, it is useful to remember the three rules of thumb shown in Table 2.3, which make it easier to perform some common decibel calculations. These rules help you calculate how much power you have after the indicated decibel gain or loss.

TABLE 2.3 Rules of thumb

Decibel	Loss	Gain
3dB	1/2	× 2
7dB	4/5	× 5
10dB	9/10	× 10

For example, a loss of 3 decibels means that you have about ½ of the original power. A gain of 3 decibels means that you have about twice the original power. These rules are intended for rough calculations only, because a 3 decibel loss actually leaves 50.1187 percent of the original power, and a 7 decibel loss leaves 19.953 percent of the original power.

Because decibels can be algebraically added and subtracted, you can use combinations of the decibel values to determine total gains or losses. For example, the rules of thumb can be applied to find the power output from an amplifier with an input of 10μW and a gain of 17dB. First, apply the 10dB rule and multiply 10μW by 10. Then apply the 7dB rule and multiply the result of the first calculation by 5. That value is the power remaining as shown in the following equations:

$$P_{\text{out}} = 10\mu\text{W} \times 10$$

$$P_{\text{out}} = 100\mu\text{W}$$

$$P_{\text{out}} = 100\mu\text{W} \times 5$$

$$P_{\text{out}} = 500\mu\text{W}$$

The optical power at the output to the amplifier is 500μW.

Keep in mind that the dB rules of thumb cannot calculate the answer for every scenario. If the gain for the amplifier had been 18dB, we would not have been able to accurately calculate the optical power output of the amplifier. However, we would have been able to estimate that value by calculating for the closest value possible 17dB. The estimated output would be greater than 500μW but less than 1mW. The 1mW value is the result of solving the above equation a gain of 20dB, the closest value above 18dB that can be calculated using the dB rules of thumb.

Absolute Power

Taken by itself, the decibel is only a relative number, and it has meaning only when it is referenced to a known input or output power. In many cases, however, it is important to have an absolute value to use for comparison or for equipment specifications. When such a value is required in fiber optics, the decibel is referenced to 1mW. When the decibel is referenced to

1mW, a lowercase “m” or sometimes “mW” is inserted after dB. The decibel (dB) referenced to 1mW is written as 0dBm.

When using dBm, power levels below 1mW are negative and power levels 1mW or greater are positive. The positive sign is typically implied and only the negative sign is used. As an example, a value of -10dBm means that 1mW of power has been attenuated by 10dB, so only 10 percent of its power, or 100 μ W, remains. A value of 10dBm means that 1mW of power has been increased by 10dB, which has a value of 10mW. Table 2.4 shows how dBm values convert to power measurements.

TABLE 2.4 Converting power to dBm

Optical power in watts	Optical power in dBm
100mW	20dBm
20mW	13dBm
10mW	10dBm
8mW	9dBm
5mW	7dBm
4mW	6dBm
2mW	3dBm
1mW	0dBm
500 μ W	-3dBm
250 μ W	-6dBm
200 μ W	-7dBm
125 μ W	-9dBm
100 μ W	-10dBm
50 μ W	-13dBm
25 μ W	-16dBm
20 μ W	-17dBm

Optical power in watts	Optical power in dBm
12.5μW	-19dBm
10μW	-20dBm
5μW	-23dBm
1μW	-30dBm
500nW	-33dBm

The formula used to convert dBm to a power measurement is:

$$\text{dBm} = 10\text{Log}_{10}(P \div 1\text{mW})$$

This equation is similar to the equation for finding a decibel value from an input power and an output power, except that in this case the input power is fixed at 1mW. Let's apply this equation to calculate optical power from a dBm value. In this example the power meter measures -11dBm. To calculate the power in watts, the equation would be written as follows:

$$-11 = 10\text{Log}_{10}(P \div 1\text{mW})$$

$$-1.1 = \text{Log}_{10}(P \div 1\text{mW})$$

$$0.0794 = P \div 1\text{mW}$$

$$P = 0.0794 \times 1\text{mW}$$

$$P = 0.0794\text{mW}$$

We can also convert a power measurement into dBm. Remember that when using dBm, power levels below 1mW are negative and power levels 1mW or greater are positive. The formula used to convert a power measurement to dBm:

$$\text{dBm} = 10\text{Log}_{10}(P \div 1\text{mW})$$

Let's apply this equation to calculate dBm from an optical power value of 10μW. To calculate dBm from the power in watts, the equation would be written as follows:

$$\text{dBm} = 10\text{Log}_{10}(10\mu\text{W} \div 1\text{mW})$$

$$\text{dBm} = 10\text{Log}_{10}(0.01)$$

$$\text{dBm} = 10 \times -2$$

$$\text{dBm} = -20$$

The power value is -20dBm.

This answer can be checked by locating 10μW in the first column of Table 2.4.



Real World Scenario

Getting Lost in Losses

You may find yourself interchanging dB and dBm; this happens to everyone from time to time. Remember that loss and gain are expressed in dB and absolute power is expressed in dBm. If you were to say that a link had a loss of 7dBm, you would actually be saying that the link has lost 5mW. However, you were probably trying to say the link lost 80 percent of its energy.

Summary

This chapter discussed the basic components that transmit, receive, and carry the optical signal. It introduced the decibel and described how it is used to express power gain or power loss relative to a known value. In addition, it explained absolute power when the decibel is referenced to 1mW.

Exam Essentials

Know the components of the fiber-optic link. Make sure you understand the four components that make up any fiber optic link. Also, you should know the function of each component within the link.

Understand the importance of decibels (dB). Know why it is important to measure energy loss through a fiber optic link. Make sure you understand how decibels relate to the percentage of power gained or lost from the original signal.

Know the difference between relative power and absolute power. Know that taken by itself, the decibel is only a relative number, and it has meaning only when it is referenced to a known value. Understand that when the decibel is referenced to 1mW it is written as 0dBm and can be used to describe an absolute power value.

Review Questions

1. Which of the following groups describes the components in a fiber-optic link?

- A. Transmitter, receiver, optical fiber, decoder
- B. Transmitter, receiver, optical fiber, connectors
- C. Transmitter, receiver, optical fiber
- D. Transmitter, receiver, optical fiber, cabling

Hint: Send signal, carry signal, capture signal

2. In a fiber-optic link, which component has an input of electrical energy and an output of light?

- A. Connector
- B. Optical fiber
- C. Receiver
- D. Transmitter

Hint: It sends the signal.

3. If an optical signal loses 10dB, what percentage of the original signal is remaining?

- A. 90%
- B. 80%
- C. 50%
- D. 10%

Hint: It could be calculated using the rules of thumb.

4. In a fiber-optic link, which component has an output of electrical energy and an input of light?

- A. Connector
- B. Optical fiber
- C. Receiver
- D. Transmitter

Hint: It captures the light.

5. If the output power of a transmitter is $100\mu\text{W}$ and the power measured at the input to the receiver is $12.5\mu\text{W}$, the loss is _____.

- A. 9.03dB
- B. 8dB

- C. 0.125dB
- D. 12.5dB

Hint: Transmitter output ÷ receiver input

6. The loss for a length of optical fiber is 4dB, the input power is 250μW, and the output power is _____.
- A. 1mW
 - B. 62.5uW (use the micro μ symbol)
 - C. 125uW (use the micro μ symbol)
 - D. 99.5μW

Hint: Solve for the loss in power first.

7. Which dB rules of thumb should be used to calculate the output power of an attenuator with a loss of 23dB?
- A. 3dB and 20dB
 - B. 3dB and 10dB
 - C. 7dB and 10dB
 - D. 3dB and 5dB

Hint: They must add up to the loss.

8. An output power of -8dBm is equal to _____.
- A. 15.8μW
 - B. 15.8mW
 - C. 158.5μW
 - D. 158.5mW

Hint: The decibel referenced to 1mW is written as 0dBm.

9. If a signal has lost 30 decibels, how much of the original signal is left?
- A. 30 %
 - B. 0.3 %
 - C. 0.01 %
 - D. 0.1 %

Hint: A 20dB loss divides the power by a factor of 100.

10. An output power of 4mW is equal to _____.
- A. 3dB
 - B. 3dBm
 - C. 6dB
 - D. 6dBm

Hint: Output power ÷ 0dBm

Chapter Exercises

Calculate the Decibel Value of a Gain or Loss in Power

The decibel is used to express gain or loss relative to a known value. In fiber optics, the decibel is most commonly used to describe signal loss through the link after the light has left the transmitter.

The output power of a transmitter is $100\mu\text{W}$ and the power measured at the input to the receiver is $12.5\mu\text{W}$. Calculate the loss in dB.

To calculate the decibel value of a gain or loss in signal power, use the following equation:

$$\text{dB} = 10\text{Log}_{10} (P_{\text{out}} \div P_{\text{in}})$$

$$\text{dB} = 10\text{Log}_{10} (12.5 \div 100)$$

$$\text{dB} = 10\text{Log}_{10} 0.125$$

$$\text{dB} = -9.03$$

$$\text{Loss} = 9.03\text{dB}$$

Calculate the Gain or Loss in Power from a Known Decibel Value

If the gain or loss in power is described in dB, the gain or loss can be calculated from the dB value. If the input power is known, the output power can be calculated, and vice versa.

The loss for a length of optical fiber is 4dB. Calculate the loss in power from the optical fiber and the output power of the optical fiber for an input power of $250\mu\text{W}$.

First solve for the loss, then for the output power. The equation would be written as follows:

$$(P_{\text{out}} \div P_{\text{in}}) = \text{antilog} (\text{dB} \div 10)$$

$$(P_{\text{out}} \div 250\mu\text{W}) = \text{antilog} (-4 \div 10)$$

$$(P_{\text{out}} \div 250\mu\text{W}) = \text{antilog} -0.4$$

$$(P_{\text{out}} \div 250\mu\text{W}) = 0.398$$

$$P_{\text{out}} = 0.398 \times 250\mu\text{W}$$

$$P_{\text{out}} = 99.5\mu\text{W}$$

Calculate the Gain or Loss in Power Using the dB Rules of Thumb

When calculating gain or loss in a system, it is useful to remember the three rules of thumb shown in Table 2.3.

A fiber-optic link has 7dB of attenuation. The output power of the transmitter is 100μW. Using the dB rules of thumb, calculate the power at the input to the receiver.

To calculate the optical power at the input to the receiver, apply the 7dB rule by dividing 100μW by 5 as shown in the equation below:

$$P_{in} = 100\mu W \div 5$$

$$P_{in} = 20\mu W$$

The optical power at the input to the receiver is 20μw.

Convert dBm to a Power Measurement

In fiber optics, dBm is referenced to 1mW of power.

A fiber-optic transmitter has an output power of -8dBm. Calculate the output power in watts.

To calculate the power in watts, the equation would be written as:

$$-8 = 10\text{Log}_{10} (P \div 1\text{mW})$$

$$-0.8 = \text{Log}_{10} (P \div 1\text{mW})$$

$$0.158 = P \div 1\text{mW}$$

$$P = 0.158 \times 1\text{mW}$$

$$P = 158.5\mu W$$

Convert a Power Measurement to dBm

In fiber optics, dBm is referenced to 1mW of power.

A fiber-optic transmitter has an output power of 4mW. Calculate the output power in dBm.

To calculate the power in dBm, the equation would be written as:

$$\text{dBm} = 10\text{Log}_{10} (4\text{mW} \div 1\text{mW})$$

$$\text{dBm} = 10\text{Log}_{10} 4$$

$$\text{dBm} = 10 \times 0.6$$

$$\text{dBm} = 6$$

The power value is 6dBm.

This answer can be checked by locating 4mW in the first column of Table 2.4.

Chapter 3

Basic Principles of Light

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe light as electromagnetic energy.
- ✓ Describe light as particles and waves.
- ✓ Describe the electromagnetic spectrum and locate light frequencies within the spectrum in relation to radio and microwave communication frequencies.
- ✓ Describe the refraction of light.
- ✓ Explain how the index of refraction is used to express the speed of light through a transparent medium.
- ✓ Explain reflection to include angle of incidence, critical angle, angle of refraction, and total internal reflection.
- ✓ Explain Snell's law and how it is used to calculate the critical angle of incidence.
- ✓ Explain Fresnel reflections and how they can impact the performance of a fiber-optic communication system.





Even the simplest fiber-optic link is a triumph of innovative design, manufacturing precision, and technical creativity. One of the factors contributing to the high data rates and long transmission distances of fiber optics is knowledge of the principles of light. While the optical fiber is the transmission medium, light is the carrier on which the data is imposed. Its small wavelengths and high transmission velocity make possible bandwidths that would be unimaginable using other forms of transmission. To get the best performance from every part of the fiber-optic link, you must understand the characteristics of light and the factors within the fiber-optic link that affect it.

This chapter describes the basic characteristics of light, especially the type of light used in fiber-optic communications. It also discusses the principles and materials that make fiber-optic communications possible, along with some of the problems that must be overcome.

In this chapter, you will learn to

- Convert various wavelengths to corresponding frequencies
- Convert various frequencies to corresponding wavelengths
- Calculate the amount of energy in a photon using Planck's constant
- Calculate the speed of light through a transparent medium using its refractive index
- Use Snell's law to calculate the critical angle of incidence
- Calculate the loss in decibels from a Fresnel reflection

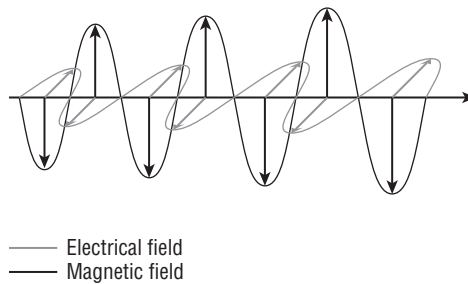
Light as Electromagnetic Energy

Whether it comes from the sun, an electric bulb, or a laser, all light is a form of electromagnetic energy. Electromagnetic energy is emitted by any object that has a temperature above absolute zero (-273.15°C), which means that the atoms in the object are in motion. The electrons orbiting the atoms pick up energy from the motion, and the energy causes them to move to higher orbits, or energy levels. As they drop back to their original energy levels, they release the energy again. The energy takes two forms: an electrical field and a magnetic field, formed at right angles to each other and at right angles to their path of travel, as shown in Figure 3.1.

The combination of these electrical and magnetic fields is an electromagnetic wave, which travels through a vacuum or air at approximately 300,000km/s (kilometers per second).

It is important to understand the three-dimensional nature of electromagnetic waves because different types of fiber-optic transmission take advantage of these characteristics. How this is done will be discussed in later chapters.

FIGURE 3.1 The three-dimensional nature of electromagnetic energy



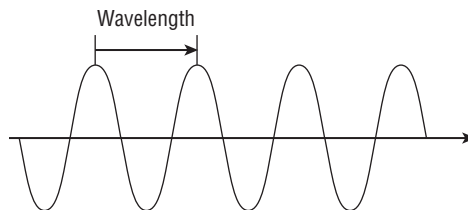
When describing the speed of electromagnetic energy through a medium, the term “free space” is sometimes used interchangeably with “vacuum.”

Wavelength and Frequency

In fiber optics, wavelength (λ) and frequency (f) are used to describe electromagnetic energy. Wavelength is typically used to describe the output of the light source and frequency is used to describe the channel spacing in dense wavelength division multiplexing (DWDM) systems. The wavelengths of commonly used light sources are described in detail in Chapter 10, “Fiber-Optic Light Sources and Transmitters.”

Although electromagnetic energy exists three-dimensionally, it is often represented as a two-dimensional sine wave, where the wavelength is the distance between corresponding points on two consecutive waves as shown in Figure 3.2. Depending on its wavelength, electromagnetic energy can occur as radio waves, microwaves, light waves, X-rays, and more.

FIGURE 3.2 Electromagnetic energy is often shown as a sine wave.



Wavelength is important because it allows us to calculate the electromagnetic energy’s frequency, which is the number of waves that pass a given point in one second. Frequency is measured in cycles per second, or hertz (Hz).

A way to express the relationship between wavelength and frequency is that wavelength equals the velocity of the wave (v) divided by its frequency, or

$$\lambda = v \div f$$

Note that v is usually the speed of light in a vacuum or in air, approximately 300,000km/s.

Likewise, you can calculate the frequency using the wavelength with the equation

$$f = v \div \lambda$$

For example, one of the infrared light sources used in fiber optics has a wavelength of 1310 nanometers (billionths of a meter), which translates to a frequency of 229 terahertz (THz), or 229 trillion Hz.

Some Useful Terms

Many of the terms used to express electromagnetic wavelengths and frequencies describe large multiples of cycles or small fractions of meters. To understand these terms, it is helpful to know the prefixes used with them.

Each prefix expresses a multiple of 10 or 1 divided by a multiple of 10 (as in 1/1000) applied to the measurement unit, such as meters or cycles. Some terms may already be familiar, such as *kilometer*, for a thousand meters, or *centimeter*, for a hundredth of a meter.

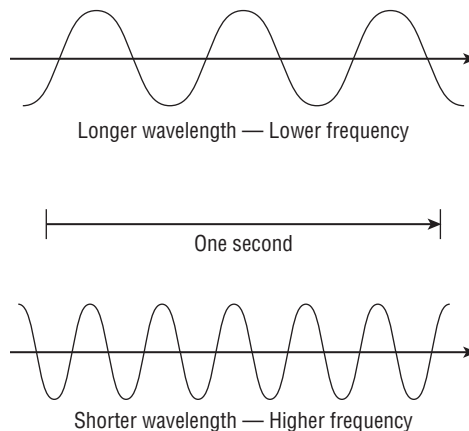
Here is a list of prefixes, along with their translations and mathematical equivalents, in descending order of magnitude.

Prefix	Meaning	Magnitude	Symbol
peta-	Quadrillion	10^{15}	P
tera-	Trillion	10^{12}	T
giga-	Billion	10^9	G
mega-	Million	10^6	M
kilo-	Thousand	10^3	k
centi-	Hundredth	10^{-2}	c

milli-	Thousandth	10^{-3}	m
micro-	Millionth	10^{-6}	μ
nano-	Billionth	10^{-9}	n
pico-	Trillionth	10^{-12}	p
femto-	Quadrillionth	10^{-15}	f

Note in Figure 3.3 that as the wavelength becomes smaller, more waves will occur in one second, which means that the frequency will increase as wavelength decreases. However, the reverse happens if the wavelength were to become longer. The frequency will decrease as the wavelength increases.

FIGURE 3.3 As wavelength decreases, frequency increases.



To illustrate this, the signal from an AM radio station transmitting at 890 kilohertz (kHz), or 890,000Hz, has a wavelength of 337.1 meters. If we choose a higher frequency, such as 960kHz, the wavelength decreases to 312.5 meters and if we choose a lower frequency such as 560kHz the wavelength increases to 535.7 meters.

Characteristics of Electromagnetic Radiation

Electromagnetic radiation, like most radiated energy, has characteristics of both waves and particles. As a wave, it propagates through a medium and transfers energy without

permanently displacing the medium. However, as a massless particle, called a *photon*, it travels in a wavelike pattern moving at the speed of light.

A photon, which is emitted from an electron as it changes energy levels, is the basic unit, or *quantum*, of electromagnetic energy. The amount of energy in each photon, however, depends on the electromagnetic energy's frequency: the higher the frequency, the more energy in the particle.

To express the amount of energy in a photon, we use the equation

$$E = hf$$

where E is the energy expressed in joules, h is Planck's constant, or 6.626×10^{-34} joule-seconds, and f is the frequency of the electromagnetic energy.

So, to find the energy of a photon of infrared light at 229 THz

$$(2.29 \times 10^{14})(6.626 \times 10^{-34}) = 1.517 \times 10^{-19} \text{ W}$$



Much of what we take for granted in the field of fiber optics comes from work done by pioneers in the field of physics in the late 1800s and early 1900s. The *joule* is a unit of energy named for James Prescott Joule, who studied the relationship between heat and mechanical work. One joule is equal to the amount of work required to produce 1 watt in 1 second. Planck's constant was defined in 1899 by Nobel Laureate Max Planck, who is regarded as the founder of quantum theory.

The Electromagnetic Spectrum

In 1964, scientists at a Bell Laboratories facility in New Jersey accidentally discovered electromagnetic radiation associated with the very beginnings of our universe. The radiation, predicted by certain cosmological theories, was emitted by hydrogen atoms at a temperature of just about 3° on the *Kelvin* (K) scale, or -270.15° C, and is practical evidence that anything with a temperature above absolute zero puts out electromagnetic energy.

The wavelengths emitted by the cosmic hydrogen atoms, about 0.5mm to 5mm, are in the *microwave* range of the electromagnetic spectrum, just above radio waves in frequency, and just below infrared light.

Some characteristics of electromagnetic energy in free space are dependent on wavelength. Free space typically implies the medium is a vacuum; however, in free-space optical communication, it could also be referring to air or some similar medium. As we have learned, optical communication can also take place in a controlled space such as an optical fiber. In a free space transmission, you may not be able to control the medium as you can in a controlled space transmission. For example, the maximum transmitting distance for the optical signal on a foggy day will be less than a clear day. However, the transmission through an optical fiber is not affected by weather.

Longer wavelengths require less energy for propagation than shorter wavelengths of the same amplitude, making them useful for long-distance communication. By the same rule, particles of higher frequency electromagnetic radiation have more energy than particles from lower frequency emissions of the same amplitude. This fact is important when determining which wavelengths to use in fiber-optic transmissions.

Using Prefixes

Very wide ranges of numbers are used in fiber optics. You may be describing the length of a cable in kilometers and the wavelength of a light source in nanometers. The following is a list of the most common prefixes used in fiber optics and an example of how they may be used to describe something.

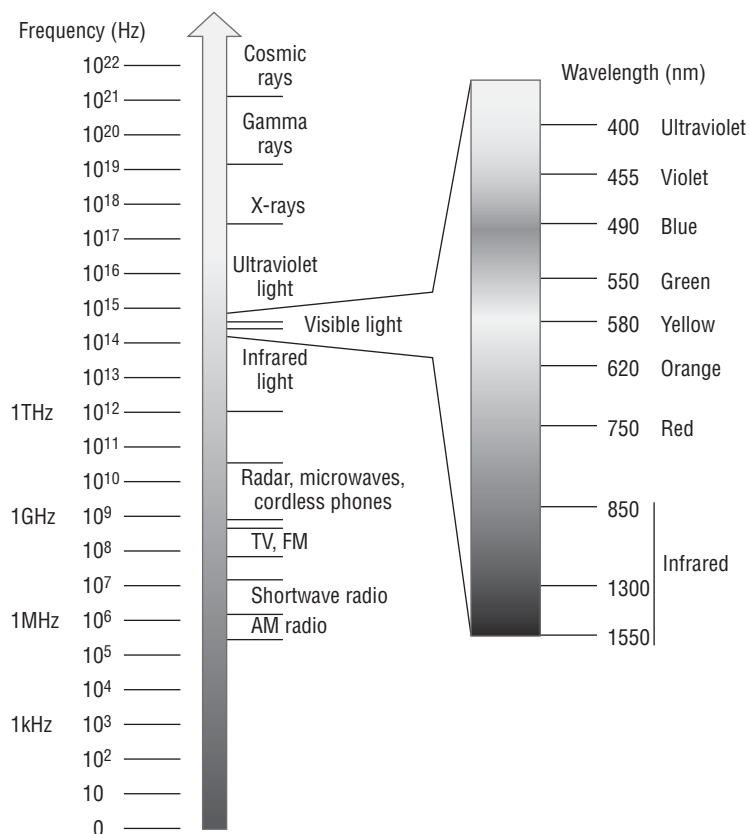
Prefix	Used to describe
tera-	The frequency of light in terahertz (THz)
giga-	The bandwidth of a link in gigabits per second (Gbps)
mega-	The bandwidth of an optical fiber in megahertz (MHz)
kilo-	The length of a fiber-optic cable in kilometers (km)
centi-	The bend radius of a fiber-optic cable in centimeters (cm)
milli-	The diameter of a fiber-optic cable in millimeters (mm)
micro-	The diameter of the optical fiber in micrometers (μm)
nano-	The wavelength of light in nanometers (nm)
pico-	The dispersion of light in picoseconds (ps)

These transmissions consist of turning the carrier, or the light, on and off at high switching rates. At the higher rates desired for communications, higher frequencies of light are needed if an extremely short “on” cycle is to have enough energy to be detected.

Remember also that higher frequencies can carry more data, because more waves per second allow more bits per second to be carried. This fact extends the principle that had been applied first to radio and then to television, which relied on ever higher frequencies to carry greater amounts of information.

What we commonly call light is just one small part of the electromagnetic spectrum, shown in Figure 3.4. Visible light is an even smaller component, bordered by infrared, with longer wavelengths, and ultraviolet, with shorter wavelengths.

FIGURE 3.4 The electromagnetic spectrum



The wavelengths most commonly used for fiber optics are in the infrared range, at *windows* of 850 nanometers (nm), 1300nm, and 1550nm. The spectrum range of these wavelengths provides an important combination of characteristics: it is high enough to make high data rates possible, but low enough to require relatively low power for transmission over long distances.



Transmission windows were created as a way of standardizing useful wavelengths in fiber optics, making it easier to build interoperable equipment.

The specific wavelength windows have been selected because they provide the best possible characteristics for transmission. Even within the range between 850nm and 1550nm, certain regions have high losses due to materials in the fiber, such as stray water molecules, absorbing light at a wavelength of 1380nm. Other wavelengths, such as 1550nm, are favored because they have a low loss, allowing longer transmission distances. On the other hand, wavelengths near 1300nm suffer less from *dispersion*, which will be discussed at length in Chapter 5, “Optical Fiber Characteristics.”

Refraction

Refraction, or the change in the direction of light as it changes speeds passing from one material into another, is a key component in fiber-optic transmissions. The principles that cause an object in water to look like it is bent, as seen in Figure 3.5, are the same principles that keep light contained within the core of an optical fiber even through curves.

FIGURE 3.5 Refraction of light through water

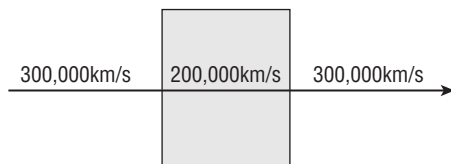


What Causes Refraction?

Refraction occurs when light waves change speed as they travel between two materials, each with a different refractive index or *index of refraction* (n). We commonly think of the speed of light as constant at about 300,000km/s (299,792.458km/s, to be precise), but that figure is actually the theoretical top speed of light, which only applies to its speed in a vacuum. In reality, light travels at lower velocities in various materials or media such as the earth's atmosphere, glass, plastic, and water. A *medium* is any material or space through which electromagnetic radiation can travel.

How quickly light travels through a medium is determined by its refractive index. Light travels more slowly in a material with a high refractive index and faster in a material with a low refractive index. It is important to note that even though light slows down when passing from a medium with a lower refractive index into a medium with a higher refractive index, it speeds up again when it passes into a medium with a lower refractive index (see Figure 3.6).

FIGURE 3.6 Light changes speeds as it passes through mediums with different refractive indexes.



When light changes velocity as it travels from one medium into another, refraction occurs. The amount of refraction is determined by the relative difference in the light's velocity in each of the media. The greater the difference, the greater the refraction.

In Figure 3.5, shown earlier, the portion of the oar in the water looks bent because the light rays reflected from it are bent as they pass through the boundary between the water and the air, as seen in Figure 3.7. Notice also that the portion of the oar that is underwater is distorted because the light rays reflected from it have been bent.

To explain refraction, we have to look at the wave nature of light. Recall that light consists of two perpendicular waves, and that the light travels in a path that forms right angles with both waves.

As the light wave meets the boundary between media with different refractive indexes, that portion of the light wave changes velocity or it experiences a *phase velocity* change, while the rest of the light wave maintains its original velocity. A change in the phase velocity of a light wave typically causes the light wave to change direction as described by Snell's law, which is discussed in detail later in this chapter. Figure 3.8 shows light waves changing direction because of a change in phase velocity as multiple light waves cross from a lower refractive index into a higher refractive index.

FIGURE 3.7 Light rays from the oar are refracted as they pass from water into the air.

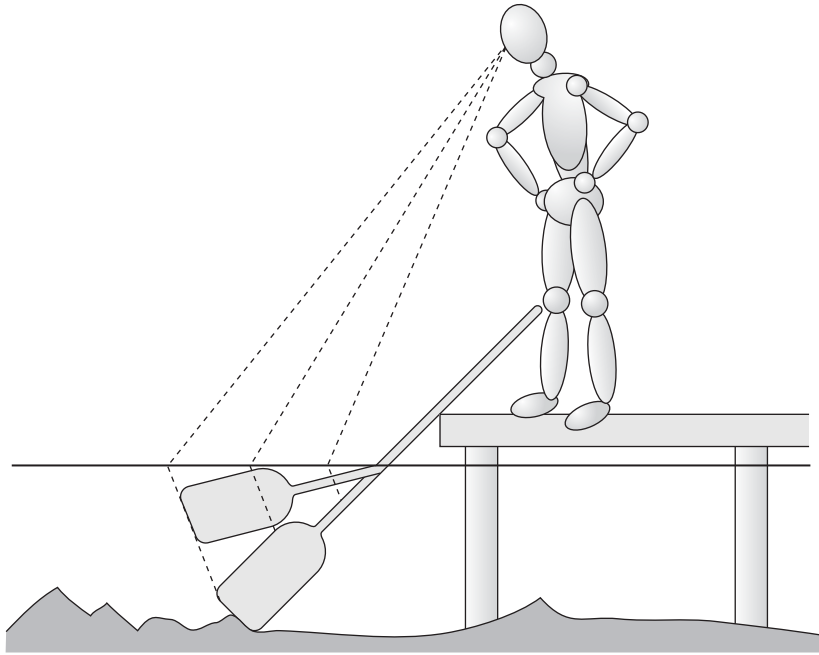
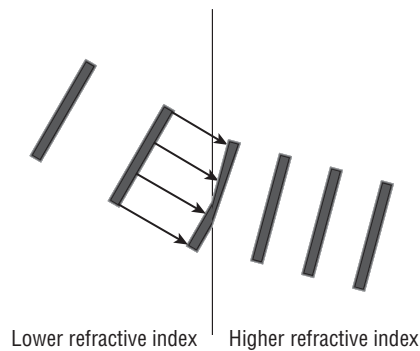


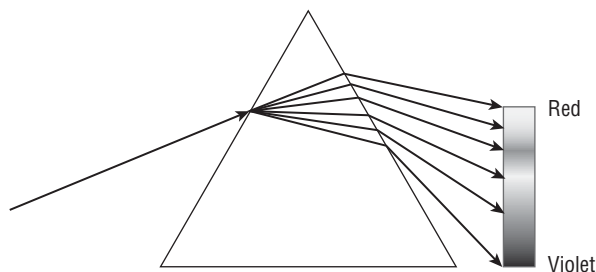
FIGURE 3.8 Light waves changing direction because of a change in phase velocity



The velocity of light through different media such as glass depends on the light's wavelength. One of the most common ways to observe this is by seeing how white light is refracted through a prism and broken up into its component wavelengths, as shown in Figure 3.9. Notice that violet, which has the shortest visible wavelength, is refracted more

than red, which has the longest visible wavelength. In this example, the red light changed direction the least and is traveling the fastest through the prism, while the violet light changed direction the most and is traveling the most slowly through the prism.

FIGURE 3.9 Refraction of white light into component colors



It is important to remember that different wavelengths of light travel at different speeds in an optical fiber.

Calculating the Index of Refraction

We know the velocity of light can change and its velocity depends on its wavelength and the index of refraction of the medium it is passing through. We also know that the higher the refractive index, the slower the light travels. In other words, the speed of light (c) in a medium is inversely proportional to its index of refraction.

It is important to be able to assign an index of refraction to different materials. Standard refractive index measurements are taken at a wavelength of 589nm. The refractive index of light wavelengths above or below 589nm will vary slightly. Table 3.1 lists the speed of light and index of refraction for some common materials.

TABLE 3.1 Refractive indexes of different materials

Material	Speed of light traveling through material (km/s)	Index of refraction
Vacuum	300,000	1.0000
Air	300,000	1.0003
Water	225,056	1.333

Material	Speed of light traveling through material (km/s)	Index of refraction
Ethyl alcohol	220,588	1.36
Optical fiber cladding	205,479	1.46
Optical fiber core	204,082	1.47
Glass	200,000	1.5

The index of refraction is a relative value and is based on the speed of light in a vacuum, which has an index of refraction of 1. We can calculate the index of refraction with the equation

$$n = c/v$$

where n is the index of refraction of the material, c is the velocity of light through vacuum, and v is the light's velocity through the material. So if light passes through a theoretical material at 230,000km/s, the material's index of refraction is

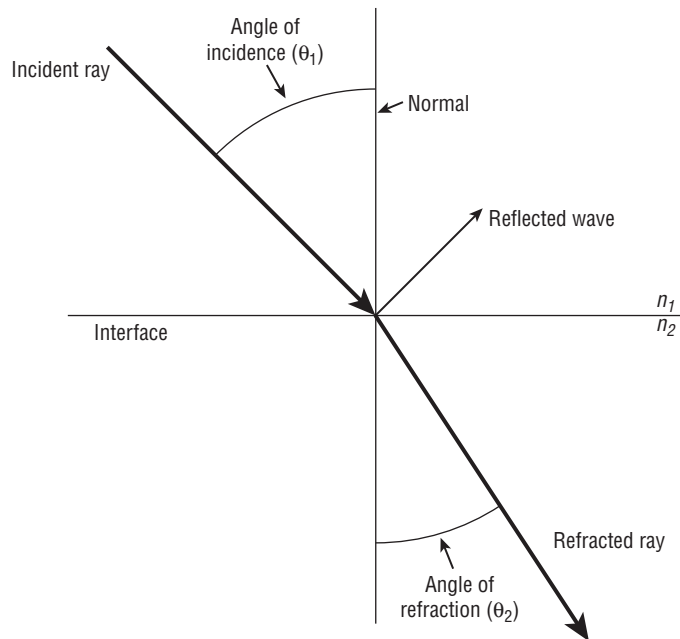
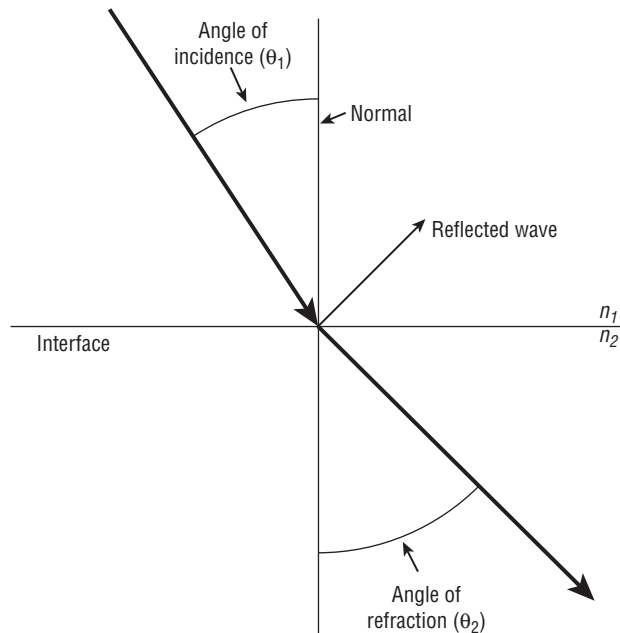
$$n = 300000 \div 230000 = 1.3$$

To calculate the amount of refraction that will take place when light passes from one material to another, we'll need a model and some basic terms. The model, shown in Figure 3.10, illustrates light passing from a medium with a lower index of refraction (n_1) to a medium with a higher index of refraction (n_2). The *interface* is represented by a horizontal line. A path perpendicular to the interface is known as the *normal*. Light traveling along the normal will change speed but will not change direction.

The *angle of incidence* represents the angle between the incident ray and normal. The *angle of refraction* represents the angle between the refracted ray and normal. Note that a small amount of light is also reflected from the interface at an angle equal to the angle of incidence.

Total Internal Reflection

As shown in Figure 3.10, light passing from a lower index of refraction to a higher index of refraction refracts toward normal. This is illustrated by the angle of refraction (θ_2), which is smaller, than the angle of incidence (θ_1). When light passes from a higher index of refraction to a lower index of refraction, as shown in Figure 3.11, the light refracts away from normal.

FIGURE 3.10 Model used to calculate refraction**FIGURE 3.11** Light passing from higher index of refraction to lower index of refraction refracts away from normal.

We can calculate the amount of refraction using Snell's law, which shows the relationship between incident light and refracted light

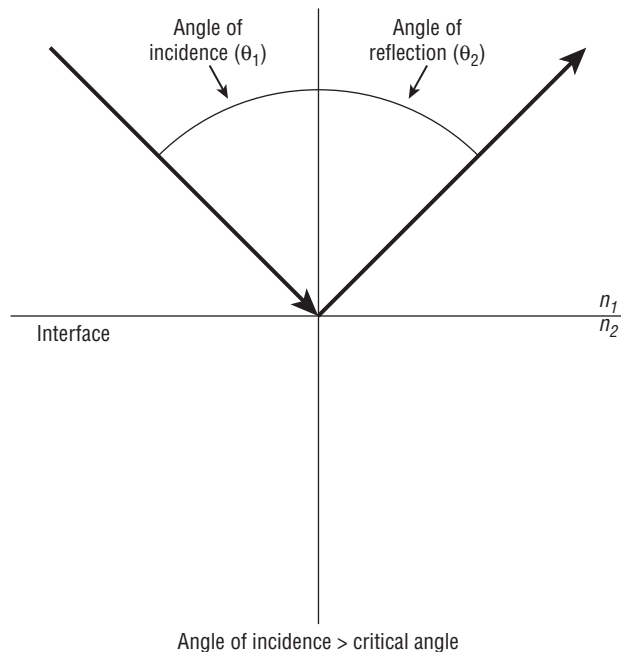
$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1 and n_2 are the index of refraction values of each material, θ_1 is the angle of incidence, and θ_2 is the angle of refraction.

Recall that the phenomenon that makes fiber-optic transmission possible, *total internal reflection* (TIR), is caused by the same principles that cause refraction. In the case of TIR, the light is passing from a higher index of refraction to a lower index of refraction at an angle that causes all of the light to be reflected. What has happened is that the angle of refraction has exceeded 90° from normal.

The incident angle required to produce a refracted angle of 90° is called the *critical angle*. As the incident ray moves from normal toward the critical angle, less and less of the incident ray's energy is carried into the refracted ray. At the critical angle all of the incident ray's energy is refracted along the interface. As the incident angle exceeds 90° , the light is reflected, as shown in Figure 3.12.

FIGURE 3.12 Reflection of light exceeding critical angle



To find the critical angle of two materials, we can use a modified version of Snell's equation:

$$\theta_c = \arcsin(n_2 \div n_1)$$

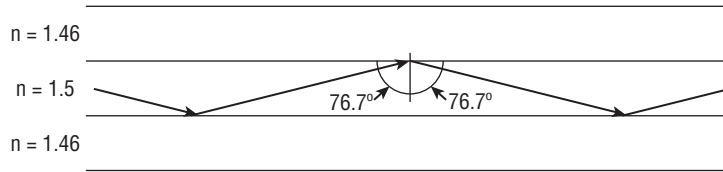
where θ_c produces a refractive angle of 90° from normal.

So if we want to know the critical angle of an optical fiber having a core index of refraction of $n_1 = 1.5$ and a cladding index of refraction of $n_2 = 1.46$, we solve this equation:

$$\theta_c = \arcsin(1.46 \div 1.5) = 76.7^\circ$$

In this example, if light is passing through the core at an angle greater than 76.7° , it will be reflected from the interface with the cladding. As long as the interface remains parallel, the light will continue to reflect at the same angle, as shown in Figure 3.13.

FIGURE 3.13 Total internal reflection



Fresnel Reflections

Recall that when light passes from one medium into another, refraction occurs as described by Snell's law. However, a reflection also occurs. This reflection is known as a *Fresnel reflection*. A Fresnel reflection occurs when light changes speed as it moves from a material with a given index of refraction into a material with a different index of refraction. The greater the difference in the index of refraction between the two materials, the greater the amount of light reflected. You experience Fresnel reflection whenever you look through a window and see a faint reflection of yourself in the glass.

Augustin-Jean Fresnel determined how to calculate the amount of light lost through a Fresnel reflection when light passes from one medium into another with the following equation:

$$\rho = ((n_1 - n_2) \div (n_1 + n_2))^2$$

where ρ is the amount of light reflected and n is the index of refraction of the medium.

If we calculate the reflection value for the interface between air, with an index of refraction of 1, and glass, with an index of refraction of 1.5, we get the following result:

$$\rho = ((1.5 - 1) \div (1.5 + 1))^2 = 0.2^2 = 0.04$$

To calculate the loss in decibels, we use the equation

$$\text{dB} = 10\text{Log}_{10} (1 - \rho)$$

plugging in the Fresnel reflection value from earlier,

$$\text{dB} = 10\text{Log}_{10} 0.96 = 10 \times -0.018 = -0.18$$

which gives us a loss of 0.18dB.

It is important to understand that a Fresnel reflection occurs whenever there is a velocity change. It does not matter whether the light is increasing speed or decreasing speed. Light traveling in the core of an optical fiber travels slower than it does in air. If there were an air gap at an interconnection, a Fresnel reflection would occur as light left the core of the optical fiber and entered the air. A Fresnel reflection of equal magnitude would also occur as light traveling in the air entered the core of the other optical fiber. The reflected light is lost and never makes it to its intended destination. Some of the reflected light may travel in the opposite direction in the core of the optical fiber back toward the source. Depending on the light source, this could cause a problem such as a shift in wavelength or a reduction in output power.



Real World Scenario

Reducing Fresnel Reflections and Making the Interconnection Work

The connector endface is the part of the fiber-optic connector where the optical fiber is exposed. From time to time, the optical fiber at the connector endface is damaged, and typically the person causing the damage is unaware of it. The damage may occur because there was debris on one connector endface when it was mated with another connector. It may occur when a damaged connector is mated with an undamaged connector. Regardless of how it occurs, the end result is typically a fiber-optic link that does not function properly.

Critical fiber-optic links need to be restored as quickly as possible. This means that replacing the damaged connector(s) at this time may not be an option. Although it is not a common practice to use index matching gel in an interconnection, using it to reduce the Fresnel reflections can sometimes restore a damaged or poorly performing interconnection. This could mean the difference between a successful and a failed restoration. (*Index matching gel* is a substance that fills the air space within a connection with a material matching the refractive index of the fiber core.)

Summary

This chapter covered the properties of light and their effects on fiber-optic transmission. We explored the nature of light as an electromagnetic wave and as a moving particle, including its generation and transmission.

We also described the characteristics of light, including speed, frequency, and wavelength, that influence its behavior as it passes through different materials. In addition,

we discussed the principles and equations that can be used to predict light's refraction, reflection, and loss.

Exam Essentials

Understand light as electromagnetic energy. Make sure you understand the nature of light as a form of electromagnetic energy. Also, be able to describe light as a three-dimensional wave and its characteristics as both a wave and a particle. Understand the relationship between frequency and wavelength.

Understand the electromagnetic spectrum. Make sure you understand the makeup of the electromagnetic spectrum. Be able to associate wavelength and frequency ranges with different types of electromagnetic energy. Understand the relationship of light, including infrared, visible, and ultraviolet, within the spectrum.

Understand refraction. Make sure you understand what refraction is and what causes it. Also, be able to explain how the refractive index is used to express the speed of light through a transparent medium. Be able to explain reflection in terms of angle of incidence, critical angle, angle of refraction, and total internal reflection.

Know what Snell's law is. Be able to explain Snell's law as it applies to fiber-optic systems. Be able to use Snell's law to determine the amount of refraction and the critical angle.

Understand Fresnel reflection. Be able to describe a Fresnel reflection and the effect it has on a fiber-optic communication system.

Review Questions

1. What relation do the electrical field, magnetic field, and the direction of travel have in light?
 - A. They are parallel to one another.
 - B. They are at right angles to one another.
 - C. They are multiples of one another.
 - D. They are unrelated.

Hint: 90-degrees

2. At approximately how many kilometers per second does an electromagnetic wave travel through free space or air?
 - A. 300,000
 - B. 225,056
 - C. 204,082
 - D. 200,000

Hint: It has the shortest wavelength.

3. Light travels in the form of photons and can travel through a medium without permanently displacing it. Because of this, we consider light to
 - A. Be neither a wave nor a particle.
 - B. Be primarily a wave.
 - C. Have characteristics of both a wave and a particle.
 - D. Be primarily a particle.

Hint: Photons travel in a wavelike pattern.

4. In fiber optics, _____ is typically used to describe the output of a light source.
 - A. Frequency
 - B. Wavelength
 - C. Hertz
 - D. Kilometers

Hint: Nanometers

5. Which of the following forms of electromagnetic energy has a longer wavelength than the other three?
 - A. Infrared light
 - B. Visible light

- C. Gamma rays
- D. Ultraviolet light

Hint: It is below visible light on the electromagnetic spectrum.

6. Which wavelength windows are most commonly used for fiber optics?
- A. 400nm, 750nm, and 850nm
 - B. 550nm, 580nm, and 620nm
 - C. 620nm, 750, and 850nm
 - D. 850nm, 1300nm, and 1550nm

Hint: The wavelengths are in the infrared range.

7. Refraction is caused by _____.
- A. Light changing color as it passes from one medium into another
 - B. Light changing speed as it passes from one medium into another
 - C. Light changing frequency as it passes from one medium into another
 - D. Light changing direction within a single medium

Hint: The wavelength will not change.

8. Light would travel the fastest through a medium with a refractive index of _____.
- A. 1.0003
 - B. 1.33
 - C. 1.47
 - D. 1.5

Hint: Inversely proportional

9. If a beam of light passing from medium A into medium B refracts toward normal, what do we know about the two media?
- A. Medium A is thicker than medium B.
 - B. Medium A has a higher n than medium B.
 - C. Medium A has a lower n than medium B.
 - D. Medium A has an n of 1.

Hint: Light travels slower through medium B.

10. If we want to create total internal reflection, we have to pass light from _____.
- A. A lower n to a higher n
 - B. A higher n to a lower n
 - C. An n of 1 to a higher n
 - D. A higher n to an n of 1

Hint: Light travels slower through the core of an optical fiber.

11. The critical angle is the angle of incidence required to produce a refracted angle of _____.

- A. 180°
- B. 0°
- C. 45°
- D. 90°

Hint: Right angle

12. The angle of refraction can be calculated using _____ law.

- A. Snell's
- B. Fresnel's
- C. Watt's
- D. Ohm's

Hint: A Dutch mathematician

13. A Fresnel reflection will occur because light changes _____ as it moves from a material with a given n into a material with a different n .

- A. direction
- B. speed
- C. the critical angle
- D. the index of refraction

Hint: Refraction also occurs

14. The loss caused by a Fresnel reflection is greater when the difference in the n between the two materials is _____.

- A. less
- B. equal
- C. greater
- D. unknown

Hint: The amount of loss depends on the velocity change.

15. Light traveling across a small air gap between two optical fibers will produce _____.

- A. total internal reflection
- B. electromagnetic radiation
- C. a Fresnel reflection
- D. the Index of Refraction

Hint: It occurs twice.

Chapter Exercises

Convert Various Wavelengths to Corresponding Frequencies

The relationship between wavelength and frequency can be expressed with the formula $\lambda = v \div f$.

If an electromagnetic wave has a frequency of 94.7MHz, what is its wavelength?

By dividing the speed of the electromagnetic wave, 300,000km/s, by the frequency, 94.7MHz, we arrive at the wavelength, 3.17 meters.

Convert Various Frequencies to Corresponding Wavelengths

The relationship between frequency and wavelength can be expressed with the formula $f = v \div \lambda$.

If an electromagnetic wave has a wavelength of 0.19 meters, what is its frequency?

By dividing the speed of the electromagnetic wave, 300,000km/s, by the wavelength, 0.19 meters, we arrive at the frequency, 1.58GHz.

Calculate the Amount of Energy in a Photon Using Planck's Constant

The amount of energy in each photon depends on the electromagnetic energy's frequency: the higher the frequency, the more energy in the particle. To express the amount of energy in a photon, we use the equation $E = hf$.

Calculate the energy of a photon at a frequency of 193.55THz.

Using Planck's constant, 6.626×10^{-34} joule-seconds, and the formula $E = hf$, you can solve for the energy as shown in this equation:

$$(6.626 \times 10^{-34}) \times (1.9355 \times 10^{14}) = 1.282 \times 10^{-19} \text{ W}$$

Calculate the Speed of Light through a Transparent Medium Using Its Refractive Index

The index of refraction is a relative value, and it is based on the speed of light in a vacuum, which has an index of refraction of 1. The index of refraction can be calculated using the equation $n = c/v$.

If light is passing through a medium with a refractive index of 1.33, what will the velocity of the light through that medium be?

To find the speed of light in a medium with a known refractive index, we divide the speed of light, 300,000km/s, by the medium's refractive index, in this case, 1.33. The result is a speed of 225,564km/s.

Use Snell's Law to Calculate the Critical Angle of Incidence

The incident angle required to produce a refracted angle of 90° is called the critical angle. To find the critical angle of two materials, we can use a modified version of Snell's equation:

$$\theta_c = \arcsin(n_2 \div n_1)$$

Calculate the critical angle for two materials where n_2 is 1.45 and n_1 is 1.47.

The critical angle can be found by dividing n_2 by n_1 , then finding the arcsin of the result as shown in the following equation.

$$\theta_c = \arcsin(1.45 \div 1.47) = 80.54^\circ$$

Calculate the Loss in Decibels from a Fresnel Reflection

Augustin-Jean Fresnel determined how to calculate the amount of light lost from a Fresnel reflection using the equation $\rho = ((n_1 - n_2) \div (n_1 + n_2))^2$.

Calculate the loss in decibels from a Fresnel reflection that is the result of light passing from a refractive index of 1.51 into a refractive index of 1.45.

To calculate the loss in decibels from a Fresnel reflection, we need to use two equations shown here.

$$\rho = ((1.51 - 1.45) \div (1.51 + 1.45))^2 = 0.02^2 = 0.0004$$

To calculate the loss in decibels, we use the equation

$$\text{dB} = 10\text{Log}_{10} (1 - 0.0004)$$

plugging in the Fresnel reflection value from above,

$$\text{dB} = 10\text{Log}_{10} 0.9996 = -0.0017$$

This gives us a loss of 0.0017dB.

Chapter 4

Optical Fiber Construction and Theory

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe the basic parts of an optical fiber.
- ✓ List the major standards organizations that publish standards that define the performance of optical fibers used in the telecommunications industry.
- ✓ List the different materials that can be used to construct an optical fiber.
- ✓ Describe the tensile strength of an optical fiber.
- ✓ Describe optical fiber manufacturing techniques.
- ✓ Describe mode in an optical fiber.
- ✓ Describe how the number of modes in an optical fiber is defined by core diameter and wavelength.
- ✓ Describe the refractive index profiles commonly found in optical fiber.
- ✓ Explain the propagation of light through a multimode step index optical fiber.
- ✓ Explain the propagation of light through a multimode graded index optical fiber.
- ✓ Explain the propagation of light through a single-mode optical fiber.
- ✓ Describe the location and function of an optical trench.
- ✓ Describe the advantages of single-mode and multimode bend insensitive optical fiber.





Optical fibers are called on to operate in a wide variety of conditions. Some of them must carry high volumes of data over many miles, whereas others carry smaller amounts of data inside an office building or aboard an aircraft. The type of job an optical fiber will do determines the type of fiber you'll choose to install. It is important to understand the types of fibers that are available and the ways in which they are built so that you can select and use them properly.

This chapter describes the construction of optical fibers and the components that make them up. We will discuss some of the important factors that must be considered in the manufacture and use of optical fibers, as well as the designs used to optimize them for different types of data transmission. Finally, the chapter will introduce you to some of the commonly used commercial optical fibers and describe their features.

In this chapter, you will learn to

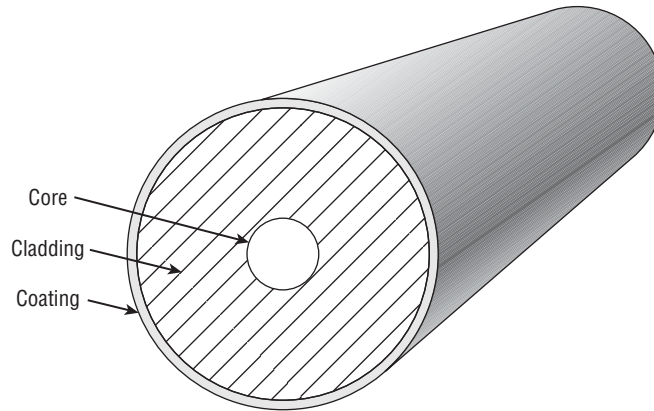
- Select the proper optical fiber coating for a harsh environment
- Identify an industry standard that defines the specific performance characteristics on an optical fiber
- Identify an optical fiber from its refractive index profile
- Calculate the numerical aperture of an optical fiber
- Calculate the number of modes in an optical fiber

Optical Fiber Components

Today's standard optical fiber is the product of precision manufacturing techniques and exacting standards. Make no mistake: even though it is found in almost any data or communications link, optical fiber is a finely tuned instrument requiring care in its production, handling, and installation.

As shown in Figure 4.1, a typical optical fiber comprises three main components: the core, which carries the light; the cladding, which surrounds the core with a lower refractive index and contains the light; and the coating, which protects the fragile fiber within.

FIGURE 4.1 Optical fiber components include the core, cladding, and coating.



Let's look at these components individually.

Core

The *core*, which carries the light, is the smallest part of the optical fiber. The optical fiber core is usually made of glass, although some are made of plastic. The glass used in the core is extremely pure silicon dioxide (SiO_2), a material so clear that you could look through miles of it as though you were looking through a household window.

In the manufacturing process, dopants such as germania, phosphorus pentoxide, and alumina are used to raise the refractive index under controlled conditions.

Optical fiber cores are manufactured in different diameters for different applications. Typical glass cores range from as small as $3.7\mu\text{m}$ up to $200\mu\text{m}$. Core sizes commonly used in telecommunications are $9\mu\text{m}$, $50\mu\text{m}$, and $62.5\mu\text{m}$. Plastic optical fiber cores can be much larger than glass. A popular plastic core size is $980\mu\text{m}$.

Cladding

Surrounding the core, and providing the lower refractive index to make the optical fiber work, is the *cladding*. When glass cladding is used, the cladding and the core are manufactured together from the same silicon dioxide-based material in a permanently fused state. The manufacturing process adds different amounts of dopants to the core and the cladding to maintain a difference in refractive indexes between them of about 1 percent. A typical core may have a refractive index of 1.49 at 1300nm while the cladding may have a refractive index of 1.47. These numbers, however, are wavelength dependent. The core of the same fiber will have a different refractive index at a different wavelength. At 850nm the core refractive index will increase slightly.

Like the core, cladding is manufactured in standard diameters. The two most commonly used diameters are 125 μm and 140 μm . The 125 μm cladding typically supports core sizes of 9 μm , 50 μm , 62.5 μm , and 85 μm . The 140 μm cladding typically has a 100 μm core.

Coating

The *coating* is the true protective layer of the optical fiber. The coating absorbs the shocks, nicks, scrapes, and even moisture that could damage the cladding. Without the coating, the optical fiber is very fragile. A single microscopic nick in the cladding could cause the optical fiber to break when it's bent. Coating is essential for all-glass fibers, and they are not sold without it.

The coating is solely protective. It does not contribute to the light-carrying ability of the optical fiber in any way. The outside diameter of the coating is typically either 250 μm or 500 μm . The coating is typically colorless. In some applications, however, the coating is colored, as shown in Figure 4.2, so that individual optical fibers in a group of optical fibers can be identified.

FIGURE 4.2 The fiber coating can be color-coded to help identify it.



The coating found on an optical fiber is selected for a specific type of performance or environment. One of the most common types of coatings is *acrylate*. This coating is typically applied in two layers. The *primary coating* is applied directly on the cladding. This coating is soft and provides a cushion for the optical fiber when it is bent. The *secondary coating* is harder than the primary coating and provides a hard outer surface. Acrylate, however, is limited in temperature performance. A typical acrylate coating has a maximum operating temperature of 100° C while some high-temperature acrylates may perform at temperatures up to 125° C.

Silicone, *carbon*, and *polyimide* are coatings that may be found on optical fibers that are used in harsh environments such as those associated with avionics, aerospace, and space. They may also be used on optical fibers designed for mining, or oil and gas drilling.

Silicone is a soft material with a higher temperature capability than acrylate, typically 200° C. Like acrylate, silicone cushions the optical fiber during bending, but it is too soft to be the sole coating. It must be used in combination with a buffer or harder thermoplastic

and is generally thicker than acrylate, with diameters up to 500 μm . Silicone has other advantages besides a high-temperature capability; it has a high resistance to water absorption and low flammability.

Silicone or acrylate coatings must be removed prior to terminating the optical fiber. Both of these coatings can leave a residue, so extra care must be used when removing silicone. Terminating optical fibers is discussed in depth in Chapter 9, “Connectors.”

Polyimide is a popular coating for aerospace applications because it can operate in temperatures up to 350° C. This coating is much thinner than acrylate or silicone, typically only 15 μm thick. Beside a very high operational temperature, polyimide is resistant to abrasion and chemicals. However, polyimide is difficult to remove. If it must be removed before termination, hot sulfuric acid or a portable polyimide-stripping machine is required.

Carbon is the thinnest coating of all, typically only a fraction of a micron thick. Carbon hermetically seals the glass surface, protecting it from moisture and increasing the optical fiber's fatigue resistance. Carbon cannot be used by itself; it must be combined with another coating such as silicone or polyimide. Unlike the other coatings discussed, carbon coating does not have to be removed before termination.

Standards

While many combinations of core and cladding sizes are possible, standards are necessary to ensure that optical fibers, connectors, splices, and equipment are interchangeable between manufacturers. This is especially important when dealing with components as small as those used in fiber optics, where even slight misalignments or differences can render the entire system useless.

There are many standards related to optical fiber. This chapter focuses on the standards that define the performance of optical fibers used in the telecommunications industry published by the Telecommunications Industry Association (TIA), the International Telecommunications Union (ITU), the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC). While TIA, ITU, ISO, and IEC publish many standards, the key standards that you should be familiar with are ANSI/TIA-568-C, ITU-T G.651.1, ITU-T G.652, ITU-T G.655, ITU-T G.657, ISO/IEC 11801, IEC 60793-2-10, and IEC 60793-2-50.

The ANSI/TIA-568-C standards and the ISO/IEC 11801 standard are applicable to *premises* cabling components. Premises are a telecommunications term for the space occupied by a customer or an authorized/joint user in a building on continuous or contiguous property that is not separated by a public road or highway. Premises cabling standards are typically optimized for cabling distances up to 2000 meters. These standards can define the performance of both multimode and single-mode optical fibers.

The ITU standards are applicable to multimode and single-mode optical fiber and cable. Here are their descriptions:

- ITU-T G.651.1: Characteristics of a 50/125 μm multimode graded index optical fiber cable for the optical access network
- ITU-T G.652: Characteristics of a single-mode optical fiber and cable

- ITU-T G.655: Characteristics of a non-zero dispersion-shifted single-mode optical fiber and cable
- ITU-T G.657: Characteristics of a bending-loss insensitive single-mode optical fiber and cable for the access network

The IEC 60793-2 series standards address optical fiber product specifications. Here are their descriptions:

- IEC 60793-2-10: Category A1 multimode optical fiber product specifications
- IEC 60793-2-50: Class B single-mode optical fiber product specifications

These standards contain important information that defines the performance of the optical fiber, fiber-optic cable, and components such as connectors and splices. Subsequent chapters will discuss these standards in detail.

Getting to Know the Standards

If you are a sports fan, you know the statistics of your favorite team and players. If you plan to be the person everyone turns to when there is a question about optical fiber, you need to know the standards just as well as the sports fan. The nice thing about standards, however, is they do not change nearly as fast as your favorite team's record does. Standards can take years to develop and typically have a five-year review cycle.

Materials

Optical fibers are commonly made with a glass core and glass cladding, but other materials may be used if the fiber's performance must be balanced with the cost of installing the fiber, fitting it with connectors, and ensuring that it is properly protected from damage. In some cases, fibers run only a short distance, and the benefits of high-quality all-glass fibers become less important than saving money or ease of use. There are also circumstances in which the fibers are exposed to harsh conditions, such as vibration, extreme temperature, repeated handling, or constant movement. Different fiber classifications have evolved to suit different conditions, cost factors, and performance requirements. Here are the major fiber classifications by material:

Glass fibers These have a glass core and glass cladding. They are used when high data rates, long transmission distances, or a combination of both are required. As you will see in later chapters, connectors for glass fibers must be built with a high degree of precision and can take time to apply correctly. Glass fibers are the most fragile of the various types

available, and as a result they must be installed in environments where they will not be subjected to a great deal of abuse, or they must be protected by special cables or enclosures to ensure that they are not damaged.

Glass fibers are commonly found in long-distance data and interbuilding and interoffice networking applications.

Glass, glass, polymer (GGP) These fibers have a glass core and a hybrid cladding. The inner cladding is glass and is identical to a glass optical fiber. However, the outer cladding is a polymer that is bonded to the inner cladding. They can be manufactured to offer the same optical performance as glass fibers. The outer cladding polymer coating is far more resistant to mechanical damage than an all-glass fiber. This unique property prevents breakage during the termination process.

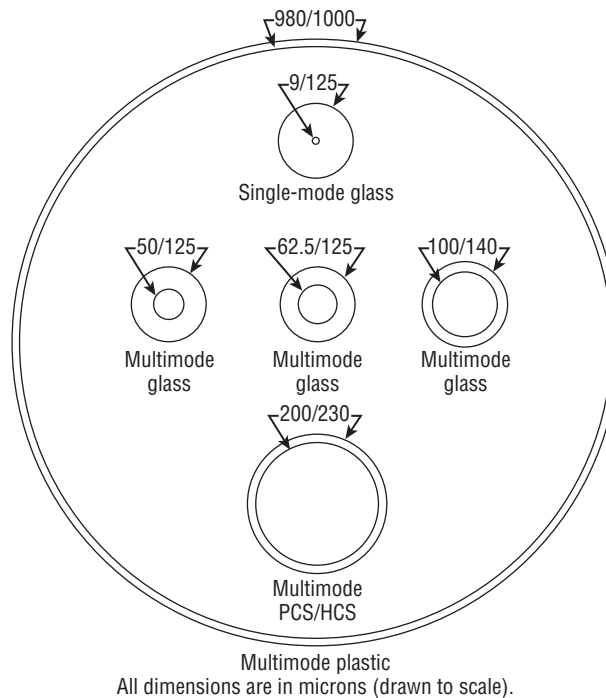
GGP fibers can be used for long-distance data and premises applications.

Plastic-clad silica (PCS) These fibers have a glass core and plastic cladding. The core is larger than all-glass fiber—typically, 200 μm with a cladding thickness of 50 μm . Like a silicone-coated glass optical fiber, the plastic coating of a PCS optical fiber is commonly used with a thermoplastic buffer that surrounds the plastic cladding. A typical PCS fiber specification would be 200/300 μm . The plastic cladding also serves as a protective layer for the glass core, so the coating normally found on all-glass fiber is not included on PCS fibers. PCS fibers are generally used for industrial sensing applications and medical/dental applications.

Hard-clad silica (HCS) These fibers are similar to PCS fiber but they have a glass core with cladding made of a hard polymer or other material, typically stronger than other cladding materials. Hard-clad silica fiber is commonly used in locations where ruggedness is a prime consideration, such as manufacturing, factory automation, and other areas where shock and vibration would render standard glass fibers unreliable.

HCS optical fibers are typically much larger than glass optical fibers. A very popular size is 200/230 μm .

Plastic fiber These fibers have a plastic core and plastic cladding. They are selected for their low cost, ruggedness, and ease of use, and are installed where high bandwidth and long transmission distances are not required. While plastic fibers are unsuited for long-distance, high-performance transmissions, they can still carry signals with useful data rates over distances of less than 100m. Plastic fibers are much larger in diameter than glass fibers, as shown in the comparison in Figure 4.3. A very popular size is 980/1000 μm . Plastic fiber is typically designed for visible wavelengths in the 650nm range. Some typical locations for plastic fiber include home entertainment systems, automotive, and manufacturing control systems. They may also be used in links between computers and peripherals and in medical equipment.

FIGURE 4.3 Comparison of fiber core and cladding diameters

Real World Scenario

The Advantages of Large Core Plastic Optical Fiber

It is easy to get excited about the high bandwidth and long-distance transmission capabilities of glass optical fiber. It clearly outperforms any other medium. However, many applications do not require a high bandwidth over great distances. There are many applications for optical fiber in your home. You may already have a home entertainment system that uses plastic optical fiber, or you may own a car that uses plastic optical fiber to connect audio devices or a DVD changer. None of these applications requires high bandwidth over great distances. These applications are ideal for large core plastic optical fiber.

As you learned in this chapter, plastic optical fiber is typically designed to operate at a visible wavelength around the 650nm range. Being able to see the light as it exits the optical fiber has a significant advantage; no expensive test equipment is required. A power meter is needed to measure the light exiting a glass optical fiber operating in the infrared range. Power meters can cost more than your home entertainment system.

The large core of the plastic optical fiber has another advantage over small glass fibers: it is easy to align with another fiber or a light source or detector. Imagine aligning two human hairs so that the ends touch and are perfectly centered. Now imagine doing the same thing with two uncooked spaghetti noodles. Which do you think would be easier?

Plastic optical fiber is a great choice for audio and video electronics being integrated into homes and vehicles.

Tensile Strength

In addition to the coating, optical fibers may have several layers of protection such as buffers, jackets, armor, and other materials designed to protect the fiber and keep it together in cables. Like copper, though, optical fiber is still subject to hazards caused by handling, installation, careless digging, and bad weather.

One characteristic of optical fiber that deserves special attention is its tensile strength, or the amount of stress from pulling that the optical fiber can handle before it breaks. Optical fiber has a high tensile strength and resists stretching.

Tensile strength is important for several reasons. It affects the way fiber must be handled during installation, the amount of curvature it can take, and the way it will perform throughout its lifespan.

To understand optical fiber's tensile strength, let's look at a standard piece of plate glass. To cut the glass, all you have to do is scribe a sharp line through the surface layer. Once the strength of that layer is compromised, the glass snaps easily along the scribe, even if it follows a curve.

Optical fiber follows the same rule. The outer layer of the cladding provides much of the fiber's tensile strength, which is often measured in thousands of pounds per square inch (kpsi). A typical breaking strength for an optical fiber is 100kpsi. That means that a typical fiber with a cladding thickness of 125 μ m can withstand a pull of about 1.9 pounds. However, the maximum pull should never be applied when removing the buffer or coating. It is a good practice to limit the amount of pull to around a half of pound while removing the buffer or coating.

Once the outer layer is scratched or cracked, however, the tensile strength is gone at that location. Like a scribed line, a scratch or crack compromises the integrity of the glass and allows the fiber to break more easily under stress. Scratching or cracking can occur due to mishandling, sharp blows, or bending beyond the fiber's *minimum bend radius*, especially if the bending takes place while the fiber is under tension. The minimum bend radius is the minimum radius the optical fiber can be bent without damaging it.

Manufacturing Optical Fiber

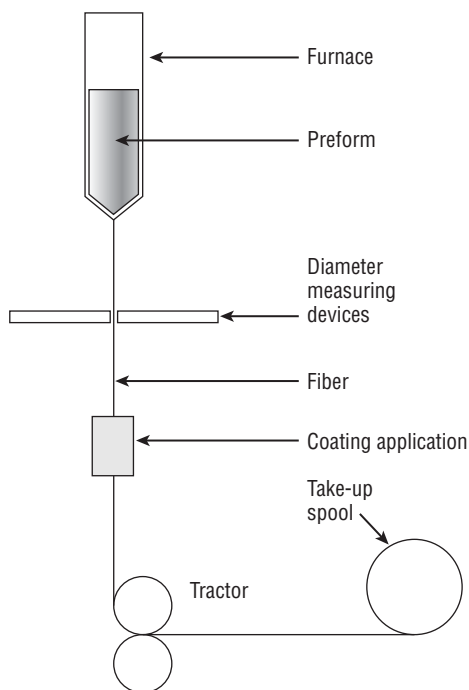
Optical fiber is manufactured to very high standards, because the core diameter and refractive indexes of the core and cladding must remain consistent over stretches of up to 80km.

Four methods are commonly used to make optical fiber:

- Modified chemical vapor deposition (MCVD)
- Outside vapor deposition (OVD)
- Vapor axial deposition (VAD)
- Plasma chemical vapor deposition (PCVD)

Each method uses variations on a process to create a *preform*, a short, thick glass rod that has a similar cross section to the final fiber. As shown in the block diagram in Figure 4.4, the preform is heated to 1900° C in a drawing tower until it begins to melt and a blob falls from the end, drawing a small thread of glass after it. The thread contains the core and cladding of the optical fiber, their relative thicknesses preset by the preform's construction.

FIGURE 4.4 Drawing the preform into fiber



This thread is taken up by a pulling machine, or *tractor*, at a constant rate to maintain a consistent thickness. A thinner, longer fiber can be created by speeding up the draw rate. The entire process is closely monitored by laser measuring devices to ensure that the thickness remains consistent over the entire length of the fiber, which can be anywhere from 10 to 80km, depending on the size of the preform.

The fiber is then drawn through another process, which deposits and cures the coating. The coating is cured using ultraviolet lamps or thermal ovens. After the coating is applied, measurements are taken and the optical fiber is taken up on a spool.

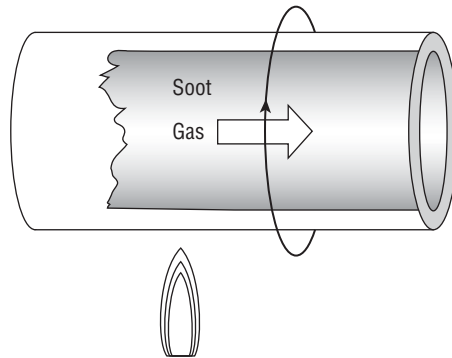
Each of the preceding methods is best suited for different types of fiber, depending on the type of signals it will carry, the distance it will cover, and other factors, which will be discussed later in this chapter.

Let's look at the differences in the manufacturing methods.

Modified Chemical Vapor Deposition (MCVD)

Fibers manufactured using MCVD begin with a hollow glass tube about 2.5cm in diameter, as shown in Figure 4.5.

FIGURE 4.5 Modified chemical vapor deposition



The cladding is created first by placing the hollow tube, or *bait*, on a lathe and spinning it rapidly. As the bait is spinning, it is heated by an oxygen/hydrogen torch passing lengthwise underneath as a gaseous mixture of vaporized silicon dioxide is introduced into the tube. The ultra-pure gas, mixed with carefully controlled impurities or *dopants*, forms a soot, which fuses to the inside of the bait in successive layers as the heat passes beneath it. The dopants are used to increase the refractive index of the fused material.

Every time the flame passes beneath the tube to heat it, another thin layer of soot adheres to the inside, building a thicker and thicker layer of glass.

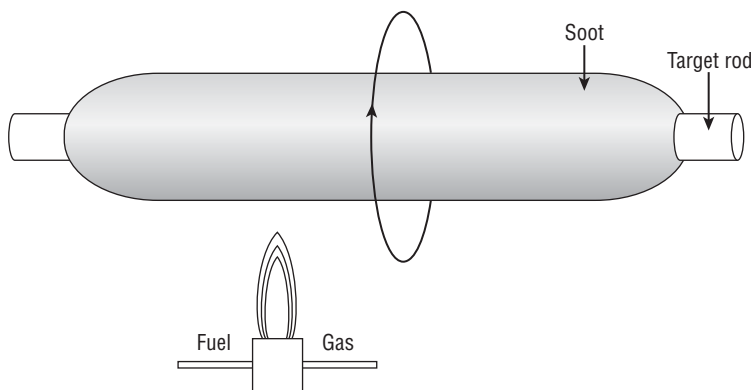
The core is formed next by changing the gas/dopant mixture to create a vapor with a higher refractive index. This, too, is introduced into the spinning, heated tube, inside the first layers that were laid down. When the deposited material has reached the desired thickness, the tube is heated to an even higher temperature to consolidate the soot into glass without melting it, a process known as *sintering*. A drying gas is passed through the core to remove water contamination, and the tube is compressed into a preform and taken to the drawing tower.

Outside Vapor Deposition (OVD)

The OVD method is similar to MCVD, except that the fiber preform is built from the inside out.

In the OVD method, a glass target rod serves as the bait, as shown in Figure 4.6. The rod is placed on the lathe and spun as heat is applied to it. The core is laid down first by introducing a gas mixture between the heat and the rod. When the core layer is thick enough, the mixture is changed and the cladding is laid down.

FIGURE 4.6 Outside vapor deposition



After the layers have been deposited, the rod is removed and the layers are dried and sintered and then collapsed into the preform.

Vapor Axial Deposition (VAD)

As with MCVD and OVD, the VAD process uses a heated glass bait to collect a soot of silicon dioxide and dopant. With VAD, however, the glass rod is suspended vertically and the heat source is at the lower end, as shown in Figure 4.7.

The gas is introduced between the end of the rod and the heat source, and the soot builds up in a radial pattern. This method gives the manufacturer a great deal of control over not only the refractive index of the various layers, but also the pattern in which they are laid down.

As with OVD, the rod is then removed, and the layers are dried, sintered, and collapsed into the preform before being taken to the drawing tower.

Plasma Chemical Vapor Deposition (PCVD)

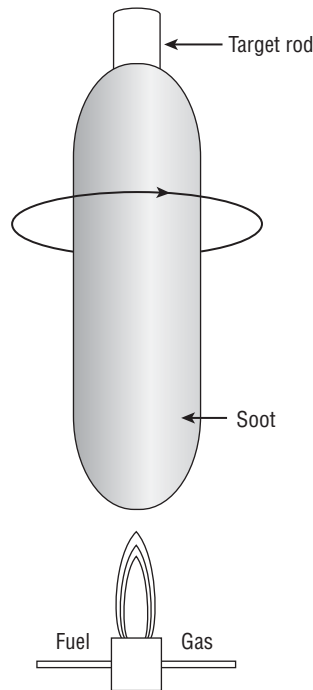
PCVD is similar to MCVD, but allows a finer layer of material to be deposited.

In PCVD, the gas particles are heated with microwaves, causing plasma to form inside the tube. The plasma contains electrons with extremely high energy levels, approximating

those found in a gas heated to $60,000^{\circ}\text{C}$, even though the temperature inside the tube is no higher than it is with MCVD—about 1200°C . As the high-energy electrons meet the soot forming inside the tube, they transfer their energy to it in the form of heat, causing it to fuse into the final glass form on the tube walls, making the step of consolidation unnecessary.

As with the other methods, the tube can then be compressed and taken to the drawing tower.

FIGURE 4.7 Vapor axial deposition



Mode

One of the most important characteristics used to distinguish types of fiber is the number of potential paths light can take through it. It may seem that light would go straight through the fiber core, following all of its curves, until it comes out the other end. However, the light itself is a complex combination of electrical and magnetic waves, and their wavelengths can be many times smaller than the core of the fiber. Like a rubber ball shot through a sewer pipe, the light actually has more than one path or *mode*. There are many potential paths or modes

it can follow, depending on the size of the core, wavelength of the light source, and *numerical aperture*.

To understand modes in optical fibers, let's take another look at the basis for fiber-optic transmission: total internal reflection.

Recall that total internal reflection depends on the principle that light passing through a medium of a given refractive index will be reflected off the interface with a medium of a lower refractive index if it hits the interface at or above the critical angle, which is determined by the difference between the refractive indexes of the two media.

To satisfy this requirement, optical fiber is manufactured with a core having a refractive index slightly higher than that of the cladding surrounding it. As a result, light entering the end of the fiber will be reflected off the interface with the cladding and guided through the fiber. By definition, each reflection will be at the same angle as the angle of incidence.

The number of modes possible in a length of fiber depends on the diameter of the core, the wavelength of the light, and the core's numerical aperture.

The Use of Mode and Path in Fiber Optics

In fiber optics, the terms "mode" and "path" are often used interchangeably. For example, ANSI/TIA-568-C.3 defines mode as "a path of light in an optical fiber." ARINC Report 805 defines mode as "a term used to describe a light path through a fiber, as in multimode or single-mode."

Calculating the Numerical Aperture and Modes

You can find the numerical aperture using the following formula:

$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{cladding}}^2}$$

For example, if the core has a refractive index of 1.485 and the cladding has a refractive index of 1.47, we can calculate:

$$NA = \sqrt{1.485^2 - 1.47^2} = 0.211$$

Note that the numerical aperture has no dimension. It is intended as a relative indication of the light-gathering capacity of the fiber core.

To find the number of modes, we can use the equation

$$M = (D \times \pi \times NA / \lambda)^2 / 2$$

where D is the diameter of the core and λ is the wavelength of the light. For example, let's use our previously derived value for the numerical aperture with a core diameter of 50 μm and a light wavelength of 850nm:

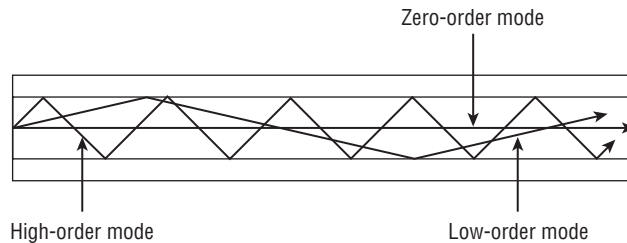
$$M = (50 \times 10^{-6} \times \pi \times 0.211 / 850 \times 10^{-9})^2 / 2 = 760.2$$

Because light cannot have part of a mode, we must round down to the nearest whole number if we come up with a decimal. So the answer is 760 modes or potential paths for the light to follow.

Refractive Index Profiles

What happens when the light rays entering a fiber take slightly different paths, some entering at sharper angles, some at shallower angles? As shown in Figure 4.8, the light can follow modes ranging from a straight line through the fiber (*zero-order mode*) to a low number of reflections (*low-order mode*) to a high number of reflections (*high-order mode*). Depending on a fiber's construction and the wavelength of the light source, the fiber may support one mode or more than a million modes (in fibers with large cores such as those found in plastic optical fibers). The path or paths that light takes through an optical fiber can be understood by examining the fiber's *refractive index profile*.

FIGURE 4.8 The three types of light modes

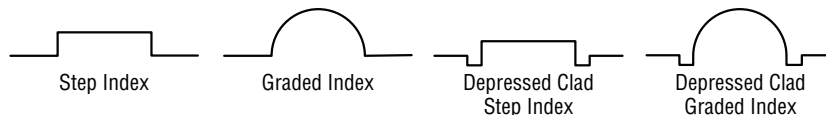


The refractive index profile graphically represents the relationship between the refractive index of the core and that of the cladding. Several common profiles are found on optical fibers used in telecommunications, avionics, aerospace, and space applications:

- Step index
- Graded index
- Depressed clad

These refractive index profiles are shown in Figure 4.9.

FIGURE 4.9 Refractive index profiles



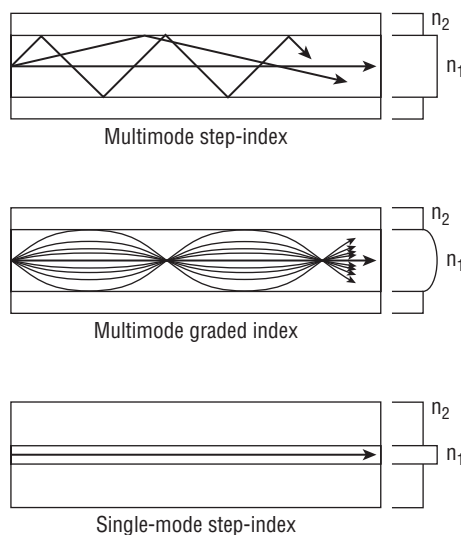
So far in this text, we have described a simple relationship in which the core has one refractive index and the cladding another, and there is a single interface between the two.

This is known as *step-index fiber*. If a step index fiber has a small core and only permits one path for light, it is referred to as a *single-mode* optical fiber. However, if the fiber has a large enough core diameter with room for the light to take a number of different possible paths, or modes, by way of reflection, it is referred to as *multimode* fiber.

Multimode Step-Index Fiber

Also known simply as step-index fiber, *multimode step-index fiber*, as already discussed, has a core with a single refractive index and a cladding with another, slightly lower refractive index. These are separated by a single interface, which reflects light that hits it at any angle greater than its critical angle. The large core of a step-index multimode optical fiber allows a ray of light to take many possible paths, as shown in Figure 4.10.

FIGURE 4.10 Light propagation profiles



The result is that even though all of the rays pass through the same length of fiber, the reflections create a longer path for the light to follow. The more reflections, the longer the path. You would get the same result if you and a friend walked down the same road at the same rate of speed, with you walking down the middle of the road and your friend zigzagging from one side of the road to the other. Even if your friend started out before you did, you would eventually pass him, because he is taking a longer path. As a result, you would arrive at the end of the path before your friend, even though you started out together.

The same effect occurs as light is reflected through the fiber. Even though light rays enter the fiber in the order 1, 2, 3, they may arrive in the order 3, 1, 2, or even overlap each other

as they arrive, causing the information they carry to become muddled and useless. This effect is called *modal dispersion*, and is an important factor limiting bandwidth in optical fiber.

Typically, modal dispersion in step-index fiber offsets light rays by about 15 to 30ns/km, depending on the diameter of the fiber core and the wavelength of the light. In other words, if one ray takes the most direct path through the fiber, and another ray takes the longest possible path by reflecting off the fiber walls, the ray taking the longest path will follow the ray taking the shortest path by anywhere from 15 to 30 nanoseconds (15 to 30 billionths of a second) for every kilometer of fiber length. That means that for every kilometer of fiber length, each bit would have to be separated from the one before it by at least 30ns, or roughly 6m, to ensure that the bits would not overlap. This results in a data rate limit of 33.33MHz for a 1km fiber, 16.67MHz for a 2km fiber, 11.11MHz for a 3km fiber, and so on—far below the capabilities of transmitters, receivers, and processors, and completely unsuited for long-distance fiber-optics transmission. For this reason multimode step-index optical fibers are not used in telecommunications and their performance is not addressed in any of the standards discussed in this text. PCS, HCS, and plastic optical fibers are typically step-index.

Two methods used to reduce modal dispersion are multimode graded-index fiber and single-mode step-index fiber.

Multimode Graded-Index Fiber

Multimode graded-index fiber tackles the problem of modal dispersion by increasing the speed of the high-order mode light rays, allowing them to keep up with the low-order mode rays.

The fiber accomplishes this by using the very principles upon which fiber optics is based: the laws of refraction.

Remember that when light passes from a higher refractive index medium to a lower refractive index medium, it gains velocity. A *graded-index* fiber core actually consists of many concentric glass layers with refractive indexes that decrease with the distance from the center. Viewed on end, these layers resemble the rings of a tree.

Any light that passes straight through the fiber travels at a constant speed. If a light ray enters at an angle, however, it passes through the graded layers. As it does, two things happen.

First, the light is refracted away from normal, because the refractive index has decreased. Second, the light propagation velocity increases. This happens in every new layer the light traverses, until it has reached the cladding or has been bent to the critical angle for one of the layers. At this point, the light is reflected and begins its new direction, this time refracting toward normal and slowing down until it reaches the highest refractive index, at the center of the optical fiber core. It then passes through the center and begins the cycle again on the opposite side of the fiber core. The path resembles a segmented sine wave, as shown in Figure 4.10.

Note that as the light follows a longer path, its average velocity increases, offsetting the extra distance it must travel. The time it takes to move from one end of the fiber to the other is now much closer to that of light following a straight path through the fiber. In fact, modal dispersion in graded-index fiber can be reduced to as little as 1ns/km.

Single-Mode Step-Index Fiber

Single-mode step-index fiber uses another approach to reduce modal dispersion. It doesn't give the light enough space to follow anything but a single path through the fiber.

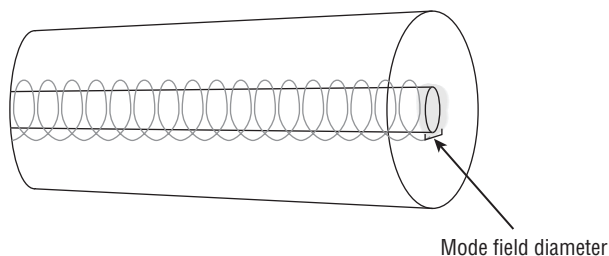
Single-mode fiber uses a core so small that light can only travel in one mode. The diameter of the single-mode fiber core is typically only about 8 to 10 μm , but its performance also depends on the wavelength of the light.

For example, if a fiber is designed to carry a wavelength of 1310nm in only one mode, there is room for light at 850nm to travel in several modes, so the fiber has become a multimode fiber at the shorter wavelength. The shortest wavelength at which the fiber carries only one mode is called the *cutoff wavelength*. The cutoff wavelength for a fiber designed to carry 1310nm in single mode is about 1260nm.

Even in single-mode fiber, light does not follow a straight path. Because of the way light propagates, it follows a corkscrew-like path through the fiber core and propagates in a portion of the cladding, as shown in Figure 4.11. As a result, manufacturers must make single-mode fiber with cladding that will carry the light with less attenuation.

The unique light propagation characteristics of single-mode fiber also make it necessary to take the light propagated in the cladding into account when matching fiber sizes for connections, so the light in the cladding is not lost. Single-mode fibers are typically specified by their *mode field diameter* rather than their core and cladding size. As you can see in Figure 4.11, the mode field diameter is the real estate used by the light within the core and the cladding as it propagates.

FIGURE 4.11 Light propagation in single-mode fiber



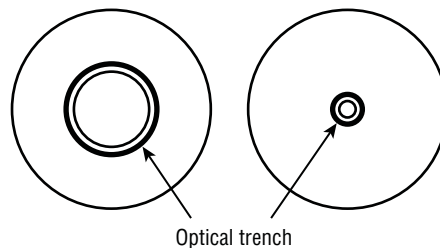
Depressed-Clad Fiber

Multimode and single-mode depressed clad optical fibers are commonly referred to as *bend insensitive*. Bringing fiber to the home or an apartment complex can require the cable to be bent sharply. Bending an optical fiber can cause loss, and single-mode optical fibers can experience considerable loss at longer wavelengths such as 1550nm.

To reduce attenuation from bending, fiber manufacturers created multimode and single-mode optical fibers that have very little loss when bent sharply. These optical fibers feature proprietary refractive index profiles similar to the depressed clad refractive index profiles shown in Figure 4.9.

These fibers are manufactured with an optical trench as shown in Figure 4.12. The *optical trench* surrounds the core of the optical fiber. It has a lower refractive index than the core or the cladding. The low refractive index acts as a barrier and prevents light from escaping the core of the optical fiber when it is bent, reducing the amount of attenuation. The exact location, size, and refractive index value of the trench is typically proprietary and not published.

FIGURE 4.12 The optical trench surrounding the core of multimode and single-mode optical fibers



Single-mode depressed-clad optical fiber has three levels of refractive indexes. Like all optical fibers, the core has the highest refractive index. However, the cladding has two refractive indexes, the refractive index of the cladding and the refractive index of the optical trench. This refractive index profile resists attenuation from bends with a small radius better than step index.

Multimode depressed-clad optical fiber is typically referred to as *bend insensitive multimode optical fiber* (BIMMF). Like single-mode depressed-clad optical fiber, an optical trench surrounds the core. These optical fibers typically offer a tenfold reduction in bending loss without any impact on bandwidth performance.

Summary

This chapter covered the structure and components of optical fiber. We discussed the different configurations and materials used in optical fibers, and the benefits and drawbacks of each.

We also discussed how optical fiber is manufactured, paying attention to different manufacturing techniques leading to the final production process.

Finally, we looked at the principles behind the construction of different types of optical fiber, and the importance of selecting the correct optical fiber for the requirements that it must meet.

Exam Essentials

Describe the main optical fiber components and their basic function. Be able to describe the three components of an optical fiber. Understand the purpose of each component and their physical attributes.

Know the major standards organizations. Know the names and acronyms for the major standards organizations that define the performance of the optical fiber, fiber-optic cable, and components such as connectors and splices.

Describe the major optical fiber classifications by material. Know that optical fibers are typically made with a glass core and glass cladding. Be able to describe the other materials used to manufacture an optical fiber. In addition, be able to list the different fiber classifications that have evolved to suit different conditions, cost factors, and performance requirements.

Understand tensile strength. Be able to define tensile strength and explain how it affects the way fiber must be handled during installation.

Understand the manufacturing of optical fiber. Understand the different processes used in the first stages of manufacturing optical fiber, as well as the final process of drawing the fiber. Know the basic terms used in the manufacturing process.

Understand modes in an optical fiber. Understand what a mode is and know that the number of modes in an optical fiber varies depending on the diameter of the core, the wavelength of the light source, and the numerical aperture.

Describe the refractive index profiles and propagation of light through multimode and single-mode optical fibers. Understand that a refractive index profile graphically represents the relationship between the refractive index of the core and that of the cladding. Describe the common profiles found on optical fibers used in telecommunications. Be able to describe how light propagates through multimode and single-mode optical fibers.

Describe the location, function, and advantage of an optical trench. Understand that the optical trench surrounds the core of the optical fiber. Know that it has a lower refractive index than the core or the cladding and that it acts as a barrier preventing light from escaping the core when the optical fiber is bent, thereby reducing the amount of attenuation.

Review Questions

1. Listed from innermost to outermost, the components of an optical fiber are

- A. Cladding, core, coating
- B. Core, cladding, coating
- C. Cladding, coating, core
- D. Core, coating, cladding

Hint: The center component has the highest refractive index.

2. An optical fiber's tensile strength refers to

- A. How much pulling it can handle
- B. How much light it can carry
- C. The wavelength of light it can carry
- D. How much heat it can resist

Hint: The ability to resist stretching

3. Which standard is applicable to premises cabling components?

- A. ITU-T G.652
- B. ITU-T G.655
- C. IEC 60793-2-10
- D. ANSI/TIA-568-C

Hint: Defines the performance of both multimode and single-mode optical fibers

4. In most manufacturing processes, the actual fiber starts out as

- A. Sand
- B. Soot
- C. Cloth
- D. Powder

Hint: It is deposited on the mold or form.

5. Which refractive index profile is found on bend insensitive multimode or single-mode optical fibers?

- A. Depressed clad
- B. Single
- C. Step
- D. Graded

Hint: Bend insensitive optical fibers have an optical trench.

6. The number of modes possible in a length of optical fiber depends on the diameter of the core, the wavelength of the light source, and the _____.
- A. Wavelength
 - B. Numerical aperture
 - C. Refractive index
 - D. Frequency

Hint: The cone of acceptance

7. Modes can be described as
- A. Potential paths for light
 - B. Potential wavelengths of light
 - C. Potential paths for optical fiber
 - D. Potential uses for optical fiber

Hint: They decrease as the core diameter decreases.

8. Which of these numbers could describe the number of modes in a fiber?
- A. 27.3
 - B. 425.1
 - C. 675
 - D. 218.75

Hint: There are no partial mode optical fibers.

9. A refractive index profile expresses
- A. The n of the core
 - B. The n of the cladding
 - C. The sum of the core n and the cladding n
 - D. The relationship between the core n and the cladding n

Hint: It is presented graphically.

10. Modal dispersion is caused when
- A. Light escapes into the cladding from the core.
 - B. Light rays follow different paths along the fiber core.
 - C. A fiber is split into several parts.
 - D. Light encounters a break in the fiber.

Hint: The more reflections there are, the longer the path.

11. Multimode graded-index fiber overcomes modal dispersion by
- A. Increasing the velocity of the low-order mode light rays
 - B. Increasing the amplitude of the low-order mode light rays
 - C. Increasing the velocity of the high-order mode light rays
 - D. Increasing the amplitude of the high-order mode light rays

Hint: Light rays that travel the furthest travel the fastest.

12. Single-mode fibers overcome modal dispersion by
- A. Routing each mode to a different receiver
 - B. Assigning each mode a different wavelength
 - C. Eliminating one mode
 - D. Only allowing one mode to pass

Hint: Has a smaller core than multimode

13. A single-mode fiber can become a multimode fiber if
- A. The wavelength is short enough.
 - B. The light is bright enough.
 - C. The data rate is high enough.
 - D. The fiber is long enough.

Hint: There is room for more than one mode at 850nm.

14. In a single-mode fiber, the mode field diameter describes
- A. The diameter of the core
 - B. The diameter of the cladding
 - C. The diameter of the light
 - D. The diameter of the entire fiber

Hint: The light travels in the core and part of the cladding.

15. The optical trench surrounds the _____ of multimode or single-mode bend insensitive optical fibers.
- A. Core
 - B. Cladding
 - C. Coating
 - D. Buffer

Hint: It has the lowest refractive index.

Chapter Exercises

Select the Proper Optical Fiber Coating for a Harsh Environment

The coating is the true protective layer of the optical fiber; it absorbs the shocks, nicks, scrapes, and even moisture that could damage the cladding. Without the coating, the optical fiber is very fragile. The coating found on an optical fiber is selected for a specific type of performance or environment.

Choose a coating for an optical fiber that must operate in temperatures as high as 175° C.

Coatings found on optical fibers include acrylate, silicone, carbon, and polyimide. Of these four, acrylate has the lowest operating temperature while carbon and polyimide have the highest. Silicone has a much higher operating temperature than acrylate—up to 200° C. While polyimide can operate at temperatures up to 350° C, it is not required for this application. Remember carbon cannot be used by itself and is always combined with another coating.

Identify an Industry Standard that Defines the Specific Performance Characteristics on an Optical Fiber

There are many standards related to optical fiber. This chapter focused on the standards that define the performance of optical fibers used in the telecommunications industry published by TIA, ITU, ISO, and the IEC.

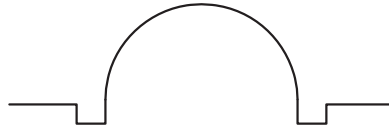
You have been placed in charge of selecting a contractor to install fiber-optic cabling in a new building. Which standard or standards should you review prior to selecting the best contractor?

The ANSI/TIA-568-C standards and the ISO/IEC 11801 standard are applicable to premises cabling components and the new building falls under the definition of premises.

Identify an Optical Fiber from Its Refractive Index Profile

The refractive index profile graphically represents the relationship between the refractive index of the core and that of the cladding. Several common profiles are found on optical fibers used in telecommunications, avionics, aerospace, and space applications.

The refractive index profile of an optical fiber is shown in the following graphic. Determine the type of optical fiber.



The optical fiber in the graphic has a graded index with an optical trench. The graded index indicates multimode and the optical trench indicates bend insensitive; this is a BIMMF.

Calculate the Numerical Aperture of an Optical Fiber

To find the numerical aperture of an optical fiber, you need to know the refractive index of the core and the cladding.

The core of an optical fiber has a refractive index of 1.51 and the cladding has a refractive index of 1.46. Calculate the numerical aperture.

Using the formula shown here, square the value for the refractive index of the core and then do the same for the cladding. Subtract the cladding value from the core value and take the square root of that value. Your answer should be 0.385.

$$NA = \sqrt{1.51^2 - 1.46^2} = 0.385$$

Calculate the Number of Modes in an Optical Fiber

To find the number of modes in an optical fiber, you need to know the diameter of the core, the wavelength of the light source, and the numerical aperture.

Calculate the number of possible modes in an optical fiber with the following specifications:

Core diameter = 62.5μm

Core refractive index = 1.48

Cladding refractive index = 1.46

Wavelength of the light source = 1300nm

First, determine the numerical aperture of the fiber's core using this equation:

$$NA = \sqrt{1.48^2 - 1.46^2} = 0.242$$

Next, use the equation $M = (D \times \pi \times NA/\lambda)^2/2$ to find the number of modes.

$$M = (62.5 \times 10^{-6} \times \pi \times 0.242/1300 \times 10^{-9})^2/2 = 667.9 \text{ modes}$$

Since you cannot have a fraction of a mode, the number of possible modes is 667.

Chapter 5

Optical Fiber Characteristics

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe dispersion in an optical fiber.
- ✓ Describe modal dispersion and its effects on the bandwidth of an optical fiber.
- ✓ Describe material dispersion and its effects on the bandwidth of an optical fiber.
- ✓ Explain waveguide dispersion in a single-mode optical fiber.
- ✓ Explain chromatic dispersion in an optical fiber.
- ✓ Explain polarization mode dispersion in a single-mode optical fiber.
- ✓ Describe how dispersion affects bandwidth in an optical fiber.
- ✓ Describe the causes of attenuation in an optical fiber.
- ✓ Describe attenuation versus wavelength in an optical fiber.
- ✓ Describe a microbend in an optical fiber.
- ✓ Describe a macrobend in an optical fiber.
- ✓ Explain the difference between a bend sensitive and bend insensitive single-mode optical fiber.
- ✓ Explain the difference between a bend sensitive and bend insensitive multimode optical fiber.
- ✓ Describe the numerical aperture of an optical fiber.



- ✓ Explain how optical fibers are designated in ISO/IEC 11810.
- ✓ Explain how optical fibers are designated in IEC 60793-2-10 and IEC 60793-2-50.
- ✓ Describe how optical fibers are designated in ANSI/TIA-568-C.
- ✓ Describe how the International Telecommunications Union (ITU) designates optical fibers.
- ✓ Describe the performance characteristics of ANSI/TIA-568-C.3 and ISO/IEC 11801-recognized optical fibers.
- ✓ Describe the performance characteristics of ITU-T G.652, ITU-T G.655, and ITU-T G.657 single-mode optical fibers.
- ✓ Describe the attenuation and bandwidth characteristics of HCS/PCS and plastic optical fibers.





In addition to factors such as construction, materials, and size as discussed in the previous chapter, optical fibers have certain performance or operational characteristics that define them.

These characteristics may describe limitations or features of the fiber with regard to its light-carrying ability under various conditions, and are generally affected by its physical properties.

In this chapter, we describe the characteristics of optical fiber that affect the way it is selected, handled, installed, and used. We'll cover in detail how these characteristics change a fiber's ability to carry light, as well as the methods used to take advantage of some characteristics while minimizing the effects of others.

In this chapter, you will learn to

- Calculate the attenuation in dB for a length of optical fiber
- Calculate the usable bandwidth for a length of optical fiber
- Calculate the total macrobending loss in a single-mode fiber-optic cable
- Calculate the total macrobending loss in a multimode fiber-optic cable
- Calculate the acceptance angle for an optical fiber
- Determine the latest published revision of a standard using the Internet

It All Adds Up

If you've ever looked at your bank balance and wondered where it all went, you have some idea of why it's important to understand optical fiber characteristics in great detail.

Chances are, when you took another look at your account, you remembered that some of it went to necessities, some to entertainment, some to service charges, and so on. It didn't all go away at once: it was spent little by little, in many different places. Any single expenditure may have gone almost unnoticed, but over time all of those expenses can add up to a significant amount.

This is typically the fate of light as it passes through an optical fiber. By the time it reaches the other end, it is diminished in several ways—most notably in its power and its ability to carry a signal. In spite of the purity of the materials that go into fibers, they are still not perfect—in part by design, and in part because there are factors that simply cannot be overcome with current technology.

Going back to your bank account, if you know where the money goes, you can take steps to make it last longer or make sure that when it is spent, you get more use out of it.

The same goes for our light beam. If you understand the characteristics in an optical fiber that take away from its ability to carry a signal at high speed and over long distances,

it is possible to overcome them or use them to your advantage. Remember, though, that propagation of light within a fiber is a complex mix of influences, and each of the characteristics that we'll be discussing affects different portions of this mix.

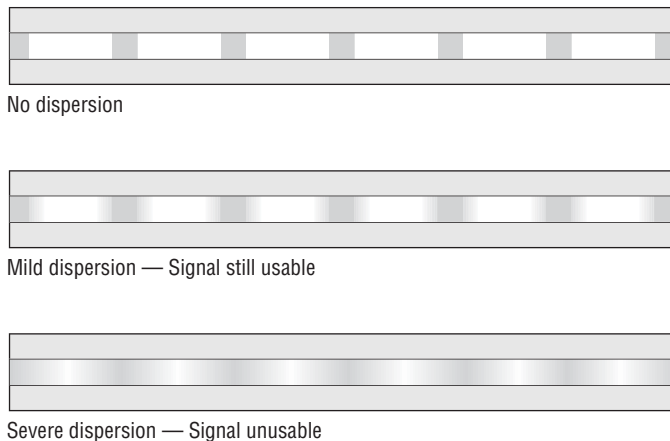
Some aspects of these characteristics have been covered in previous chapters to describe why fiber is constructed in certain ways. Now we are going to look at them more closely.

Dispersion

In general, *dispersion* is the spreading of light as it travels away from its source. The light spreads because different components of it travel at slightly different velocities or take slightly different paths, depending on the conditions in the medium through which it is traveling and the wavelengths that make up the light. There are different kinds of dispersion, however, and the kind that is taking place depends on several factors in the optical fiber and in the light itself.

The greatest effect of dispersion is that as the light spreads, it can degrade or destroy the distinct pulses of the digital signals in the light by making them overlap each other, as shown in Figure 5.1, blurring and blending them to the point that they are unusable. The effect grows more pronounced as the distance the light travels increases.

FIGURE 5.1 The effects of dispersion on a signal



The effect is similar to looking into a hallway through a frosted glass window. If people are moving through the hallway close together, the glass spreads their images so much that they merge with one another and look like a single mass rather than individuals. If they spread out far enough from each other, however, you can see each person moving past the window. The images are still spread out, but the space between each person is great enough to see.

To prevent signal degradation due to dispersion, it is necessary to keep the pulses far enough apart to ensure that they do not overlap. This limits the signals to a bit rate that is low enough to be only minimally affected.

Dispersion limits the optical fiber's *bandwidth*, or the amount of information it can carry. Fortunately, optical fiber manufacturers and standards organizations provide information about the bandwidth or the information-carrying capacity of optical fibers at specific wavelengths.

The types of dispersion that affect optical fiber performance are

- Modal dispersion
- Material dispersion
- Waveguide dispersion
- Chromatic dispersion
- Polarization-mode dispersion

Let's look at these types of dispersion more closely.

Modal Dispersion

We mentioned modal dispersion in the previous chapter to explain why fibers are classified as multimode or single-mode. It will help to review some of the important points.

Modal dispersion results from light taking different paths, or modes, as it passes through the optical fiber. The number of potential modes the light can take is determined by the diameter of the optical fiber core, the refractive indices of the fiber core and cladding, and the wavelength of the light.

A mode can be a straight line through the fiber, or the light can follow an angular path, resulting in reflections every time the light meets the interface between the core and the cladding. The more reflections, the longer the path through the fiber, and the longer the light takes to pass through it.

Depending on the mode, some parts of the light will pass through the fiber in less time than others. The difference in travel time can cause parts of the light pulses to overlap each other, or in extreme cases, to arrive in a different order from the order in which they were transmitted. If this happens, the signal is no longer usable.

Methods for overcoming modal dispersion include

Lower bit rate Lowering the bit rate increases the gap between bits in the signal. Although dispersion will still affect the bits, they will not overlap one another, and the signal will still be usable.

Graded-index fiber *Graded-index fiber* gradually reduces the refractive index of the fiber core from the center toward the cladding, allowing the light that follows a more angled path to speed up as it leaves the center and causing it to slow down again as it reaches the center. This effect reduces the difference in travel time between modes and allows greater bandwidths and higher bit rates. Graded-index fiber is a moderately priced solution that allows wider bandwidths than multimode step-index fiber. In addition, the gradual change of indexes as the light heads for the cladding causes the light to curve back into the core of the fiber before it has a chance to approach the cladding at a penetrating angle.

Reducing the core size or increasing the wavelength In the previous chapter, we calculated the number of potential modes in a multimode optical fiber. For a given wavelength, a smaller core optical fiber will have fewer modes than a larger core optical fiber. For a given core size, a multimode optical fiber will have fewer modes at a longer wavelength than a shorter wavelength.

Single-mode fiber Single-mode fiber has a core that is small enough for only one mode to propagate, eliminating the problems caused by multiple modes. This type of fiber requires more expensive connectors and equipment because of the small core size and is typically used when very high data rates over long distances justify the cost.

Material Dispersion

Material dispersion occurs when different wavelengths of light travel at different velocities in the optical fiber. Even the light from a laser is made up of different wavelengths within a narrow range called the *spectral width*, which varies depending on the light source. Depending on the wavelength, a *light-emitting diode (LED)* may have a spectral width as narrow as 20nm or as wide as 170nm. A laser's spectral width is much narrower than that of the LED. Depending on the laser type, the spectral width can range from 0.1nm to 3nm.

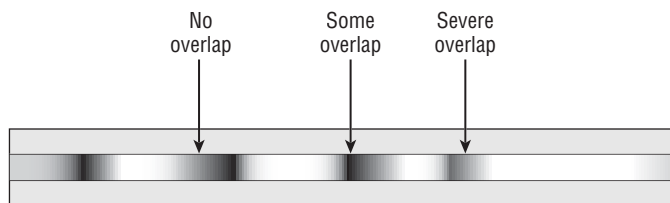
Recall the formula for determining the refractive index of a material:

$$n = c/v$$

where n is the refractive index, c is the speed of light in a vacuum, and v is the speed of the wavelength of light through the material. In this equation, n changes with the wavelength of the light passing through the material. Remember that this is why white light breaks into its component colors in a prism.

When the different wavelengths travel at different velocities, the slower wavelengths begin to lag behind as the light travels down the optical fiber core, causing the light to spread as shown in Figure 5.2. If the light pulses must travel a great distance, the lag in the slower wavelengths can cause them to overlap the faster wavelengths of the bits following them. As with modal dispersion, these overlaps can degrade and ultimately destroy the signal.

FIGURE 5.2 Material dispersion in fiber causes some wavelengths to travel more slowly than others.



Because the light sources used in fiber-optic transmissions have a narrow spectral width, material dispersion takes place on a much smaller scale than modal dispersion.

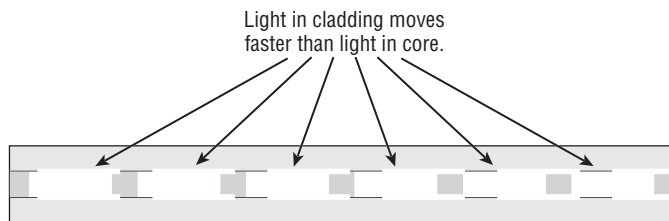
Its effects in a fiber are measured in picoseconds per nanometer of spectral width per kilometer (ps/nm/km) and are insignificant in a multimode fiber when compared to the effects of modal dispersion.

Material dispersion becomes a problem only when modal dispersion is overcome with single-mode fiber and higher data rates are used over long distances.

Waveguide Dispersion

Waveguide dispersion occurs in single-mode fiber as the light passes through not only the core, but also part of the cladding, as shown in Figure 5.3. By design, the core has a higher refractive index than the cladding, so the light will travel more slowly through the core than through the cladding.

FIGURE 5.3 Waveguide dispersion in optical fiber



While the difference in refractive indexes of a single-mode fiber's core and cladding are minuscule, they can still become a factor over great distances. In addition, waveguide dispersion can combine with material dispersion to create another problem for single-mode fiber: chromatic dispersion (discussed in the next section).

Various tweaks in the design of single-mode fiber can be used to overcome waveguide dispersion, and manufacturers are constantly refining their processes to reduce its effects.

Chromatic Dispersion

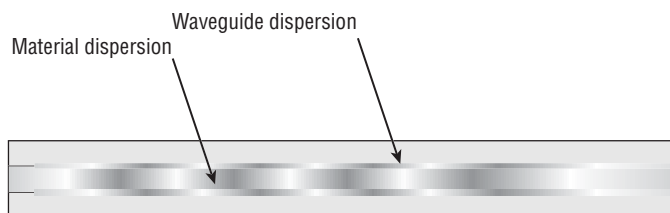
Chromatic dispersion occurs in single-mode optical fiber and results from the combination of effects from material dispersion and waveguide dispersion.

When chromatic dispersion occurs, the effects of material dispersion, as shown in Figure 5.4, compound the effects of waveguide dispersion. At lower bit rates and in multimode fiber, the effects of chromatic dispersion are so small as to be unnoticed, especially when buried under modal dispersion. It is mostly in single-mode fiber carrying high bit data over long distances that the detrimental effects build up.

One way to reduce chromatic dispersion is to take advantage of the fact that the relationship between wavelength, refractive index, and velocity is not linear. In the infrared range of most fiber-optic transmissions, the light's velocity through the medium drops as the wavelength increases until it reaches the range between 1300nm and 1550nm.

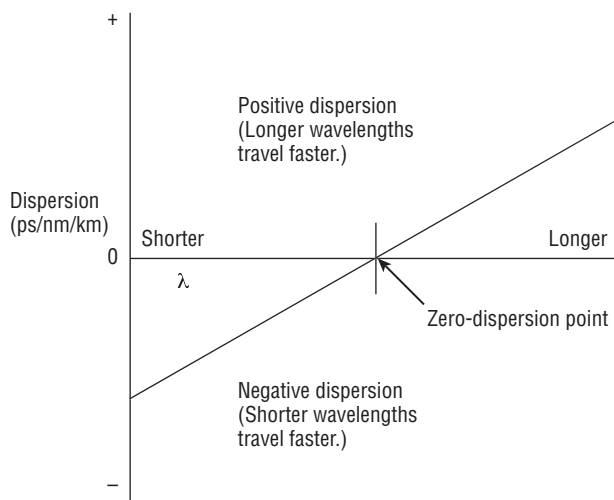
At wavelengths greater than 1550nm, the longer wavelengths have a higher velocity. Somewhere in the 1300nm to 1550nm range there is a crossover where, depending on the specific composition of the fiber, the refractive index is the same for the wavelengths within the narrow spectral width of the light being transmitted. In other words, as shown in Figure 5.5, as the wavelength approaches this range, dispersion drops to zero. This *zero-dispersion point* normally occurs at 1300nm in a standard single-mode fiber. Unfortunately, other characteristics of the optical fiber attenuate the signal at this wavelength, making it unusable for long-distances.

FIGURE 5.4 Waveguide dispersion and material dispersion combine to create chromatic dispersion.



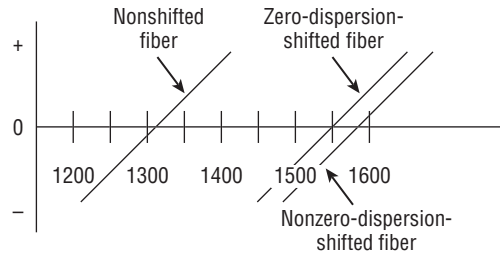
There are two approaches to reduce chromatic dispersion. One approach is to use *dispersion-shifted fiber* and the other is to reduce the spectral width of the light source. The spectral width of the light source is discussed in Chapter 10, “Fiber-Optic Light Sources and Transmitters.”

FIGURE 5.5 Dispersion profile of a typical optical fiber



Dispersion-shifted fiber, sometimes called *zero-dispersion-shifted fiber*, is made so that the zero-dispersion point is shifted from 1300nm to 1550nm, where the attenuation is lowest, as shown in Figure 5.6. Shifting the zero-dispersion point to 1550nm allows longer transmission distances because of less attenuation. Dispersion-shifted fiber is used when high data rates over long distances are required.

FIGURE 5.6 Zero-dispersion points of nonshifted, dispersion-shifted, and non-zero-dispersion-shifted fibers

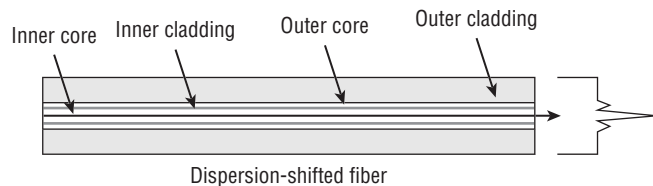


A variation of the dispersion-shifted fiber, *non-zero-dispersion-shifted fiber (NZ-DSF)*, is used when multiple wavelengths are being used to send more than one channel through the fiber, a process known as *wavelength division multiplexing (WDM)*. When WDM takes place at the zero-dispersion point, an effect known as *four-wave mixing* occurs, in which multiple wavelengths combine to form new wavelengths. These new wavelengths can interfere with the signal carrying wavelengths, reducing their power and introducing noise into the system.

To reduce four-wave mixing, NZ-DSF is used to move the dispersion point just off the 1550nm mark. The slight amount of dispersion introduced minimizes four-wave mixing while preserving most of the benefits of dispersion-shifted fiber.

The refractive index profile of dispersion-shifted fiber is shown in Figure 5.7. The peak in the center of the profile reveals an inner core that has its highest refractive index at the center. The refractive index gradually decreases toward a thin inner layer of cladding. The smaller peaks represent a ring of glass with a higher refractive index surrounding the inner cladding, slowing the light that would normally increase its velocity in the cladding. This effect reduces waveguide dispersion, which in turn reduces chromatic dispersion.

FIGURE 5.7 Refractive index profile of dispersion-shifted fiber





Real World Scenario

Mixing It Up—The Problem of Four-Wave Mixing

It seems that nobody can leave a good thing alone. Once engineers overcame the problem of chromatic dispersion in single-mode optical fiber using dispersion shifting, they decided to squeeze all the use they could out of it by piling on different wavelengths to create multiple transmission channels. The idea behind this is that different wavelengths can actually occupy the same space but remain distinct from one another until they are sorted out at the other end of the fiber link.

It seemed to make very good sense, but then another problem cropped up. The wavelengths used in the multiple channels must stay near the zero-dispersion range of 1550nm, so you end up with individual channels only 2nm apart, typically at 1546, 1548, 1550, and 1552nm, for example. It's difficult for any two things to be only 2nm apart and not interact, so interact they do. In fact, they create new wavelengths that can interfere with the wavelengths that are part of the transmission.

The problem gets exponentially worse as the number of wavelengths being transmitted increases. The formula for predicting the number of new waves created is

$$FWM = (n^2 (n - 1))/2$$

where FWM is the number of waves created and n is the number of wavelengths being transmitted through the fiber. So if 2 wavelengths are being used, an extra 2 wavelengths will appear. That's not too bad in itself. But if you are transmitting 4 original wavelengths, 24 extra wavelengths are created:

$$FWM = (16 \times 3)/2 = 24$$

and 8 original wavelengths will produce 224 extra wavelengths:

$$FWM = (64 \times 7)/2 = 224$$

The solution to four-wave mixing actually involves creating just enough dispersion in the fiber to render the newly created wavelengths harmless to the signals while leaving the original signal clear enough to use. The fiber created for this purpose is *non-zero-dispersion-shifted fiber (NZ-DSF)*. Non-zero-dispersion shifting moves the zero-dispersion point slightly away from the wavelength used for the transmission, usually about 10nm above or below the transmission wavelength, so there is sufficient dispersion to keep the effects of four-wave mixing to minimum.

Reduced Spectral Width

Because material dispersion is caused by an overabundance of wavelengths in the optical signal, the simplest solution is to reduce the number of wavelengths by reducing the spectral width.

Recall that dispersion is expressed as picoseconds per nanometer of spectral width per kilometer of fiber, so any reduction in spectral width will have a significant effect on the amount of material dispersion.

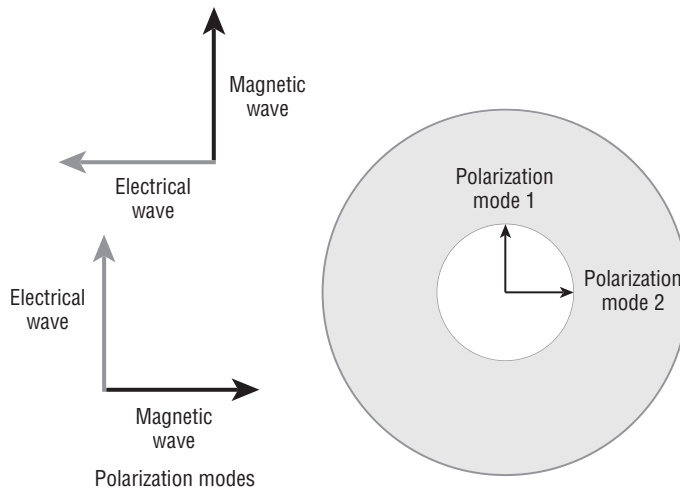
Polarization-Mode Dispersion

Polarization-mode dispersion (PMD) is masked by other forms of dispersion until the bit rate exceeds 2.5Gbps. In order to understand PMD, we must look at an information pulse more closely.

Recall that light is an electromagnetic wave, consisting of an electrical and a magnetic wave traveling at right angles to one another. The orientation of the two waves along the path of propagation determines the light's *polarization mode*, or *polarity*.

As shown in Figure 5.8, it is possible to have different polarities of light traveling through the fiber in a signal, occupying different parts of the fiber as they pass through it. Because no fiber is perfect, there will be obstacles in one part of the fiber that are not present in another. As a result, the light having one polarity may pass an area without interference, while light having another polarity may pass through a defective region, slowing it down.

FIGURE 5.8 Polarized light shown in a cross section of optical fiber



Polarization-mode dispersion is not so much a function of the fiber's overall characteristics as it is a result of irregularities, damage, or environmental conditions such as temperature. Small areas of damage called *microbends* can cause PMD, as can fiber that is not perfectly round or concentric.

Because PMD is caused by specific conditions within the fiber, and not the fiber's overall characteristics, it is difficult to assign a consistent PMD value to a length of fiber.

The exact amount of PMD changes with external conditions, the physical condition of the fiber, and the polarization state of the light passing through it at any given moment. For this reason, PMD is measured in terms of the total difference in the travel time between the two polarization states, referred to as the *differential group delay (DGD)* and measured in picoseconds (ps).

How Dispersion Affects Bandwidth

While different types of dispersion have different causes and are measured different ways (ns/km, ps/nm/km), all of them have one effect in common: they place a limit on the bandwidth of optical fibers. The bandwidth of a specific brand of optical fiber can be found in the manufacturer's datasheet. The bandwidth of a type of optical fiber can also be found in an industry standard. Remember, an industry standard defines the minimum performance levels; optical fibers are available that exceed industry standards.

The product of bandwidth and length (MHz · km) expresses the information carrying capacity of a multimode optical fiber. Bandwidth is measured in megahertz (MHz) and the length is measured in kilometers. The MHz · km figure expresses how much bandwidth the fiber can carry per kilometer of its length. The fiber's designation must always be greater than or equal to the product of the bandwidth and the length of the optical fiber.

The minimum bandwidth-length product for a 50/125μm laser optimized OM4 multimode optical fiber as defined in ANSI/TIA-568-C.3 is 3500MHz · km with an overfilled launch at a wavelength of 850nm. To find the usable bandwidth for this optical fiber at a length of 1250 meters, or 1.25km, you must change the equation around and solve as shown here:

$$\text{Bandwidth-length product} = 3500\text{MHz} \cdot \text{km}$$

$$\text{Optical fiber length} = 1.25\text{km}$$

$$\text{Bandwidth in MHz} = 3500 \div 1.25$$

$$\text{Bandwidth} = 2800\text{MHz}$$

As you can see, the bandwidth decreases as the optical fiber length increases. However, the opposite is true when the length of the optical fiber decreases as shown here:

$$\text{Bandwidth-length product} = 3500\text{MHz} \cdot \text{km}$$

$$\text{Optical fiber length} = 0.125\text{km}$$

$$\text{Bandwidth in MHz} = 3500 \div 0.125$$

$$\text{Bandwidth} = 28000\text{MHz}$$

Bandwidth-length product is defined in ANSI/TIA-568-C.3 for multimode optical fiber, but not for single-mode optical fiber. The bandwidth for a length of a single-mode optical fiber is specified by dispersion for a specific wavelength. The spectral width of the light source will affect the bandwidth. For a given length of optical fiber, a laser with a wide spectral width will have less bandwidth than a laser with a narrower spectral width.

Spectral width can vary tremendously among lasers. Laser sources are covered in detail in Chapter 10.

Attenuation

The *attenuation* of a fiber-optic signal is the loss of optical power as the signal travels through the fiber. There are two types of attenuation: intrinsic and extrinsic. *Intrinsic attenuation* occurs because no manufacturing process can produce a perfectly pure fiber. Either by accident or by design, the fiber will always have some characteristics that attenuate the signal passing through it. *Extrinsic attenuation* is caused by external mechanisms that bend the optical fiber. These bends are broken into two categories: *microbends* and *macrobends*, which are discussed in detail later in this chapter.

The wavelength of the light passing through the fiber also affects attenuation. In general, attenuation decreases as wavelength increases, but there are certain wavelengths that are more easily absorbed in plastic and glass fibers than others. One of the reasons for establishing standard operating wavelengths of 850nm, 1300nm, and 1550nm in glass fiber and in the visible range of 650nm for plastic is because the wavelengths in between are considered high-loss regions. Specifically, these wavelengths for glass optical fiber are in the range of 730, 950, 1250, and 1380nm.



Attenuation provides a good example of the superiority of fiber over copper for carrying signals. When an electrical signal is carried through copper wire, attenuation increases with the data rate of the signal, requiring an increase in transmission power or, more often, the use of repeaters. Attenuation per unit length in an optical signal for an optical fiber of a given type is constant no matter what the data rate.

Remember from Chapter 2, “Principles of Fiber-Optic Transmission,” that attenuation is measured in *decibels* (dB). Decibels help us account for the loss of power when we are measuring attenuation in an optical fiber. In glass optical fiber, attenuation is lowest at long wavelengths and highest at the short wavelengths. This is not true for plastic optical fiber. Plastic optical fiber is typically designed for visible light sources with a center wavelength of 650nm. Attenuation increases dramatically above or below this wavelength.

Attenuation values for a length of optical fiber can be calculated using the attenuation coefficient for a specific type of optical fiber. This information can be found in the manufacturer’s datasheet or in a standard. The “Fiber Specifications and Standards” section, later in this chapter, provides attenuation information for multimode and single-mode optical fibers.

You can calculate the maximum attenuation for a length of optical fiber using this formula.

Attenuation for the length of optical fiber = Attenuation coefficient $\times$ Optical fiber length

Here we calculate the maximum attenuation for a 1.75km length of optical fiber with and attenuation coefficient of 1.5dB/km.

Attenuation coefficient = 1.5dB/km

Optical fiber length = 1.75km

Attenuation for the length of optical fiber = $1.75 \times 1.5 = 2.625$

Maximum attenuation for 1.75km of optical fiber is 2.625dB

Remember that the attenuation is wavelength dependent. The shorter wavelengths will attenuate more and the longer wavelengths less. Also remember that the attenuation coefficient is a “not to exceed” number. In other words, the attenuation should be no greater than that. The measured attenuation will typically be less.

While decibels are useful in measuring *total attenuation*, we can also divide attenuation into two types: *absorption* and *scattering*.

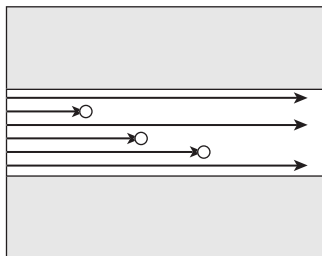
Absorption

The absorption of light is a form of intrinsic attenuation. Absorption accounts for a very small percentage of attenuation in an optical fiber—typically between 3 and 5 percent.

All materials, even the clearest glass, absorb some light. The amount of absorption depends on the type of material and the wavelength of the light passing through it. You can see absorption easily in sunglasses. Even on the brightest days, only a fraction of the light energy passes through the tinted lenses. The wavelengths that do not pass through are mostly absorbed by impurities that have been placed in, or coated on, the lens material.

In an optical fiber, absorption occurs when impurities such as water or ions of materials such as copper or chromium absorb certain wavelengths, as shown in Figure 5.9. By keeping these impurities as low as possible, manufacturers can produce fibers with a minimum of attenuation.

FIGURE 5.9 Absorption in optical fiber

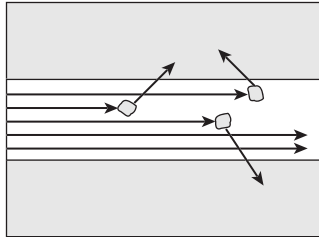


Scattering

The scattering of light is a form of intrinsic attenuation. Scattering accounts for the greatest percentage of attenuation in an optical fiber—typically between 95 to 97 percent.

Scattering is caused by atomic structures and particles in the fiber redirecting light that hits them, as shown in Figure 5.10. The process is called *Rayleigh scattering*, for Lord Rayleigh, a British physicist who first described the phenomenon in the late nineteenth century.

FIGURE 5.10 Scattering in optical fiber

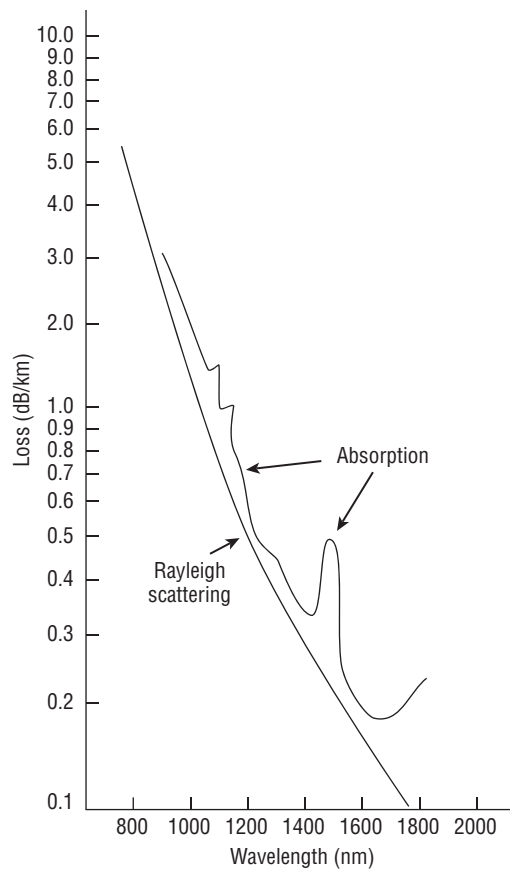


Rayleigh scattering is also the answer to the age-old question “Why is the sky blue?” The blue that we see is actually the more prevalent blue wavelengths of light from the sun being scattered by particles in the atmosphere. As the sun moves toward the horizon and the light must pass through more of the atmosphere, the scattering increases to the point where the blue light is almost completely attenuated, leaving the red wavelengths, which are less affected by the scattering for reasons that we’ll see shortly.

Rayleigh scattering depends on the relationship between wavelength and the size of the structures in the fiber. Scattering decreases as the wavelength of the light approaches the size of the structures, which means that in fiber-optic communications, shorter wavelengths are more likely to be scattered than the longer wavelengths. This is one of the main reasons that infrared wavelengths are used in fiber optics. The relatively long wavelengths of infrared are less subject to scattering than visible wavelengths. It also explains why the sun turns red on the horizon. The shorter blue wavelengths are more likely to be scattered by the similarly sized particles in the atmosphere than are the red wavelengths.

Total Attenuation

Total attenuation is the combination of the intrinsic effects of absorption and scattering in a fiber. Figure 5.11 shows a typical attenuation curve for an optical fiber with the effects of absorption and scattering combined. Note that the general curve is caused by the effects of scattering, while the irregularities in the plot are caused by specific impurities, such as hydroxyl molecules, absorbing light in those wavelengths.

FIGURE 5.11 An optical fiber's attenuation curve

Note also the windows at the 850, 1300, and 1550nm ranges. Remember that while the 1300nm range is better in terms of dispersion, it still has a higher attenuation than the 1550nm range, which is the reason for dispersion-shifted fiber.

Bending Losses

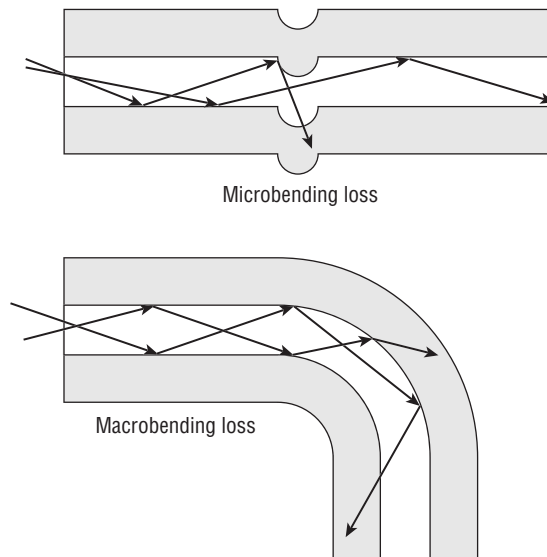
In addition to intrinsic losses within the optical fiber, the actual condition of the fiber can lead to losses. These losses are referred to as *extrinsic losses* and are caused by bends in the optical fiber. These bends, depending on their radius, are referred to as *microbends* or *macrobends*.

Microbends

Microbends are small distortions of the boundary layer between the core and cladding caused by crushing or pressure. Microbends are very small and may not be visible when looking at the fiber-optic cable.

Microbends change the angle of incidence within the fiber, as shown in Figure 5.12. Changing the angle of incidence forces high-order light rays to reflect at angles that prevent further reflection, causing them to be lost in the cladding and absorbed.

FIGURE 5.12 Losses caused by microbending and macrobending



Microbends can be caused by the coating, cabling, packaging, and installation of the cable. Remember from Chapter 4, “Optical Fiber Construction and Theory,” that one of the most common types of coatings is *acrylate*. Acrylate is limited in temperature performance. Exceeding the operating temperature range of the acrylate coating can cause microbending, which can result in significant attenuation.

Macrobends

Compared to a microbend, a macrobend has a much larger radius. Macrobends occur when the fiber is bent around a radius that can be measured in millimeters. As shown in Figure 5.12, these tight radii change the angle of incidence within the fiber, causing some of the light rays to reflect outside of the fiber and, as with microbending, be lost in the cladding and absorbed.

At long wavelengths such as 1550nm to 1625nm, single-mode optical fiber can experience severe attenuation from macrobends. This is not the case with multimode optical fiber. A multimode optical fiber is less sensitive to bending loss than single-mode. However, bend-insensitive single-mode and multimode optical fibers are available for installations where tight bends may be required.

As we learned in Chapter 4, multimode and single-mode depressed clad optical fibers are commonly referred to as bend insensitive. These optical fibers are manufactured with an optical trench that surrounds the core and feature proprietary refractive index profiles.

Single-Mode Macrobending Performance

As you will learn in the “Fiber Specifications and Standards” section later in this chapter, single-mode optical fiber can be broken into two groups: bend sensitive and bend insensitive. There is a considerable difference in the bending attenuation of the two fiber types. ITU Standards G.652 and G.655 define macrobending loss for bend-sensitive optical fiber, whereas G.657 defines macrobending loss for bend-insensitive fiber.

The macrobending loss in each standard is for an optical fiber that has not been placed in a cable. The macrobending loss will vary when the optical fiber is placed in a cable and that information must be obtained from the cable manufacturer. However, to demonstrate how macrobending loss is calculated, it is assumed that optical fiber values listed in the standards are the same as the cable loss.

ITU-T G.652 allows for a maximum attenuation of 0.1dB at 1550nm with 100 turns around a 30mm radius mandrel. The 30mm radius of the mandrel is slightly less than the radius of a tennis ball. Reducing the radius of the bend will increase the attenuation. Optical fiber with a macrobending specification like this is not a good choice for installations that may require several sharp bends, especially if they are much less than 30mm.

ITU-T G.657 defines the macrobending loss for bend-insensitive single-mode optical fiber. This type of optical fiber is typically employed in *passive optical networks (PONs)* that run to your home or residence. Installing a fiber-optic cable in the wall of a home or apartment may require one or more sharp bends. A single sharp 90-degree bend with an optical fiber that is not bend insensitive could cause enough attenuation to prevent the link from functioning.

Macrobending loss is defined several ways in ITU-T G.657. Tables 5.7 through 5.10 in the section “Fiber Specifications and Standards” list the parameters. In Table 5.7, the maximum macrobending loss for one turn around a 10mm radius mandrel at a wavelength of 1550nm is 0.75dB. With this information, it is possible to estimate the macrobending loss for a typical installation. If the number of bends and the radius of each bend are known, the macrobending losses can be calculated.

Let’s look at a single-mode ITU-T G.657 Category A2 bend-insensitive fiber-optic cable installed in a home with three small radius 90 bends. Two bends have a radius of 10mm and one bend has a radius of 7.5mm. The wavelength of the light source is 1550nm.

Referring to Table 5.8, we see that the loss for each single-turn 10mm bend is 0.1dB and the loss for the single-turn 7.5mm bend is 0.5dB.

The total macrobending loss (MBL) will be the sum of the three losses; however, we cannot simply add up the dB values from Table 5.8 because each bend is not a full 360° turn. Both 10mm bends are 90° and the 7.5mm bend is 180°. First you have to calculate the loss for the partial turns using the following formula:

$$(\text{Bend in degrees} \div 360) \times \text{dB loss per turn} = \text{MBL}$$

$$(90 \div 360) \times 0.1\text{dB} = 0.025\text{dB}$$

$$(180 \div 360) \times 0.5\text{dB} = 0.25\text{dB}$$

Then we calculate the total loss using this formula:

$$\text{MBL 1} + \text{MBL 2} + \text{MBL 3} = \text{Total MBL}$$

$$0.025\text{dB} + 0.025\text{dB} + 0.25\text{dB} = 0.3\text{dB}$$

$$\text{Total macrobending loss} = 0.3\text{dB}$$

When the bending loss is not listed for a single turn but a number of turns, you must first establish the single turn loss using the following formula:

$$\text{MBL} \div \text{number of turns} = \text{dB loss per turn}$$

Multimode Macrobending Loss

Unlike single-mode optical fiber, there are currently no standards for bend-insensitive multimode optical fiber (BIMMF). However, as of this writing they are in development and offered by multiple manufacturers with extremely low bending loss at 850nm and 1300nm. Table 5.1 describes the bending loss performance of a generic 50/125μm BIMMF. The macrobending loss is defined for a number of turns of optical fiber around a mandrel at a specified radius and wavelength.

TABLE 5.1 Generic 50/125μm BIMMF bending loss

Radius (mm)	Number of turns	Attenuation (dB) at 850nm	Attenuation (dB) at 1300nm
37.5	100	≤0.05	≤0.15
15	2	≤0.1	≤0.3
7.5	2	≤0.2	≤0.5

Non-bend-insensitive multimode optical fiber macrobending loss is defined in ITU-T G.651.1 and IEC 60793-2-10. It will be covered in greater detail in the “Fiber Specifications and Standards” section later in this chapter.

Numerical Aperture

As we saw in Chapter 4, many of a fiber's characteristics are determined by the relationship between the core and the cladding. The *numerical aperture (NA)* expresses the light-gathering ability of the fiber. The NA is a dimensionless number, meaning that it is to be used as a variable in determining other characteristics of the fiber, or as a means of comparing two fibers.

As discussed earlier, the numerical aperture is determined by the refractive indexes of the core and the cladding:

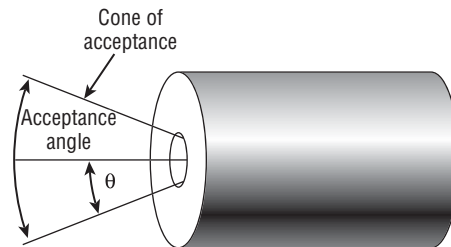
$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{cladding}}^2}$$

where n_1 is the refractive index of the core and n_2 is the refractive index of the cladding.

Recall that in order for light to be contained within a fiber it must stay above the *critical angle*, or the angle at which it reflects off the boundary between the core and the cladding, rather than penetrating the boundary and refracting through the cladding.

To maintain the critical angle, light must enter within a specified range defined by the *acceptance cone* or *cone of acceptance*. As shown in Figure 5.13, this region is defined by a cone extending outside the optical fiber's core. Light entering the core from outside of the cone will either miss the core or enter at an angle that will allow it to pass through the boundary with the cladding and be lost. The *acceptance angle* defines the acceptance cone. Light entering the core of the optical fiber at an angle greater than the acceptance angle may not propagate the length of the optical fiber. For light to propagate the length of the optical fiber, it must enter the core at an angle that does not exceed the acceptance angle.

FIGURE 5.13 The cone of acceptance



The cone of acceptance is determined using the numerical aperture:

$$NA = \sin\theta$$

where θ is the angle defining the cone of acceptance.

To determine the acceptance angle of a fiber with a core refractive index of 1.48 and a cladding refractive index of 1.47, we must determine the NA of the fiber:

$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{cladding}}^2} = \sqrt{(2.19 - 2.16)} = \sqrt{0.03} = 0.17$$

Next, we can use the NA to determine the acceptance angle. Remember that the acceptance angle is the value of θ .

$$NA = \sin\theta$$

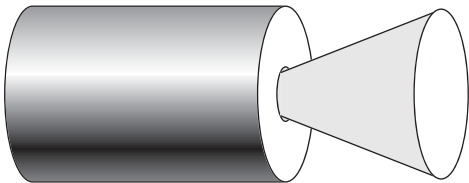
So

$$\theta = \arcsin NA = \arcsin 0.17 = 9.79$$

The acceptance angle is 9.79° .

The acceptance angle is also used to determine how light emerges from a fiber. The light that comes out of a fiber end is the light that has not been scattered, absorbed, or lost in the cladding, so with the exception of a small percentage of light that has propagated in the boundary between the core and the cladding, what emerges is coming out at an angle equal to or less than the acceptance angle, as shown in Figure 5.14.

FIGURE 5.14 Light emerges from the fiber in the cone of acceptance.



The numerical aperture varies depending on the optical fiber type. Single-mode optical fibers have the smallest NA and narrowest acceptance cone whereas large core plastic optical fibers have the largest NA and widest acceptance cone. NA is not defined in ANSI/TIA-568-C.3, ITU-T G.652, ITU-T G.655, or ITU-T G.657. However, it is defined in ITU-T G.651.1 and IEC 60793-2-10 for some multimode optical fibers. Table 5.2 lists some of the typical NA values for various single-mode and multimode optical fibers.

TABLE 5.2 Optical fiber numerical aperture (NA)

Fiber type	NA
Single-mode	0.14
Multimode 50/125 μm	0.20
Multimode 62.5/125 μm	0.275
Multimode 100/140 μm	0.29
Multimode 200/230 μm	0.37
Multimode 980/1000 μm	0.47

Equilibrium Mode Distribution

Equilibrium mode distribution (EMD) is a condition in which the light traveling through the fiber populates the available modes in an orderly way.

When light energy first enters a fiber, it does not automatically fill every available mode with an equal amount of energy. Some modes will carry a high amount of light energy, whereas others may carry nothing at all. Over distance, this condition balances out as light transfers between modes because of imperfections or bends in the fiber. Until EMD is achieved, however, light may be traveling in inefficient modes that will eventually lose energy. These modes may include paths that carry the light through the cladding, or high-order modes that produce a high number of reflections through the fiber core.

Let's back up to where the light enters the fiber and see how EMD is reached.

A light source rarely, if ever, perfectly matches the numerical aperture of the fiber. Typically, the beam will either be larger or smaller than the NA. If it is larger, as is the case with most LEDs, the fiber will be overfilled. This means that the light energy will occupy most of the modes, including those that are less efficient and are doomed to loss somewhere down the length of the fiber. The *overfilled launch* is discussed in detail in Chapter 10.

If the source has a smaller beam, as with most lasers, the fiber will be underfilled. The light energy will occupy only a few of the available modes until twists, turns, and imperfections in the fiber distribute some of the light energy into higher-order modes.

The effect is similar to traffic entering a highway. If eight lanes of cars are trying to get onto a six-lane highway, you'll see cars jockeying for position in the available lanes, and possibly even a few driving on the shoulder until they are attenuated by the highway patrol. This condition will persist until the drivers settle down into their new pattern. On the other hand, if a single lane of cars enters the same six-lane highway, they will first occupy only a single lane but eventually spread out to a more evenly distributed pattern.

Even when EMD is achieved, the effect is short-lived due to minute changes in the fiber characteristics and the effects of connectors. It does not take much to throw the modes out of equilibrium.

The distance required to achieve EMD depends on the fiber material. Light passing through plastic fibers reaches EMD in a few meters, but light passing through glass fibers may reach equilibrium only after several kilometers.

It is important to understand EMD because it affects the measurement of light energy in a fiber. In an overfilled fiber, light that has not yet reached EMD still has much of the energy traveling in the inefficient or high-order modes, and this energy can be measured over a short distance. Over a longer distance, however, once EMD has been reached, the energy in the less efficient modes has been lost, and the energy reading may drop significantly.

Once EMD has been reached, however, energy loss drops off, because the light is traveling in more efficient modes and is less likely to be lost. For this reason, loss before EMD is proportional to the length of the fiber, and loss after EMD has been reached is proportional to the square root of the fiber length. In other words, if the energy loss before EMD is

0.2dB/km, the loss over 2km would be 0.4dB. After reaching EMD, however, the loss would be $0.2 \times \sqrt{2}$.

Fiber Specifications and Standards

As you learned in Chapter 1, “History of Fiber Optics and Broadband Access,” the first full-scale commercial application of fiber-optic communication systems occurred in 1977, when both AT&T and GTE began using fiber-optic telephone systems for commercial customers. Twenty-three years later, as published in The National Economic Council report entitled “Four Years of Broadband Growth,” 4.4 percent of the households in America had a broadband connection to their home with a download speed of just 200kbps. By 2013, basic broadband download speed was 3Mbps; however, more than 94 percent of the homes in America exceed 10Mbps and more than 3 percent enjoy download speeds greater than 1Gbps.

In just 13 years, download speeds increased 5000 times. A download in the year 2000 that would take one hour can be accomplished today in less than one second. Creating a global fiber-optic infrastructure that supports this type of rapid change requires coordinated global efforts that standardize physical and optical properties.

As download rates have evolved since the turn of the century, so have optical fibers, transmitters, and receivers. In fact, there has been so much change it is difficult even for those working in the industry to keep pace with the changes and new developments, let alone a novice or someone just learning about fiber optics.

In this section, the major industry standards that define optical fiber performance are presented. Keep in mind that standards define a minimum level of performance and many manufacturers offer products that exceed the minimum performance levels. Often the working groups within standards organizations are playing catch-up, creating standards for products that currently exist and may have been in widespread use for several years or more.

In Chapter 4, some of the many standards related to optical fiber were introduced along with four major standards organizations that publish standards that define the performance of optical fibers used in the telecommunications industry. The four major standards organizations are the Telecommunications Industry Association (TIA), the International Telecommunications Union (ITU), the International Organization for Standardization (ISO), and the International Electrotechnical Commission (IEC). These organizations publish many different standards on optical fiber, fiber-optic cable, and testing.

The TIA standards covered in this text were prepared in accordance with the policies and procedures of the American National Standards Institute (ANSI). The mission of ANSI is to “enhance both the global competitiveness of U.S. business and the U.S. quality of life by promoting and facilitating voluntary consensus standards and conformity assessment systems, and safeguarding their integrity.” Because these TIA standards were prepared in accordance with ANSI policies and procedures, each standard contains the ANSI prefix.

Revisions and Addendums

Standards can take many years to develop, and the organizations that publish the standards typically do not give them away. The cost for a standard can be significant. For example, as of this writing, the cost for the ANSI/TIA-568-C.3 base document is \$219. It is for this reason that companies that require access to many standards typically pay a significant annual fee for an online service that allows access to standards published by many different organizations.

The ANSI/TIA-568-C series of standards contains documents C.0, C.1, C.2, C.3, and C4. The letter “C” represents the major revision. When the “C” revisions were published, they replaced the “B” revisions. As of this writing, the “D” revisions are in development. In addition to the revision letter, the date the standard was approved can be found in the upper-right corner of the cover page. The cover page will also list the month and year the standard was published.

When a TIA standard is updated without a major revision, an addendum is included. The addition of the addendum also includes the addition of a suffix. For example, ANSI/TIA-568-C.3 was published in June 2008; Addendum 1 was published in December 2011. The addendum is known as ANSI/TIA-568-C.3-1.

It is important to be aware of standards that have been superseded. Companies typically do not rip out their old cabling just because a standard changed. Fiber-optic cabling installed at the turn of the century would have complied with ANSI/TIA/EIA-568-B.3. More than likely that cabling is still in use today. You will need access to the ANSI/TIA/EIA-568-B.3 optical fiber cable performance parameters to determine compatibility with newer fiber-optic cables, transmitters, and receivers.

The ISO, IEC, and ITU standards covered in this section do not use a letter to represent the revision; each has a date on the cover page. The IEC standards also state the edition number on the cover page just as the publisher of this book does.

Unless you work with industry standards on a regular basis and keep track of the progress of the many standards working groups, it is difficult to know which standard is the most current. The best way to determine this is to visit each organization’s website. The following links will assist you in staying up to date:

www.tiaonline.org

www.iec.ch

www.itu.int

www.iso.org

www.ansi.org

Mirroring of Performance Specifications

It is not uncommon to find the performance specifications for an optical fiber mirrored in another standard; after all, the goal is inter-compatibility. While it may be the intent of a standards organization to produce specifications that can be utilized globally, they may not be the recognized organization for specific geographic locations. The TIA headquarters is located in Arlington, Virginia and the IEC central office is located in Geneva, Switzerland.

When ANSI/TIA/EIA-568-B.3 was first published in March 2000, there were no references to ISO/IEC optical fiber cable performance parameters in Section 4, Table 1. However, when ANSI/TIA-568-C.3 was published in June 2008 the ISO/IEC 11801 fiber designations and optical fiber cable performance parameters were included in Section 4, Table 1. This is an example of the performance specifications for optical fibers being mirrored in multiple standards.

Fiber Designations

While optical fiber performance parameters may be mirrored in different standards, the fiber designations typically are not. ISO/IEC 11810 uses the acronym “OM” to describe multimode optical fiber and “OS” to describe single-mode. IEC 60793-2-10 uses the letter “A” to describe multimode optical fiber and IEC 60793-2-50 uses the letter “B” to describe single-mode. In the ANSI/TIA-568-C standard optical fiber is designated using the ANSI/TIA-492 series of documents where the letter “A” after 492 designates multimode optical fiber and “C” designates single-mode. ITU does not use different letters, only numbers; G.651.1 is a multimode standard while G.652, G.655, and G.657 are single-mode. These designations are broken out in Table 5.3.

TABLE 5.3 Fiber designation by industry standard

Industry standard	Multimode	Single-mode
ISO/IEC 11810	OM	OS
IEC 60793-2-10/50	A	B
ANSI/TIA	TIA-492A	TIA-492C
ITU	G.651.1	G.652, G.655, G.675

Premises Standards

The ANSI/TIA-568-C standards and the ISO/IEC 11801 standard are applicable to premises cabling components; this includes copper and fiber-optic cabling. However, in this section only optical fiber cable transmission performance parameters will be addressed. Optical fiber cable transmission performance parameters can be found in Section 4 of ANSI/TIA-568-C.3, Section 9 of ISO/IEC 11801, and the annex’s of IEC-60793-2-10. To simplify presentation, the performance parameters for all three standards are combined into Table 5.4.

Table 5.4 describes the attenuation and bandwidth characteristics for multimode and single-mode optical fibers. Unlike ANSI/TIA-568-C.3 and ISO/IEC 11801, IEC-60793-2-10 does not list a minimum value for attenuation and bandwidth performance, it lists a value range. In Table 5.4, IEC-60793-2-10 value ranges have been correlated with the ANSI/TIA-568-C.3 and ISO/IEC 11801 designations.

Two laser-optimized multimode optical fibers that were not previously described in ANSI/TIA/EIA-568-B.3 are introduced; they will be covered in detail later in this chapter. New bandwidth requirements for multimode optical fiber are also defined. Bandwidth is defined with an overfilled launch (OFL) and a restricted mode launch (RML), also known as effective laser launch in ISO/IEC 11801.

Up until 1996 when TIA formed the TIA FO-2.2.1 Task Group on Modal Dependence of Bandwidth to study the interaction between multimode optical fiber and a laser source, multimode optical fiber bandwidth was defined with an overfilled launch. However, as laser technology improved and IEEE began developing a Gigabit Ethernet Standard (IEEE 802.3z) for high-speed local area networks (LANs) using lasers, a better indicator of bandwidth performance in a laser-based system was required.

Since 1996 the FO-2.2.1 Task Group has developed two Fiber-Optic Test Procedures (FOTPs). FOTP 203 defines a standard procedure for measuring the launch power distribution of a laser-based multimode fiber-optic transmitter. FOTP 204 describes the methods used to measure the information-carrying capacity of a multimode optical fiber using an OFL or RML. As you learned earlier, bandwidth-length product (MHz · km) is used to define the information-carrying capacity of a multimode optical fiber.

TABLE 5.4 Characteristics of ANSI/TIA-568-C.3 and ISO/IEC 11801-recognized optical fibers

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
62.5/125μm	850	3.5	200	Not Required
Multimode OM1	1300	1.5	500	Not Required
Type A1b				
TIA 492AAAA				
50/125μm	850	3.5	500	Not Required
Multimode OM2	1300	1.5	500	Not Required
TIA 492AAAB				
Type A1a.1				
850nm Laser-Optimized	850	3.5	1500	2000
50/125μm Multimode OM3	1300	1.5	500	Not Required
Type A1a.2				
TIA 492AAAC				

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
850nm Laser-Optimized 50/125µm Multimode OM4 Type A1a.3 TIA 492AAAD	850	3.5	3500	4700
	1300	1.5	500	Not Required
Single-mode Indoor-Outdoor OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1310	0.5	N/A	N/A
	1550	0.5	N/A	N/A
Single-mode Inside Plant OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1310	1.0	N/A	N/A
	1550	1.0	N/A	N/A
Single-mode Outside Plant OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1310	0.5	N/A	N/A
	1550	0.5	N/A	N/A

As noted in Table 5.4, laser optimized multimode optical fiber offers the greatest bandwidth performance at 850nm. This is because it is designed to be used with the vertical-cavity surface-emitting lasers (VCSELs). VCSEL technology has improved tremendously since the turn of the century and is the light source of choice for gigabit-plus LANs. The VCSEL and other light sources are covered in detail in Chapter 10.

As you can see in Table 5.4, a bandwidth-length product is not defined for single-mode optical fiber. As you learned earlier, the bandwidth of a single-mode optical fiber depends on the dispersion for a given length at a given wavelength. It is also impacted by the spectral width of the light source. A light source with a narrow spectral width will provide a greater bandwidth over a given distance than a light source with a wide spectral width.

Single-Mode ITU Standards

ITU-T G.652, ITU-T G.655, and ITU-T G.657 are three standards typically used to describe the performance characteristics of single-mode optical fiber:

- ITU-T G.652 defines the characteristics of a single-mode optical fiber and cable.
- ITU-T G.655 defines the characteristics of a non-zero-dispersion-shifted single-mode optical fiber and cable.
- ITU-T G.657 defines the characteristics of a bending loss-insensitive single-mode optical fiber and cable for the access network.

ITU-T G.652 contains several tables that list the performance characteristics of single-mode optical fiber and cable. The values in each table are identical for many of the fiber and cable attributes. Table 5.5 lists commonly used optical and geometric characteristics for single-mode optical fiber and cable.

TABLE 5.5 ITU-T G.652, single-mode optical fiber and cable

G.652.A, G.652.B, G.652.C, and G.652.D		
Fiber attributes	Detail	Value
Mode field diameter	Wavelength	1310nm
	Range of nominal values	8.6–9.5µm
	Tolerance	±0.6µm
Cladding diameter	Nominal	125.0µm
	Tolerance	±1µm
Core concentricity error	Maximum	0.6µm

G.652.A, G.652.B, G.652.C, and G.652.D

Fiber attributes	Detail	Value
Cladding noncircularity	Maximum	1%
Cable cut-off wavelength	Maximum	1260nm
Macrobend loss	Radius (mm)	30
	Number of turns	100
	Maximum at 1550nm or 1625nm	0.1dB
Cable attributes, G.652.A		
Attenuation coefficient	Maximum at 1310nm	0.5dB/km
	Maximum at 1550nm	0.4dB/km
Cable attributes, G.652.B		
Attenuation coefficient	Maximum at 1310nm	0.4dB/km
	Maximum at 1550nm	0.35dB/km
	Maximum at 1625nm	0.4dB/km
Cable attributes, G.652.C		
Attenuation coefficient	Maximum from 1310nm to 1625nm	0.4dB/km
	Maximum at 1550nm	0.3dB/km
Cable attributes, G.652.D		
Attenuation coefficient	Maximum from 1310nm to 1625nm	0.4dB/km
	Maximum at 1550nm	0.3dB/km

ITU-T G.655 contains several tables that list the characteristics of a non-zero-dispersion-shifted single-mode optical fiber and cable. Unlike ITU-T G.652, several of the fiber attributes are not similar among the cable types. Table 5.6 lists commonly used optical and geometric characteristics for non-zero-dispersion-shifted single-mode optical fiber and cable.

TABLE 5.6 ITU-T G.655.C, G.655.D, and G.655.E, non-zero-dispersion-shifted single-mode optical fiber and cable

		G.655.C	G.655.D and G.655.E
Fiber attributes	Detail	Value	Value
Mode field diameter	Wavelength	1550nm	1550nm
	Range of nominal values	8–11 μ m	8–11 μ m
	Tolerance	$\pm 0.7\mu$ m	$\pm 0.6\mu$ m
Cladding diameter	Nominal	125.0 μ m	125.0 μ m
	Tolerance	$\pm 1\mu$ m	$\pm 1\mu$ m
Core concentricity error	Maximum	0.8 μ m	0.6 μ m
Cladding noncircularity	Maximum	2%	1%
Cable cut-off wavelength	Maximum	1450nm	1450nm
Macrobend loss	Radius (mm)	30	30
	Number of turns	100	100
	Maximum at 1625nm	0.5dB	0.1dB
Cable attributes, G.655.C, G.655.D, and G.655.E			
Attenuation coefficient	Maximum at 1550nm	0.35dB/km	
	Maximum at 1625nm	0.4dB/km	

ITU-T G.657 defines the characteristics of two category “A” and “B” bending loss-insensitive single-mode optical fibers and cables for the access network. Like ITU-T G.655, several of the fiber attributes are not similar among the cable types, primarily in the area of bend radius. Table 5.7 through Table 5.10 list commonly used optical and geometric

characteristics for bending loss-insensitive single-mode optical fiber and cable for the access network.

TABLE 5.7 ITU-T G.657, specifications for G.657 category A1 bending loss-insensitive single-mode optical fiber and cable

G.657 category A1			
Fiber attributes	Detail	Value	Value
Mode field diameter	Wavelength	1310nm	
	Range of nominal values	8.6–9.5 μ m	
	Tolerance	$\pm 0.4\mu$ m	
Cladding diameter	Nominal	125.0 μ m	
	Tolerance	$\pm 0.7\mu$ m	
Core concentricity error	Maximum	0.5 μ m	
Cladding noncircularity	Maximum	1%	
Cable cut-off wavelength	Maximum	1260nm	
Macrobend loss	Radius (mm)	15	10
	Number of turns	10	1
	Maximum at 1550nm	0.25dB	0.75dB
	Maximum at 1625nm	1.0dB	1.5dB
Cable attributes			
Attenuation coefficient	Maximum from 1310 to 1625nm	0.4dB/km	
	Maximum at 1550nm	0.3dB/km	

TABLE 5.8 ITU-T G.657, specifications for G.657 category A2 bending loss-insensitive single-mode optical fiber and cable

G.657 category A2				
Fiber attributes	Detail	Value	Value	Value
Mode field diameter	Wavelength	1310nm		
	Range of nominal values	8.6–9.5 μ m		
	Tolerance	$\pm 0.4\mu$ m		
Cladding diameter	Nominal	125.0 μ m		
	Tolerance	$\pm 0.7\mu$ m		
Core concentricity error	Maximum	0.5 μ m		
Cladding noncircularity	Maximum	1%		
Cable cut-off wave-length	Maximum	1260nm		
Macrobend loss	Radius (mm)	15	10	7.5
	Number of turns	10	1	1
	Maximum at 1550nm	0.03dB	0.1dB	0.5dB
	Maximum at 1625nm	0.1dB	0.2dB	1.0dB
Cable attributes				
Attenuation coefficient	Maximum from 1310 to 1625nm	0.4dB/km		
	Maximum at 1550nm	0.3dB/km		

TABLE 5.9 ITU-T G.657, specifications for G.657 category B2 bending loss-insensitive single-mode optical fiber and cable

G.657 category B2				
Fiber attributes	Detail	Value	Value	Value
Mode field diameter	Wavelength	1310nm		
	Range of nominal values	8.6–9.5 μ m		
	Tolerance	$\pm 0.4\mu$ m		
Cladding diameter	Nominal	125.0 μ m		
	Tolerance	$\pm 0.7\mu$ m		
Core concentricity error	Maximum	0.5 μ m		
Cladding noncircularity	Maximum	1%		
Cable cut-off wavelength	Maximum	1260nm		
Macrobend loss	Radius (mm)	15	10	7.5
	Number of turns	10	1	1
	Maximum at 1550nm	0.03dB	0.1dB	0.5dB
	Maximum at 1625nm	0.1dB	0.2dB	1.0dB
Cable attributes				
Attenuation coefficient	Maximum at 1310nm to 1625nm	0.4dB/km		
	Maximum at 1550nm	0.3dB/km		

TABLE 5.10 ITU-T G.657, specifications for G.657 category B3 bending loss-insensitive single-mode optical fiber and cable

G.657 category B3				
Fiber attributes	Detail	Value	Value	Value
Mode field diameter	Wavelength	1310nm		
	Range of nominal values	8.6–9.5μm		
	Tolerance	±0.4μm		
Cladding diameter	Nominal	125.0μm		
	Tolerance	±0.7μm		
Core concentricity error	Maximum	0.5μm		
Cladding noncircularity	Maximum	1%		
Cable cut-off wavelength	Maximum	1260nm		
Macrobend loss	Radius (mm)	10	7.5	5
	Number of turns	1	1	1
	Maximum at 1550nm	0.03dB	0.08dB	0.15dB
	Maximum at 1625nm	0.1dB	0.25dB	0.45dB
Cable attributes				
Attenuation coefficient	Maximum at 1310nm	0.4dB/km		
	Maximum at 1550nm	0.3dB/km		

Multimode ITU and IEC Standards

As you have learned in this chapter, ITU-T G.651.1 and IEC 60793-2-10 describe the performance characteristics of multimode optical fiber and many of the optical performance specifications are covered in Table 5.4. This section of the chapter describes only the macrobending performance of these optical fibers in Tables 5.11 and 5.12.

TABLE 5.11 ITU-T G.651.1 multimode optical fiber macrobending loss

Radius (mm)	Number of turns	Maximum attenuation at 850nm	Maximum attenuation at 1300nm
15	2	1dB	1dB

TABLE 5.12 IEC 60793-2-10 types A1a and A1b multimode optical fiber macrobending loss

Radius (mm)	Number of turns	Maximum attenuation at 850nm	Maximum attenuation at 1300nm
37.5	100	0.5dB	0.5dB

Specialty Optical Fibers

Table 5.13 describes the typical attenuation and bandwidth characteristics for plastic fibers and HCS (hard-clad silica) and PCS (plastic-clad silica) fibers. These optical fibers are sometimes referred to as specialty optical fibers. The performance characteristics of these optical fibers are not defined in the TIA, ISO, IEC, or ITU standards covered in this text. The data contained in this table was obtained from manufacturers' datasheets and can vary from manufacturer to manufacturer.

TABLE 5.13 Characteristics of HCS/PCS and plastic optical fibers

Optical fiber cable type	Wavelength (nm)	Maximum attenuation	Maximum information transmission capacity for overfilled launch (MHz · km)
200/230μm HCS/PCS	650	10dB/km	17
	850	8dB/km	20
1mm plastic fiber	650	0.19dB/m	1.5

Summary

This chapter covered the characteristics of optical fiber relating to their ability to carry light and information. It described conditions in the fiber that limit both bit rate and signal strength, as well as specific methods used to overcome some of the conditions.

The chapter described how optical fibers are designated by the four major standards organizations. In addition, it discussed and defined optical fiber performance characteristics.

Exam Essentials

Understand dispersion in optical fiber. Make sure you understand the different types of dispersion that occur in optical fiber. Be able to explain the effects dispersion has on bandwidth.

Understand attenuation in optical fiber. Make sure you understand the different causes of attenuation and how it is affected by wavelength.

Be familiar with bending loss. Be able to explain the difference between a microbend and a macrobend. Understand the bending loss differences between bend-insensitive and bend-sensitive optical fiber.

Be familiar with optical fiber designations. Be able to describe how multimode and single-mode optical fibers are designated by the four major standards organizations.

Know optical fiber cable performance characteristics. Be able to describe multimode and single-mode optical fiber cable transmission performance characteristics as defined by the four major standards organizations.

Review Questions

1. What is dispersion?

- A. The weakening of light
- B. The spreading of light
- C. The coloring of light
- D. The wavelength of light

Hint: Taking a different path or traveling at a different speed

2. What does dispersion limit in an optical fiber?

- A. Bandwidth
- B. Attenuation
- C. Acceptance angle
- D. Scattering

Hint: The information carrying capacity

3. What causes modal dispersion?

- A. Light rays change wavelength as they pass through the optical fiber.
- B. Light rays change speed as they pass through the optical fiber.
- C. Light rays take different paths through the optical fiber.
- D. Light rays attenuate as they travel through the optical fiber.

Hint: It is influenced by the diameter of the optical fiber core and the wavelength of the light source.

4. Material dispersion is partly dependent on

- A. The power of the light source
- B. The bandwidth of the light source
- C. The manufacturer of the light source
- D. The spectral width of the light source

Hint: Different wavelengths of light travel at different velocities in an optical fiber.

5. Where does waveguide dispersion take place?

- A. Only in the core
- B. Only in the cladding
- C. In the core and the cladding
- D. In the cladding and the coating

Hint: Light in the cladding moves faster than light in the core.

6. Chromatic dispersion is
- A. Another name for modal dispersion
 - B. Another name for polarization-mode dispersion
 - C. Another name for waveguide dispersion
 - D. A combination of waveguide dispersion and material dispersion

Hint: It can be a problem when single-mode optical fiber carries high data rates over long distances.

7. How does dispersion-shifted optical fiber reduce chromatic dispersion?
- A. It shifts the zero-dispersion point to the wavelength that attenuates the least.
 - B. It adds enough modal dispersion to counteract the chromatic dispersion.
 - C. It shifts the wavelength of the light passing through it.
 - D. It shifts the light from the core to the cladding.

Hint: It takes advantage of the fact that the relationship between wavelength, refractive index, and velocity are not linear.

8. Another solution for chromatic dispersion is to
- A. Reduce the power of the light source.
 - B. Reduce the wavelength of the light source.
 - C. Reduce the spectral width of the light source.
 - D. Shift to plastic optical fiber.

Hint: Use fewer colors.

9. _____ accounts for the greatest percentage of attenuation in an optical fiber.
- A. Scattering
 - B. Absorption
 - C. Dispersion
 - D. Bandwidth

Hint: It also explains why the sun turns red on the horizon.

10. Polarization-mode dispersion takes place when
- A. Light travels in opposite directions in the fiber.
 - B. Two polarities of light are affected differently by conditions in the optical fiber.
 - C. Light passes near a magnet, splitting it into north and south polarities.
 - D. Light passes through an electrical field, splitting it into positive and negative components.

Hint: It is a result of irregularities, damage, or environmental conditions such as temperature.

11. Which industry standard organization does not use different letters to designate optical fibers?
- A. Telecommunications Industry Association (TIA)
 - B. International Telecommunications Union (ITU)
 - C. International Organization for Standardization (ISO)
 - D. International Electrotechnical Commission (IEC)

Hint: G.651.1 is a multimode standard.

12. For an optical fiber of a given bandwidth, as the length of the fiber decreases, the bandwidth
- A. Increases
 - B. Decreases
 - C. Does not change
 - D. Is independent of the length

Hint: The relationship is inversely proportional.

13. The maximum attenuation allowable for OM1 fiber at 850nm is _____.
- A. 0.5 dB/km
 - B. 1.0 dB/km
 - C. 1.5 dB/km
 - D. 3.5 dB/km

Hint: It is the same for all recognized multimode optical fibers.

14. An optical fiber will experience the least attenuation at _____.
- A. 650nm
 - B. 850nm
 - C. 1300nm
 - D. 1550nm

Hint: It has the lowest frequency.

15. Which optical fiber designation offers a minimum effective modal bandwidth-length product of 4700MHz · km?
- A. OM1
 - B. OM2
 - C. OM3
 - D. OM4

Hint: It is also known as type A1a.3 or TIA 492AAAD.

16. The numerical aperture of an optical fiber expresses its

- A.** Core diameter
- B.** Cladding diameter
- C.** Light gathering ability
- D.** Bandwidth

Hint: The cone of acceptance

17. Which standard defines the characteristics of a bending loss-insensitive single-mode optical fiber and cable for the access network?

- A.** ANSI/TIA-568-C.3
- B.** ISO/IEC 11801
- C.** ITU-T G.652
- D.** ITU-T G.657

Hint: It does not apply to premises cabling components.

18. What type of bending loss may not be visible when looking at the fiber optic cable?

- A.** Macrobend
- B.** Microbend
- C.** Intrinsic
- D.** Attenuation

Hint: The coating can cause them.

19. Which type of optical fiber offers the best macrobending performance?

- A.** Bend-sensitive
- B.** Bend-insensitive
- C.** Zero-dispersion-shifted
- D.** Laser-optimized

Hint: Macrobends occur when the optical fiber is bent around a radius that can be measured in millimeters.

20. The attenuation and bandwidth performance of a plastic optical fiber is typically defined at _____.

- A.** 650nm
- B.** 850nm
- C.** 1310nm
- D.** 1550nm

Hint: It is visible.

Chapter Exercises

Calculate the Attenuation in dB for a Length of Optical Fiber

The attenuation values for a length of optical fiber cable can be calculated using the attenuation coefficient for a specific type of optical fiber cable. This information can be found in the manufacturer's data sheet or in a standard.

Calculate the maximum attenuation for a 22.5km ITU-T G.657.B.3 optical fiber cable at 1550nm.

First, locate the attenuation coefficient for an ITU-T G.657.B.3 optical fiber cable at 1550nm in Table 5.10. Then multiply the attenuation coefficient by the length as shown here:

$$\text{Attenuation for 22.5km of optical fiber} = 22.5 \times 0.3 = 6.75$$

The maximum attenuation for 22.5km of ITU-T G.657.B.3 optical fiber at 1550nm is 6.75dB.

Calculate the Usable Bandwidth for a Length of Optical Fiber

The product of bandwidth and length (MHz · km) expresses the information carrying capacity of a multimode optical fiber. Bandwidth is measured in megahertz (MHz) and the length is measured in kilometers. The MHz · km figure expresses how much bandwidth the fiber can carry per kilometer of its length. The fiber's designation must always be greater than or equal to the product of the bandwidth and the length of the optical fiber.

Refer to Table 5.4 and calculate the usable bandwidth for 425 meters of 50/125µm OM3 multimode optical fiber using an OFL at a wavelength of 850nm.

The minimum bandwidth-length product for a 50/125µm OM3 multimode optical fiber at 850nm using an OFL is 1500MHz · km. To find the usable bandwidth for this optical fiber at a length of 425 meters, you must change the equation around and solve as shown here:

$$\text{Bandwidth-length product} = 1500\text{MHz} \cdot \text{km}$$

$$\text{Optical fiber length} = 0.425\text{km}$$

$$\text{Bandwidth in MHz} = 1500 \div 0.425$$

$$\text{Bandwidth} = 3529.4\text{MHz}$$

Calculate the Total Macrobending Loss in a Single-Mode Fiber-Optic Cable

The macrobending loss is defined for a number of turns of optical fiber around a mandrel at a specified radius and wavelength.

A single-mode ITU-T G.657.A2 bend-insensitive fiber-optic cable is installed in a home with two small 15mm 90° radius bends. The wavelength of the light source is 1625nm.

The macrobending loss in Table 5.8 is for optical fiber that has not been placed in a cable. The macrobending loss will vary when the optical fiber is placed in a cable and that information must be obtained from the cable manufacturer. However, to demonstrate how macrobending loss is calculated, it is assumed that optical fiber values listed in the standards are the same as the cable loss.

Referencing Table 5.8, we see that the loss for ten 15mm radius turns at 1625nm is 0.1dB. The total macrobending loss (MBL) will be the sum of the two losses. However, we cannot simply add up the dB values from Table 5.8 because each bend is not a full 360° turn; both 15mm radius bends are 90° and the bending loss is not listed for a single turn but a number of turns. First, we must establish the single turn loss using the following formula:

$$\text{MBL} \div \text{number of turns} = \text{dB loss per turn}$$

$$0.1\text{dB} \div 10 = 0.01\text{dB}$$

Then we have to calculate the loss for the partial bends using the following formula:

$$(\text{Bend in degrees} \div 360) \times \text{dB loss per turn} = \text{MBL}$$

$$(90 \div 360) \times 0.01\text{dB} = 0.0025\text{dB}$$

Then we calculate the total loss using this formula:

$$\text{MBL 1} + \text{MBL 2} = \text{Total MBL}$$

$$0.0025\text{dB} + 0.0025\text{dB} = 0.005\text{dB}$$

The total macrobending loss = 0.005dB.

Calculate the Total Macrobending Loss in a Multimode Fiber-Optic Cable

The macrobending loss is defined for a number of turns of optical fiber around a mandrel at a specified radius and wavelength.

A generic multimode bend-insensitive fiber-optic cable is installed in a home with two small 7.5mm radius, 120° bends. The wavelength of the light source is 850nm.

The macrobending loss in Table 5.1 is for a generic optical fiber that has not been placed in a cable. The macrobending loss will vary when the optical fiber is placed in a cable and that information must be obtained from the cable manufacturer. However, to demonstrate

how macrobending loss is calculated, it is assumed that optical fiber values listed are the same as the cable loss.

Referencing Table 5.1, we see that the loss for two 7.5mm radius turns at 850nm is $\leq 0.2\text{dB}$. The total macrobending loss (MBL) will be the sum of the two losses. However, we cannot simply add up the dB values from Table 5.1 because each bend is not a full 360° turn; both 7.5mm radius bends are 120° and the bending loss is not listed for a single turn but a number of turns. First, we must establish the single turn loss using the following formula:

$$\begin{aligned}\text{MBL} \div \text{number of turns} &= \text{dB loss per turn} \\ \leq 0.2\text{dB} \div 2 &= \leq 0.1\text{dB}\end{aligned}$$

Then we have to calculate the loss for the partial bends using this formula:

$$\begin{aligned}(\text{Bend in degrees} \div 360) \times \text{dB loss per turn} &= \text{MBL} \\ (120 \div 360) \times 0.1\text{dB} &= 0.033\text{dB}\end{aligned}$$

Then we calculate the total loss using this formula:

$$\begin{aligned}\text{MBL 1} + \text{MBL 2} &= \text{Total MBL} \\ 0.033\text{dB} + 0.033\text{dB} &= 0.066\text{dB}\end{aligned}$$

The total macrobending loss = 0.066dB.

Calculate the Acceptance Angle for an Optical Fiber

The acceptance angle defines the acceptance cone. Light entering the core of the optical fiber at an angle greater than the acceptance angle will not propagate the length of the optical fiber. For light to propagate the length of the optical fiber, it must be at an angle that does not exceed the acceptance angle.

Determine the acceptance angle of an optical fiber with a core refractive index of 1.48 and a cladding refractive index of 1.45.

To determine the acceptance angle of an optical fiber with a core refractive index of 1.48 and a cladding refractive index of 1.45, we must determine the NA of the fiber using this formula:

$$\text{NA} = \sqrt{[(n_1)^2 - (n_2)^2]}$$

The acceptance angle is the value of θ .

$$\text{NA} = \sin\theta$$

So

$$\theta = \arcsin \text{NA} = \arcsin 0.2965 = 17.25$$

The acceptance angle is 17.25° .

Determine the Latest Published Revision of a Standard Using the Internet

It is important to be aware of standards that have been superseded.

Using the Internet, determine the latest revision and publication date for ANSI/TIA-568-C.3.

Unless you work with industry standards on a regular basis and keep track of the progress of the many standards working groups, it is difficult to know which standard is the most current. The best way to determine this is to visit the TIA website at www.tiaonline.org/ and click on the Standards tab. In the Standards tab, click Buy TIA Standards and enter **TIA-568-C.3** in the search box. As of this writing the latest revision was ANSI/TIA-568-C.3-1, published December 2011.

Chapter 6

Safety

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Cite the government agency that publishes and enforces regulations on safety in the workplace.
- ✓ Explain how to safely handle and dispose of fiber-optic cable and bare optical fiber.
- ✓ List the laser hazard classifications of fiber-optic light sources and describe the dangers associated with each.
- ✓ Describe potential chemical hazards in the fiber-optic environment and explain the purpose of the material safety data sheet (MSDS).
- ✓ Describe potential electrical hazards in the fiber-optic installation environment.
- ✓ Describe typical work place hazards in the fiber-optic environment.





Whether you work as a technician or an installer, your work with fiber optics can expose you to several workplace hazards that are defined and regulated by the *Occupational Safety and Health Administration (OSHA)*. OSHA has published numerous regulations on workplace hazards ranging from laser light sources to ladders, and employers are required to be familiar with these regulations and follow them to keep the workplace safe.

You are responsible for your own safety as well as for the safety of your co-workers. It is up to you to know and incorporate safe work practices in everything you do.

This chapter describes the types of hazards that you will encounter as you work with fiber optics. Some of the hazards are unique to fiber optics work, but others are more common. This chapter discusses the dangers that these hazards create and explains different methods of working safely around them.

This chapter provides a general overview on safety; it is not the catch-all for safety. Each company and individual is responsible for maintaining a safe environment. The availability of safety information is also the responsibility of the company and individual.

In this chapter, you will learn to

- Classify a light source based on optical output power and wavelength
- Identify the symptoms of exposure to solvents
- Calculate the proper distance the base of a ladder should be from a wall

Basic Safety

Whenever you work in a hazardous environment, such as a construction site, lab, or production facility, you must always be aware of the potential dangers you face. Your workplace is required by law to provide you with equipment and facilities that meet standards set by OSHA, but you will have to be an active participant in your own safety.

You can use three lines of defense to help you get through the day safely: engineering controls, personal protective equipment (PPE), and good work habits.

Engineering Controls

Engineering controls are the mechanisms that your facility has established to make a hazardous situation safer. They may include ventilation in the form of exhaust fans or hoods, special cabinets for storing flammables, or workstations that minimize the hazards of specialized work, such as cutting optical fibers.

Do not ignore or try to get around the engineering controls set up in your workplace. By doing so, you only endanger yourself and others. Make sure that fans and ventilation systems are working properly. If they are not, report any problems to your facility supervisor immediately.

Do not try to alter or modify the engineering controls unless the modifications have been approved by your safety officer. Improper modifications could reduce the effectiveness of the controls and create a greater hazard.

Personal Protective Equipment (PPE)

Personal protective equipment (PPE) consists of anything that you would wear to protect yourself from materials or situations. It can include protective gloves and eyewear for cutting and grinding operations, respirators for working with chemicals that put out harmful vapors, and specialized goggles for working with lasers.

Your PPE protects you not only from short-term accidents, such as cuts or flying shards of glass, but also from damage that can build up over time. Such damage may include dust from construction operations such as drywall sanding that can build up over time in your lungs and cause diseases such as silicosis, or exposure to chemicals such as solvents that can have harmful long-term or chronic effects as well as harmful short-term effects.

Whenever you use PPE, inspect it carefully to ensure that it is in good condition. Look for cuts, tears, or other signs of damage in protective outerwear such as gloves or aprons. Inspect eyewear for cracks or pitting. If you use goggles designed to protect you from certain light wavelengths, make sure they are clean and free from scratches that could reduce their effectiveness.

If you wear contact lenses, be sure your facility allows them in your work area. If you work with adhesives or solvents, you should avoid wearing them anyway, because splashed chemicals could be trapped by the lens and be more difficult to wash out. You may be able to obtain safety goggles with prescription lenses if you have to use them on a regular basis.

If you work with a respirator, test it every time you put it on. Cover the canisters with your hands and try to inhale, then cover the exhaust port and try to exhale. The respirator should form a good seal with your face, and no air should leak through the canisters or exhaust.

Some construction areas may require hardhats. Do not take these warnings lightly. Even a small hand tool dropped from a few feet can injure or kill you if you are not protected. Hardhats are designed to absorb the shock from falling objects so your head doesn't

have to. To make sure the hardhat fits properly, adjust the inner band so that it fits snugly against your forehead and does not allow the hat to move around on your head. Make sure there is enough room between the suspension and the hardhat shell to absorb any blows.

Good Work Habits

Good work habits are in some ways the simplest and most effective means to working safely. Good work habits can help you prevent accidents and spot potential problems in time to correct them.

Here are some general rules for working safely:

- Keep a clean workspace. Clean up at the end of your work day and store tools properly. A “rat’s nest” can hide problems and add to confusion.
- Observe your surroundings. Look up from what you are doing once in a while to make sure everything around you is the way it should be.
- Use tools for the job they were designed to perform. Misuse of tools is one of the most common causes of accidents in the workplace.
- Do not eat or drink in the work area. In addition to accidentally drinking from the wrong bottle, you could accidentally ingest glass fiber or other dangerous materials that might get mixed in with your food.
- Report problems or injuries immediately. Let your facility supervisors know about hazards so they can correct them as soon as possible.
- Know how to reach emergency personnel. Have emergency numbers posted by the nearest telephone so you don’t waste time fumbling through a directory in an emergency.
- Put your emergency contact information in your cell phone.

Let’s look at some of the hazards directly related to your work with fiber optics.

Light Sources

Even though most lasers and LEDs used in fiber optics operate in the near-infrared and infrared (IR) wavelengths and are invisible to the eye, they can still cause damage if they are delivered at high intensity or if the exposure is long enough. The possibility of damage is even greater because you cannot see the beam, and in many cases, the damage is done before you know it.

A laser can be especially dangerous because it can concentrate a great amount of power into a small beam of coherent light. Many lasers used in fiber optics operate below dangerous levels, but some, such as those used for transmission over long distances, put out enough power to cause damage in a very short time.

Injuries from the infrared wavelengths output by lasers used in fiber-optic systems and test equipment include but are not limited to cataracts and corneal and retinal burns.

Federal Regulations and International Standards

Federal regulations and international standards have been created to prevent injuries from laser radiation.

In Chapter 4, “Optical Fiber Construction and Theory,” TIA and ITU were introduced as organizations that publish standards on the performance of optical fibers used in the telecommunications industry. There is a difference between a standard and a federal regulation. A federal regulation is a law. If there are federal regulations for a specific type of product sold in the United States, such as a laser, that product must meet federal regulations. However, it does not need to meet any standards. A standard only provides guidance; it is not a law.

Federal Regulations

The U.S. Food and Drug Administration (FDA) Center for Devices and Radiological Health (CDRH) is responsible for eliminating unnecessary human exposure to man-made radiation from medical, occupational, and consumer products. The FDA has had performance standards for light-emitting products since 1976. These performance standards are described in the Code of Federal Regulations 21 CFR, Subchapter J.

Fiber-optic equipment sold in the United States that contains a laser must meet 21 CFR, Subchapter J. However, it is not required by 21 CFR, Subchapter J to meet any standard. Many fiber-optic products that contain lasers only meet 21 CFR, Subchapter J.

21 CFR, Subchapter J encompasses all consumer products that contain a laser. In this chapter, only the areas of 21 CFR, Subchapter J typically required for fiber-optic communication and test equipment will be addressed; detailed information can be obtained from the FDA website:

www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?FR=1040.10

The laser classifications described in 21 CFR, Subchapter J are based on emission duration and emission limits for specific wavelengths. There are two wavelength windows that apply to fiber-optic communication and test equipment:

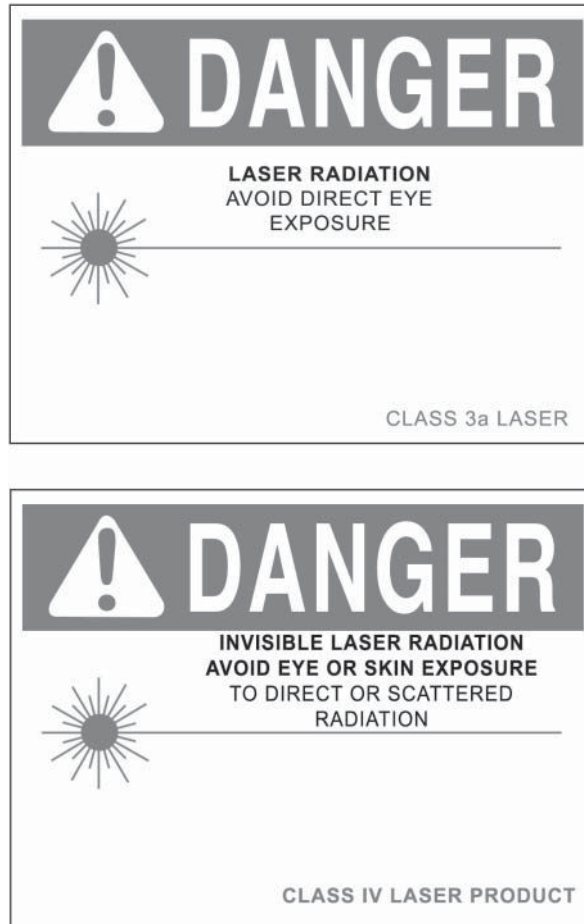
- >400nm and $\leq$ 1400nm
- >1400nm and $\leq$ 2500nm

CFR, Subchapter J divides laser products into classes and defines the labeling requirements based on the optical output power:

- Class I products do not require a label.
- Class IIa products require a label; however, there is no logotype for this label.
- Class II products require a “Caution” label affixed bearing the warning logotype “A.”

- Class IIIa with an irradiance less than 2.5mW must have a “Caution” label affixed bearing the warning logotype “A.”
- Class IIIa with an irradiance greater than 2.5mW, Class IIIb, and Class IV laser products must have a “Danger” label affixed bearing the warning logotype “B” similar to the labels, as shown in Figure 6.1.

FIGURE 6.1 Warning placards for lasers





The FDA warning logotypes are available for download from the FDA website.

The classification tables for accessible emission limits for laser radiation found in 21 CFR, Subchapter J are very detailed and cover all consumer laser products. The following is a list of the hazards associated with each class. Many of the products used in fiber-optic communication systems and test equipment are Class I; however, some equipment may be Class IV.

Class I These levels of laser radiation are not considered to be hazardous.

Class IIa These levels of laser radiation are not considered to be hazardous if viewed for any period of time less than or equal to 1,000 seconds but are considered to be a chronic viewing hazard for any period of time greater than 1,000 seconds. A label is required bearing the wording “Class IIa Laser Product—Avoid Long-Term Viewing of Direct Laser Radiation.”

Class II These levels of laser radiation are considered to be a chronic viewing hazard. A caution label is required that states “DO NOT STARE INTO THE BEAM” and “CLASS II LASER PRODUCT.”

Class IIIa These levels of laser radiation are considered to be, depending on the irradiance, either an acute intrabeam viewing hazard or chronic viewing hazard, and an acute viewing hazard if viewed directly with optical instruments. A caution label is required with an irradiance less than or equal to 2.5mW. The caution label should state “LASER RADIATION—DO NOT STARE INTO THE BEAM OR VIEW DIRECTLY WITH OPTICAL INSTRUMENTS” and “CLASS IIIa LASER PRODUCT.”

A danger label is required with an irradiance greater than 2.5mW. The danger label should state “AVOID DIRECT EYE EXPOSURE” and “CLASS IIIa LASER PRODUCT.”

Class IIIb These levels of laser radiation are considered to be an acute hazard to the skin and eyes from direct radiation. A danger label is required that states “LASER RADIATION—AVOID DIRECT EXPOSURE TO BEAM” and “CLASS IIIb LASER PRODUCT.”

Class IV These levels of laser radiation are considered to be an acute hazard to the skin and eyes from direct and scattered radiation. A danger label is required that states “LASER RADIATION—AVOID EYE OR SKIN EXPOSURE TO DIRECT OR SCATTERED RADIATION” and “CLASS IV LASER PRODUCT.”

Intrabeam Viewing Conditions

Intrabeam viewing conditions exist when the eye is exposed to the direct or specularly reflected laser beam. A *specular reflection* as defined in IEC 60825-1 is a reflection from a surface that maintains angular correlation between incident and reflected beams of radiation, as with reflections from a mirror. To avoid intrabeam viewing conditions, do not directly view the beam or view reflections from the surfaces hit by the beam.

Standards

Two organizations have published recognized laser safety standards: the International Electrotechnical Commission (IEC) and the American National Standards Institute (ANSI). ANSI Z136.2 addresses the safe use of optical fiber communication systems using laser diode and LED sources. IEC 60825-2 addresses the safety of optical fiber communication systems. The objective of both of these standards is to protect people from optical radiation released by an optical communication system.

These organizations have also published other laser standards. ANSI Z136.1-200 addresses the safe use of lasers in general and is not specific to optical fiber communication systems. IEC 60825-1 addresses the safety of laser products in general, not specific to optical fiber communication systems.

Test equipment or an optical fiber communication system (OFCS) containing a laser sold in the United States should have an FDA classification and the manufacturer may advertise compliance to an ANSI or IEC standard. Any OFCS or test equipment sold outside the United States may or may not have an FDA classification and may or may not comply with an ANSI or IEC standard.

OSHA OFCS Service Groups

OSHA also classifies an OFCS. Section IV, Chapter 6 of the OSHA Technical Manual describes the Optical Fiber Service Group (SG) Designations. It is based on ANSI Z136.2. These SG designations relate to the potential for ocular hazards to occur only when the OFCS is being serviced. “Being serviced” can involve something as simple as removing a connector from a receptacle. The SGs outlined in the OSHA Technical Manual are described next, and you can get detailed information from the OSHA:

https://www.osha.gov/dts/osta/otm/otm_iii/otm_iii_6.html

SG1 An OFCS in this SG has a total output power that is less than the accessible emission limit (AEL) for Class I (400nW), and there is no risk of exceeding the maximum permissible irradiance (MPI) when viewing the end of a fiber with a microscope, an eye loupe, or the unaided eye.

SG2 An OFCS is in this SG only if wavelengths between 400nm and 700nm are emitted. Such lasers are potentially hazardous if viewed for more than 0.25 second.

SG3A An OFCS in this SG is not hazardous when viewed with the unaided eye and is hazardous only when viewed with a microscope or eye loupe.

SG3B An OFCS in this SG does not meet any of the previous criteria.

OSHA Laser Hazard Classes

If the total power for an OFCS is at or above 0.5W, it does not meet the criteria for an optical fiber SG designation and should be treated as a standard laser system. The OSHA laser hazard classes are based on ANSI Z136.1 and summarized next:

Class 1 These lasers cannot emit laser radiation at known hazard levels, typically 400nW for a continuous wave at visible wavelengths. Users of these lasers are generally exempt from radiation hazard controls during operation and maintenance, but not necessarily during service.

Class II These lasers only emit visible wavelengths above Class I but below 1mW. These lasers can cause damage if you look directly at them for more than 0.25 second, but they have limited requirements for protection. It is assumed that the normal aversion to pain will cause anyone looking at the bright light to turn away or close their eyes before any damage can take place.

Class IIIA These intermediate power lasers have a continuous wave output from 1–5mW. Directly viewing the beam can damage the eye.

Class IIIB These moderate power lasers have a continuous wave output in the 5–500mW range (up to +27dBm) and can cause damage to the eye, even if the beam is reflected.

Class IV These high-power lasers have a continuous wave output above 500mW, or +27dBm, and they can burn almost any living tissue they contact under any viewing condition. They also pose a fire hazard.

IEC Laser Hazard Classifications

IEC standard 60825-1 addresses the safety of laser products and is not specific to an OFCS. However, components in the OFCS may be classified by this standard. Classifications that appear in this standard are summarized here:

Class 1 These levels of laser radiation are safe under normal operating conditions, including the use of optical instruments such as a microscope or eye loupe.

Class 1M These levels of laser radiation are safe under normal operating conditions; however, they may be hazardous when viewed with optical instruments such as a microscope or eye loupe.

Class 2 These levels of laser radiation at wavelengths between 400nm and 700nm are not hazardous if viewed for less than 0.25 second. It is assumed that the normal aversion to pain will cause anyone looking at the bright light to turn away or close his or her eyes (blink).

Class 2M These levels of laser radiation at wavelengths between 400nm and 700nm are not hazardous if viewed for less than 0.25 second. It is assumed that the normal aversion to pain will cause anyone looking at the bright light to turn away or close his or her eyes (blink). However, they may be hazardous when viewed with optical instruments such as a microscope or eye loupe.

Class 3R These levels of laser radiation are potentially hazardous when viewed directly.

Class 3B These levels of laser radiation are normally hazardous when viewed directly.

Class 4 These levels of laser radiation are considered to be an acute hazard to the skin and eyes from direct beam viewing or diffuse reflections.

IEC Laser Hazard Levels for OFCS

IEC 60825-2 addresses the safety of an OFCS. This standard describes hazard levels associated with an OFCS. It also defines labeling requirements and the location type for each hazard level. There are three location types:

Unrestricted This is a location where access to the protective housing is unrestricted. Examples of this type of access include domestic premises and any public areas.

Restricted This is a location where access to the protective housing is restricted; it is not open to the public. Examples of this type of access include secured areas within industrial and business/commercial premises not open to the public; general areas within switching centers; delimited areas not open to the public on trains, ships, or other vehicles; overhead fiber-optic cables and cable drops to a building; and optical test sets.

Controlled This is a location where access to the protective housing is controlled and accessible only to authorized personnel who have been trained in laser safety and the system they are servicing. Examples of this type of access include cable ducts, street cabinets, manholes, and dedicated and delimited areas of network operator distribution centers.

The hazard level labeling requirements for each location type that appear in this standard are summarized here:

Hazard level 1 There are no labeling requirements for unrestricted, restricted, or controlled location types.

Hazard level 1M Labels are required for unrestricted, restricted, or controlled location types.

Hazard level 2 and 2M Labels are required for unrestricted, restricted, or controlled location types.

Hazard level 3R This hazard level is not permitted in an unrestricted location type. Labels are required for restricted or controlled location types.

Hazard level 3B These hazard levels are not permitted in an unrestricted or restricted location type. A label is required for a controlled location type.

Hazard level 4 These hazard levels are not permitted in unrestricted, restricted, or controlled location types.

Laser Safety

Because most lasers used in fiber-optic systems emit IR radiation, you cannot see the beam, no matter how powerful it is. As a result, you will not be able to tell if the system is powered, especially if you are working with a piece of fiber far from the transmitter.

You should treat an optical fiber coupled to a laser with the same caution that you would treat electrical cables connected to a breaker panel. Do not assume that the system is turned off, especially if you have to use a microscope to look at the connector endface. Do not take anyone else's word that the system is off or that the fiber is uncoupled from the laser. You will have to endure the results for the rest of your life.

Unless you can be sure that the fiber is not coupled to a laser, do not look at the connector endface without some kind of protection. Use filters and protective eyewear that block out the specific wavelengths used by the lasers. Use a video microscope when available instead of a handheld optical microscope to view the endface of a connector.

Hazardous laser areas should be clearly identified with warning placards and signs stating that access is limited to personnel with proper safety gear and authorized access, as shown in Figure 6.1. Do not ignore these signs or think that they don't apply to you. They are there for your protection and for the safety of those working inside the restricted areas.

If the lab has a separate door and a hazardous laser is operating inside, the door should have interlocks to kill the laser before the door is opened. Some of these doors may have separate combination locks to prevent unauthorized entry.

Handling Fiber

In spite of optical fiber's flexibility, remember that it is glass. In short pieces, it is stiff enough to pierce your skin or eyes and cause discomfort, pain, or damage. If the pieces become airborne, you may even accidentally inhale or swallow them, risking damage to your throat or respiratory system. You can protect yourself with correct procedures and the right PPE.

When you cut, cleave, scribe, or accidentally break optical fibers, the ends can get lost easily, either by becoming airborne or by rolling along a surface. These ends can have extremely sharp edges, and if they are mishandled, they can lodge in your skin or eyes. If they are not removed immediately, the pieces can work themselves in further, increasing the risk of damage or infection. As shown in Figure 6.2, always work over a nonreflective black surface, which makes it easier to keep track of cut fiber ends. Also, keep a separate labeled container with a lid nearby for cut fiber ends.

FIGURE 6.2 A nonreflective black surface, a fiber waste container, and safety glasses can help prevent injury from fiber ends.



To prevent injury to your hands, always handle cut pieces of fiber with tweezers. To prevent eye injury, always wear proper eye protection. It takes only one piece of glass to damage your vision permanently. If you have been handling fiber, do not rub your eyes or put your hands near them until you have washed your hands. If you do get a piece of fiber in your skin, remove it immediately with a pair of tweezers or seek medical attention.

You may not always have the convenience of a laboratory or workshop environment for your fiber work. Work areas for splicing, building connectors, or other tasks may include basements, crawlspaces, underground vaults, an attic, or the back of a van. Don't take shortcuts just because you don't have the luxury of a full workshop at your disposal. Make sure you have an appropriate work surface and the proper tools and safety equipment before you start working.

Additional details on splicing and connectorization safety for fiber-optic cables and connectors can be found in Chapter 8, "Splicing," and Chapter 9, "Connectors."



Real World Scenario

Removing Fiber from Your Diet

Whenever you are cleaving fiber, you are going to wind up with ends. No matter how short or long they are, rest assured that they will be hard to spot when it's time to clean up. Your first line of defense against these hard-to-spot fibers is immediate response. In other words, as soon as you cleave, pick up and dispose of the end.

If this discipline doesn't work for you or if you find yourself with stray ends anyway, you'll still have to round them up and dispose of them properly. There are some low-tech and high-tech strategies to help you, as long as you can prepare them beforehand.

First, to prevent fiber ends from traveling beyond your work area, create a barrier around your work surface. This will keep bouncers and fliers confined to the work surface, where you can find them more easily.

If you want to make sure you've collected all of your strays or if you know that you created more ends than you have collected, use a bright light source, such as an LED flashlight. To find the fibers, place the light at the level of the work surface and radiate along the surface. The fiber ends will pick up the light and Fresnel reflection will cause them to shine brightly.

As a last, low-tech measure, double up a piece of adhesive tape and pat it all over the work surface. This will pick up any minute fiber particles that you could not spot or retrieve using other methods.

Chemicals

In your work with fiber optics, you might use several types of chemicals, including 99 percent isopropyl alcohol for cleaning components, solvents for removing adhesives and other materials, and anaerobic epoxy for making connectors.

Each of these chemicals poses a number of hazards and should be handled carefully. Each chemical that you use is accompanied by a *material safety data sheet (MSDS)*, which provides important information on the chemical's properties, characteristics such as appearance and odor, and common uses. The MSDS also gives you information on specific hazards posed by each chemical and ways to protect yourself through specific handling procedures, protective clothing and equipment, and engineering controls such as ventilation. Finally, the MSDS describes emergency procedures, including first aid for exposure to the chemical, methods for fighting fires in the case of flammable chemicals, and cleanup procedures for spills.

Even if you think you're familiar with the chemicals you handle, take time to read the MSDS. The information you gain could help prevent an accident or save valuable time in an emergency.

Let's look at some of the most important hazards associated with the chemicals you'll be handling.

Isopropyl Alcohol

Even though alcohol is commonly used in the home and the lab, its hazards should not be ignored or taken lightly. Alcohol vapors escape into the air easily, and they can cause

damage to your liver and kidneys if they are inhaled. The vapors are also highly flammable and can ignite if exposed to a spark or flame in high enough concentrations.

Alcohol can also cause irritation to your eyes, skin, and mucous membranes (nose and mouth) if it comes in direct contact with them.

Always use alcohol in a proper dispenser, shown in Figure 6.3. Store and transfer it carefully to avoid spills and excess evaporation. If you do spill any alcohol, clean it up with a dry cloth and dispose of the cloth in a container designed for flammable waste materials. As with all flammables, do not use alcohol in areas where sparks, open flames, or other heat sources will be present.

FIGURE 6.3 An alcohol dispenser helps reduce the risk of spills and fire.



It's important to remember that alcohol flames are almost invisible, and spills could lead to a mad scramble as you try to dodge flames you cannot see. Alcohol fires can be extinguished with water or a Class A fire extinguisher.

Solvents

Many solvents have similar properties that require that you handle them with great care. Like alcohol, solvents are very volatile and sometimes flammable. Their primary danger, however, is their hazard to your health.

One of the health hazards posed by solvents comes from the fact that they can cause excessive drying in your skin and mucous membranes, and the resultant cracking of the surface layers can leave you open to infection. The hazard is more serious if you inhale the solvent vapors, as they can damage your lungs and respiratory system.

Solvents can also cause organ damage if inhaled or ingested. The molecules that make up most solvents can take the place of oxygen in the bloodstream and find their way to the brain and other organs. As the organs are starved of oxygen, they can become permanently damaged. One of the first signs that this kind of damage is taking place is dizziness or a reaction similar to intoxication. If you feel these symptoms, get to fresh air immediately.

Solvents may come in glass or plastic containers. Make sure that the container you are using is properly marked for the solvent it contains. Do not leave solvent in an unmarked container.

If you are carrying solvent in a glass bottle, use a rubber cradle to carry the bottle. The cradle protects the bottle from breaking if it falls.

Never leave a solvent container open. Keep the top off just long enough to transfer the amount necessary for the job, and replace it firmly to prevent the vapors from escaping.

Anaerobic Epoxy

The two-part epoxy used for making connectors is typically used in small quantities and does not present any immediate health hazards. If you are working in an enclosed space, however, such as the back of a van or an access area, vapors from the adhesive portion can irritate your eyes, nose, and throat. If you feel any of these symptoms, get to an open space immediately. The adhesive can also irritate your skin or eyes on contact. If you get any of the adhesive on you, wash the area immediately. If any material splashes in your eyes, flush them for 15 minutes at an eyewash station or sink.

Use caution when working with the primer portion of these adhesives. With a flash point of -18°C , it is highly flammable. Do not use anaerobic epoxy where there is an open spark or flame or where heating components or elements are being used.

As with solvents, do not leave epoxy containers open any longer than necessary to dispense the amount you are using.

Site Safety

Many of the locations for fiber-optic components may be in areas that require special safety precautions. These may include construction sites, enclosed areas, locations near high-voltage power lines, or areas requiring access by ladder or scaffold.

Always follow the on-site safety requirements and observe all warning signs. Here are some general safety rules to help you.

Electrical

When fiber-optic cabling runs through the same area as electrical wiring, use extreme caution with tools and ladders. One wrong move can send enough current through your body

to kill you. Remember that electrical fields can exist beyond a cable's insulation if high voltages are present, so use wooden ladders to reduce the possibility of exposure to induced voltages. Use care with cutters and other tools to avoid accidental contact with electrical wires, and report any hazardous conditions that may exist.

Remember that high voltage causes most of its damage by making muscles seize up, including the heart and lungs. The greatest chance of damage comes when current passes through your heart to get the ground, such as when you touch a wire with your hand and your opposite leg provides the path to ground, or when the current passes from one hand to the other.

If you accidentally grab a live wire, the current may keep your hand clenched, making it nearly impossible to release the wire. If you see someone who is in this situation, do not try to pull them away with your hands. You may be caught up in the circuit as well. Instead, use a nonconducting stick, such as a wooden or fiberglass broom handle, to knock the victim away from the voltage source.

If the victim is not breathing, artificial respiration may be necessary to get the heart and lungs operating again.

Ladders

You may often find that you need a ladder to reach a work area. When choosing a ladder, make sure you select one that matches your requirements. Self-supporting ladders, such as stepladders, should be used only if the work area is not near a vertical support such as a wall and the floor beneath the work area is even and firm.

Non-self-supporting ladders, such as extension ladders, are useful when there is a firm vertical support near the work area and there is a stable, nonslip surface on which to rest the ladder. When setting up a non-self-supporting ladder, it is important to place it at an angle of $75\frac{1}{2}^\circ$ to support your weight and be stable. A good rule for finding the proper angle is to divide the working height of the ladder (the length of the ladder from the feet to the top support) by 4, and place the feet of the ladder that distance from the wall. For example, if the ladder is 12 feet tall, the bottom should be 3 feet from the wall.

Be sure that the ladder you select can carry your weight along with the weight of any tools and equipment you are carrying. Read the labels and warnings on the ladder you select to make sure that it is the right one for the work you are performing.

If you are working near live electrical systems, be sure to use a nonconducting ladder of wood or fiberglass. If you are working near high heat, select an aluminum ladder to avoid scorching or melting your only means of support.

In the work area, place the ladder so that you can work comfortably without having to reach too high or too far to the side. Overreaching can cause you to lose your balance and fall or place too much weight above or to the side of the ladder, causing the entire ladder to come down. If your work area extends beyond a comfortable reach, climb down the ladder and move it. Do not try to “walk” the ladder to a new work area.

Inspect ladders before using them. Make sure the rungs and rails are in good shape and are not split, broken, or bent. Make sure all fittings and fasteners are secure and that all locking mechanisms are working properly.

When carrying a ladder to the worksite, reach through the rail and balance it on your shoulder. Be aware of obstacles and corners as you carry it, and make sure others are aware that you have an awkward load, especially if you are walking through hallways or other limited visibility areas.

Trenches

If you are working on a fiber system in a trench, be sure the trench is properly dug and shored before entering it. Never work in a trench without someone else around, in case of a collapse.

If you have never worked in a trench before, learn the proper way to enter and exit a trench. Always use a ladder. Never jump into a trench or try to climb down the sides. You could trigger a collapse.

If you witness a collapse and others are trapped but not in immediate danger, do not try to dig them out yourself. You risk making the problem worse. Get help immediately. Special training is required to recover victims from a trench collapse, so leave it to the experts.

Emergencies

It takes only one slip-up to create an emergency. It could come from a moment of carelessness, an attempt at taking a shortcut, or ignorance of the proper procedures. Whatever the cause, the first response is always the same. Remain calm. Panic can cause even more damage and complicate matters beyond repair.

The best way to handle emergencies is to accept the fact that they will occur and be ready for them. Make sure you know what can go wrong with the materials and chemicals you handle and what you can do to minimize the damage.

Injury

Injuries can be caused by misuse of tools, fibers penetrating your skin or eyes, burns, falls, or any number of other mishaps. Make sure that you and your co-workers know the location of first-aid kits in your work area. Also, make sure you know how to reach emergency personnel. If you are on a new job site, make it a priority to familiarize yourself with emergency procedures and contact information.

Chemical Exposure

Accidental chemical exposure can result in anything from temporary discomfort to permanent injury or death. The first few seconds of an emergency involving chemical exposure can be critical in the victim's recovery.

If the chemical is splashed on the skin or in the eyes, flush the affected area with clean water immediately, using a shower or eyewash station if available. Continue flushing the

area for at least 15 minutes. This washes the chemical away but also dilutes its effects if it has been absorbed by the skin or eyes.

In case of inhalation, move the victim to fresh air immediately and call for medical attention. If a chemical has been accidentally swallowed, induce vomiting unless the chemical is corrosive and could damage the esophagus and throat as it comes back up. Use a neutralizing liquid such as milk to dilute corrosive chemicals if they have been swallowed, and seek medical attention immediately.

Fire

If a fire breaks out in your work area, it may be small enough for you to handle alone. If it is small, you can smother it with a damp cloth. If it is larger, but contained in a trash can or other enclosure, use the appropriate fire extinguisher for the material that is burning.

To use a fire extinguisher properly, remember the acronym PASS as you use the following procedure:

Pull Pull the pin from the fire extinguisher trigger.

Aim Aim the extinguisher at the base of the fire.

Squeeze Squeeze the handle firmly to activate the extinguisher.

Sweep Sweep the extinguisher discharge at the base of the fire until the flames are out.

Do not give the fire a chance to trap you. If you think that the extinguisher will not put out the fire completely and there is a risk that your exit will be cut off, leave immediately and call for help. You can do more good with a phone call than you can in a failed attempt at being a hero.

Summary

This chapter covered basic safety guidelines for your work with fiber-optic systems. It described hazards specifically associated with fiber optics work, as well as general safety concerns.

It also described the different types of lasers that you may encounter and the ways in which they are classified. In addition, it discussed methods for working safely with optical fiber in the lab and in various work situations.

Exam Essentials

Understand basic safety. Know the government agency that publishes and enforces safety regulations in the workplace. Make sure you understand the three lines of defense to help you safely get through the day. Be sure you know the importance of using engineering

controls and personal protective equipment properly. Be able to describe good work practices to help keep you safe.

Be familiar with light sources. Make sure you understand the dangers of working with lasers. Be able to list the laser hazard classifications and describe the dangers associated with each. Be sure you know how to protect yourself from the hazards of laser light.

Know the hazards of chemicals. Be familiar with the hazards of the chemicals you will use in working with fiber optics. Be able to describe their effects and the ways in which you can protect yourself. Also, make sure you understand what a material safety data sheet is and why it is important to your safety.

Understand site safety. Be able to describe the hazards of different sites where you may work. Make sure you can discuss how to work safely under different conditions.

Know emergency procedures. Be able to describe emergency procedures and the proper responses for different types of emergencies. Understand the importance of being prepared for emergencies.

Review Questions

1. Which government agency publishes and enforces safety regulations in the workplace?
 - A. TIA
 - B. OSHA
 - C. NASA
 - D. FBI

Hint: They also designate optical fiber service groups.

2. How often should you inspect your personal protective equipment?
 - A. Annually
 - B. Monthly
 - C. Weekly
 - D. Whenever you use it

Hint: It could never be too often.

3. Because infrared lasers cannot be seen, they _____.
 - A. Cannot hurt your eyes
 - B. Are not useful in fiber optics
 - C. Are especially dangerous to your eyes
 - D. Require special instruments

Hint: You cannot tell if they are on.

4. Lasers in Service Group 1 are considered _____.
 - A. The most hazardous
 - B. The least hazardous
 - C. Hazardous only under certain conditions
 - D. Hazardous only to living tissue

Hint: The total output power cannot exceed 400nW.

5. A Class IIIA laser as defined by OSHA operates at power levels of _____.
 - A. 400 nW
 - B. Below 1 mW
 - C. 1–5 mW
 - D. Above 5 mW

Hint: They are considered intermediate power lasers.

6. Which IEC laser hazard classification is safe to view with a microscope or eye loupe under normal operating conditions?
- A. Class 1
 - B. Class 1M
 - C. Class 2
 - D. Class 2M

Hint: The output may be viewed continuously with no protection.

7. You can learn how to properly handle or store a chemical by reading the _____.
- A. Hazard level
 - B. MSDS
 - C. Classification level
 - D. OSHA

Hint: You should have one for each chemical that you use.

8. One of the symptoms of exposure to solvent vapors is _____.
- A. The appearance of intoxication
 - B. Yellowing skin
 - C. Irritability
 - D. Hunger

Hint: You might feel dizzy.

9. The primer portion of anaerobic epoxy is _____.
- A. A powder
 - B. Flammable
 - C. Safe to use around a flame
 - D. Colorless

Hint: Has a flash point of -18°C .

10. If you see someone who has accidentally grabbed a live wire, you should
- A. Pull them away with your hands.
 - B. Pry the wire out of their hand with a screwdriver.
 - C. Knock the victim away with a wooden stick.
 - D. Knock the victim away with a stream of water.

Hint: Choose a tool that is nonconductive.

Chapter Exercises

Classify a Light Source Based on Optical Output Power and Wavelength

The OSHA Technical Manual describes the Optical Fiber Service Group (SG) Designations based on ANSI Z136.2. These SG designations relate to the potential for ocular hazards to occur only when the OFCS is being serviced. OSHA also classifies standard laser systems.

An OFCS does not fall into any of the SG designations defined in the OSHA Technical Manual because during servicing it is possible to be exposed to laser emissions greater than 750mW. What classification would be assigned to this OFCS based on the OSHA Technical Manual?

If the total power for an OFCS is at or above 0.5W, it does not meet the criteria for an optical fiber SG designation and should be treated as a standard laser system. A Class IV laser hazard as defined in the OSHA Technical Manual has a continuous wave output above 500mW, or +27dBm. This OFCS would be Class IV.

Identify the Symptoms of Exposure to Solvents

Solvents can cause organ damage if inhaled or ingested. The molecules that make up most solvents can take the place of oxygen in the bloodstream and find their way to the brain and other organs. As the organs are starved of oxygen, they can become permanently damaged.

Your co-worker spilled a bottle of liquid several minutes ago and now appears impaired. What has your co-worker potentially been exposed to?

Because solvent molecules can displace oxygen molecules in the blood, they can starve the brain of oxygen and give the appearance of intoxication. Your co-worker may have been exposed to a solvent.

Calculate the Proper Distance the Base of a Ladder Should Be from a Wall

Remember non-self-supporting ladders, such as extension ladders, require a firm vertical support near the work area and a stable, nonslip surface on which to rest the ladder. When setting up a non-self-supporting ladder, it is important to place it at an angle of $75\frac{1}{2}^{\circ}$ to support your weight and be stable.

A non-self-supporting ladder is 10' in length. How far should the feet of the ladder be from the wall?

A good rule for finding the proper angle is to divide the working height of the ladder (the length of the ladder from the feet to the top support) by 4, and place the feet of the ladder that distance from the wall. This ladder was 10' in length so the ladder's feet should be 2.5' from the wall.

Chapter 7

Fiber-Optic Cables

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Explain the purpose of each component displayed in a cross section view of a fiber-optic cable.
- ✓ Explain why and where loose buffer fiber-optic cable is used.
- ✓ Describe the difference between gel-filled and gel-free loose buffer fiber-optic cables.
- ✓ Describe tight buffer fiber-optic cable.
- ✓ Compare common strength members found in fiber-optic cables.
- ✓ Name common jacket materials found in fiber-optic cables.
- ✓ Describe simplex and duplex cordage and explain the difference between cordage and cable.
- ✓ Describe the characteristics of the following:
 - Distribution cable
 - Breakout cable
 - Armored cable
 - Messenger cable
 - Ribbon cable
 - Submarine cable
 - Hybrid cable
 - Composite cable
- ✓ Explain how and when a fan-out kit is used.
- ✓ Explain how and when a breakout kit is used.



- ✓ Describe the National Electrical Code (NEC) fiber-optic cable types.
- ✓ Describe the NEC listing requirements for fiber-optic cables.
- ✓ Explain the difference between a listed and nonlisted fiber-optic cable.
- ✓ List the types of markings typically found on the jacket of a fiber-optic cable.
- ✓ Describe the TIA-598-C color-coding scheme for individual fibers bundled in a fiber-optic cable.
- ✓ Describe the TIA-598-C color-coding scheme for premises cable jackets.
- ✓ Explain how numbering is used to identify the individual fibers bundled in a fiber-optic cable.
- ✓ Describe how to use sequential markings to determine fiber-optic cable length.





So far we have studied the principles and characteristics of individual optical fibers. While they are certainly adequate for the job of carrying signals from one place to another, they are not rugged enough to withstand the rigors of handling, transportation, and installation. In addition, some installations require multiple optical fibers for sending and receiving or for routing to a number of locations.

For an optical fiber to be suitable for everyday use, it must be incorporated into cables that provide standardized fiber groupings, protection from the environment, and suitable size for handling.

In this chapter, we will describe standard and harsh environment fiber-optic cables used in many types of installations. We will detail different types of fiber-optic cables and the uses for which they were designed. We will also describe some of the basic requirements for handling and installation.

In this chapter, you will learn to

- Determine the cable type from the NEC markings
- Identify the fiber number from the color code
- Determine the fiber number from the cable markings
- Identify the optical fiber type from the color code
- Determine the optical fiber type from the cable markings
- Determine the cable length using sequential markings

Basic Cable

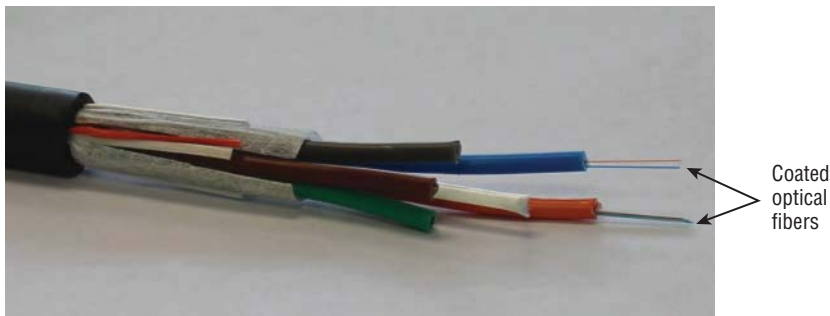
You may already be familiar with cables used for electrical wiring. These cables typically consist of two or more insulated wires bundled together and surrounded by a protective outer covering called the *jacket* or *sheath*. In addition to holding the wires in place, the jacket or sheath protects the wires from the environment and damage that may occur during handling or installation.

Some of the largest cables, used for telephone transmissions, can be several inches in diameter and contain hundreds of wires, as shown in Figure 7.1. These cables are very heavy, difficult to bend, and expensive.

FIGURE 7.1 Cable with hundreds of wires

Photo courtesy of Roger Crider

Fiber-optic cables like the one shown in Figure 7.2 do not need to be as large as the cable shown in Figure 7.1 because the bandwidth of an optical fiber over a long distance is many times greater than the bandwidth of a wire. Greater bandwidth means fewer optical fibers are required to carry the same information the larger cable is carrying. This results in a small, low-cost fiber-optic cable that is much easier to handle and terminate than the cable shown in Figure 7.1. Chapter 13, “Fiber-Optic System Advantages,” compares the performance of fiber-optic and copper cables.

FIGURE 7.2 Typical fiber-optic cable

Optical fibers are used in many different configurations and environments; manufacturers have created a wide variety of cable types to meet the needs of almost any application. The type of signal being carried and the number of optical fibers required are just two of the many considerations when selecting the right cable for an application. Other factors include the following:

- Tensile strength
- Temperature range
- Bend radius
- Flammability
- Buffer type
- Structure type
- Jacket type
- Weight
- Armor
- Crush resistance

The exact combination of these factors varies, depending on the application and operational environment. A cable installed inside an office building, for example, will be subject to less extreme temperatures than one installed outdoors or in an aircraft. Cabling installed in a manufacturing facility may be exposed to abrasive dusts, corrosive chemicals, or hotwork, such as welding, requiring special protection. Some cables may be buried underground, where they are exposed to burrowing or chewing animals, whereas others may be suspended between poles, subject to their own weight plus the weight of animals and birds who think the cables were put there for them.

Cable Components

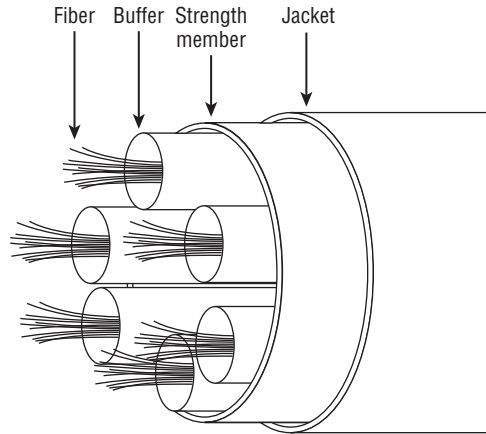
Whether a cable contains a single optical fiber, several optical fibers, or hundreds of optical fibers, it has a basic structure in common with other cables. As shown in Figure 7.3, a typical fiber-optic cable consists of the optical fiber (made up of the core, cladding, and coating), a buffer, a strength member, and an outer protective jacket or sheath. Let's look at these components individually.

Buffer

As you learned in Chapter 4, “Optical Fiber Construction and Theory,” all optical fiber has a coating. The coating is the optical fiber's first protective layer. The protective layer placed

around the coating is called the buffer. There are two buffer types: *loose buffer* and *tight buffer*.

FIGURE 7.3 Fiber-optic cable components



Loose Buffer

Loose buffer is also referred to as loose tube buffer or loose buffer tube. A loose buffer consists of a buffer layer or tube that has an inner diameter much larger than the diameter of the coated optical fiber, as shown in Figure 7.4.

The loose buffer provides room for additional optical fiber when the cable is manufactured. This extra room also allows the optical fiber to move independently of the buffer and the rest of the cable components. This is an important factor when the cable is subjected to temperature extremes. A variety of materials are used to make up a fiber-optic cable. These materials may not expand or contract the same way as the optical fiber during temperature changes. The additional fiber in the loose buffer will allow the cable to expand (increase in length) or contract (decrease in length) without placing tensile forces on the optical fiber.

Loose-buffered cable may be single-fiber or multifiber (not to be confused with single-mode and multimode fiber), meaning that it may have one or many optical fibers running through each buffer tube. In addition, a cable may contain a number of loose buffers grouped together, as shown in Figure 7.5. In such cables, loose-buffered tubes are grouped around a central core that provides added tensile strength and a resistance to bending.

Loose-buffered cables are designed for indoor and outdoor applications. For outdoor applications, the buffer tubes may be filled with a gel. The gel displaces or blocks water and prevents it from penetrating or getting into the cable. A gel-filled loose-buffered cable is typically referred to as a *loose tube, gel-filled (LTGF) cable*.

Fully waterblocked gel-free loose-tube cables are an alternative to gel-filled cables. A gel-free cable replaces the gel with a water-swellaable tape. When the water-swellaable tape makes contact with water, it quickly turns into a gel and swells. The swelling tape fills up the free space within the cable and blocks the water protecting the cable against damage from water ingress.

FIGURE 7.4 Loose-buffered cable has a buffer diameter greater than the fiber diameter.

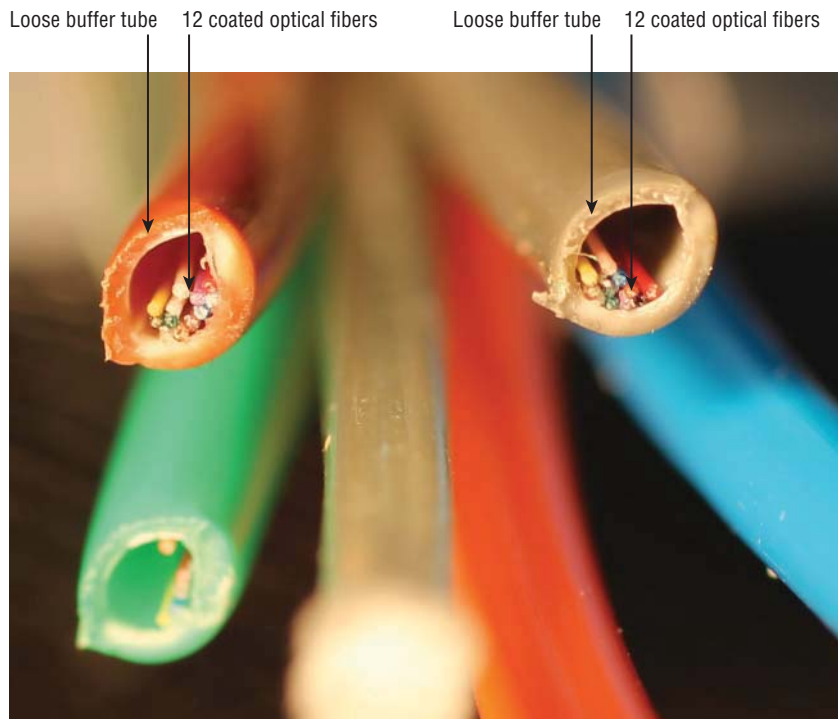
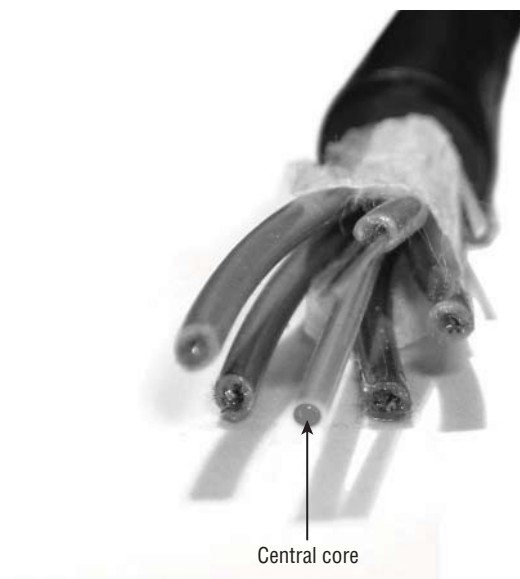


FIGURE 7.5 Loose-buffered tubes in a cable



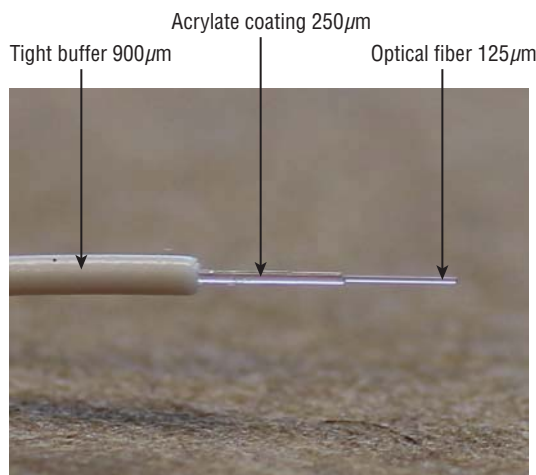
Loose-buffered cables are ideal for direct burial and aerial installations. They are also very popular for indoor/outdoor applications.

Tight Buffer

Tight-buffered cable is typically used in more controlled environments where the cable is not subjected to extreme temperatures or water. In short, tight-buffered cable is generally used for indoor applications rather than outdoor applications.

As shown in Figure 7.6, tight-buffered cable begins with a 250 μm coated optical fiber. The buffer is typically 900 μm in diameter and is applied directly to the outer coating layer of the optical fiber. In this way, it resembles a conventional insulated copper wire. The buffer may have additional strength members running around it for greater resistance to stretching.

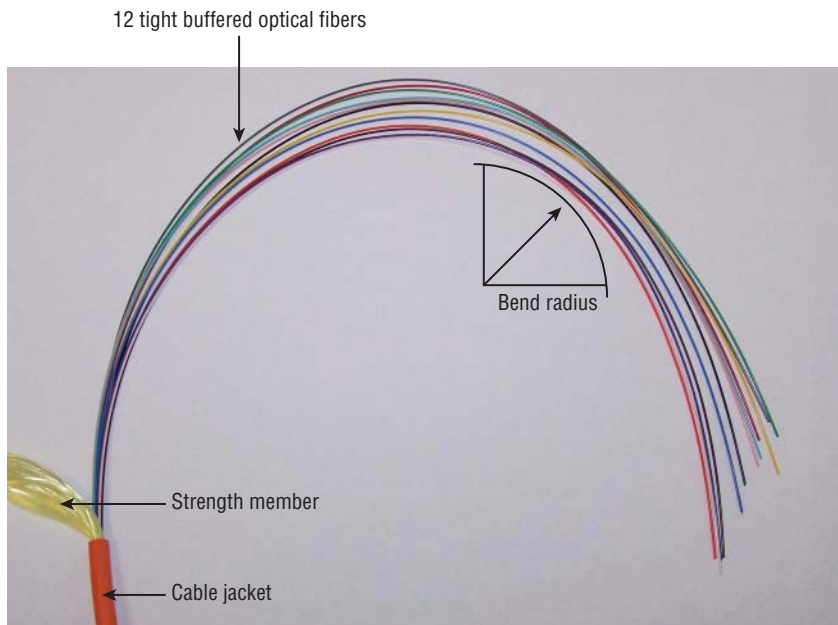
FIGURE 7.6 Tight-buffered cable uses a buffer attached to the fiber coating.



One of the benefits of tight-buffered cable is that the buffered optical fiber can be run outside of the larger cable assembly for short distances, as shown in Figure 7.7. This makes it easier to attach connectors. The standard diameter of the buffer allows many different types of connectors to be applied.

Tight-buffered cables have some advantages over loose-buffered cables. Tight-buffered cables are generally smaller in diameter than comparable loose-buffered cables. The minimum bend radius of a tight-buffered cable is typically smaller than a comparable loose-buffered cable. These two advantages make tight-buffered cable the choice for indoor installations where the cable is routed through walls and other areas requiring tight bends.

FIGURE 7.7 Tight-buffered optical fibers extending out of the cable assembly



Strength Members

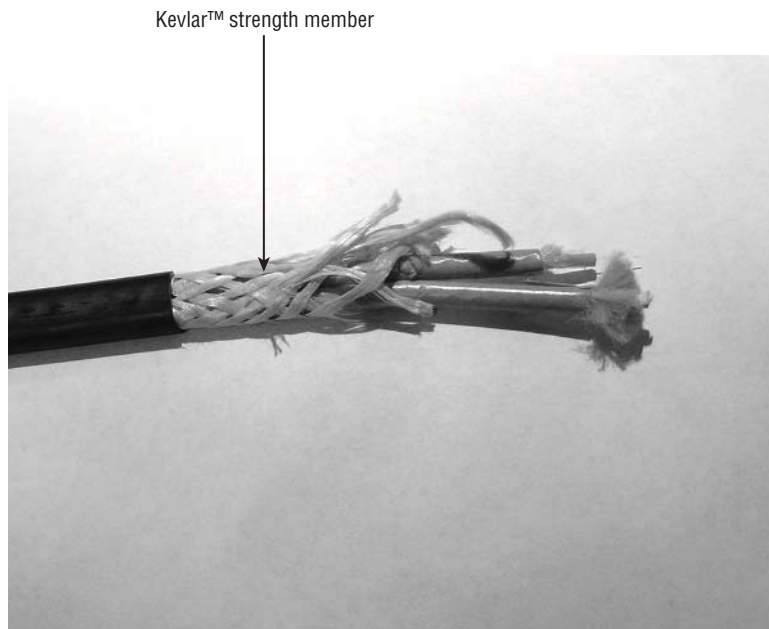
You've already learned a bit about how strength members help increase a cable's tensile strength. The primary importance of these members is to ensure that no tensile stress is ever placed on the optical fiber. The installation of a fiber-optic cable may require a great deal of pulling, which places considerable tensile stress on the strength members. The strength members need to be able to support the installation tensile stresses and the operational tensile stresses.

Strength members may run through the center of a fiber-optic cable, or they may surround the buffers just underneath the jacket. Combinations of strength members may also be used depending on the application and the stress the cable is designed to endure. Some of the most common types of strength members are made of

- Aramid yarns, usually Kevlar™
- Fiberglass rods
- Steel

Aramid yarns, as shown in Figure 7.8, are useful when the entire cable must be flexible. These fine, yellowish or gold yarns are made of the same material used in high-performance sails, bulletproof vests, and fire protection gear. They have the advantages of being light, flexible, and quite strong. Aramid yarns may be used in cable subgroups if they will be bundled into larger cables.

FIGURE 7.8 Aramid (Kevlar) yarns used as strength members



Larger cables and most loose-buffered cables typically have a central member of either fiberglass or steel added, as shown in Figure 7.9. This central member provides tensile strength and a resistance to bending. This resistance to bending prevents the cable from being over-bent and kinking the loose-buffered tubes.

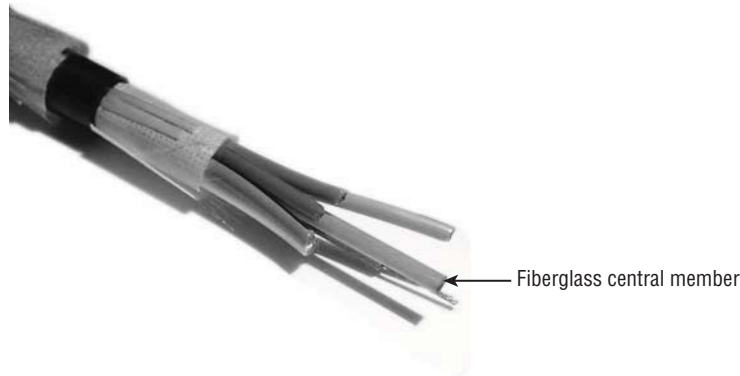
Although the tensile strength of steel is greater than the tensile strength of fiberglass, many applications require a *dielectric*, or nonconductive cable. Fiberglass is nonconductive whereas steel is not. Steel running through a cable outdoors makes an excellent lightning rod.

Jacket

The jacket is the cable's outer protective layer. It protects the internal components from the outside world. The jacket may be subject to many factors such as sunlight, ice, animals,

equipment accidents, and ham-handed installers. The jacket must also provide protection from abrasion, oil, corrosives, solvents, and other chemicals that could destroy the components in the cable.

FIGURE 7.9 A fiberglass central member adds tensile strength and a resistance to bending.



Jacket materials vary depending on the application. Typical jacket materials include **Polyvinyl chloride (PVC)** PVC is used primarily for indoor cable runs. It is fire retardant and flexible, and it's available in different grades to meet different conditions. PVC is water resistant; however, PVC does not stand up well to solvents. In addition, it loses much of its flexibility at low temperatures.

Polyethylene This material is typically used outdoors. It offers excellent weather and sun resistance in addition to excellent water resistance and flexibility in low temperatures. Specially formulated polyethylene offers low-smoke, zero-halogen performance.

Polyvinyl difluoride (PVDF) This material is chosen for its low-smoke and fire-retardant properties for use in cables that run through airways or *plenums* in a building. PVDF cables are not as flexible as other types, so their fire safety properties are their primary draw.

Polytetrafluoroethylene (PTFE) This material is primarily used in aerospace fiber-optic cables. It has an extremely low coefficient of friction and can be used in environments where temperatures reach as high as 260° C.

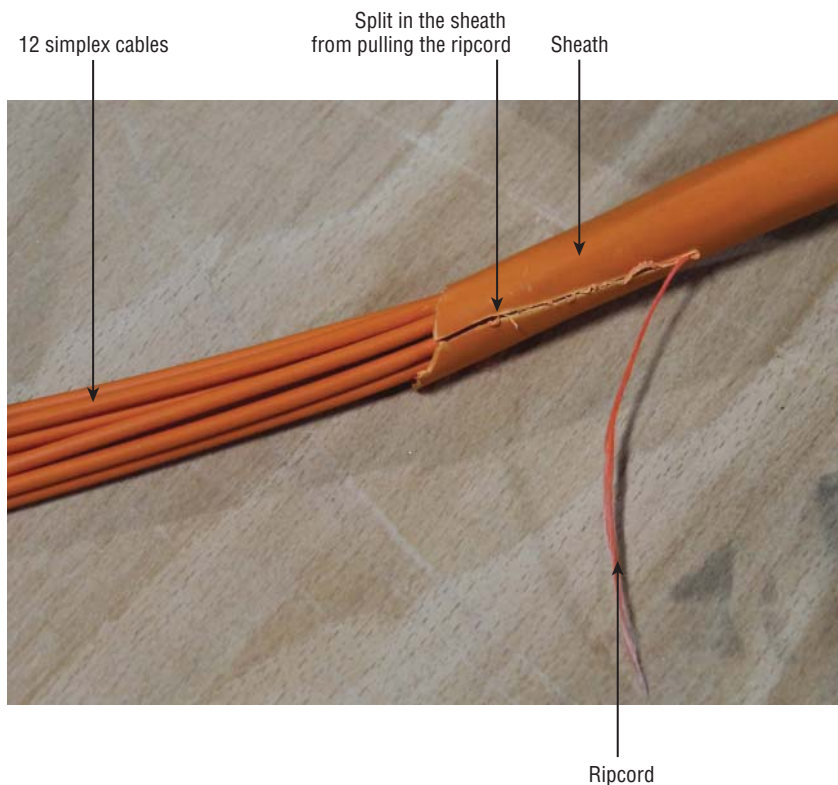
Some cables contain multiple jackets and strength members, as shown in Figure 7.10. In such cables, the outermost jacket may be referred to as the sheath; the inner protective layers are still called jackets.

FIGURE 7.10 The sheath is the outer layer of a cable with multiple jackets.



The *ripcord* is a piece of strong thread running through the cable just under the jacket or sheath as shown in Figure 7.11. When the ripcord is pulled, it splits the jacket easily to allow the fibers to be separated for connectorization or splicing. The ripcord reduces the risk incurred when cutters or knives are used to split the jacket. The ripcord will not penetrate the jackets of the individual simplex cables as a cutter or knife may when the blade depth is not accurately controlled.

FIGURE 7.11 The ripcord splitting the cable sheath



Cable Types

As uses for optical fiber have become more varied, manufacturers have begun producing cables to meet specific needs. Cable configurations vary based on the type of use, the location, and future expansion needs, and it is likely that more will be created as future applications emerge.

Bear in mind that different cable arrangements are variations on a theme. Different combinations of buffer type, strength members, and jackets can be used to create cables to meet the needs of a wide variety of industries and users.

Let's look at some of the commonly available optical fiber cables.

Cordage

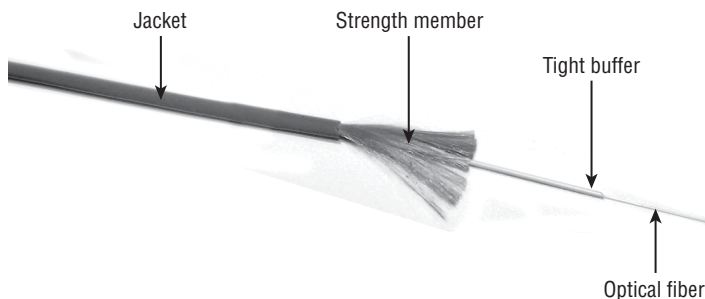
The simplest types of cables are called *cordage* and are used in connections to equipment and patch panels. They are typically made into patch cords or jumpers. The major difference between cordage and cables is that cordage has only one optical fiber/buffer combination in a jacket, whereas cables may have multiple optical fibers inside a jacket or sheath.

The two common types of cordage are *simplex* and *duplex*.

Simplex Cordage

Simplex cordage, shown in Figure 7.12, consists of a single optical fiber with a tight buffer, an aramid yarn strength member, and a jacket.

FIGURE 7.12 Simplex cordage



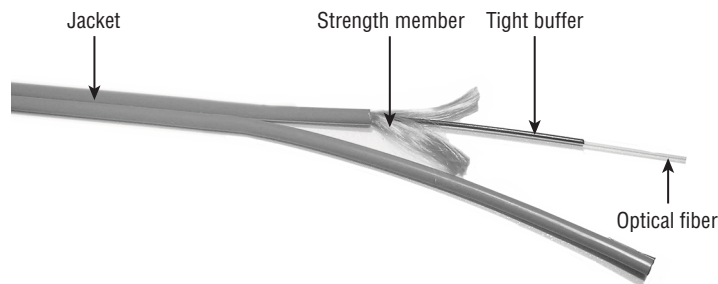
Simplex cordage gets its name from the fact that, because it is a single fiber, it is typically used for one-way, or simplex, transmission, although bidirectional communications are possible using a single fiber.

Duplex Cordage

Duplex cordage, also known as *zipcord*, is similar in appearance to household electrical cords, as you can see in Figure 7.13. Duplex cordage is a convenient way to combine two

simplex cords to achieve duplex, or two-way, transmissions without individual cords getting tangled or switched around accidentally.

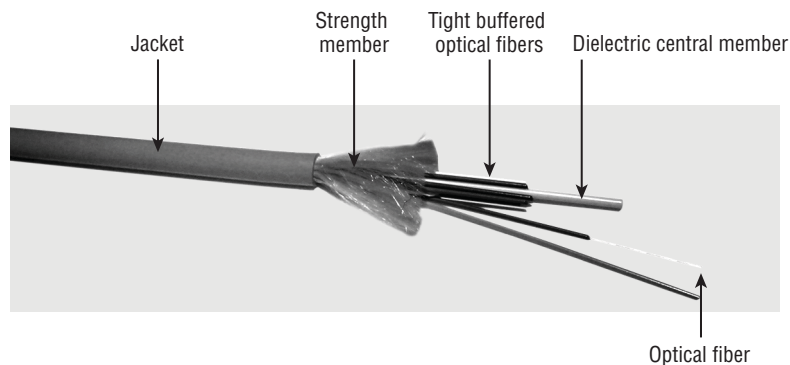
FIGURE 7.13 Duplex cordage



Distribution Cable

When it is necessary to run a large number of optical fibers through a building, distribution cable is often used. Distribution cable consists of multiple tight-buffered fibers bundled in a jacket with a strength member. These cables, like the one shown in Figure 7.14, may also feature a dielectric central member to increase tensile strength, resist bending, and prevent the cable from being kinked during installation.

FIGURE 7.14 Distribution cable with dielectric central member



Distribution cables are ideal for inter-building routing. Depending on the jacket type they may be routed through plenum areas or riser shafts to telecommunications rooms,

wiring closets, and workstations. The tight-buffered optical fibers are not meant to be handled much beyond the initial installation, because they do not have a strength member and jacket.

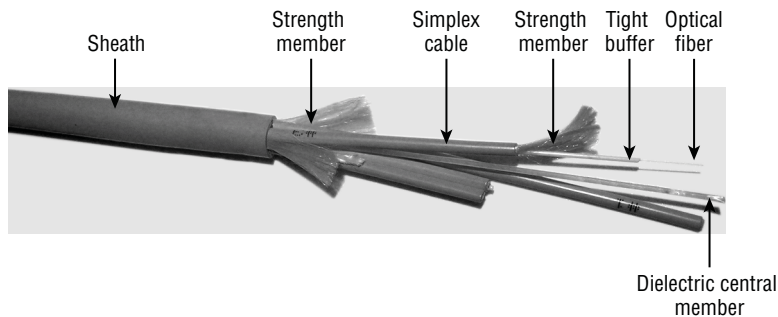
Distribution cables may carry up to 144 individual tight-buffered optical fibers, many of which may not be used immediately but allow for future expansion.

Breakout Cable

Breakout cables are used to carry optical fibers that will have direct termination to the equipment, rather than being connected to a patch panel.

Breakout cables consist of two or more simplex cables bundled with a strength member and/or central member covered with an outer sheath, as shown in Figure 7.15. These cables are ideal for routing in exposed trays or any application requiring an extra rugged cable that can be directly connected to the equipment.

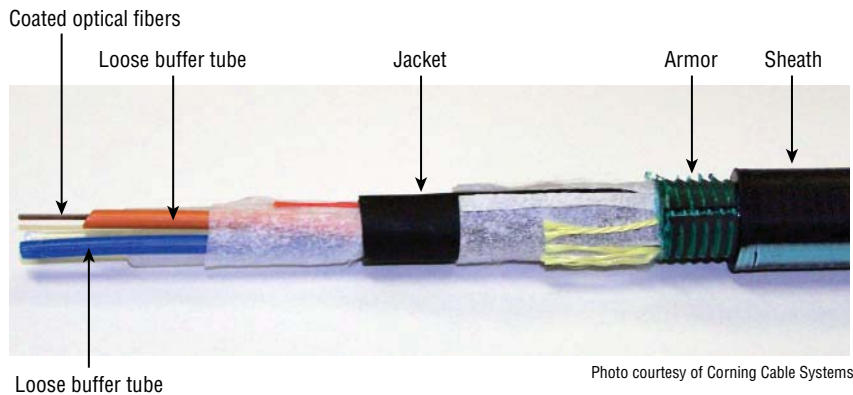
FIGURE 7.15 Breakout cable containing simplex cables bundled with a central strength member



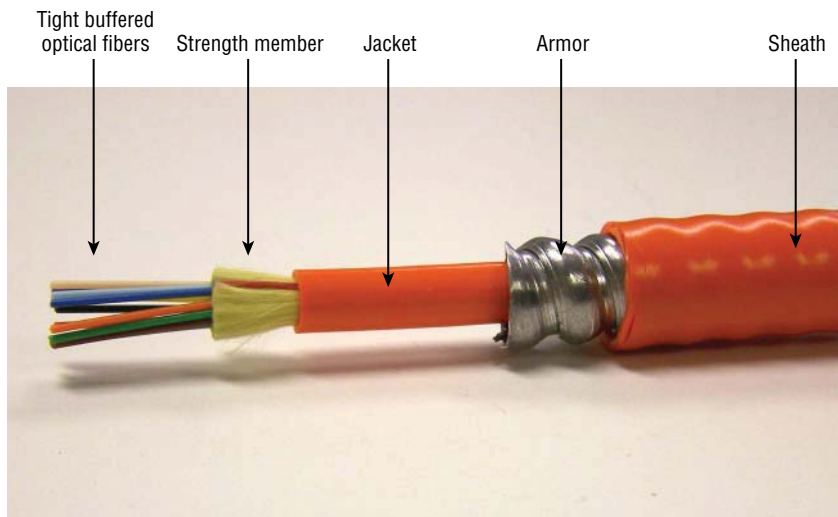
Armored Cable

Armored cable can be used for indoor applications and outdoor applications. An armored cable typically has two jackets. The inner jacket is surrounded by the armor and the outer jacket or sheath surrounds the armor.

An armored cable used for outdoor applications, shown in Figure 7.16, is typically a loose tube construction designed for direct burial applications. The armor is generally a corrugated aluminum tape surrounded by an outer polyethylene jacket. This combination of outer jacket and armor protects the optical fibers from gnawing animals and the damage that can occur during direct burial installations.

FIGURE 7.16 Armored cable for outdoor applications

Armored cable used for indoor applications may feature tight-buffered or loose-buffered optical fibers, strength member(s), and an inner jacket. The inner jacket is commonly surrounded by a spirally wrapped interlocking metal tape armor. This type of armor, shown in Figure 7.17, is rugged and provides crush resistance. These cables are used in heavy traffic areas and installations that require extra protection, including protection from rodents.

FIGURE 7.17 Armored cable for indoor applications

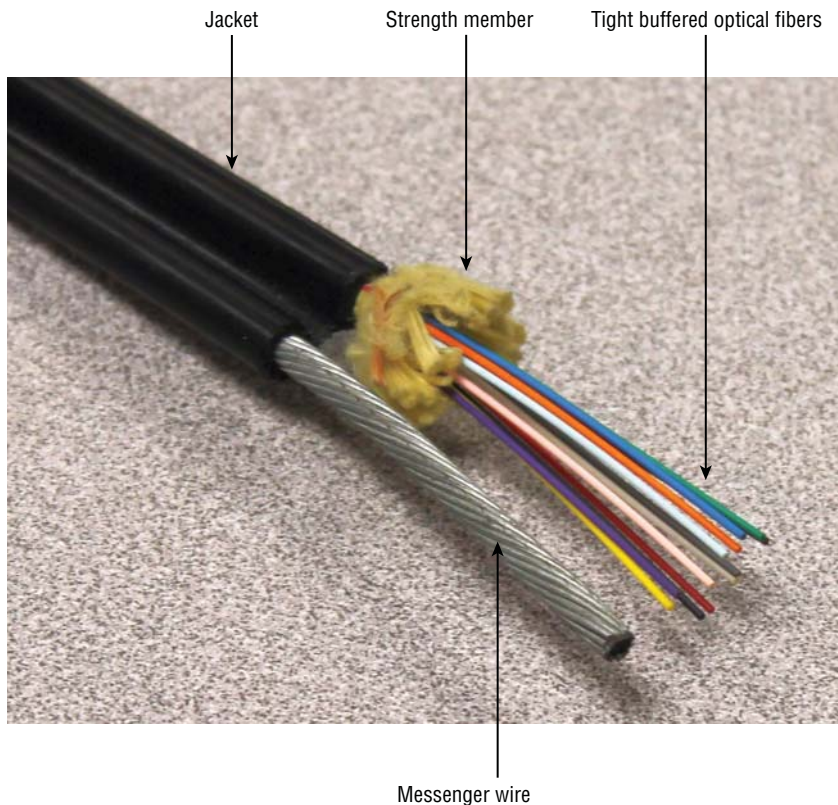
Messenger Cable

When a fiber-optic cable must be suspended between two poles or other structures, the strength members alone are typically not enough to support the weight of the cable and any

additional forces that may be placed on the cable. For aerial installations a messenger wire is required. The messenger wire can be external to the fiber-optic cable or integrated into the cable.

When the messenger wire is integrated into the fiber-optic cable, the cable is typically referred to as a *messenger cable*. These cables typically feature a 0.25" stranded steel messenger wire. The messenger wire, sometimes referred to simply as the messenger, is integrated into the outer jacket of the cable, as shown in Figure 7.18.

FIGURE 7.18 Messenger cable used for aerial installations



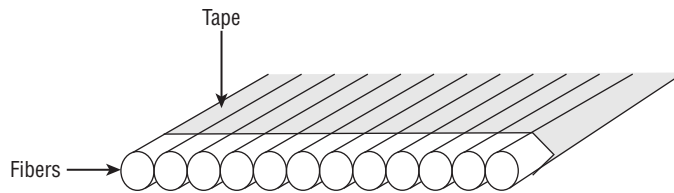
Also called *figure 8 cable* for the appearance of its cross section, messenger cable greatly speeds up an aerial installation because it eliminates the need to lash the fiber-optic cable to a pre-run messenger wire.

If a messenger wire is not incorporated into the cable assembly used, the cable will have to be lashed to messenger wire. A messenger wire is a steel or aluminum wire that supports the fiber-optic cable in an aerial installation. Either way, cables in aerial installations must be able to withstand loading from high winds, ice, birds and climbing animals, and even windblown debris such as branches.

Ribbon Cable

Ribbon cable is a convenient solution for space and weight problems. The cable contains fiber ribbons, which are actually coated optical fibers placed side by side, encapsulated in Mylar tape (see Figure 7.19), similar to a miniature version of wire ribbons used in computer wiring. A single ribbon may contain 4, 8, or 12 optical fibers. These ribbons can be stacked up to 22 high.

FIGURE 7.19 Ribbon cables consist of parallel fibers held together with Mylar tape.



Because the ribbon contains only coated optical fibers, this type of cable takes up much less space than individually buffered optical fibers. As a result, ribbon cables are denser than any other cable design. They are ideal for applications where limited space is available, such as in an existing conduit that has very little room left for an additional cable.

As shown in Figure 7.20, ribbon cables come in two basic arrangements. In the *loose tube ribbon cable*, fiber ribbons are stacked on top of one another inside a loose-buffered tube. This type of arrangement can hold several hundred fibers in close quarters. The buffer, strength members, and cable jacket carry any strain while the fiber ribbons move freely inside the buffer tube.

The *jacketed ribbon cable* looks like a regular tight-buffered cable, but it is elongated to contain a fiber ribbon. This type of cable typically features a small amount of strength member and a ripcord to tear through the jacket.

While ribbon fiber provides definite size and weight savings, it does require special equipment and training to take advantage of those benefits. Connectors, strippers, cleavers, and fusion splicers must all be tailored to the ribbon fiber. For these reasons, ribbon fiber may not be the best solution in all situations.

Submarine Cable

Submarine cable is specially designed for carrying optical fiber underwater. Not all submarine cable is the same, however. Depending on the distance it will span and the type of service it will provide, submarine cable can take many different forms.

Submarine cable may be laid in trenches under the bottom of waterways where shipping or fishing activities threaten to snag or damage the cable, or it may be laid directly on the bottom of less-traveled waterways, or on the deep ocean floor where such activities do not penetrate. Figure 7.21 shows a multi-optical-fiber submarine cable with armor and metal strength members.

FIGURE 7.20 Armored loose tube ribbon cable (bottom) and jacketed ribbon cable (top)

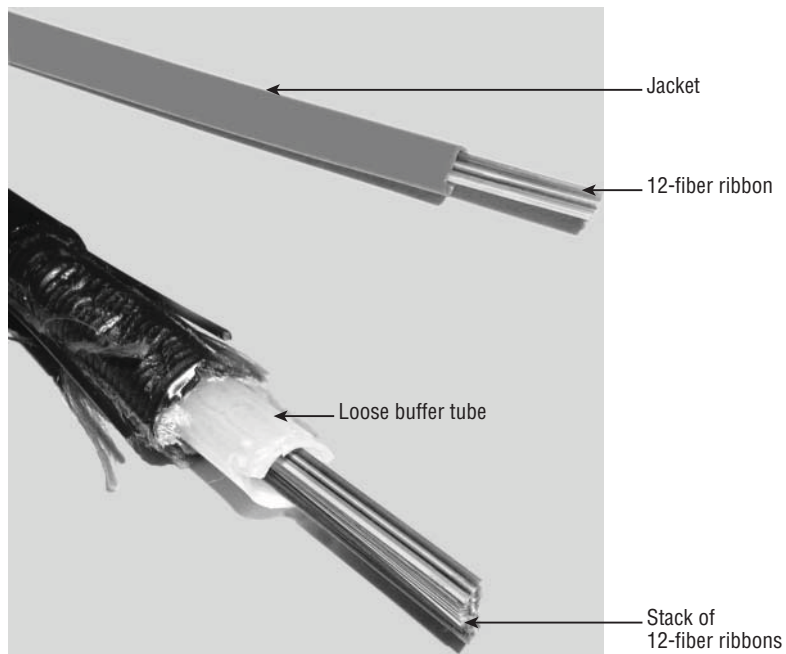
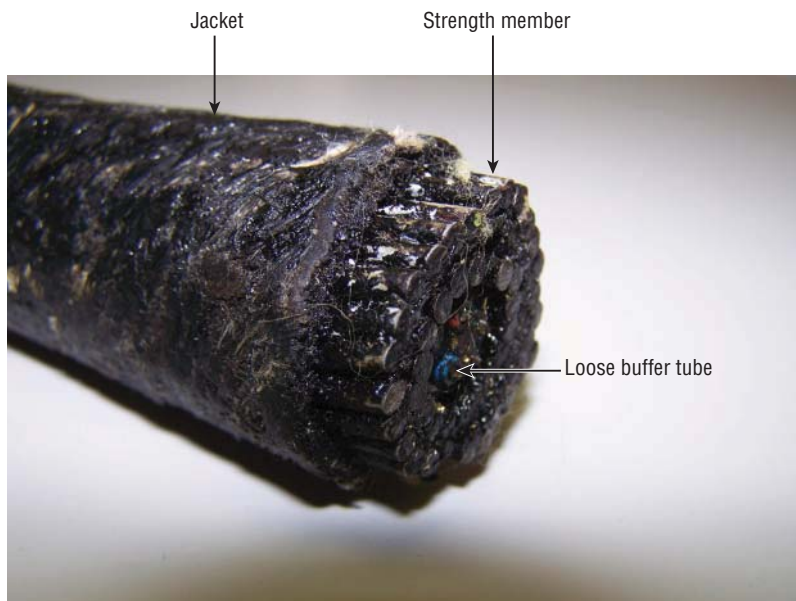


FIGURE 7.21 Submarine cable with armor and metal strength members





Real World Scenario

More Underwater Hazards

While fishing operations, anchoring, and other hazards pose a threat to underwater cables of all types, a fiber-optic cable running in the Grand Canary Islands encountered a special danger soon after it was laid.

The cable was part of a fiber-optic test system run by AT&T in 1985. Only three weeks after the cable was laid about 3/4 of a mile below the surface, the cable stopped operating and was out of commission for a week. Technicians first thought that the cable had separated because of abrasion due to scraping across the ocean floor, but when the cable was examined more closely, it was found to have small shark teeth embedded in it. The shark attacks were repeated twice over the next several months.

Why the repeated shark bites on fiber-optic cable? Sharks are known to be drawn to prey by their electromagnetic emissions, and attacks on standard coaxial cable would have been much more likely if the cable were not shielded against electromagnetic interference. But the fiber did not carry any electricity, only light, so the sharks should have ignored it.

The answer lay in the fact that light does have an electromagnetic signature, but because the optical fiber is not subject to electromagnetic interference, no shielding was applied to the cable. The small sharks were drawn to the weak electrical fields put out by the light traveling through the fiber and in the dark water attacked it, thinking it was a meal.

Aerospace Cable

Aerospace cables are designed to be installed in aircraft and spacecraft. These cables are designed to operate in extreme temperature environments. In addition, they must protect the optical fiber from the shock and vibration associated with aircraft and spacecraft. While the optical fiber used in many aerospace cables may be the same optical fiber used in other cable types, the coating, buffer, and jacket are typically different.

Structure Type

A variety of aerospace fiber-optic cables are available for use in different locations on aircraft or spacecraft. These cables can typically be separated into two structure types: tight and loose. Structure types should not be confused with buffer types. A tight-buffered optical fiber may be used in a loose structure fiber-optic cable.

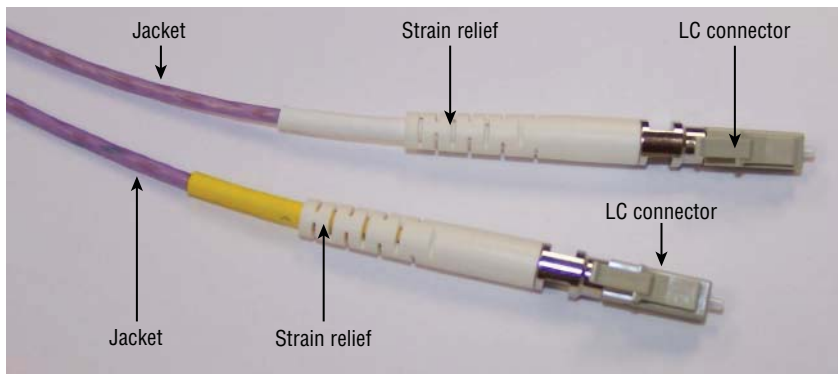
ARINC Report 802, Fiber Optic Cables defines the construction requirements for loose and tight structure cables for aerospace and avionic applications. A loose structure cable

allows limited movement of the buffered fiber (usually 900 μ m in diameter) with respect to the outer jacket and strength member. A tight structure cable does not allow any movement of the buffered fiber with respect to the outer jacket and strength member.

Applications

In Chapter 1, “History of Fiber Optics and Broadband Access,” you learned that since the turn of the century, fiber optics has helped to increase broadband download speeds by a factor of 5000. Just as the telecommunications industry is incorporating more and more optical fiber, so are the aerospace and avionics industries. The Boeing 787 Dreamliner has 110 fiber-optic links and over 1.7km of fiber-optic cable. The aerospace cable assembly shown in Figure 7.22 is similar to one that could be found on the Dreamliner.

FIGURE 7.22 Aerospace fiber-optic cable assembly with LC connectors



Drawing courtesy of Carlisle Interconnect Technologies

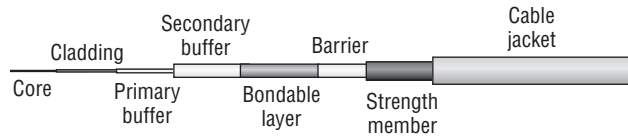
Aerospace tight or loose structure cables can have a very wide temperature range based on the location where they are installed. Cables designed to be used inside avionics boxes or cabinets typically have a temperature range of -40°C to $+85^{\circ}\text{C}$. Cables designed to be used between cabinets typically have a temperature range of -65°C to $+125^{\circ}\text{C}$, whereas those cables used by the engines have a temperature range of -65°C to $+260^{\circ}\text{C}$.

Cable Components

Figure 7.23 shows the components of a typical aerospace fiber-optic cable. This cable features a primary and secondary buffer. The primary buffer is the coating applied directly over the cladding of the optical fiber. Aerospace cables may use a variety of coatings depending on the temperature range required. A typical acrylate coating has a maximum operating temperature of 100°C whereas some high-temperature acrylates may perform at temperatures up to 125°C . Silicone, carbon, and polyimide coatings have greater

temperature ranges than acrylate. Detailed information on optical fiber coatings can be found in Chapter 4.

FIGURE 7.23 Typical aerospace fiber-optic cable



Drawing courtesy of Carlisle Interconnect Technologies

The secondary buffer shown in Figure 7.24 is polytetrafluoroethylene (PTFE). The bondable layer surrounding the secondary layer is a polyimide tape, and the barrier layer is unsealed PTFE. The strength member is braided Kevlar, and the jacket shown in Figure 7.25 is extruded FEP (fluorinated ethylene propylene). This jacket material has a service temperature range of -100°C to $+200^{\circ}\text{C}$ as well as excellent chemical and UV resistance properties; it is also flame retardant.

FIGURE 7.24 PTFE secondary buffer

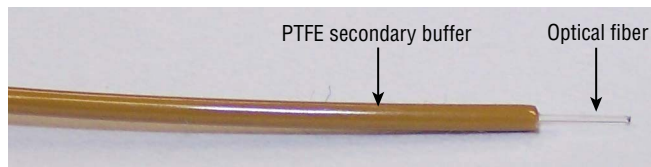


Photo courtesy of Carlisle Interconnect Technologies

Hybrid Cable

Hybrid cable, as applied to fiber optics, combines multimode and single-mode optical fibers in one cable. Hybrid cable should not be confused with composite cable, although the terms have been used interchangeably in the past.

Composite Cable

Composite cable, as defined by the National Electrical Code (NEC), is designed to carry both optical fiber and current-carrying electrical conductors. These cables may also contain non-current-carrying conductive members such as metallic strength members, metallic vapor barriers, metallic armor, or a metallic sheath.

The composite cable shown in Figure 7.26 consists of optical fibers along with twisted-pair wiring typical of telephone wiring. This arrangement is convenient for networks that carry fiber-optic data and conventional telephone wiring to the same user. Composite cable

also provides installers with a way to communicate during fiber installation and provides electrical power to remote equipment, such as repeaters, along the fiber's route.

FIGURE 7.25 FEP jacket and braided Kevlar strength member

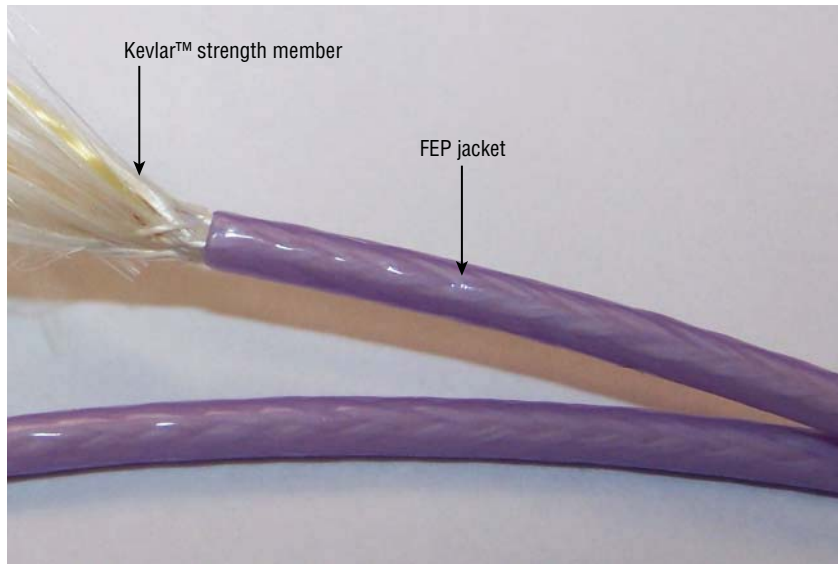
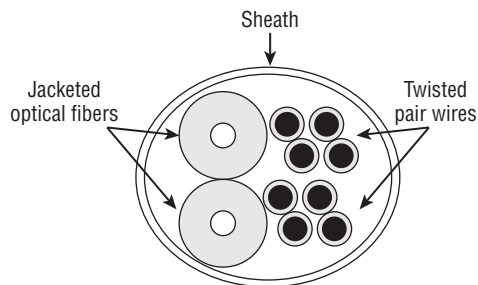


Photo courtesy of Carlisle Interconnect Technologies

FIGURE 7.26 Composite cable carries fiber and wiring in the same run.



Cable Duty Specifications

The various combinations of strength members, jacket materials, and fiber arrangements are determined by the specific requirements of an installation. Among the factors

considered are the amount of handling a cable will take, the amount of stress the cable must endure in normal use, and the locations where it will run.

Cable duty specifications are typically divided into two basic types:

Light-duty cables These are designed for basic protection of the fiber within and minimal handling. A good example of a light-duty fiber cable is a simplex cord, which consists of a tight-buffered fiber with a strength member and a jacket. Simplex cordage is not engineered to withstand excessive pulling forces, and the jacket is engineered for flexibility and ease of handling, not harsh environments.

Heavy-duty cables These are designed for more and rougher handling, with additional strength members and jacketing around the fiber. They are made for harder pulling during installation, and they protect the fiber within from damage in exposed or extreme environments.

Cable Termination Methods

Some fibers, such as those found in simplex and duplex cords and breakout cable, are already set up to receive connectors and can be handled easily. Others, including loose-buffered cables, must be prepared for connectors and handling with special kits.

These kits, known as *fanout kits* and *breakout kits*, are designed to adapt groups of coated fibers for connectors by separating them and adding a tight buffer (fanout kit) or a tight buffer, strength member, and jacket (breakout kit) to each one.

Fanout Kit

The fanout kit, shown in Figure 7.27, converts loose-buffered fibers into tight-buffered fibers ready for connectors. A typical fanout kit contains an enclosure sometimes called a *furcation unit*. The furcation unit attaches to the loose-buffer tube. Hollow tight-buffer tubes 900µm in diameter are applied over the optical fibers and passed into the furcation unit, which is then closed, locking the tight buffers in place on the fibers. After the fibers have the buffers applied, connectors can be attached for use in a patch panel or other protected enclosure.

Breakout Kit

The breakout kit, shown in Figure 7.28, is similar to the fanout kit in that it spreads the fibers from the loose-buffer tube through a furcation kit and provides 900µm tight buffers to be applied over the optical fiber. The breakout kit, however, is designed to allow the optical fiber to be connected directly to equipment with standard connectors.

FIGURE 7.27 A fanout kit adds a tight buffer to individual fibers.

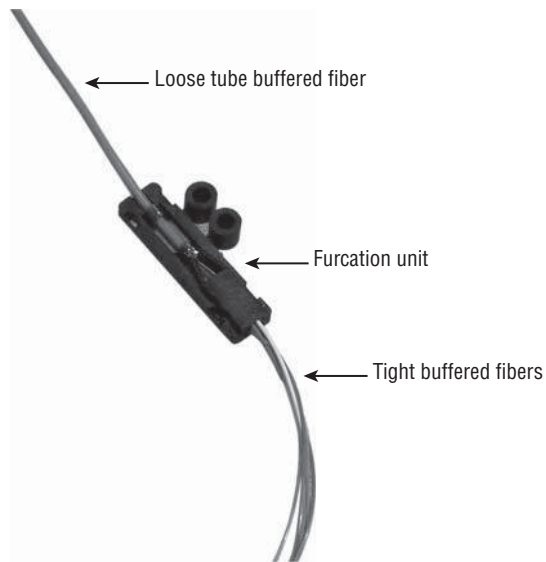
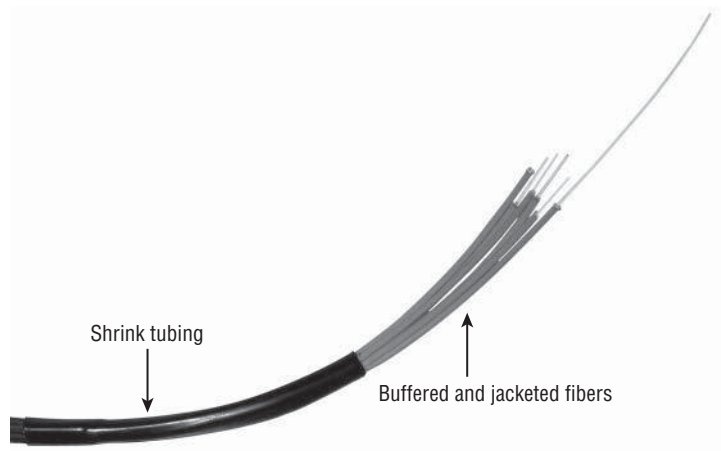


FIGURE 7.28 A breakout kit adds a tight buffer, strength member, and a jacket to individual optical fibers.



In addition to buffer material, the breakout kit provides a 3mm diameter jacket with an aramid strength member that slips over the optical fiber. Heat-shrink tubing and epoxy is used to join the individual jackets to the cable.

Breakout kits and fanout kits are available to match the number of fibers in the cable being used, and some companies offer kits that can be used with ribbon cable. Different length buffer tubes are available for fanout kits.

Note that different vendors may use a variety of terms or trade names to refer to breakout kits and fanout kits.

Blown Fiber

What if you knew that you wanted to run fiber in a building, but you didn't know exactly what equipment it would be serving or what the requirements for the fiber were? One solution to this dilemma is *blown fiber*.

Blown fiber installation starts with a hollow tube about 5mm in diameter. This tube is installed just like a cable and acts as a loose tube buffer for fibers that will run through it later. The tube may also be part of a cable assembly, which may carry up to 19 tubes and is available in configurations similar to other cables, including armored, all-dielectric, plenum, and riser.

To run the fibers through the tube, you simply lead the fibers into one end of the tube and blow pressurized air through the tube. The air carries the fibers with it through the tube, similar to the way that pneumatic carriers at drive-through tellers shoot your deposits into the bank building.

Blown fiber can be used, as already described, when the fiber needs in a building are not yet known or when fiber needs to be repaired or upgraded. Blown fiber is also useful when a company cannot afford to install all of the fiber that it might potentially use. As a business grows and the new fiber can be economically justified, it can be blown in as needed.

One advantage of blown fiber, even when it is installed all at once, is that cuts or damaged sections can be repaired quickly. In case of an accidental cut from power equipment, for example, the damaged section can be cut out, a new section spliced in, and new fiber blown in. This can be a great time-savings over splicing multiple broken fibers in a cable. Blown fiber can be run 1000' or more through a building, around curves, and uphill, in a very short time.

Blown fiber is covered in detail in Chapter 12, "Cable Installation and Hardware."

NEC Provisions for Fiber-Optic Cables and Raceways

The National Electrical Code (NEC) is published by the National Fire Protection Association (NFPA) and contains provisions that are considered necessary for safety. NEC Article 90 states that the purpose of the code is the practical safeguarding of persons and

property from hazards arising from the use of electricity. It also states that the NEC is not intended as a design specification or an instruction manual for untrained persons. Local and state governments typically adopt it, making compliance mandatory.

It provides specific guidance for running fiber-optic cable within buildings. The NEC requires cables to be tested for fire resistance and smoke characteristics, establishes which types of cables may be run in different areas of a building, and specifies the types of raceways that can be used.

Be sure you stay current with the latest NEC provisions. The NEC is updated every 3 years. The information in this chapter can be found in NEC Article 770, “Optical Fiber Cables and Raceways.” To ensure this chapter contains the latest information, the 2014 NEC was used for the updates.

Like the industry standards, Article 770 in the NEC has increased in page count since the turn of the century. It required approximately five pages in the 1999 NEC. That page count doubled in the 2014 NEC.

NEC Fiber-Optic Cable Types

The NEC recognizes three types of fiber-optic cables in Article 770:

Nonconductive Cables containing no electrically conductive materials.

Conductive Cables containing non-current-carrying conductive members, including strength members, armor, or sheath.

Composite Cables containing optical fiber and current-carrying electrical conductors. These cables may also contain non-current-carrying conductive members such as metallic strength members, metallic vapor barriers, metallic armor, or a metallic sheath.

Listed and Nonlisted

The NEC classifies fiber-optic cables according to their electrical and fire safety characteristics and contains rules for the use of each cable type to minimize hazards. Any fiber-optic cabling run indoors must be listed as suitable for the purpose. Any outside plant nonlisted fiber-optic cabling that is brought into a structure must be terminated after no more than 15m or 50' and connectorized or spliced to a listed cable.

A listed fiber-optic cable has had its performance verified by Underwriters Laboratory (UL). The outer surface of the cable or a marker tape should be clearly marked at intervals not to exceed 40" with the following information:

- Cable type-letter designation such as OFN, OFC, OFNR, OFNP, etc.
- Manufacturer's identification
- UL symbol or the letters “UL”

Optical fiber cables installed within buildings shall be listed as being resistant to the spread of fire. Three listed types of cables are recognized in Article 770 for indoor installations: plenum cable, riser cable, and general-purpose cable.

Plenum Cable

Plenum cables, whether conductive or nonconductive, are suitable for use in ducts, plenums, and other spaces used for environmental air. These cables will have fire resistance and low smoke-producing characteristics.

Riser Cable

Riser cables, whether conductive or nonconductive, are suitable for a vertical run in a shaft or from floor to floor. These cables will have fire-resistance characteristics capable of preventing the carrying of a fire from floor to floor.

General-Purpose Cable

General-purpose cables, whether conductive or nonconductive, are resistant to the spread of fire. However, these cables are not suitable for plenum or riser applications.

Table 7.1 shows the cable types along with the markings each cable will contain. Markings should be clearly visible on the cable, as shown in Figure 7.29, and no cable should be used indoors unless it listed and contains the NEC cable type marking on it.

FIGURE 7.29 NEC cable type marking



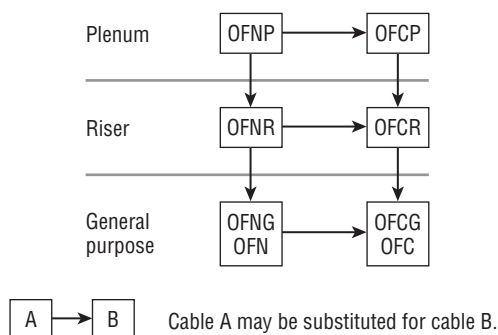
TABLE 7.1 NEC cable types and description

Marking	Type	Location	Permitted substitutions
OFNP	Nonconductive optical fiber plenum cable	Ducts, plenums, other air spaces	None
OFCP	Conductive optical fiber plenum cable	Ducts, plenums, other air spaces	OFNP
OFNR	Nonconductive optical fiber riser cable	Risers, vertical runs	OFNP
OFCR	Conductive optical fiber riser cable	Risers, vertical runs	OFNP, OFCP, OFNR
OFNG	Nonconductive optical fiber general-purpose cable	General-purpose use except for risers and plenums	OFNP, OFNR

Marking	Type	Location	Permitted substitutions
OF CG	Conductive optical fiber general-purpose cable	General-purpose use except for risers and plenums	OFNP, OFCP, OFNR, OFCR, OFNG, OFN
OFN	Nonconductive optical fiber general-purpose cable	General-purpose use except for risers, plenums	OFNP, OFNR
OFC	Conductive optical fiber general-purpose cable	General-purpose use except for risers, plenums	OFNP, OFCP, OFNR, OFCR, OFNG, OFN

All indoor cables must be resistant to the spread of fire, and those listed as suitable for environmental air spaces such as plenums must also have low smoke-producing characteristics. Note in the substitutions list that nonconductive cables may always be substituted for conductive cables of an equal or lower rating, but conductive cables may never be substituted for nonconductive cables. Figure 7.30 shows the order in which cables may be substituted for one another.

FIGURE 7.30 NEC cable substitution guide



The NFPA 262 describes test methods for flame travel and smoke of cables for use in air-handling spaces.

The NEC also describes the *raceways* and *cable trays* that can be used with fiber-optic cables and the conditions under which each type of cable may be used. This information is covered in Chapter 12.

Cable Markings and Codes

In addition to the NEC cable marking, optical fiber cables typically have a number of other markings and codes. Markings that appear on the jacket of the cable help identify what is inside the cable and where it originated. They may also include aids for measuring the cable length.

Inside the cable, the fiber buffers are usually color-coded with standard colors to make connections and splices easier. However, some manufacturers use numbers instead of colors.

External Markings

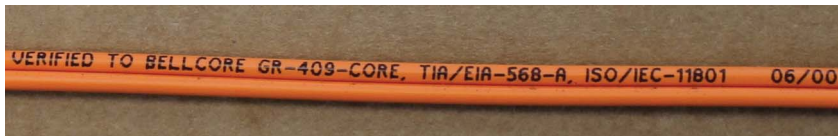
A cable's external markings typically consist of the manufacturer's information, including the manufacturer's name and phone number, cable part number or catalog number, the date the cable was manufactured, and occasionally the industry standards the optical fiber complies to. Information about the cable itself includes the NEC cable marking, the fiber type (multimode/single-mode or core/cladding size), and sequential cable length markings in meters or feet.

Date, Industry Standards, and Fiber Type

The date the cable was manufactured can be very helpful in determining the performance standards of the optical fiber within the cable. As you learned in Chapter 5, "Optical Fiber Characteristics," there have been significant changes in the performance of multimode optical fiber since the turn of the century. The cable shown in Figure 7.31 was manufactured at the turn of the century. In addition, the cable markings identify three standards that define the optical and geometrical performance of the optical fiber. These standards are

- Bellcore GR-409-CORE
- ISO/IEC 11801
- TIA/EIA-568-A

FIGURE 7.31 Duplex cordage manufactured in June 2000



The cable shown in Figure 7.31 was manufactured in June 2000. Optical Fiber Cabling Component Standard TIA/EIA-568-B.3 superseded the Commercial Building

Telecommunications LAN/WAN Cabling Standard TIA/EIA-568-A in April 2000. This is an example of cable being manufactured with optical fiber that did not comply with the latest published standards. As it has been mentioned before in this book, when a standard changes, old cable is not ripped out and replaced with new. A cable manufacturer with reels of TIA/EIA-568-A optical fiber on the shelf is not going to waste that inventory just because a standard changed.

The cable shown in Figure 7.32 was manufactured in May 2008. There are no standards markings on this cable, so determining the industry standard(s) that define the optical and geometrical performance of the optical fibers in this cable is not as straightforward as with the cable in Figure 7.31. However, there is a clue: the word “laser” is printed on the jacket. This implies that the optical fiber was optimized for a laser light source. The 50µm optical fiber core size is also printed on the jacket. Combining both pieces of data indicates that the optical fiber in the cable is OM3.

FIGURE 7.32 Laser-optimized 6-fiber breakout cable manufactured in May 2008



OM3 optical fiber cannot be found in TIA/EIA-568-B.3. However, it can be found in ANSI/TIA-568-C.3, which was published June 2008, one month after the cable shown in Figure 7.31 was manufactured. This is an example of a cable that contains an optical fiber that exceeds the transmission performance parameters of a current published standard. As mentioned previously in this book, standards define a minimum level of performance and many manufacturers offer products that exceed the minimum performance levels. Oftentimes the working groups within standards organizations are playing catch-up, creating standards for products that currently exist and may have been in widespread use for several years or more.

Remember, you can always contact the cable manufacturer using the phone number printed on the cable to determine the optical and geometrical performance of the optical fiber.

Color Codes

As with copper wiring, optical fibers running in cables must have some way of being distinguished from one another so that they can be connected properly at each end. TIA-598-C provides color-coding schemes for premises jackets and optical fibers within a fiber-optic cable. Table 7.2 shows the color-coding scheme for individual fibers bundled in a cable.

TABLE 7.2 TIA-598-C fiber color code

Fiber number	Base color/tracer	Fiber number	Base color/tracer
1	Blue	19	Red/Black
2	Orange	20	Black/Yellow
3	Green	21	Yellow/Black
4	Brown	22	Violet/Black
5	Slate	23	Rose/Black
6	White	24	Aqua/Black
7	Red	25	Blue/Double Black
8	Black	26	Orange/Double Black
9	Yellow	27	Green/Double Black
10	Violet	28	Brown/Double Black
11	Rose	29	Slate/Double Black
12	Aqua	30	White/Double Black
13	Blue/Black	31	Red/Double Black
14	Orange/Black	32	Black/Double Yellow
15	Green/Black	33	Yellow/Double Black
16	Brown/Black	34	Violet/Double Black
17	Slate/Black	35	Rose/Double Black
18	White/Black	36	Aqua/Double Black

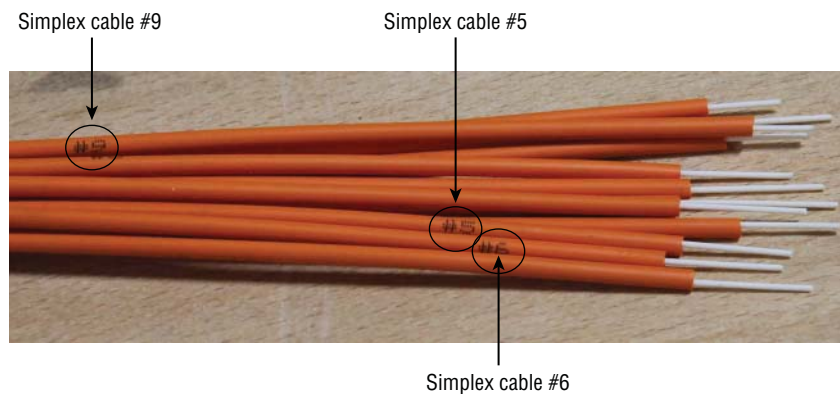
Table 7.3 shows the color coding used on premises cable jackets to indicate the type of fiber they contain, if that is the only type of fiber they contain.

TABLE 7.3 TIA-598-C premises cable jacket colors

Fiber type	Jacket color for nonmilitary applications	Jacket color for military applications
Multimode (50/125 μ m)	Orange	Orange
Multimode (50/125 μ m) Laser-optimized	Aqua	
Multimode (62.5/125 μ m)	Orange	Slate
Multimode (100/140 μ m)	Orange	Green
Single-mode (NZDS)	Yellow	Yellow
Polarized maintaining Single-mode	Blue	

Cable Numbers

The color-coding schemes for premises jackets and optical fibers within a fiber-optic cable described in TIA-598-C are not always used. For example, the jacket colors of every simplex cable shown in Figure 7.33 are the same. You can identify a specific simplex cable by locating the number that is printed on the jacket. The simplex cable number is printed every 6" along the entire length of the cable.

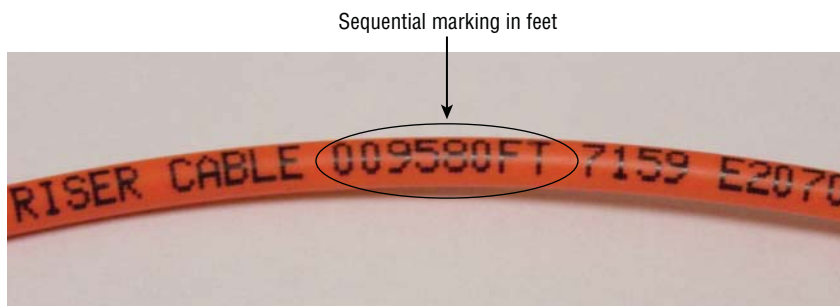
FIGURE 7.33 Breakout cable with 12-simplex cables, each with printed cable numbers

With all numbering or coloring schemes, there are advantages and disadvantages. In an area that is not lit well, it can be difficult to correctly identify colors. In addition, how well color differences are perceived can vary from person to person. Numbering eliminates color perception problems.

Sequential Markings

A cable's external markings may include sequential markings, as shown in Figure 7.34. Sequential markings are numbers that appear every 2' or 1m. These markings are useful in determining how much cable is left on a reel, measuring off large runs of cable, or simply determining the length of a piece of cable without pulling out a tape measure.

FIGURE 7.34 Sequential marking in feet on a fiber-optic cable



The numbers themselves indicate cable length, not optical-fiber length. To measure the length of the cable using the sequential markings, first determine the measurement standard that is being used. Next, subtract the number at the low end from the number at the high end. The difference between the two is the length.

Because some measurements dealing with fiber optics use meters rather than feet, you may need to convert any measurements in feet to the metric system. The formula for converting length is

$$1 \text{ foot} = 0.3048 \text{ meter}$$

Let's practice measuring out a length of cable using sequential markings. To find the length in meters of a cable that has sequential markings of 846 at one end and 2218 at the other end, and the markings are measuring the cable in feet, you first determine the distance between the markings using this formula:

$$2218 - 846 = 1372 \text{ feet}$$

Now, convert feet to meters:

$$1372 \times 0.3048 = 418.2 \text{ meters}$$

Remember that the length of the cable is not necessarily the length of the optical fiber inside of it. In loose tube cable, the fiber is actually slightly longer than the cable. This fact

becomes important if fault location procedures tell you that there is a fault at a specific distance from the end of the fiber. That distance could be short of the measured distance on the outside of the cable since the fiber wanders inside of the buffer tube.

Bend Radius Specifications

Throughout the installation process, optical fiber's light-carrying abilities are threatened by poor handling, damage from tools or accidents, and improper installation procedures.

One of the installation hazards that can cause attenuation or damage to the optical fiber is extreme bending. When an optical fiber is bent too far, the light inside no longer reflects off the boundary between the core and the cladding, but passes through to the coating.

To reduce the risk of excessive bending during installation, manufacturers specify a minimum installation and operational bend radius for their optical fiber cables. Following the manufacturer's guidelines reduces the risk of damage to the cable and optical fiber during and after installation. Chapter 12 provides detailed information on cable bend radius.

Summary

This chapter covered the types of cables used to carry optical fibers while protecting them and keeping large numbers of fibers bundled together. It described the materials used in common cables and the different types of cables used to meet different needs.

The chapter described the ways in which fiber can be arranged within cables, along with the methods used to protect the fibers from stretching, damage, and excessive handling. It also described some of the locations in which cables commonly run.

Finally, the chapter discussed NEC fiber-optic cable types and classifications. It described the different types of cable markings and color codes. It also explained how to use sequential markings and where to find cable bend radius information.

Exam Essentials

Understand optical fiber cable construction. Make sure that you understand the basic components of a fiber optic cable. Be able to describe the materials commonly used in fiber optic cables. Be able to explain the differences between some types of cable components.

Describe the types of fiber optic cables. Be able to describe the different types of fiber optic cables. Be able to explain how different types of fiber optic cables are used. Be familiar with the advantages and drawbacks of different types of cable.

Know the NEC cable types and listing requirements. Make sure you know the NEC fiber-optic cable types and classifications. Be able to explain the difference between a listed and nonlisted cable. Be familiar with the NEC cable substitution guide.

Describe typical cable jack markings. Be able to list the external markings typically found on the jacket of a fiber-optic cable.

Know the color-coding and numbering schemes. Make sure that you know the TIA-598-C color-coding schemes for premises jackets and optical fibers within a fiber-optic cable.

Understand sequential markings. Be able to explain how sequential markings are used to determine how much cable is left on a reel, measure off large runs of cable, or simply determine the length of a piece of cable without pulling out a tape measure.

Review Questions

1. Which of the following describes the four components of a fiber optic cable?

- A. Core, cladding, coating, buffer
- B. Optical fiber, buffer, strength member, jacket
- C. Core, buffer, jacket, coating
- D. Optical fiber, buffer, strength member, coating

Hint: The outer layer protects the internal components from the outside world.

2. A buffer tube that has an inner diameter much larger than the diameter of the coated optical fiber is called a

- A. Loose buffer
- B. Tight buffer
- C. Coating
- D. Tensile buffer

Hint: Sometimes it is filled with gel.

3. One benefit of tight-buffered fiber is that it

- A. Has greater tensile strength
- B. Has less attenuation
- C. Can run for short distances outside the cable
- D. Carries a signal farther

Hint: The standard diameter of the buffer allows many different types of connectors to be applied.

4. Aramid yarn is used as a strength member because it is

- A. Flexible, light, and strong
- B. Rigid, heavy, and strong
- C. Conductive
- D. Only used in combination with fiberglass rods

Hint: It is also sold under the trade name Kevlar.

5. The outer protective layer of a fiber-optic cable is called the

- A. Coating
- B. Buffer
- C. Jacket
- D. Ripcord

Hint: It may be referred to as the sheath.

6. Duplex cordage is convenient for applications that require
- A. One-way transmission
 - B. Two-way transmission
 - C. Electrical power
 - D. Two transmission speeds

Hint: It combines two simplex cords.

7. Breakout cables contain
- A. Two or more individual tight-buffered fibers
 - B. Two or more loose buffer tubes
 - C. Two or more simplex cables
 - D. A mix of loose buffer and tight-buffered fibers

Hint: Each optical fiber has a jacket.

8. Distribution cables contain
- A. Two or more individual tight-buffered fibers
 - B. Two or more loose buffer tubes
 - C. Two or more simplex cables
 - D. A mix of loose buffer and tight-buffered fibers

Hint: There is only a single jacket.

9. An armored cable protects the optical fibers from
- A. Temperature changes
 - B. Sunlight
 - C. Gnawing animals
 - D. Magnetic fields

Hint: It also protects the optical fibers from damage that can occur during direct burial installations.

10. A messenger cable would typically be
- A. Suspended between two poles or other structures
 - B. Laid in trenches next to the road
 - C. Laid in trenches under the bottom of waterways
 - D. Installed in aircraft and spacecraft

Hint: Also called a figure 8 cable

11. Hybrid cable as applied to fiber optics combines
- A. Loose and tight buffers
 - B. Single-mode and multimode optical fibers
 - C. Optical fibers and current-carrying electrical conductors
 - D. Plastic and glass optical fibers

Hint: It should not be confused with composite cable.

12. Composite cable, as defined by the NEC, is designed to carry
- A. Optical fiber and conductive strength members
 - B. Single-mode and multimode optical fibers
 - C. Optical fibers and current-carrying electrical conductors
 - D. Plastic and glass optical fibers

Hint: They can provide installers with a way to communicate during installation.

13. A _____ converts loose-buffered fibers into tight-buffered fibers.
- A. Breakout kit
 - B. Buffer kit
 - C. Coating kit
 - D. Fanout kit

Hint: Typically contains a furcation unit.

14. Which type of NEC recognized fiber-optic cable contains non-current-carrying conductive members?
- A. Nonconductive
 - B. Conductive
 - C. Composite
 - D. Hybrid

Hint: It may also have armor.

15. Which type of NEC listed fiber-optic cable is suitable for use in ducts and other spaces used for environmental air?
- A. Plenum
 - B. Riser
 - C. General-purpose
 - D. Hybrid

Hint: These cables have low smoke-producing characteristics.

16. Which NEC marking describes a nonconductive general-purpose fiber-optic cable?

- A.** OFCG
- B.** OFNP
- C.** OFCR
- D.** OFN

Hint: It has two possible markings.

17. Which of the following can be substituted for a cable marked “OFN”?

- A.** OFC
- B.** OFCP
- C.** OFNP
- D.** OFCR

Hint: It must be the same NEC listed cable type.

18. What is the jacket color of laser-optimized simplex cordage?

- A.** Orange
- B.** Slate
- C.** Blue
- D.** Aqua

Hint: There is no TIA-598-C jacket color for military applications.

19. Color-coding schemes for optical fibers within a fiber-optic cable are sometimes replaced by _____.

- A.** Sequential markings
- B.** Numbers
- C.** Letters
- D.** Symbols

Hint: It is printed every 6”.

20. _____ are useful in determining how much cable is left on a reel.

- A.** Color codes
- B.** Cable numbers
- C.** Sequential markings
- D.** NEC listing

Hint: They appear every 2’ or 1m.

Chapter Exercises

Determine the Cable Type from the NEC Markings

Article 770 of the NEC states that optical fiber cables installed within buildings shall be listed as being resistant to the spread of fire. These cables are also required to be marked.

You have been asked to install the two cables shown later in a riser space. From the markings on the cables, determine the cable types and determine if each cable can be installed in a riser space.



The cable marked OFNR is a nonconductive riser cable and the cable marked OFNP is a nonconductive plenum cable. Per the NEC cable substitution guide in Figure 7.30, both cables can be installed in a riser space. However, only the cable marked OFNP can be installed in the plenum space.

Identify the Fiber Number from the Color Code

TIA-598-C defines a color code for optical fibers within a cable assembly.

You have been asked to identify the fourth and seventh tight-buffered optical fibers in the 12-fiber cable shown here. There are no numbers on the tight buffers; however, each buffer is colored as defined in TIA-598-C. What are the colors of the fourth and seventh tight-buffered optical fibers?



Table 7.2 lists the optical fiber color codes as defined in TIA-598-C. The color for fiber number 4 is brown and the color for fiber number 7 is red.

Determine the Fiber Number from the Cable Markings

The color-coding schemes for premises jackets and optical fibers within a fiber-optic cable described in TIA-598-C are not always used.

You have been asked to identify simplex cable number nine in the 12-fiber cable shown here. Describe the location of that cable.



Simplex cable number 9 is located in the upper-left corner.

Identify the Optical Fiber Type from the Color Code

Premises cable jacket colors are defined in TIA-598-C.

It is your first day on the job and your supervisor has asked you to get some laser-optimized multimode optical fiber from the back of the van. The van contains several

different cables, each with a different jacket color. What is the jacket color of the laser-optimized cable?

Table 7.3 list the premises jacket colors as defined in TIA-598-C. The color for the laser-optimized cable is aqua.

Determine the Optical Fiber Type from the Cable Markings

Premises cable jacket colors are defined in TIA-598-C. However installed cables may not comply with revision C of this standard.

You have been asked to identify the optical fiber type in the cable shown here. The jacket color is slate and this cable was not intended for military applications. What type of optical fiber is used in this cable?



This cable was manufactured in 1999, 6 years before TIA-598-C was published. It does not comply with this standard. However, the optical fiber type is printed on the jacket. The optical fiber in this simplex cable is 62.5/125 μ m multimode.

Determine the Cable Length Using Sequential Markings

Many markings are found on a cable. Sequential markings typically appear every 2 feet or every 1m.

You have been asked to determine the length of fiber-optic cable coiled up on the floor. The sequential markings for this cable are meters. The marking on one end of the cable is 0235 and the marking on the other end is 0485. How long is this cable?

To find the length of the cable, determine the distance between the markings as shown here:

$$485\text{m} - 235\text{m} = 250\text{m}$$

Chapter

8

Splicing

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Describe the intrinsic factors that affect splice performance.
- ✓ Describe the extrinsic factors that affect splice performance.
- ✓ Explain how a mechanical splice creates a low loss interconnection.
- ✓ Describe how to assemble a mechanical splice.
- ✓ Explain how a fusion splicer creates a low loss interconnection.
- ✓ Describe the basic operation of a fusion splicer.
- ✓ Describe the different alignment techniques that can be used to align the optical fibers.
- ✓ Explain how to assemble and protect a fusion splice.
- ✓ List the ANSI/TIA-568-C inside plant splice performance requirements.
- ✓ List ANSI/TIA-758-B outside plant splice performance requirements.
- ✓ Describe the Telcordia GR-765 required and objective fusion splice insertion loss requirements for passive and active alignment splicers.





In this chapter, we will discuss fiber-optic splices and examine the factors that affect splice performance. We will also describe tools and methods used to splice optical fibers as well as standards used to gauge splice performance.

In this chapter, you will learn to

- Determine if the splice loss is from an intrinsic factor
- Determine if the splice loss is from an extrinsic factor
- Calculate the potential splice loss from a core diameter mismatch
- Troubleshoot a fusion splice

Why Splice?

Up to this point, we have not addressed how to connect or mate two optical fibers together. Nor have we discussed why we would want to do this. This chapter discusses splicing. Splicing is one way to join two optical fibers together so the light energy from one optical fiber can be transferred into another optical fiber.

A splice is a permanent connection of two optical fibers. Once the two optical fibers are joined with a splice, they cannot be taken apart and put back together as they can if you join them using connectors. A splice is typically employed for one of four reasons: to repair a damaged cable, to extend the length of a cable, to join two different cable types, or to attach a pigtail. Pigtails are covered in-depth in Chapter 9, “Connectors.”

Splice Performance

How well a splice performs depends on many variables. These variables can be broken into two groups: *intrinsic factors* and *extrinsic factors*.

As you have learned in earlier chapters, optical fibers are not perfect and variations between optical fibers can affect splice performance. These variations are referred to as intrinsic factors.

The performance of a splice can also be affected by alignment and optical fiber mating issues that have nothing to do with the optical fiber. The factors that affect the alignment and/or mating of the optical fibers are referred to as extrinsic factors.

Intrinsic Factors

Even when fibers are manufactured within specified tolerances, slight variations still exist from one optical fiber to another. These variations can affect the performance of the splice even though the optical fibers are perfectly aligned when mated. The variations between two optical fibers that affect splice performance are referred to as intrinsic factors.

Let's look at the most common types of variations.

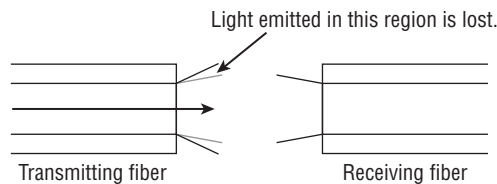
Numerical Aperture (NA) Mismatch

A numerical aperture (NA) mismatch occurs when the NA of one optical fiber is different from the NA of the other optical fiber. If the NA of the transmitting fiber is larger than the NA of the receiving optical fiber, a loss may occur. However, a loss will not occur if the NA of the transmitting optical fiber is less than the NA of the receiving optical fiber.

In Chapter 5, “Optical Fiber Characteristics,” we showed you how to calculate the NA of an optical fiber using the refractive indexes of the core and cladding. Recall that in order for light to be contained within a multimode optical fiber it must reflect off the boundary between the core and the cladding, rather than penetrating the boundary and refracting through the cladding. Light must enter within a specified range defined by the acceptance cone or cone of acceptance.

Light entering the core of an optical fiber from outside of the acceptance cone will either miss the core or enter at an angle that will allow it to pass through the boundary with the cladding and be lost, as shown in Figure 8.1. Light entering the core of the optical fiber at an angle greater than the acceptance angle will not propagate the length of the optical fiber. For light to propagate the length of the optical fiber, it must enter at an angle that does not exceed the acceptance angle.

FIGURE 8.1 When NA mismatch loss occurs, the receiving optical fiber cannot gather all of the light emitted by the transmitting fiber.



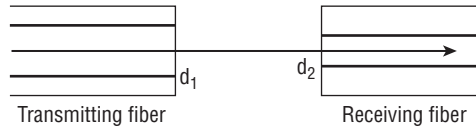
The exact loss from an NA mismatch is difficult to calculate. Factors such as light source type, light source launch condition, optical fiber length, and bends in the optical fiber all affect the potential loss. It is possible to have an NA mismatch between two optical fibers and no loss resulting from the mismatch.

Core Diameter Mismatch

Core diameter mismatch occurs when there is a difference in the core diameters of the two optical fibers. A core diameter mismatch loss results when the core diameter of the

transmitting optical fiber is greater than the core diameter of the receiving optical fiber, as shown in Figure 8.2. A loss occurs when light at the outer edge of the transmitting optical fiber core falls outside the diameter of the receiving optical fiber core. This light is lost in the cladding of the receiving optical fiber. Core diameter mismatch loss is typically only a concern with multimode optical fiber.

FIGURE 8.2 Core diameter mismatch loss is the result of the transmitting optical fiber having a larger core diameter than the receiving optical fiber.



It is not uncommon for two multimode optical fibers with different core diameters to be spliced together. As you saw in Chapter 5, ANSI/TIA-568-C.3 recognizes multimode optical fibers with 50 μm cores and 62.5 μm cores. We can calculate the worst-case loss percentage for a splice that joins a 50 μm core and 62.5 μm core using this formula:

$$\text{Loss} = (d_1^2 - d_2^2) / d_2^2$$

where d_1 is the diameter of the transmitting core and d_2 is the diameter of the receiving core.

$$62.5^2 - 50^2 = 1406.25$$

$$1406.25 / 50^2 = 0.5625 \text{ or } 56.25\%$$

Using the following formula, the decibel loss can be calculated:

$$\text{dB} = 10\text{Log}_{10}(P_{\text{out}} / P_{\text{in}})$$

Note that the decibel formula is a ratio where the power output (P_{out}) is divided by the power input (P_{in}). In this case, the power input is the power being put into the receiving optical fiber. We know that the receiving optical fiber will not accept 56.25 percent of the light from the transmitting optical fiber. Subtracting 56.25 percent from 100 percent, as shown next, results in a difference of 43.75 percent, which can be written as 0.4375.

$$100\% - 56.25\% = 43.75\% \text{ or } 0.4375$$

$$10\text{Log}_{10}(0.4375 / 1) = -3.59\text{dB}$$

$$\text{Loss} = 3.59\text{dB}$$

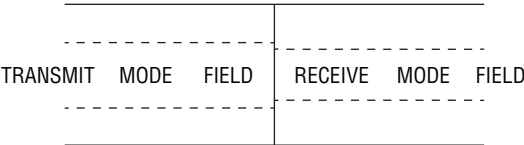
The 3.59dB loss that we just calculated is the maximum loss possible from the core diameter mismatch. The actual loss may vary depending on factors such as light source type, light source launch condition, optical fiber length, and bends in the optical fiber.

Mode Field Diameter Mismatch

Mode field diameter is only a concern in single-mode optical fibers. It describes the diameter of the light beam traveling through the core and a portion of the inner cladding. Values for the mode field diameter are defined by wavelength with longer wavelengths typically having a larger diameter than shorter wavelengths. A typical mode field diameter nominal range is 8.6 to 9.55μm at 1310nm and 8 to 11μm at 1550nm.

A mode field diameter mismatch occurs when there is a difference in the mode field diameters of two single-mode optical fibers. A mode field diameter mismatch loss results when the mode field diameter of the transmitting optical fiber is greater than the mode field diameter of the receiving optical fiber, as shown in Figure 8.3. A loss occurs when optical fiber with the smaller mode field diameter will not accept all of the light from the optical fiber with the larger mode field diameter.

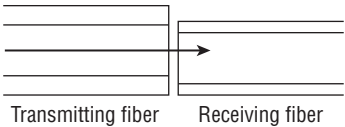
FIGURE 8.3 Mode field diameter mismatch loss is the result of the transmitting optical fiber having a larger mode field diameter than the receiving optical fiber.



Cladding Diameter Mismatch

Cladding diameter mismatch occurs when the cladding diameters of the transmit and receive optical fibers are not the same. Cladding diameter mismatch loss occurs when the cores of the optical fiber are not aligned because of the cladding diameter mismatch, as shown in Figure 8.4. A cladding diameter mismatch can cause the light exiting the core of the transmitting optical fiber to enter the cladding of the receiving optical fiber. The light entering the cladding is lost, causing attenuation.

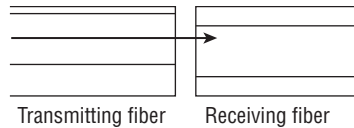
FIGURE 8.4 Cladding diameter mismatch loss results from differing cladding diameters.



Concentricity

Ideally, the core and cladding of an optical fiber are perfectly round and *concentric*, which means that they share a common geometric center. However, optical fibers are not perfect and there will be concentricity variations. These concentricity variations can cause the optical fiber cores to misalign, as shown in Figure 8.5, causing a loss when the light exiting the core of the transmitting optical fiber enters the cladding of the receiving optical fiber.

FIGURE 8.5 Off-center fiber cores cause concentricity loss.

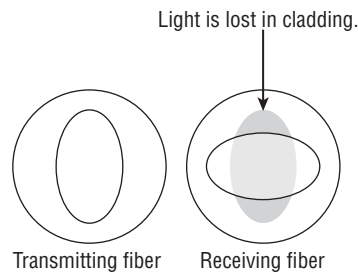


The illustration in Figure 8.5 is greatly exaggerated to clearly show how a concentricity loss may occur. The core and cladding concentricity differences are typically less than 1 μ m.

Noncircularity

Just as the core and cladding of an optical fiber may not be perfectly concentric, they may also not be perfectly circular. The noncircularity of the core will cause a loss when light from the core of the transmitting optical fiber enters the cladding of the receiving optical fiber. In Figure 8.6, the core is elliptical to show how core noncircularity loss occurs.

FIGURE 8.6 Noncircularity loss takes place when the ellipticities of two optical fibers do not match exactly.



Cladding noncircularity may cause loss when it causes part of the core of the transmitting optical fiber to align with the cladding of the receiving optical fiber. Any light that enters the cladding of the receiving optical fiber will be lost, causing attenuation.

Note that the amount of loss depends on the alignment of the ellipticities of the two cores. Maximum loss occurs when the long or major axes of the cores are at right angles to one another, and minimum loss occurs when the axes of the cores are aligned. After reading about all the possible intrinsic losses, you may find it hard to believe that two optical fibers can be spliced together with virtually no loss. Optical fiber manufacturing produces

optical fibers with little variation, as shown in Tables 8.1, 8.2, 8.3, and 8.4. These tables list the attributes and tolerances for different single-mode optical fibers as defined in ITU standards ITU-T G.652, ITU-T G.655, and ITU-T G.657.

TABLE 8.1 ITU-T G.652, single-mode optical fiber attributes

G.652.A, G.652.B, G.652.C, and G.652.D		
Fiber attributes	Detail	Value
Mode field diameter	Wavelength	1310nm
	Range of nominal values	8.6–9.5 μ m
	Tolerance	$\pm$ 0.6 μ m
Cladding diameter	Nominal	125.0 μ m
	Tolerance	$\pm$ 1 μ m
Core concentricity error	Maximum	0.6 μ m
Cladding noncircularity	Maximum	1%

TABLE 8.2 ITU-T G.655.C, non-zero-dispersion-shifted single-mode optical fiber and cable

ITU-T G.655.C		
Fiber attributes	Detail	Value
Mode field diameter	Wavelength	1550nm
	Range of nominal values	8–11 μ m
	Tolerance	$\pm$ 0.7 μ m
Cladding diameter	Nominal	125.0 μ m
	Tolerance	$\pm$ 1 μ m
Core concentricity error	Maximum	0.8 μ m
Cladding noncircularity	Maximum	2%

TABLE 8.3 ITU-T G.655.D, non-zero-dispersion-shifted single-mode optical fiber and cable

ITU-T G.655.D		
Fiber attributes	Detail	Value
Mode field diameter	Wavelength	1550nm
	Range of nominal values	8–11 μ m
	Tolerance	$\pm 0.6\mu$ m
Cladding diameter	Nominal	125.0 μ m
	Tolerance	$\pm 1\mu$ m
Core concentricity error	Maximum	0.6 μ m
Cladding noncircularity	Maximum	1%

TABLE 8.4 ITU-T G.657, bending loss-insensitive single-mode optical fiber and cable

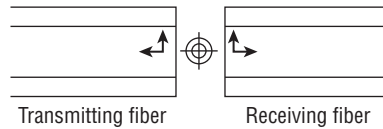
ITU-T G.657 Categories A and B		
Fiber attributes	Detail	Value
Mode field diameter	Wavelength	1310nm
	Range of nominal values	8.6–9.5 μ m
	Tolerance	$\pm 0.4\mu$ m
Cladding diameter	Nominal	125.0 μ m
	Tolerance	$\pm 0.7\mu$ m
Core concentricity error	Maximum	0.5 μ m
Cladding noncircularity	Maximum	1%

Extrinsic Factors

Extrinsic factors that affect optical fiber splice performance are factors related to the condition of the splice itself, external to the optical fiber. In an ideal splice, the optical fibers

are identical and they are aligned so that cores are perfectly centered on each other and the core axes are perpendicular to the endfaces being joined, as shown in Figure 8.7. However, there is no such thing as an ideal splice, only a real splice. In a real splice, intrinsic and extrinsic factors affect splice performance.

FIGURE 8.7 Conditions for an ideal splice

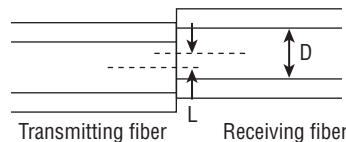


In this section, we will examine common extrinsic factors that affect splice performance. As you read about these extrinsic factors, keep in mind that many times they are caused by dirt and contamination. Microscopic particles of dirt can cause the misalignment of one or both optical fibers, creating a high-loss splice.

Lateral Misalignment

Lateral misalignment occurs when the two optical fibers are offset, as shown in Figure 8.8. Lateral misalignment loss occurs when light from the core of the transmitting optical fiber enters the cladding of the receiving optical fiber, creating a loss. As the lateral misalignment increases, less light from the core of the transmitting optical fiber makes its way into the core of the receiving optical fiber, increasing the loss of the splice.

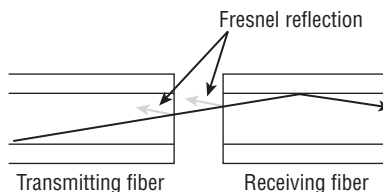
FIGURE 8.8 Lateral misalignment of the optical fibers



End Separation

Even if the optical fibers are perfectly aligned, the splice still may still experience loss from *end separation*. End separation is simply a gap between the transmitting and receiving optical fibers, as shown in Figure 8.9.

FIGURE 8.9 End separation is a gap between fiber ends in a mechanical splice.



Two different types of losses can be generated from end separation. The first is through Fresnel reflection, which takes place when light passes from the higher refractive index of the core in the transmitting optical fiber into the lower refractive index of the air, and then back into the core of the receiving optical fiber. Each change in the refractive index causes a certain amount of light to be reflected and therefore lost.

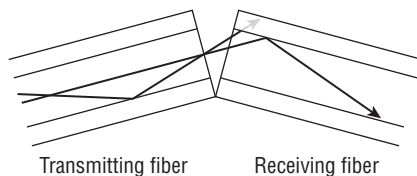
One way to overcome the effects of Fresnel reflections in separated optical fibers is to use an *index matching gel*, which is a transparent gel having a refractive index close to that of the core of the optical fibers being spliced. The gel fills the gap and reduces or eliminates the Fresnel reflection. Index matching gel is typically used in all mechanical splices.

End separation also causes loss because when light exits the transmitting optical fiber it spreads out like the light from a flashlight. Some of the light leaving the core of the transmitting optical fiber may enter the cladding of the receiving optical fiber, causing a loss. How much the light spreads out depends on several variables, including the distance between the optical fibers, light source type, launch type, length and NA of the transmitting optical fiber, and the bends in the transmitting optical fiber.

Angular Misalignment

If the optical fibers in a splice meet each other at an angle, a loss from *angular misalignment* may occur, as shown in Figure 8.10. The amount of loss depends on the severity of the angular misalignment and the acceptance cones of the transmitting and receiving optical fibers. Because the NA of a multimode optical fiber is greater than the NA of a single-mode optical fiber, multimode splices tolerate angular misalignment better than single-mode splices.

FIGURE 8.10 Angular misalignment results when fiber ends are not parallel to each other.



The loss from angular misalignment occurs when light from the core of transmitting optical fiber enters the cladding of the receiving optical fiber or enters the core of receiving optical fiber at an angle exceeding the acceptance angle. Light entering the core of the receiving optical fiber at an angle exceeding the acceptance angle may not propagate the length of the receiving optical fiber.

Angular misalignment also prevents the endfaces from contacting each other, resulting in Fresnel reflections exactly like those described in the “End Separation” section earlier in this chapter. Again, this is why index-matching gel is used with every mechanical splice.

Splicing Safety

As discussed in Chapter 6, “Safety,” you are responsible for your own safety as well as for the safety of your co-workers. It is up to you to know and incorporate safe work practices when splicing optical fibers. Remember, good work habits can help you prevent accidents and spot potential problems in time to correct them.

Here are some general safety rules that are discussed in detail in Chapter 6.

- Keep a clean workspace.
- Observe your surroundings.
- Use tools for the job they were designed to perform.
- Do not eat or drink in the work area.
- Report problems or injuries immediately.
- Know how to reach emergency personnel.
- Put your emergency contact information in your cell phone.

Splicing Hazards

Let’s look at some of the hazards directly related to splicing.

Light Sources

Most lasers used in fiber-optic systems emit infrared radiation; you cannot see the beam, no matter how powerful it is. As a result, you will not be able to tell if the system is powered, especially if you are working on a piece of fiber far from the transmitter. Always treat an optical fiber as if it is coupled to a laser unless you can be sure that the fiber is not coupled to a laser.

Handling Fiber

Optical fiber may be flexible; however, it is rigid enough to pierce your skin or eyes and cause discomfort, pain, or damage. If it becomes airborne, you may even accidentally

inhale or swallow it, risking damage to your throat or respiratory system. Protect yourself with proper procedures and the right personal protective equipment.

When you cut, cleave, or accidentally break optical fibers, the ends can get lost easily, either by becoming airborne or by rolling along a surface. Whenever possible, work over a nonreflective black mat; it will enable you to more easily identify fiber fragments. In addition, keep track of cut fiber ends and dispose of them in a separate labeled container.

To prevent eye injury, always wear proper eye protection. To prevent injury to your hands, always handle cut pieces of fiber with tweezers. If you have been handling fiber, do not rub your eyes or put your hands near them until you have washed your hands. If you do get a piece of fiber in your skin, remove it immediately with a pair of tweezers or seek medical attention.

Chemicals

In your work with fiber optics, you will use various types of chemicals; each poses a number of hazards and should be handled carefully. Every chemical that you use should be accompanied by a material safety data sheet (MSDS), which provides important information on the chemical's properties and characteristics such as appearance, odor, and common uses. The MSDS also gives you information on specific hazards posed by each chemical and ways to protect yourself through specific handling procedures, protective clothing and equipment, and engineering controls such as ventilation. Finally, the MSDS describes emergency procedures, including first aid for exposure to the chemical, methods for fighting fires in the case of flammable chemicals, and cleanup procedures for spills.

Even if you think you're familiar with the chemicals you handle, take time to read the MSDS. The information you gain could help prevent an accident or save valuable time in an emergency.

Electrical

Unlike mechanical slicing, fusion splicing uses a high-voltage electric arc between two electrodes to heat and melt the fiber ends as shown in Figure 8.11. Touching the electrodes while the arc is present can cause severe electrical shock. Remember that high voltage causes most of its damage by making muscles seize up, including the heart and lungs.

Fusion splicers like the one shown in Figure 8.12 have a cover that prevents access to the splicing area during the splicing process. This device typically has a safety interlock that interrupts the splicing process if lifted. Do not attempt to gain access to the splicing area until after the arc on the fusion splicer display shown in Figure 8.13 is extinguished. The arc is extinguished when it no longer appears on the display, as shown in Figure 8.14.

FIGURE 8.11 Close-up of a high-voltage electric arc melting the fiber ends during the splicing process

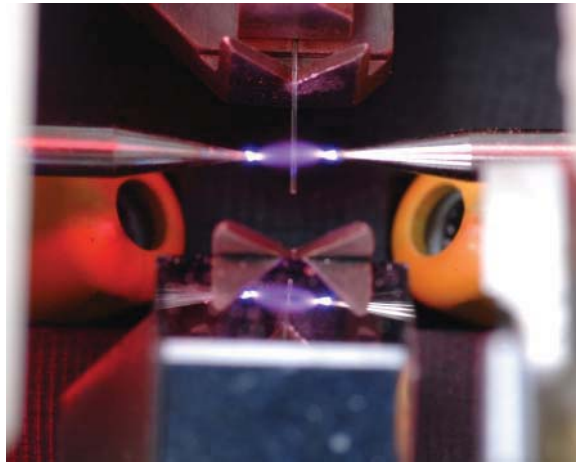


Photo courtesy of Aurora Optics

FIGURE 8.12 Automatic fusion splicer with the cover closed over the splicing area



Photo courtesy of Aurora Optics

FIGURE 8.13 Electric arc melting the fiber ends during the splicing process shown on the fusion splicer display



Photo courtesy of Aurora Optics

Explosion

Prior to fusion splicing, ensure you are not in a potentially explosive atmosphere. Remember the arc generated during the splicing process is essentially a continuous spark capable of igniting flammable vapors such as those produced by an open container of isopropyl alcohol or the refueling of a vehicle. The ignition of these vapors will cause an explosion.

If you are using isopropyl alcohol during fusion splicing, keep it a safe distance from the fusion splicer and work in a well-ventilated area. Do not initiate the fusing process in the presence of a high concentration of isopropyl alcohol vapors.

In Chapter 7, “Fiber-Optic Cables,” you learned that the aerospace and avionics industries are incorporating more optical fiber in aircraft. This is evident with the Boeing 787 Dreamliner, which has 110 fiber-optic links and over 1.7km of fiber-optic cable. As more commercial and military aircraft install optical fiber, there will be a need for fusion splicing.

A fueled aircraft is a hazardous environment for fusion splicing and special precautions must be taken prior to splicing. Only a fusion splicer like the one shown in Figure 8.15, developed specifically for explosion-proof operation, can be used in hazardous environments such as near a fueled aircraft.

FIGURE 8.14 Fusion splicer display after the arc is extinguished

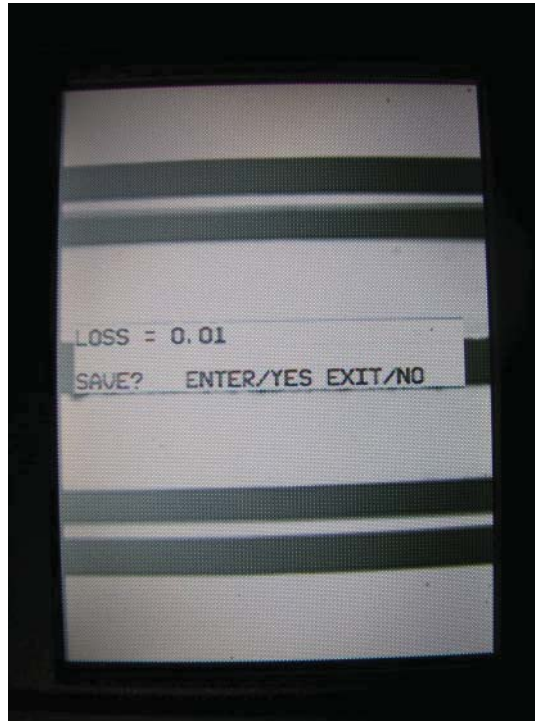


Photo courtesy of Aurora Optics

FIGURE 8.15 Fusion splicer developed specifically for use in hazardous environments



Photo courtesy of Aurora Optics

Splicing Equipment

The goal of any splice is to join two optical fiber endfaces together with as little loss as possible. This can be accomplished through mechanical or fusion splicing. This section describes the equipment required to perform a mechanical or fusion splice.

Cleaning Materials

One of the keys to performing a successful fiber-optic splice is cleanliness. The optical fiber needs to be properly cleaned prior to cleaving and splicing. In addition, the splicing area should be kept as clean as possible. As discussed earlier in the chapter, a microscopic piece of dirt could cause lateral displacement, end separation, or angular misalignment.

Prior to starting any splice, ensure you have lint-free wipes designed for optical fiber; a solvent designed to clean and remove the coating residue from the optical fiber; and if you will be fusion splicing, swabs to clean the V-groove. Lint-free wipes like those shown in Figures 8.16 and 8.17 are engineered to lift away oils, grime, and dust from the surface of the optical fiber.

FIGURE 8.16 A container of lint-free wipes engineered for cleaning optical fibers



Photo courtesy of MicroCare

When handling a lint-free wipe, be sure to only touch one side of the wipe because the wipe will absorb the oil from your skin. Use the side of the wipe that you did not touch for cleaning the optical fiber. You do not want to contaminate the optical fiber during the cleaning process. Cleaning of the optical fiber is covered in depth later in the “Splicing Procedures” section.

When you select an optical fiber cleaning solvent, ensure it has been engineered to clean optical fibers. The goal is to select a solvent that leaves little or no residue on the surface of the optical fiber. If you choose isopropyl alcohol, make sure that it is virtually free of water and has been filtered to remove impurities, like the isopropyl alcohol shown in Figure 8.18. Do not use the isopropyl alcohol commonly available at your local pharmacy; this alcohol contains a large percentage of water and will leave a residue on the optical fiber.

FIGURE 8.17 Packages of lint-free wipes engineered for cleaning optical fibers



Photo courtesy of ITW Chemtronics

FIGURE 8.18 Isopropyl alcohol for cleaning optical fibers

Photo courtesy of MicroCare

Isopropyl alcohol poses a problem for air travel because of its flammability. If the splicing job requires air travel, you will not be able to bring isopropyl alcohol on the aircraft. For jobs requiring air travel, you need to select a cleaning fluid that is approved for air travel. Remember from Chapter 6 that isopropyl alcohol vapors are highly flammable and can ignite if exposed to a spark or flame in high enough concentrations.

Cleaning fluids like the one shown in Figure 8.19 are approved for air travel and engineered specifically to clean the optical fibers. These cleaning fluids typically contain little or no isopropyl alcohol and have been filtered to remove microscopic contaminants.

FIGURE 8.19 Optical fiber cleaning fluid approved for air travel

Photo courtesy of MicroCare

As will be discussed later in this chapter, many fusion splicers have a precision V-groove that holds the optical fiber. Dirt or contamination in a V-groove can cause the optical fibers to misalign resulting in a high-loss fusion splice. When the V-grooves become contaminated, they can be cleaned using a fiber-optic swab, as shown in Figure 8.20.

FIGURE 8.20 Cleaning the V-groove with a fiber-optic swab

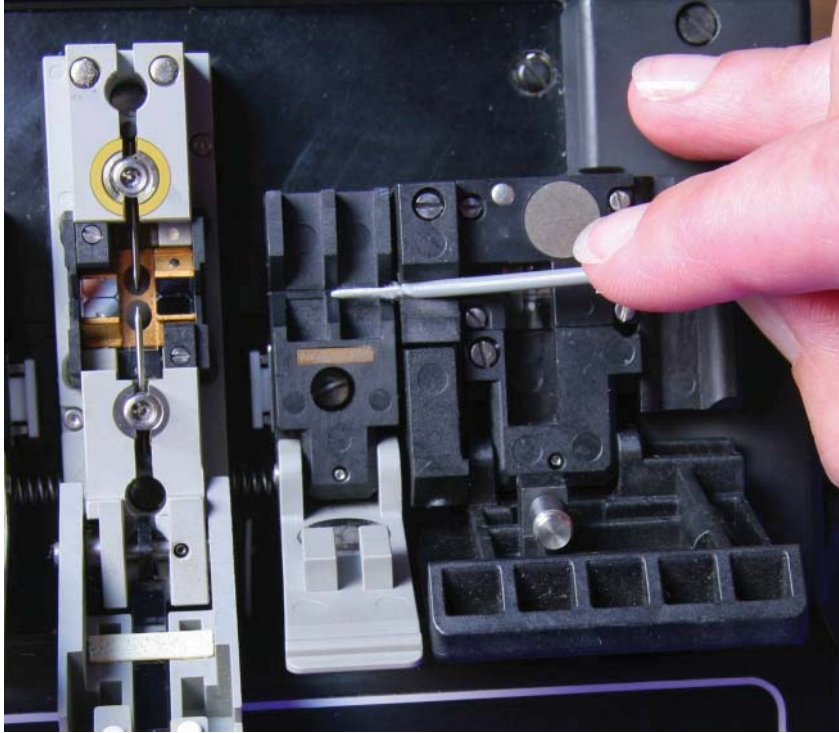


Photo courtesy of ITW Chemtronics

MicroCare, ITW Chemtronics, Corning Cable Systems, and AFL Telecommunications manufacture some of the most commonly used cleaning materials.

MicroCare www.microcare.com

ITW Chemtronics www.chemtronics.com

Corning www.corningcablesystems.com

AFL Telecommunications www.afltele.com

Cleavers

A cleaver is required for all splice types. Cleavers are available for single optical fibers or both single and ribbon fibers. Cleavers that support both single and ribbon fibers have removable adapters designed specifically for single fiber or ribbon fiber cleaving. An adapter

is not inserted in the cleaver shown in Figure 8.21. A single fiber adapter is inserted in the cleaver shown in Figure 8.22.

FIGURE 8.21 Single or ribbon fiber cleaver without the adapter

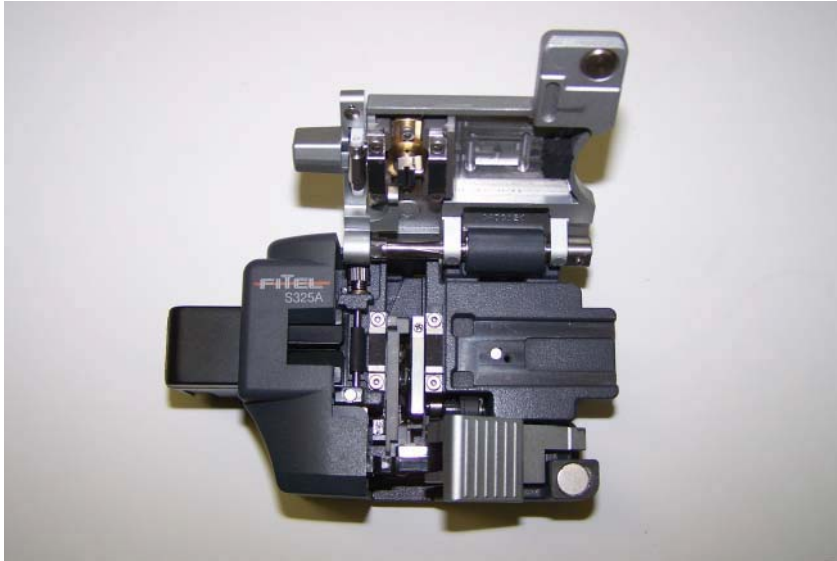


Photo courtesy of KITCO Fiber Optics

FIGURE 8.22 Single or ribbon fiber cleaver with the single fiber adapter inserted

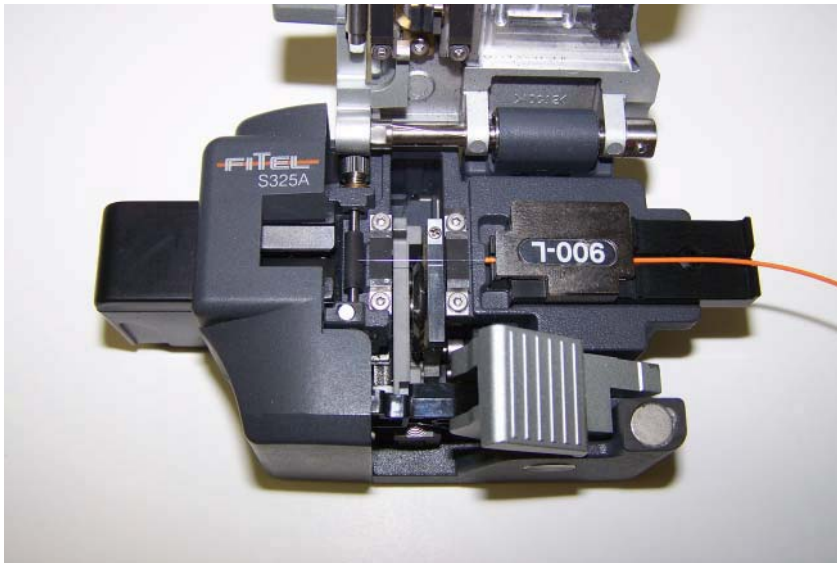
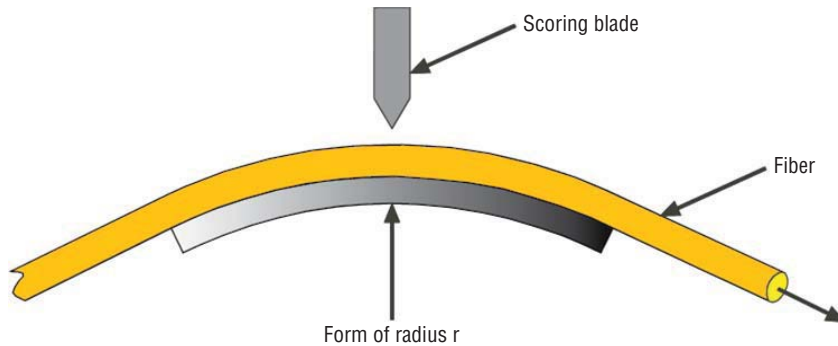


Photo courtesy of KITCO Fiber Optics

Cleaving is typically a two-step process, and it takes place after the optical fiber has been properly prepared and cleaned. The first step in the process is scoring the optical fiber.

This is accomplished with a scoring blade. The scoring blade should be held perpendicular to the optical fiber, as shown in Figure 8.23. When the cleaver is operated, the scoring blade lightly contacts the optical fiber, creating a small surface flaw.

FIGURE 8.23 Scoring blade ready to score the optical fiber



Drawing courtesy of Corning Cable Systems

After the optical fiber is scored, the next step is to bend the optical fiber. The optical fiber is bent slightly, as shown in Figure 8.23. Slightly bending the optical fiber places tensile stress on the surface of the optical fiber where it has been scored. This causes the small surface flaw to open and split the fiber.

After the optical fiber is cleaved, the endface should be perpendicular to the optical fiber without any surface defects and the cleave angle should not exceed 1.0° . Cleave angles can be evaluated only with a microscope. This is typically only done when a fusion splice is performed. Many fusion splicers will measure the cleave angle of the optical fiber prior to splicing.

Figure 8.24 is a photograph of a cleaved optical fiber with defects. In an ideal cleave, the mirror area would cover the entire endface. However, in this example the mirror area represents only a small portion of the endface. The rest of the endface looks like it was hacked with a sharp object many times. These irregular defects are commonly referred to as *hackles*.

FIGURE 8.24 Cleaved endface with hackles

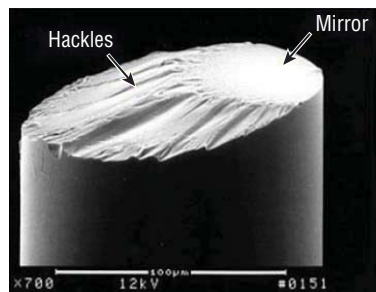


Photo courtesy of Corning Cable Systems

Mechanical Splice

Many manufacturers offer mechanical splices. Mechanical splices like the ones shown in Figures 8.25 and 8.26 are typically permanent. They align the cleaved optical fibers and hold them in place. Index matching gel inside the mechanical splice reduces or eliminates Fresnel reflections.

Mechanical splices can be used for both multimode and single-mode optical fiber. Mechanical splices do not outperform fusion splices. However, they will outperform mated connector pairs. The key advantage to a mechanical splice over a fusion splice is the low cost of the equipment required to perform the mechanical splice.

FIGURE 8.25 3M Fibrlok II universal mechanical splice



FIGURE 8.26 Corning CamSplice mechanical splice



Photo courtesy of Corning Cable Systems

The assembly tool that holds the optical fibers and the mechanical splice is relatively inexpensive when compared to the price of a fusion splicer. However, the actual mechanical splices shown in Figures 8.25 and 8.26 cost considerably more than the protective sleeve required for a fusion splice.

Specialty Mechanical Splice

Up to this point, we have only discussed splicing optical fibers that have been cleaved so the endface is perpendicular to the optical fiber. However, angled mechanical splices that offer better *return loss* performance than traditional mechanical splices are being introduced.

Angled mechanical splices require a special cleaver, angle cleaved fiber holders, and a special mechanical splice. The special cleaver like the one shown in Figure 8.27 places a 7° angle on the fiber endface while the angle cleaved fiber holders maintain the orientation of the cleave. Maintaining the orientation of the cleaved fibers is critical so that they align properly inside the mechanical splice.

FIGURE 8.27 3M fiber optic angle cleaver



Photo courtesy of 3M

The angled mechanical splice is an alternative to fusion splicing for analog video transmission, which typically requires lower return loss than a digital transmission. The 7° angle on the fiber endface reflects light into the cladding where it is absorbed. This reduces the amount of light energy that could be reflected back toward the light source, minimizing the back reflection.

Like the traditional mechanical splice, the angled mechanical splice is filled with index matching gel. The combination of the 7° angle and the index matching gel create a mechanical splice that meets or exceeds the return loss requirements for analog video transmission.

3M and Corning Cable Systems manufacture some of the most commonly used mechanical splices.

3M www.3m.com

Corning www.corningcablesystems.com

Fusion Splice

As you learned earlier in this chapter, fusion splicing uses a high-voltage electric arc between two electrodes to heat and melt the fiber ends as shown in Figure 8.11. This creates a permanent splice that is very fragile and must be protected from the outside environment and bending. This is typically accomplished using heat-shrink tubing and a small metal rod commonly referred to as a protective sleeve.

Prior to performing the splice, the optical fiber is passed through the center of the protective sleeve and the sleeve is positioned so it will not interfere with the fusion splicing process. After the fusion splice is successfully completed, the optical fiber is removed from the fusion splicer and the protective sleeve is positioned directly over the splice area.

The protective sleeve is then placed in an electric oven that is typically built into the fusion splicer. The oven heats the tubing, shrinking it around the fusion splice. The end result looks like the protected fusion splice shown in Figure 8.28.

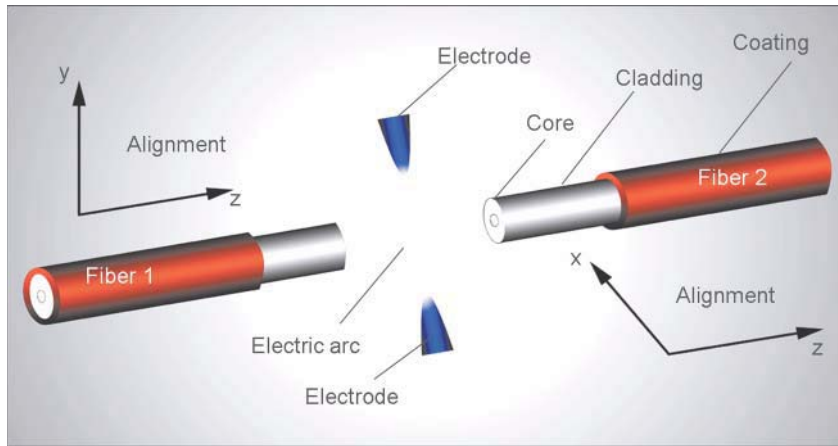
FIGURE 8.28 A fusion splice covered with heat shrink held rigid with a metal rod



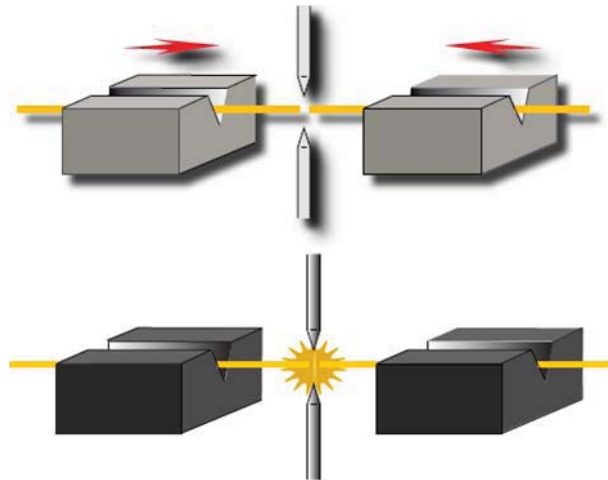
Fusion splicers are more expensive than the assembly tool required for a mechanical splice. However, they provide the lowest-loss splice possible. In addition, fusion splices do not produce Fresnel reflections.

Fusion splicing is the most accurate and durable method for joining two optical fibers. After the optical fibers are stripped and cleaved, they are placed into the fusion splicer, where the optical fibers are aligned between two electrodes, as shown in Figure 8.29. Alignment may be accomplished several ways depending on the type of fusion splicer used.

Several alignment techniques are used to align the optical fibers. A common alignment technique is to use a fixed V-groove, as shown in Figure 8.30. Each optical fiber is placed in a precision-machined V-groove. The V-grooves align the cladding of the optical fibers. The fusion splicer moves the optical fibers axially into the electric arc, melting the optical fibers together. This type of fusion splicer may be referred to as a *passive alignment fusion splicer* or a *passive v-groove fusion splicer*.

FIGURE 8.29 Optical fibers aligned between the fusion splicer electrodes

Drawing courtesy of Corning Cable Systems

FIGURE 8.30 Optical fibers in a V-groove aligned between the fusion splicer electrodes

Drawing courtesy of Corning Cable Systems

Fusion splicers with this type of alignment system have the lowest cost and are typically the smallest and lightest. However, they align the cladding of the optical fiber instead of the core and can create higher loss splices than fusion splicers that align the cores of the optical fibers.

A fusion splicer that aligns the cores of the optical fibers will produce the lowest-loss splice. However, aligning the cores of the optical fibers is more difficult than aligning two optical fibers placed in V-grooves. To align the cores, the fusion splicer must be able to

detect the cores and have the ability to move the optical fibers on each axis. This type of fusion splicer may be referred to as an *active alignment fusion splicer*.

Several patented technologies are available to detect the cores of optical fibers. Because they are patented, they will not be discussed in detail in this chapter.

Fusion splicers on the market today typically feature a display that allows you to see the optical fibers on two different axes. The cameras in the fusion splicer magnify the optical fibers so that the endfaces can be evaluated. You cannot look at an optical fiber with the naked eye and evaluate the endface. Without magnification, optical fiber with a perfect cleave will look just like an optical fiber that has been broken. Figure 8.31 is a photograph of a fusion splicer display showing a two-axis view of properly cleaved and aligned optical fibers. Shown in Figure 8.32 is a two-axis view of two broken optical fibers. This photograph clearly shows that a broken optical fiber can have a jagged endface.

FIGURE 8.31 Properly cleaved and aligned optical fibers viewed on the display of a fusion splicer

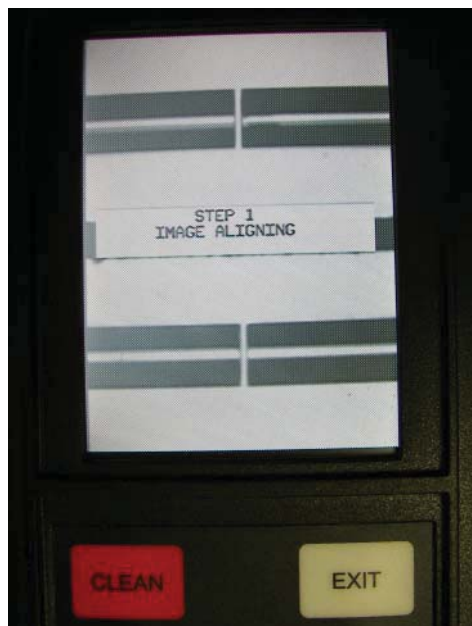


Photo courtesy of Aurora Optics

Most fusion splicers available today also have the ability to approximate the loss for the splice after it is completed. Fusion splicers are also available that have the ability to measure the loss for the splice after it is completed. Figure 8.33 is a photograph of a fusion splicer display showing the estimated splice loss. The splice loss is displayed on the bottom of the screen, and different axis views of the spliced optical fiber are shown at the top of the screen. Note that the spliced optical fibers appear as a single optical fiber with no flaws.

FIGURE 8.32 Optical fibers with jagged endfaces viewed on the display of a fusion splicer

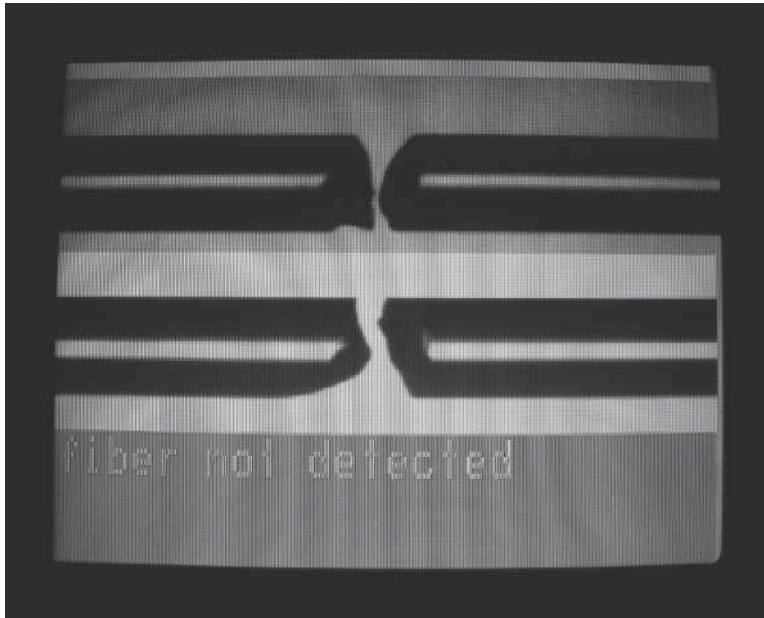
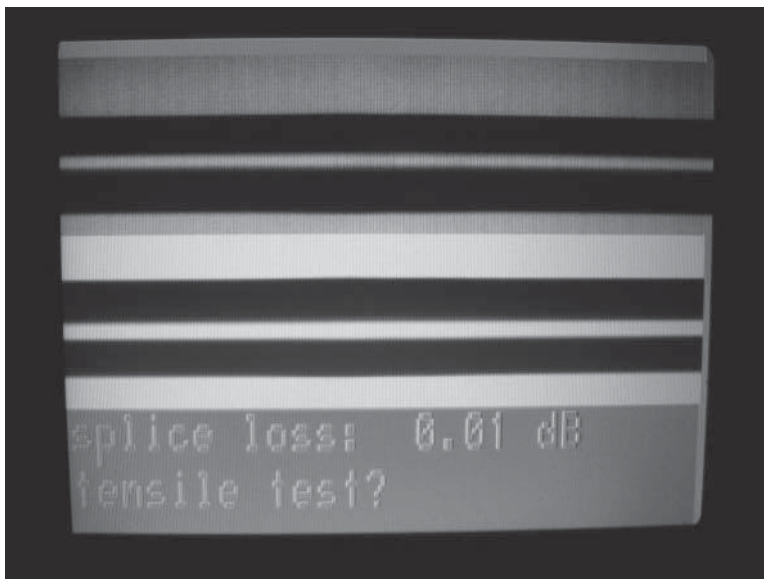


FIGURE 8.33 Fusion splicer displaying the estimated splice loss



You can find out more about specific features of fusion splicers through their manufacturers' websites. Some of the major manufacturers of fusion splicers are as follows:

Corning www.corningcablesystems.com

Aurora Optics www.aurora-optics.com

AFL Telecommunications www.afltele.com

Fitel www.fitel.com

Next let's look at the procedures used with each type of splicer.

Splicing Procedures

This section will familiarize you with the basic procedures used for mechanical or fusion splicing. While manufacturers' models will vary, many of the same principles and requirements apply. Be sure to read and follow the directions for your particular equipment carefully.



Whenever handling optical fiber, remember to follow the safety precautions described in Chapter 6.

Mechanical Splicing Procedure

To prepare for mechanical splicing, make sure that the work area is clean, dry, and well lit. Do all your work over a fiber-optic mat and place any scrap optical fibers in their proper container, as described in Chapter 6. Assemble the following tools before you begin:

- Mechanical splice assembly tool
- Mechanical splice
- Buffer and coating removal tool
- Optical fiber cleaning fluid
- Lint-free wipes
- Cleaver

Once your materials are assembled, clean the cleaver as directed by the manufacturer, and then proceed with the following steps:

1. Remove the 3M Fibrlok II mechanical splice from its protective packaging and load the splice into the assembly tool by pressing firmly at the ends of the splice. Do not depress the raised section on the mechanical splice.
2. Strip approximately 3cm of buffer and/or coating from the optical fiber using a stripper, as shown in Figure 8.34.

FIGURE 8.34 Stripping the buffer and coating from an optical fiber



3. Clean the optical fiber by pulling the fiber through a lint-free wipe soaked in optical fiber cleaning fluid, as shown in Figure 8.35.

FIGURE 8.35 Clean the optical fiber by pulling it through the lint-free wipe soaked in optical fiber cleaning fluid.



4. Place the optical fiber in the cleaver, as shown in Figure 8.36, to the length specified by the mechanical splice manufacturer.

FIGURE 8.36 Place the optical fiber in the cleaver at the length specified by the manufacturer.

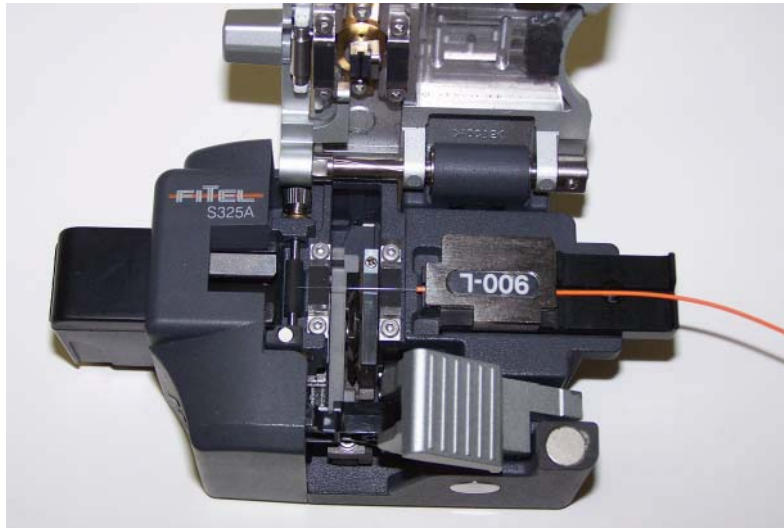


Photo courtesy of KITCO Fiber Optics

5. Cleave the optical fiber.
6. If a gauge is provided, check the cleave length with the gauge. If the optical fiber has touched the gauge, repeat step 3.
7. Repeat steps 2 through 6 for the other fiber end to be spliced.
8. Push one cleaved optical fiber into one end of the mechanical splice until it stops moving.
9. Push the other cleaved optical fiber into the other end of the mechanical splice until it stops moving.
10. Place both optical fibers in the clamping mechanisms on the opposite sides of the splicing tool, forming a modified loop in the optical fiber, as shown in Figure 8.37.
11. Pivot the splicing tool handle down until it contacts the top of the splice; then squeeze the tool handle to complete the assembly and lock the spliced ends in place.
12. Remove the optical fibers from the clamping mechanisms and lift the mechanical splice from the tool.

FIGURE 8.37 Mechanical splice tool properly set up to perform a mechanical splice



Mechanical Splicing Troubleshooting

Following the mechanical splicing procedures outlined earlier typically results in a low-loss splice that outperforms the attenuation requirements for inside or outside plant splices. However, occasionally you will encounter a splice that does not meet the attenuation requirement. As you learned earlier, how well a splice performs depends on many variables. These variables can be broken into two groups: intrinsic factors and extrinsic factors.

Intrinsic factors are the result of slight variations from one optical fiber to another. These variations can affect the performance of the splice even though the optical fibers are perfectly aligned when mated. However, the optical fibers available today have very little variation, which means that a small percentage of the splice loss may be from intrinsic factors.

Extrinsic factors are related to the condition of the splice itself, external to the optical fiber. Many times, they are caused by dirt and contamination. Microscopic particles of dirt can cause the misalignment of one or both optical fibers, creating a high-loss splice.

Many mechanical splices can be opened or unlocked so they can be used more than one time in case the initial splice does not meet the attenuation or back reflection requirements. Next we will examine some of the common causes for a high-loss mechanical splice.

Cleave Angle

As you've learned, the endface of a properly cleaved optical fiber should be perpendicular to the optical fiber without any surface defects and the cleave angle should not exceed 1.0°.

Cleave angles can be evaluated only with a microscope. Unfortunately, mechanical splicing tools are not equipped with a 2-axis microscope as most fusion splicers are. If you suspect a bad cleave, perform the following steps:

1. Open or unlock the splice and remove the optical fibers.
2. Prepare the optical fibers for cleaving adhering to the manufacturer's instructions.
3. Ensure the optical fiber and cleaver is clean.
4. Re-cleave both optical fibers.
5. Verify the cleave dimensions as stated in the manufacturer's instructions.
6. Reassemble the mechanical splice as described in the manufacturer's instructions.

Cleave Length

Cleave length is just as important as cleave angle. Each mechanical splice manufacturer defines how much bare optical fiber should be exposed after the cleaving process is completed. This dimension is typically provided in millimeters with maximum and minimum values. Exposing too much optical fiber or not enough optical fiber can create a high-loss mechanical splice. If you suspect improper cleave length, follow these steps:

1. Open or unlock the splice and remove the optical fibers.
2. Prepare the optical fibers for cleaving adhering to the manufacturer's instructions.
3. Ensure the optical fiber and cleaver is clean.
4. Re-cleave both optical fibers.
5. Verify the cleave dimensions as stated in the manufacturer's instructions.
6. Reassemble the mechanical splice as described in the manufacturer's instructions.

Contamination

Contamination on the optical fiber or cleaver that is invisible to the unaided eye can cause a mechanical splice to exceed attenuation and back reflection requirements. Because you cannot see the contamination, make sure that you are using the types of cleaning products and the cleaning process required by the splice and cleaver manufacturer or described in this chapter. Never let a cleaned optical fiber touch any surface after cleaving. If you have any cleaning related questions, you can always reach out to a cleaning product manufacturer such as MicroCare or ITW Chemtronics for help.

MicroCare www.microcare.com

ITW Chemtronics www.chemtronics.com



Real World Scenario

Can't Afford a Fusion Splicer?

If you want the joy of fusion splicing without the purchase price, you can rent or lease a fusion splicer. Companies such as Fiber Instrument Sales, Inc. (www.fiberinstrumentsales.com) provide rental arrangements for fusion splicers and other high-end fiber equipment if you know you won't need it on a long-term or recurring basis.

Rental rates typically run 1/10 to 1/12 of the purchase price of a new fusion splicer per month. Bear in mind, though, that the rental period starts the day that the splicer is shipped to you and ends the day that it is received back at the rental office. In other words, you're going to pay for the days that it is in transit.

If you still want to buy a fusion splicer but don't want to pay full price, consider a pre-owned or reconditioned unit. These are occasionally available through distributors and still have years of life left in them.

Fusion Splicing Procedure

The equipment required for fusion splicing is far more complex and expensive than the equipment required for mechanical splicing. However, many of the steps required to perform a fusion splice are identical to the steps required to perform a mechanical splice.

Many fusion splicers contain a feature that automatically positions the fiber ends in proper relationship with each other and with the electrodes for the best possible splice. All that is required of the operator is to prepare the fibers properly and place them in the fusion splicer as outlined by the manufacturer.

The fusion-splicing procedures outlined in this chapter describe the typical steps required to fusion-splice single optical fibers and ribbon optical fibers. These steps are not product specific and modifications may be required depending on the fusion splicing equipment being used. Always review the manufacturer's operating documentation prior to performing a splice.

To prepare for fusion splicing, as with mechanical splicing, make sure that the work area is clean, dry, and well lit. Do all your work over a fiber-optic mat and place any scrap

optical fibers in their proper container, as described in Chapter 6. Assemble the following tools before you begin:

- Fusion splicer
- Buffer and coating removal tool
- Optical fiber cleaning fluid
- Lint-free wipes
- Cleaver
- Heat-shrink protective covering

Once your materials are assembled, clean the cleaver and the fusion splicer as directed by the manufacturer, and then proceed with the following steps. First, we'll look at single optical fiber fusion splicing:

1. Power on the fusion splicer and select the appropriate splicing program or profile for the optical fiber you will be splicing.
2. Slide the protective heat-shrink tubing over one optical fiber end and move it far enough up the optical fiber to place it out of the way.
3. Strip approximately 3cm of buffer and/or coating from the optical fiber using a stripper, as shown in Figure 8.38.

FIGURE 8.38 Stripping the buffer and coating from an optical fiber



4. Clean the optical fiber by pulling the fiber through a lint-free wipe soaked in optical fiber cleaning fluid, as shown in Figure 8.39.

5. Place the optical fiber in the cleaver, as shown in Figure 8.40, to the length specified by the mechanical splice manufacturer.

FIGURE 8.39 Clean the optical fiber by pulling it through the lint-free wipe soaked in optical fiber cleaning fluid.



FIGURE 8.40 Place the optical fiber in the cleaver at the length specified by the manufacturer.

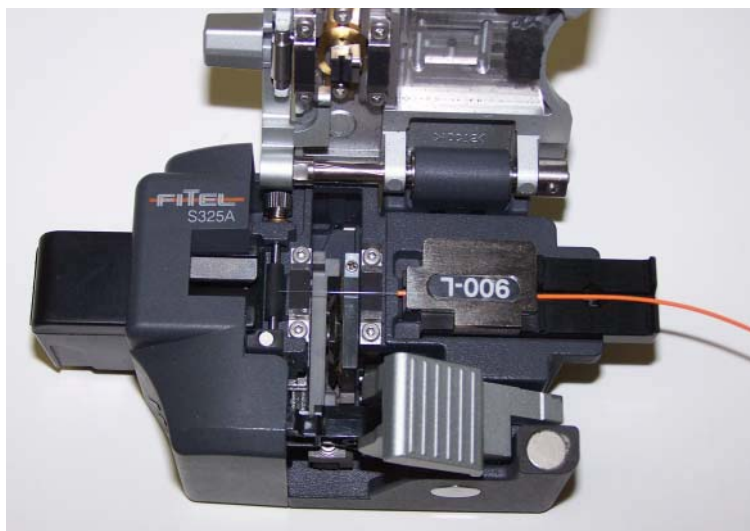


Photo courtesy of KITCO Fiber Optics

6. Cleave the optical fiber.
7. Place the optical fiber in the fusion splicer following the manufacturer's instructions. Position the endface of the optical fiber between the electrodes.
8. Repeat steps 3 through 7 for the other fiber end to be spliced. The properly placed fibers should be slightly separated between the electrodes, as shown in Figure 8.41.
9. Close the electrode cover.
10. Begin the fusion-splicing process.
11. Carefully remove the splice and position the heat-shrink tubing from step 2 over it. Place the splice and tubing in the heat-shrink oven to seal and protect the splice.

Next we'll look at ribbon fiber fusion splicing:

1. Power on the fusion splicer and select the appropriate splicing program or profile for the optical fiber you will be splicing.
2. Slide the protective heat-shrink tubing over one ribbon fiber end and move it far enough up the optical fiber to place it out of the way.
3. Strip approximately 3cm of Mylar tape and coating from the ribbon fiber, as shown in Figure 8.42.

FIGURE 8.41 Properly placed fibers almost touch each other between the electrodes.

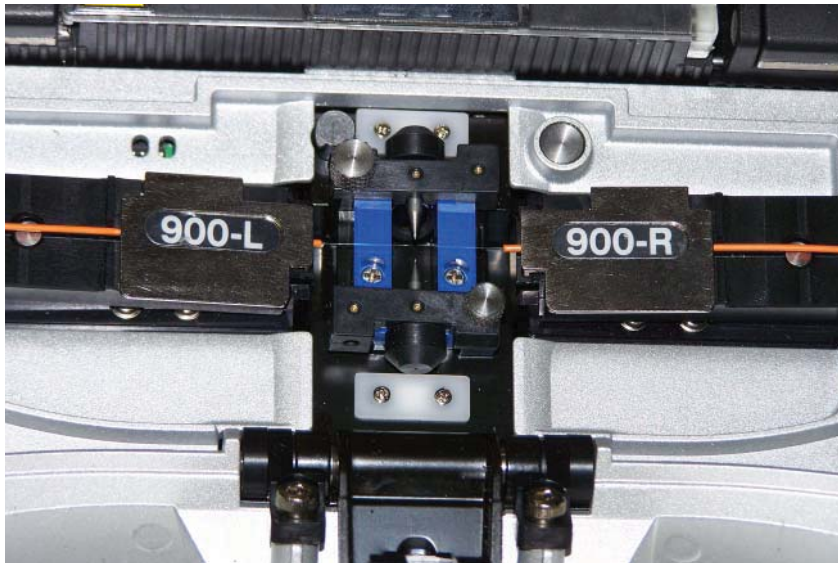


Photo courtesy of KITCO Fiber Optics

4. Clean the ribbon fiber by pulling the fiber through a lint-free wipe soaked in optical fiber cleaning fluid, as shown in Figure 8.43.
5. Place the ribbon fiber in the cleaver, as shown in Figure 8.44, to the length specified by the manufacturer of splicer you are using.
6. Cleave the ribbon fiber.
7. Place the ribbon fiber in the fusion splicer, as shown in Figure 8.45. Position the ribbon fiber as described in the manufacturer's instructions.
8. Lock the ribbon fiber in place as described in the manufacturer's instructions.
9. Repeat steps 3 through 8 for the other fiber end to be spliced.
10. Close the electrode cover.
11. Begin the fusion-splicing process.
12. Carefully remove the splice and position the heat-shrink tubing from step 2 over it. Place the splice and tubing in the heat-shrink oven to seal and protect the splice.

FIGURE 8.42 Ribbon fiber stripper using heat to remove the Mylar tape and coating



Photo courtesy of MicroCare

FIGURE 8.43 Ribbon fiber being pulled through a lint-free wipe soaked in optical cleaning fluid



Photo courtesy of MicroCare

FIGURE 8.44 Ribbon fiber placed in the cleaver for cleaving

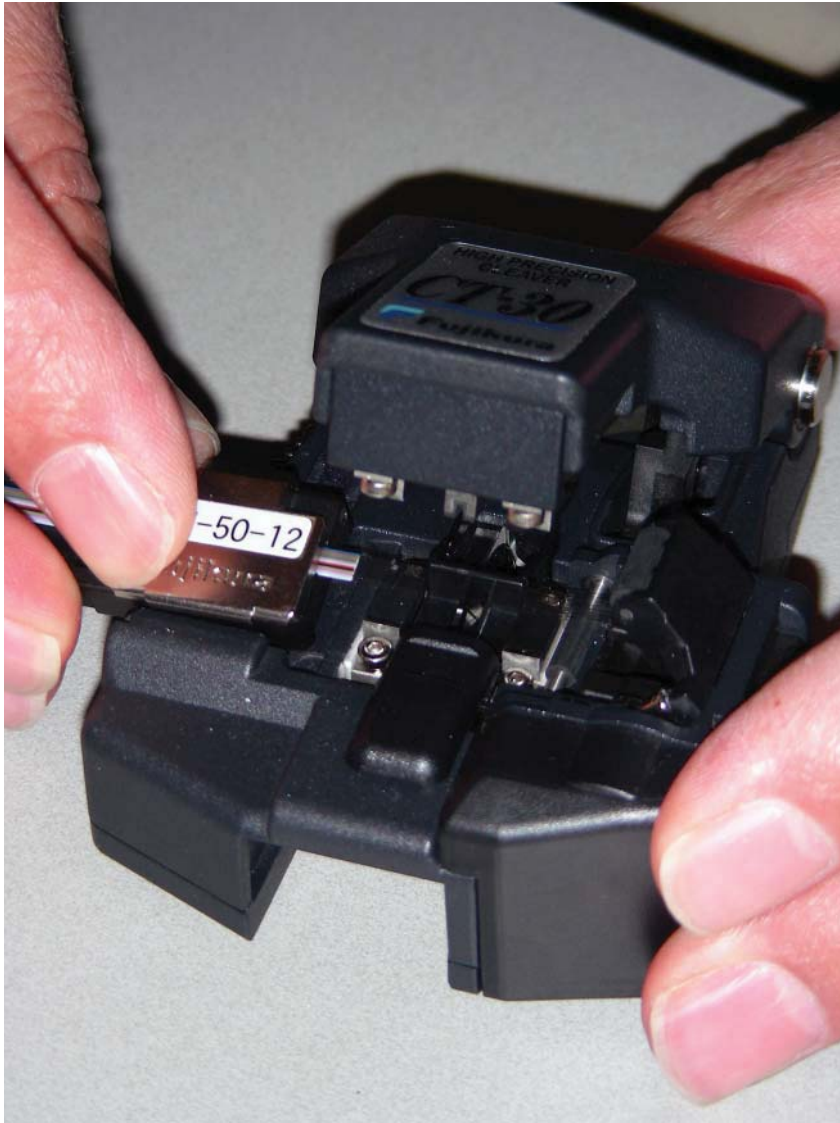


Photo courtesy of MicroCare

FIGURE 8.45 Ribbon fiber placed in the fusion splicer for splicing

Photo courtesy of MicroCare

Fusion Splicing Troubleshooting

Following the fusion splicing procedures outlined earlier typically results in a low-loss splice that outperforms the attenuation requirements for inside or outside plant splices. However, occasionally you will encounter a splice that exceeds maximum attenuation

requirements. As you've learned, how well a splice performs depends on many variables. As with mechanical splicing, these variables can be broken into two groups: intrinsic factors and extrinsic factors.

Intrinsic factors are the result of slight variations from one optical fiber to another. These variations can affect the performance of the splice even though the optical fibers are perfectly aligned when mated. However, the optical fibers available today have very little variation, which means that a small percentage of the splice loss may be from intrinsic factors.

Extrinsic factors are related to the condition of the splice itself, external to the optical fiber. Many times, they are caused by dirt and contamination. Microscopic particles of dirt can cause the misalignment of one or both optical fibers, creating a high-loss splice.

Next we will examine some of the common causes for a high-loss fusion splice.

Cleave Angle

As discussed earlier, the endface of a properly cleaved optical fiber should be perpendicular to the optical fiber without any surface defects and the cleave angle should not exceed 1.0° . Cleave angles can be evaluated only with a microscope. Fortunately many fusion splicers are equipped with a 2-axis microscope that allows you to observe the cleave and the measured cleave angle values. If you or the fusion splicer detects a bad cleave, as shown in Figure 8.46, perform the following steps:

1. Remove the optical fibers from the fusion splicer.
2. Prepare the optical fibers for cleaving adhering to the manufacturer's instructions.
3. Ensure the optical fiber and cleaver is clean.
4. Re-cleave both optical fibers.
5. Verify the cleave dimensions as stated in the manufacturer's instructions.
6. Place the optical fibers in the fusion splicer following the manufacturer's instructions. Position the endface of the optical fibers between the electrodes.
7. Perform the fusion splice.

FIGURE 8.46 Improperly cleaved optical fibers viewed on the display of a fusion splicer

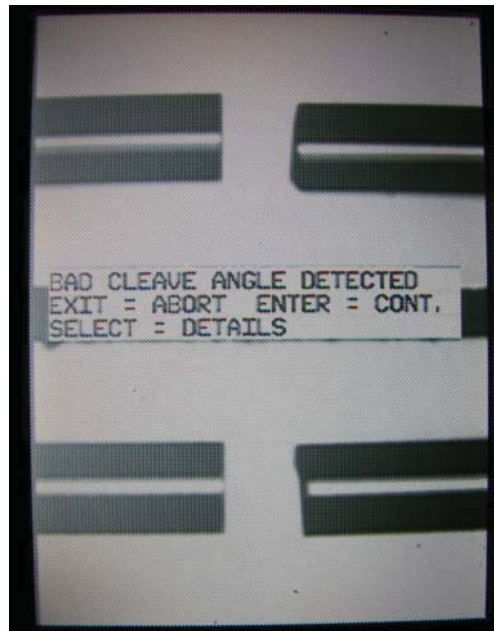


Photo courtesy of Aurora Optics

Cleave Length

Cleave length is just as important as cleave angle. Each fusion splice manufacturer defines how much bare optical fiber should be exposed after the cleaving process is completed. This dimension is typically provided in millimeters with maximum and minimum values. Exposing too much optical fiber or not enough optical fiber can create a high-loss fusion splice. If you suspect improper cleave length, follow these steps:

1. Remove the optical fibers from the fusion splicer.
2. Prepare the optical fibers for cleaving, adhering to the manufacturer's instructions.
3. Ensure the optical fiber and cleaver is clean.
4. Re-cleave both optical fibers.
5. Verify the cleave dimensions as stated in the manufacturer's instructions.
6. Place the optical fibers in the fusion splicer, following the manufacturer's instructions. Position the endface of the optical fibers between the electrodes.
7. Perform the fusion splice.

Bubbles

Dirt or entrapped air may create a bubble or bubbles, resulting in a high-loss fusion splice like the one shown in Figure 8.47. To prevent bubbles

1. Ensure you are using the types of cleaning products and the cleaning processes required by the fusion splicer and cleaver manufacturer or described in this chapter.
2. Verify that you have selected the appropriate splicing program or profile for the optical fiber you are splicing.

FIGURE 8.47 Bubbles in a high-loss fusion splice viewed on the display of a fusion splicer

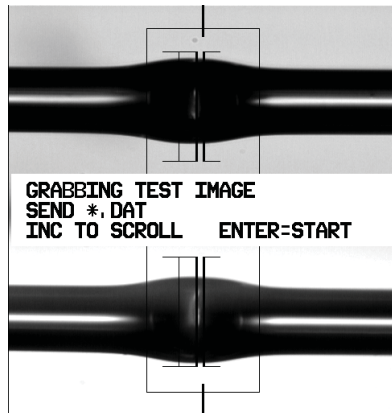


Photo courtesy of Aurora Optics

Necking

Necking is a term used to describe a fusion splice where the diameter of the fused optical fiber is smaller near the electrodes than it was prior to being fused, as shown in Figure 8.48. Necking is typically the result of too much heat during the prefuse, causing glass to be transported away from the splice area and thus producing a high-loss splice. To prevent necking

1. Ensure you are using the types of cleaning products and the cleaning processes required by the fusion splicer and cleaver manufacturer or described in this chapter.
2. Verify that you have selected the appropriate splicing program or profile for the optical fiber you are splicing.
3. Ensure the optical fibers are properly cleaved.

Contamination

Contamination on the optical fiber or cleaver that is invisible to the unaided eye can cause fusion splices to exceed attenuation requirements. Because you cannot see the contamination, make sure that you are using the types of cleaning products and the cleaning process required by the

fusion splicer and cleaver manufacturer or described in this chapter. Never let a cleaned optical fiber touch any surface after cleaving. If you have any cleaning-related questions, you can always reach out to a cleaning product manufacturer such as MicroCare or ITW Chemtronics for help.

MicroCare www.microcare.com

ITW Chemtronics www.chemtronics.com

FIGURE 8.48 A high-loss fusion splice caused by necking viewed on the display of a fusion splicer

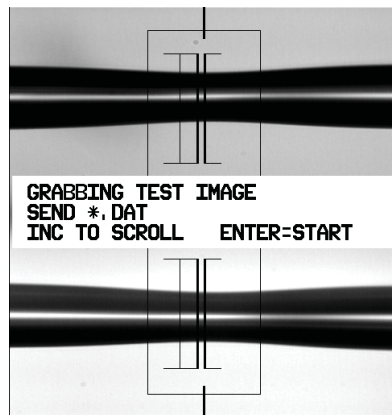


Photo courtesy of Aurora Optics

Splice Performance Requirements

As with connectors, standards specify a maximum permissible loss for splices, depending on their location. ANSI/TIA-568-C.3 addresses the performance of fusion or mechanical splices for premises applications, also known as inside plant applications. The standard states that a fusion or mechanical splice shall not exceed a maximum optical insertion loss of 0.3dB when measured in accordance with ANSI/EIA/TIA-455-34-A, Method A (factory testing), or ANSI/TIA-455-78-B (field-testing).

ANSI/TIA-568-C.3 also defines the minimum return loss for mechanical or fusion optical fiber splices. Multimode mechanical or fusion splices shall have a minimum return loss of 20dB while single-mode mechanical or fusion splices shall have a minimum return loss of 26dB when measured in accordance with ANSI/TIA/EIA-455-107-A. The minimum single-mode return loss for broadband analog video CATV applications is 55dB when measured in accordance with ANSI/TIA/EIA-455-107-A.

ANSI/TIA-758-B specifies the minimum performance requirements for mechanical or fusion splices for outside plant applications. It states that the splice insertion loss shall not exceed 0.1dB mean, with a maximum of 0.3dB, as measured with an optical time-domain reflectometer (described in Chapter 14, “Test Equipment and Link/Cable Testing”).

Return loss performance requirements for mechanical or fusion splices are also defined in ANSI/TIA-758-B. Either type of splice shall have a return loss greater than or equal to 45.0dB mean, 40.0dB minimum, when measured in accordance with ANSI/TIA/EIA-455-8.

SAE International published AS6479 in January 2014. This aerospace standard defines fusion splicers acceptable for aerospace applications. It also defines splice insertion loss requirements. This standard is different from ANSI/TIA-568-C.3 and ANSI/TIA-758-B because it defines the performance of the fusion splicer, not the performance of the as installed splice.

Under laboratory conditions, a fusion splicer compliant with AS6479 must be able to splice single-mode optical fibers with a mean insertion loss of 0.03dB or better. It must also be able to splice multimode optical fibers with a mean insertion loss of 0.02dB or better. These mean insertion loss values apply to passive and active alignment fusion splicers.

Telcordia GR-765 is a telecommunications document that provides requirements for single fiber, single-mode optical fiber splicing. Like AS6479, this document contains criteria governing the ability of fusion splicing equipment to assemble complete splices. It specifies required and objective splice insertion loss values for passive alignment and active alignment fusion splicers.

For passive alignment fusion splicers, the GR-765 required mean insertion loss is $\leq 0.15\text{dB}$ and the objective mean insertion loss is $\leq 0.10\text{dB}$. For active alignment fusion splicers, the required mean insertion loss is $\leq 0.10\text{dB}$ and the objective mean insertion loss is $\leq 0.05\text{dB}$.



Real World Scenario

Fusion or Mechanical?

It may seem like an easy choice between mechanical and fusion splicing: fusion splices are superior to mechanical splices because they essentially make one optical fiber out of two. The cost per fusion splice is virtually nil, whereas mechanical splices typically cost from \$7 to \$25 per splice. In short, if you have a fusion splicer, there is really no need to even look at mechanical splicing.

So why bother with mechanical splices at all?

The one key factor that favors mechanical splicing is the initial setup cost. Mechanical splicing and fusion splicing both require a cleaver, hand tools, and cleaning supplies. However, only a mechanical splice assembly tool is required to assemble the mechanical splice. This tool typically costs around \$100 whereas many fusion splicers cost more than \$10,000.

If your application requires only a few splices, mechanical splicing is the most economical way to go. However, if your application requires many splices, fusion splicing may be the most economical way to go.

Summary

This chapter examined the intrinsic and extrinsic factors that can affect splice performance. It described the hazards associated with mechanical and fusion splicing. In addition, it introduced the tools and equipment required to perform splicing.

We also learned how to use the tools and equipment to assemble a mechanical or fusion splice. In addition, we learned how to troubleshoot a splice and the maximum permissible insertion and return loss values for inside plant and outside plant splices.

Exam Essentials

Understand the intrinsic factors affecting splice performance. Make sure you understand how optical fiber manufacturing variations can affect splice performance. Be able to describe the different intrinsic factors.

Understand the extrinsic factors affecting splice performance. Make sure you understand how the physical relationship between the optical fibers can affect splice performance. Be able to describe the different extrinsic factors.

Be familiar with the different types of splices. Make sure you can explain the difference between a mechanical splice and a fusion splice. Be able to describe the equipment used to assemble them. In addition, be able to describe the benefits and drawbacks of each type of splice.

Be familiar with splicing procedures. Make sure you understand the basic steps for assembling each type of splice. Be able to describe the tools and procedures required for each type of splice.

Understand splice performance specifications. Make sure you know the maximum permissible insertion and return loss for inside and outside plant mechanical and fusion splices.

Review Questions

1. NA mismatch loss occurs when
 - A. The numerical aperture of the transmitting optical fiber is smaller than that of the receiving fiber.
 - B. The numerical aperture of the transmitting optical fiber is larger than that of the receiving fiber.
 - C. The numerical apertures of both optical fibers are larger than normal.
 - D. The numerical apertures of both optical fibers are smaller than normal.

Hint: Light enters at an angle that exceeds the acceptance angle.
2. Core diameter mismatch loss occurs when
 - A. The diameter of the transmitting core is greater than that of the receiving core.
 - B. The diameter of the transmitting core is less than that of the receiving core.
 - C. The diameter of the transmitting core is not precisely aligned with the diameter of the receiving core.
 - D. The diameter of the transmitting core is equal to the diameter of receiving core.

Hint: Light will be lost in the cladding of the receiving optical fiber.
3. Cladding diameter mismatch loss will not occur when
 - A. The cladding diameter of the transmitting fiber is larger than the cladding of the receiving fiber.
 - B. The cladding diameter of the transmitting fiber is smaller than the cladding of the receiving fiber.
 - C. The cladding diameters of both optical fibers are identical.
 - D. The cladding diameters of both optical fibers are not the same.

Hint: Light will be lost in the cladding of the receiving optical fiber.
4. Which is not an intrinsic factor that affects splice performance?
 - A. Noncircularity
 - B. Mode field diameter mismatch
 - C. Concentricity
 - D. Lateral misalignment

Hint: It is external to the optical fibers.
5. Which is not an extrinsic factor that affects splice performance?
 - A. End separation
 - B. Concentricity

- C. Angular misalignment
- D. Lateral misalignment

Hint: The core and cladding should share a common geometric center.

6. A mechanical splice typically uses _____ to reduce or eliminate Fresnel reflections.
- A. Epoxy
 - B. A high-voltage electric arc
 - C. Index matching gel
 - D. Isopropyl alcohol

Hint: It should have the same refractive index as the optical fiber core.

7. A fusion splicer uses _____ to produce a low loss splice.
- A. Epoxy
 - B. A high-voltage electric arc
 - C. Index matching gel
 - D. An angled cleaver

Hint: It generates heat.

8. Cleaving is typically a _____ process.
- A. One-step
 - B. Two-step
 - C. Three-step
 - D. Four-step

Hint: Bending the optical fiber places tensile stress on the cladding surface.

9. When should the optical fiber be passed through the center of the protective sleeve?
- A. Before fusion splicing
 - B. While fusion splicing
 - C. During cable installation
 - D. After fusion splicing

Hint: It should be positioned directly over the splice area.

10. ANSI/TIA-568-C.3 states that optical fiber splices, whether fusion or mechanical, will have a maximum attenuation of _____.
- A. 0.1dB
 - B. 0.3dB
 - C. 0.5dB
 - D. 0.75dB

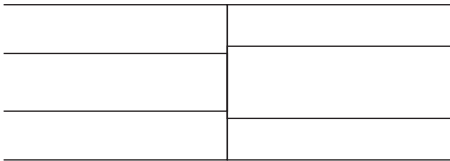
Hint: It is less than the maximum loss for a mated pair.

Chapter Exercises

Determine if the Splice Loss Is from an Intrinsic Factor

Even when fibers are manufactured within specified tolerances, there are still slight variations from one optical fiber to another. These variations can affect the performance of the splice even though the optical fibers are perfectly aligned when mated.

Two optical fibers are about to be fusion spliced together, as shown here. Determine what type of problem exists with the optical fibers about to be spliced.

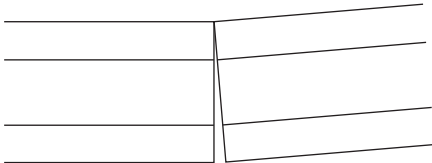


A close inspection of the graphic reveals that the core diameter of the left optical fiber is slightly smaller than the right optical fiber. This core diameter mismatch is the result of an intrinsic factor.

Determine if the Splice Loss Is from an Extrinsic Factor

Extrinsic factors that affect optical fiber splice performance are factors related to the condition of the splice itself, external to the optical fiber.

Two optical fibers are about to be fusion spliced together, as shown here. Determine what type of problem exists with the optical fibers about to be spliced.



A close inspection of the graphic reveals that there is some angular misalignment. The angular misalignment is the result of an extrinsic factor.

Calculate the Potential Splice Loss from a Core Diameter Mismatch

A core diameter mismatch loss results when the core diameter of the transmitting optical fiber is greater than the core diameter of the receiving optical fiber.

Calculate the worst-case loss for a splice where the transmitting optical fiber has a core diameter of $51\mu\text{m}$ and the receiving optical fiber has a core diameter of $49.5\mu\text{m}$.

We can calculate the worst-case loss percentage for a splice that joins a 50.5µm core and 49.5µm core using the following formula:

$$\text{Loss} = (d_1^2 - d_2^2) / d_2^2$$

where d1 is the diameter of the transmitting core and d2 is the diameter of the receiving core.

$$50.5^2 - 49.5^2 = 100$$

$$100/49.5^2 = 0.0408 \text{ or } 4.08\%$$

Using this formula, the decibel loss can be calculated:

$$\text{dB} = 10\text{Log}_{10}(P_{\text{out}} / P_{\text{in}})$$

Note that the decibel formula is a ratio, where the power output is divided by the power input. We know that the receiving optical fiber will not accept 4.08 percent of the light from the transmitting optical fiber. Subtracting 4.08 percent from 100 percent, as shown here, results in a difference of 95.92 percent, which can be written as 0.9592.

$$100\% - 4.08\% = 95.92\% \text{ or } 0.9592$$

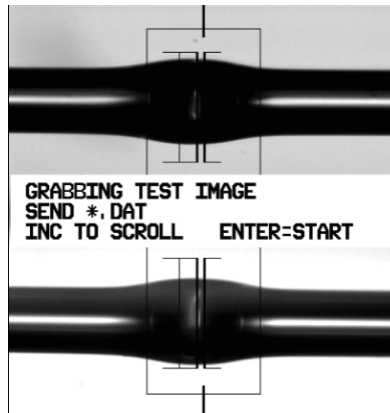
$$10\text{Log}_{10}(0.9592 / 1) = -0.181\text{dB}$$

$$\text{Worst-case loss} = 0.181\text{dB}$$

Troubleshoot a Fusion Splice

How well a splice performs depends on many variables. These variables can be broken into two groups: intrinsic factors and extrinsic factors.

Two optical fibers have been fused together as shown here. Determine the type of problem and potential causes.



A close inspection of the of the fusion splicer display reveals that the optical fiber ends that have been joined are larger in diameter than the optical fiber. This is typically caused by dirt or entrapped air creating a bubble or bubbles. The result is a high-loss fusion splice. To prevent bubbles, verify that you have selected the appropriate splicing program or profile for the optical fiber you are splicing and ensure you are properly cleaning the fusion splicer and cleaver.

Chapter 9

Connectors

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Describe the basic components of a fiber-optic connector.
- ✓ Describe common connector ferrule materials.
- ✓ List the intrinsic factors that affect connector performance.
- ✓ List the extrinsic factors that affect connector performance.
- ✓ Describe the following endface geometries:
 - Flat
 - Curved
 - Angled
 - Lensed
- ✓ Describe return or back reflections, return loss, and reflectance in an interconnection.
- ✓ Explain how endface geometry affects return loss and reflectance.
- ✓ Describe how an interferometer is used in the evaluation of endface geometry.
- ✓ Describe the following critical parameters that are required by Telcordia GR-326 to evaluate connector endface geometry for single-mode connectors and jumper assemblies:
 - Radius of curvature
 - Apex offset
 - Fiber undercut or protrusion





- ✓ Explain the difference between a contact and noncontact connector.
- ✓ Describe the ANSI/TIA-568 recognized connectors.
- ✓ Describe small form factor connectors.
- ✓ Describe MPO connectors.
- ✓ Describe a pigtail and the potential advantages it offers over field terminations.
- ✓ Describe the steps involved in an oven cured epoxy connector termination and polish.
- ✓ Describe the steps involved in an anaerobic epoxy connector termination and polish.
- ✓ Explain how machine polishing produces a better and more consistent endface than hand polishing.
- ✓ Describe pre-polished connector termination.
- ✓ Explain how to properly clean a connector endface using dry cleaning techniques.
- ✓ Explain how to properly clean a connector endface using wet-dry cleaning techniques.
- ✓ Explain how to examine the endface of a connector per ANSI/TIA-455-57-B and IEC 61300-3-35.
- ✓ List the ANSI/TIA-568-C.3 maximum insertion and return loss values for multimode and single-mode mated connector pairs.
- ✓ List the ITU-T G.671 maximum insertion loss and reflectance values for single-mode single-fiber mated connector pairs.
- ✓ Explain how to use the ANSI/TIA-568-C.3 color code to identify multimode and single-mode connectors and adapters.



The ideal traveling environment for a pulse of light is an unbroken optical fiber. At some point, however, that optical fiber must connect to a piece of equipment or join another optical fiber in order to extend its length or change the type of cable being used.

One of the most common methods for *terminating* an optical fiber, or making its end useful, is to use a connector. A connector is a device that supports the end of the optical fiber while allowing it to be quickly and reliably joined to equipment or other optical fibers.

Connectors are often used instead of splices to join two optical fibers together because they allow the optical fibers to be disconnected and reconnected easily. Splices, on the other hand, are permanent connections between two optical fibers. Connectors can be useful when network assignments must be changed, when equipment must be removed/replaced, or when expansion is anticipated.

This chapter describes common connectors used in optical fiber termination. It investigates the factors that affect connector performance and methods used to improve performance. The chapter also discusses methods used to install connectors so they meet industry performance standards.

In this chapter, you will learn to

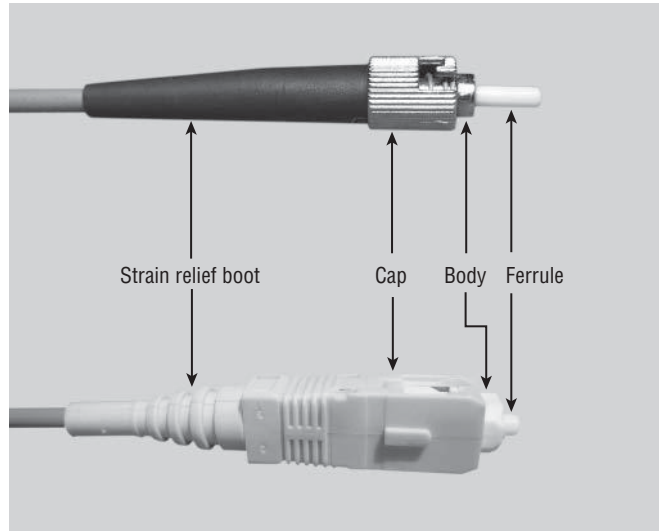
- Evaluate connector endface polishing
- Evaluate connector endface cleaning
- Evaluate connector endface geometry
- Identify an optical fiber type from the color of the connector strain relief

The Fiber-Optic Connector

The job of a fiber-optic connector is to couple an optical fiber end mechanically to a piece of equipment or to another optical fiber so that the cores line up accurately and produce the smallest amount of loss. Inherent in this requirement is the need for the connector to protect the fiber from repeated handling during connection and disconnection, align the fiber end precisely with its counterpart in the interconnection, and prevent strain on the fiber itself. The strength members in the fiber-optic cable normally provide strain relief, but once the fiber itself is attached to the connector, the connector must perform that job.

Although there are several types of connectors recognized by industry standards, they all contain common components, shown in Figure 9.1.

FIGURE 9.1 Fiber-optic connector components



Ferrule

Beginning at the working end of the connector, the *ferrule* holds the fiber in place. The ferrule must hold the fiber exactly centered in its endface for the best possible connection, so its construction is critical. Not only must the hole for the optical fiber be accurately placed; it must also be sized precisely to receive the exact diameter of the optical fiber cladding. The optical fiber's coating is stripped away prior to inserting the optical fiber into the ferrule.

Ferrules typically are made of metal, ceramic, or plastic with a selection of hole or bore diameters ranging from slightly larger than the optical fiber diameter to slightly smaller to allow for minute variations in the manufactured optical fiber cladding diameters. For example, for a 125 μ m optical fiber, ferrules might be available with hole sizes ranging from 124 μ m to 127 μ m. Because the ferrule must align the optical fiber end precisely, it must meet several important criteria:

- The ferrule must be strong enough to withstand many cycles of connection and disconnection without bending, cracking, or breaking.
- The ferrule must maintain dimensional stability to ensure proper alignment of the optical fiber.
- The ferrule must be of the right shape and have material properties to ensure a low-loss interconnection.

Ceramic materials such as aluminum oxide and zirconium oxide are among the best materials for ferrules, offering the best combination of characteristics. They are hard enough to protect the fiber end, and their *coefficient of thermal expansion*, the measure of how much a material expands and contracts with temperature changes, is about the same as the optical fiber itself.

Metal ferrules, typically made of stainless steel, are stronger than ceramic, but they are less dimensionally stable. Plastic ferrules are less expensive than metal or ceramic, but they are neither as strong nor as stable as the other materials.

When a connector is assembled, which we'll describe in detail later in this chapter, the optical fiber is typically epoxied into the ferrule with the end protruding slightly beyond the endface of the ferrule. The optical fiber end is later trimmed and polished with the ferrule endface for a precise fit.

The ferrule fits inside the body of the connector. The body, which can be either metal or plastic, holds the ferrule, the optical fiber, and the cable in place and transfers any strain placed on the connector to the cable rather than the optical fiber.

Cap

The cap, sometimes called the coupling nut, fits over the body of the connector and provides a means to secure the connector, as shown in Figure 9.2. The cap can be a locking mechanism, a threaded ring, or a snap attachment, depending on the type of connector.

FIGURE 9.2 Connector cap types



The strength member of the fiber-optic cable is secured to the connector body. This allows all the tensile stresses placed on the connector to be handled by the cable's strength member instead of the optical fiber. There are several ways to accomplish this; crimping a ring or band around the strength member, securing it to the connector body, is a very popular method. Figure 9.3 shows the band of an LC connector securely attaching the strength member to the connector body.

FIGURE 9.3 Crimped band of an LC connector securely attaching the cable strength member to the connector body



Depending on the connector type, the cable jacket may or may not be securely fastened to the connector body. Sometimes a band or ring is crimped around the cable jacket to secure it to the connector body, as shown in Figure 9.4. Other times a piece of shrink tubing is used to secure the jacket to the connector body, as you can see in Figure 9.5.

FIGURE 9.4 Crimped band of an SC connector securely attaching the cable jacket to the connector body



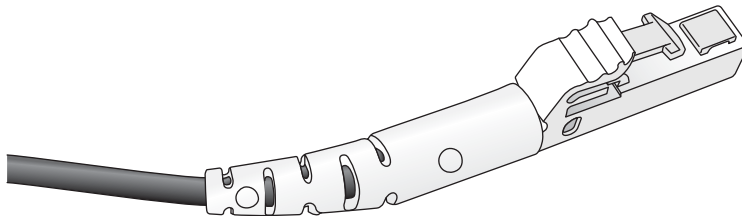
FIGURE 9.5 Shrink tubing attaching the cable jacket to an LC connector body



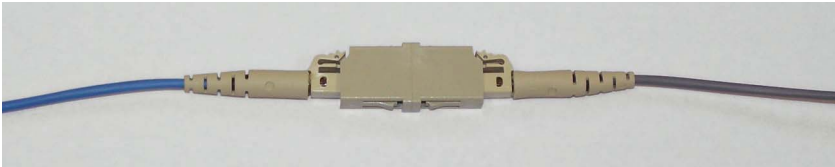
Strain Relief

A strain relief, or boot, is typically placed over the cable jacket and secured to the connector body. The boot is generally made of an elastic material that slides over the connector body. Friction between the boot and the connector body holds the boot in place. The boot prevents the cable from being pulled at too great an angle against the connector, as shown in Figure 9.6.

FIGURE 9.6 Boot or strain relief preventing the fiber-optic cable from being over bent



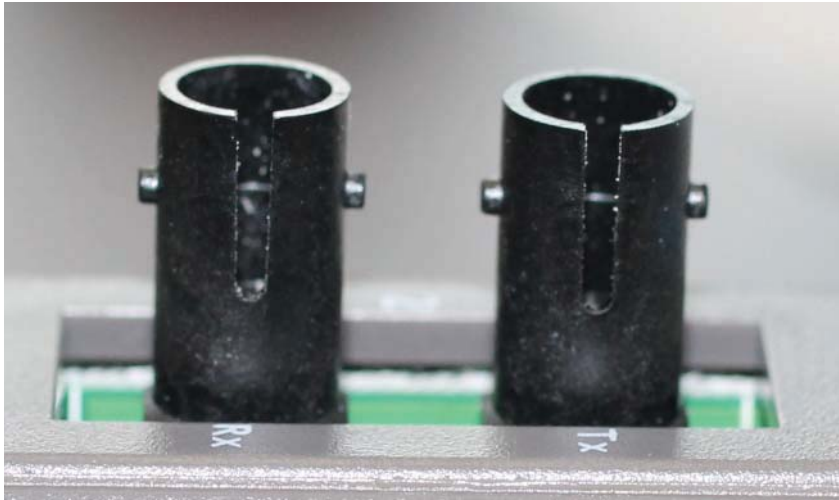
Fiber-optic connectors, commonly referred to as *plugs*, mate with other connectors or receptacles. The ferrule of the connector is the plug, and when two connectors mate, their ferrules are aligned with a sleeve that is typically called a *mating sleeve* or an *alignment sleeve*. Figure 9.7 is a photograph of an LC alignment sleeve that mates two LC connectors.

FIGURE 9.7 LC mating or alignment sleeve with connectors

Connectors also mate with equipment or device receptacles. The device may be a transmitter or receiver, as discussed in Chapter 10, “Fiber-Optic Light Sources and Transmitters,” and Chapter 11, “Fiber-Optic Detectors and Receivers.” The receptacle is similar to one-half of a mating sleeve. Figure 9.8 shows an ST mating sleeve and Figure 9.9 shows ST transceiver receptacles. Looking at these two photographs, you can see how closely they match. You can also see a slot at the receiving end of the receptacle and both ends of the sleeve. This slot is for the key on the connector body.

FIGURE 9.8 ST mating or alignment sleeve without connectors

The terms *alignment sleeve*, *mating sleeve*, and *adapter* are interchangeable. ANSI/TIA-568-C defines an adapter as a mechanical device designed to align and join two optical fiber connectors (plugs) to form an optical connection. However, sometimes the term *adapter* is used when referring to a device that mates two different connector types such as an SC and ST. Every adapter contains an alignment sleeve that aligns the connector ferrules or plugs.

FIGURE 9.9 ST transceiver receptacles

Body

The connector body can take many forms; however, most connector body types fall into one of four categories:

- Round
- Square
- Rectangular
- D-shaped

The connector body may or may not have a key to orient the connector. The key ensures the connector only goes into the receptacle or alignment sleeve one way, and it prevents the ferrule from rotating and damaging the optical fiber when it makes contact with another optical fiber. Connectors without a key cannot make contact with the optical fiber of another connector when they are mated. Contact and noncontact connectors are discussed in detail later in this chapter.

The ST connector is an example of a round body type contact connector; the key typically has a pin-like shape, as shown in Figure 9.10, that must be aligned with a slot in the receptacle or the alignment sleeve, as shown in Figures 9.8 and 9.9 earlier. If the key is not aligned, you will not be able to seat the connector and a high-loss interconnection will result.

The LC connector is an example of a square body type with a rectangular key at the top of the connector as shown in Figure 9.11. The MTP connector shown in Figure 9.12 is an example of a rectangle body type with a rectangular key at the top.

The SC connectors shown in Figure 9.13 are an example of a D-shaped body that serves as the key to align it properly as it is inserted into the cap during assembly. The cap is rectangular with a rectangular key on the top that ensures it is not plugged in upside down.

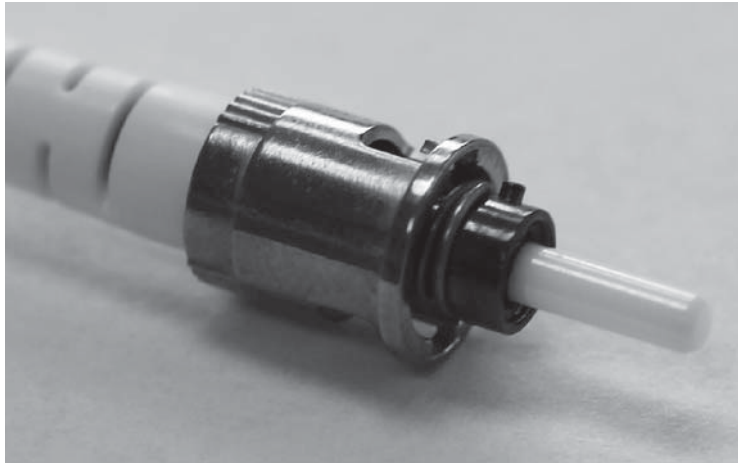
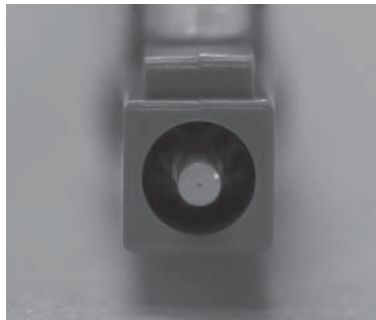
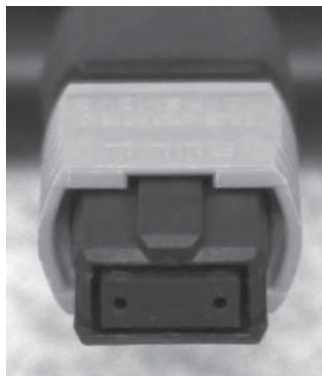
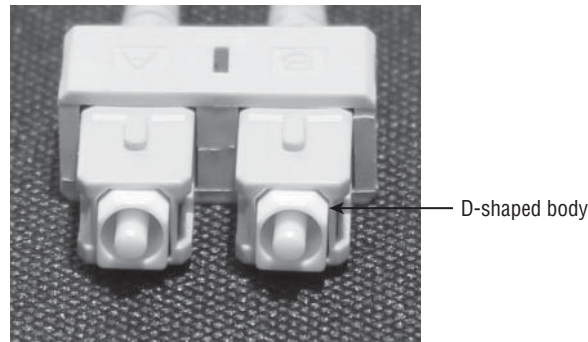
FIGURE 9.10 ST connector with key visible**FIGURE 9.11** LC connector with key visible**FIGURE 9.12** MTP connector with key visible

FIGURE 9.13 SC connectors with the D-shaped body visible



The key design in each of these connectors ensures that they can be inserted into a receptacle or alignment sleeve in only one way. All of these connectors are discussed in detail in the “Connector Types” section later in this chapter.

Connection Performance

As described earlier, the connector mates with either another connector or the equipment. The performance of two connectors mated together, also referred to as a mated connector pair, can be measured. This is described in detail in Chapter 14, “Test Equipment and Link/Cable Testing.”

The performance of a connector mating with a piece of equipment such as a transceiver or passive optical device requires specialized test equipment that is described in detail in Chapter 14. Because of the specialized equipment required to perform these measurements, they typically are not performed in the field.

Connection performance describes the performance of an interconnection. In this section, we are going to examine the factors that affect the performance of an interconnection. This interconnection may consist of two connectors and a mating sleeve or a connector and a receptacle.

Connection performance is dependent on several factors, including intrinsic factors, extrinsic factors, geometry, endface finish, and cleanliness. Intrinsic factors, extrinsic factors, and cleanliness apply to the connector, mating sleeve, and the receptacle. Endface finish and geometry apply only to the connector. Endface finish and the cleaning of the connector endface are discussed at the end of the chapter after connector assembly.

Intrinsic Factors

As discussed in detail in Chapter 8, “Splicing,” even when optical fibers are manufactured within specified tolerances, there are still slight variations from one optical fiber to another.

The same is true for the ferrule and mating sleeve. Variations in these components can affect the performance of the connection.

Let's briefly review the most common types of intrinsic variations:

NA mismatch A numerical aperture (NA) mismatch occurs when the NA of one optical fiber is different from the NA of the other optical fiber. A loss may occur if the NA of the transmitting optical fiber is larger than the NA of the receiving optical fiber.

Core diameter mismatch Core diameter mismatch occurs when there is a difference in the core diameters of the two optical fibers. A loss may occur when the core diameter of the transmitting optical fiber is greater than the core diameter of the receiving optical fiber.

Mode field diameter mismatch A mode field diameter mismatch occurs when there is a difference in the mode field diameters of two single-mode optical fibers. A loss may occur when the mode field diameter of the transmitting optical fiber is greater than the mode field diameter of the receiving optical fiber.

Cladding diameter mismatch Cladding diameter mismatch occurs when the cladding diameters of the two optical fibers are not the same. A loss may occur when the cores of the optical fibers are not aligned because of the cladding diameter mismatch.

Ferrule bore diameter mismatch Ferrule bore diameter mismatch occurs when the bore or hole diameter of the transmit ferrule is different from the bore diameter of the receive ferrule. A loss may occur if the bore diameter mismatch prevents the cores of the optical fibers from properly aligning.

Mating sleeve diameter mismatch Mating sleeve diameter mismatch occurs when the diameters of the transmit side of the mating sleeve and the receive side are not the same. A loss may occur when the cores of the optical fibers are not aligned because of the mating sleeve diameter mismatch.

Optical fiber concentricity The core and cladding of an optical fiber should be round and concentric, which means that they share a common geometric center. Concentricity variations can cause the optical fiber cores to misalign, creating a loss when the light exiting the core of the transmitting optical fiber enters the cladding of the receiving optical fiber.

Ferrule concentricity Concentricity variations in the ferrule can also cause a loss. The ferrule and the bore or hole through the ferrule that accepts the optical fiber should have the same geometric center just like the core and cladding of the optical fiber. Variations in this geometric center can cause the optical fiber cores to misalign when two connectors are mated, creating a loss when light from the core of the transmit optical fiber enters the cladding of the receive optical fiber.

Mating sleeve concentricity Concentricity variations in the transmit side and receive side of the mating sleeve can also cause a loss. The transmit and receive sides of the mating sleeve should have the same geometric center just like the core and cladding of the optical fiber. Variations in the geometric center of the two sides of the sleeve can cause the optical fiber cores to misalign when two connectors are mated, which may create a loss when light from the core of the transmit optical fiber enters the cladding of the receive optical fiber.

Optical fiber noncircularity In an ideal optical fiber, the core and the cladding are perfectly circular. Noncircularity of the core or cladding can cause the cores of the transmit and receive fibers to misalign. This misalignment will cause a loss when light from the core of the transmitting optical fiber enters the cladding of the receiving optical fiber.

Ferrule noncircularity Just as the core and cladding of an optical fiber may not be perfectly circular, the ferrule and the bore through the ferrule may also not be perfectly circular. The noncircularity of the ferrule or the bore through the ferrule may cause the core of the transmit and receive optical fibers to misalign. A loss will occur when light from the core of the transmitting optical fiber enters the cladding of the receiving optical fiber.

Mating sleeve noncircularity Noncircularity in either side of the mating sleeve may prevent the transmit and receive ferrules from properly aligning, which may cause the cores of the optical fibers to misalign. A loss will occur if the misalignment from the noncircularity causes light from the core of the transmitting optical fiber to enter the cladding of the receiving optical fiber.

Extrinsic Factors

Extrinsic factors that affect connection performance are factors related to the condition of the interconnection itself or factors that affect the interconnection. An ideal interconnection would have identical optical fibers, ferrules, and mating sleeve ends. These components would allow the cores of the transmitting and receiving optical fibers to align perfectly, resulting in a low-loss interconnection. However, there is no such thing as a perfect interconnection, only an imperfect interconnection. In a real interconnection, intrinsic and extrinsic factors affect connection performance.

Extrinsic factors and cleanliness go hand in hand. Dirt or contamination can cause any or all of the extrinsic factors described in this chapter. This section briefly reviews the common extrinsic factors that affect connection performance. Cleaning is covered later in the chapter and extrinsic factors are covered in detail in Chapter 8.

Lateral misalignment Lateral misalignment occurs when the two optical fibers are offset laterally. A loss occurs when the lateral misalignment causes light from the core of the transmitting optical fiber to enter the cladding of the receiving optical fiber. As the lateral misalignment increases, the loss increases because less light from the core of the transmitting optical fiber makes its way into the core of the receiving optical fiber.

In an interconnection, lateral misalignment can be caused by intrinsic variations in the optical fiber, ferrule, or mating sleeve. It may also be the result of dirt, contamination, or wear.

End separation End separation is simply an unintended air gap between the endfaces of the transmitting and receiving optical fibers. The air gap will cause a loss and produce a Fresnel reflection. End separation may be caused by dirt, contamination, an improperly polished connector endface, or wear. However, it is more often the result of not fully inserting and securing the connector in the receptacle or mating sleeve.

Figure 9.14 shows two ST connectors in a mating sleeve. The connector on the right has been inserted into the mating sleeve; however, it has not been latched like the connector on the left of the mating sleeve. This happens often and will cause end separation.

FIGURE 9.14 ST connectors in a mating or alignment sleeve: the connector on the left is latched and the connector on the right is not latched.



Angular misalignment Angular misalignment occurs when the optical fibers in an interconnection meet each other at an angle. Loss from angular misalignment occurs when light from the core of the transmit optical fiber enters the cladding of the receive optical fiber or enters the core of the receive optical fiber at an angle exceeding the acceptance angle. Light entering the core of the receive optical fiber at an angle exceeding the acceptance angle typically does not propagate the length of the receive optical fiber.

As with end separation and lateral misalignment, angular misalignment may be caused by dirt, contamination, or wear.

Geometry

Geometry refers to the shape of the ferrule endface. It seems at first that the ideal shape would be a flat surface that would mate with the flat surface of another ferrule, but this geometry actually presents the most potential problems. Endface geometry can be broken into four categories:

- Flat
- Curved
- Angled
- Lensed

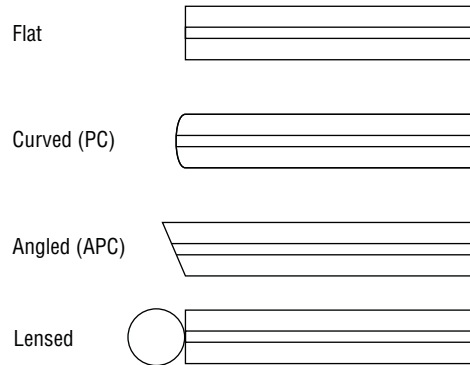
This section provides an overview of each of these geometries.

Flat

Flat endfaces like the one shown in Figure 9.15 will always have some polishing irregularities; they will never be perfectly flat across. When two precision flat metal pieces such as the head and cylinder block of a car engine are brought together, a gasket is required.

The gasket fills the small voids between the head and cylinder block, compensating for the irregularities of the metal surfaces. Without the gasket, the car would not run properly.

FIGURE 9.15 Endface geometry configurations



The slightest variation in the flat endface surface will keep the fiber ends far enough apart to cause end separation loss due to Fresnel reflection. In addition, some of the reflected light may return down the core of the optical fiber to the transmitter as *return reflection* or *back reflection*, which can interfere with the operation of the light source. Flat endface geometry typically offers the poorest back reflection performance.

Return Loss and Reflectance

The terms *return loss* and *reflectance* are both expressed in decibels (dB). *Return loss* as defined in ANSI/TIA-568-C.3 is a ratio of the power of the outgoing signal to the power of the reflected signal. In a mated connector pair, the outgoing signal is the light from the source and the reflected signal is the light that travels through the core of the optical fiber back toward the source. Return loss values are positive numbers. For example, ANSI/TIA-568-C.3 states that 55dB is the acceptable minimum return loss for a single-mode mated connector pair used for broadband analog video (CATV) applications.

Reflectance as defined in ITU-T G.671 is the ratio of reflected power to incident power. In a mated connector pair, reflected power is the light that travels through the core of the optical fiber back toward the source and incident power is the light from the source. Reflectance values are negative numbers. For example, ITU-T G.671 states that the maximum reflectance value for a single-mode single-fiber mated connector pair is -35dB.

Return loss and reflectance can be confusing. When working with return loss, the minimum value is always stated and a higher value is better than a lower value. For example, if the return loss were 10dB, 10% of the light energy would be reflected into the core of the optical fiber and travel back toward the source. However, if the return loss were 20dB,

only 1% of the light energy would be reflected into the core of the optical fiber and travel back toward the source.

When working with reflectance, maximum value is always stated and a lower value is better than a higher value. For example, if the reflectance were -40dB , 0.01% of the light energy would be reflected into the core of the optical fiber and travel back toward the source. However, if the reflectance were -20dB , 1% of the light energy would be reflected into the core of the optical fiber and travel back toward the source. Since reflectance values are negative numbers, -20dB is greater than -40dB .

Curved

To ensure *physical contact* (PC) between optical fiber ends, the best endface geometry is a convex curve. This curved, or *PC finish*, shown in Figure 9.15, ensures that the highest feature or *apex* on the endface is the center of the optical fiber end. When the optical fiber ends are in direct physical contact, the light behaves as if the connected ends are a continuous piece of optical fiber and passes through with very little loss and back reflection.

The prefixes super or ultra are often used when describing PC connector return loss or reflectance performance. A super PC or SPC connector will offer better (typically 10dB better) return loss or reflectance performance than a PC connector will. The ultra PC or UPC connector will offer better (typically 10dB better) return loss or reflectance performance than an SPC connector will. You may also find SPC defined as super-polished connector and UPC defined as ultra-polished connector.

Angled

Another method of reducing back reflection is with an *angled PC (APC) finish*. This type of finish, also shown in Figure 9.15, puts an angle of about 8° on the endface, with the intent that it will mate with a similarly angled endface when properly aligned. With the endface and the fiber end angled, any light that is reflected is sent into the cladding and absorbed by the coating, rather than traveling back through the core of the optical fiber toward the light source. Angled endface geometry offers the best back reflection performance.

Lensed

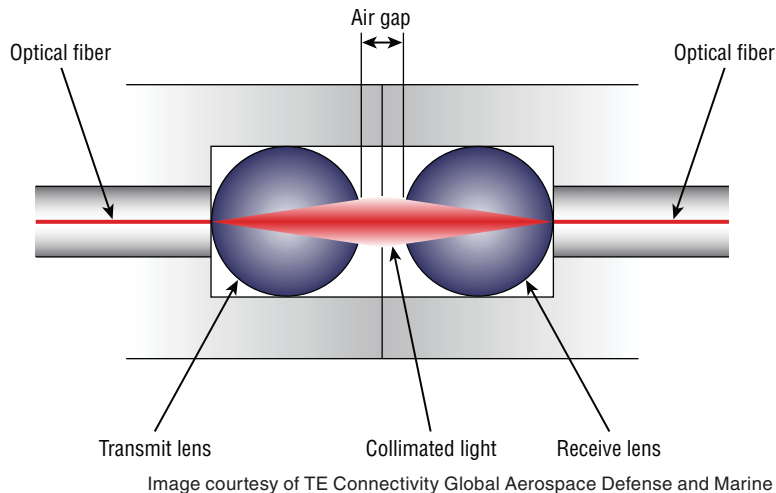
The *lensed ferrule* endface is more commonly known as the *expanded beam*. This endface is not a new concept; it has been in use for over three decades. However, until recently it was not very well known.

Unlike the other geometries that have been discussed, the lensed geometry literally has a lens at the end of the ferrule, as shown in Figure 9.15. One of two different lens types can be used: the collimating lens and the imaging lens. Most expanded beam connectors feature the collimating lens.

When the light traveling through the optical fiber enters the transmit lens, it is expanded and *collimated*, as shown in Figure 9.16. The collimated light travels through the air gap between the two lenses and enters the receive lens. The receive lens collects the collimated

light and focuses it to the point where it is within the cone of acceptance of the receiving optical fiber, as you can see in Figure 9.16.

FIGURE 9.16 The basic components and operation of an expanded beam interconnection



Expanded beam offers some advantages over the other geometries that make it ideal for harsh environments such as those associated with avionics, aerospace, space, mining, and drilling. As of this writing, SAE International is preparing to publish their first expanded beam standard AIR6112, A Guideline for Aerospace Platform Fiber Optic Expanded Beam Interconnect Technology. This standard discusses the operational advantages expanded beam has over physical contact optical interfaces, such as the following:

- Greater tolerance to particulate contamination
- Ease of cleaning
- Protection of the optical fiber core
- Non-physically contacting interface
- Lower mating forces

Lensed geometry has better back reflection than flat geometry and can compete with PC geometry; however, it does not offer the back reflection performance of SPC, UPC, or APC.

Interferometer

An interferometer like the one shown in Figure 9.17 is a device that uses light to take precision measurements of the fiber-optic connector endface. The measurements are used to generate a three-dimensional (3D) map of the surface and quantitatively define three critical

parameters: *radius of curvature*, *apex offset*, and *fiber undercut or protrusion*. These critical parameters are required by Telcordia GR-326 to evaluate connector endface geometry for single-mode connectors and jumper assemblies. Telcordia GR-326 is an industry standard that defines generic requirements for single-mode connectors and jumpers.

FIGURE 9.17 Interferometer with a connector inserted for evaluation



Photo courtesy of PROMET International Inc.

The interferometer shown in Figure 9.17 also has the capability to perform a two-dimensional (2D) optical analysis as you would with an inspection microscope. Two-dimensional analysis is discussed in detail in the “Endface Cleaning” section later in this chapter.

Three-Dimensional (3D) Analysis

When two connectors with a PC finish are brought together in a mating sleeve, springs are used to provide compressive forces that force the endfaces to touch under pressure. These compressive forces maintain glass on glass contact, ideally producing a low-loss and low-reflection interconnection. The geometry of the connector endface is critical to ensure a high-performance interconnection and can be evaluated only with a 3D analysis from an interferometer.

In addition to being able to quantitatively define the radius of curvature, apex offset, and fiber undercut or protrusion, 3D analysis can be used to evaluate surface roughness and contamination. The surface of both the ferrule and optical fiber appear very rough in Figure 9.18, whereas the surface of the ferrule in Figure 9.19 is smooth, although the optical fiber is scratched in several places. The endface shown in Figure 9.20 features a smooth ferrule and optical fiber, but both surfaces are covered with debris.

FIGURE 9.18 3D endface image of surface roughness on the ferrule and the optical fiber

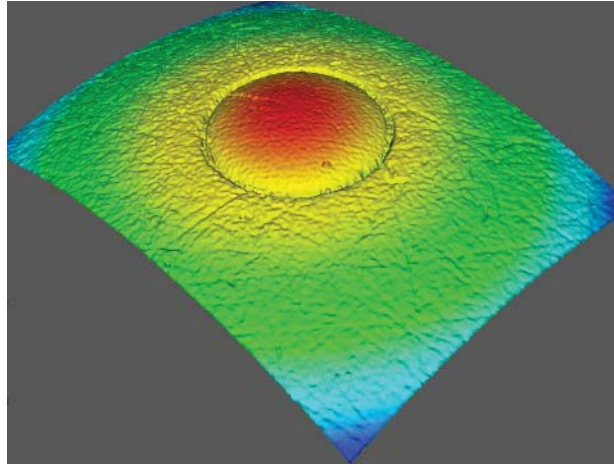


Image courtesy of PROMET International Inc.

FIGURE 9.19 3D endface image of an optical fiber with scratches and a smooth ferrule

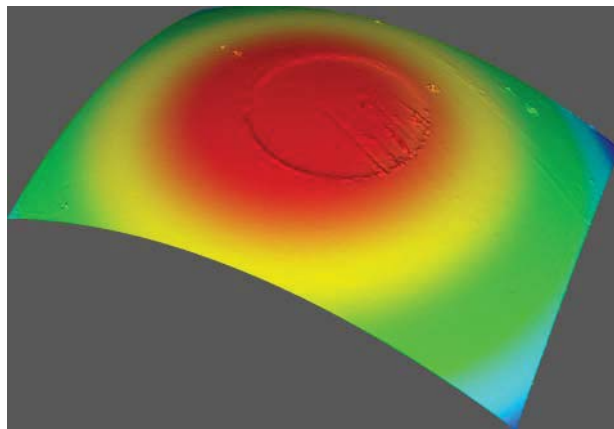


Image courtesy of PROMET International Inc.

FIGURE 9.20 3D endface image of a smooth ferrule and optical fiber both covered with debris

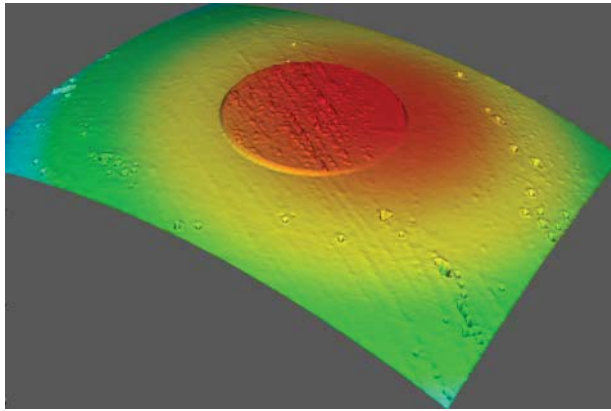
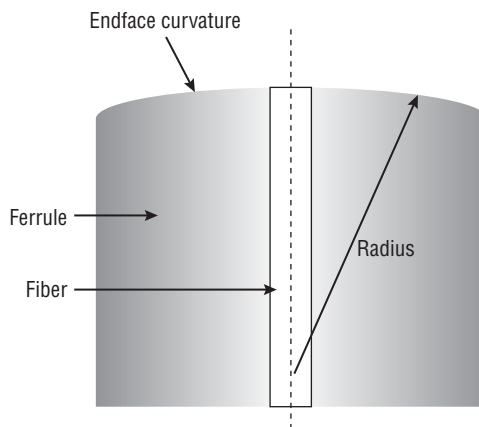


Image courtesy of PROMET International Inc.

Radius of curvature The radius of curvature describes the roundness of the connector endface, and it is measured from the center axis of the connector ferrule, as shown in Figure 9.21. The minimum and maximum values for the radius of curvature as defined in Telcordia GR-326 are 7mm and 22mm. Values below or above this range increase the risk of optical fiber damage. They also increase the possibility of reflection and insertion loss.

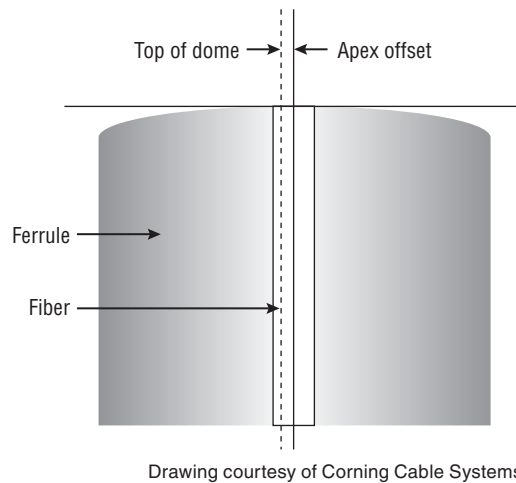
FIGURE 9.21 Connector radius of curvature



Drawing courtesy of Corning Cable Systems

Apex offset The apex of the connector endface is the highest point of the rounded endface. Ideally, the apex would be in the center of the optical fiber. Not having the apex in the center of the optical fiber can lead to the lack of physical contact between the two optical fiber cores in the mating sleeve. Apex offset is the displacement between the apex of the endface and the center of the optical fiber core, as shown in Figure 9.22. Telcordia GR-326 states that apex offset cannot exceed $50\mu\text{m}$.

FIGURE 9.22 Connector apex offset



The image captured by an interferometer in Figure 9.23 uses grayscale shading to display the apex. The darkest shade of gray is the apex. This shaded section should be centered directly over the core of the optical fiber; however, it is offset by roughly the radius of the optical fiber, or $62.5\mu\text{m}$. The apex in this image is directly over the epoxy ring, exceeding the Telcordia GR-326 maximum of $50\mu\text{m}$.

Fiber undercut or protrusion Optical fiber undercut or protrusion describes the distance the optical fiber endface is below or above the rounded connector endface, as shown in Figure 9.24. When the optical fiber protrudes too far from the ferrule endface, as shown in Figure 9.25, the compressive forces on the optical fiber increase and so does the chance that the optical fiber will be damaged. When the optical fiber sits too far below the ferrule endface, as shown in Figure 9.26, an air gap will result, increasing the loss and reflection. Telcordia GR-326 states that an undercut value cannot exceed $\pm 55\text{nm}$.

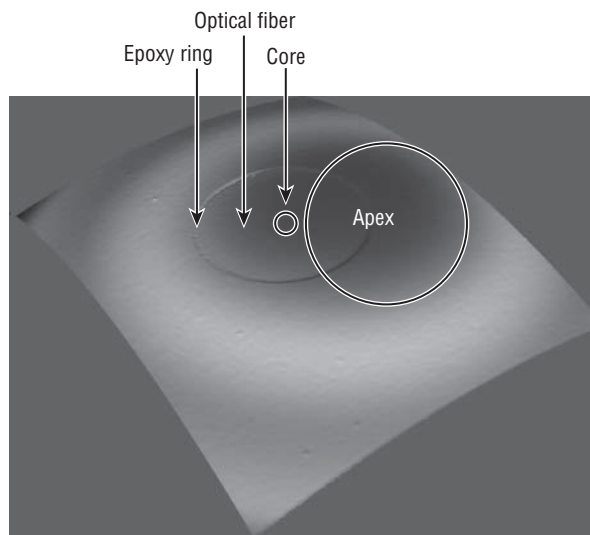
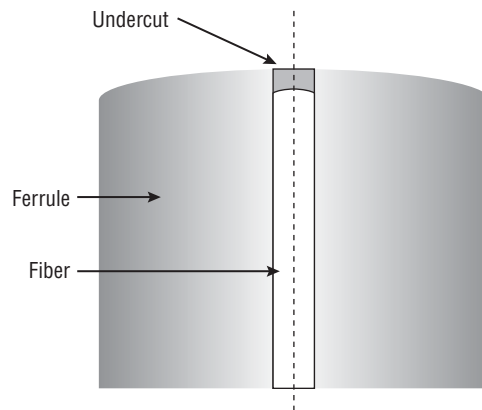
FIGURE 9.23 3D endface image showing an apex offset

Image courtesy of PROMET International Inc.

FIGURE 9.24 Connector with fiber undercut

Drawing courtesy of Corning Cable Systems

FIGURE 9.25 3D endface image showing excessive fiber protrusion

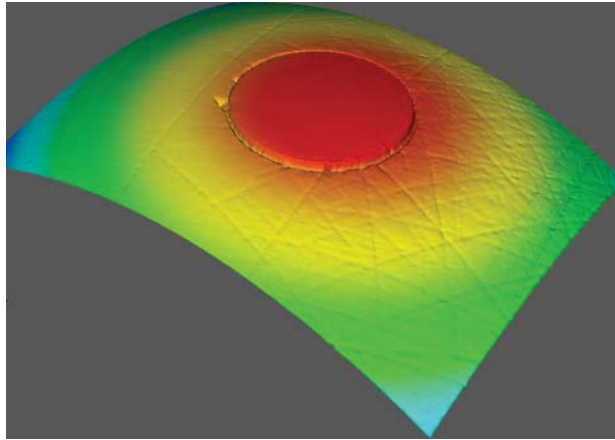


Image courtesy of PROMET International Inc.

FIGURE 9.26 3D endface image showing fiber undercut

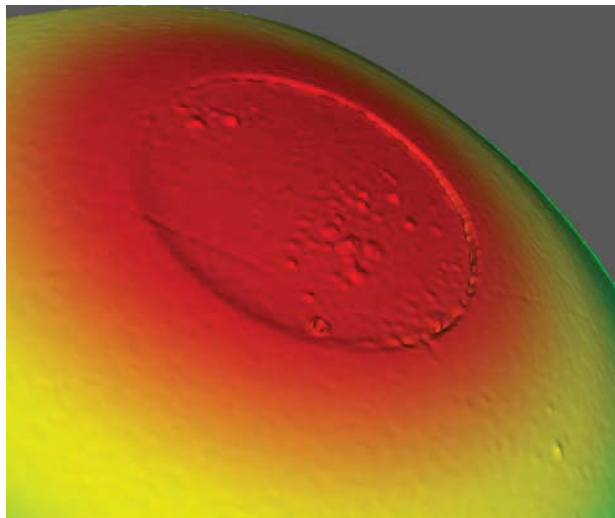


Image courtesy of PROMET International Inc.

A connector endface is like a snowflake—no two are exactly alike. It is for this reason that Telcordia GR-326 provides ranges for critical geometric values. An interferometer can typically display information several different ways to help with the 2D and 3D evaluation of the endface. The manner in which the information is displayed on the monitor and in a report will vary from manufacturer to manufacturer.

The computer screen capture shown in Figure 9.27 features side-by-side 3D and 2D images and analysis of the connector endface. A metrology report below the 3D image provides geometric measurement data and a pass/fail indication. The measurement data includes radius of curvature, fiber height, apex offset, fiber diameter, ferrule angle, key angle, and epoxy width. A defects report below the 2D image displays the number of chips, particles, scratches, total defects, and a pass/fail indication. This connector has a fiber height greater than 55nm and therefore fails to meet the Telcordia GR-326 requirement.

FIGURE 9.27 Interferometer screen capture with 3D and 2D views, along with metrology data and defect analysis

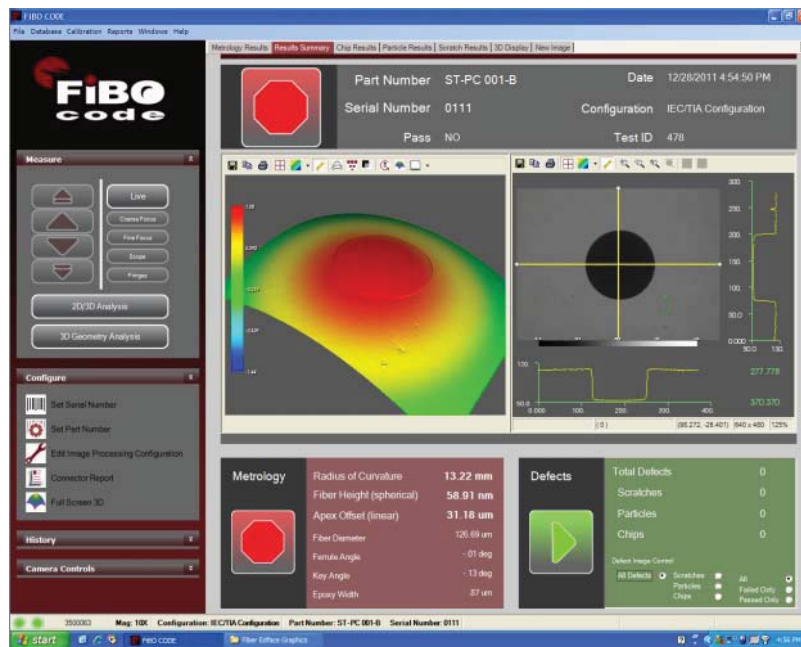


Image courtesy of PROMET International Inc.

Connector Types

The purpose of all fiber-optic connectors is the same—to align a terminated optical fiber end with another terminated optical fiber end or a piece of equipment and hold it firmly in place. The manner in which this has been accomplished, however, has continued to evolve as technology improves and as requirements change. As a result, manufacturers have created many types of connectors. Most of the connectors are still in use, though some have become more common whereas others exist only in older or legacy systems.

Connectors can be broken into two categories: single-fiber connectors and multifiber connectors. Whether the endfaces make contact or do not make contact when mated defines them as a contact or noncontact connector.

Single-fiber connectors are designed for only one optical fiber, but they may be used to break out multiple optical fibers from a cable for individual termination at a patch panel or piece of equipment. Multifiber connectors are available in many configurations and may be used to connect paired fibers in duplex systems, or to connect up to 72 optical fibers in a single ferrule array.

The majority of the connectors covered in this chapter are contact type connectors. The endfaces of contact type connectors, regardless of the endface geometry, physically touch when mated. However, the endfaces of noncontact type connectors, regardless of their geometry, do not physically touch when mated. These types of interconnections will always have an air gap.

Because there are so many types of connectors, the Telecommunications Industry Association (TIA) has not attempted to recognize or create standards for all of the available types. In the ANSI/TIA-568-C standard, however, they recognize performance standards for connectors, which will be described at the end of this chapter.

Let's look at the various types of connectors.

Single-Fiber Contact Connectors

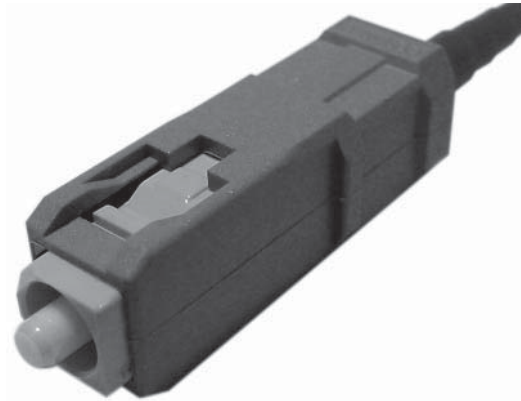
Single-fiber contact connectors have a wide variety of connection methods. Some, including the earliest types, are engaged by pushing and twisting or by using a threaded sleeve to draw the connector tight. However, many of the newer form factors are square or rectangular snap-in connectors. These are inserted with a simple push that engages a locking mechanism.

SC Connectors

The SC (subscriber connector), shown in Figure 9.28, is among the most widely used connectors. Originally developed by Nippon Telephone and Telegraph (NTT), the SC has a

standard-sized 2.5mm ferrule and a snap-in connection, which was created as an alternative to connectors that required turning or twisting to keep them in place. In addition, SC connectors can be installed in only one orientation, making them suitable for an APC endface.

FIGURE 9.28 The SC connector is one of the most common types in use.



ST Connectors

The ST (straight tip) connector, shown in Figure 9.29, was developed by AT&T as a variation on a design used with copper coaxial cables. This connector has a metal connector cap that must be twisted to lock into place. The ST is considered a legacy connector, as it has been around for quite some time and can still be found in many installations today.

FIGURE 9.29 An ST connector must have its connector cap twisted to lock in place.



FC Connectors

An FC (face contact) connector is a rugged metal connector with a screw-on connector cap and a 2.5mm ferrule. Like the SC connector, the FC, shown in Figure 9.30, is used in connections where proper polarization must be maintained. Because the connector is cylindrical, it must be aligned with a built-in key. Note, however, that there are several different variations for the size of the key, meaning that the connector must be properly matched with its adapter or receptacle.

FIGURE 9.30 FC connector with screw-on connector cap

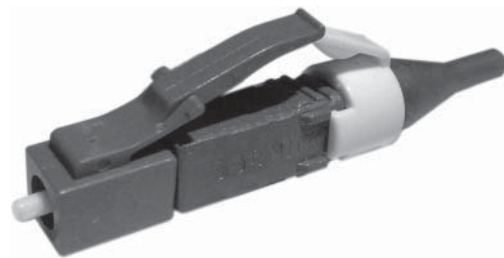


Photo courtesy of Corning Cable Systems

LC Connectors

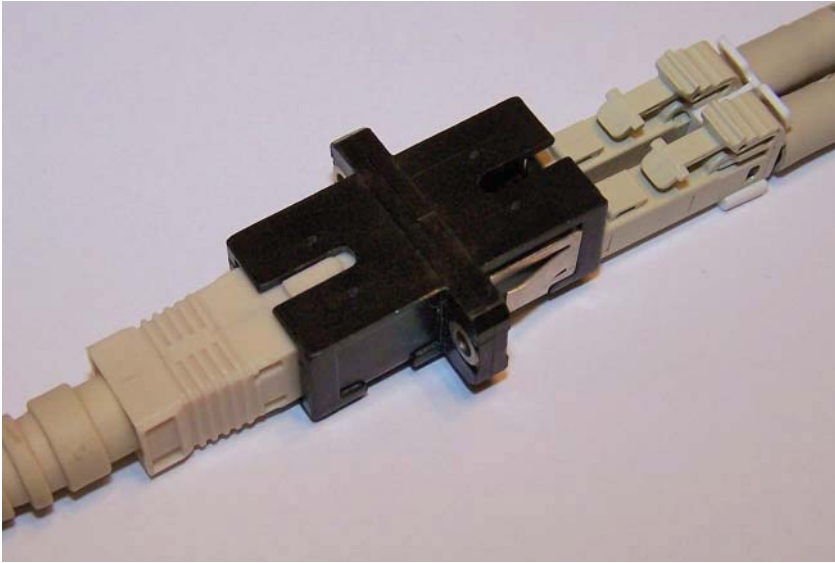
The LC connector, shown in Figure 9.31, is a *small form factor* connector. Developed by Lucent Technologies, this snap-in connector is considered to be a smaller version of the SC connector and is sometimes referred to as a mini-SC. The small form factor connector has a 1.25mm ferrule, half the diameter of an SC connector ferrule.

FIGURE 9.31 The LC connector is a small form factor version of the SC connector.



The smaller form factor allows two LC connectors to fit into roughly the same size space as a single SC connector. Figure 9.32 shows two LC connectors on one side of an SC mating sleeve and an SC connector plugged into the other side of the mating sleeve. The two LC connectors require no more room than a single SC.

FIGURE 9.32 SC mating sleeve with an SC connector on the left side and two LC connectors on the right side



D4 Connectors

The D4 connector is an older-style connector with a 2.5mm ferrule and a threaded connector cap that must be screwed on to secure the connector. It is similar in function to the FC connector, but its profile is slightly smaller, allowing it to be used in smaller spaces than the FC connector. D4 connectors are keyed and the optical fibers make contact when mated.

Mini-BNC Connectors

The mini-BNC, shown in Figure 9.33, is similar in appearance to its counterpart made for copper coaxial cables. The mini-BNC (short for either Bayonet Nut Connector or Bayonet Neill-Concelman, the inventor of the original BNC) is a metal twist-lock connector with a 2.5mm ferrule. The mini-BNC is considered a legacy connector, so though it may never be installed in new a system, you may find it on older installations.

FIGURE 9.33 The mini-BNC connector resembles the BNC connector used in RF applications.



Single-Fiber Noncontact Connectors

Single-fiber noncontact connectors are typically engaged by pushing and twisting or by using a threaded sleeve to draw the connector tight.

SMA Connectors

The SMA connector was developed by Amphenol from its line of microwave connectors known as SubMiniature A. The connector has a 3mm stainless steel ferrule and a connector cap that is threaded on the inside. Because it was originally developed before the invention of single-mode optical fiber, the SMA connector does not provide as precise a connection as more recent designs. In addition, they do not have an alignment key and can rotate during the mating process. To prevent damage to the optical fiber, they are noncontact. SMA connectors can still be found in military applications and in the delivery of high-power laser light.

Biconic Connectors

The biconic connector shown in Figure 9.34 may feature a stainless steel or ceramic cone-shaped ferrule that helps in aligning fiber ends during connection. Biconic connectors do not have an alignment key and can rotate during the mating process. It is for this reason that they are noncontact.

FIGURE 9.34 The biconic connector has a cone-shaped ferrule.



The biconic is threaded on the outside for screw-in placement, and although they are considered legacy connectors, they are still used in military applications. The biconic connector ferrule is also incorporated into the tactical fiber-optic cable assembly (TFOCA) connector.

Multiple-Fiber Contact Connectors

Multiple-fiber contact connectors are mostly designed to support duplex operations and ribbon fibers, allowing a greater number of connections in a smaller space. Some duplex connectors resemble larger or ganged versions of single-fiber connectors.

FDDI Connectors

The FDDI connector was developed for use in Fiber-Distributed Data Interface (FDDI) networks. This plastic duplex connector is also called FDDI MIC for *medium interface connector*, a connector that is used to link electronics and fiber transmission systems. It features protective shields around the two 2.5mm ferrules to prevent damage. The connector locks into place with latches on the sides and can be keyed to ensure proper orientation of the two fiber ends.

ESCON Connectors

The ESCON connector was developed for IBM's ESCON (Enterprise Systems Connection) architecture in the early 1990s. It is similar to the FDDI, but it has a shroud that retracts from the ferrules when the connector is engaged.

SC Duplex Connectors

The SC duplex connector, shown in Figure 9.35, is actually two single SC connectors joined with a plastic clip. This arrangement can be extended into even more plugs, if necessary, but is most commonly applied in the duplex configuration.

FIGURE 9.35 The SC duplex is made of two single SC connectors joined together with a clip.

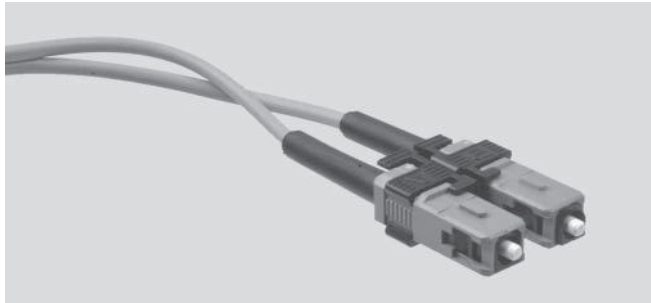


Photo courtesy of Norfolk Wire & Electronics

LC Duplex Connectors

The LC duplex connector, shown in Figure 9.36, is actually two single LC connectors joined with a plastic clip. The clip keeps the transmit and receive optical fibers for a single channel together.

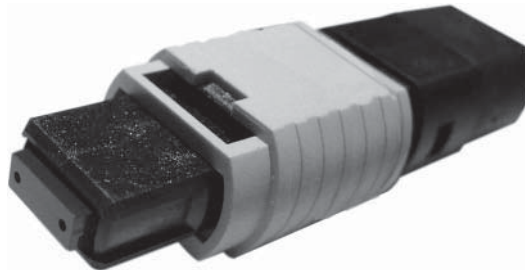
FIGURE 9.36 The LC duplex is made of two single LC connectors joined together with a clip.



MPO Connectors

The MPO (Multifiber Push On) connector, shown in Figure 9.37, is built on the MT ferrule. The MT (mechanical transfer) ferrule is rectangular and the endface measures 6.4mm $\times$ 2.5mm. It is designed to hold from 4 to 72 fibers. The MT ferrule is ideally suited for use with ribbon fiber because the optical fibers are arranged in rows of 4, 8, or 12 optical fibers. The optical fibers are spaced 0.25mm apart, as shown in the 12-fiber ferrule in Figure 9.38. The MT ferrule can support up to six rows of 12 optical fibers.

FIGURE 9.37 The MPO connector packs 4 to 72 fibers into a single ferrule.



The MT ferrule can accept two 0.7mm in diameter precision-machined guide pins that maintain the alignment necessary for connecting 4 to 72 fibers at once. These guide pins can be arranged as necessary between the mating connectors depending on the way they

will be used. The 12-fiber ribbon patch cord shown in Figure 9.39 features one MT ferrule with guide pins and one without.

FIGURE 9.38 12-fiber MT ferrule with 0.25mm spacing

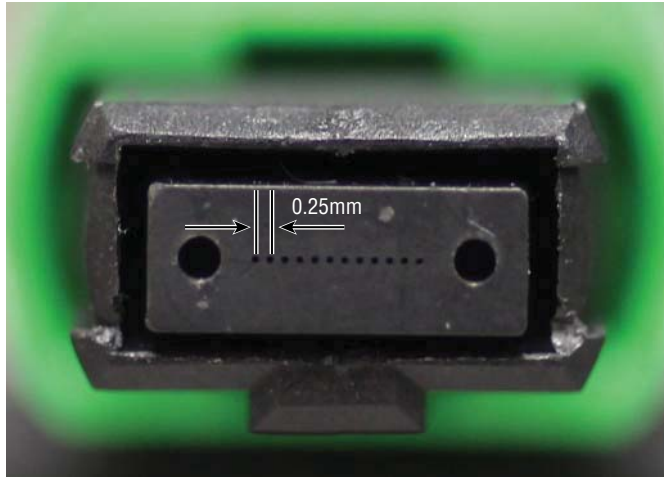


FIGURE 9.39 12-fiber ribbon patch cord featuring one MT ferrule with guide pins and one without



The MT ferrule is available with a flat or angled endface. The 8-degree angled endface is typically used for single-mode optical fiber to reduce back reflections. The MT ferrule endface cannot be polished with a radius because the optical fibers in the center of the ferrule would extend further out than the optical fibers away from the center. This would create a large air gap between the outside fibers when mated with another connector.



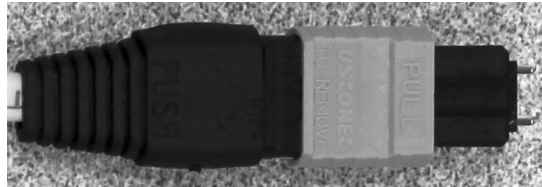
The MPO connector has a plastic body that is spring-loaded like many other fiber-optic connectors to keep the connectors together. International standard IEC 61754-7 defines the standard interface dimensions for the different types of MPO connectors available.

Single ferrule connectors such as the MPO that contain multiple optical fibers arranged in a row or in rows and columns are also known as array connectors.

MTP Connectors

The US Conec MTP® connector shown in Figure 9.40 is built around the MT ferrule. It is designed as a high-performance version of the MPO and will interconnect with MPO connectors. Details on the performance differences between MPO and MTP connectors are explained on the US Conec website at www.usconec.com.

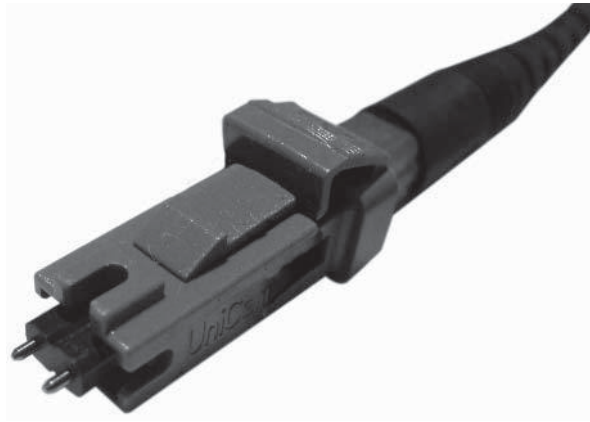
FIGURE 9.40 MTP connector with alignment pins



MT-RJ Connectors

The MT-RJ connector, shown in Figure 9.41, was developed to emulate the functionality of the RJ-45 modular connector, which is used in most office networking. It provides a snap-in, duplex connection in a housing that resembles the familiar modular network plug. In fact, the connector is designed to fit in the same physical opening as the RJ-45, so many networking fixtures and wall plates can be retained after the hardware inside them has been upgraded for fiber.

FIGURE 9.41 The MT-RJ connector is designed to emulate the RJ-45 modular plug-in fit and ease of use.



The MT-RJ's single ferrule holds two fibers in a housing smaller than most single-fiber connectors, so it is attractive in many applications where size matters. The connector is built like a mechanical splice, so the fibers are inserted into the ferrule and the cam is rotated to hold the fibers in place without the need for epoxy.

MT-RJ connectors may be used as most connectors are, with a sleeve that joins two connectors together, as shown in Figure 9.42, or in a plug-and-receptacle arrangement with a transceiver.

FIGURE 9.42 Two MT-RJ connectors and a mating sleeve

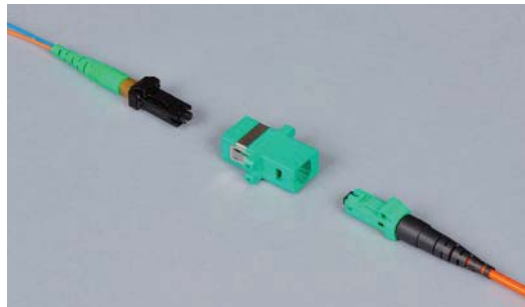


Photo courtesy of Corning Cable Systems

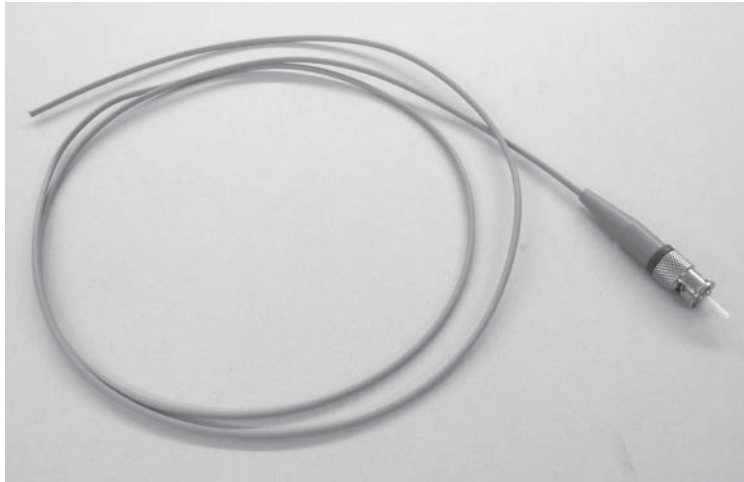
Many other connector types are in use, and more are being added to support new applications and higher bandwidths in a variety of environments. Table 9.1 gives you a quick reference to the connectors we've discussed and their key features.

TABLE 9.1 Quick reference for fiber-optic connectors

Connector	No. of fibers	Connection	Shape	Ferrule size
Single-fiber				
SC	1	Push-in	Rectangular/ D-shaped	2.5 mm
ST	1	Twist-lock	Round	2.5 mm
FC	1	Screw-on	Round	2.5 mm
LC	1	Push-in	Square	1.25 mm
D4 (DIN)	1	Screw-on	Round	2.5 mm
SMA	1	Screw-on	Round	3 mm
Biconic	1	Screw-on	Round	Conical
Mini-BNC	1	Twist-lock	Round	2.5 mm
Multifiber				
FDDI	2	Push-in	Rectangular	(2) 2.5 mm
ESCON	2	Push-in	Rectangular	(2) 2.5 mm
SC Duplex	2	Push-in	Rectangular	(2) 2.5 mm
MPO	4 to 74	Push-in	Rectangular	6.4 mm × 2.5 mm
US Conec MTP	4 to 72	Push-in	Rectangular	6.4 mm × 2.5 mm
MT-RJ	2	Push-in	Rectangular	5 mm × 2.5 mm

Pigtails

Pigtails are fiber ends with connectors factory-attached for future splicing into a system, as shown in Figure 9.43. Typically, a pigtail starts as a manufactured patch cord or jumper with a connector at each end. You can then cut the jumper in half and have two pigtails ready to splice.

FIGURE 9.43 Use a pigtail to splice a connector to an existing fiber.

Pigtails are available with a variety of connectors, depending on your needs. These products have advantages over field terminations because the connectors are factory-installed and polished to exacting standards. This can save time over attaching a connector to the end of an optical fiber, and it typically produces a better connection. They are especially useful in situations where many connectors have to be added to cables in a relatively short time, or in a location where it is easier to make a splice than it is to add a connector.

On the downside, pigtails require hardware to protect the splice and an investment in mechanical or fusion splicing equipment.

Specialized Harsh Environment Connectors

While standard commercial connectors are used in most networks and communications links, fiber-optic connectors are also available in specialized configurations for harsh environments such as those associated with avionics, aerospace, space, mining, and underground drilling. They feature durable cases, heavy-duty connector bodies, and multiple-fiber connections, such as the TFOCA III connector shown in Figure 9.44. This connector is often found in military applications where heavy use and destructive environments are common.

The connectors shown in Figure 9.45 are typically used in aerospace applications. These are military qualified MIL-DTL-38999 four-optical-fiber connectors cabled for an aerospace application.

FIGURE 9.44 This 24-channel TFOCA III connector is a specialized connector for harsh environment applications.



Photo courtesy of MicroCare

FIGURE 9.45 Two MIL-DTL-38999 connectors cabled for an aerospace application



Photo courtesy of Carlisle Interconnect Technologies



The vocabulary associated with harsh environment connectors contains some terms that are typically not used when working with the other types of connectors described in the text. A contact describes the ferrule and all the other physical components required to complete the mechanical assembly. A single contact may be referred to as a *terminus* and multiple contacts may be referred to as *termini*. A terminus can be a socket or a pin. The socket terminus would typically have an alignment sleeve.

As mentioned earlier in this chapter, expanded beam offers some advantages over the other geometries that make it ideal for harsh environments. It is for this reason that SAE International created AIR6112, A Guideline for Aerospace Platform Fiber Optic Expanded Beam Interconnect Technology. Shown in Figure 9.46 is a detailed drawing of a lensed socket terminus and a lensed pin terminus. The actual disassembled socket and pin lensed termini represented in that drawing are shown in Figure 9.47. These termini can be used in multi-contact connectors like the ones shown in Figure 9.48.

FIGURE 9.46 Detailed drawing of a socket and pin lensed termini

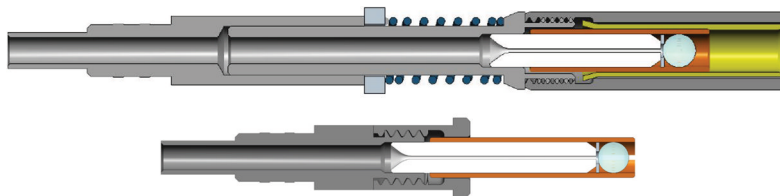


Photo courtesy of TE Connectivity Global Aerospace Defense and Marine

FIGURE 9.47 Photograph of a disassembled socket and pin lensed termini



Photo courtesy of TE Connectivity Global Aerospace Defense and Marine

FIGURE 9.48 Multi-contact expanded beam connectors broken out to individual LC connectors



Photo courtesy of TE Connectivity Global Aerospace Defense and Marine

Connector Termination

One of the most exacting jobs you will encounter in working with fiber optics is connector termination. If you do the job properly, an optical fiber the size of a human hair will mate with another optical fiber or a piece of hardware and transfer a large percentage of the light passing through it. If you do the job wrong, the light could stop at the connector and go no further, or so little of the light could transfer that it would essentially be useless. With those encouraging words urging us forward, let's look at three ways in which an optical fiber can be terminated with a connector.



There are many precision mechanical tools used for fabricating connections, but the actual process is largely an art that must be practiced. Do not make the mistake of making your first connection in a real installation; practice before every installation.

Tools

Regardless of the method you choose to terminate the optical fiber, tools will be required. In this section, we discuss three termination methods. Each of these methods requires some of the tools discussed in this section.

Shears Even though most cabling materials are fairly easy to cut, the aramid or Kevlar strength member is strong, fibrous, and loose fitting, making it difficult to cut. You will need a good pair of sharp shears like the pair shown in Figure 9.49 to cut through the strength member quickly and cleanly.

FIGURE 9.49 Shears designed to cut the Kevlar strength member



Photo courtesy of KITCO Fiber Optics

Stripper A stripper is used to remove the outer jacket, tight buffer, and coating so that only the optical fiber itself is exposed. Some strippers are designed to only remove the tight buffer and coating, like the one shown in Figure 9.50. Other strippers may be designed to remove the jacket, tight buffer, and coating, like the one shown in Figure 9.51. Some are adjustable, like the one shown in Figure 9.52, and can be used to remove the jacket, tight buffer, or coating. However, they need to be adjusted for each application.

FIGURE 9.50 Stripper designed to remove only the tight buffer and coating



FIGURE 9.51 Stripper designed to remove the jacket, tight buffer, and coating



Photo courtesy of KITCO Fiber Optics

FIGURE 9.52 Adjustable stripper that can be set to remove the jacket, tight buffer, or coating



Scribe The scribe, shown in Figure 9.53, is used for precision work in removing the fiber end once it has been adhered inside the ferrule. To use it, you only need to score or nick one side of the optical fiber lightly. This breaks the surface of the cladding, which provides the fiber's tensile strength. You'll need some practice with a scribe to get the right results, since it requires just the right touch to keep from damaging or breaking the optical fiber.

FIGURE 9.53 Scribe used to score or nick the optical fiber

Photo courtesy of KITCO Fiber Optics

Polishing fixture (puck) The polishing fixture, or puck as it is typically referred to, is shown in Figure 9.54. The puck is used to ensure that the ferrule stays perpendicular to the polishing film during the polishing process; see Figure 9.55. Even the slightest variation in the polishing angle can affect the endface finish of the connector, which in turn could impact the performance of the connector.

FIGURE 9.54 Polishing fixture (puck) for a 2.5mm ferrule

Photo courtesy of KITCO Fiber Optics

Lint-free wipes Lint-free wipes like those shown in Figure 9.56 are engineered to lift away oils, grime, and dust from the surface of the optical fiber. When handling a lint-free wipe, only touch one side of the wipe because the wipe will absorb the oil from your skin. Use the side of the wipe that you did not touch for cleaning the optical fiber. You do not want to contaminate the optical fiber during the cleaning process.

Cleaning fluid Cleaning fluids like those shown in Figure 9.57 are approved for air travel and engineered specifically to clean the optical fibers without leaving a residue. These cleaning fluids typically contain little or no isopropyl alcohol and have been filtered to remove microscopic contaminants.

FIGURE 9.55 Polishing fixture (puck) holding an LC connector perpendicular to the polishing film

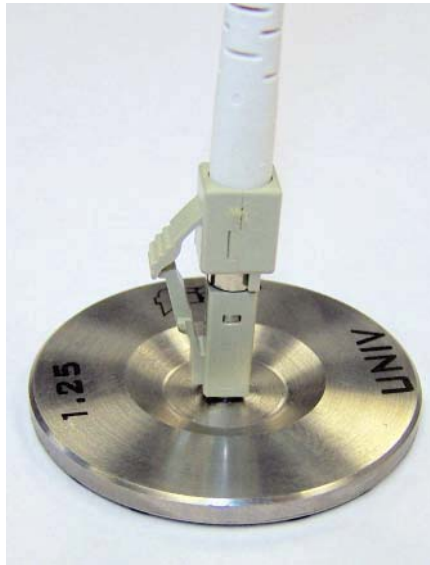


FIGURE 9.56 Lint-free wipes engineered for cleaning optical fibers



Photo courtesy of MicroCare

FIGURE 9.57 Optical fiber cleaning fluids approved for air travel

Photo courtesy of MicroCare

Alcohol Isopropyl alcohol is used to remove the residual coating from the optical fiber prior to termination. While it is widely used, it does leave a residue and you may want to consider a cleaning fluid that does not. If you choose isopropyl alcohol, make sure that it is virtually free of water and has been filtered to remove impurities, like the isopropyl alcohol shown in Figure 9.58. Do not use the isopropyl alcohol commonly available at your local pharmacy; this alcohol contains a large percentage of water and will leave a residue on the optical fiber. While you are using the isopropyl alcohol, ensure it is kept covered when not in use. Doing so will not only prevent spills; it will also prevent the isopropyl alcohol from absorbing the moisture in the atmosphere and help to minimize contamination.

FIGURE 9.58 Isopropyl alcohol for cleaning optical fibers

Photo courtesy of MicroCare

Cleaver A cleaver is only required for prepolished connectors. Cleavers like the one shown in Figure 9.59 score and break the optical fiber, leaving a perpendicular finish. Cleaving is typically a two-step process and it takes place after the optical fiber has been properly prepared and cleaned. Cleaving is described in detail in Chapter 8.

FIGURE 9.59 Optical fiber cleaver

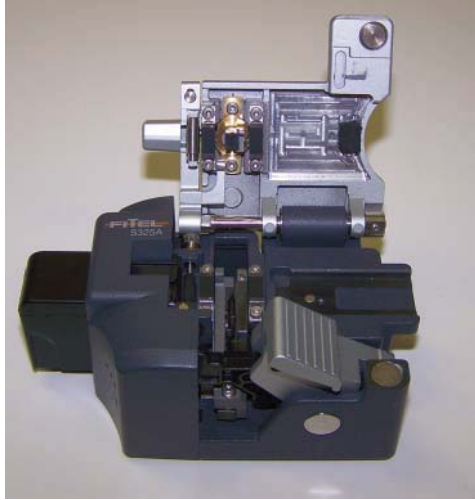


Photo courtesy of KITCO Fiber Optics

Nonreflective mat Always work over a nonreflective black surface like the one shown in Figure 9.60. The nonreflective black mat makes it easier to keep track of cut optical fiber ends. Keep in mind that the small fiber pieces are sharper and smaller than a hypodermic needle and can painlessly penetrate your skin.

FIGURE 9.60 A nonreflective black mat makes it easier to see the bare optical fiber.



Optical fiber disposal container Always use a labeled container with a lid like the one shown in Figure 9.61 to dispose of bare optical fibers.

FIGURE 9.61 Container designed and labeled for optical fiber waste



Safety glasses To prevent eye injury, always wear proper eye protection like the safety glasses shown in Figure 9.62.

FIGURE 9.62 Safety glasses designed to prevent eye injuries



Tweezers To prevent injury to your hands from the optical fiber, you may want to handle pieces of bare optical fiber with tweezers, like the pair shown in Figure 9.63.

FIGURE 9.63 Tweezers can be used to pick up the pieces of optical fiber.



Crimper The crimper like the one shown in Figure 9.64 is typically used to secure the strength member and jacket to the connector.

FIGURE 9.64 Crimpers are used to secure the strength member and jacket to the connector.



Photo courtesy of KITCO Fiber Optics

Eye loupe An eye loupe like the one shown in Figure 9.65 provides moderate magnification and is used to view the connector ferrule after the scribing process.

FIGURE 9.65 An eye loupe provides moderate magnification for viewing the endface of the connector ferrule.



Photo courtesy of KITCO Fiber Optics

Rubber pad The rubber pad like the one shown in Figure 9.66 is placed under the polishing film. It provides a cushioned surface for PC connectors.

FIGURE 9.66 The rubber pad provides a cushioned surface for the polishing film.



Photo courtesy of KITCO Fiber Optics

Cure adapter The cure adapter like the one shown in Figure 9.67 is placed over the ferrule of the connector. It helps transfer heat from the curing oven and protects the optical fiber during the curing process.

FIGURE 9.67 The cure adapter slides over the connector ferrule and helps transfer heat from the curing oven to the connector.



Photo courtesy of KITCO Fiber Optics

Curing oven The curing oven is only required for epoxies that need to be heated to a specific temperature to cure. The curing oven like the one shown in Figure 9.68 accepts the cure adapter.

FIGURE 9.68 Curing oven used to heat the connector and cure the epoxy



Photo courtesy of W.R. Systems

Epoxy

This section of the chapter discusses oven-cured and anaerobic epoxies. There are advantages and disadvantages to each type. The epoxy should be chosen based on the application. This section provides a general overview on epoxy. You should always consult the epoxy data sheet for specific information. If you have specific questions about the type of epoxy to use for your application, consult the manufacturer of the connector or epoxy.

Oven-Cured Epoxy

Oven-cured epoxy is probably the best type you can use, but it is also the most cumbersome. The epoxy itself consists of a resin and a hardener, which must be mixed in the right proportions. Once it is mixed, it has a limited *pot life* before it begins to harden, and any epoxy that is unused must be discarded. On the other hand, once a batch is mixed, several connectors can be assembled at the same time, so it is useful in making a large number of terminations at one location.

Once the connector has been assembled, you have to insert it into a specially built oven to cure the epoxy and then let the assembly cool before you can begin polishing it. The oven will hold a number of connectors, so as each one is built it can be inserted for curing.

While oven-cured epoxy is time-consuming and equipment-intensive, it produces a hard, fully cured bead around the base of the optical fiber, as shown in Figure 9.69. This bead reduces the risk that the fiber will break inside the ferrule during cleaving, which could ruin the connector.

FIGURE 9.69 The bead of oven-cured epoxy protects the fiber during cleaving.

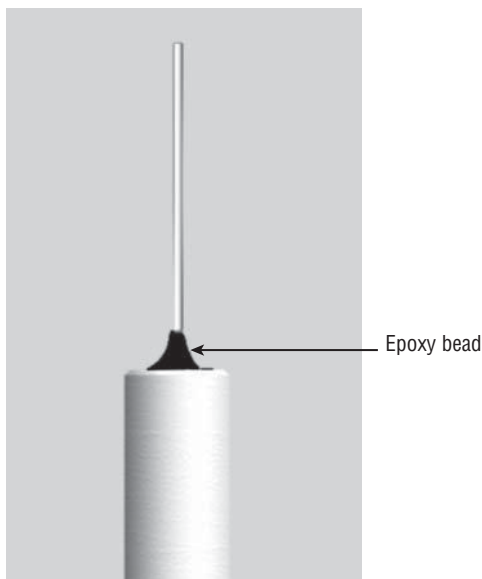


Photo courtesy of W.R. Systems

Anaerobic Epoxy

Anaerobic epoxy is typically used where there is no power available or when time is at a premium. While the adhesive still comes in two parts, each part is applied separately to the fiber and the ferrule. When the two are joined, the epoxy hardens and cures in about 10 seconds.

Although this may seem to be an ideal solution for all situations, there are some drawbacks to anaerobic epoxy. For one or two connectors, there is a significant time-savings. However, because of its quick hardening time, anaerobic epoxy does not lend itself to be used in batches as oven-cured epoxy does. You typically prepare and work on only one connector at a time. In addition, this epoxy does not form a hard bead the way oven-cured epoxy does, so greater skill is required for cleaving and polishing the optical fiber. Finally, this epoxy is not approved for air travel.

Abrasives

Abrasives are used to polish the optical fiber end. Depending on the application, different types of abrasives are needed. Essentially, they are like very fine sandpaper, but the abrasive material is adhered or fixed to a Mylar film backing instead of paper. Two popular abrasive materials are aluminum oxide and diamond.

Aluminum Oxide

Aluminum oxide is an abrasive that is harder than the optical fiber but softer than the ceramic ferrule. Because of this, it will polish down the optical fiber but not the ferrule. While this abrasive prevents the person polishing the connector from altering the endface geometry of the ferrule, it does not produce a very attractive finish when compared to diamond abrasive, and it chips the optical fiber at the epoxy ring. (The epoxy ring is the area between the optical fiber and the ferrule where the epoxy rests.) This abrasive will produce an endface similar to the one shown in Figure 9.70.

FIGURE 9.70 Multimode endface polished with aluminum oxide polishing film

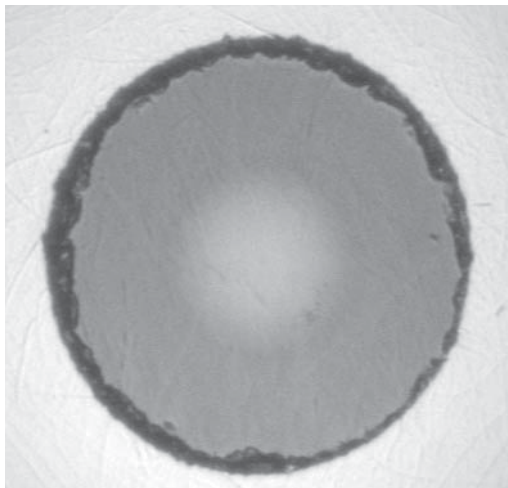


Photo courtesy of W.R. Systems

While the cosmetic finish of an endface polished with aluminum oxide may not look as attractive as an endface polished with diamond, like the one shown in Figure 9.71, aluminum oxide polishing films are significantly less expensive than diamond. The cosmetic finish does not always tell the whole story when it comes to connector performance. The endface shown in Figure 9.70 will offer insertion loss performance that exceeds the ANSI/TIA-568-C.3 requirements for multimode interconnections.

FIGURE 9.71 A single-mode endface that was polished with diamond polishing film and is absent of scratches or chips



Photo courtesy of MicroCare

Diamond

Diamond is an abrasive that is harder than both the optical fiber and the ceramic ferrule. Because of this, it will polish down the optical fiber and the ceramic ferrule. This abrasive does not prevent the person polishing the connector from altering the ferrule's endface geometry. However, it does produce an attractive finish and does not chip the optical fiber at the epoxy ring. This abrasive will produce an endface like the one shown in Figure 9.71.

Although the cosmetic finish of an endface polished with diamond may be very attractive, its geometry may also be altered. When polishing with a diamond abrasive, you have to be careful not to overpolish. As discussed earlier in this chapter, an interferometer can be used to evaluate the geometry of the connector endface. However, an interferometer is expensive and not always available.

You can use the inspection microscope to determine if you are polishing the ferrule endface and possibly changing its geometry. When you look at Figure 9.70, which shows a connector endface polished with aluminum oxide, you can see some texture markings on the ferrule. The same type of texture marking exists on the unpolished connector ferrule shown in Figure 9.72. This clearly shows that the aluminum oxide abrasive did not polish down the ferrule endface. However, when you look at the endface polished with a diamond abrasive shown in Figure 9.71, you see there are no texture markings on the ferrule because they have been removed with the diamond abrasive.

The risk associated with polishing the ferrule is that you may alter the shape of the ferrule endface. When you buy a connector, the endface meets the geometric specifications you

selected. Polishing the connector with a diamond abrasive may alter those specifications. It is easier to control the geometry with a polishing machine than with hand polishing.

FIGURE 9.72 An unpolished connector endface without an optical fiber

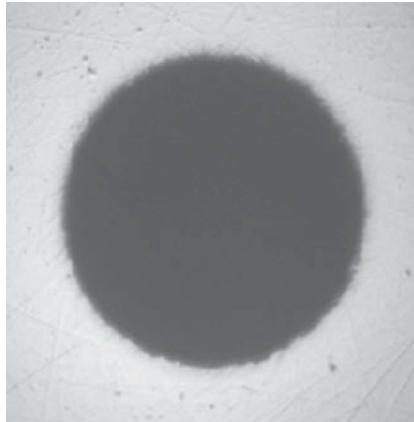


Photo courtesy of W.R. Systems

Hand Polishing

Some of the procedures involved may vary, but the main task is the same: you must insert a bare optical fiber into a hole in the ferrule so that it sticks out slightly beyond the ferrule endface and secure it there; then you must *score*, or *cleave*, the excess optical fiber and polish the ferrule endface and the optical fiber together to achieve the proper profile and finish.

Although this task sounds simple, it requires some careful planning and execution. Remember that the hole in the ferrule is almost microscopic, since it must be only slightly larger than the optical fiber itself.

There are many different fiber-optic connectors on the market from a variety of manufacturers. Each manufacturer has its own polishing process. This section discusses two basic connector assembly and polishing processes using oven-cured epoxy and anaerobic epoxy. The connectors will be terminating simplex cordage.

Assembling the Connector

Assemble your tools before you start so you don't have to look for them in the middle of the process. It may seem like a fussy detail, but your work will proceed more smoothly and efficiently if you have prepared all of your tools and materials and laid them out where you can reach them easily when you need them.

Cable and Optical Fiber Preparation

The following steps describe how to prepare simplex cordage for connectorization:

1. Cut the cable 2 " longer than you'll need.
2. Install the strain relief boot on the cable end, small diameter first. Important: The strain relief boot will not fit over the connector, so it must be placed on the cable first.
3. Install the crimp sleeve if one was provided with the connector.
4. Locate the appropriate strip chart from the connector manufacturer.
5. Mark the cable jacket as shown on the strip chart, cut the jacket with the stripper, and then twist the jacket and pull it away from the cable to remove it.
6. Pull the strength member back; it will be cut to length in the last step.
7. Strip away the tight-buffer and coating to the length shown on the strip chart in 1/4 " increments to reduce the chance of breaking the optical fiber.
8. Clean the fiber with a lint-free wipe soaked in optical fiber cleaning fluid; this will remove the coating residue so that the epoxy in the ferrule can bond with the cladding surface.
9. Using the shears, cut the strength member to the length shown on the strip chart, and then distribute it evenly around the tight buffer.

Oven-Cured Epoxy Application, Connector Assembly, and Polishing

The following procedures apply to oven-cured epoxies. Remember these are general procedures and you should always refer to the epoxy and connector manufacturer data sheets for detailed information.

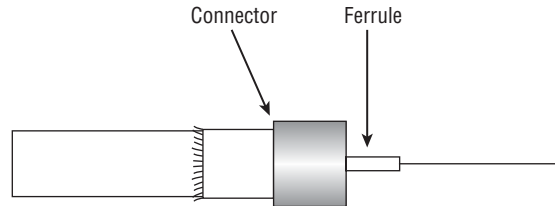
Oven-Cured Epoxy Application and Connector Assembly

This procedure describes how to prepare the epoxy, inject it into the connector, and assemble the connector for polishing:

1. Turn on the curing oven and set the temperature as recommended by the epoxy manufacturer.
2. Mix the epoxy according to the manufacturer's instructions.
3. Secure the needle to the syringe. Load the syringe with epoxy and ensure that the syringe is free of air pockets.
4. Insert the needle tip into the back of the connector until it bottoms out against the ferrule. Maintain pressure and slowly inject the epoxy until a bead appears on the end of the ferrule tip. Continue to inject epoxy until the bead covers about a third of the ferrule diameter end.
5. Release the pressure on the plunger, wait 5 seconds, and then remove the needle.
6. If required, place the cure adapter over the connector ferrule.

7. Gently feed the optical fiber through the connector, as shown in Figure 9.73, making sure the strength member folds back as you push the optical fiber forward.

FIGURE 9.73 Feed the fiber through the connector.



8. Crimp the connector using the crimper as described in the manufacturer's data sheet.
9. Place the connector in the curing oven and cure the epoxy according to the manufacturers cure schedule.
10. After the epoxy has cured, carefully remove the connector and allow it to cool. Then proceed to polishing the connector, described in the next section.

Oven-Cured Epoxy Connector Polishing

This procedure describes how to polish the endface of the connector after the epoxy has cured and the connector has cooled. This polishing process is only for PC finishes. The same process can be applied for a flat finish by substituting a glass plate for the rubber pad.

The following procedure is a general polishing procedure. You should always refer to the connector manufacturer data sheet for detailed information.

1. Score the fiber with the scribe where it exits the epoxy bead, and then gently pull the optical fiber, lifting it from the connector endface. A small optical fiber nub should be protruding from the connector endface. You can use the eye loupe to view the connector endface.
2. Polish down or de-burr the fiber end by holding the connector so that it is facing up. Arch a piece of 5 μ m polishing film over it. Lightly rub the film in a circle over the connector until you no longer feel the fiber end "grab" the film. Then stop.
3. Place a clean piece of 5 μ m aluminum oxide polishing film over the rubber pad.
4. Insert the connector into the puck and place the puck on the polishing film.
5. Slowly move the puck in a figure-8 pattern over the polishing film to begin removing the epoxy bead. After about five or ten figure-8 strokes, you should start to feel less resistance as the epoxy bead polishes down. Stop polishing before you have removed all of the epoxy.
6. Remove the 5 μ m polishing film and place a 1 μ m polishing film on the rubber pad. Slowly move the puck in a figure-8 pattern over the polishing film to begin removing what is left of the epoxy bead. After about five or ten figure-8 strokes, the polishing

puck should glide smoothly over the polishing film. Stop polishing and go to the next step for single-mode applications.

7. For single-mode applications, remove the $1\mu\text{m}$ polishing film and place a 0.3 or $0.1\mu\text{m}$ diamond polishing film on the rubber pad. Slowly move the puck in a figure-8 pattern over the polishing film and stop after ten figure-8 strokes. Polishing is complete.



An aluminum oxide abrasive works well for most non-laser-optimized multimode applications; however, a diamond abrasive is recommended for single-mode and laser-optimized multimode applications. Do not overpolish when using a diamond abrasive.

Anaerobic Epoxy Application, Connector Assembly, and Polishing

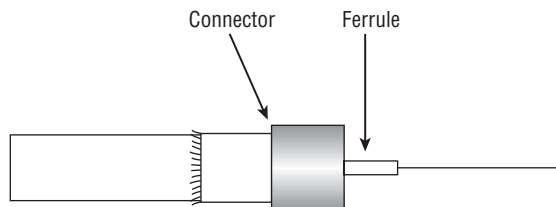
The following procedure applies to anaerobic epoxies. Remember this is a general procedure and you should always refer to the epoxy and connector manufacturer data sheets for detailed information.

Anaerobic Epoxy Application and Connector Assembly

This procedure describes how to inject the adhesive into the connector, apply the primer to the optical fiber, and assemble the connector:

1. Place a needle on top of the anaerobic adhesive and insert the needle tip into the back of the connector until it bottoms out against the ferrule. Maintain pressure and slowly inject the epoxy until a bead appears on the end of the ferrule tip. Continue to inject epoxy until the bead covers about a third of the ferrule diameter end.
2. Open the primer. Using the small brush on the cap of the primer, coat the bottom two-thirds of the exposed optical fiber with the primer. Ensure that you place the lid back on the primer immediately after use; the primer evaporates quickly and will melt the nonreflective mat if you spill it on the mat.
3. Gently feed the optical fiber through the connector, as shown in Figure 9.74, making sure the strength member folds back as you push the optical fiber forward. Wait 10 seconds for the epoxy to cure before going to the next step.

FIGURE 9.74 Feed the fiber through the connector.





Do not pull back on the optical fiber; doing so may break the optical fiber.

4. Crimp the connector as described in the manufacturer's data sheet, and then proceed to the next section.

Anaerobic Epoxy Connector Polishing

This procedure describes how to polish the endface of the connector after the anaerobic epoxy has cured. This polishing process is only for PC finishes. The same process can be applied for a flat finish by substituting a glass plate for the rubber pad.

The following procedure is a general polishing procedure. You should always refer to the connector manufacturer data sheet for detailed information.



The anaerobic epoxy bead is not hard like the oven-cured epoxy bead and does not offer any support to the optical fiber.

1. Score the optical fiber with the scribe where it exits the connector endface, and then gently pull the optical fiber, lifting it from the connector endface. A small optical fiber nub should be protruding from the connector endface. You can use the eye loupe to view the connector endface.
2. Polish down or de-burr the optical fiber end by holding the connector so that it is facing up. Arch a piece of 5 μ m polishing film over it. Lightly rub the film in a circle over the connector until you no longer feel the fiber end “grab” the film. Then stop.
3. Place a clean piece of 1 μ m polishing film on the rubber pad.



An aluminum oxide abrasive works well for most non-laser-optimized multimode applications; however, a diamond abrasive is recommended for single-mode and laser-optimized multimode applications. Do not overpolish when using a diamond abrasive.

4. Insert the ferrule into the puck and place the puck on the polishing film.
5. Slowly move the puck in a figure-8 pattern over the polishing film to begin polishing down the optical fiber. After about five or ten figure-8 strokes, you should start to feel less resistance and the puck should glide across the polishing film. Stop polishing and go to the next step for single-mode applications.
6. For single-mode applications, remove the 1 μ m polishing film and place a 0.3 or 0.1 μ m diamond polishing film on the rubber pad. Slowly move the puck in a figure-8 pattern over the polishing film and stop after ten figure-8 strokes. Polishing is complete.

Machine Polishing

Machine polishing is typically performed at the factory. Depending on the polishing machine, one or more connectors can be polished at a time. Machine polishing produces the best connector endface and produces more consistent results than hand polishing. Most single-mode connectors are polished with a machine.

The same steps required to assemble a connector for hand polishing are also required to assemble a connector for machine polishing. The polishing machine is substituted for the hand polishing. Connectors are inserted into the polishing machine after they have been cleaved and de-burred. When the polishing process is complete, remove the connectors and inspect them.

Pre-polished Connectors

Pre-polished connectors have been around for many years and do not require hand or machine polishing. A pre-polished connector is similar to a mechanical splice because a cleaved piece of optical fiber is inserted into the rear of the connector and mated with a small piece of optical fiber that was bonded to the connector ferrule at the factory. The gap between the two optical fibers is filled with index matching gel to reduce Fresnel reflections. The small piece of optical fiber bonded to the connector ferrule has a cleaved perpendicular finish on the side mating with the other optical fiber. The other end of the optical fiber is polished with a machine to produce an exacting endface.

Pre-polished connectors are available for single or multiple fiber applications. When using a pre-polished connector, you must ensure that the optical fiber in the connector is the same size and type of optical fiber you are terminating. You cannot terminate a multimode optical fiber that has a 50 μ m core with a pre-polished connector that has a 62.5 μ m core, and vice versa. You should not terminate a non-laser-optimized multimode optical fiber that has a 50 μ m core with a pre-polished connector that has a laser-optimized 50 μ m core, and vice versa.

Three popular brands of pre-polished connectors are the UniCam from Corning Cable Systems, the OptiCam from Panduit, and the Light Crimp Plus from TE Connectivity. They all feature a cam-type locking mechanism that holds the optical fiber firmly in place throughout the life of the connector. If a problem occurs during termination, the cam-type locking mechanism can be unlocked to remove the optical fiber and reattempt the termination.

Shown in Figure 9.75 is a UniCam 50 μ m laser-optimized multimode LC connector and in Figure 9.76 is a single-mode SC connector. Shown in Figure 9.77 is an OptiCam 62.5 μ m multimode SC connector and in Figure 9.78 is a single-mode LC connector. The Light Crimp Plus multimode SC connector kit is shown in Figure 9.79 and the single-mode LC is shown in Figure 9.80.

FIGURE 9.75 Corning Cable Systems' UniCam 50 μ m laser-optimized multimode LC connector



Photo courtesy of Corning Cable Systems

FIGURE 9.76 Corning Cable Systems' UniCam single-mode SC connector



Photo courtesy of Corning Cable Systems

FIGURE 9.77 Panduit OptiCam 62.5 μ m multimode SC connector

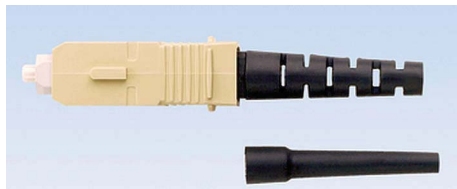


Photo courtesy of Panduit

Cleaning and Inspection

Good cleaning and inspection skills are essential for low-loss and low back reflection interconnections. Often the cause of a problem is contamination, which can be discovered through inspection and repaired by cleaning. Contamination can take many forms, and there is no spot on a fiber-optic connector that contamination cannot find.

Until recently, cleaning and inspection focused on the endface. However, that school of thought has changed and when working in fiber optics, you need to be aware that contamination anywhere on the ferrule or alignment sleeve can affect the performance of an interconnection. Also keep in mind that contamination can migrate, which means it can move from one location to another both during and after the cleaning process. It is very easy to pull contamination from the side of the ferrule onto the endface during cleaning.

There are many ways that contamination can migrate or move from one location on the ferrule to another. Some of the ways are more obvious than others. A typical ferrule is made of ceramic; both ceramic and optical fiber are dielectrics. A dielectric is another name for an insulator.

As a dielectric, both the ferrule and optical fiber are capable of holding an electrical charge. This electrical charge may be referred to as an electrostatic charge that is commonly known as static electricity. Because the ferrule and optical fiber can hold a charge, they are capable of attracting particles (contamination) with an opposite charge. It is for this reason that most fiber-optic cleaning fluids are engineered to dissipate those charges.

Endface Cleaning

After a connector is polished or before it is inserted into a receptacle or mating sleeve, the endface should be cleaned and inspected. Mating a clean connector with a dirty connector will result in two dirty connectors. Figure 9.84 shows what a formerly clean connector endface looks like after it has been mated with a connector that has skin oil on the endface. You can tell that this connector had been mated because the oil is distributed in a crater-like pattern.

FIGURE 9.84 Skin oil distributed in a crater-like pattern—the result of mating a clean and dirty connector

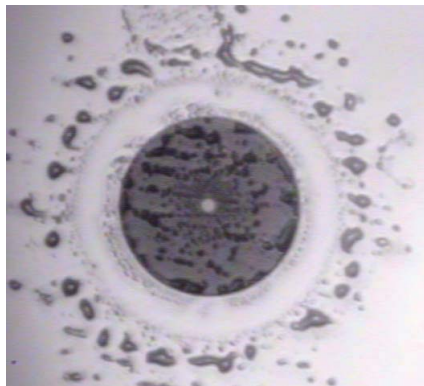


Photo courtesy of MicroCare

Cleaning the endface of a connector is a simple process; however, it does require cleaning products engineered for that purpose. You need only two basic products: lint-free wipes and a cleaning fluid engineered to clean the optical fiber endface.

In August 2012, SAE International published AIR6031, Fiber Optic Cleaning. That standard states that manual cleaning can be done using a wet-dry process or a dry wipe process. This section will describe dry and wet-dry cleaning techniques.

Dry Cleaning Techniques

When a cleaning fluid is not required to clean a connector endface, the connector endface can be cleaned with a dry, lint-free wipe. Multiple manufacturers offer lint-free wipes developed specifically for fiber-optic cleaning applications. They are typically packaged in different ways, as shown in Figure 9.85, for different applications or environments. These wipes have been engineered to lift oils, dirt, and contamination from the endface of the connector.

FIGURE 9.85 Lint-free wipes packaged in a round container, box, and sealed in a foil package



The foil packaged wipes shown in Figure 9.85 are sealed to prevent the wipe from being contaminated prior to use. This wipe is glued to the foil on one side; that way, you can handle the wipe without getting oil and dirt from your hand on the wipe. To clean the endface with this wipe, open the package and place the wipe in the palm of your hand, as shown in Figure 9.86, and move the connector in a straight line from top to bottom several times, each time starting in a different location. Be sure to lift the connector from the wipe before

it reaches the end of the wipe. If the connector comes in contact with your skin or hair, it will pick up the oil, as shown in Figure 9.87.

FIGURE 9.86 Cleaning the connector endface with a lint-free wipe that was sealed in a foil package

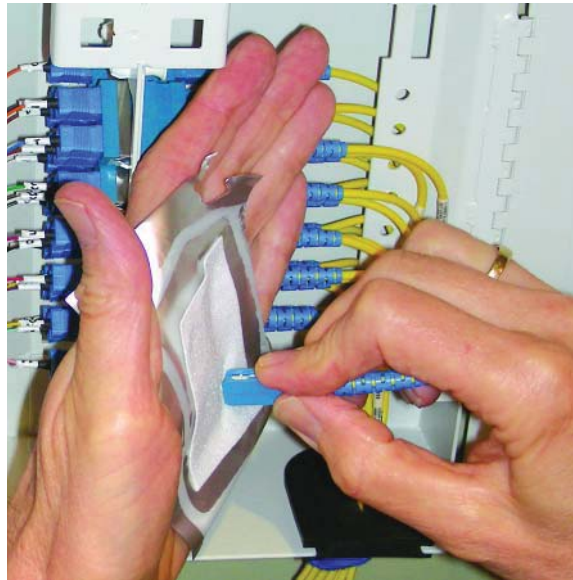


Photo courtesy of MicroCare

FIGURE 9.87 Oil on the endface of a connector

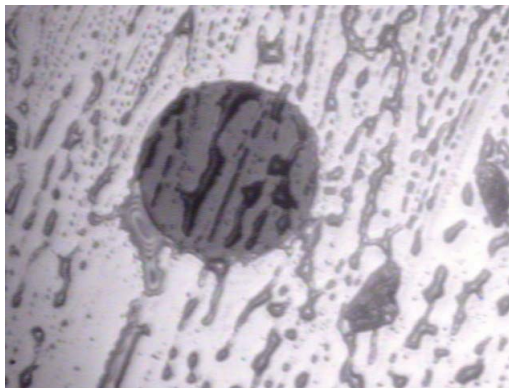


Photo courtesy of MicroCare

Lint-free wipes are also available in round containers, as shown in Figure 9.85. These are the same wipes found in the sealed foil package. When using these wipes, be careful to touch only one side of the wipe with your hand. Only use the side that did not touch your hand to clean the connector endface.

To clean the endface with this wipe, place the wipe in the palm of your hand, as shown in Figure 9.88, and move the connector in a straight line from top to bottom several times, each time starting in a different location. Be sure to lift the connector from the wipe before it reaches the end of the wipe. If the connector comes in contact with your skin or hair, it will pick up the oil, as you saw in Figure 9.87.

FIGURE 9.88 Lint-free wipe properly placed in the palm of a hand for cleaning a connector endface

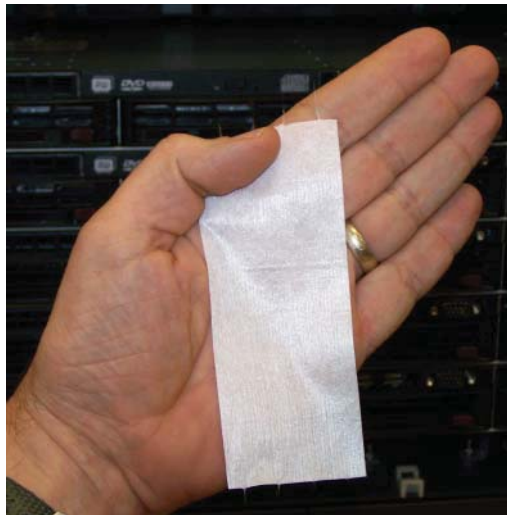


Photo courtesy of MicroCare

Lint-free wipes are also available in boxes, also shown in Figure 9.85. These are the same wipes found in the sealed foil package; however, depending on the manufacturer, they may be packaged so the wipe can be placed over a slotted stencil that prevents the endface from touching sections of the wipe that were contaminated during the cleaning process. When using these wipes, be careful to touch only one side of the wipe with your hand while positioning it over the stencil.

To clean the endface with this wipe, place the endface at the top of a slot, as shown in Figure 9.89, and move the connector in a straight line from top to bottom several times, using a different slot each time. Be sure to lift the connector from the wipe before it reaches

the end of the wipe. If the connector comes in contact with your skin, it will pick up the oil, as you saw in Figure 9.87.

FIGURE 9.89 Connector properly placed in the slot for cleaning the endface

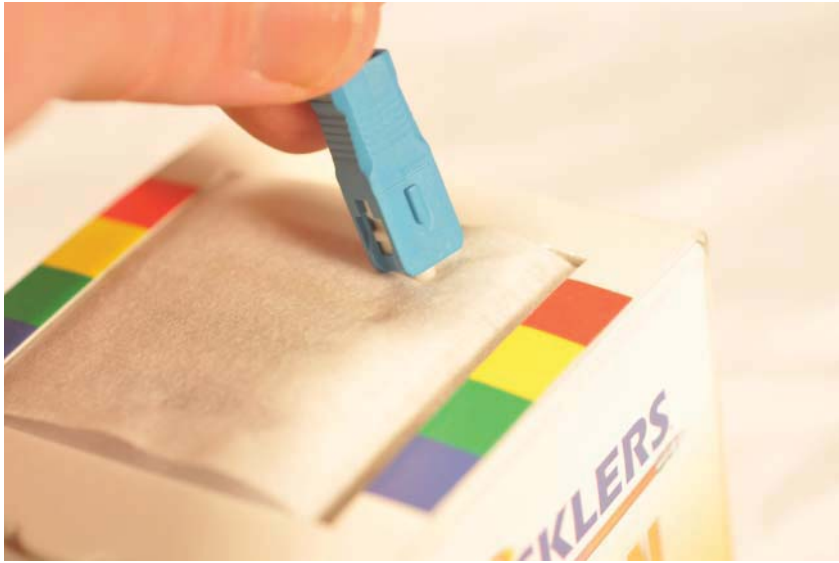


Photo courtesy of MicroCare

Wet-Dry Cleaning Techniques

If cleaning the endface of the connector with a dry, lint-free wipe does not remove the contamination, a wet-dry clean technique is required. As discussed earlier, several manufacturers offer lint-free wipes developed specifically for fiber-optic cleaning applications packaged in different ways for different applications or environments, as shown earlier in Figure 9.85. These wipes have been engineered to lift oils, dirt, and contamination from the endface of the connector.

The foil packaged wipes shown in Figure 9.85 are sealed to prevent the wipe from being contaminated prior to use. To clean the endface with this wipe, open the package and wet the wipe with an optical fiber cleaning fluid. Place the wipe in the palm of your hand, as shown in Figure 9.90, and move the connector in a straight line from top to bottom several times, each time starting in a different location. Be sure to lift the connector from the wipe before it reaches the end of the wipe. If the connector comes in contact with your skin or hair, it will pick up the oil, as you saw in Figure 9.87.

FIGURE 9.90 Cleaning the connector endface with a wet-dry lint-free wipe that was sealed in a foil package

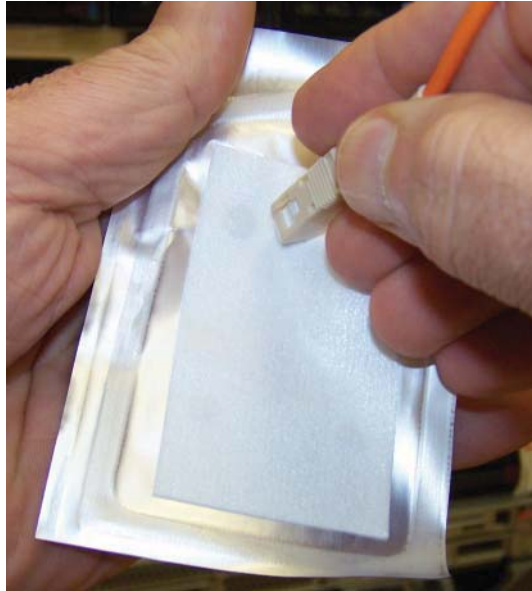


Photo courtesy of MicroCare

Lint-free wipes are also available in round containers, as shown in Figure 9.85. These are the same wipes found in the sealed foil package. When using these wipes, be careful to touch only one side of the wipe with your hand. Use only the side that did not touch your hand to clean the connector endface.

To clean the endface with this wipe, wet the wipe with an optical fiber cleaning fluid. To do this, fold the wipe over, place it in the cleaning fluid dispenser, and pump the dispenser, as shown in Figure 9.91. Place the wipe in the palm of your hand, as shown in Figure 9.92, and move the connector in a straight line from top to bottom several times, each time starting in a different location. Be sure to lift the connector from the wipe before it reaches the end of the wipe. If the connector comes in contact with your skin or hair, it will pick up the oil, as you saw in Figure 9.87.

Lint-free wipes are also available in boxes, as shown in Figure 9.85. These are the same wipes found in the sealed foil package; however, depending on the manufacturer they may be packaged without a stencil. When using these wipes, be careful to touch only one side of the wipe with your hand while positioning it over the stencil.

To clean the endface with this wet-dry wipe, moisten the wipe, as shown in Figure 9.93. Place the endface at the top, as shown in Figure 9.94, and move the connector in a straight line from top to bottom several times, using a different location each time. Be sure to lift the connector from the wipe before it reaches the end of the wipe. If the connector comes in contact with your skin, it will pick up the oil, as you saw in Figure 9.87.

FIGURE 9.91 Wetting the lint-free wipe with a cleaning fluid



Photo courtesy of MicroCare

FIGURE 9.92 Wet-dry lint-free wipe properly placed in the palm of a hand for cleaning a connector endface



Photo courtesy of MicroCare

FIGURE 9.93 Wetting the lint-free wipe with a cleaning fluid

Photo courtesy of ITW Chemtronics

FIGURE 9.94 Connector properly placed on the lint-free wipe for cleaning the endface

Photo courtesy of ITW Chemtronics

A connector that has been cleaned properly will be free of any dirt or contamination, as shown in Figure 9.95. Only after the connector has been cleaned and inspected should it be mated with another connector or receptacle.

FIGURE 9.95 Properly cleaned single-mode connector endface



Photo courtesy of MicroCare

Special kits developed just for cleaning fiber-optic components are available from several manufacturers, among them the ones shown in Figures 9.96 and 9.97.

FIGURE 9.96 Sticklers fiber-optic cleaning kit



Photo courtesy of MicroCare

FIGURE 9.97 ITW Chemtronics complete fiber-optic cleaning kit

MicroCare, ITW Chemtronics, Corning Cable Systems, and AFL Telecommunications manufacture some of the most commonly used cleaning materials.

MicroCare www.microcare.com

ITW Chemtronics www.chemtronics.com

Corning www.corningcablesystems.com

AFL Telecommunications www.afltele.com

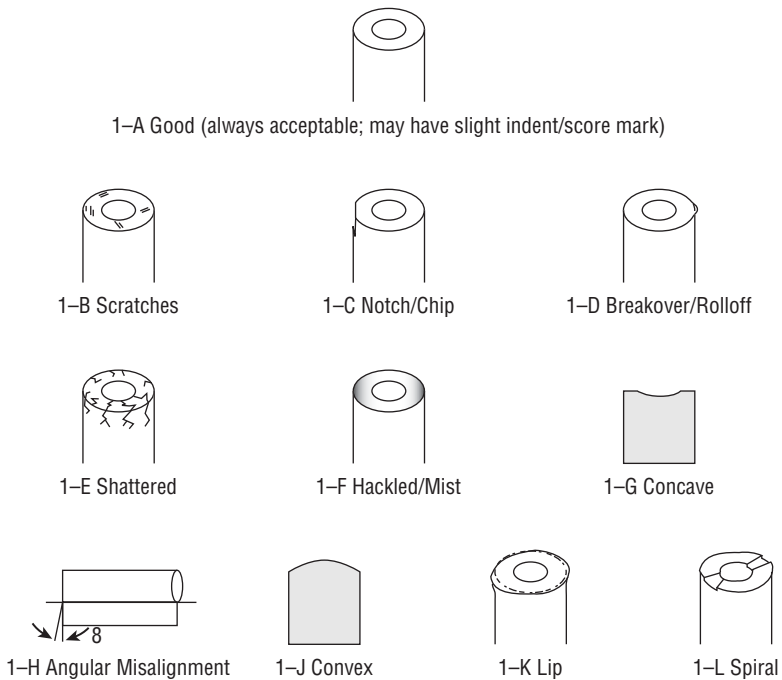
Endface Inspection

Various standards have been developed to assess the endface of polished fiber-optic connectors. This section provides an overview of two standards, ANSI/TIA-455-57-B and IEC 61300-3-35.

ANSI/TIA-455-57-B

ANSI/TIA-455-57-B was published in 1994 and reaffirmed in 2000, 2005, and 2013. When a standard is reaffirmed, it is not changed or updated. This standard provides a guideline for examination of an optical fiber endface. Figure 9.98 contains possible core and cladding fiber endface results. Each endface in Figure 9.98 is labeled with a number and a letter. The number and letter combinations are listed in Table 9.2. This table is used to identify always acceptable, usually acceptable, and often acceptable endface results.

FIGURE 9.98 ANSI/TIA-455-57-B core and cladding endface results



Typical fiber endface results with core and cladding

TABLE 9.2 ANSI/TIA-455-57-B core and cladding endface results

Appearance acceptability			
Figure	Always acceptable	Usually acceptable ^{1, 2}	Often acceptable ¹
9.98	1-A	1-B, 1-C, 1-D, 1-G, 1-H	1-E, 1-F, 1-J, 1-K, 1-L
1. Tighter or looser limits may be specified by the fiber-optic test procedure (FOTP).			
2. Unless defect extends into the core			

Looking at Table 9.2, you will see that one endface result—endface finish 1-A—is always acceptable. Note that 1-A is drawn to have no imperfections, or to be “cosmetically perfect.” However, a slight indent or score mark is acceptable.

IEC 61300-3-35

IEC 61300-3-35 is a standard that provides methods for quantitatively evaluating the endface quality of a polished fiber optic connector. This standard breaks the endface into four zones—core, cladding, adhesive or epoxy ring, and contact zone, as shown in Figure 9.99. The diameters of the zones are defined in Table 9.3.

FIGURE 9.99 Approximation of the four endface zones

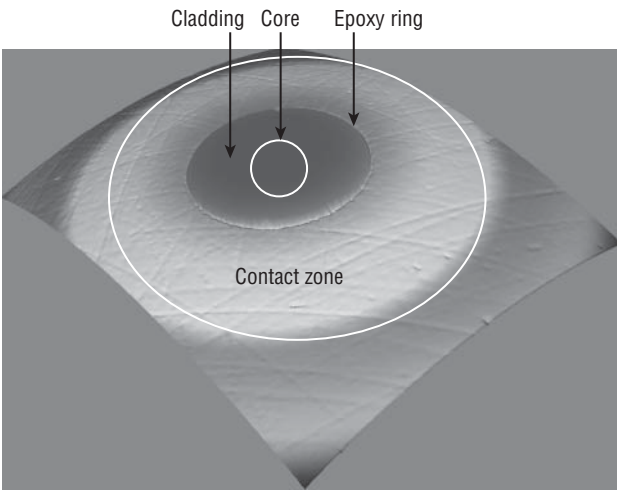


TABLE 9.3 IEC 61300-3-35 endface measurement regions

Zone	Single-mode diameter	Multimode diameter
Core	0 to 25µm	0 to 65µm
Cladding	25 to 120µm	65 to 120µm
Epoxy ring	120 to 130µm	120 to 130µm
Contact zone	130 to 250µm	130 to 250µm

This standard uses an inspection procedure flowchart and various pass/fail criteria for different optical fiber and endface types. Only a portion of these will be examined in this text. Imperfections are broken into two groups: scratches and defects. A scratch is a

permanent linear surface feature whereas a defect may be anything that is not a scratch. Regardless of the optical fiber type or endface, scratches and defects are permitted in the 10 μ m wide epoxy ring zone.

No defects are permitted in the core zone of APC single-mode or PC single-mode connectors with a return loss of 45dB. PC single-mode connectors with a return loss of 26dB are permitted to have two defects 3 μ m in size. PC single-mode connectors with a return loss of 45dB may not have any scratches in the core zone either.

As of this writing, this standard was under revision. It is unknown whether an additional zone will be included in the next revision. In this book, however, zone five, the side of the ferrule, will be discussed. In addition, this section will discuss the tools required to inspect the connector endface and examine common defects and their causes.

Inspection Microscopes

To evaluate the endface of a fiber-optic connector, you need a microscope designed for this application. This type of microscope is typically referred to as an *inspection microscope*. Many different inspection microscopes are available, and they can be grouped into two basic categories: optical microscopes and video microscopes. Video microscopes can be further broken into two groups: bench top and video probe.

An optical inspection microscope allows the direct viewing of the connector endface.



Care should be taken to ensure that there is no light source on the other end of the connector endface being viewed. As you have learned in this book, fiber-optic light sources are infrared and are not visible to the naked eye. Viewing a connector endface that is radiating infrared light with an optical microscope can temporarily or permanently damage your eye.

Optical inspection microscopes are typically handheld and are available with magnification levels from 100X to 400X. Three different light sources are also available to illuminate the endface: incandescent lamp, LED, and laser. The 400X handheld inspection microscope shown in Figure 9.100 uses a laser to illuminate the connector endface, and the 400X model shown in Figure 9.101 uses an LED.

FIGURE 9.100 400X microscope with a laser for illumination



Photo courtesy of KITCO Fiber Optics

FIGURE 9.101 400X microscope with an LED for illumination and an LC connector inserted for inspection



The key advantage to the LED light source is long battery life, and the LED should never need replacing. The laser light source requires more power than the LED; therefore battery life is reduced. However, the laser does make it easier to identify light scratches and defects in the optical fiber endface, and it too should never need replacing. Incandescent lamps are typically found in lower-cost 100X microscopes like the one shown in Figure 9.102.

FIGURE 9.102 100X microscope with an incandescent lamp for illumination



Video inspection microscopes allow you to view the connector endface indirectly. This eliminates any possibility of your eye being damaged from infrared light while inspecting the endface with the microscope. Figure 9.103 is a photograph of a 200X video probe microscope.

Typically, a multimode connector can be evaluated with a 100X microscope, whereas a single-mode connector requires a minimum of 200X magnification. A 400X microscope, of course, works for both multimode and single-mode.

FIGURE 9.103 200X video probe microscope

Photo courtesy of KITCO Fiber Optics

Higher magnification microscopes allow a close inspection of the core, cladding, epoxy ring, and contact zones. However, they will not show as much of the endface as a lower power microscope and may miss contamination.

Contaminated endfaces at different magnifications are shown in Figures 9.104 and 9.105. Less endface is visible in the higher magnification image shown in Figure 9.104 when compared to the lower magnification image shown in Figure 9.105.

FIGURE 9.104 Alcohol residue contaminated endface with higher power magnification

Photo courtesy of MicroCare

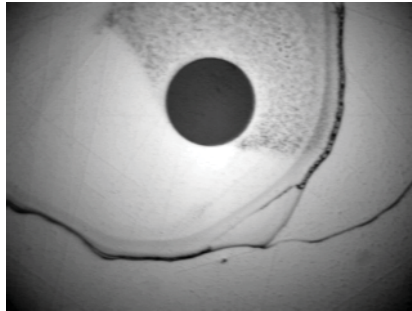
FIGURE 9.105 Alcohol residue contaminated endface with lower power magnification

Photo courtesy of MicroCare

As mentioned earlier in this chapter, zone five includes the side of the ferrule and contamination on the side of the ferrule can affect the performance of the interconnection. It is not uncommon to find cured epoxy on the side of the ferrule. Cured epoxy will not wipe off, and because it is on the side of the ferrule, it will not be visible with an inspection microscope.

Examining the side of the ferrule requires little or no magnification. It is a good practice to perform this as part of your inspection routine. An eye loupe like the one shown in Figure 9.106 provides moderate magnification and is helpful in locating contamination on the ferrule.

FIGURE 9.106 An eye loupe provides moderate magnification for viewing the ferrule.

Photo courtesy of KITCO Fiber Optics

Examples of Endface Inspection

This section will examine different endface finishes. Some finishes will be a hand polish with an aluminum oxide abrasive, and some will be a machine polish with a diamond abrasive. Each endface will be evaluated with a video microscope. As mentioned earlier in the chapter, this type of evaluation is a 2D evaluation that cannot quantitatively define radius of curvature, apex offset, fiber undercut, fiber protrusion, the depth of scratches, or the height of debris as can be done with an interferometer.

Good Quality, Clean Endfaces

The multimode endface shown in Figure 9.107 was hand polished with a $5\mu\text{m}$ aluminum oxide abrasive to remove the epoxy, followed with a $1\mu\text{m}$ aluminum oxide. The light scratches on the core and the cladding are typical of a $1\mu\text{m}$ aluminum oxide abrasive. The chipping and pitting around the epoxy ring is also typical for this abrasive. This endface is an example of a good hand polish with an aluminum oxide abrasive.

FIGURE 9.107 Good hand polish with an aluminum oxide abrasive viewed at 400X

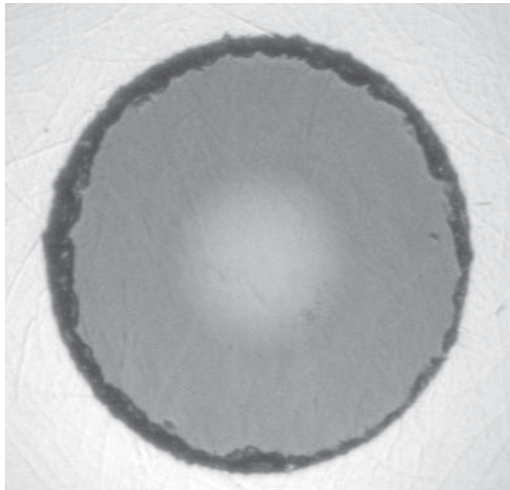


Photo courtesy of W.R. Systems

The single-mode endface shown in Figure 9.108 was machine polished. The final polishing step used a $0.1\mu\text{m}$ diamond abrasive. This endface is cosmetically perfect; there are no visible scratches on the optical fiber and no chipping or pitting near the epoxy ring. This endface is an example of a good machine polish with a diamond abrasive.

FIGURE 9.108 Good machine polish with a diamond abrasive viewed at 400X



Photo courtesy of MicroCare

Good Quality, Dirty Endfaces

The multimode endface shown in Figure 9.109 was hand polished with a 5 μ m aluminum oxide abrasive to remove the epoxy, followed by a 1 μ m aluminum oxide. The light scratches on the core and the cladding are typical of a 1 μ m aluminum oxide abrasive. Note that the endface is covered with skin oil and needs to be cleaned.

FIGURE 9.109 Skin oil covering a good hand polish with an aluminum oxide abrasive viewed at 400X

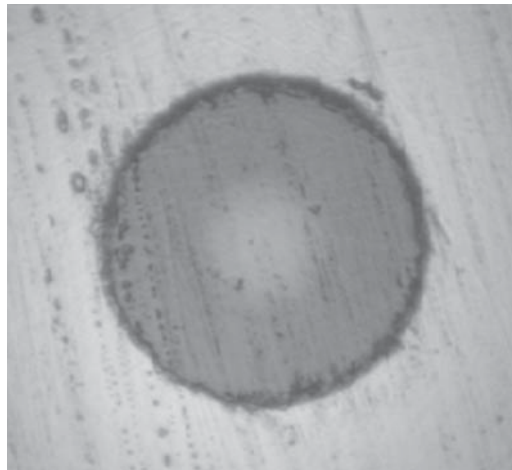


Photo courtesy of W.R. Systems

The single-mode endface shown in Figure 9.110 was machine polished. The final polishing step used a 0.1 μ m diamond abrasive. This endface is cosmetically perfect; however, it is covered with hair oil. You can typically tell the difference between hair oil and skin oil by the size of the oil deposits. Skin oil as shown in Figure 9.109 typically leaves smaller deposits than hair oil, and the oil tends to look it was applied in streaks. This connector needs to be cleaned.

FIGURE 9.110 Hair oil covering a good single-mode machine polish with a diamond abrasive viewed at 400X

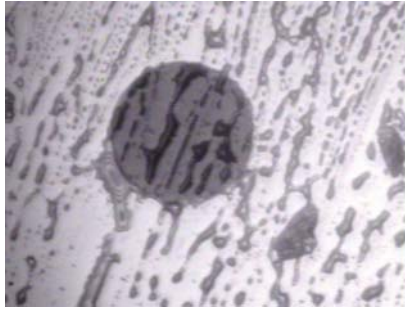


Photo courtesy of MicroCare

The single-mode endface shown in Figure 9.111 was machine polished. The final polishing step used a $0.1\mu\text{m}$ diamond abrasive. This endface is cosmetically perfect; however, it was cleaned with a shirttail instead of a lint-free wipe. Debris from the shirttail was deposited on the endface of the connector. This connector needs to be cleaned with a lint-free wipe.

FIGURE 9.111 Debris from a shirttail covering a good single-mode machine polish with a diamond abrasive viewed at 200X



Photo courtesy of MicroCare

Broken or Damaged Endfaces

It is impossible to tell how a connector was broken or damaged if it is inspected only after polishing is complete. If damage happens more than once, stop after every step and inspect the endface to determine which step is causing the problem.

The multimode endface shown in Figure 9.112 was hand polished with a $5\mu\text{m}$ aluminum oxide abrasive to remove the epoxy, followed by a $1\mu\text{m}$ aluminum oxide. Somewhere in the polishing process, the optical fiber was broken and a majority of the break extended below the surface of the endface.

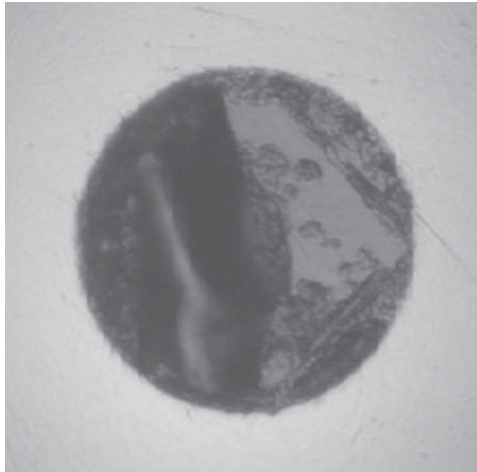
FIGURE 9.112 Broken multimode endface viewed at 400X

Photo courtesy of W.R. Systems

The multimode endface shown in Figure 9.113 was hand polished with a 5 μ m aluminum oxide abrasive to remove the epoxy, followed by a 1 μ m aluminum oxide. Somewhere in the polishing process, the optical fiber was broken; however, only part of the break extended below the surface of the endface.

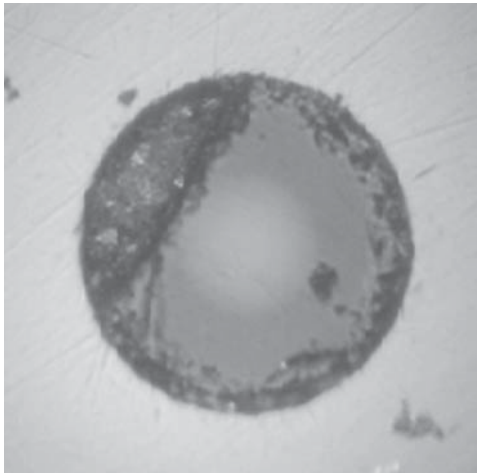
FIGURE 9.113 Damaged multimode endface viewed at 400X

Photo courtesy of W.R. Systems

Connector Performance

If you have assembled, polished, cleaned, and inspected the connector properly, there is a good chance that your connector will provide a low-loss interconnection. Appendix A of ANSI/TIA-568-C.3 defines fiber-optic connector performance. The maximum insertion loss allowed by ANSI/TIA-568-C.3 for a multimode or single-mode mated connector pair is 0.75dB.

Return loss as defined in ANSI/TIA-568-C.3 is the ratio of the power of the outgoing optical signal to the power of the reflected signal. This ratio is expressed in dB. The minimum return loss for a multimode interconnection is 20dB. This means that light energy from the light source traveling through the interconnection is 20dB greater than the light energy being reflected back by the interconnection toward the source.

The minimum return loss for a single-mode interconnection is 26dB unless that interconnection is part of a broadband analog video (CATV) application. Interconnections used in CATV applications must have a minimum return loss of 55dB. Optical return loss testing is discussed in Chapter 14.

ITU-T G.671 defines the performance of single and multiple-fiber connectors used in long-haul networks and access networks. While the standard does not mention an optical fiber type, it is assumed to be single-mode because multimode optical fiber is not used for long-haul or access networks. The maximum insertion loss allowed by ITU-T G.671 for a single-fiber mated connector pair is 0.5dB and 1.0dB for a multiple-fiber mated connector pair.

Reflectance as defined in ITU-T G.671 is the ratio of reflected power to incident power expressed in dB. In a mated connector pair, reflected power is the light that travels through the core of the optical fiber back toward the source and incident power is the light from the source. The maximum reflectance permitted for single-fiber or multiple-fiber mated connector pairs is -35dB.

Connector Color Code

Multimode and single-mode connectors and adapters can be identified using the color code in Section 5.2.3 of ANSI/TIA-568-C.3. The color code in Table 9.4 should be used unless color-coding is used for another purpose. This table lists the strain relief and adapter housing color for different optical fiber types.

TABLE 9.4 Multimode and single-mode connector and adapter identification

Optical fiber type or connector type	Strain relief and adapter housing color
850nm laser-optimized 50/125μm optical fiber	Aqua
50/125μm optical fiber	Black
62.5/125μm optical fiber	Beige

TABLE 9.4 Multimode and single-mode connector and adapter identification *(continued)*

Optical fiber type or connector type	Strain relief and adapter housing color
Single-mode optical fiber	Blue
Angled contact ferrule single-mode connectors	Green

Summary

This chapter covered the use of connectors as a means of terminating optical fibers. It described the typical construction of connectors and the materials used in connector ferrules.

This chapter discussed factors that affect connector performance and the methods used to improve connector performance through endface geometry. It described different types of connectors and their typical uses, along with the use of pigtails and multifiber connectors.

This chapter described methods for preparing and assembling connectors using different types of epoxies and polishing methods. It also described how to perform a quantitative endface evaluation using an inspection microscope. Finally, it described the performance requirements that connectors must meet to conform to TIA and ITU standards and the color codes that describe optical fiber or connector types.

Exam Essentials

Be familiar with fiber-optic connectors. Make sure that you know the components of a fiber-optic connector and understand each component’s function. Be able to describe connector construction and materials. Be able to describe the intrinsic and extrinsic factors affecting connector performance.

Understand the differences between connectors. Be able to describe the four endface geometries and their return loss or reflectance performance. Be able to describe different types of connectors and their uses. Make sure that you understand the use of pigtails as a means of terminating a fiber-optic cable.

Be familiar with how an interferometer is used to evaluate endface geometry. Be able to describe the three critical parameters used to evaluate endface geometry.

Be familiar with connector termination. Make sure that you can describe the steps in preparing a fiber for termination. Be able to describe the preparation and assembly of a connector. Be able to describe the proper polishing techniques for each type of epoxy.

Be familiar with cleaning and inspecting connectors. Make sure that you understand how to clean, inspect, and evaluate a connector. Know the two industry standards typically used for endface inspection and evaluation.

Know connector performance values. Make sure that you are familiar with the two standards typically referenced for connector performance. Be able to cite the insertion loss, return loss, and reflectance values for multimode and single-mode mated connector pairs.

Know the connector color code. Make sure that you are familiar with the ANSI/TIA-568-C.3 color code used to identify multimode and single-mode connectors and adapters.

Review Questions

1. Which part of the connector holds the fiber in place?

- A. The ferrule
- B. The cap
- C. The boot
- D. The body

Hint: The fiber must be exactly centered in its endface for the best possible connection.

2. The best material for a ferrule is _____.

- A. Metal
- B. Plastic
- C. Ceramic
- D. Glass

Hint: Zirconium oxide

3. Intrinsic factors in connector performance are determined by

- A. The geometry of the ferrule
- B. The production of the optical fiber, ferrule, and mating sleeve
- C. The relationship of the fiber to the buffer
- D. The type of connector being used

Hint: They are manufactured within specified tolerances

4. Which endface geometry ensures a physical contact (PC) between optical fiber ends?

- A. Flat
- B. Lensed
- C. Curved
- D. None

Hint: It ensures that the highest feature or apex on the endface is the center of the optical fiber end.

5. What is the ferrule diameter of the SC connector?

- A. 1.25mm
- B. 2.5mm
- C. 5mm
- D. 6.4mm × 2.5mm

Hint: It is the same as the ST connector.

6. The LC, which has a ferrule diameter half the size of the SC ferrule, is what kind of connector?
- A. Small form factor
 - B. Multiple-fiber
 - C. Screw-on
 - D. Twist-lock

Hint: It is sometimes referred to as a mini-SC.

7. A connector populated with twelve fibers in a single MT ferrule is called a _____.
- A. Multimode connector
 - B. Small form factor connector
 - C. Multifiber push-on connector
 - D. MT-RJ

Hint: It is designed to hold from 4 to 72 fibers.

8. A length of fiber with one factory-installed connector is called a _____.
- A. Quick-connect
 - B. Rat tail
 - C. Pigtail
 - D. Splice-on

Hint: Hardware is required to protect the splice.

9. The best type of epoxy for use in connectors is _____.
- A. Anaerobic
 - B. UV
 - C. One-part
 - D. Oven-cured

Hint: It is useful in making a large number of terminations at one location.

10. A drawback of using oven-cured epoxy is
- A. It is useful in making a large number of terminations.
 - B. It has a limited pot life.
 - C. The hard bead reduces the risk that the fiber will break inside the ferrule during cleaving.
 - D. It requires a special connector.

Hint: Any epoxy that is unused must be discarded.

11. When working with oven-cured epoxy, a _____ is typically placed over the ferrule of the connector.
- A. Cure adapter
 - B. Crimper
 - C. Scribe
 - D. No-Nik

Hint: It helps transfer heat from the curing oven.

12. Anaerobic epoxy has as its main advantage
- A. A quick cure time.
 - B. A hard bead.
 - C. A long pot life.
 - D. It is approved for air travel.

Hint: One part is applied to the fiber, the other to the ferrule.

13. What is a scribe used to do?
- A. Mark the buffer for stripping.
 - B. Mark the ferrule for polishing.
 - C. Score the fiber for removing the end.
 - D. Mark the fiber to measure the proper length.

Hint: It breaks the surface of the cladding.

14. The _____ is a device that uses light to take precision measurements of the fiber-optic connector endface.
- A. Interferometer
 - B. Video inspection probe
 - C. Optical inspection microscope
 - D. Video inspection microscope

Hint: It can generate a three-dimensional (3D) map of the surface.

15. The puck is used to
- A. Keep a firm grip on the connector for polishing.
 - B. Hold down the abrasive material during polishing.
 - C. Keep the abrasive surface clean.
 - D. Keep the ferrule perpendicular to the polishing surface.

Hint: It is also known as a polishing fixture.

16. Of the items below, which is the first step you need to perform prior to installing a fiber-optic connector?
- A. Remove the buffer.
 - B. Install the crimp sleeve.
 - C. Install the strain relief on the cable end.
 - D. Clean the fiber.

Hint: It will not fit over the connector.

17. The radius of curvature describes the _____ of the connector endface.
- A. Apex offset
 - B. Fiber undercut
 - C. Fiber protrusion
 - D. Roundness

Hint: The minimum and maximum values for the radius of curvature are defined in Telcordia GR-326.

18. The IEC 61300-3-35 standard breaks the endface into _____ zones.
- A. Three
 - B. Four
 - C. Five
 - D. Six

Hint: It provides methods for quantitatively evaluating the endface quality.

19. Per ANSI/TIA-568-C.3 the strain relief for a single-mode connector should be which color?
- A. Aqua
 - B. Black
 - C. Blue
 - D. Beige

Hint: An APC connector strain relief is green.

20. According to ANSI/TIA-568-C.3, what is the maximum insertion loss allowed for a mated connector pair?
- A. 0.1dB
 - B. 0.5dB
 - C. 0.75dB
 - D. 1.0dB

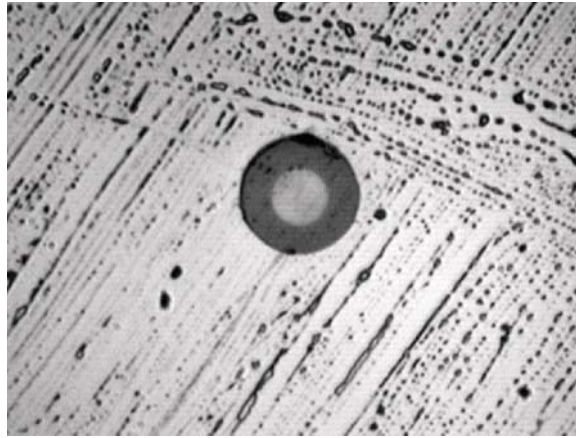
Hint: It is the same for multimode or single-mode mated connector pairs.

Chapter Exercises

Evaluate Connector Endface Polishing

To evaluate the endface of a fiber-optic connector, you need a microscope designed for this application.

You are inspecting the multimode endface shown here with a 200X inspection microscope. Is there anything wrong with this connector endface?



This connector has oil from a fingertip on the endface; however, the endface has a good polish. After cleaning and re-inspection, this connector will be ready to be put to use.

Evaluate Connector Endface Cleaning

A connector that has been cleaned properly will be free of any dirt or contamination.

You are inspecting the connector endfaces shown in these two graphics. Which of these connectors has oil on the endface?

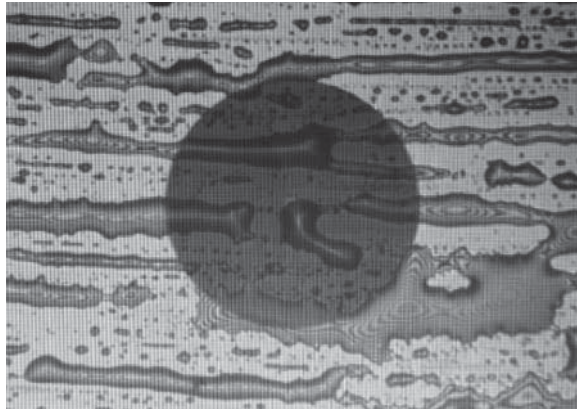


Photo courtesy of MicroCare

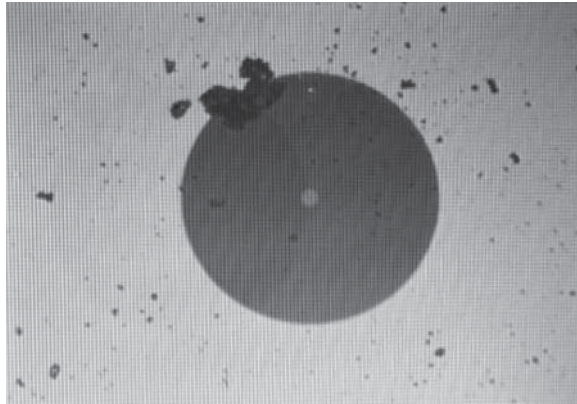


Photo courtesy of MicroCare

The connector endface shown in the first graphic has oil on it, and the connector endface shown in the second graphic has debris from a cotton shirt on it. Both of these connectors need to be cleaned and re-inspected.

Evaluate Connector Endface Geometry

The key measurements provided by the interferometer are radius of curvature, apex offset, and fiber undercut or protrusion. These critical parameters are required by Telcordia GR-326 to evaluate connector endface geometry.

You are evaluating a machine polished endface with the interferometer. The radius of curvature measured by the interferometer is 25mm. Does this endface radius of curvature fall within the range specified by Telcordia GR-326?

The minimum and maximum radius of curvature values as defined in Telcordia GR-326 are 7mm and 22mm. Values below or above this range increase the risk of optical fiber damage; 25mm exceeds the maximum acceptable value.

Identify an Optical Fiber Type from the Color of the Connector Strain Relief

Multimode and single-mode connectors and adapters can be identified using the color code in ANSI/TIA-568-C.3.

You need to mate two 850nm laser-optimized 50/125 μ m fiber-optic cables together with a jumper at a patch panel. What color strain reliefs would the correct jumper have?

850nm laser-optimized 50/125 μ m optical fiber should have an aqua-colored strain relief.

Chapter 10

Fiber-Optic Light Sources and Transmitters

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe the basic operation and types of LED light sources used in fiber-optic communications.
- ✓ Describe the basic operation and types of laser light sources used in fiber-optic communications.
- ✓ Describe LED performance characteristics.
- ✓ Describe laser performance characteristics.
- ✓ Describe the performance characteristics of an LED transmitter.
- ✓ Describe the performance characteristics of a laser transmitter.
- ✓ Explain the difference between a serial and parallel laser transmitter.
- ✓ Describe the laser types and wavelengths associated with serial and parallel laser transmitters.
- ✓ Describe the optical fiber types associated with VCSEL serial and parallel laser transmitters.
- ✓ Explain the safety classifications of the light sources used in fiber-optic communication.





As discussed in Chapter 1, “History of Fiber Optics and Broadband Access,” the idea of transmitting information with light is not new—only the technology that makes it easily possible. Like optical fiber, light source technology has improved rapidly over the decades. These technological advances have greatly increased data transmission rates and reduced costs. Fiber-optic transmitters are available to support every standardized network with a variety of connector choices.

This chapter discusses current fiber-optic light source and transmitter technology as it applies to common telecommunication network standards, industrial control systems, and general-purpose systems. The performance standards that we discuss in this chapter do not represent the highest levels achievable in the lab or on the test bench. They represent commonly available parts for standardized networks. All performance values were obtained from manufacturers’ data sheets or networking standards.

In this chapter, you will learn to

- Determine the minimum optical output power for an LED transmitter
- Determine the maximum optical output power for an LED transmitter
- Determine the minimum optical output power for a laser transmitter
- Determine the maximum optical output power for a laser transmitter

Semiconductor Light Sources

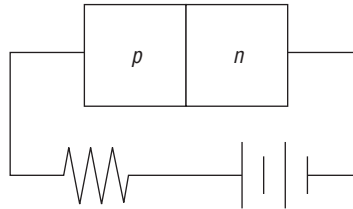
The light sources used in fiber-optic communication systems are far different from the light sources used to illuminate your home or office. Fiber-optic light sources must be able to turn on and off millions to billions of times per second while projecting a near-microscopic beam of light into an optical fiber. On top of this performance, they must be reasonably priced, highly reliable, easy to use, and available in a small package.

Semiconductor light source technology has made all this possible. Today’s fiber-optic communication systems use *light-emitting diodes (LEDs)* and *laser diodes* (from this point forward, the laser diode will be referred to as the *laser*) exclusively. These semiconductor light sources are packaged to support virtually every fiber-optic communication system imaginable.

LED Sources

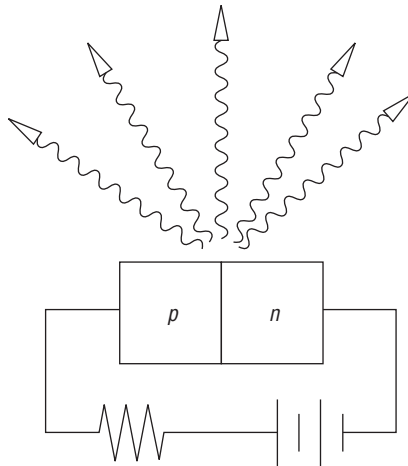
A basic LED light source is a semiconductor diode with a p region and an n region. When the LED is *forward biased* (a positive voltage is applied to the p region and a negative voltage to the n region), current flows through the LED. As current flows through the LED, the junction where the p and n regions meet emits random photons. This process is referred to as *spontaneous emission*. Figure 10.1 shows a forward-biased LED in a basic electric circuit.

FIGURE 10.1 Forward-biased LED



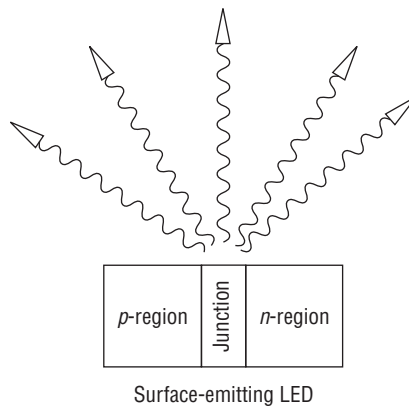
Photons emitted from the junction where the p and n regions meet are not in phase, nor are they launched in the same direction. These out-of-phase photons are called *incoherent light*. This incoherent light cannot be focused so that each photon traverses down the optical fiber. Because of this, only a percentage of the photons emitted will be coupled into the optical fiber. Figure 10.2 shows the out-of-phase photons being spontaneously emitted from the LED.

FIGURE 10.2 Radiating forward-biased LED



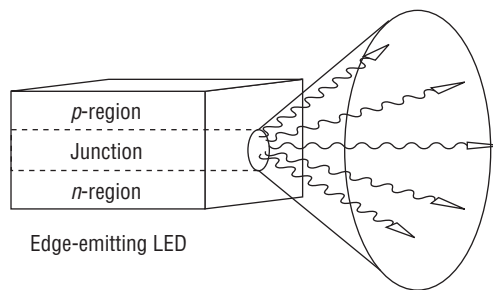
Two types of LEDs, the *surface-emitting LED* and the *edge-emitting LED*, are commonly used in fiber-optic communication systems. Surface-emitting LEDs are a *homojunction structure*, which means that a single semiconductor material is used to form the *pn* junction. Incoherent photons radiate from all points along the *pn* junction, as shown in Figure 10.3.

FIGURE 10.3 Radiating surface-emitting LED



Edge-emitting LEDs are a *heterojunction structure*, which means that the *pn* junction is formed from similar materials with different refractive indexes. The different refractive indexes are used to guide the light and create a directional output. Light is emitted through an etched opening in the edge of the LED, as shown in Figure 10.4.

FIGURE 10.4 Radiating edge-emitting LED

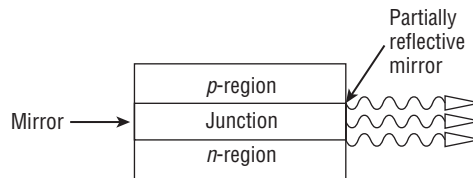


Laser Sources

The term *laser* is actually an acronym that stands for *light amplification by stimulated emission of radiation*. Like the LED, the laser is a semiconductor diode with a *p* region and an *n* region. Unlike the LED, the laser has an optical cavity that contains the emitted

photons with reflecting mirrors on each end of the diode. One of the reflecting mirrors is only partially reflective. This mirror allows some of the photons to escape the optical cavity, as shown in Figure 10.5.

FIGURE 10.5 Radiating laser



Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This process of generating light energy is called *stimulated emission*. The photons radiated from the laser have a fixed relationship that is referred to as *coherent light* or coherent radiation. Figure 10.5 shows the in-phase light waves emitted from the laser.

There are three families of lasers used in fiber-optic communication systems: *Fabry-Pérot*, *distributed feedback (DFB)*, and the *vertical-cavity surface-emitting laser (VCSEL)*. Each laser family has unique performance characteristics that will be discussed in this chapter and is designed to support a specific telecommunication application.

Light Source Performance Characteristics

This section will compare the performance characteristics of the LED and laser light sources. The performance of a light source can be judged in several areas: output pattern, wavelength, spectral width, output power, and modulation speed. These performance areas determine the type of optical fiber that the source can be coupled to, transmission distance, and data rate. Without a doubt, the laser is the hands-down winner when it comes to ultra-high-speed long-distance data transmission. However, many applications require only a fraction of the performance that a laser offers; for these applications, an LED is used.

Output Pattern

The LED and laser semiconductors used in fiber-optic light sources are packaged to couple as much light as possible into the core of the optical fiber. The output pattern or NA of the light source directly relates to the energy coupled into the core of the optical fiber. The LED has a wide output pattern compared to a laser and does not couple all its light energy into the core of a multimode optical fiber. The output pattern of a laser light source is very

narrow, allowing a majority of the light energy to be coupled into the core of a single-mode or multimode optical fiber.

LED Output Pattern

Unpackaged surface-emitting and edge-emitting LEDs have wide output patterns, as shown earlier in Figures 10.3 and 10.4. An LED light source must be assembled in a package to mate with a specific connector type and optical fiber. However, occasionally there may be a requirement for an LED to be packaged as a *pigtail*. A pigtail is a short length of tight-buffered optical fiber permanently bonded to the light source package, as shown in Figure 10.6.

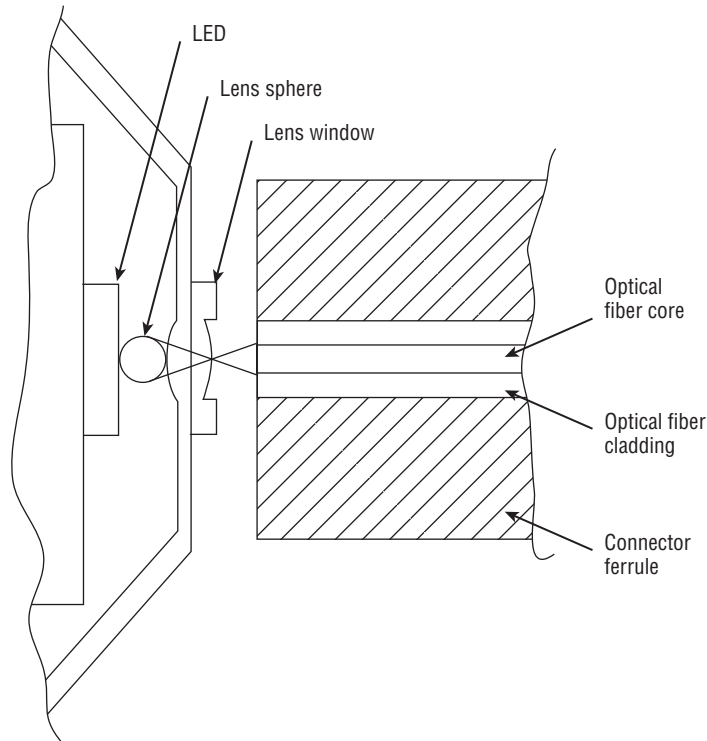
FIGURE 10.6 Pigtailed light source



To couple as much light as possible into the core of the optical fiber, the manufacturer typically mounts a micro lens in the shape of a sphere directly on top of the LED. The manufacturer may also use an additional lens to further control the output pattern. Figure 10.7 shows a surface-emitting LED with a series of lenses directing light into the optical fiber.

Even the best lens can't direct all the light energy into the core of the optical fiber. The typical LED light source overfills the optical fiber, allowing light energy to enter the cladding and the core at angles exceeding the NA. These high-order or marginal modes are attenuated over a short distance. Light energy that enters the core exceeding the acceptance angle will not propagate to the end of the optical fiber. This light energy will refract into the cladding. Figure 10.8 shows a typical LED source overfilling an optical fiber, which is illustrated by the light spot that the source projects.

FIGURE 10.7 Packaged surface-emitting LED



The process of an LED transmitter overfilling a multimode optical fiber is referred to as an *overfilled launch (OFL)*. An OFL is used to define the minimum modal bandwidth-length product of a multimode optical fiber. Multimode OFL optical fiber bandwidth requirements are defined in ANSI/TIA-568-C.3 and ISO/IEC 11801, as shown in Table 10.1.

FIGURE 10.8 Laser and LED spot sizes

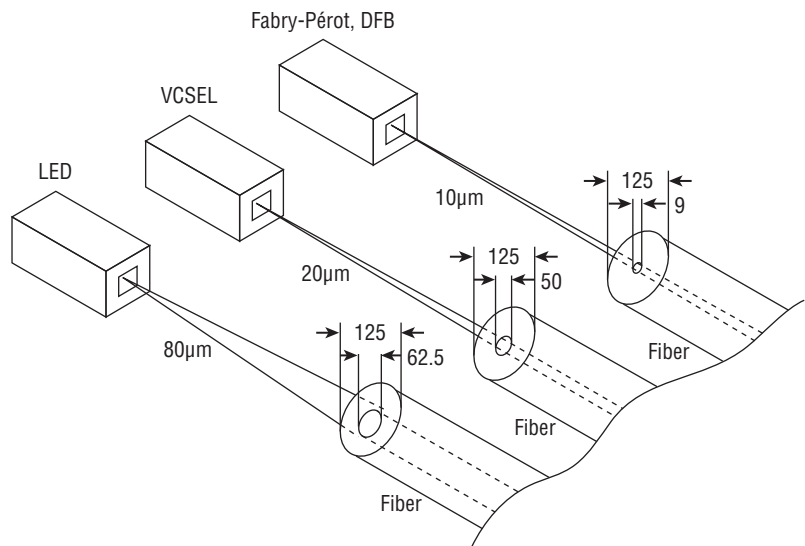


TABLE 10.1 Characteristics of ANSI/TIA-568-C.3 and ISO/IEC 11801-recognized optical fibers

Optical fiber and cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
62.5/125µm Multimode OM1	850	3.5	200	Not required
Type A1b	1300	1.5	500	Not required
TIA 492AAAA				
50/125µm Multimode OM2	850	3.5	500	Not required
TIA 492AAAB	1300	1.5	500	Not required
Type A1a.1				
850nm laser-optimized 50/125µm Multimode OM3	850	3.5	1500	2000
Type A1a.2	1300	1.5	500	Not required
TIA 492AAAC				
850nm laser-optimized 50/125µm Multimode OM4	850	3.5	3500	4700
Type A1a.3	1300	1.5	500	Not required
TIA 492AAAD				

Laser Output Pattern

Unlike the LED, the laser light source has a narrow output pattern or NA. Like the LED, the laser must be packaged to align the source with the optical fiber and couple as much energy as possible into the core. Lasers can be packaged for a specific connector or bonded to a pigtail.

Fabry-Pérot and DFB laser light sources are designed for either multimode or single-mode applications. VCSELs are currently designed only for multimode optical fiber applications. The VCSEL emits a wider output pattern and larger spot size than the Fabry-Pérot or DFB lasers. The output of the VCSEL fills only the center of the core of the multimode optical fiber, as shown earlier in Figure 10.8.

Fabry-Pérot and DFB lasers are designed for single-mode and multimode optical fiber applications. The output pattern and spot size of these lasers for single-mode applications are narrower and smaller than the VCSEL. Figure 10.8 shows the typical output pattern of the Fabry-Pérot, DFB, and VCSEL laser.

As bandwidth requirements increase, LED transmitters are being replaced with laser transmitters. However, the multimode optical fiber is not being replaced with single-mode. A laser transmitter, unlike an LED transmitter, does not overfill the multimode optical fiber. Power from a laser transmitter is distributed in the narrow center region of the multimode optical fiber core. This is very different from an LED transmitter, where power is distributed to 100 percent of the core.

Multimode optical fiber has a greater bandwidth potential with a laser transmitter than an LED transmitter. The launch from a laser transmitter results in far fewer modes than the overfilled launch from an LED transmitter. Fewer modes mean less modal dispersion, resulting in greater potential bandwidth.

While all of the laser types—Fabry-Pérot, DFB, and VCSEL—are used with multimode optical fiber, only the VCSEL is used for 850nm applications. The VCSEL is not currently available at a wavelength of 1300nm. For 1300nm or 1310nm applications, the Fabry-Pérot or DFB type is used.

Because the launch conditions for the VCSEL transmitter are different from those for the LED, ANSI/TIA-568-C.3 now defines the bandwidth of a multimode optical fiber both with an OFL and with a *restricted mode launch (RML)*. The minimum effective modal bandwidth length product of a multimode optical fiber at 850nm is defined with a restricted mode launch condition. RML applies only to laser-optimized multimode optical fiber, as shown in Table 10.1.

Source Wavelengths

The performance of a fiber-optic communication system is dependent on many factors. One key factor is the wavelength of the light source. Short wavelengths (650nm through 850nm) have more modes and greater attenuation than longer wavelengths (1300nm through 1600nm). The wavelength of the light source can determine system bandwidth and transmission distance.

LED Wavelengths

LED light sources are manufactured from various semiconductor materials. The output wavelength of the LED depends on the semiconductor material that it's manufactured from. Table 10.2 breaks down the semiconductor materials and their associated wavelengths. LED wavelengths are chosen for fiber-optic communication systems based on their application. Visible wavelengths (650nm) are typically used in short-distance, low-data-rate systems with large-core-diameter optical fiber. These systems are typically found in industrial control systems. Infrared wavelengths (820nm, 850nm, and 1300nm) are typically used for longer-distance, higher-data-rate systems with smaller-core optical fiber.

TABLE 10.2 LED semiconductor materials

Wavelength	Semiconductor materials
650nm	Aluminum (Al), gallium (Ga), arsenic (As)
820nm	Aluminum (Al), gallium (Ga), arsenic (As)
850nm	Aluminum (Al), gallium (Ga), arsenic (As)
1300nm	Indium (In), gallium (Ga), arsenic (As), phosphorus (P)

Laser Wavelengths

Just like the LED, a laser light source output wavelength is dependent on the semiconductor material that it's manufactured from, as shown in Table 10.3. As you learned in Chapter 5, “Optical Fiber Characteristics,” single-mode optical fiber performance can be defined at 1310nm, 1550nm, or 1625nm. Fabry-Pérot and DFB lasers can be manufactured to operate at or near these wavelengths.

TABLE 10.3 Laser semiconductor materials

Wavelength	Semiconductor materials
850nm	Aluminum (Al), gallium (Ga), arsenic (As)
1310nm	Indium (In), gallium (Ga), arsenic (As), phosphorus (P)
1550nm	Indium (In), gallium (Ga), arsenic (As), phosphorus (P)

As VCSEL technology improves, it will begin to replace the Fabry-Pérot for 1310nm and 1550nm applications.

The DFB laser is unique because it can be tuned to a specific wavelength. This allows many lasers to operate around a center wavelength in a multichannel system.

The DFB laser is virtually a single-wavelength laser and has the least dispersion of laser fiber-optic sources. Because of this, the DFB is the choice for long-distance, high-speed applications. An uncooled off-the-shelf 1550nm DFB laser can support a 2.5Gbps data rate at a transmission distance of 80km.

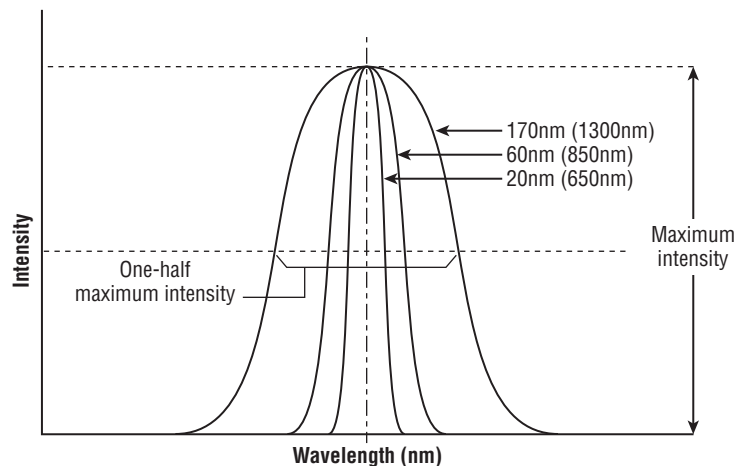
Source Spectral Output

The spectral output of a fiber-optic light source can have a significant impact on the bandwidth of a fiber-optic communication system. Different wavelengths of light travel at different speeds through an optical fiber. Because of this, a light pulse made up of more than one wavelength will disperse as each wavelength travels at a different velocity down the optical fiber. This dispersal limits the bandwidth of the system. To achieve the highest bandwidth possible, a fiber-optic light source must output the narrowest range of wavelengths possible. The range of wavelengths that a light source outputs is described by its *spectral width*.

LED Spectral Width

The spectral width of an LED light source is much wider than the spectral width of a laser. LED spectral width is typically described by the term Full Width, Half Maximum (FWHM). When the spectral width of an LED is displayed graphically, FWHM is measured at one-half the maximum intensity across the full width of the curve, as shown in Figure 10.9.

FIGURE 10.9 Typical FWHM spectral width of 650nm, 850nm, and 1300nm LEDs



LED spectral widths vary tremendously. A short wavelength 650nm visible LED may have a spectral width as narrow as 20nm. The spectral width of a long wavelength 1300nm infrared LED may be as wide as 170nm. Because an LED is used only for multimode applications, the wide spectral width has no effect on the bandwidth of the communication system. Figure 10.9 compares the spectral width of various LED wavelengths.

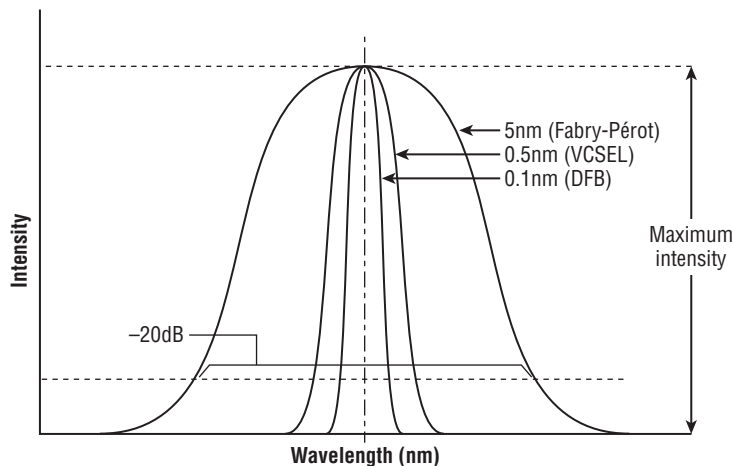
Laser Spectral Width

Laser light sources offer the highest level of performance for a fiber-optic communication system. Unlike the LED, lasers output a very narrow spectrum of light. The manufacturer may describe the spectral output width of a laser several ways. These include FWHM, Root Mean Squared (RMS), and 20dB below peak spectral density.

All laser light sources output a narrow spectrum of light. However, there are significant differences in the spectral widths of each laser family. The spectral width of the Fabry-Pérot laser may span as much as 5nm, causing dispersion problems in high-speed or long transmission distance systems. VCSELs have shorter cavities than Fabry-Pérot lasers, which reduce their spectral output width. The DFB laser is virtually a one-wavelength laser with spectral widths less than 0.1nm.

The DFB laser achieves this narrow spectral output width by using a series of corrugated ridges on the semiconductor substrate to diffract unwanted wavelengths back into the laser cavity. These corrugated ridges form an internal diffraction grating. The laser manufacturer can set the desired output wavelength with the spacing of the corrugated ridges on the internal diffraction grating. The DFB laser is used in the highest performance systems. Figure 10.10 compares the spectral widths of the three laser families.

FIGURE 10.10 Typical -20dB spectral width of Fabry-Pérot, VCSEL, and DFB laser



Source Output Power

The output power of the light sources used in fiber-optic communication systems varies dramatically depending on the application. LED light sources are typically designed to support transmission distances up to 2km whereas laser light sources may support transmission distances well in excess of 80km. Laser optical power output levels can exceed LED optical output power levels by more than 20dB for premises local area network (LAN) and storage area network (SAN) applications. This number increases to 50dB or greater for outside-plant fiber-optic communication systems that incorporate automatic reduction to reduce emissions to a lower hazard level. These systems may operate with power levels up to 2.6 watts at a 1550nm wavelength.

LED Output Power

Today's LED technology seems to be changing daily. High-power LED light sources are beginning to replace many incandescent light sources. As of this writing, single-lamp 20-watt LED white light flashlights are available with beams so powerful that viewing them directly is like viewing a flashbulb. However, the LED light sources used in fiber-optic communication output considerably less power than that flashlight.

The optical output power of a typical LED light source used in fiber-optic communications is 10dB or more below the lowest power laser light source. Output power is typically expressed by the manufacturer at the Beginning of Life (BOL) and the End of Life (EOL). An LED light source will lose some output power over its usable lifetime. The industry convention allows for a degradation of 1.5dB from BOL to EOL.

LED light sources couple only a small portion of their light energy into the core of the optical fiber. The amount of energy coupled into the optical fiber depends on the optical fiber core size and NA. The larger the core size and greater the NA, the more energy is coupled into the core.

Off-the-shelf LED light sources are available to support a wide range of glass, plastic, and PCS optical fiber. Most LED light sources used in telecommunication applications are designed to work with either 50/125 μ m or 62.5/125 μ m optical fiber. The light energy coupled into the core of a 50/125 μ m optical fiber with an NA of 0.20 is approximately 3.5dB less than the energy coupled into the core of a 62.5/125 μ m optical fiber with an NA of 0.275. Because the same LED light source may be used with different sizes of optical fiber, the manufacturer typically measures the output power at the end of one meter of optical fiber with the cladding modes removed. The cladding modes can be removed by a mode filter, which is discussed in detail in Chapter 14, "Test Equipment and Link/Cable Testing."

Laser Output Power

Laser light sources offer the highest output power available for fiber-optic communication systems. The higher output power of the laser allows greater transmission distances than with an LED. Laser output power varies depending on the application. VCSELs used in multimode applications have the lowest output power. Fabry-Pérot and DFB lasers used in single-mode applications have the greatest output power.

Like the LED, the output power of the laser will diminish over its usable lifetime. Laser manufacturers typically provide BOL and EOL minimum optical power levels. The industry convention for lasers allows for a 1.5dB reduction in power from BOL to EOL. Optical output power levels are normally expressed as the amount of light coupled into a one-meter optical fiber.

Source Modulation Speed

There are many advantages to using optical fiber as a communications medium. High bandwidth over long distances is one of them. What limits the bandwidth of today's fiber-optic communication systems is not the optical fiber but the light source. The modulation speed of a light source is just one factor that can limit the performance of a fiber-optic communication system.

LED Modulation Speed

Today's LED light sources can be modulated to support data rates greater than 400Mbps. However, most LED light sources are designed to support network standards that do not require a data rate that high. An example of this is the IEEE 802.3 Ethernet standard, which establishes network data rates at 10Mbps, 100Mbps, 1Gbps, and 10Gbps. The LED can easily support the 10Mbps and 100Mbps data rate. However, current LED technology does not support the 1Gbps or greater data rate.

Laser Modulation Speed

Laser light sources are constantly evolving. Laser manufacturers have been able to modulate all three-laser families at data rates up to 10Gbps. Today laser transmitters that support 1Gbps, 2.5Gbps, or 10Gbps are very affordable and are being integrated into more and more LANs and SANs. Laser light sources may also be used in low-data-rate applications such as 10Mbps or 100Mbps when transmission distances greater than 2,000 meters are required.

Transmitter Performance Characteristics

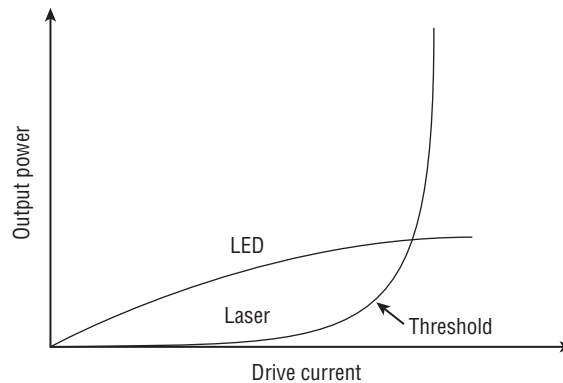
Up to this point, we have discussed only the characteristics of the unpacked and packaged LED and laser light source with no mention of the electronics required to drive these light sources. The incredible data rates we've discussed in the previous text would not be possible without integrated electronics packaged into the transmitter.

For most users, the LED or laser transmitter is a black box. The manufacturer has neatly integrated everything required to convert the electrical input signal into light energy in the smallest package possible. However, it is important to be able to interpret the manufacturer data sheet and verify the optical output of the transmitter.

LED Transmitter Performance Characteristics

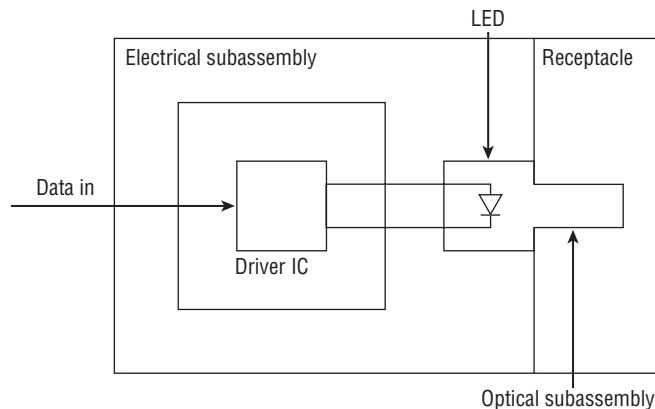
LED transmitters are designed to support multimode optical fibers with core sizes ranging from as small as $50\mu\text{m}$ to as large as 1mm. LED transmitters are directly modulated. (Direct modulation is when the drive current through the LED is varied.) The output power of the LED is directly proportional to the current flow through the LED, as shown in Figure 10.11. In a digital application, the drive current is switched on and off. In an analog application, the drive current is varied.

FIGURE 10.11 LED and laser optical output power versus drive current



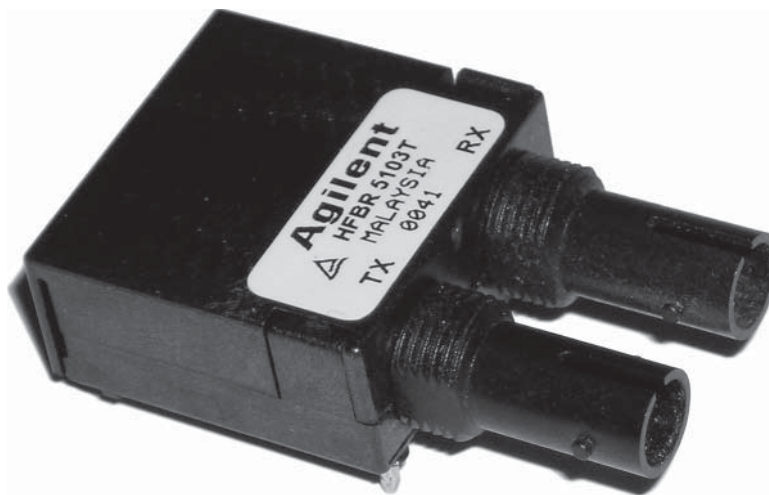
A typical LED transmitter contains an electrical subassembly, optical subassembly, and receptacle, as shown in Figure 10.12. The electrical subassembly amplifies the electrical input signal with the driver integrated circuit (IC). The driver IC provides the current to drive the LED in the optical subassembly. The optical subassembly mates with the receptacle to direct light into the optical fiber.

FIGURE 10.12 Block diagram of a typical LED transmitter



Because LED fiber-optic links are simplex, they require a transmit optical fiber and a receive optical fiber. Typically, the LED transmitter is packaged with the receiver section. Packaging the transmitter and receiver together reduces the overall space required, simplifies circuit board design, and reduces cost. Figure 10.13 is a photograph of a 1300nm transceiver with an ST receptacle.

FIGURE 10.13 A 100Mbps 1300nm LED transceiver with an ST receptacle



The term *transceiver* describes a device in which the fiber-optic transmitter and receiver are combined into a single package. Standalone transmitters and receivers are still available; however, the vast majority of standardized networks use transceivers.

The performance characteristics of the LED transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe maximum and minimum temperature and voltage ranges that the device can operate in without damage. Table 10.4 shows the typical recommended operating conditions for a 1300nm 100Mbps LED transmitter.

The electrical characteristics of the LED transmitter describe the supply current requirements, the data input requirements, and the power dissipated by the device. Table 10.5 shows the typical electrical characteristics for a 1300nm 100Mbps LED transmitter.

The optical characteristics of the LED transmitter at a minimum include output power, center wavelength, and spectral width. Table 10.6 shows the typical optical characteristics for a 1300nm 100Mbps LED transmitter.

As you can see in Table 10.6, there is a minimum, maximum, and typical value for the optical output power of an LED transmitter. These values are defined for 62.5/125 μ m

optical fiber and for 50/125 μm optical fiber. As you learned in Chapter 5, the NA of a 62.5/125 μm optical fiber is typically 0.275 and the NA of a 50/125 μm optical fiber is 0.20.

TABLE 10.4 LED transmitter recommended operating conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient Operating Temperature	T_A	0		70	$^{\circ}\text{C}$
Supply Voltage	V_{CC}	4.75		5.25	V
Data Input Voltage—Low	$V_{IL}-V_{CC}$	-1.810		-1.475	V
Data Input Voltage—High	$V_{IH}-V_{CC}$	-1.165		-0.880	V
Data and Signal Detect Output Load	R_L		50		Ω

TABLE 10.5 LED transmitter electrical characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Current	I_{CC}		145	185	mA
Power Dissipation	P_{DISS}		0.76	0.97	W
Data Input Current—Low	I_{IL}	-350	0		μA
Data Input Current—High	I_{IH}		14	350	μA

The amount of optical output power coupled into a multimode optical fiber depends on the core diameter and the NA of the fiber. As you saw earlier in this chapter, the same LED transmitter will typically couple 3.5dB more power into the 62.5/125 μm optical fiber than into the 50/125 μm . The 3.5dB difference is shown in Table 10.6. There is a 3.5dB difference between the minimum optical output power for the 62.5/125 μm and 50/125 μm optical fiber.

Earlier in this chapter, we stated that light sources age, and as they age, their output power decreases. The industry convention allows for 1.5dB of aging. However, the difference between the BOL value and EOL for the LED transmitter described in Table 10.6 is only 1dB. The difference between BOL and EOL output power will vary from manufacturer to manufacturer. Always obtain this data from the manufacturer's data sheet.

TABLE 10.6 LED transmitter optical characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power BOL 62.5/125 μ m, NA = 0.275 Fiber EOL	P_0	-19 -20	-16.8	-14	dBm avg.
Optical Output Power BOL 50/125 μ m, NA = 0.20 Fiber EOL	P_0	-22.5 -23.5	-20.3	-14	dBm avg.
Optical Extinction Ratio			0.001 -50	0.03 -35	% dB
Optical Output Power at Logic "0" State	P_0 ("0")			-45	dBm avg.
Center Wavelength	λ_c	1270	1308	1380	nm
Spectral Width—FWHM	$\Delta\lambda$		137	170	nm
Optical Rise Time	t_r	0.6	1.0	3.0	ns
Optical Fall Time	t_f	0.6	2.1	3.0	ns
Duty Cycle Distortion Contributed by the Transmitter	DCD		0.02	0.6	ns _{p,p}
Data-Dependent Jitter Contributed by the Transmitter	DDJ		0.02	0.6	ns _{p,p}

The output power of most fiber-optic transmitters is very close to the typical value. However, the output power can vary as defined by the minimum and maximum values in Table 10.6. To find the minimum optical output power for a specific optical fiber, locate the Optical Output Power row for that optical fiber in Table 10.6. In the Min. column, you will see a BOL and an EOL value. The higher number is the BOL value, and the lower number is the EOL value.

To find the maximum optical output power for a specific optical fiber, locate the Optical Output Power row for that optical fiber in Table 10.6. The number in the Max. column is the value for the maximum optical output power. The optical output of the fiber-optic transmitter should never exceed this value. There is no BOL or EOL value for the maximum optical output power.

LED Transmitter Applications

LED transmitters are often designed to support one or more network standards. A 100Mbps 1300nm transmitter could support 100Base-FX applications or 155Mbps ATM applications.

Table 10.7 lists the data rates, wavelengths, and optical fiber types for various LED transmitters used in IEEE 802.3 Ethernet communication systems.

TABLE 10.7 LED transmitters for Ethernet applications

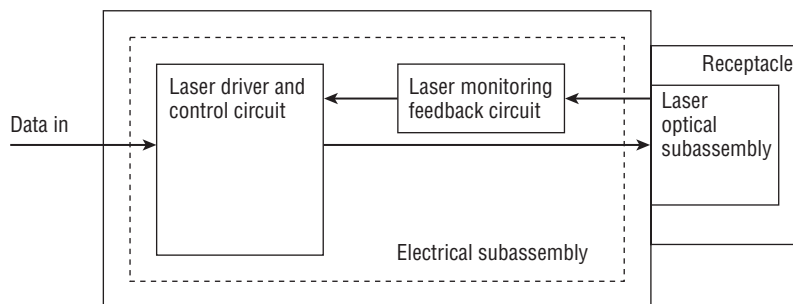
Network	Data rate	Wavelength and media
10BASE-FL	10Mbps	850nm, multimode fiber
100BASE-FX	100Mbps	1300nm, multimode fiber
10/100BASE-SX	10/100Mbps	850nm, multimode fiber

Laser Transmitter Performance Characteristics

Laser transmitters are designed to support either single-mode optical fiber systems or multimode optical fiber systems. The single-mode transmitter is designed to interface with 8–11 μ m optical fiber cores. Until recently, all single-mode transmitters were from the Fabry-Pérot or DFB families. However, manufacturers keep improving VCSEL technology and the VCSEL is beginning to emerge as an alternative 1300nm or 1550nm single-mode optical fiber transmitter.

Like the LED transmitter, the laser transmitter can be broken into three sections: the electrical subassembly, optical subassembly, and receptacle or pigtail, as shown in Figure 10.14. The electrical subassembly amplifies the input signal and provides the drive current for the laser. Unlike the LED, the laser emits very few photons until the drive current passes a threshold level, as shown earlier in Figure 10.11. A feedback circuit that monitors laser output intensity typically controls the drive current provided to the laser to ensure a constant output level.

FIGURE 10.14 Block diagram of a typical laser transmitter



The optical subassembly contains a photodiode that receives part of the energy output by the laser. The electrical output of the photodiode is monitored in the electrical subassembly to help control drive current. The optical subassembly may mate with an external modulator, receptacle, or pigtail.

Laser transmitters may or may not be packaged with the receiver section. How the transmitter is packaged depends on the application. Laser transmitters operating up to 2.5Gbps typically do not require coolers and are often packaged with a receiver section. Laser transmitters operating at 10Gbps or greater produce enough heat to require an integrated cooler to help the laser maintain a constant temperature. Temperature variations will affect the laser's threshold current, output power, and frequency.

The VCSEL is used in multimode transmitters. Currently VCSEL multimode transmitters support only 850nm operation with 50/125 μm or 62.5/125 μm optical fiber. The VCSEL can support data rates as high as 10Gbps through multimode optical fiber.

The VCSEL, Fabry-Pérot, or DFB laser is capable of direct modulation up to 10Gbps. To achieve data rates greater than 10Gbps, the laser must typically be indirectly modulated. Indirect modulation does not vary the drive current as direct modulation does. Maintaining a constant drive current prevents the output wavelength from changing as the electron density in the semiconductor material changes. This is commonly referred to as *laser chirp*, which causes dispersion and limits bandwidth over long transmission distances.



Real World Scenario

Problems Associated with Core Diameter Mismatch

LED transmitters are typically designed to support different core size multimode optical fiber. For example, a 100BASE-FX transmitter is designed to operate with 50/125 μm or 62.5/125 μm optical fiber, as shown in Table 10.6. If the LED transmitter receptacle is designed to accept an SC connector, any SC connector regardless of optical fiber size or type can be plugged into the transmitter.

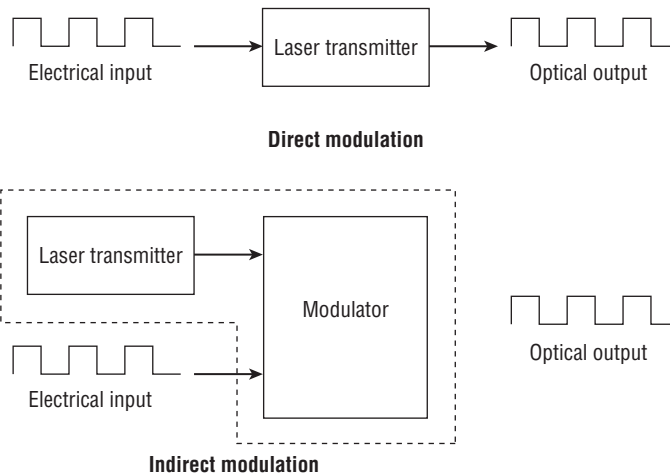
So what happens when a 62.5/125 μm patch cord from the LED transmitter is mated with a 50 μm horizontal cable? Back reflections and attenuation. As you may remember from the “LED Output Power” section, we mentioned that an LED light source couples roughly 3.5dB more light energy into a 62.5 μm core than a 50 μm core. So when you take the 62.5 μm core from the transmitter and mate it with the 50 μm core horizontal cable, there is roughly a 3.5dB loss at that connection. As mentioned in Chapter 2, “Principles of Fiber-Optic Transmission,” a loss of 3dB means that half of the power is lost. So if less than half the photons from a light pulse get coupled into the horizontal cable, where do the rest of the photons go?

Well, some of the photons will be absorbed by the coating of the 50 μm optical fiber. Others will be reflected back into the cladding of the 62.5 μm optical fiber and absorbed by the coating. However, some of the photons will be reflected back into the core of the 62.5 μm optical fiber. Many of those photons will travel back to the transmitter, where they will be reflected off the transmitter window or lens assembly back into the core. These photons will travel the length of the optical fiber to the receiver. There is a good possibility that the receiver will detect these photons and convert their light energy into electrical energy. Now the circuit decoding the electrical energy is being bombarded with good and bad data that prevents the system from operating.

The fix for this problem is simple: use a 50 μm patch cord. The bottom line is that you should pay careful attention to core diameter to prevent core diameter mismatch. You can't look at only the color of the patch cord to determine the core diameter. There are orange patch cords with a 62.5 μm core and there are orange patch cords with a 50 μm core. You must read the cable markings from the manufacturer to find out the actual core diameter.

Indirect modulation requires a constant drive current to the laser. The laser light source outputs a continuous wave (CW) that is modulated by an external device. The external modulation device is typically integrated into the laser transmitter package. Indirect modulation of a laser allows higher modulation rates (up to 40Gbps or greater) and less dispersion than direct modulation. Figure 10.15 shows a functional block diagram of direct and indirect modulation.

FIGURE 10.15 Block diagram of direct and indirect modulation



The performance characteristics of the laser transmitter are typically broken up into four groups: recommended operating conditions, electrical characteristics, optical characteristics, and data rate. The recommended operating conditions describe the maximum and minimum temperatures and voltage ranges that the device can operate in without damage. Table 10.8 shows the typical recommended operating conditions for a 1300nm, 2.5Gbps laser transmitter.

TABLE 10.8 Laser transmitter recommended operating conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Ambient Operating Temperature	T_A	0		+70	°C
Supply Voltage	V_{CC}	3.1		3.5	V
Power Supply Rejection	PSR		100		mVPP
Transmitter Differential Input Voltage	V_D	0.3		2.4	V
Data Output Load	R_{DL}		50		Ω
TTL Signal Detect Output Current—Low	I_{OL}			1.0	mA
TTL Signal Detect Output Current—High	I_{OH}	−400			μA
Transmit Disable Input Voltage—Low	T_{DIS}			0.6	V
Transmit Disable Input Voltage—High	T_{DIS}	2.2			V
Transmit Disable Assert Time	T_{ASSERT}			10	μS
Transmit Disable Deassert Time	$T_{DEASSERT}$			50	μS

The electrical characteristics of the laser transmitter describe the supply current requirements, the data input requirements, and the power dissipated by the device. Table 10.9 shows the typical electrical characteristics for a 1300nm, 2.5Gbps laser transmitter.

TABLE 10.9 Laser transmitter electrical characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Current	I_{CCR}		115	140	mA
Power Dissipation	P_{DISS}		0.38	0.49	W
Data Output Voltage Swing (single-ended)	$V_{OH} - V_{OL}$	575		930	mV

Parameter	Symbol	Min.	Typ.	Max.	Unit
Data Output Rise Time	t_r		125	150	ps
Data Output Fall Time	t_f		125	150	ps
Signal Detect Output Voltage—Low	V_{OL}			0.8	V
Signal Detect Output Voltage—High	V_{OH}	2.0			V
Signal Detect Assert Time (OFF to ON)	AS_{MAX}			100	μs
Signal Detect Deassert Time (ON to OFF)	ANS_{MAX}			100	μs

The optical characteristics of the laser transmitter at a minimum include output power, center wavelength, spectral width, and back-reflection sensitivity. Excessive back reflections can interfere with the operation of the laser. Table 10.10 shows the typical optical characteristics for a 1300nm, 2.5Gbps laser transmitter.

TABLE 10.10 Laser transmitter optical characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power 9 μm SMF	P_{OUT}	−10	−6	−3	dBm
Center Wavelength	λ_c	1260		1360	nm
Spectral Width—rms	$\Delta\lambda$		1.8	4	nm rms
Optical Rise Time	t_r		30	70	ps
Optical Fall Time	t_f		150	225	ps
Extinction Ratio	E_R	8.2	12		dB
Optical Output Eye	Compliant with eye mask Telecordia GR-253-CORE				
Back-Reflection Sensitivity				−8.5	dB
Jitter Generation	pk to pk			70	mUI
	RMS			7	mUI

As you can see in Table 10.10, there is a minimum, maximum, and typical value for the optical output power of a laser transmitter. These values are defined for a single-mode optical fiber with a core diameter of 9 μ m. However, the NA is not stated as it was in the LED transmitter data sheet. Recall from Chapter 5 that single-mode optical fibers have the smallest NA and narrowest acceptance cone. A typical NA for a single-mode optical fiber is 0.14.

Like the LED, the laser ages and its output power decreases over time. In Table 10.10, there is no BOL or EOL value. If the manufacturer does not provide these values, use the industry convention of 1.5dB and assume that the values stated for optical output power are the BOL values.

To find the minimum optical output power, locate the Optical Output Power row in Table 10.10. In the Min. column, there is a value. Since BOL and EOL are not stated, assume this value is the BOL value. To approximate the EOL value, subtract 1.5dB.

To find the maximum optical output power, locate the Optical Output Power row in Table 10.10. The number in the Max. column is the value for the maximum optical output power. The optical output of the fiber-optic transmitter should never exceed this value. There is no BOL or EOL value for the maximum optical output power.

Laser Transmitter Applications

Laser transmitters are available for serial transmission applications and parallel transmission applications. A typical transmitter may support multiple data rates and multiple network standards such as IEEE 802.3 and Fibre Channel. IEEE 802.3 is an international standard for LANs and metropolitan area networks (MANs). Fibre Channel is the foundation for over 90 percent of all SAN installations globally.

Serial Laser Transmitters

A serial laser transmitter is designed to be used with network standards such as IEEE 802.3 or Fibre Channel. It transmits data in a serial format at a single wavelength over a single optical fiber. These transmitters can be designed to support a broad range of data rates over multimode or single-mode optical fiber. Tables 10.11 and 10.12 list the data rates, wavelengths, and optical fiber types for various serial laser transmitters used in IEEE 802.3 Ethernet communication systems and Fibre Channel communication systems.

Parallel Laser Transmitters

Up to this point in the chapter only single-fiber transmitters that transmit serial data at a single wavelength have been discussed. These transmitters have limitations to the data rates they can achieve, and that limit, as defined by IEEE 802.3 and Fibre Channel, is approximately 10Gbps. To achieve higher data rates, you can take one of two approaches. One approach is to combine the output from multiple transmitters, each transmitting at a different wavelength, into a single optical fiber; this is called *wavelength division multiplexing* (WDM). WDM is ideal for high-data-rate systems over long distances using single-mode optical fiber.

TABLE 10.11 Serial fiber laser transmitters for Ethernet applications

Network	Data rate	Wavelength and media
100BASE-LX	100Mbps	1310nm FP, single-mode fiber
1000BASE-SX	1Gbps	850nm VCSEL, multimode fiber
1000BASE-LX	1Gbps	1310nm FP, multimode or single-mode fiber
10GBASE-S	10Gbps	850nm VCSEL, multimode fiber
10GBASE-L	10Gbps	1310nm DFB, single-mode fiber

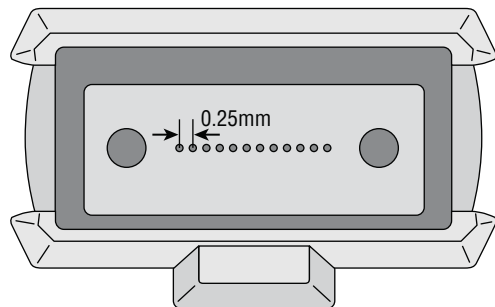
TABLE 10.12 Serial fiber laser transmitters for Fibre Channel applications

Network	Data rate	Wavelength and media
100-MX-SN-I	1062Mbaud	850nm VCSEL, multimode fiber
100-SM-LC-L	1062Mbaud	1310 FB, single-mode fiber
200-MX-SN-I	2125Mbaud	850nm VCSEL, multimode fiber
200-SM-LC-L	2125Mbaud	1310 FP, single-mode fiber
1200-MX-SN-I	10512Mbaud	850nm VCSEL, multimode fiber
1200-SM-LL-L	10512Mbaud	1310nm DFB, single-mode fiber

Another approach to achieving high data rates over short distances using multimode optical fiber is parallel optics. With parallel optics, multiple transmitters each operating at the same wavelength are used. The difference between parallel optics and WDM is that parallel optics uses one optical fiber for each transmitter vice, combining outputs of multiple transmitters into a single optical fiber. Since each transmitter is dedicated to a single optic fiber, all the transmitters can operate at the same wavelength.

The light source of choice for parallel optics is the VCSEL. VCSEL technology has matured to where multiple 10Gbps transmitters can be manufactured in the form of an array that physically aligns with the fibers in an MT ferrule. As you learned in Chapter 9, “Connectors,” the optical fibers in an MT ferrule are spaced 0.25mm apart, as shown in Figure 10.16.

FIGURE 10.16 MT ferrule optical fiber spacing is 0.25mm.



IEEE Standard 802.3-2012, Section 6 introduces 40Gbps and 100Gbps Ethernet. It defines the maximum transmission distances for each data rate using single-mode optical fiber, multimode optical fiber, or copper cabling for the physical layer. When multimode OM3 or OM4 optical fiber is used for the physical layer, multiple transmitters operating in parallel are required, as described in Table 10.13. Achieving a 40Gbps data rate requires four 10Gbps transmitters operating in parallel. A 100Gbps data rate is achieved using ten 10Gbps transmitters operating in parallel.

As of this writing, the maximum transmission distance is 100m for systems using OM3 optical fiber and 150m for systems using OM4 optical fiber.

TABLE 10.13 Parallel laser transmitters for Ethernet applications

Network	Data rate	Data rate per channel	Number of transmitters	Wavelength and media
40GBASE-SR4	40Gbps	10Gbps	4	850nm VCSEL, multimode fiber
100GBASE-SR10	100Gbps	10Gbps	10	850nm VCSEL, multimode fiber

To support parallel operation, the transmitters must be physically arranged to accept an MPO female plug like the one shown in Figure 10.17. For 40GBASE-SR-4 networks, the transmitters use the four leftmost optical fibers or optical transmit lanes and the receivers use the four rightmost optical fibers or receive lanes, as shown in Figure 10.18. The four optical fibers in the center are not used. This arrangement allows for 40Gbps full duplex operation using a single 12-fiber ribbon.

For 100GBASE-SR10 networks the optical lanes can be configured for a single-receptacle using Option A shown in Figure 10.19; this approach is recommended in IEEE 802.3. They can also be configured using the two-receptacle Option B, shown in Figure 10.20 or Option C, shown in Figure 10.21; these options are not recommended and considered alternatives.

FIGURE 10.17 MPO female plug with 12 optical fibers

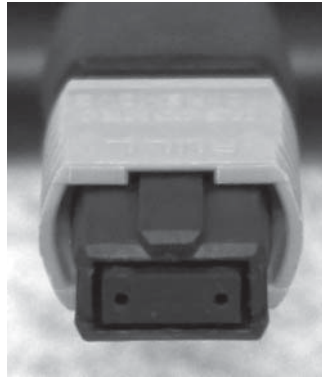


FIGURE 10.18 40GBASE-SR4 optical lane assignments

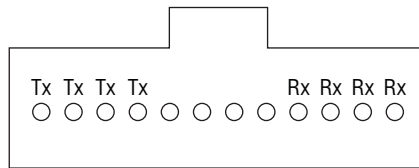


FIGURE 10.19 100GBASE-SR10 Option A optical lane assignments

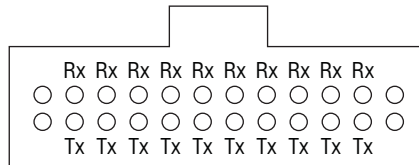


FIGURE 10.20 100GBASE-SR10 Option B optical lane assignments

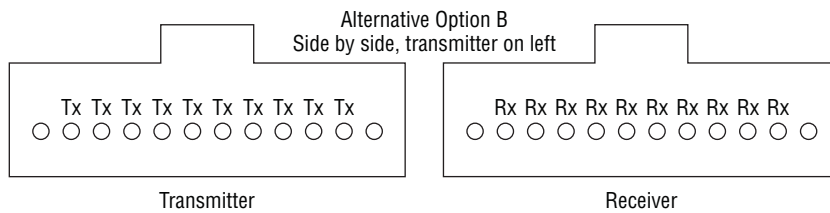
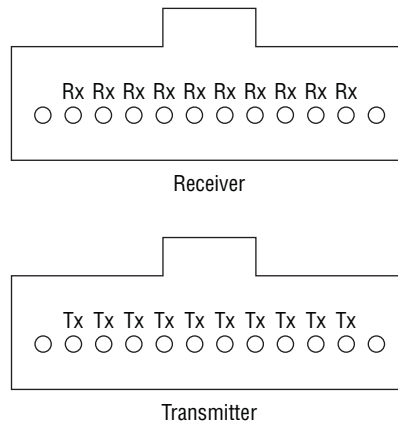


FIGURE 10.21 100GBASE-SR10 Option C optical lane assignments

Higher Power Transmitters

LANs and SANs are not limited to premises applications. These networks are being used in spacecraft, aircraft, on the battlefield, and in numerous other harsh environments. In many of the applications, the number of interconnections required to connect the transmitter to the receiver is typically greater than a premises-based system. In addition, the connectors may experience greater loss than is typically experienced in a premises-based system.

To compensate for the greater number of interconnections and the increased loss at each interconnection, a higher power transmitter may need to be selected. For example, the transceiver shown in Figure 10.22 will support a 1000BASE-SX LAN. The manufacturer offers it with a standard output power or a higher output power for applications with a greater number of interconnections or higher loss interconnections.

IEEE 802.3 states that the minimum average launch power for a 1000BASE-SX transmitter is -9.5dBm . The transceiver shown in Figure 10.22 can be purchased to meet this requirement, or a higher power version can be purchased that has a minimum average launch power of -5dBm . The guaranteed extra 4.5dB compensates for the additional losses caused by a greater number of interconnections and/or higher loss interconnections while still meeting the Class I eye safety requirements. You can obtain additional information about higher power transmitters at www.cotsworks.com.

FIGURE 10.22 Higher-power 850nm VCSEL multi-rate transceiver

Light Source Safety

The light sources used in fiber-optic communication systems typically operate at low power levels. Unlike an incandescent lamp, their light energy is distributed over a narrow spectrum, typically the infrared spectrum. This narrow spectrum light energy is often invisible, and it can be a danger to the fiber-optic installer or fiber-optic technician. The level of danger depends on the classification of the light source and the amount of energy coupled into the optical fiber. Laser light sources pose the greatest risk because of their narrow spectral width and coherent light.

Classifications

We have learned that fiber-optic light sources are very different from the lightbulbs used to illuminate our home or office. We shop for lightbulbs by their wattage. The wattage shown on the package of the lightbulb tells us how much energy the bulb requires to operate. A 60-watt incandescent lightbulb does not output 60 watts of light energy; it requires 60 watts of power to operate. Much of the energy required to operate the bulb is actually emitted as heat, not visible light. We all know how hot a 60-watt incandescent lightbulb can get.

That same 60-watt lightbulb emits a very broad spectrum of visible light. In other words, the bulb emits many different wavelengths of visible light that combine together to create white light. Each wavelength of light radiating from the bulb represents only a

small fraction of the overall light energy output by the bulb. Unlike the laser, the visible light that radiates from the bulb is incoherent. Because the broad visible spectrum of light radiating from the bulb is incoherent and the light energy is spread out over many wavelengths, the light from the bulb will not damage your eye. The coherent light from the laser and its lack of divergence is what can cause eye damage. An ordinary incandescent lightbulb is not an eye hazard like a laser. However, this doesn't mean that you should stare at 60-watt light bulbs.

Fiber-optic light sources are classified by their ability to cause damage to your eye. As you learned in Chapter 6, "Safety," there are standards and federal regulations that classify the light sources used in fiber-optic transmitters. Every laser transmitter sold in the United States must be classified as defined in 21CFR, Chapter 1, Subchapter J. Only these classifications are covered in this chapter; additional information can be found in Chapter 6. Listed here are the hazards associated with each class:

Class I These levels of laser radiation are not considered to be hazardous.

Class IIa These levels of laser radiation are not considered to be hazardous if viewed for any period of time less than or equal to 1,000 seconds, but are considered to be a chronic viewing hazard for any period of time greater than 1,000 seconds.

Class II These levels of laser radiation are considered to be a chronic viewing hazard.

Class IIIa These levels of laser radiation are considered to be, depending on the irradiance, either an acute intrabeam viewing hazard or a chronic viewing hazard, and an acute viewing hazard if viewed directly with optical instruments.

Class IIIb These levels of laser radiation are considered to be an acute hazard to the skin and eyes from direct radiation.

Class IV These levels of laser radiation are considered to be an acute hazard to the skin and eyes from direct and scattered radiation.

Safe Handling Precautions

There are some fundamental precautions that you should always observe when working with a fiber-optic communication system.

- Assume that the fiber-optic cable assembly you are handling is energized.
- Never directly view the end of an optical fiber or the endface of a fiber-optic connector without verifying that the optical fiber is dark.

Summary

In this chapter, you gained an understanding of LED and laser light sources. The chapter looked at the different performance characteristics of LED and laser transmitters. It discussed laser transmitter applications and reviewed serial and parallel performance for standardized networks. In addition, it also reviewed light source safety.

Exam Essentials

Describe the types of LED light sources used in fiber-optic communication and their basic operation. Make sure you understand the basic theory behind the operation of the LED light sources. Be able to describe how the light is emitted from each LED type and the phase relationship between the photons.

Know the types of laser light sources used in fiber-optic communication and their basic operation. You should know the three laser families used for fiber-optic light sources. Make sure you understand the basic theory of operation of the laser. Be able to describe how light energy is emitted from the laser and the phase relationship between the photons.

List the performance characteristics of an LED light source. Make sure you understand that the unpackaged LED has a wide output pattern. The LED is packaged to direct as much light as possible into the optical fiber. However, the packaged LED still overfills the optical fiber. You need to know the infrared LED wavelengths used in fiber-optic light sources. Make sure you can describe the spectral widths of the infrared LED light sources. You should know the maximum data rate that an LED light source can transmit.

Know the performance characteristics of a laser light source. Make sure you understand that the laser has a narrow output pattern. You need to know the infrared wavelengths used in each laser family. Make sure you can describe the spectral widths of each family too. You should know the maximum data rate that a laser light source can transmit.

Describe the performance characteristics of an LED transmitter. You should know that LED transmitters are designed to support multimode optical fibers with core sizes ranging from as small as 50 μ m to as large as 1mm. Make sure you understand that LED transmitters are typically designed to support network standards and rarely support data rates above 100Mbps.

Know the performance characteristics of a laser transmitter. Remember that only laser transmitters are designed to support either single-mode optical fiber systems or multimode optical fiber systems. Make sure you understand that the VCSEL is currently only used in transmitters supporting 850nm operation with multimode optical fiber. You should know that the direct modulation of a laser can support data rates up to 10Gbps.

Know the differences between a serial and parallel laser transmitter. Remember that a serial laser transmitter transmits data at a single wavelength over a single optical fiber. Know that parallel laser transmitters use one optical fiber for each transmitter and all the transmitters operate at the same 850nm wavelength. Remember only the VCSEL is used in a parallel transmitter with multimode OM3 or OM4 fiber types.

Explain the safety classifications of the light sources used in fiber-optic communication. You should know the safety classifications for fiber-optic light sources, which classification is eye safe, and how to view light sources that are not eye safe. You must also know the fundamental safety precautions that should be observed when working around a fiber-optic communication system.

Review Questions

1. LEDs emit photons that are out of phase; this is referred to as _____ light.

- A. Coherent
- B. Incoherent
- C. Stimulated
- D. Spontaneous

Hint: They do not have a fixed relationship.

2. LEDs emit random photons through a process called _____ emission.

- A. Coherent
- B. Incoherent
- C. Stimulated
- D. Spontaneous

Hint: It is caused by current flow through the LED.

3. Lasers emit photons that are in phase through a process called _____ emission.

- A. Coherent
- B. Incoherent
- C. Stimulated
- D. Spontaneous

Hint: Every photon that escapes the optical cavity is a duplicate of the first photon to escape.

4. The photons emitted from a laser have the same wavelength and are in phase; this is referred to as _____ light.

- A. Coherent
- B. Incoherent
- C. Stimulated
- D. Spontaneous

Hint: They have a fixed relationship.

5. LEDs have a _____ output pattern when compared to the output pattern of a laser.

- A. Narrow
- B. Wide
- C. Coherent
- D. Stimulated

Hint: A micro lens is typically used to direct the light into the optical fiber.

6. With multimode optical fiber, only the VCSEL is used for _____ applications.
- A. 1550nm
 - B. 1310nm
 - C. 850nm
 - D. 650nm

Hint: Works best with laser-optimized multimode optical fiber.

7. The spectral width of a laser is _____ than the spectral width of an LED.
- A. Wider
 - B. Narrower
 - C. Shorter
 - D. Longer

Hint: It is typically less than 5nm.

8. The optical output power of a laser is _____ than the optical output power of an LED.
- A. Wider
 - B. Greater
 - C. Narrower
 - D. Less

Hint: LED optical power output levels vary depending on the core size and NA of the optical fiber.

9. Laser light sources have a _____ modulation speed than LEDs.
- A. Wider
 - B. Slower
 - C. Narrower
 - D. Higher

Hint: The LED can easily support the 10Mbps and 100Mbps data rate.

10. The _____ is the laser primarily used in multimode transmitters.
- A. DFB
 - B. VCSEL
 - C. Fabry-Pérot
 - D. LED

Hint: Currently supports only 850nm operation.

11. The light source of choice for parallel optics is the _____.

- A. LED
- B. DFB
- C. Fabry-Pérot
- D. VCSEL

Hint: It can support data rates as high as 10Gbps through multimode optical fiber.

12. As defined in IEEE Standard 802.3-2012, Section 6, achieving a 40Gbps data rate requires _____ transmitter(s) operating in parallel.

- A. 1
- B. 4
- C. 6
- D. 12

Hint: Full duplex operation requires a single 12-fiber ribbon.

13. Which types of optical fibers should be used for the physical layer when multiple transmitters are operating in parallel?

- A. OS1, OS2
- B. OS3, OS4
- C. OM1, OM2
- D. OM3, OM4

Hint: The maximum transmission distance is 100m or 150m.

14. Only a _____ laser is considered eye safe.

- A. Class I
- B. Class II
- C. Class III
- D. Class IV

Hint: These levels of laser radiation are not considered to be hazardous.

15. The end of an optical fiber _____ be viewed directly without verifying that the optical fiber is dark.

- A. May always
- B. Should never
- C. May sometimes
- D. Must always

Hint: Assume that the fiber-optic cable assembly you are handling is energized.

Chapter Exercises

Determine the Minimum Optical Output Power for an LED Transmitter

The minimum optical output power of an LED transmitter should be defined in the manufacturer's data sheet.

Refer to the following table to determine the BOL and EOL values for the minimum optical output power of a 50/125 μm optical fiber.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power BOL 62.5/125 μm , NA = 0.275 Fiber EOL	P_0	-22 -23	-19.8	-17	dBm avg.
Optical Output Power BOL 50/125 μm , NA = 0.20 Fiber EOL	P_0	-25.5 -26.5	-23.3	-17	dBm avg.
Optical Extinction Ratio			0.001 -50	0.03 -35	% dB
Optical Output Power at Logic "0" State	P_0 ("0")			-45	dBm avg.
Center Wavelength	λ_c	1270	1308	1380	nm
Spectral Width—FWHM	$\Delta\lambda$		137	170	nm
Optical Rise Time	t_r	0.6	1.0	3.0	ns
Optical Fall Time	t_f	0.6	2.1	3.0	ns
Duty Cycle Distortion Contributed by the Transmitter	DCD		0.02	0.6	ns _{p-p}
Data-Dependent Jitter Contributed by the Transmitter	DDJ		0.02	0.6	ns _{p-p}

To find the minimum optical output power for a specific optical fiber, locate the Optical Output Power row for that optical fiber in the table. In the Min. column, there is a BOL and EOL value. The higher number, -25.5 dBm, is the BOL value and the lower number, -26.5 dBm, is the EOL value.

Determine the Maximum Optical Output Power for an LED Transmitter

The manufacturer's data sheet should contain the maximum value for the optical output power of an LED transmitter.

Refer to the following table to determine the maximum optical output power of a 62.5/125 μm optical fiber.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power BOL 62.5/125 μm , NA = 0.275 Fiber EOL	P_0	-25 -25	-21.8	-19	dBm avg.
Optical Output Power BOL 50/125 μm , NA = 0.20 Fiber EOL	P_0	-27.5 -28.5	-25.3	-19	dBm avg.
Optical Extinction Ratio			0.001 -50	0.03 -35	% dB
Optical Output Power at Logic "0" State	P_0 ("0")			-45	dBm avg.
Center Wavelength	λ_c	1270	1308	1380	nm
Spectral Width—FWHM	$\Delta\lambda$		137	170	nm
Optical Rise Time	t_r	0.6	1.0	3.0	ns
Optical Fall Time	t_f	0.6	2.1	3.0	ns
Duty Cycle Distortion Contributed by the Transmitter	DCD		0.02	0.6	ns _{p-p}
Data-Dependent Jitter Contributed by the Transmitter	DDJ		0.02	0.6	ns _{p-p}

To find the maximum optical output power for a specific optical fiber, locate the Optical Output Power row for that optical fiber in the table. The number in the Max. column, -19 dBm, is the value for the maximum optical output power.

Determine the Minimum Optical Output Power for a Laser Transmitter

The manufacturer's data sheet should contain the minimum, maximum, and typical value for the optical output power of a laser transmitter.

Refer to the following table to determine the BOL and EOL values for the minimum optical output power.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power 9 μ m SMF	P _{OUT}	-6	0	+3	dBm
Center Wavelength	λ_c	1260		1360	nm
Spectral Width—rms	$\Delta\lambda$		1.8	4	nm rms
Optical Rise Time	t _r		30	70	ps
Optical Fall Time	t _f		150	225	ps
Extinction Ratio	E _R	8.2	12		dB
Optical Output Eye	Compliant with eye mask Telecordia GR-253-CORE				
Back-Reflection Sensitivity				-8.5	dB
Jitter Generation	pk to pk			70	mUI
	RMS			7	mUI

To find the minimum optical output power, locate the Optical Output Power row in the table. In the Min. column, there is a value; since BOL and EOL are not stated, assume this value is the BOL value. To approximate the EOL value, subtract 1.5dB. The BOL value is -6dBm and the EOL value is -7.5dBm.

Determine the Maximum Optical Output Power for a Laser Transmitter

The manufacturer's data sheet should contain the maximum value for the optical output power of a laser transmitter.

Refer to the following table to determine the maximum optical output power.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical Output Power 9 μ m SMF	P _{OUT}	-12	-8	-5	dBm
Center Wavelength	λ_c	1260		1360	nm
Spectral Width—rms	$\Delta\lambda$		1.8	4	nm rms
Optical Rise Time	t _r		30	70	ps
Optical Fall Time	t _f		150	225	ps
Extinction Ratio	E _R	8.2	12		dB
Optical Output Eye	Compliant with eye mask Telecordia GR-253-CORE				
Back-Reflection Sensitivity				-8.5	dB
Jitter Generation	pk to pk			70	mUI
	RMS			7	mUI

To find the maximum optical output power, locate the Optical Output Power row in the table. The number in the Max. column is the value for the maximum optical output power. There is no BOL or EOL value for the maximum optical output power. The maximum value is -5dBm.

Chapter 11

Fiber-Optic Detectors and Receivers

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Describe the basic operation of a photodiode.
- ✓ Describe the basic components in a fiber optic receiver.
- ✓ Explain dynamic range and operating wavelength.
- ✓ Explain why an optical attenuator may be used in a communication system.





In Chapter 1, “History of Fiber Optics and Broadband Access,” we introduced you to the fiber-optic receiver. The job of the receiver is to take light energy from the optical fiber and convert it to electrical energy. In this chapter, we will explain the basic components that make up the fiber-optic receiver, starting with the photodiode. You will learn about the effects of optical input power on the performance of the receiver and the role of an optical attenuator in a fiber-optic communication system.

In this chapter, you will learn to

- Calculate attenuator values

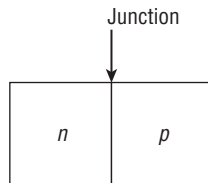
Photodiode Fundamentals

A *photodiode* in a fiber-optic receiver is like the tire on your car. The photodiode is where the rubber meets the road. Light energy from the optical fiber stops at the photodiode. It’s the job of the photodiode to convert the light energy received from the optical fiber into electrical energy. Just as there are different performance-level tires that you can put on your car, so there are different performance-level photodiodes that can be incorporated into a receiver. This section of the chapter discusses the fundamentals of basic photodiode operation and the various types of photodiodes that may be used in a receiver.

The best way to imagine a photodiode is to think about a *solar cell*. Most of us have seen the exhibits at museums where a solar cell or a group of solar cells powers a small boat or car with the light energy from a lightbulb. Maybe you own a solar charger for your boat battery or have decorative outdoor lighting that uses solar cells to recharge the batteries.

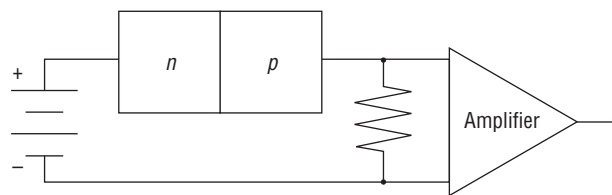
The solar cell takes the light energy it receives and converts it into electrical energy. In other words, the photons absorbed by the solar cell cause electrons to flow within the solar cell. This electron flow is called a *current*. The current from the solar cell flowing through the motor of the small boat or car causes the motor to rotate. The more current, the faster the motor rotates and the faster the boat or car travels.

Like the LED you learned about in Chapter 10, “Fiber-Optic Light Sources and Transmitters,” the basic photodiode is a semiconductor diode with a *p* region and an *n* region, as shown in Figure 11.1. Photons absorbed by the photodiode excite electrons within the photodiode in a process called *intrinsic absorption*. When stimulated with an outside bias voltage, these electrons produce a current flow through the photodiode and the external circuit providing the bias voltage.

FIGURE 11.1 PN photodiode

The *PN photodiode* is *reverse biased* when used in an electrical circuit. This is the opposite of how the LED is used in an electrical circuit. Reverse bias means that the *n* region of the photodiode is connected to a positive electrical potential and the *p* region is connected to a negative electrical potential.

In an electrical circuit, as shown in Figure 11.2, light that is absorbed by the photodiode produces current flow through the entire external circuit. As current flows through the *resistor*, it produces a voltage drop across the resistor. This voltage drop is input to an amplifier for amplification.

FIGURE 11.2 PN photodiode in an electrical circuit

You may be wondering why the output of the photodiode needs to be amplified since there is no amplification used with a solar cell. The solar cell is typically supplied with photons by a powerful light source such as the sun or a very bright lamp. The photodiode used in a fiber-optic receiver gets all of its light energy from the optical fiber connected to it. As you have learned in this book, the core of an optical fiber is extremely small and carries very little light energy in comparison to the energy that a solar cell receives. A photodiode that is stimulated by the light energy in an optical fiber does not produce a great amount of electrical current flow. This is why amplification is required. It's also a reason why different photodiodes have been developed for various fiber-optic receiver applications.

Other Types of Photodiode

The basic photodiode is called a PN photodiode, but there are two other types: the PIN photodiode and the avalanche photodiode. This section describes how they work.

PIN Photodiode

The *PIN photodiode* works like a PN photodiode; however, it is manufactured to offer better performance. The better performance comes in the form of improved efficiency and greater speed. Improved efficiency means that it has a better photon-to-carrier conversion ratio. If the same amount of light energy hit a PN photodiode and a PIN photodiode, the PIN photodiode would generate more current flow through an external circuit.

Greater speed means that the diode can turn on and off faster. Remember that in fiber optics the light pulses being sent by the transmitter happen at a very fast rate. The photodiode needs to be able to stop and start electron flow fast enough to keep up with the incoming light pulses.

The PIN photodiode shown in Figure 11.3 is constructed a little differently than the PN photodiode. An *intrinsic* layer is used to separate the *p* region and the *n* region. This creates a large depletion region that absorbs the photons with improved efficiency when compared to the PN photodiode.

FIGURE 11.3 PIN photodiode

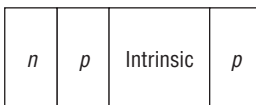


Avalanche Photodiode

The *avalanche photodiode (APD)* works just as its name suggests. On a snow-covered mountain, a small vibration can trigger an avalanche of snow. With the APD, a small bundle of photons can trigger an avalanche of electrons. The APD accomplishes this through a process called photomultiplication.

The APD is constructed with one more *p* region than the PIN photodiode, as shown in Figure 11.4. When the APD is biased very close to its breakdown voltage, it acts like an amplifier with a multiplication factor, or *gain*. An APD with a multiplication factor of 50 sets free, on average, 50 electrons for each photon absorbed. The free electrons produce current flow through the electrical circuit connected to the APD. The APD is typically used in receivers that operate with lower optical input power levels than those associated with PIN diodes. These receivers typically have data rates ≤ 1 Gbps.

FIGURE 11.4 Avalanche photodiode



Photodiode Responsivity, Efficiency, and Speed

This section describes the factors determining how responsive a photodiode is.

Responsivity

The *responsivity* of the photodiode describes how well the photodiode converts a wavelength or a range of wavelengths of optical energy into electrical current. It's the ratio of the photodiode electrical output current to its optical input power. The greater the responsivity, the greater the electrical current output for a given amount of optical input power.

Responsivity is described in amperes/watt (A/W). However, a photodiode in a fiber-optic receiver will never generate an ampere of electrical current. That's not to say that a photodiode *couldn't* be built to generate an ampere of electrical current, but remember that a photodiode in a fiber-optic receiver receives a very small amount of light energy. A typical LED receiver works well with an optical input power as low as one microwatt. One microwatt is one millionth of a watt.

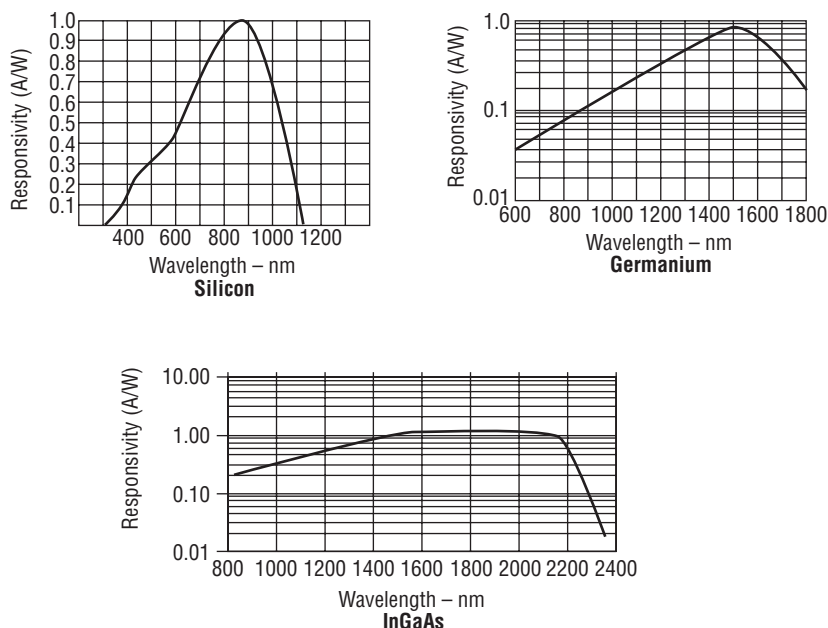
The overall responsivity of a photodiode depends on three factors:

- Semiconductor material makeup
- Wavelength
- Diode construction

Photodiodes are constructed for specific wavelengths. Some semiconductor materials perform better at longer wavelengths and some perform better at shorter wavelengths. Silicon photodiodes, for example, perform best in the visible and short infrared wavelengths. Germanium and indium gallium arsenic (InGaAs) photodiodes perform best at long infrared wavelengths, as shown in Figure 11.5. Diode construction also plays a large role in responsivity. The responsivity of an APD photodiode may be 100 or more times greater than a PIN photodiode.

Quantum Efficiency

The responsivity of a photodiode depends on its quantum efficiency. Quantum efficiency describes how efficiently a photodiode converts light energy into electrical energy—that is, photons into free electrons. It is typically expressed as a percentage. A quantum efficiency of 47 percent means that for every 100 photons absorbed, 47 current-generating electrons will be created.

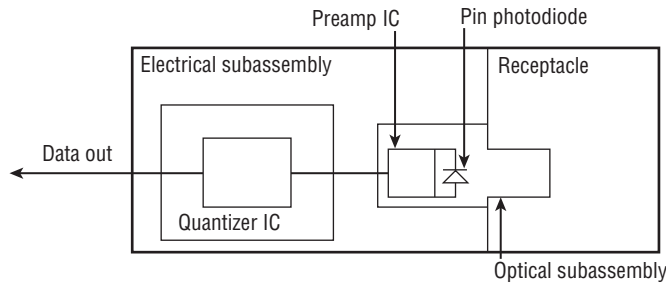
FIGURE 11.5 Photodiode semiconductor responsivity

Switching Speed

It has been stressed many times in this book that fiber-optic systems transmit and receive millions or billions of light pulses per second. When data is being moved at this rate, there is little time for the photodiode to switch on and off. Switching speed depends on physical size, construction, and electrical biasing. Photodiodes in fiber-optic receivers are small in size and biased to produce the fastest possible switching time.

Fiber-Optic Receiver

Many different fiber-optic receiver designs are in use today. The complexity of these receivers varies by application. The job of the receiver is to take the light energy from the optical fiber and convert it into electrical energy. The output of the receiver is designed to interface with electronics that handle the information after the light-to-electricity conversion. A typical receiver can be broken into three components: receptacle, optical subassembly, and electrical subassembly, as shown in Figure 11.6.

FIGURE 11.6 Block diagram of a typical LED receiver

Receptacle

Regardless of the receiver packaging, the *receptacle* is the part of the fiber-optic receiver that accepts the connector. Its job is to provide alignment for the ferrule or terminus so that the optical fiber is perpendicular to the photodiode in the optical subassembly.

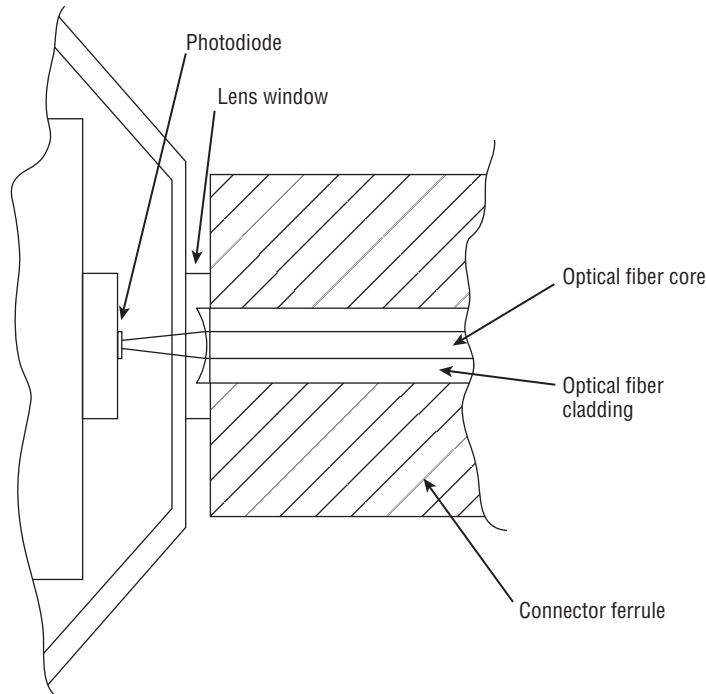
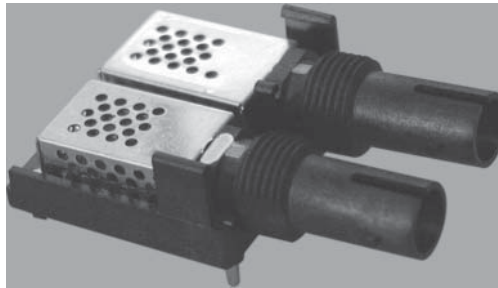
Optical Subassembly

The *optical subassembly* guides the light energy from the optical fiber to the photodiode. Depending on the receiver packaging, a window or terminus endface in the optical subassembly makes contact with the optical fiber endface. The window may be shaped like a lens to focus the light energy onto the photodiode, as shown in Figure 11.7, or a lens may be placed between the window and the photodiode.

In many receivers, the photodiode and preamplifier are housed in the optical subassembly. Sometimes the optical subassembly is referred to as the photodiode preamplifier subassembly. As we learned earlier in this chapter, the photodiode used in a receiver does not produce a substantial electrical signal. Mounting the preamplifier as physically close as possible to the photodiode allows the preamplifier to receive the strongest electrical signal.

Electrical Subassembly

The *electrical subassembly* is typically built on a multilayer printed circuit board. This printed circuit contains the *quantizer IC* and other passive components required to complete the electrical circuit. The photodiode and preamplifier contained in the optical subassembly are electrically connected to the printed circuit board. A metal shield, shown in Figure 11.8, is typically placed over the electrical subassembly. The shield serves two purposes; it protects the electrical subassembly from external electromagnetic radiation, and it reduces the amount of electromagnetic radiation from the electrical subassembly.

FIGURE 11.7 Photodiode and window**FIGURE 11.8** Transceiver module with EMI shield exposed

The photodiode converts the light pulses from the optical fiber to electrical current pulses, as discussed earlier in this chapter. The *transimpedance amplifier*, or preamplifier, amplifies the electrical current pulses from the photodiode and outputs voltage pulses to the quantizer IC. The *limiting amplifier* in the quantizer IC amplifies the voltage pulses and provides a binary decision. It determines whether the electrical pulses received represent a binary 1 or a binary 0.

The quantizer IC also measures the received optical energy. It sets the signal detect line when there is adequate signal strength to convert the light energy into electrical energy. This prevents the electronics from trying to decode a weak signal or noise.

Receiver Optical Performance Characteristics

This section examines fiber-optic receiver dynamic range and operating wavelength. The data used in this section was extracted from manufacturer datasheets and represents typical performance characteristics of readily available laser or LED fiber-optic receivers.

Dynamic Range

Fiber-optic receivers are limited in the amount of optical input power they can receive. Too much optical input power will saturate the photodiode. Saturating the photodiode prevents the photodiode from turning off after the light pulse has been absorbed. This can cause electrical output pulses of the receiver to overlap, creating a *bit error*.

The dynamic range of the receiver is measured in decibels. It is the difference between the maximum and minimum optical input power that the receiver can accept. If the maximum optical input power is -14dBm and the minimum optical input power is -32dBm , the dynamic range would be the difference between the two values, or 18dB .

The receiver will generate a minimum number of errors when the optical input power is kept within the minimum and maximum values. All receivers generate errors. Error generation is typically described by the receiver's *bit error rate* (BER). A typical receiver may have a BER of one error in a billion to one error in a trillion. Typically this would be written as 10^{-9} or 10^{-12} , respectively.

Operating Wavelength

Fiber-optic receivers are designed to operate within a range of wavelengths. A typical 1300nm receiver may have an operating wavelength range from 1270nm to 1380nm . This is because fiber-optic transmitters output optical energy within a wavelength range. The receiver must be able to accept a 1300nm transmitter that has a center wavelength of 1275nm or 1375nm . A receiver designed for 1300nm may not perform well or may not perform at all when connected to an 850nm or 1550nm transmitter because the responsivity of the photodiode changes over a range of wavelengths.

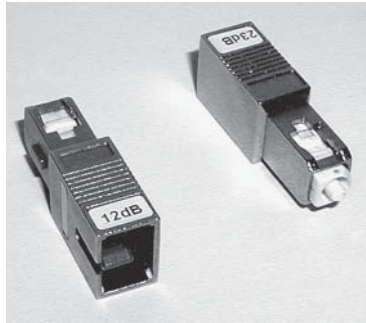
Optical Attenuators

An *optical attenuator* is a passive device that is used to reduce the power level of an optical signal. The attenuator circuit will allow a known source of power to be reduced by a predetermined factor, which is usually expressed as decibels. Optical attenuators are generally used in single-mode long-haul applications to prevent optical overload at the receiver.

Optical attenuators typically come in two forms of packaging. The bulkhead optical attenuator shown in Figure 11.9 can be plugged into the receiver receptacle. The inline

attenuator resembles a patch cord and is typically used between the patch panel and the receiver.

FIGURE 11.9 Bulkhead optical attenuator



Types of Attenuators

The types of attenuators generally used are fixed, stepwise variable, and continuously variable.

Fixed Attenuators

Fixed attenuators are designed to have an unchanging level of attenuation. They can theoretically be designed to provide any amount of attenuation that is desired. The output signal is attenuated relative to the input signal. Fixed attenuators are typically used for single-mode applications.

Stepwise Variable Attenuators

A stepwise variable attenuator is a device that changes the attenuation of the signal in known steps such as 0.1dB, 0.5dB, or 1dB. The stepwise attenuator may be used in applications dealing with multiple optical power sources—for example, if there are three inputs available, there may be a need to attenuate the signal at a different level for each of the inputs.

Conversely, the stepwise attenuator may also be used in situations where the input signal is steady, yet the output requirements change depending on the device that the signal is output to.

The stepwise attenuator should be used in applications where the inputs, outputs, and operational configurations are known.

Continuously Variable Attenuators

A continuously variable attenuator is an attenuator that can be changed on demand. These attenuators generally have a device in place that allows the attenuation of the signal to change as required. A continuously variable attenuator is used in uncontrolled environments where the input characteristics and/or output needs continually change. This allows the operator to adjust the attenuator to accommodate the changes required quickly and precisely without any interruption to the circuit.

Calculating the Attenuation Value

The key to choosing the appropriate attenuation value is to know the maximum optical input power the receiver can tolerate while maintaining a low BER. In this example, let's assume that the maximum optical input power is -6dBm . If the input power exceeds this power level, the receiver will be overloaded. The transmitter, which is located 10km from the receiver, has an output power of 3dBm . The loss for the 10km of optical fiber, including interconnections, is 5dB .

To calculate the minimum attenuation required to prevent the receiver from being overloaded, we need to subtract all the known losses from the output power of the transmitter as shown here:

Transmitter power (TP) = 3dBm

Receiver maximum optical input power (MP) = -6dBm

Total losses (TL) = 5dB

Minimum attenuation required = $\text{MP} + \text{TL} - \text{TP}$

$-6\text{dBm} + 5\text{dB} - 3\text{dBm} = -4\text{dB}$

At a minimum, a 4dB attenuator is required. However, an attenuator with a larger value could be used as long as it did not overattenuate the signal.

Summary

This chapter covered photodiode fundamentals and the basic optical, electrical, and mechanical components found in a typical fiber-optic receiver. We examined the basic optical performance characteristics of the receiver and learned when an optical attenuator is required.

Exam Essentials

Describe the basic operation of a photodiode. Be able to describe the basic operation of a photodiode. Remember that the job of the photodiode is to convert optical energy into electrical energy.

Describe the basic components in a fiber-optic receiver. Be able to describe the functionality of the three basic components in a fiber-optic receiver. Remember that a receiver is typically made up of an electrical subassembly, optical subassembly, and receptacle.

Describe receiver optical performance characteristics. You should know that the optical characteristics of the receiver at a minimum include minimum optical input power, maximum optical input power, and operating wavelength. Be able to explain when an optical attenuator is required to prevent optical overload at the receiver.

Review Questions

1. Fiber-optic receivers use a(n) _____ to convert the light energy from the optical fiber into electrical energy.
 - A. Photoresistor
 - B. Photodiode
 - C. Phototransistor
 - D. LED

Hint: It is reverse biased when used in an electrical circuit.

2. When photons are absorbed by a photodiode, they excite electrons within the photodiode in a process called _____ absorption.
 - A. Intrinsic
 - B. Extrinsic
 - C. Internal
 - D. External

Hint: The electrons produce a current flow through the photodiode.

3. A small number of photons absorbed by the _____ photodiode can cause a large flow of electrons.
 - A. PN
 - B. PIN
 - C. Avalanche
 - D. Unbiased

Hint: Uses a process called photomultiplication.

4. _____ range is the difference between the maximum and minimum optical input power that the receiver can accept.
 - A. Maximum
 - B. Minimum
 - C. Dynamic
 - D. Quantum

Hint: It is measured in decibels.

5. Error generation in a fiber-optic receiver is typically described by the receiver's _____.
 A. BRE
 B. BER
 C. RER
 D. FORER

Hint: A typical receiver may have one error in a billion.

6. The _____ is the part of the fiber-optic receiver that accepts the connector.
 A. Receptacle
 B. Outlet
 C. Sleeve
 D. Cap

Hint: It also aligns the ferrule.

7. In a fiber-optic communication system, a(n) _____ may be required to reduce the optical power level at the receiver.
 A. Photodiode
 B. Resistor
 C. Voltage drop
 D. Attenuator

Hint: They are generally used in single-mode long-haul applications.

Chapter Exercises

Calculate the Dynamic Range for an LED Receiver

The dynamic range of the LED receiver is the difference between the minimum value for the maximum optical input power and the maximum value for the minimum optical input.

Refer to the following table and calculate the dynamic range for the LED receiver.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Optical input power minimum at window edge	$P_{IN\ Min}$		-30.5	-29	dBm avg.
Optical input power maximum	$P_{IN\ Max}$	-17	-13.8		dBm avg.

To calculate the dynamic range, subtract the minimum value for the maximum optical input power ($P_{IN\ Max}$) from the maximum value for the minimum optical input power ($P_{IN\ Min}$) as shown in this formula:

$$P_{IN\ Max} - P_{IN\ Min} = \text{dynamic range}$$
$$-17\text{dBm} - -29\text{dBm} = 12\text{dB}$$
$$\text{Dynamic range} = 12\text{dB}$$

Calculate the Dynamic Range for a Laser Receiver

The dynamic range of the laser receiver is the difference between the minimum value for the maximum optical input power and the maximum value for the minimum optical input.

Refer to the following table and calculate the dynamic range for the laser receiver.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Receiver sensitivity	$P_{IN\ Min}$		-20	-18	dBm avg.
Receiver overload	$P_{IN\ Max}$	-3	+3		dBm avg.

To calculate the dynamic range, subtract the minimum value for the maximum optical input power ($P_{IN\ Max}$) from the maximum value for the minimum optical input power ($P_{IN\ Min}$) as shown in this formula:

$$P_{IN\ Max} - P_{IN\ Min} = \text{dynamic range}$$
$$-3\text{dBm} - -18\text{dBm} = 15\text{dB}$$
$$\text{Dynamic range} = 15\text{dB}$$

Chapter 12

Cable Installation and Hardware

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Explain manufacturer installation cable specifications.
- ✓ Discuss ANSI/TIA-568-C performance specifications for the optical fiber cables recognized in premises cabling standards to include:
 - Inside plant cable
 - Indoor-outdoor cable
 - Outside plant cable
 - Drop cable
- ✓ Explain the static and dynamic loading on a fiber-optic cable during installation.
- ✓ Describe commonly used installation hardware.
- ✓ Summarize the following types of preparation:
 - Patch panel
 - Racks and cable
 - Splice enclosure
- ✓ Describe the following types of installations:
 - Tray and duct
 - Conduit
 - Direct burial
 - Aerial
 - Blown fiber
 - Wall plate



- ✓ Explain entrance cable bonding and grounding per NEC Articles 250, 770.93, and 770.100.
- ✓ Recognize that ANSI/TIA-606-B concisely describes the administrative record keeping elements of a modern telecommunications infrastructure.
- ✓ Explain that administration includes basic documentation and the timely updating of drawings, labels, and records.
- ✓ Explain why proper polarity is required to ensure the operation of bidirectional fiber-optic communication systems.
- ✓ Explain the roles of the following:
 - National Electrical Code (NEC)
 - Canadian Electrical Code (CEC)
 - National Electrical Safety Code (NESC)





Under the right conditions, an optical fiber will carry light great distances almost instantly. If the cable is not installed properly, though, the light it is supposed to carry may not even

travel from one part of an office building to another.

Proper installation depends on a thorough knowledge of the strengths and limits of a fiber-optic cable, as well as the methods for protecting the cable from damage both during the installation and over its lifetime. The cable must endure the pulling force that is necessary for it to be put in place; environmental conditions that threaten to freeze, soak, or otherwise damage it; and the daily stresses that result from its location or position.

The rules governing fiber-optic cable installation are designed to minimize the short-term and long-term stresses on the cable as well as ensure that the installation conforms to codes governing fire safety. Some of these requirements were discussed in Chapter 7, “Fiber-Optic Cables,” with regard to the structure of the cable itself. The rules governing cable construction can only be effective, though, if the cable is installed properly.

This chapter describes the requirements for a successful fiber-optic cable installation. I discuss the conditions affecting fiber-optic cable and the methods for installing the cable so that the effects of those conditions are minimized. I describe regulations for electrical safety in cable installations as well as methods for routing cables in different situations. The chapter also examines the ways in which cables are enclosed and terminated, including proper labeling methods.

In this chapter, you will learn to

- Determine the minimum cable bend radius
- Determine the maximum cable tensile rating
- Determine the fill ratio for a multiple-cable conduit installation

Installation Specifications

Many fiber-optic cables are created for specific types of service and must be installed in accordance with their manufacturer’s specifications. Though some specifications are concerned with the particular duty or job a cable will perform, others apply to traits shared by all cables. Two of these specifications are *bend radius* and *tensile strength* or *pull strength*. Both specifications apply to conditions faced by fiber-optic cable while it is being installed and, once it has been installed, to its normal working conditions.

Two standards that apply to cable installation are ANSI/TIA-568-C.0 and ANSI/TIA-568-C.3. Section 5 of ANSI/TIA-568-C.0 addresses copper and optical fiber cabling installation requirements. The installation requirements for optical fiber cabling

are described in Section 5.4; minimum bend radius and maximum pulling tension requirements are defined in Section 5.4.1.

Section 4 of ANSI/TIA-568-C.3 defines performance standards for optical fiber cables recognized in premises cabling standards. The physical requirements for optical fiber cables are defined in Section 4.3. Four cable types are covered in this section and in Section 5.4.1 of ANSI/TIA-568-C.0: *inside plant*, *indoor-outdoor*, *outside plant*, and *drop cable*.

Inside plant cables Designed for installation in the building interior.

Indoor-outdoor cables Designed for installation in the building interior or exterior to the building. These cables are not designed for long-haul outdoor applications. They are typically used to provide connectivity between two buildings. Unlike an outside plant cable, these cables resist the spread of fire as defined in Article 770 of the National Electrical Code (NEC).

Outside plant cables Designed for outdoor installations. These cables typically do not resist the spread of fire as defined in Article 770 of the NEC. Because of this, the length of outside plant cable within a building cannot exceed 50' (15.2m) from the point of entrance into the building. To comply with Article 770 of the NEC, these cables must also be terminated in an enclosure. This is discussed in detail in the section “Fire Resistance Bonding and Grounding” later in this chapter.

Drop cables Designed to link a drop terminal to a premise terminal. A terminal is a device that is capable of sending and/or receiving information over a communications channel that, in this case, uses an optical fiber. The service provider typically owns the drop terminal.

Bend Radius

As you have seen in previous chapters, optical fiber depends on the maintenance of total internal reflection to carry an optical signal. Macrobends, or very small radius bends in the optical fiber, can change the light's angle of incidence enough to cause some or a majority of the light to pass into the cladding, severely attenuating the signal or cutting it off completely.

Manufacturers often specify a minimum installation and operational bend radius for their optical fiber cables. The minimum installation bend radius is the short-term bend radius, and the minimum operational bend radius is the long-term bend radius. The minimum short-term bend radius is actually the larger of the two because of the tensile stresses that may be placed on the cable during installation.

Following the manufacturer's guidelines reduces the risk of damage to the cable and optical fiber during and after installation. It also reduces the risk of macrobends. Keep in mind that bends approaching the minimum bend radius might cause some attenuation. However, how much attenuation occurs depends on multiple factors, including cable construction, optical fiber type, and the wavelength of the light source.

Another, more fundamental reason for the minimum bend radius is that the optical fibers, though flexible, are not indestructible. Bending beyond the minimum radius may cause the fiber to break inside the cable, which would require repair at the very least and replacement of the entire cable as a worst-case scenario. In addition, overbending may

cause the optical fiber to fail days, months, or years after the installation. It is impossible to predict when an overbending failure may occur. Environmental factors such as temperature, shock, and vibration stress an optical fiber, and typically the greater the stress, the sooner the optical fiber will fail.

Both ANSI/TIA-568-C.0 and ANSI/TIA-568-C.3 define the bend radius requirements for inside plant, indoor-outdoor, outside plant, and drop cables.

Inside plant cables are broken into three groups:

- Cables with four or fewer optical fibers intended for Cabling Subsystem 1
- Cables with four or fewer fibers intended to be pulled through *pathways* during installation
- All other inside plant cables

Inside plant cables intended for Cabling Subsystem 1 run from an *equipment outlet* to a *distributor* in the *hierarchical star topology*. This distributor may be an optional connection facility, intermediate connection facility, or central connection facility. These cables must support a bend radius of 25mm (1") when not subject to a tensile load. In other words, 25mm is the minimum bend radius for this cable type.

Inside plant cables with four or fewer fibers intended to be pulled through pathways during installation must support a bend radius of 50mm (2") while under a pull load of 220N (50 pound-force [lbf]). For all other inside plant cables, the bend radius is based on the cable diameter and the loading conditions. Under no tensile load or pull load, these cables must support a bend radius of 10 times the cable's outside diameter. At the maximum tensile load rating or pull load defined by the manufacturer, these cables must support a bend radius of 20 times the cable's outside diameter.

Indoor-outdoor cables, outside plant cables, and drop cables under no tensile load or pull load must support a bend radius of 10 times the cable's outside diameter. At the maximum tensile load rating or pull load defined by the manufacturer, these cables must support a bend radius of 20 times the cable's outside diameter.



While ANSI/TIA-568-C.0 and ANSI/TIA-568-C.3 provide bend radius requirements, not all cables comply with these requirements. You should always refer to the manufacturer's datasheet for the bend radius and pull load performance for each cable being installed.

Tensile Rating

Recall that there are different types of strength members in fiber-optic cables. The job of the strength member is to ensure that no tensile stress is placed on the optical fiber during and after installation. The strength member does have physical limitations. Tensile forces that exceed the physical limitations the cable was designed to handle can damage the cable and possibly break the optical fiber. Excessive tensile loading may also create macrobends, causing the optical fiber to attenuate a signal.

There are two types of tensile loads:

- A *static load* is a tensile load on a cable that does not change. A static load is often referred to as the *operational load*.
- A *dynamic load* is a changing tensile load. Dynamic load is referred to as the *installation load*.

ANSI/TIA-568-C.3 Section 4.3 defines the minimum pull strength requirements for inside plant, indoor-outdoor, outside plant, and drop cables. *Pull load*, *pull strength*, and *tensile load* are all the same type of load on a cable, and the terms may be interchanged.

Inside plant cables As you learned in the section “Bend Radius” earlier in the chapter, inside plant cables are broken into three groups. The minimum pull load requirements only apply to cables with four or fewer fibers intended to be pulled through pathways during installation. The minimum pull load for these cables is 220N (50lbf).

Indoor-outdoor cables Indoor-outdoor cables with 12 or fewer fibers must have a minimum pull strength or tensile load rating of 1335N (300lbf). Cables with more than 12 fibers must have a minimum tensile load rating or pull strength of 2670N (600lbf).

Outside plant cables The minimum tensile load rating or pull strength for an outside plant cable is not dependent on the fiber count. All outside plant cables must have a minimum pull strength of 2670N (600lbf).

Drop cables The minimum pull strength for a drop cable depends on how the cable is installed. A drop cable that is directly buried, placed by trenching, or blown into a duct must have a minimum pull strength 440N (100lbf). If the cable is installed by pulling, the minimum pull strength is 1335N (300lbf).

When installing any cable type, always refer to the manufacturer’s datasheet for physical requirement specifications. Some cables may or may not meet the minimum physical requirements defined in ANSI/TIA-568-C.0 and C.3. Other cables may exceed these requirements. The maximum tensile load and minimum bend radius specifications for all the optical fiber cable types defined in ANSI/TIA-568-C.0 and C.3 are summarized in Table 12.1.

TABLE 12.1 ANSI/TIA-568-C.0 and C.3 maximum tensile load and minimum bend radius summary

Cable type	Maximum load: short term	Minimum bend radius: short term	Minimum bend radius: long term
Inside plant cable with four or fewer optical fibers intended for Cabling Sub- system 1	50lbf 220N	2" 50mm	1" 25mm
Inside plant cable with four or fewer fibers intended to be pulled through path- ways during installation	50lbf 220N	2" 50mm	1" 25mm

Cable type	Maximum load: short term	Minimum bend radius: short term	Minimum bend radius: long term
All other inside plant cables	Per manufacturer	20 times the cable's outside diameter	10 times the cable's outside diameter
Indoor-outdoor cables with 12 or fewer fibers	300lbf 1335N	20 times the cable's outside diameter	10 times the cable's outside diameter
Indoor-outdoor cables with more than 12 fibers	600lbf 2670N	20 times the cable's outside diameter	10 times the cable's outside diameter
Outside plant cables	600lbf 2670N	20 times the cable's outside diameter	10 times the cable's outside diameter
Drop cable installed by pulling	300lbf 1335N	20 times the cable's outside diameter	10 times the cable's outside diameter
Drop cable that is directly buried, placed by trench- ing, or blown into a duct	100lbf 440N	20 times the cable's outside diameter	10 times the cable's outside diameter

Let's examine the physical requirements specification for two fiber-optic cables. Table 12.2 lists the tensile load and bend radius specifications from a manufacturer's datasheet for two fiber-optic cables. This table defines the minimum bend radius and the maximum tensile load or pull load for each cable. These values are defined for short- and long-term operation.

The cables described in Table 12.2 are inside plant cables. The single-fiber cable is shown in Figure 12.1.

FIGURE 12.1 Single fiber cable with ST connectors



The 12-fiber cable is shown in Figure 12.2. Both cables use a tight-buffered fiber surrounded by aramid yarn strength members with a flexible jacket.

TABLE 12.2 Inside plant cable physical requirements specification

Cable Type	Diameter	Weight	Minimum bend radius: short term	Minimum bend radius: long term	Maximum load: short term	Maximum load: long term
Single-fiber cable	0.114", 2.9mm	6lb/kft, 9kg/km	1.8", 4.5cm	1.2", 3.0cm	50lbf, 220N	15lbf, 66N
12-fiber cable	0.26", 6.6mm	27lb/kft, 40kg/km	3.9", 9.9cm	2.6", 6.6cm	300lbf, 1,334N	90lbf, 396N

FIGURE 12.2 12-fiber cable



Note that the weight of the each cable over distance is included in Table 12.2. The reason for including the weight is that it can add to the tensile load if the fiber is to be suspended or hung vertically for long distances. In the single-fiber cable, for example, the long-term load limit is 15 pounds, while its weight is 6 pounds per thousand feet. If the cable must support itself for a length of 500', it already has 3 pounds of tensile loading on it.



When you're working with cable diameter and bend radius data, it is easy to mistake bend diameter for bend radius. Remember bend diameter is twice the bend radius. If a cable has a diameter of 1" and the bend radius is 10 times the diameter, the bend radius is 10" and the bend diameter is 20". Confusing the two may lead to overbending the cable.

Note also that the manufacturers' specified bend radius for installation of the 12-fiber cable is about 15 times the cable diameter, exceeding the ANSI/TIA-568-C.0 and C.3 minimum requirements. The long-term bend radius is about 10 times the cable diameter, meeting the ANSI/TIA-568-C.0 and C.3 minimum requirements. The installation tensile load is about three times as much as the long-term tensile load. Remember that the figures used in this example are for a cable that is installed inside a building. Cables that will be installed outside of a building will typically have a larger minimum bend radius.

Installation Hardware

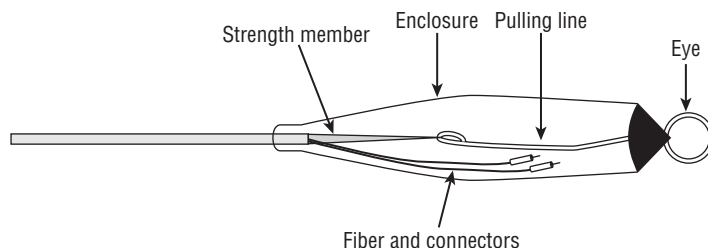
Fiber-optic cables and the optical fibers within them have a number of specialized requirements for their installation and termination. From cable-pulling tools to cable-protecting enclosures, installation hardware has been specially designed and built to meet the needs of fiber-optic cables in almost any environment and situation.

Let's take a look at some of the hardware commonly used in fiber-optic installation.

Pulling Eye

Sooner or later, you will need to run a cable through a wall, conduit, or other inaccessible space. An indispensable tool for this job is the pulling eye, shown in Figure 12.3. This device is specially designed to attach to the cable's strength member at one end and a pulling line at the other. The pulling line is fed through the space to be occupied by the cable. The line is then used to pull the eye through the space with the cable attached.

The pulling eye also uses a sheath that encloses the fiber ends to protect them from damage while the cable is being pulled.

FIGURE 12.3 The pulling eye is used to pull cable through the conduit.

Pull Box

Optical fiber is typically small enough, light enough, and flexible enough to be relatively easy to pull through conduits. However, friction is always a concern. Friction on the cable as it is pulled through the conduit will increase the tensile loading. Increases in friction as the cable is pulled through a turn or several turns can cause the tensile load to exceed its maximum.

To make the cable easier to pull and to ease the tensile load on it, pull boxes are installed at intervals in the conduit. Typically, pull boxes are installed after long straight runs and every time a set of turns totals 180° or more. The purpose of a pull box is to create an intermediate opening for pulling the cable to reduce the length that is being pulled through the conduit and to reduce the number of turns through which the cable must be pulled at any one time.

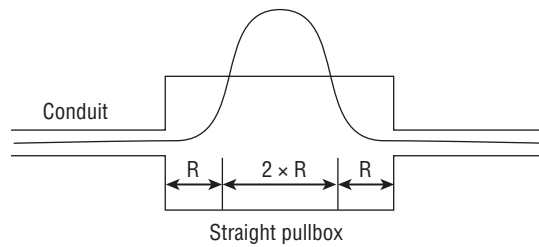
To use a pull box, pull the fiber-optic cable through the box and out the large opening. Be very conscious of how much the cable is being bent while you are doing this. Also be careful not to place a bend in the cable that has a radius less than the minimum bend radius defined by the manufacturer. Once the cable has been pulled as far as it will go, feed it into the other side of the box and down the conduit, again paying close attention to the bend radius. This process minimizes the stress placed on the cable.

There are two types of pull boxes, as shown in Figure 12.4: straight and corner.

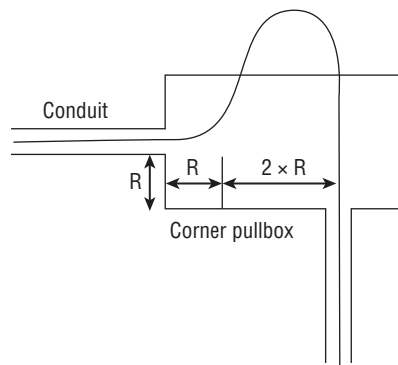
Straight pull box Straight pull boxes are installed in-line in the conduit and should have an opening at least four times the minimum bend radius of the cable being pulled. This length prevents the cable from exceeding its minimum bend radius as the last of it is pulled through the box.

Corner pull box Corner pull boxes are installed at angles in the conduit and typically require a length of three times the cable's bending radius and a depth that equals or exceeds the bending radius. This requirement prevents the cable from dragging against a sharp turn when it is pulled through.

FIGURE 12.4 Straight and corner pull boxes ease tensile loading on fiber during installation.



R = Manufacturer's specified minimum bend radius



Splice Enclosures

Any time you have a splice in an optical fiber, whether it is a mechanical or a fusion splice, you must protect it from exposure and strain. *Splice enclosures* take many forms, depending on their location and specific application. Some have been adapted from electrical splice enclosures used in the telecommunications industry for aerial and underground cable, whereas others are designed specifically for optical fibers and are used for indoor installations.

Splice enclosures can typically be placed into two categories:

Radial splice enclosure With a radial splice enclosure, cables enter and exit the enclosure through the same side, as shown in Figure 12.5.

Axial splice enclosure With an axial splice enclosure, cables enter and exit the enclosure at opposite ends, as shown in Figure 12.6. Both types can be used for an aerial or direct burial installation. However, the radial is typically used with a pedestal enclosure.

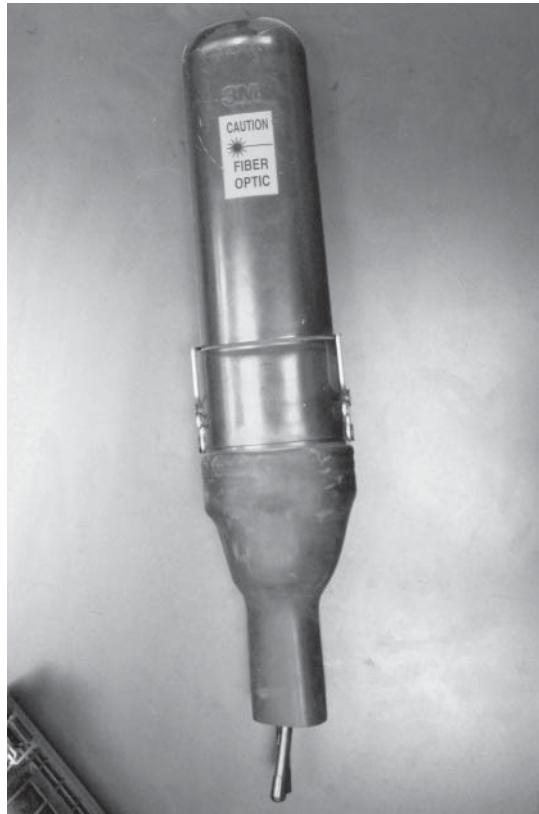
FIGURE 12.5 Radial splice enclosure

Photo Courtesy Of Kitco Fiber Optics

The *pedestal enclosure* shown in Figure 12.7 is used when the fiber-optic cable has been buried underground. The pedestal enclosure is placed over the cables as they enter or exit the ground.

Splice enclosures designed for outdoor applications like those shown in Figures 12.5 and 12.6 are environmentally sealed. However, indoor splice enclosures may or may not be environmentally sealed. The rack-mounted splice enclosure shown in Figure 12.8 is for indoor applications and is not environmentally sealed.

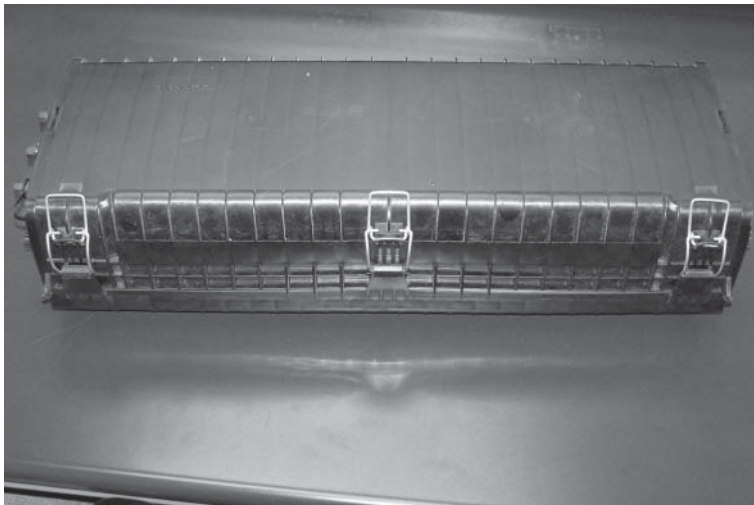
FIGURE 12.6 Axial splice enclosure

Photo Courtesy Of Kitco Fiber Optics

FIGURE 12.7 Fiber-optic pedestal

Photo Courtesy Of Kitco Fiber Optics

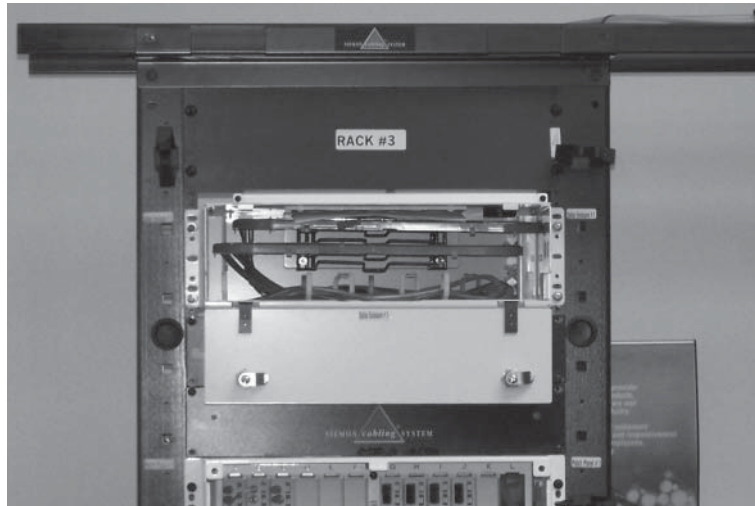
FIGURE 12.8 Rack-mounted splice enclosure

Photo Courtesy Of Kitco Fiber Optics

Typically, splice enclosures will incorporate the following features:

- A strain-relief system that ensures the strength member will carry all of the tensile loading.
- Clips incorporated into a panel or tray, or bonded to the enclosure. The clips hold the actual splices in an orderly fashion.
- Space for looping the extra optical fiber required to perform the splice outside of the enclosure.

The splice enclosure shown in Figure 12.9 has three mechanical splices and three fusion splices. The clips that hold the splices are bonded to the enclosure. The mechanical splices are held in the top three clips and the fusion splices in the bottom three. Notice that this splice enclosure allows sufficient space for the optical fiber to be bent without exceeding its minimum bend radius.

Patch Panels

A *patch panel* is an interconnection point for fiber-optic cables. They allow signals to be routed from one cable to another with a patch cord or jumper. Patch panels are available in many different shapes and sizes. Patch panels are often mounted in a rack, as shown in Figure 12.10.

FIGURE 12.9 A splice enclosure with three mechanical and three fusion splices

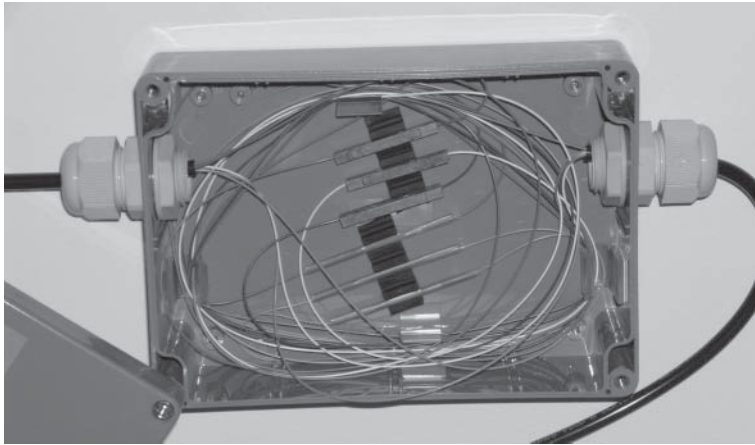


FIGURE 12.10 A rack-mounted patch panel

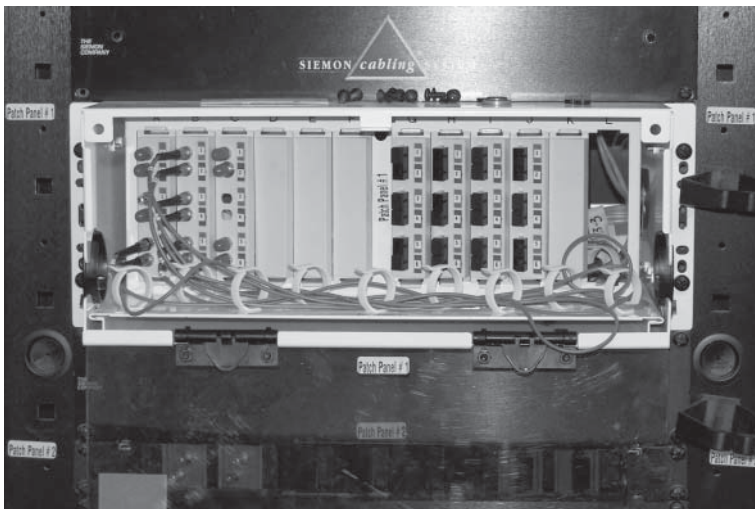


Photo Courtesy Of Kitco Fiber Optics

However, patch panels may be contained within an enclosure, as shown in Figure 12.11. The patch panel contained within the wall-mounted enclosure shown in Figure 12.11

allows the interconnection of the four fiber-optic cables entering the box from the top. Signals from one cable can be routed to another simply by making a connection between the two optical fibers with a jumper. The jumpers to the right of the patch panel provide an optical path from one fiber-optic cable to another.

FIGURE 12.11 An enclosure-mounted patch panel

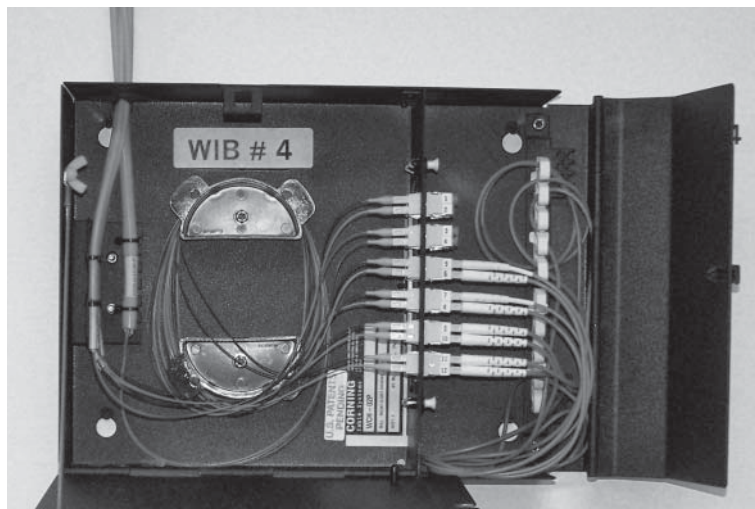


PHOTO COURTESY OF KITCO FIBER OPTICS

Patch panels use a bulkhead-mounted *mating sleeve* or adapter to make the interconnection. These mating sleeves can feature identical connector receptacles or different receptacles. Figure 12.12 shows an ST-to-ST mating sleeve with connectors on each end. Figure 12.13 shows an SC-SC mating sleeve with connectors on each end, and Figure 12.14 shows an LC-LC mating sleeve with connectors on each end.

FIGURE 12.12 ST-to-ST mating sleeve with connectors

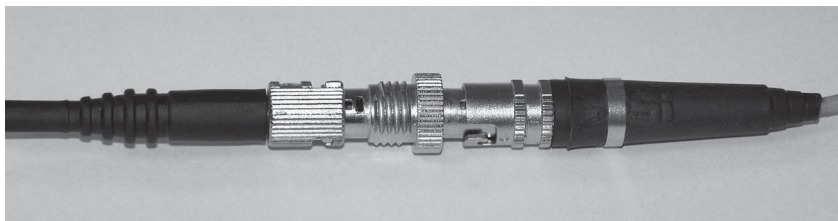


FIGURE 12.13 SC-to-SC mating sleeve with connectors

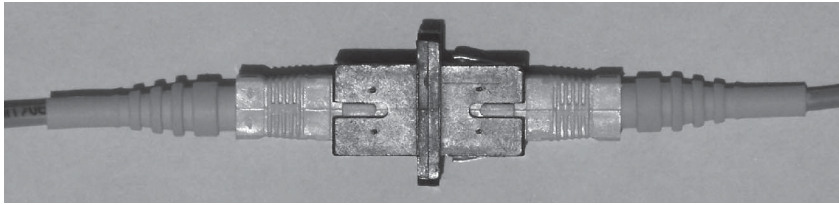
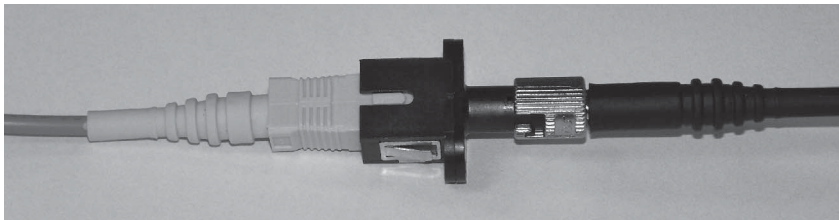


FIGURE 12.14 LC-to-LC mating sleeve with connectors



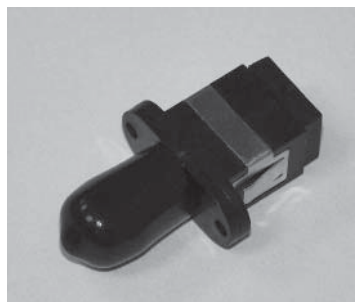
Figure 12.15 shows a mating sleeve that mates an ST connector with an SC connector. Mating sleeves with different connector receptacles are often referred to as *hybrid adapters*.

FIGURE 12.15 ST-to-SC mating sleeve or hybrid adapter with connectors



Like connectors, receptacles have dust caps to prevent contamination. Figure 12.16 shows an ST-to-SC hybrid adapter with the dust caps in place.

FIGURE 12.16 ST-to-SC hybrid adapter with dust caps installed



Installation Methods

Optical fiber has already reached into most of the places that once only knew copper cable. As technology, regulations, and pricing permit, fiber may ultimately replace copper for most signal-carrying applications, even into the home.

Many fiber installations resemble those used for copper wiring and have built on the lessons learned from it. There are some installation requirements and methods unique to fiber, however, that are required to protect the cable and ensure the highest-quality transmission.

Let's look at a typical application for optical fiber that uses a variety of installation methods. In our example, shown in Figure 12.17, a manufacturing plant is using fiber to carry instrumentation and control signals between the production building and another building several hundred meters away (called the control building in the diagram). The simplex cables carrying the signals must be collected in a central area and routed into a multi-fiber cable that runs through trays and ductwork to the outside. The cable then runs from the building and is strung across several poles until it reaches a road. There, the cable runs underground until it enters the next building and is distributed to data collection and control systems.

FIGURE 12.17 A sample fiber installation scenario



Let's look at installation for each of these situations.

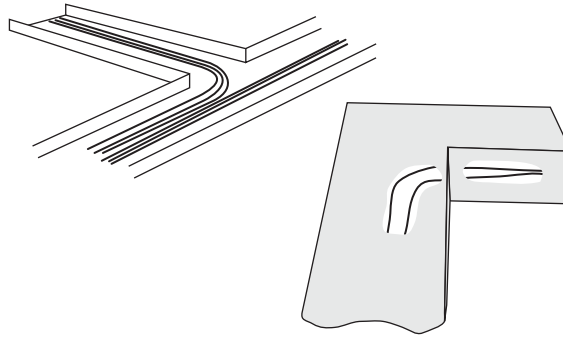
Tray and Duct

Tray and duct installation is used inside structures and is similar to installation methods used for electrical wiring. Because many optical fiber cables are nonconductive, some of the requirements and restrictions for copper cables do not apply to fiber.

When the optical fiber cable rests in trays or horizontal ductwork, as shown in Figure 12.18, the weight of the cable is usually not a factor as long as the runs remain on the same vertical level. If optical fiber cabling is run vertically, however, the cable will have

to support itself or be secured using either cable clamps or hangers. Be sure to follow the cable manufacturer's specifications for vertical cable rise.

FIGURE 12.18 Tray and duct installation



There are two methods for clamping cables for vertical runs:

- One type of clamp, which can also be used to secure cables horizontally, secures the cable directly to the surface by placing pressure against it. These should be installed carefully to prevent crushing the cable with excessive force.
- The second type of vertical clamp is a tight wire mesh that wraps around the cable and is secured to a hanger. This method has the advantage of reducing pressure against the cable itself while still supporting it. It also allows the cable to be removed more easily, since the mesh wrap is slipped over the hanger, and not permanently attached.

When installing cable in a tray, be aware of whether the tray will also be occupied by other cables, especially electrical cables. Aside from safety considerations surrounding the use of conductive or composite optical fiber cables, the weight of the much larger copper cables can cause macrobends if they are piled on top of the fiber. If there is a risk of this happening, choose an optical fiber cable that will protect optical fiber inside from the crushing weight of other cables.

If you are laying the cable into the tray, rather than pulling it, you can use the manufacturer's minimum long-term bend radius specification. If you will be pulling it, however, be sure to leave some extra radius to account for the tensile loading.

Whether you are installing in a tray or in a duct, do not allow the fiber to contact any sharp edges or exceed the minimum bend radius. Ideally, you should be working with

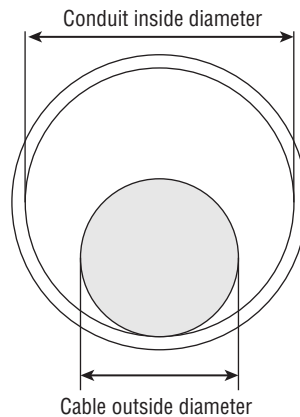
components that do not have these hazards built in; however, if these hazards exist you should keep cable tension as low as possible during installation.

Conduit

Conduit installation uses dedicated conduits for the cable runs, which may be installed by feeding pulling lines through the conduit with a fish tape, attaching the cable's strength member to a pulling eye, and then pulling the cable through the conduit.

Conduit may be run inside structures or underground, and in many cases, conduits may already be in place for other applications, such as power or telephone lines. When you are installing cable in conduits within a structure, be sure that you have allowed enough room in the conduit for the cable or cables you are installing, as shown in Figure 12.19. If possible, account for future expansion by installing a cable with extra optical fibers.

FIGURE 12.19 The conduit must leave room for fiber to be pulled.



When preparing for a conduit installation, you must be aware of the *fill ratio* for the conduit. If you are installing cable in existing conduit already occupied by one or more cables containing electrical conductors, you will need to determine the fill ratio of the conduit prior to selecting your cable. The NEC specifies the following maximum fill ratios by cross-sectional area for conduit:

- 1 cable: 53 percent
- 2 cables: 31 percent
- 3 or more cables: 40 percent

To determine the fill ratio, use the following formula:

$$\text{Fill Ratio} = (OD_{\text{cable}_1} + OD_{\text{cable}_2} \dots) / ID_{\text{conduit}}$$

where OD is the outside diameter of the cable and ID is the inside diameter of the conduit.

After you have determined the fill ratio of the existing cables in the conduit, select your cable. You need to ensure that the cable you select does not cause the NEC fill ratios to be exceeded. Fortunately, optical fiber is available in many different cable types. If the existing fill ratio will only permit a fiber-optic cable with a very small cross-sectional area, use a ribbon cable.



When you're installing fiber-optic cabling in conduit, the NEC fill ratios apply only when they are placed in conduit with electrical power cables or when the fiber-optic cable contains metallic members. A metallic member might be a wire, metal strength, or metal shield.

Let's determine the conduit fill ratio with four existing cables in place. The inside diameter of the conduit is 1.5". The cable to be installed has an outside diameter of 0.35". The outside diameters of the existing cables are 0.45", 0.37", 0.25", and 0.55". To determine the fill ratio of the conduit before the fiber-optic cable is installed, use the following formula:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 + OD^2\text{cable}_3 + OD^2\text{cable}_4) / ID^2\text{conduit}$$

$$\text{Fill Ratio} = (0.45^2 + 0.37^2 + 0.25^2 + 0.55^2) / 1.5^2$$

$$\text{Fill Ratio} = (0.2025 + 0.1369 + 0.0625 + 0.3025) / 2.25$$

$$\text{Fill Ratio} = 0.7044 / 2.25$$

$$\text{Fill Ratio} = 0.313 \text{ or } 31.3\%$$

The next step is to determine the conduit fill ratio with the addition of the fiber-optic cable. To determine that fill ratio, use the following formula:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 + OD^2\text{cable}_3 + OD^2\text{cable}_4 + OD^2\text{cable}_5) / ID^2\text{conduit}$$

$$\text{Fill Ratio} = (0.45^2 + 0.37^2 + 0.25^2 + 0.55^2 + 0.35^2) / 1.5^2$$

$$\text{Fill Ratio} = (0.2025 + 0.1369 + 0.0625 + 0.3025 + 0.1225) / 2.25$$

$$\text{Fill Ratio} = 0.8269 / 2.25$$

$$\text{Fill Ratio} = 0.3675 \text{ or } 36.75\%$$

The calculated fill ratio is less than 40 percent, so the conduit is large enough as defined by the NEC.

If a large conduit is already in place and contains other cables or if it is likely that other cables will be run through the conduit in the future, you may want to consider installing an *innerduct*, a conduit sized for the optical fiber cable that will protect it from other activities in the larger conduit.

Direct Burial

To run an outdoor fiber-optic cable out of sight, we can install it by *direct burial*. As the name implies, this method can be as simple as placing a suitable cable directly in the ground. Direct burial methods also include placing a cable within a protective pipe or conduit and burying it.

Direct burial has some advantages. Cables that are buried underground are not visible and do not obstruct the scenery like cables that span telephone poles or buildings. Burying a cable keeps it out of the wind, rain, snow, and ice. The wind from a tornado or hurricane will not damage a buried cable. Unlike aerial cables, a buried cable will not break from the excessive weight of ice from an ice storm.

However, direct burial of fiber-optic cables also has some disadvantages. Fiber-optic cables buried underground are difficult to locate since they do not emit any electromagnetic energy the way a copper cable such as a power line does. Animals that burrow can damage cables buried underground. A direct burial cable typically features an armor layer that protects it from burrowing animals and damage that may be caused when rock is used to cover the cable as is placed in the ground.

To place a direct burial cable in the ground, a backhoe or shovel can be used to dig a trench in which the cable is laid and then covered up. This approach is typically used for short runs. For longer runs that require more efficient methods, a cable-laying plow is available. This device is designed to open a trench, lay the cable, and cover it up again while on the move. It is a more complicated machine, but it is useful when longer distances must be covered.

When using direct burial methods, be sure to dig the trench deep enough to be below the frost line. In some areas, this can be as much as 30" deep.

Remember that you must contact the local utilities before you begin digging or trenching operations. The location of existing underground cables must be known before you dig.

Aerial

As our cable leaves the production building, it's going to be spliced to a cable with a messenger wire, which will be strung along a series of poles in an aerial installation. These poles already carry power lines, but they will not affect the data on the optical fiber itself because it is an insulator.

If a messenger wire is not incorporated into the cable assembly used, the cable will have to be lashed to messenger wire. A messenger wire is a steel or aluminum wire running between the poles to support the fiber-optic cable. Either way, cables in aerial installations must be able to withstand loading from high winds, ice, birds and climbing animals, and even windblown debris such as branches.

Blown Fiber

Blown fiber is an alternative to installing cables. In new construction or renovation, it is a good idea to consider blown fiber as an alternative to traditional methods.

The three steps to installing blown fiber are as follows:

1. Install the special tubing or conduit.
2. Blow the optical fiber through the tubes from location to location.
3. Terminate the optical fiber.

The optical fiber used in a blown fiber installation is the same optical fiber found in fiber-optic cables. However, the strand or strands of optical fiber will be coated with a protective layer that creates friction as dry gas passes over it. The friction generated by the dry gas as it flows through the tubing pulls the optical fiber or optical fiber bundle through the tubing. During this process very little tensile stress is placed on the optical fiber.

The tubing is about 5mm in diameter. Typically, a jacket is applied around two or more tubes, forming what looks like a cable. This cable-like bundle of tubes is installed without the optical fiber. The optical fiber is blown through these tubes after the installation. Figure 12.20 shows two different bundles of tubes and a single tube that is spliced in the center.

FIGURE 12.20 Blown fiber tubing assemblies



Blown optical fiber is used in buildings, between buildings, and on board ships. The U.S. Navy uses blown optical fiber on aircraft carriers and other types of ships. The life of a ship can exceed 50 years. Removing and installing cables on board a ship is expensive, and there is always the risk that this process may cause damage to other cables or equipment.

The tubing installed for blown optical fiber never needs to be removed because old optical fiber can be pulled out and new fiber blown in. The process of pulling out the old or damaged fiber and blowing in new fiber takes very little time in comparison to the time it would take to remove and reinstall a new cable. Figure 12.21 is a photograph of shipboard enclosure where blown optical tubes are interconnected with a splice.

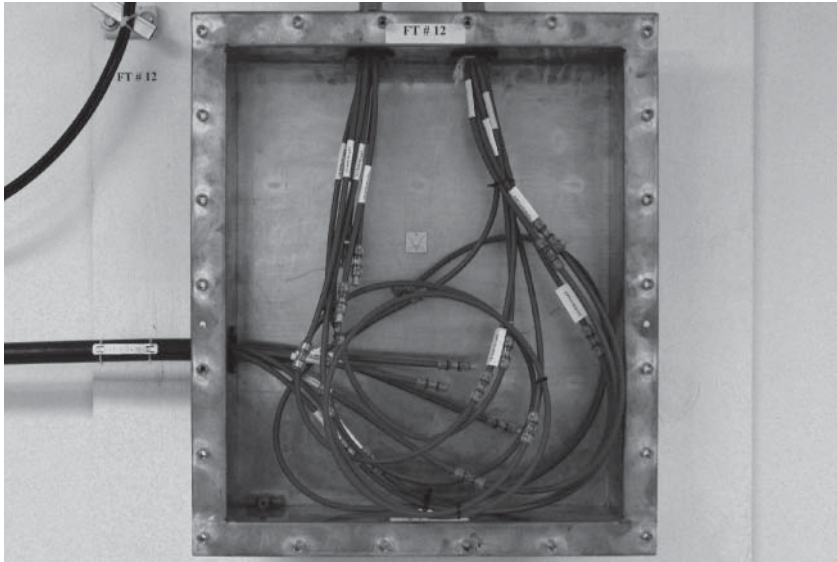
FIGURE 12.21 Interconnected blown fiber tubes

Photo Courtesy Of Kitco Fiber Optics

Figure 12.22 is a photograph of a shipboard enclosure where the optical fibers in the tubes are terminated.

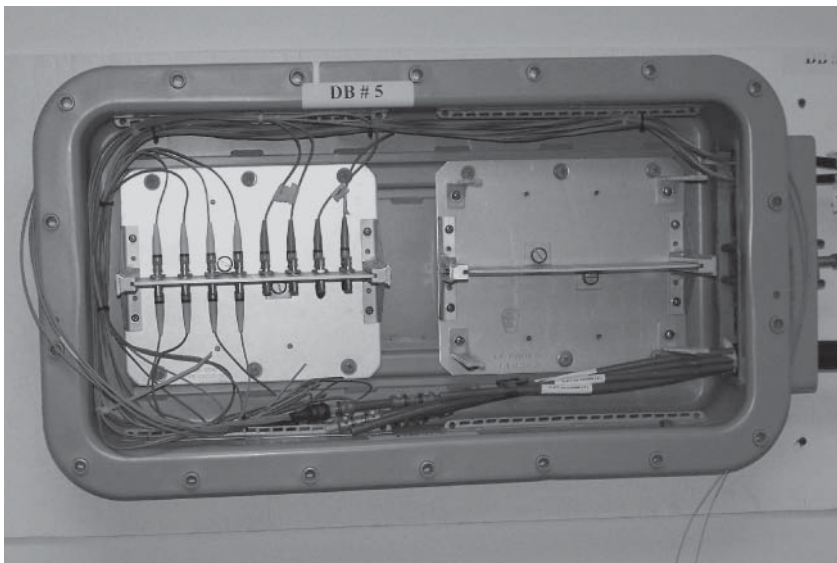
FIGURE 12.22 Terminated blown optical fibers

Photo Courtesy Of Kitco Fiber Optics



Real World Scenario

Leaving Slack for Repairs

No matter where you install cable, you'll want to leave some slack for the inevitable repairs that will have to be performed. How much slack should you leave? That depends on the situation. Look at the location; the number of turns that the fiber takes as it passes through conduits, trays, or ductwork; and the likelihood that the fiber will be disturbed after you have installed it. Consider the following cautionary tale.

During the construction of a strip mall in Virginia, cable installers ran cable underground in an area where construction was ongoing. After the cable was laid, backhoe operators accidentally pulled up the conduit carrying the cable seven times, requiring the cable to be spliced and re-laid in the ground. After the sixth event, all of the slack in the cable had been used up in repairs and the cable had to be replaced.

For the record, the location of the cable was clearly marked.

Wall Plate

A wall plate may also be called a wall outlet, workstation outlet, or station outlet. Wall plates are available in many different styles, types, brands, and colors. When choosing a location for a wall plate, make sure the distance from the workstation to the wall plate does not exceed 5m as defined in ANSI/TIA-568-C.1.

When deciding the vertical location of the wall plate, the NEC suggests that you place the top of the plate no more than 18" from the subfloor. If the wall plate is intended to service a countertop, the top plate should be no more than 48" from the subfloor.

Cable Slack

How much slack an installation should have and where the slack should be stored depends on the installation. However, there are some basic guidelines for slack.

When a cable is brought into a building from outside and the cable is not a listed indoor-outdoor type cable, the slack should be outside the building. Typically, 10 meters of slack is sufficient for most installations.

When a cable is brought into a building from outside and the cable is a listed indoor-outdoor type cable, the slack should be inside the building. Typically, 10 meters of slack is sufficient for most installations.

Cables that span the inside of a building between cross-connects should have slack on each end of the cable. As with the other installations, 10 meters of slack is typically sufficient.

All cable slack should be stored in the form of a loop. For an indoor installation, it should be either hung from the wall or placed under a subfloor. For an outdoor installation, it should be placed in the closest suitable outdoor access point.

Fire Resistance Bonding and Grounding

As you learned in Chapter 7, Article 770 of the NEC places requirements on fiber-optic cables and their installation within buildings. One requirement is that the cable must be able to resist the spread of fire. This requirement applies to all fiber-optic cables installed in a building. Other requirements address the metallic members in some cable types.

This section will address cable selection to meet fire resistance requirements defined in Article 770 of the NEC. It will also address the installation of fiber-optic cables that contain metallic members.



This chapter provides an overview of Article 770 and is not a substitute for the NEC. The National Fire Protection Association (NFPA) updates the NEC every three years. The person performing the installation is responsible for compliance with the current version of the NEC. In addition, the installer needs to be aware of local, state, and federal codes that apply to the installation.

Fire Resistance

Article 770 of the NEC states that optical fiber cables installed within buildings shall be listed as being resistant to the spread of fire. These cables are also required to be marked in accordance with Table 12.3.

TABLE 12.3 NEC cable markings, types, locations, and permitted substitutions

Marking	Type	Location	Permitted substitutions
OFNP	Nonconductive optical-fiber plenum cable	Ducts, plenums, other air spaces	None
OFCP	Conductive optical-fiber plenum cable	Ducts, plenums, other air spaces	OFNP
OFNR	Nonconductive optical-fiber riser cable	Risers, vertical runs	OFNP
OFCR	Conductive optical-fiber riser cable	Risers, vertical runs	OFNP, OFCP, OFNR
OFNG	Nonconductive optical-fiber general-purpose cable	General-purpose use except for risers and plenums	OFNP, OFNR
OF CG	Conductive optical-fiber general-purpose cable	General-purpose use except for risers and plenums	OFNP, OFCP, OFNR, OFCR, OFNG, OFN

Marking	Type	Location	Permitted substitutions
OFN	Nonconductive optical-fiber general-purpose cable	General-purpose use except for risers, plenums, and spaces used for environmental air	OFNP, OFNR
OFC	Conductive optical-fiber general-purpose cable	General-purpose use except for risers, plenums, and spaces used for environmental air	OFNP, OFCP, OFNR, OFCR, OFNG, OFN

A listed fiber-optic cable has had its performance verified by Underwriters Laboratory (UL). The outer surface of the cable or a marker tape should be clearly marked at intervals not to exceed 40" with the following information:

- Cable type-letter designation such as OFN, OFC, OFNR, OFNP, etc.
- Manufacturer's identification
- UL symbol or the letters UL

There are two exceptions to installing listed fiber-optic cables:

- Unlisted conductive and nonconductive outside plant optical fiber cables may be installed in building spaces, other than risers, ducts used for environmental air, plenums used for environmental air, and other spaces used for environmental air, as long as the length of the cable within the building, measured from its point of entrance, does not exceed 15 meters (50') and is terminated in an enclosure.
- A nonconductive outside plant optical fiber cable is not required to be listed or marked when the cable enters the building from the outside and is run in any of the following raceways:
 - Intermediate metal conduit (IMC)
 - Rigid metal conduit (RMC)
 - Rigid polyvinyl chloride conduit (PVC)
 - Electrical metallic tubing (EMT)

Three listed cable types are recognized in Article 770 for indoor installations: plenum cable, riser cable, and general-purpose cable.

Plenum cable Plenum cables, whether conductive or nonconductive, are suitable for use in ducts, plenums, and other space used for environmental air. These cables will have fire resistance and low smoke-producing characteristics.

Riser cable Riser cables, whether conductive or nonconductive, are suitable for a vertical run in a shaft or from floor to floor. These cables will have fire resistance characteristics capable of preventing the carrying of a fire from floor to floor.

General-purpose cable General-purpose cables, whether conductive or nonconductive, are resistant to the spread of fire. However, these cables are not suitable for plenum or riser applications.

Bonding and Grounding

Even though optical fiber cable itself does not carry electrical power, there may be some circumstances in which you will have to contend with electricity. You should be aware that Article 250, Grounding and Bonding, covers the general requirements for the grounding and bonding of electrical installations, while Article 770, Optical Fiber Cables and Raceways, covers the installation of optical fiber cables, raceways, and cable routing assemblies.

If a cable contains conductive components such as armor or metal strength members and comes in contact with electrical circuits, there is a chance that the metal could become a path for an electrical current. If you happen to touch the cable or anything to which the cable is connected, that path could be through you. Depending on the voltage and current involved, you could face severe injury or even death.



The NEC defines the term *bonding* as connecting to establish electrical continuity and conductivity. It defines the term *grounding* as connecting to ground or to a conductive body that extends the ground connection. The term *ground* is defined as the earth.

For this reason, Article 770.93 places a grounding or isolation requirement on any fiber-optic cable entering a building that contains electrically conductive materials and that may be exposed to or come in contact with electrical light or power conductors. The metallic member of the cable must be grounded as close to the point of entrance as practicable. An alternative to grounding is placing an insulating joint or equivalent device as close to the point of entrance as practicable. The insulating joint interrupts the flow of current through the conductive materials in the cable.

Article 770.100 describes bonding and grounding methods for fiber-optic cables with electrically conductive non-current-carrying metallic members. The bonding conductor or grounding electrode conductor must be copper or some other corrosion-resistant conductive material such as stainless steel. The conductor should be as short as practicable. It can be solid or stranded and range in size from 14 American wire gauge (AWG) to 6 AWG.

Cable Types

Article 770 of the NEC groups fiber-optic cables into three types:

- Nonconductive cables contain no electrically conductive materials.
- Conductive cables have conductive components such as metallic strength members, vapor barriers, or armor. These conductive components were not designed into the cable to conduct current and are thus referred to as *non-current-carrying conductive members*.
- Composite cables contain optical fibers and electrical conductors designed to carry current. These cables may also have metallic members that were not designed to carry current such as metallic strength members, vapor barriers, or armor.

Hardware Management

Eventually, fiber-optic cables end up in some kind of cabinet, panel, rack, or enclosure. Whether it's a patch panel, rack, splice enclosure, or wall outlet, the preparation of these items is crucial to the performance of the entire network. In these locations, the fiber or cable is terminated and connected in some way to another fiber, a connector, or a piece of hardware, and this process involves a risk of mistakes and mismanagement.

Poor hardware management can lead to confusion, excess strain on cables and fibers, and inefficient troubleshooting. In addition, a poorly organized hardware space reflects badly on the installer.

In this section, we will look at some guidelines for good hardware management.

Cleanliness

A clean working environment is essential to fiber work. Whether you are splicing fibers, terminating cables, or mounting cables in cabinets, it's important to keep dust, trash, water, and other contaminants away from your work area.

If possible, block off the space in which you are working to prevent exposure to contaminants. If required, use air filters to draw dust and grit out of the air, and try to limit the amount of nonessential work traffic in the area where you are working. Crews have been known to park their work trucks over manholes to keep dirt from the street from falling inside while they work. In other areas, it may be enough to use good housekeeping practices to ensure that the work area won't be contaminated.

Organization

Once a cable enters the telecommunications room, it is likely to be joined by anywhere from one to dozens of other cables. If these cables are not organized from the very beginning, they can quickly become a rat's nest of tangled, interwoven lines.

Many manufacturers offer cable management devices. These devices will help you organize cables for a neater, more user-friendly layout. Often the shortest distance between two points is not the best way to route a cable. Cables should only be routed vertically and horizontally. A cable should never be routed diagonally.

Secure cables in place when required. This helps relieve strain on the cable and keeps it from assuming new and interesting configurations after the cabinet is closed. Figure 12.23 shows cables routed to the top of a 19" rack. These cables are secured to minimize stress and maintain a configuration.

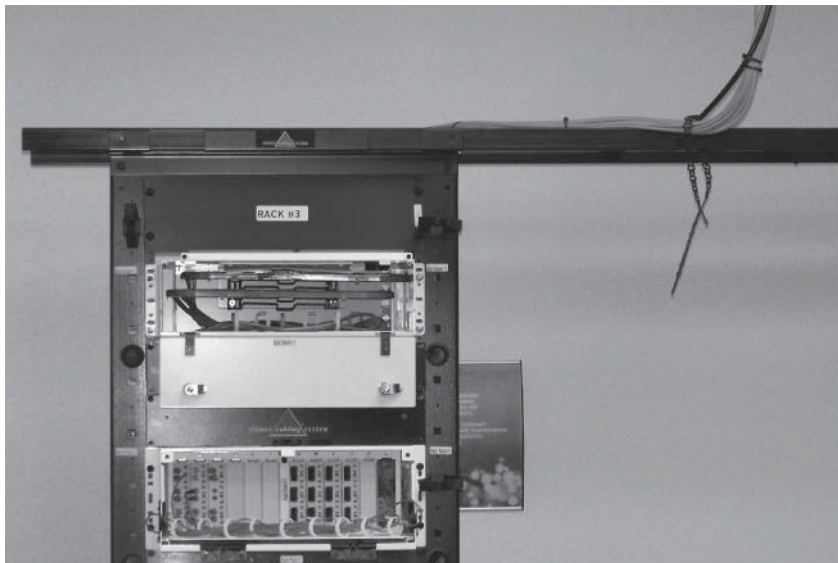
FIGURE 12.23 Secured cables routed to a 19" rack

Photo Courtesy Of Kitco Fiber Optics

Organization and neatness are also essential for efficient troubleshooting. It's much easier to trace connections and links if the cables are organized and neatly bundled than if they are spread all over the cabinet and tangled together.

Clamps and Cable Ties

Sometimes the best-planned installation can be destroyed by a simple cable tie or clamp. Earlier in the chapter, we discussed the bend radius of a fiber-optic cable. We learned that exceeding the bend radius can cause attenuation or damage to the cable or optical fiber. A cable tie or clamp, if improperly applied, can place a sharp bend in a cable. This sharp bend can produce a bend radius small enough to cause attenuation or damage.

When you are installing a cable tie or clamp, be very conscious of the pressure that is being placed on the cable. Look closely at the jacket of the cable to see if you are compressing or crushing it. You do not have to be very strong to damage a cable with a clamp or cable tie. Clamps and cable ties should be installed so they do not place unnecessary pressure on the cable jacket.

It is not always easy to tell that you have placed too much pressure on the jacket with a clamp or cable tie. One way to avoid problems is to test a piece of cable with a light source on one end of the cable and a power meter on the other. Apply the clamp or cable tie to the center of the cable under test and watch the power meter to see if there are any changes in power. If the power meter shows a reduction in power as you apply the clamp or cable tie,

the pressure is too great. Testing with a light source and power meter is covered in detail in Chapter 14, “Test Equipment and Link/Cable Testing.”

Another way to test, without a light source and power meter, is to apply the clamp or cable tie to the desired pressure. Let it sit for several minutes and then remove it. Carefully look at the cable after the clamp or cable tie is removed to see if an imprint was left in the jacket. If no imprint was left in the jacket, the pressure was not too great. If an imprint was left, the pressure should be reduced.

Labeling Requirements and Documentation

Even the simplest fiber-optic network can become confusing if it is improperly or insufficiently labeled to show where cables originate, how they are connected, and where they go.

It may be tempting to come up with your own system for labeling, but unless you can predict all of the changes that will take place as the system grows, you may soon find yourself with a mess on your hands. In addition, if you don't clearly document your labeling system, it will be difficult if not impossible for others to decipher it.

ANSI/TIA-606-B is the administration standard for telecommunications infrastructures. This document contains detailed information on conventions for labeling optical fiber connections within buildings, datacenters, and between buildings that share a telecommunication network. The standard specifies the methods used to determine how optical fiber channels are identified using a numbering/lettering system. It is also backward compatible with ANSI/TIA-606-A Addendum 1, and it is compatible with the following identifiers in ISO/IEC/TR 14763-2-1:

- Premises identifiers
- Space identifiers
- Closure identifiers
- Closure port and closure termination point identifiers
- Cabling identifiers
- Patch cord and jumper identifiers
- Pathway system identifiers
- Cabinet and frame bonding conductor identifiers

Identifiers are used to label telecommunications infrastructure components such as cables, racks, telecommunications rooms, equipment rooms, pathways, and outlets. Each identifier should be a unique set of numbers, letters, or an alphanumeric combination. Identifiers should be tracked in an administrative database. This database may be a simple spreadsheet depending on the number of users.

ANSI/TIA-606-B requires that a label be attached to each end of the cable assembly. These labels need to be clearly visible and durable. The labels should be designed to last as long as the cable assembly and must be printed with a mechanical device designed for this purpose. Handwritten labels are not acceptable.

Most patch panels for copper cross connects have provisions for inserting labels between the wiring blocks, these labels do not need to be color coded as defined in ANSI/TIA-606-A. The use of colored labels is recommended in ANSI/TIA-606-B, but not required. When colored labels are used, the color of the label identifies the origin of the cables. Labels may also contain additional information about the connection such as the cabinet number and port number. Table 12.4 lists the termination type color-coding scheme.

Fiber-optic patch panels like the one shown in Figure 12.24 do not use color-coded labels because there is no provision for inserting a label due to the limited amount of space. The label would be secured to a nearby location such as the enclosure door.

TABLE 12.4 ANSI/TIA-606-A/B patch panel color-coding scheme

Termination type	Label color	Typical application type
Demarcation point	Orange	Central office or entrance facility terminations
Network connection	Green	User side of central office or entrance facility connection, equipment room
Common equipment	Purple	Connections to a mainframe, LAN, or multiplexer
Key system	Red	Connections to key telephone systems
First-level backbone	White	Main cross-connect to intermediate cross-connect terminations
Second-level backbone	Gray	Intermediate cross-connect to telecommunications room cable terminations
Campus backbone	Brown	Inter-building backbone terminations
Horizontal	Blue	Horizontal cable terminations
Miscellaneous	Yellow	Alarms, security, or energy management

ANSI/TIA-606B was published in June 2012, and prior to that publication date, labels were generated in accordance with ANSI/TIA-606-A with and without Addendum 1. Addendum 1 added identifiers for datacenter racks and cabinets and was incorporated into ANSI/TIA-606-B. Depending on the size of the datacenter, X and Y coordinates that relate to floor tiles or rows may be used to generate a unique alphanumeric identifier for each location. However, a smaller datacenter may use the number of rows and racks or cabinets to develop the identifier for each location.

FIGURE 12.24 SC-SC patch panel inside an enclosure

Photo Courtesy Of Microcare

All telecommunication spaces require a unique FS.XY identifier. The first character, F, is numeric and identifies the floor of the building. The second character, S, is alphabetical and defines the space. The third character, X, is the X coordinate and is an alphabetical character. The last character, Y, is the Y coordinate and is a numerical character.

Keep in mind that the FS.XY identifier may be longer than four characters with a decimal point in the center. The first character that identifies the floor may require

three numbers. The second character that defines the space is typically one letter. The X coordinate is typically two letters, and the Y coordinate is typically two numbers. For example, 15B.AJ08 would be on the 15th floor in room B. The cabinet or rack would be in grid location AJ08.

Additional information is required to locate the patch panel and the specific port within the datacenter. This information follows the FS.XY identifier and is separated by a hyphen and a colon. For example, 1A.AB02-20:03 would be on the first floor in room A. The cabinet or rack would be in grid location AB02. The patch panel would be 20 rack units from the bottom of the rack or cabinet and the specific port would be 03.

An installer or technician should also be familiar with identifiers prior to Addendum 1. As an example of labeling according to ANSI/TIA-606-A without Addendum 1, imagine working with a wall outlet that has a port labeled as follows:

1A-B002

This particular number, known as a horizontal link identifier, immediately communicates the following information:

1: The fiber originates in a telecommunications room (TR) on the first floor of the building.

A: The TR is designated as A.

B: The fiber originates in patch panel B in TR A.

002: The fiber originates in port 002 of the patch panel.

Using this labeling system, we can very quickly trace the fiber to a specific connection point, even in a very large datacenter or building. Horizontal link identifiers are also described in ANSI/TIA-606B.

While the previous examples are basic, they show that by using a consistent labeling system, it is possible to communicate fiber routings accurately. A copy of ANSI/TIA-606-B may be obtained from the TIA website: www.tiaonline.org.

Documentation

Properly labeling cables, racks, telecommunications rooms, equipment rooms, pathways, and outlets is not enough; you must take the time to document this information and any supporting information in a database. Depending on the complexity, this database may be a simple spreadsheet or custom software package. Many companies define documentation requirements in their quality manual. Good documentation can speed up troubleshooting, system repair, and modification.

Polarity

Polarity is required to ensure the proper operation of bidirectional fiber-optic communication systems that use separate transmit and receive optical fibers. Polarity is discussed in ANSI/TIA-568-C.0 and C.3; however, neither document provides a definition. Absent a standardized definition, this text defines polarity in a fiber-optic network as the positioning of connectors and adapters to ensure that there is an end-to-end transmission path between the transmitter and the receiver of a channel. A channel as defined in ANSI/TIA-568-C.0 is an end-to-end transmission path between two points at which application-specific equipment is connected.

ANSI/TIA-568-C.3 defines two polarity schemes for duplex patch cords, A-to-A and A-to-B. The most commonly deployed scheme is A-to-B; therefore this section focuses on that scheme. An A-to-B duplex patch cord should be oriented so that position A connects to position B on one fiber and position B connects to position A on the other fiber, as shown in Figure 12.25.

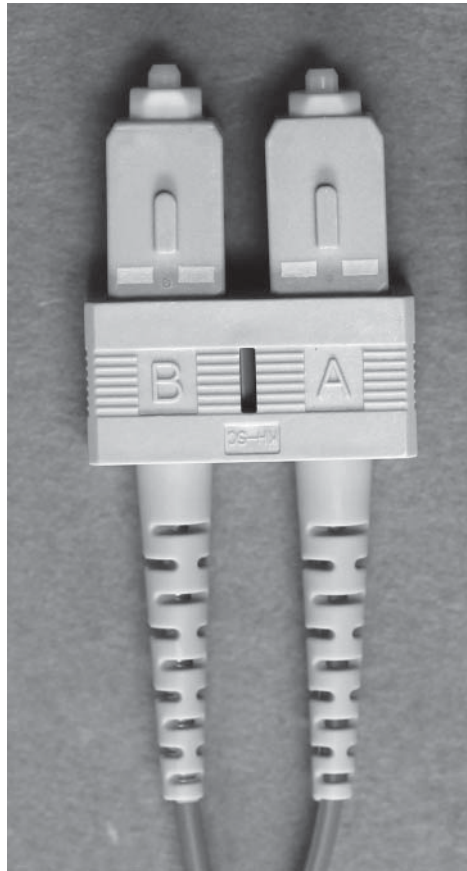
FIGURE 12.25 A-to-B duplex patch cord



ANSI/TIA-568-C.0 Annex B outlines methods that are used to maintain polarity in a fiber-optic cabling system. The A-to-B polarity scheme would implement a method known as consecutive fiber positioning, and the A-to-A polarity scheme would implement the reverse-pair positioning method. As you learned earlier, A-to-B is the most commonly deployed polarity scheme, so this section will focus on consecutive fiber positioning. Information about the reverse-pair positioning and array polarity systems covered in Annex B may be obtained from the TIA website: www.tiaonline.org.

When employing either polarity scheme, connector positions at the ends of a duplex patch cord must be identified as position A and position B. This is typically accomplished with raised lettering on the latch that holds the two connectors at each end of the patch cord together. A raised letter latch holding two SC connectors together is shown in Figure 12.26. Because of the physical size of an SC connector, the raised lettering is easy to read. However, this is not the case with small form factor connectors, such as the latched LC pair shown in Figure 12.27.

When employing the A-to-B polarity scheme, the position A connector is always plugged into the receive port on the transceiver and the position B connector is always plugged into the transmit port, as shown in Figure 12.28. Be aware that a manufacturer may use text or a symbol to mark the transmit and receive ports. The transmit port may be marked with the text “TX” and the receive port with the text “RX.” When symbols are used, they typically look like an arrowhead, such as those shown in Figure 12.28. The arrow points outward on the transmit port and inward on the receive port.

FIGURE 12.26 A raised letter latch holding two SC connectors together

To understand how to employ consecutive-fiber positioning on a pair of channels, a cabling configuration drawing similar to the one shown in Figure 12.29 must be developed. Working from the switch receive port to the workstation transmit port, the position B connector always mates with the position A connector. It is the opposite for the switch transmit port; the position A connector always mates with the position B connector.

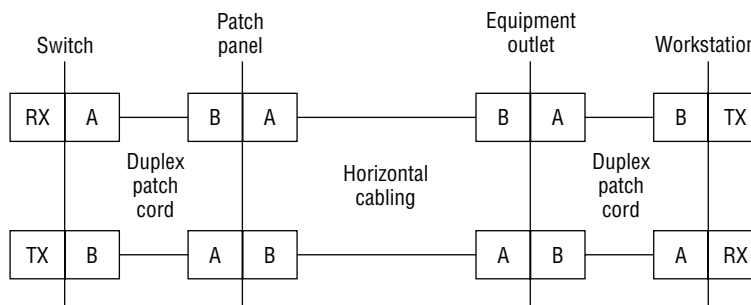
Working from the workstation receive port back to the switch transmit port, the position B connector always mates with the position A connector. It is the opposite for the workstation transmit port; the position A connector always mates with the position B connector. Regardless of which end of the channel you start from, the position A connector will always mate with the position B connector, and vice versa.

FIGURE 12.27 A raised letter latch holding two LC connectors together



FIGURE 12.28 Arrows on a transceiver identify the transmit and receive ports.



FIGURE 12.29 Cabling configuration for a basic fiber-optic network

Electrical Codes

Various organizations publish codes for the purpose of safeguarding persons and property from electrical hazards. This section will briefly examine three codes that are associated with the installation of electrical and fiber-optic cables.

The National Electrical Code (NEC)

The National Electrical Code (NEC) is published by the National Fire Protection Association (NFPA). The NFPA is an international nonprofit organization located in Quincy, Massachusetts; it was established in 1896. Its mission is to reduce the worldwide burden of fire and other hazards on the quality of life by providing and advocating consensus codes and standards, research, training, and education.

NEC Article 90 states that the purpose of the code is the practical safeguarding of persons and property from hazards arising from the use of electricity. It also states that the NEC is not intended as a design specification or an instruction manual for untrained persons. Local and state governments typically adopt it, making compliance mandatory.

It provides specific guidance for running fiber-optic cable within buildings. The NEC requires cables to be tested for fire resistance and smoke characteristics, establishes which types of cables may be run in different areas of a building, and specifies the types of raceways that can be used.

Be sure you stay current with the latest NEC guidelines. The NEC is generally updated every three years.

The Canadian Electrical Code

The Canadian Electrical Code (CEC) is published by the Canadian Standards Association (CSA) Group. The CSA Group is an independent, not-for-profit, member-based association dedicated to advancing safety, sustainability, and social good. It will publish the 23rd Edition of the CEC in 2015. Like the NEC, the CEC serves as a guide for safe electrical installation and maintenance. It provides specific guidance for running electrical and fiber-optic cables within buildings.

The National Electrical Safety Code (NESC)

The Institute of Electrical and Electronics Engineers, Inc. (IEEE) publishes the National Electrical Safety Code (NESC) every five years. The purpose of the NESC is to set the ground rules for the practical safeguarding of persons during the installation, operation, or maintenance of electric supply and communication lines and associated equipment.

Summary

This chapter covered equipment and methods used in installing fiber-optic cable. It described specifications used by manufacturers to ensure that their cables continue to operate properly and the ways to meet those specifications during installation.

The chapter discussed hardware commonly used in cable installation to make installation easier and reduce the potential for damage to the cable. It also described installation methods used in some common scenarios.

The chapter described safety measures to be employed when installing cable with conductive components and the reasons for these measures.

It described techniques for ensuring proper installation of cable in various enclosures and discussed the importance of organization and preparation of these enclosures.

It described the labeling requirements for telecommunications infrastructure components such as cables, racks, telecommunications rooms, equipment rooms, pathways, and outlets. In addition it described polarity schemes to ensure the proper operation of bidirectional fiber-optic communication systems.

Exam Essentials

Be familiar with installation specifications. Make sure that you understand the manufacturer's specifications for minimum bend radius and maximum tensile load. Be able to describe methods for maintaining these requirements during installation. Know the bend radius and tensile strength requirements for inside plant, indoor-outdoor, outside plant, and drop cables.

Be familiar with installation hardware. Be able to describe the hardware used during the installation process and as part of the permanent installation. Make sure that you understand how the hardware is used and what role it serves during and after installation.

Understand installation methods. Be able to describe different installation methods and the requirements for each type of installation.

Understand the importance of grounding and bonding. Be able to describe grounding and bonding. Understand how they relate to the installation of fiber-optic cables. Be able to describe the conditions that require grounding or bonding.

Be familiar with hardware preparation. Understand the importance of preparing hardware for cable installation. Understand methods used to keep the work environment clean and organized.

Understand the importance of administrative record keeping. Be able to describe the labeling requirements for telecommunications infrastructure components such as cables, racks, telecommunications rooms, equipment rooms, pathways, and outlets.

Review Questions

1. A cable's minimum bend radius specification is greater
 - A. During installation
 - B. After installation
 - C. When the temperature is higher
 - D. When it is in a cable tray

Hint: It is the short-term bend radius.

2. A cable's maximum tensile load rating
 - A. Is lower for installation than for long-term use
 - B. Is higher for installation than for long-term use
 - C. Is the same for all conditions
 - D. Does not apply during installation

Hint: A short-term load

3. Tensile loads encountered during installation are also called _____.
 - A. Static loads
 - B. Duct loads
 - C. Horizontal loads
 - D. Dynamic loads

Hint: It is a changing tensile load.

4. A pulling eye is attached to
 - A. The cable's outer jacket
 - B. The cable's strength member
 - C. The fiber
 - D. The armor

Hint: A pulling line is at the other end.

5. A pullbox is used to
 - A. Pull multiple cables through a conduit
 - B. Pull conduit into place
 - C. Relieve tensile loads on cable during pulling
 - D. Hold cable taut until pulling can resume

Hint: It provides an intermediate pulling location.

6. When is a splice enclosure used?
- A. Whenever a fiber has been spliced
 - B. When a splice must be placed underground
 - C. When a splice must be placed underwater
 - D. Splice enclosures are optional.

Hint: It protects the splice from exposure and strain.

7. What is the purpose of a patch panel?
- A. It takes the place of a splice enclosure.
 - B. It fills a hole where fiber has been installed.
 - C. It is used to route signals between cables.
 - D. It provides a permanent link between two pieces of hardware.

Hint: It is an interconnection point.

8. A fiber cable in a tray may incur a macrobend if
- A. The tray is not horizontal.
 - B. The fiber has to be pulled through the tray.
 - C. The tray is metal.
 - D. It is sharing the tray with electrical cables.

Hint: The problem is caused by weight.

9. The most common method for installing cable in a conduit is _____.
- A. Laying
 - B. Pulling
 - C. Pushing
 - D. Pre-installing

Hint: A line is typically used.

10. Installing a cable by opening a trench and laying it in the ground is also called _____.
- A. Trenching
 - B. Grounding
 - C. Layering
 - D. Direct burial

Hint: The cable must be protected from damage and from the elements.

11. The type of cable commonly used in aerial installations is called _____.
- A. Messenger cable
 - B. Aerial cable

C. Strung cable

D. Flying cable

Hint: It may be strung between poles.

12. Blown fiber installation is an alternative to _____.

A. Direct burial

B. Aerial installation

C. Pulling cables

D. Cable trays

Hint: It uses pressurized dry air.

13. Which NEC Article describes bonding and grounding methods for fiber-optic cables with electrically conductive non-current-carrying metallic members?

A. 250

B. 770.93

C. 770.100

D. 860

Hint: It does not place a requirement for grounding on cables entering the building.

14. _____ is the administration standard for telecommunications infrastructures.

A. ANSI/TIA-568-C.3

B. ANSI/TIA-606-B

C. ANSI/TIA-568-C.0

D. Article 770

Hint: This document contains detailed information on conventions for labeling optical fiber connections.

15. _____ is required to ensure the proper operation of bidirectional fiber-optic communication systems that use separate transmit and receive optical fibers.

A. Common equipment

B. A demarcation point

C. Polarity

D. A first-level backbone

Hint: It ensures that there is an end-to-end transmission path between the transmitter and the receiver of a channel.

Chapter Exercises

Determine the Minimum Cable Bend Radius

Remember that cable manufacturers tend to define the minimum bend radius when the cable is stressed and unstressed. These numbers are typically in compliance with ANSI/TIA-568-C.0 and C.3.

Refer to the following table and determine the minimum installation bend radius and the minimum operational bend radius for the 24-fiber cable.

Cable type	Diameter	Weight	Minimum bend radius: short term	Minimum bend radius: long term	Maximum load: short term	Maximum load: long term
12-fiber cable	0.26", 6.6mm	27lb/kft, 40kg/km	3.9" 9.9cm	2.6" 6.6cm	300lb, 1334N	90lb, 396N
24-fiber cable	0.375", 9.5mm	38.2lb/kft, 56.6kg/km	5.5" 14cm	3.7" 9.4cm	424lb, 1887N	127lb, 560N

The short-term minimum bend radius in the table is the installation bend radius. The minimum installation bend radius for the 24-fiber cable is 5.5".

The long-term minimum bend radius in the table is the operational bend radius. The minimum operational bend radius for the 24-fiber cable is 3.7".

Determine the Maximum Cable Tensile Rating

Remember that cable manufacturers typically define a dynamic and static pull load.

Refer to the following table and determine the maximum dynamic tensile load and the maximum static tensile load for the 12-fiber cable.

Cable type	Diameter	Weight	Minimum bend radius: short term	Minimum bend radius: long term	Maximum load: short term	Maximum load: long term
12-fiber cable	0.26", 6.6mm	27lb/kft, 40kg/km	3.9", 9.9cm	2.6", 6.6cm	300lb, 1334N	90lb, 396N
18-fiber cable	0.325", 8.3mm	33.8lb/kft, 50kg/km	4.9", 12.4cm	3.25", 8.3cm	375lb, 1668N	112lb, 495N

The short-term maximum load in the table is the dynamic tensile load. The dynamic tensile load for the 12-fiber cable is 300lb.

The long-term maximum load in the table is the static tensile load. The static tensile load for the 12-fiber cable is 90lb.

Determine the Fill Ratio for a Multiple-Cable Conduit Installation

When preparing for a multiple cable conduit installation, you must be aware of the fill ratio for the conduit. If you are installing cable in existing conduit already occupied by one or more cables that contain electrical conductors, you will need to determine the fill ratio of the conduit prior to selecting your cable.

You need to install one cable into a conduit that has three existing electrical cables. The inside diameter of the conduit is 1". The cable to be installed has an outside diameter of 0.25". The outside diameters of the existing cables are 0.28", 0.32", and 0.38". Determine the fill ratio of the conduit before and after the fiber-optic cable is installed and verify that this installation will not overfill the conduit as defined by the NEC.

The NEC specifies the following fill ratios by cross-sectional area for conduit:

1 cable: 53 percent

2 cables: 31 percent

3 or more cables: 40 percent

The first step is to determine the conduit fill ratio with the existing cables. To determine that fill ratio, use the following formula:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 + OD^2\text{cable}_3) / ID^2\text{conduit}$$

$$\text{Fill Ratio} = (0.28^2 + 0.32^2 + 0.38^2) / 1^2$$

$$\text{Fill Ratio} = (0.0784 + 0.1024 + 0.1444) / 1$$

$$\text{Fill Ratio} = 0.3252 / 1$$

$$\text{Fill Ratio} = 0.3252 \text{ or } 32.52\%$$

The next step is to determine the conduit fill ratio with the addition of the fiber-optic cable. To determine that fill ratio, use the following formula:

$$\text{Fill Ratio} = (OD^2\text{cable}_1 + OD^2\text{cable}_2 + OD^2\text{cable}_3 + OD^2\text{cable}_4) / ID^2\text{conduit}$$

$$\text{Fill Ratio} = (0.28^2 + 0.32^2 + 0.38^2 + 0.25^2) / 1^2$$

$$\text{Fill Ratio} = (0.0784 + 0.1024 + 0.1444 + 0.0625) / 1$$

$$\text{Fill Ratio} = 0.3877 / 1$$

$$\text{Fill Ratio} = 0.3877 \text{ or } 38.77\%$$

The calculated fill ratio is less than 40 percent, so the conduit is large enough as defined by the NEC.

Chapter 13

Fiber-Optic System Advantages

**THE FOLLOWING ETA FIBER-OPTICS
INSTALLER COMPETENCIES ARE COVERED
IN THIS CHAPTER:**

- ✓ Compare the bandwidth advantages of optical fiber over twisted pair and coaxial copper cables.
- ✓ Compare the attenuation advantages of optical fiber over twisted pair and coaxial copper cables.
- ✓ Explain the electromagnetic immunity advantages of fiber-optic cable over copper cable.
- ✓ Describe the size advantages of fiber-optic cable over copper cable.
- ✓ Describe the weight-saving advantages of fiber-optic cable over copper cable.
- ✓ Describe the security advantages of fiber-optic cable over copper cable.
- ✓ Compare the safety advantages of fiber-optic cables over copper cables.





Up to this point, you have learned about transmitters, receivers, connectors, splices, and fiber-optic cable. These pieces are some of the building blocks of a fiber-optic system.

A basic fiber-optic system contains a transmitter, receiver, fiber-optic cable, and connectors.

This chapter explores the performance advantages that optical fiber has over twisted pair and coaxial copper cable.

In this chapter, you will learn to

- Calculate the bandwidth for a length of optical fiber
- Calculate the attenuation for a length of optical fiber

The Advantages of Optical Fiber over Copper

In this chapter, we will examine seven fiber-optic system performance areas and compare the performance of optical fiber to twisted pair and coaxial copper cable. Performance in this comparison will be evaluated in the areas of bandwidth, attenuation, electromagnetic immunity, size, weight, safety, and security. The fiber-optic system will operate at an 850nm wavelength with a VCSEL source.

Performance data for the laser-optimized optical fiber will be taken from the ANSI/TIA-568-C.3 and ISO/IEC 11801 standards. It will be compared to both Category 6A twisted pair cable and RG6 coaxial cable. Performance data for the Category 6A cable will be taken from the ANSI/TIA-568-C.2 Commercial Building Telecommunication Cabling Standard Part 2: Balanced Twisted-Pair Cabling Components, and the performance data for the RG6 coaxial cable is derived by averaging the values found in several manufacturers' datasheets.

All the comparisons in this chapter are based on latest standards and datasheets that were available as of this writing.

Bandwidth

Bandwidth is a popular buzzword these days. We are bombarded with commercials advertising high-speed downloads from a *cable modem*, satellite, or fiber-to-the-home (FTTH).

The competition to sell us bandwidth is fierce. As you learned in Chapter 1, “History of Fiber Optics and Broadband Access,” while the basic broadband speed was defined with a 3Mbps download speed, more than 94 percent of the homes in America exceed 10Mbps. More than 75 percent have download speeds greater than 50Mbps, 47 percent have download speeds greater than 100Mbps, and more than 3 percent enjoy download speeds greater than one billion bits (a gigabit) per second (Gbps).

In earlier chapters, we looked at the physical properties of the optical fiber and fiber-optic light source that limit bandwidth. You learned that single-mode systems with laser transmitters offer the greatest bandwidth over distance and that multimode systems with LED transmitters offer the least bandwidth and are limited in transmission distance. You also learned that the bandwidth of the optical fiber is inversely proportional to its length. In other words, as the length of the optical fiber increases, the bandwidth of the optical fiber decreases.

So how does length affect the bandwidth of a copper cable? Well, when it comes to cable length, copper suffers just as optical fiber does. When the length of a copper cable is increased, the bandwidth for that cable decreases. So if both copper and optical fiber lose bandwidth over distance, why is optical fiber superior to copper? To explain that, we will look at the minimum bandwidth requirements defined in ANSI/TIA-568-C.3 and ISO/IEC 11801 for multimode optical fiber, ANSI/TIA-568-C.2 for unshielded twisted pair (UTP) Category 6A cable, and the datasheets for RG6 coaxial cable.

Remember that the values defined in ANSI/TIA-568-C.2, ANSI/TIA-568-C.3, and ISO/IEC 11801 are the minimum values that a manufacturer needs to achieve. There are many manufacturers offering optical fiber and copper cables that exceed these minimum requirements. However, for this comparison, only values defined in these two standards will be used.

One of the problems in doing this comparison is cable length. ANSI/TIA-568-C.2 defines Category 6A performance at a maximum physical length of 100m at frequencies up to 500MHz. ANSI/TIA-568-C.3 and ISO/IEC 11801 do not define a maximum bandwidth at a maximum optical fiber length. As you learned in Chapter 5, “Optical Fiber Characteristics,” ANSI/TIA-568-C.3 and ISO/IEC 11801 define multimode optical fiber bandwidth two ways depending on the type of light source. In this comparison the 850nm VCSEL is used with 50/125 μ m laser-optimized optical fiber (OM4, Type A1a.3, TIA 492AAAD). The minimum effective modal bandwidth-length product in MHz $\cdot$ km is 4700 for this combination, as shown in Table 13.1.



ANSI/TIA-568-C.1 states that the maximum horizontal cable length should be limited to 90m regardless of whether copper or fiber is used. The 90m distance allows for an additional 5m at each end for equipment cables or patch cords. The total combined length cannot exceed 100m.

To do this comparison, we will need to calculate the bandwidth limitations for 100m of OM4 laser-optimized multimode optical fiber, as shown here:

Bandwidth-length product = 4700 MHz · km

Optical fiber length = 0.1km

Bandwidth in MHz = $4700 \div 0.1$

Bandwidth = 47,000MHz

TABLE 13.1 Characteristics of ANSI/TIA-568-C.3 and ISO/IEC 11801-recognized optical fibers

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
62.5/125µm	850	3.5	200	Not Required
Multimode OM1	1300	1.5	500	Not Required
Type A1b				
TIA 492AAAA				
50/125µm	850	3.5	500	Not Required
Multimode OM2	1300	1.5	500	Not Required
TIA 492AAAB				
Type A1a.1				
850nm laser-optimized	850	3.5	1500	2000
50/125µm Multimode OM3	1300	1.5	500	Not Required
Type A1a.2				
TIA 492AAAC				
850nm laser-optimized	850	3.5	3500	4700
50/125µm Multimode OM4	1300	1.5	500	Not Required
Type A1a.3				
TIA 492AAAD				

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
Single-mode	1310	0.5	N/A	N/A
Indoor-outdoor OS1	1550	0.5	N/A	N/A
Type B1.1				
TIA 492CAAA				
OS2				
Type B1.3				
TIA 492CAAB				
Single-mode	1310	1.0	N/A	N/A
Inside plant OS1	1550	1.0	N/A	N/A
Type B1.1				
TIA 492CAAA				
OS2				
Type B1.3				
TIA 492CAAB				
Single-mode	1310	0.5	N/A	N/A
Outside plant OS1	1550	0.5	N/A	N/A
Type B1.1				
TIA 492CAAA				
OS2				
Type B1.3				
TIA 492CAAB				

The 100m OM4 laser-optimized multimode optical fiber has a bandwidth 94 times greater than the Category 6A cable. This clearly shows that optical fiber offers incredible bandwidth advantages over Category 6A cable.

Now that we have seen how the bandwidth of OM4 laser-optimized multimode optical fiber with an 850nm VCSEL source greatly exceeds that of Category 6A cable, let's do a comparison with RG6 coaxial cable. This cable has been chosen because it is widely used in homes and buildings for video distribution. Remember the performance data for the RG6 coaxial cable is taken from the average of several manufacturers' datasheets. ANSI/TIA-568-C does not define performance parameters for RG6 coaxial cable.

An RG6 coaxial cable with a transmission frequency of 4GHz has roughly the same 100m transmission distance characteristics as a Category 6A cable at 500MHz. We know

from the previous comparison that a 100m length of 50/125μm multimode optical fiber will support transmission frequencies up to 47GHz over that same distance with an 850nm light source. In this comparison, the laser-optimized multimode optical fiber offers a bandwidth advantage 11.75 times greater than the RG6 coaxial cable.

These comparisons have demonstrated the bandwidth advantages of optical fiber over copper cable. The comparisons were done at very short distances because at this point we have not addressed how attenuation in a copper cable changes with the transmission frequency, whereas in an optical fiber, attenuation is constant regardless of the transmission frequency.

Attenuation

All transmission mediums lose signal strength over distance. As you know, this loss of signal strength is called attenuation and is typically measured in decibels. Optical fiber systems measure attenuation using optical power. Copper cable systems typically use voltage drop across a defined load at various transmission frequencies to measure attenuation. The key difference here is not that optical fiber uses power and copper uses voltage. The key difference is that attenuation in copper cables is measured at different transmission frequencies. This is not the case with optical fiber, where attenuation is measured with a continuous wave light source.

The attenuation in a copper cable increases as the transmission frequency increases. Table 13.2 shows the maximum worst pair insertion loss for a 100m horizontal Category 5e and Category 6A cable as defined in ANSI/TIA-568-C.2. This table clearly shows the effects that transmission frequency has on a copper cable.

TABLE 13.2 Horizontal cable insertion loss, worst pair*

Frequency (MHz)	Category 5e (dB)	Category 6A (dB)
1.0	2.0	2.1
4.0	4.1	3.8
8.0	5.8	5.3
10.0	96.5	5.9
16.0	8.2	7.5
20.0	9.3	8.4
25.0	10.4	9.4
31.25	11.7	10.5
62.5	17.0	15.0
100.0	22.0	19.1

Frequency (MHz)	Category 5e (dB)	Category 6A (dB)
200.0		27.6
300.0		34.3
400.0		40.1
500.0		45.3

* For a length of 100m (328')

The maximum allowable attenuation in an optical fiber is defined in ANSI/TIA-568-C.3 and ISO/IEC 11801. Table 13.3 shows the attenuation portion of the optical fiber cable transmission performance parameters. This table defines attenuation for both multimode and single-mode optical fibers. You will notice that there is no column for transmission frequency in this table. That is because optical fiber does not attenuate as the transmission frequency increases like copper cable does.

TABLE 13.3 ANSI/TIA-568-C.3 and ISO/IEC 11801 optical fiber cable attenuation performance parameters

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)
50/125µm multimode	850	3.5
	1300	1.5
62.5/125µm multimode	850	3.5
	1300	1.5
Single-mode inside plant cable	1310	1.0
	1550	1.0
Single-mode outside plant cable	1310	0.5
	1550	0.5

Now that you know how optical fiber and copper cable attenuate, let's do a comparison. The first comparison will put optical fiber up against Category 6A cable. The distance will be 100m and the transmission frequency will be 500MHz.

Table 13.2 lists 45.3dB for the worst-case attenuation for the Category 6A cable at 500MHz. This means the Category 6A cable loses 99.997 percent of its energy over a distance of 100m at that transmission frequency. As you learned in Chapter 5, the attenuation of an optical fiber is different for each wavelength. Table 13.3 lists the maximum attenuation in dB/km. We can solve for the maximum attenuation at both wavelengths as shown here.

Attenuation coefficient at 850nm = 3.5dB/km
Optical fiber length = 0.1km
Attenuation for the length of optical fiber = $0.1 \times 3.5 = 0.35$
Maximum attenuation at 850nm for 100m of optical fiber is 0.35dB
Attenuation coefficient at 1300nm = 1.5dB/km
Optical fiber length = 0.1km
Attenuation for the length of optical fiber = $0.1 \times 1.5 = 0.15$
Maximum attenuation at 1300nm for 100m of optical fiber is 0.15dB

The equations show that multimode optical fiber loses 7.7 percent of the light energy from the transmitter over the 100m distance at 850nm. At 1300nm, the loss is only 3.4 percent. However, the Category 6A cable loses 99.997 percent of the transmitted signal over that distance. This means that the Category 6A cable loses roughly 31261 times more energy than the optical fiber 850nm and roughly 32734 times more energy at 1300nm.

This comparison clearly shows the attenuation advantages of optical fiber over Category 6A cable.

Now let's compare the same optical fiber at the same wavelengths to an RG6 coaxial cable. For this comparison, the RG6 attenuation characteristics are taken from the average of several different manufacturers' published datasheets, as shown in Table 13.4.

TABLE 13.4 RG6 cable insertion loss*

Frequency (MHz)	RG6 (dB)
1.0	0.8
5.0	1.7
100	6.4
200	9.5
500	14.0
1,000	21.3
3,000	38.8
4,500	49.0

* For a length of 100m (328')

As you look at Table 13.4, you can see that RG6 coaxial cable easily outperforms Category 5e and 6A cable. However, it does not even begin to approach the multimode optical fiber operating at 850nm or 1300nm. The RG6 coaxial cable has 14dB of attenuation at a transmission frequency of 500MHz over a distance of 100m. Whereas the Category 6A cable lost 99.997 percent of its signal strength, in the previous comparison the RG6 coaxial cable lost only 96 percent of its energy over the same distance.

These comparisons should make it clear that optical fiber has an enormous attenuation advantage over copper cable. The comparisons were done only with multimode optical fiber. Outside-plant, single-mode optical fiber greatly outperforms multimode optical fiber. Single-mode optical fiber links are capable of transmission distances greater than 80km without re-amplification. Based on the previous comparisons, Category 6A cable would require re-amplification every 100m at a transmission frequency of 500MHz. The RG6 coaxial cable would require re-amplification every 325m at a transmission frequency of 500MHz.

An 80km RG6 coaxial cable link transmitting a frequency of 500 MHz would require roughly 246 amplifiers to re-amplify the signal. However, the single-mode optical fiber link would not require re-amplification. Optical fiber links are unsurpassed in transmission distance.



Real World Scenario

Extending Copper's Range with a Media Converter

Some applications for optical fiber require media converters. During a recent radar installation on a commercial ship, it was determined that the location of one monitor greatly exceeded the distance supported by the copper Digital Video Interface (DVI). Choosing another location for the monitor was not an option; neither was relocating the radar processor. The only alternative was to replace the copper DVI cable between the radar processor and the monitor with a fiber-optic cable.

As we have learned in this chapter, optical fiber offers superior bandwidth and attenuation performance over copper cable. However, you cannot just substitute a fiber-optic cable for a copper cable to obtain these benefits; media converters are required.

Media converters work very much like a fiber-optic transmitter and receiver. One media converter plugs into the DVI receptacle on the radar processor; it converts the electrical signals into optical signals. The second media converter plugs into the DVI receptacle on the monitor; it converts optical signals into electrical signals. A single optical fiber cable connects the two media converters together. When single-mode optical fiber is used, distances of up to 1,000m are possible.

Electromagnetic Immunity

Electromagnetic interference (EMI) is electromagnetic energy, sometimes referred to as *noise*, which causes undesirable responses, degradation, or complete system failure. Systems using copper cable are vulnerable to the effects of EMI because a changing electromagnetic field will induce current flow in a copper conductor. Optical fiber is a dielectric or an insulator, and current does not flow through insulators. What this means is that EMI has no effect on the operation of an optical fiber.

Let's take a look at some examples of EMI-induced problems in a copper system. A Category 6A cable has four pairs of twisted conductors. The conductors are twisted to keep the impedance uniform along the length of the cable and to decrease the effects of EMI by canceling out opposing fields. Two of the conductor pairs may be used to transmit and two of the conductor pairs may be used to receive.

For this example, only one pair is transmitting and one pair is receiving. The other two pairs in the cable are not used. Think of each pair of conductors as an antenna. The transmitting pair is the broadcasting antenna and the receiving pair is picking up that transmission just like your car radio antenna.

The data on the transmitting pair is broadcast and picked up by the receiving pair. This is called *crosstalk*. If enough current is induced into the receiving pair, the operation of the system can be affected. This is one reason why it is so critical to maintain the twists in a Category 6A cable.

Would we have this problem with an optical fiber? Does crosstalk exist in optical fiber? Regardless of the number of transmitting and receiving optical fibers in a cable assembly, crosstalk does not occur. To have crosstalk in an optical fiber cable assembly, light would have to leave one optical fiber and enter one of the other optical fibers in the cable assembly. Because of total internal reflection, under normal operating conditions light never leaves the optical fiber. Therefore, crosstalk does not exist in optical fiber cable assemblies.

Now let's take a look at another EMI scenario. Copper cables are being routed through a manufacturing plant. This manufacturing plant houses large-scale electromechanical equipment that generates a considerable amount of EMI, which creates an EMI-rich environment.

Routing cables through an EMI-rich environment can be difficult. Placing the cables too close to the EMI-generating source can induce unwanted electrical signals strong enough to cause systems to function poorly or stop operating altogether. Copper cables used in EMI-rich environments typically require electrical shielding to help reduce the unwanted electrical signals. In addition to the electrical shielding, the installer must be aware of the EMI-generating sources and ensure that the copper cables are routed as far as possible from these sources.

Routing copper cables through an EMI-rich environment can be challenging, time-consuming, and expensive. However, optical fiber cables can be routed through an EMI-rich environment with no impact on system performance. The fiber-optic installer is free to route the optical fiber as efficiently as possible. Optical fiber is very attractive to every industry because it is immune to EMI.

Size and Weight

You must take into account the size of any cable when preparing for an installation. Often fiber-optic cables will be run through existing conduits or raceways that are partially or almost completely filled with copper cable. This is another area where small fiber-optic cable has an advantage over copper cable.

Let's do a comparison and try to determine the reduced-size advantage that fiber-optic cable has over copper cable. A common size for a coated optical fiber is typically 250 μ m in diameter. You learned that ribbon fiber-optic cables sandwich up to 12-coated optical fibers between two layers of Mylar tape. Eighteen of these ribbons stacked on top of each other form a rectangle roughly 4.5mm by 3mm. This rectangle can be placed inside a buffer and surrounded by a strength member and jacket to form a cable. The overall diameter of this cable would be 16.5mm (0.65"), only slightly larger than a bundle of four RG6 coaxial cables or a bundle of four Category 6A cables.

So how large would a copper cable have to be to offer the same performance as the 216 optical fiber ribbon cable? That would depend on transmission distance and the optical fiber data rate. Because we have already discussed Category 6A performance, let's place a bundle of Category 6A cables up against the 216 optical fiber ribbon cable operating at a modest 2.5Gbps data rate over a distance of just 100m.

A Category 6A cable contains four conductor pairs and each pair is capable of a 500MHz transmission over 100m. As you learned earlier in this chapter, a 500MHz transmission carries 1 billion symbols per second. If each symbol is a bit, the 500MHz Category 6A cable is capable of a 1Gbps transmission rate. When the performance of each pair is combined, a single Category 6A cable is capable of a 4Gbps transmission rate over a distance of 100m.

Now let's see how many Category 6A cables will be required to provide the same performance as the 216 optical fiber ribbon cable. The 216 optical fiber ribbon cable has a combined data transmission rate of 540Gbps (2.5Gbps $\times$ 216). When we divide 540Gbps by 4Gbps, we see that 135 Category 6A cables are required to equal the performance of this modest fiber-optic system.

When 135 Category 6A cables are bundled together, they are roughly 3.4" in diameter. As noted earlier in this chapter, the 216 optical fiber ribbon cable is approximately the size of four Category 6A cables bundled together. The Category 6A bundle thus has a volume roughly 27.4 times greater than the 216 optical fiber ribbon cable. In other words, Category 6A bundles need 27.4 times more space in the conduit than the 216 optical fiber ribbon cable.

The comparison we just performed is very conservative. The distance we used was kept very short and the transmission rate for the optical fiber was kept low. We can get even a better appreciation for the cable size reduction fiber-optic cable offers if we increase the transmission distance and the data rate.

In this comparison, let's increase the transmission distance to 1000m and the data transmission rate to 10Gbps. The bandwidth of a copper cable decreases as distance increases, just as with fiber-optic cables. Because we have increased the transmission distance by a factor of 10, it's fair to say that the Category 6A cable bandwidth will decrease by a factor of 10 over 1000m.

With a reduction in bandwidth by a factor of 10, we will need ten times more Category 6A cables to equal the old 2.5Gbps performance. In other words, we need 1350 Category 6A cables bundled together. In this comparison, however, the bandwidth has been increased from 2.5Gbps to 10Gbps. This means we have to quadruple the number of Category 6A cables to meet the bandwidth requirement. We now need 5400 Category 6A cables bundled together. Imagine how many cables we would need if the transmission distance increased to 80,000m. We would need a whopping 432,000 Category 6A cables bundled together.

These comparisons vividly illustrate the size advantage that fiber-optic cables have over copper cables. The advantage becomes even more apparent as distances increase. The enormous capacity of such a small cable is exactly what is needed to install high-bandwidth systems in buildings where the conduits and raceways are almost fully populated with copper cables.

Now that we have calculated the size advantages of optical fiber over Category 6A cable, let's look at the weight advantages. It is pretty easy to see that thousands or tens of thousands of Category 6A cables bundled together will outweigh a ribbon fiber-optic cable 0.65" in diameter. It's difficult to state exactly how much less a fiber-optic cable would weigh than a copper cable performing the same job—there are just too many variables in transmission distance and data rate. However, it's not difficult to imagine the weight savings that fiber-optic cables offer over copper cables. These weight savings are being employed in commercial aircraft, military aircraft, and the automotive industries, just to mention a few.

Security

We know that optical fiber is a dielectric and because of that it is immune to EMI. So why is an optical fiber secure and virtually impossible to tap? Because of total internal reflection, optical fiber does not radiate.

In Chapter 5, you learned about macrobends. Excessive bending on an optical fiber will cause some of the light energy to escape the core and cladding. This light might penetrate the coating, buffer, strength member, and jacket. This energy is detectable by means of a *fiber identifier*.

Fiber identifiers detect light traveling through an optical fiber by inserting a macrobend. Photodiodes are placed against the jacket or buffer of the fiber-optic cable on opposite sides of the macrobend. The photodiodes detect the light that escapes from the fiber-optic cable. The light energy detected by the photodiodes is analyzed by the electronics in the fiber identifier. The fiber identifier can typically determine the presence and direction of travel of the light.

If the fiber identifier can insert a simple macrobend and detect the presence and direction of light, why is fiber secure? Detecting the presence of light and determining the source of the light does not require much optical energy. However, as you learned earlier in this chapter, a fiber-optic receiver typically has a relatively small window of operation. In other words, the fiber-optic receiver typically needs at least 10 percent of the energy

from the transmitter to accurately decode the signal on the optical fiber. Inserting a macrobend in a fiber-optic cable and directing 10 percent of the light energy into a receiver is virtually impossible. A macrobend this severe would also be very easy to detect with an *optical time-domain reflectometer* (OTDR). There is no transmission medium more secure than optical fiber.



The fiber identifier and OTDR are covered in detail in Chapter 14, “Test Equipment and Link/Cable Testing.”

Safety

Electrical safety is always a concern when working with copper cables. Electrical current flowing through copper cable poses shock, spark, and fire hazards. Optical fiber is a dielectric that cannot carry electrical current; therefore, it presents no shock, spark, or fire hazard.

Because optical fiber is a dielectric, it also provides electrical isolation between electrical equipment. Electrical isolation eliminates ground loops, eliminates the potential shock hazard when two pieces of equipment at different potentials are connected together, and eliminates the shock hazard when one piece of equipment is connected to another with a ground fault.

Ground loops are typically not a safety problem; they are usually an equipment operational problem. They create unwanted noise that can interfere with equipment operation. A common example of a ground loop is the hum or buzz you hear when an electric guitar is plugged into an amplifier with a defective copper cable or electrical connection. Connecting two pieces of equipment together with optical fiber removes any path for current flow, which eliminates the ground loop.

Using copper cable to connect two pieces of equipment that are at different electrical potentials poses a shock hazard. It's not uncommon for two grounded pieces of electrical equipment separated by distance to be at different electrical potentials. Connecting these same two pieces of equipment together with optical fiber poses no electrical shock hazard.

If two pieces of electrical equipment are connected together with copper cable and one develops a ground fault, there is now a potential shock hazard at both pieces of equipment. Everyone is likely to experience, or hear of someone experiencing, a ground fault at least once in their life. A common example is when you touch an appliance such as an electric range or washing machine and experience a substantial electrical shock. If the piece of equipment that shocked you was connected to another piece of equipment with a copper cable, there is a possibility that someone touching the other piece of equipment would also be shocked. If the two pieces of equipment were connected with optical fiber, the shock hazard would exist only at the faulty piece of equipment.

Nonconductive fiber-optic cables offer some other advantages, too. They do not attract lightning any more than any other dielectric. They can be run through areas where faulty

copper cables could pose a fire or explosion hazard. The only safety requirement that Article 770 of the NEC places on nonconductive fiber-optic cables addresses the type of jacket material. When electrical safety, spark, or explosion hazards are a concern, there is no better solution than optical fiber.



Article 770 of the NEC is discussed in detail in Chapter 7, “Fiber-Optic Cables.”

Summary

This chapter explained the performance advantages that optical fiber has over copper cable. Optical fiber is the highest performance transmission medium available today, boasting incredible bandwidth with very little attenuation over great distances. Fiber-optic cables the size of a garden hose can hold hundreds of optical fibers each capable of carrying billions of bits of data per second. They also offer better security and safety.

Exam Essentials

Describe the bandwidth advantages of optical fiber over twisted pair and coaxial copper cables. Remember that single-mode systems with laser transmitters offer the greatest bandwidth and that multimode systems with LED transmitters offer the least bandwidth. The bandwidth of optical fiber and copper cable is inversely proportional to its length. In other words, as the length of the optical fiber or copper cable increases, the bandwidth of the optical fiber or copper cable decreases.

Describe the attenuation advantages of optical fiber over twisted pair and coaxial copper cables. Remember that all transmission mediums lose signal strength over distance and the loss of signal strength, or attenuation, is typically measured in decibels. Optical fiber systems measure attenuation using optical power. Copper cable systems typically use voltage drop across a defined load at various transmission frequencies to measure attenuation.

Remember that attenuation in copper cables is measured at different frequencies. This is not the case with optical fiber, where attenuation is measured with a continuous wave light source that is not modulated. The attenuation in a copper cables increases as the transmission frequency increases. This is not the case with optical fiber, where transmission frequency has no impact on attenuation.

Describe the electromagnetic advantages of optical fiber over copper cable. You need to know that EMI is electromagnetic energy that causes undesirable responses, degradation, or complete system failure. Systems using copper cable are vulnerable to the effects of EMI because a changing electromagnetic field will induce current flow in a copper conductor. Optical fiber is a dielectric and current does not flow through optical fiber. Thus EMI has no effect on the operation of an optical fiber, and fiber-optic cables do not suffer from crosstalk as copper cables do.

Describe the size advantages of fiber-optic cable over copper cable. You need to know that a copper cable and fiber-optic cable with similar performance characteristics may differ greatly in size. There is no rule of thumb for size difference; however, the fiber-optic cable may be up to hundreds of times smaller than the copper cable.

Describe the weight-saving advantages of fiber-optic cable over copper cable. You need to know that a copper cable and fiber-optic cable with similar performance characteristics may greatly differ in weight. There is no rule of thumb for weight difference; however, the fiber-optic cable may be up to hundreds of times lighter than the copper cable.

Describe the security advantages of fiber-optic cable over copper cable. You need to know that because of total internal reflection, optical fiber does not radiate, making it virtually impossible to tap. Optical fiber is the most secure transmission medium available.

Describe the safety advantages of fiber-optic cable over copper cable. You need to know that optical fiber is a dielectric that cannot carry electrical current and presents no shock, spark, or fire hazard. Optical fiber also provides electrical isolation between electrical equipment.

Review Questions

1. Optical fiber offers _____ bandwidth and _____ attenuation than twisted pair or coaxial cable.
- A. Less, more
 - B. Equal, more
 - C. Greater, more
 - D. Greater, less

Hint: Also known as information carrying capacity and loss

2. Optical fiber is a _____. Therefore it is immune to the effects of EMI.
- A. Conductor
 - B. Composite
 - C. Dielectric
 - D. Radiator

Hint: Resists the flow of current

3. Fiber-optic cables are _____ and _____ than copper cables with comparable transmission frequency performance.
- A. Larger, heavier
 - B. Smaller, lighter
 - C. Smaller, heavier
 - D. Larger, lighter

Hint: The difference in size and weight increases considerably as the transmission distance increases.

4. Fiber-optic cables do not _____ light energy; therefore they are virtually immune to being tapped, which makes them very secure.
- A. Rotate
 - B. Receive
 - C. Detect
 - D. Radiate

Hint: Total internal reflection

5. Optical fiber does not _____ electricity and offers safety advantages over copper cables.
- A. Rotate
 - B. Conduct
 - C. Detect
 - D. Radiate

Hint: It provides electrical isolation between electrical equipment.

Chapter Exercises

Calculate the Bandwidth for a Length of Optical Fiber

The minimum effective bandwidth-length product in MHz · km is defined in TIA-568-C.3 and ISO/IEC 11801.

Refer to the following table and determine the minimum effective modal bandwidth for 300 meters of OM4 optical fiber at a wavelength of 850nm.

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
62.5/125μm	850	3.5	200	Not required
Multimode OM1	1300	1.5	500	Not required
Type A1b TIA 492AAAA				
50/125μm	850	3.5	500	Not required
Multimode OM2	1300	1.5	500	Not required
TIA 492AAAB Type A1a.1				
850nm laser-optimized	850	3.5	1500	2000
50/125μm Multimode OM3	1300	1.5	500	Not required
Type A1a.2 TIA 492AAAC				

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)	Minimum overfilled modal bandwidth-length product (MHz · km)	Minimum effective modal bandwidth-length product (MHz · km)
850nm laser-optimized 50/125µm Multimode OM4 Type A1a.3 TIA 492AAAD	850	3.5	3500	4700
	1300	1.5	500	Not required
Single-mode	1310	0.5	N/A	N/A
Indoor-outdoor OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1550	0.5	N/A	N/A
Single-mode	1310	1.0	N/A	N/A
Inside plant OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1550	1.0	N/A	N/A
Single-mode	1310	0.5	N/A	N/A
Outside plant OS1 Type B1.1 TIA 492CAAA OS2 Type B1.3 TIA 492CAAB	1550	0.5	N/A	N/A

The steps to calculate the minimum effective modal bandwidth for 300 meters of OM4 optical fiber at a wavelength of 850nm are as follows:

$$\text{Bandwidth-length product} = 4700 \text{ MHz} \cdot \text{km}$$

$$\text{Optical fiber length} = 0.3\text{km}$$

$$\text{Bandwidth in MHz} = 4700 \div 0.3$$

$$\text{Bandwidth} = 15,666.67\text{MHz}$$

Calculate the Attenuation for a Length of Optical Fiber

The maximum allowable attenuation in an optical-fiber is defined in TIA-568-C.3 and ISO/IEC 11801.

Refer to the following table and determine the maximum allowable attenuation for 5.6km of outside plant single-mode optical fiber.

Optical fiber cable type	Wavelength (nm)	Maximum attenuation (dB/km)
50/125μm multimode	850	3.5
	1300	1.5
62.5/125μm multimode	850	3.5
	1300	1.5
Single-mode inside plant cable	1310	1.0
	1550	1.0
Single-mode outside plant cable	1310	0.5
	1550	0.5

The maximum attenuation for outside plant single-mode optical fiber is the same for both wavelengths. The steps to calculate the maximum attenuation are as follows:

$$\text{Attenuation coefficient at 1310 or 1550nm} = 0.5\text{dB/km}$$

$$\text{Optical fiber length} = 5.6\text{km}$$

$$\text{Attenuation for the length of optical fiber} = 5.6 \times 0.5 = 2.8$$

$$\text{Maximum attenuation for 5.6m of outside plant single-mode optical fiber is } 2.8\text{dB}$$

Chapter 14

Test Equipment and Link/Cable Testing

THE FOLLOWING ETA FIBER-OPTICS INSTALLER COMPETENCIES ARE COVERED IN THIS CHAPTER:

- ✓ Explain why test equipment calibration should be traceable to the National Institute of Standards and Technology (NIST) calibration standard.
- ✓ Describe the types of fiber-optic test equipment that can be used to test for continuity.
- ✓ Explain the use of a visual fault locator (VFL) when troubleshooting a fiber span.
- ✓ Describe the basic operation of a multimode and single-mode optical loss test set (OLTS).
- ✓ Explain the difference between a patch cord and a measurement quality jumper (MQJ).
- ✓ Define the purpose of a mode filter.
- ✓ Explain why five small-radius nonoverlapping loops around a mandrel may be required on the transmit jumper when measuring multimode link attenuation in accordance with ANSI/TIA-526-14-A.
- ✓ Explain why a single turn 30mm in diameter loop must be applied to the transmit jumper when measuring single-mode link attenuation in accordance with ANSI/TIA-526-7.
- ✓ Explain why the encircled flux requirement was developed for multimode link attenuation measurements.
- ✓ Explain why multimode insertion loss measurements being performed in accordance with ANSI/TIA-526-14-B require a modal controller on the transmit jumper.
- ✓ Describe how to measure the optical loss in a patch cord with an OLTS using the steps described in ANSI/TIA-526-14, method A, two-test jumper reference.
- ✓ Summarize the basic operation of an optical time domain reflectometer (OTDR).





This chapter describes the basic operation and application of essential fiber-optic test equipment. The basic theory, operation, and application of each piece of test equipment will be explained.

Many of the test methods described in this chapter are based on current industry standards. How to test to these standards is described in detail.

In this chapter, you will learn to

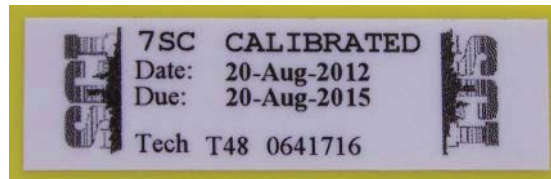
- Perform optical loss measurement testing of a cable plant
- Determine the distance to a break in the optical fiber with an OTDR
- Measure the loss of a cable segment with an OTDR
- Measure the loss of an interconnection with an OTDR
- Measure the loss of a fusion splice or macrobend with an OTDR

Calibration Requirements

The routine calibration of test equipment is essential to achieving accurate test results. Calibration is always required for test equipment that makes absolute measurements, such as an optical power meter. However, not all test equipment requires calibration. For example, calibration is not required for a continuity tester; however, it is necessary for a stabilized light source.

Optical power meters should be calibrated to the three principle wavelength regions: 850nm, 1300nm, and 1550nm. The calibration should be traceable to the National Institute of Standards and Technology (NIST) calibration standard or some other standard. A calibration sticker similar to the one shown in Figure 14.1 should be clearly visible and not expired. The calibration is expired when the posted due date has passed.

FIGURE 14.1 Calibration sticker on a piece of test equipment



NIST in general does not require or recommend any set calibration intervals. Many companies adopt their own measurement assurance programs and define calibration intervals based on various factors such as

- Equipment manufacturer recommendations
- Contractual requirements
- Regulatory requirements
- Environmental factors
- Inherent stability of the device

To access NIST publications, go to www.nist.gov.

Continuity Tester

The continuity tester is an essential piece of test equipment for every fiber-optic toolkit. It is also one of the least expensive tools in your toolkit. This low-cost tool will allow you to quickly verify the continuity of an optical fiber.

Many of the continuity testers available today are just modified flashlights. Some flashlights, such as the one pictured in Figure 14.2, have been modified to use an LED instead of an incandescent lamp. Others may just use a brighter incandescent lamp, like the tester pictured in Figure 14.3.

FIGURE 14.2 LED continuity tester



The job of the continuity tester is to project light from the LED or incandescent lamp into the core of the optical fiber. This is usually accomplished by attaching a receptacle to the lamp end of the flashlight. The receptacle is designed to center and hold the connector ferrule directly above the LED or incandescent lamp. When the connector ferrule is inserted into the receptacle, the endface is typically just above the light source.

This approach eliminates the need for a lens to direct light into the core of the optical fiber. However, it directs only a fraction of the light emitted by the lamp or LED into the core of the optical fiber.

FIGURE 14.3 Incandescent continuity tester



Because there is no lens used to focus light energy into the optical fiber, the continuity tester works best with multimode optical fiber. The measured optical output power of an LED continuity tester is typically less than -36dBm when measured at the end of one meter of $62.5/125\mu\text{m}$ optical fiber. The optical output power is reduced another $3\text{--}4\text{dB}$ when used with $50/125\mu\text{m}$ optical fiber and $16\text{--}20\text{dB}$ when used with single-mode optical fiber. The continuity tester works best with multimode optical fiber; however, it can be used with single-mode optical fiber. For the best results with single-mode optical fiber, dim the lights in the test area if possible.

The low output power of the continuity tester combined with the high attenuation of visible light by an optical fiber limit the length of optical fiber that can be tested. A multimode optical fiber will attenuate a 650nm light source roughly 7dB per kilometer. This high attenuation limits the use of the continuity tester to multimode optical fibers typically no greater than 2km in length.

LED continuity testers have a couple of advantages over incandescent lamp testers. They typically feature a red ($635\text{--}650\text{nm}$) LED that is easy to see. They require far less power from the batteries than an incandescent lamp. An LED continuity tester may provide 10 or more times longer battery life compared to an incandescent lamp. The LED lamp will never need replacing, unlike an incandescent lamp, which may last only 10 hours. LED lamps are also shock resistant.

Whether you are using an LED or an incandescent lamp continuity tester, operation and testing are identical. The continuity tester can test only for breaks in the optical fiber. It does not have sufficient power to identify the location of a break.

The first step when using the continuity tester is to clean and visually inspect the endface of the connector before inserting it into the continuity tester. You need to visually inspect the connector to verify that there is no endface damage. A shattered endface will significantly reduce the light coupled into the core of the optical fiber under test.

After the connector has been cleaned and inspected, you need to verify that the continuity tester is operating properly. Turn the continuity tester on and verify that it is emitting light. This could save you from embarrassment. You don't want to tell the customer that there is a break in their fiber-optic cable when the real problem is dead batteries in your continuity tester.

Depending on where the other end of the fiber-optic cable to be tested is located, you may need an assistant to help you. With the continuity tester turned on, insert the ferrule of the connector under test into the receptacle. If light is being emitted from the other end of the optical fiber, there is good continuity. This means only that there are no breaks in the optical fiber. It does not mean that there are no macrobends or high-loss interconnections in the fiber-optic cable or link under test.

The continuity tester is often used to verify that there are no breaks in a reel of fiber-optic cable before it is installed. There are a few ways you could approach testing the reel. One way would be to install a connector on either end of the cable. The other end of the cable should have the jacket and strength member stripped back so that the buffer is exposed. You should remove a small amount of buffer to expose the optical fiber under test. This will allow you to clearly see the light from the continuity tester, ensuring accurate results. To test the optical fiber in the reel, turn on the continuity tester and insert the ferrule of the connector under test into the receptacle. If light is being emitted from the other end of the optical fiber, there is good continuity.

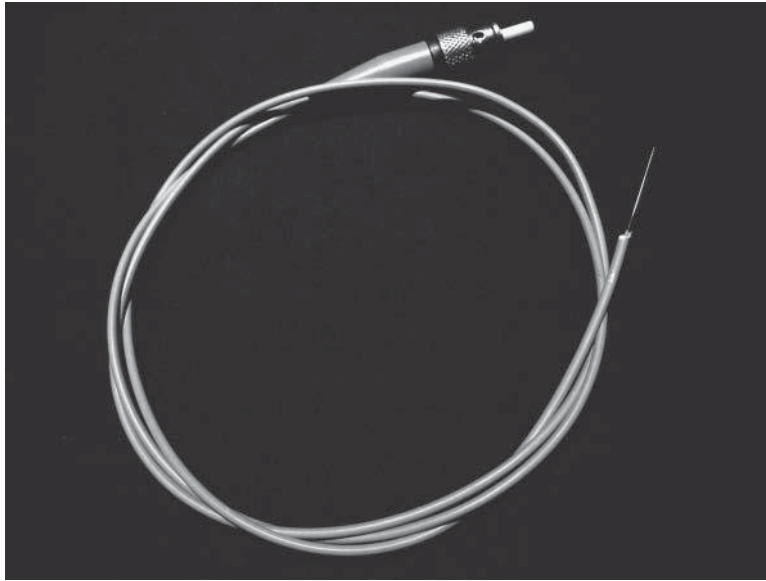
Another approach is to use a pigtail with a mechanical splice. The pigtail would have a connector on one end that will mate with the continuity tester receptacle. The other end should have the jacket, strength member, and buffer stripped back so that 10 to 15mm of optical fiber is exposed, as shown in Figure 14.4. The optical fiber should have a perpendicular cleave.

One end of the cable under test should be prepared just like the nonconnectorized end of the pigtail. The other end should have the jacket and strength member stripped back so that the buffer is exposed. A small amount of buffer should be removed to expose the optical fiber under test.

To test for continuity, insert the pigtail connector into the continuity tester receptacle. Turn on the continuity tester and verify that light is being emitted from the exposed optical fiber on the opposite end of the pigtail. Insert the exposed optical fiber from the pigtail into one side of a mechanical splice. Insert the optical fiber from the cleaved end of the cable into the other side of the mechanical splice and check for continuity. Typically, one mechanical splice can be reused for many optical fibers depending on the type of splice.

Another method that works well with multimode optical fiber is simply projecting light into the optical fiber with a continuity tester. As with the other techniques discussed, both ends of the cable should have the jacket and strength member stripped back so that the buffer is exposed. A small amount of buffer should be removed to expose the optical fiber under test and one end of the optical fiber should have a perpendicular cleave.

After the fiber-optic end is cleaved, simply insert the buffered optical fiber into the continuity tester receptacle as shown in Figure 14.5 to test the continuity of the entire length of fiber. Use caution when performing this step to ensure that the optical fiber end is not broken off.

FIGURE 14.4 Pigtail prepared for temporary mechanical splice**FIGURE 14.5** Testing the continuity of a single fiber in a 12-fiber breakout cable

While performing the steps outlined previously, always work over a nonreflective black surface like the one shown in Figure 14.5. The nonreflective black mat makes it easier to keep track of optical fiber ends. Always use a labeled container designated only for fiber-optic waste to dispose of bare optical fibers. Refer to Chapter 6, “Safety,” for additional safety information.

Visual Fault Locator

Like the continuity tester, the *visual fault locator (VFL)* is an essential tool for every fiber-optic toolkit. Unlike the continuity tester, it is not one of the least expensive tools in your toolkit. The VFL will allow you to quickly identify breaks or macrobends in the optical fiber and identify a poor fusion splice in multimode or single-mode optical fiber.

The big difference between the continuity tester and the VFL is the light source and optical output power of the light source. The VFL typically uses a red (635–650nm) laser light source. The optical output power of the laser is typically 1mW or less. Because of the high optical output power, you should never view the output of the VFL directly.

The VFL is available in different shapes and sizes. Some may look like a pen like the one shown in Figure 14.6, others may be built into an optical time-domain reflectometer (OTDR) or other piece of test equipment like the one shown in Figure 14.7, and some may look like a small test equipment box like the one shown in Figure 14.8.

FIGURE 14.6 VFL in the shape of a pen



Photo courtesy of KITCO Fiber Optics

The VFL fills the core of the optical fiber with light from the laser. The light from the laser escapes the optical fiber at a break or macrobend. The light escaping from the optical fiber will typically illuminate the buffer surrounding the optical fiber. Macrobends are not always visible through the jacket but are typically visible through the buffer. Breaks may be visible through the jacket of the fiber-optic cable depending on jacket color, thickness, number of optical fibers in the cable, and amount of strength member.

Unlike the continuity tester, the VFL is not limited to testing multimode optical fibers 2km or less in length. The VFL can be used to verify continuity of multimode or single-mode optical fiber longer than 2km. Due to attenuation of the 635–650nm laser light source by the optical fiber, macrobends may not be detectable beyond 1km in multimode optical fiber and 500m in single-mode optical fiber. The same holds true for finding breaks in the optical fiber through the jacket of the fiber-optic cable.

FIGURE 14.7 VFL built into a piece of test equipment**FIGURE 14.8** VFL with LC receptacle

The first step before using the VFL for testing an optical fiber is to measure the output power of the VFL. This can be done with a multimode test jumper and an optical power meter. Connect the output of the VFL to the test jumper and the other end of the test jumper to the power meter input. Set the power meter at the shortest wavelength possible.

This is typically 850nm. The measured optical output power of the VFL should be no less than -7dBm with a $62.5/125\mu\text{m}$ optical fiber. If the measured optical output power is less than -7dBm , replace the battery and retest. Most VFLs will have a measured optical output power greater than -3dBm with a fresh battery. Optical output power below -7dBm will limit the effectiveness of the VFL to identify breaks and macrobends.

The second step before using the VFL is to clean and visually inspect the endface of the connector on the optical fiber to be tested before inserting it into the VFL. You need to visually inspect the connector to verify that there is no endface damage. A shattered endface will not allow light to be properly coupled into the optical fiber under test. A shattered endface may also damage the VFL. Some VFLs use a short pigtail from the laser to the receptacle. The other end of the pigtail has a connector that will mate with the receptacle; this is referred to as a contact VFL. A shattered endface can damage the optical fiber in the connector on the end of the pigtail.



There are two types of VFLs: contact and non-contact. With a contact VFL, the optical fiber under test will make contact with the VFL. However, with a non-contact VFL the optical fiber under test will not touch the VFL.

To identify a macrobend in the optical fiber, visually inspect the length of the cable under test. If you locate a bend in the cable that is glowing, this is a macrobend. Figure 14.9 shows simplex cordage with a macrobend.

To identify a break in the optical fiber, visually inspect the length of the cable under test. If you locate a small spot on the cable that is emitting red light, this is a break. Figure 14.10 shows multimode cordage with a break.

The VFL can be used to test the continuity of an optical fiber in the same manner described in the “Continuity Tester” section earlier in this chapter. The VFL, however, will couple 1000 times (30dB) or more light energy into the optical fiber than the LED continuity tester. You should never view the output of the VFL or the endface of a connector being tested by the VFL directly.

Using the VFL to test for continuity can produce false results. A single break in an optical fiber does not stop all the light from the VFL cold. Some of the light will typically reach the end of the optical fiber. In most cases, so much light reaches the end of the optical fiber that the person viewing the light might believe there is good continuity when in fact the optical fiber is broken.

A break in the optical fiber will typically attenuate the light from the VFL by 20–30dB. The VFL will couple 30dB or more light energy into the optical fiber than the LED continuity tester. The maximum loss from a break in an optical fiber typically never exceeds 30dB. Therefore, the light energy from the VFL after passing through a single break with 30dB of loss will be equal to or greater than the light energy from an LED continuity tester. While the VFL can be used to test continuity, the continuity tester is typically the best tool for that job.

FIGURE 14.9 The light from a VFL identifying a macrobend in simplex cordage

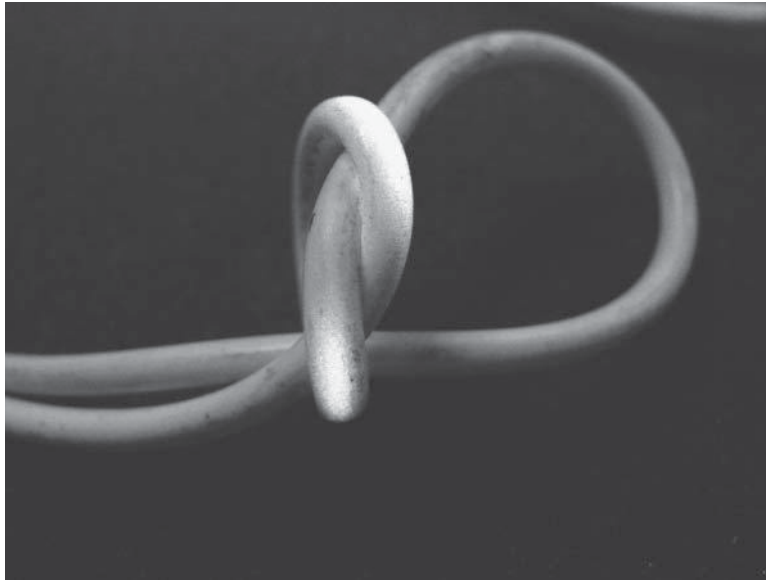


FIGURE 14.10 The light from a VFL identifying a break in simplex cordage



The VFL is often used in conjunction with an OTDR to help identify the actual location of the fault.



Real World Scenario

Good Light or Bad Light

When working with other installers or technicians, you may hear them refer to good light or bad light while using a VFL for continuity testing. The term *good light* implies there is good continuity and *bad light* implies there is not good continuity. The use of the good light/bad light technique will produce false results simply because the VFL may couple 1000 times (30dB) or more light energy into the optical fiber than the LED continuity tester. The light energy from the VFL will pass through a single break and typically look brighter than the light energy from an unbroken optical fiber being tested with an LED continuity tester. This is why a VFL is not called a continuity tester and a continuity tester is not called a VFL.

Fiber Identifier

The fiber identifier acts as the fiber-optic installer or technician's infrared eyes. By placing a slight macrobend in an optical fiber or fiber-optic cable, it can detect infrared light traveling through the optical fiber and determine the direction of light travel. Some fiber identifiers can also detect test pulses from an infrared (800–1700nm) light source or measure the power in the optical fiber.

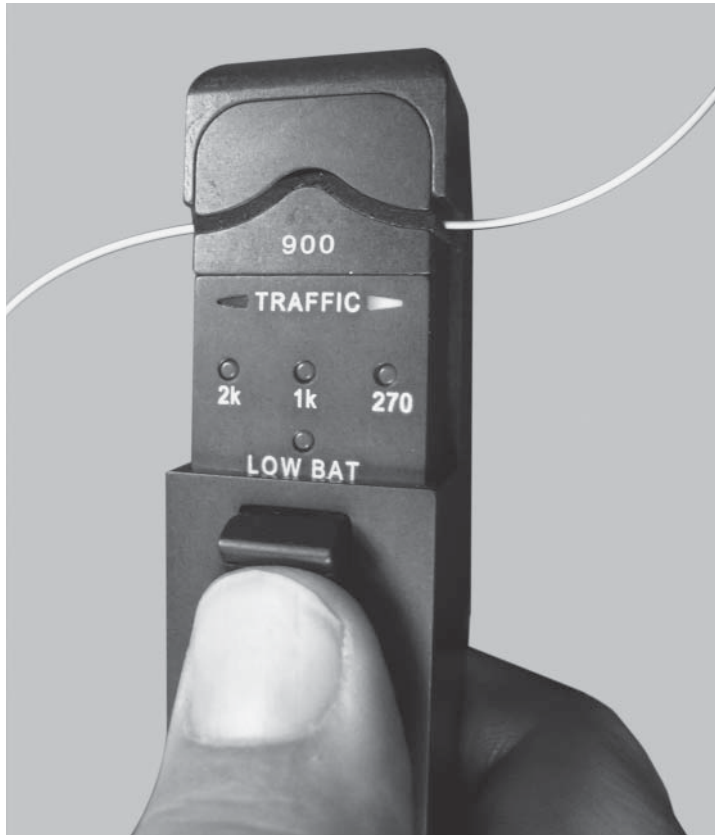
The fiber identifier typically contains two photodiodes that are used to detect the infrared light. The photodiodes are mounted so that they will be on opposite ends of the macrobend of the optical fiber or fiber-optic cable being tested; this is often referred to as the optical head. Figure 14.11 shows the location of the photodiode assemblies in the fiber identifier. The photodiode assemblies look like two small glass lenses.

The fiber identifier can typically be used with coated optical fiber, tight-buffered optical fiber, a single optical fiber cable, or a ribbon cable. Each of these must be placed in the center of the photodiodes during testing. Selecting the correct attachment for the optical fiber or optical fiber cable type under test typically does this. Figure 14.12 shows three attachments. Some fiber identifiers do not require attachments to be interchanged when working with different diameter buffers or cables.

To test an optical fiber or fiber-optic cable, select the correct attachment if required and install it on the fiber identifier. Figure 14.13 shows the fiber identifier ready to test a 900µm tight-buffered optical fiber. Center the tight-buffered optical fiber over the photodiode assemblies and insert a macrobend into the tight-buffered optical fiber by pushing the slide into the tight-buffered optical fiber and compressing it, as shown in Figure 14.14.

FIGURE 14.11 Photodiode assemblies in the fiber identifier**FIGURE 14.12** Fiber identifier optical fiber and fiber-optic cable attachments**FIGURE 14.13** Fiber identifier ready to test a 900 μ m tight-buffered optical fiber

FIGURE 14.14 Fiber identifier compressing a tight-buffered optical fiber



If there is sufficient infrared light energy traveling through the optical fiber, one of the directional arrows on the fiber identifier should illuminate. The directional arrow is pointing in the direction the light is traveling. If a test pulse is being transmitted through the optical fiber, the directional arrow and test pulse indicator should illuminate. If the fiber identifier contains a power meter, the optical power level should be displayed.

The fiber identifier works best with coated optical fiber or tight-buffered optical fiber. Like the VFL, the fiber identifier may not always work with single optical-fiber cables. How well it works with a cable depends on the amount of strength member within the cable, jacket thickness, jacket color, and amount of light energy available from the optical fiber.

The fiber identifier can be used by itself or in conjunction with an OTDR. It can be used to identify traffic in a working fiber-optic link or can be used to help identify the location of a fault.

Inline Optical Power Monitoring

Inline power monitoring is not a new concept in telecommunications. It has been deployed for decades in radio frequency (RF) applications measuring the forward and reverse power in coaxial copper cables. However, it is relatively new to fiber-optic applications. As of this writing, AIR6552/1, Inline Optical Power Monitoring, was under development within SAE International's AS-3 Fiber Optics and Applied Photonics Committee. This standard and its slash sheets are discussed in the “Emerging Testing Standards” section at the end of this chapter.



Slash sheets are typically a family of documents where the main document such as AIR6552 will refer to an overall/general aspect of the document and then the slash sheet document(s) will refer to specific items within that overall document.

There are many applications for inline optical power monitoring, such as measuring forward optical power, reverse optical power, controlling optical power in an amplifier, anticipating failures, and detecting fluctuations, to name a few.

Inline optical power monitoring is different from using a fiber identifier, which places a macrobend in the optical fiber to detect infrared light traveling through the optical fiber. It requires an inline power tap like the one shown in Figure 14.15 to be placed inline with the optical fiber.

FIGURE 14.15 Inline optical fiber power tap

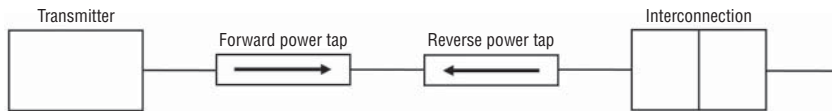


Photo courtesy of EigenLight Corporation

An inline power tap like the one shown in Figure 14.15 is a directional sensor that can only measure the light traveling in one direction. This power tap can be placed inline to measure forward power or reverse power. Two taps placed in series could be configured to measure forward and reverse power simultaneously. An example of this is shown in the block diagram in Figure 14.16, where two taps are used to measure the forward power into an interconnection and the reverse power or back reflection from the interconnection.

While other test equipment such as an OTDR can be used to evaluate the light energy being reflected back from an interconnection, only the power tap allows this to be performed in real time using the light energy from the transmitted signal.

FIGURE 14.16 Inline optical fiber power taps in series measuring forward and reverse power



Inline Optical Power Monitor

Inline optical power monitors like the one shown in Figure 14.17 are optically passive devices that provide a continuous readout of optical power either absolute or relative traveling through the optical fiber in one direction. They are typically available with pigtails like the unit shown in Figure 14.17 or receptacles that accept connectors like the unit shown in Figure 14.18. Unlike the fiber identifier, which can be used with single-mode or multimode optical fiber, the inline optical power monitor is optimized for a specific optical fiber type.

FIGURE 14.17 Inline optical power monitor with pigtail



Photo courtesy of EigenLight Corporation

This book cannot cover all the different ways an inline optical power monitor can be used. Some typical uses include measuring the output power of the transmitter or the input power of the receiver. That same meter could be reversed so light travels in the opposite direction, allowing it to measure the power being reflected back toward the transmitter.

FIGURE 14.18 Inline optical power monitor with receptacles

Photo courtesy of EigenLight Corporation

Multiple inline optical power monitors can be configured to support many types of measurements. Two common applications include measuring the gain of an optical amplifier, as shown in the block diagram in Figure 14.19, and measuring the power delivered to each receiver in a single-fiber bidirectional application, as shown in the block diagram in Figure 14.20.

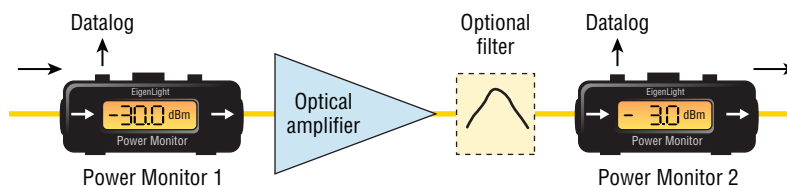
FIGURE 14.19 Block diagram of an inline optical power monitor measuring amplifier gain

Image courtesy of EigenLight Corporation

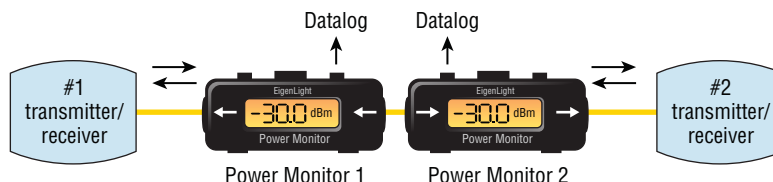
FIGURE 14.20 Block diagram of inline optical power monitors measuring receiver power

Image courtesy of EigenLight Corporation

Inline Network Sensors

Inline network sensors are similar to inline optical power monitors; however, they allow real-time access to the output of the transimpedance amplifier and typically have adjustable gain. The transimpedance amplifier in the inline network sensor converts the light energy received by the photodiode in the power tap into electrical current.

The electrical current from the transimpedance amplifier could be measured by a number of devices, such as a voltmeter, oscilloscope, or analog-to-digital converter, to name a few. The gain of each transimpedance amplifier is typically adjustable to compensate for higher power levels when measuring forward power or lower power levels when measuring reverse power. Multiple sensors may be incorporated into a single configurable rack mount enclosure, as they are in the unit shown in Figure 14.21.

FIGURE 14.21 1U rack mount enclosure with three independently adjustable inline network sensors



Photo courtesy of Engineering Development Laboratory, Inc.

The unit shown in Figure 14.21 features three power taps, each terminated at the front panel with FC receptacles. The output of each transimpedance amplifier can be measured across the banana receptacles on the front panel. The gain of each transimpedance amplifier is adjustable, so this unit can be configured to measure a wide range of forward or reverse power levels.

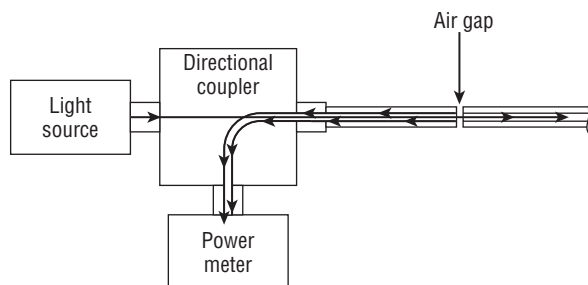
Optical Return Loss Test Set

Optical return loss (ORL) testing is performed with an optical return loss test set. ORL testing measures the amount of optical light energy that is reflected back to the transmit end of the fiber-optic cable. The energy being reflected back, or back reflections, comes from mechanical interconnections, passive devices, fiber ends, and Rayleigh scattering caused by impurities within the optical fiber.

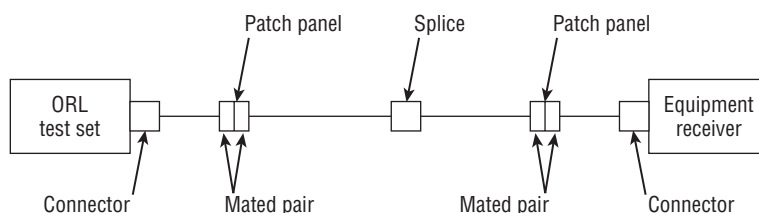
Besides reducing the amount of light transmitted, back reflections can cause various laser light source problems. They can cause the laser's output wavelength to vary. They can also cause fluctuations in the laser's optical output power and possibly permanently damage to the laser light source.

The impact that back reflections have on the laser light source can cause problems in analog and digital systems. In digital systems, they can increase the BER. In analog systems, they reduce the *signal-to-noise ratio (SNR)*.

The ORL test set measures return loss using an *optical continuous wave reflectometer (OCWR)*. A light source within the ORL test set continuously transmits light through a directional coupler, as shown in Figure 14.22. Light energy returned from the optical fiber is directed to the photodiode of a power meter. The light energy measured by the power meter is the return loss.

FIGURE 14.22 Directional coupler in an ORL test set

ORL measurements of a fiber-optic link should be taken with all patch cords and equipment cords in place. All system equipment should be turned off. The receive connector should remain plugged into the equipment receiver. The transmit connector should be unplugged from the equipment transmitter and plugged into the ORL test set after the test set has been calibrated, as shown in Figure 14.23. The test set should be calibrated as described in the manufacturer's operation manual. Before performing ORL testing, clean and inspect all connectors. Allow sufficient time for the ORL test set to warm up and stabilize.

FIGURE 14.23 ORL test set connected to fiber-optic link and system equipment

ORL test sets are available for both multimode and single-mode optical fiber. The multimode ORL test uses an LED light source with a typical output power of -20dBm . The return loss measurement range is usually between 10 and 45dB. Accuracy is normally within 0.5dB.

The single-mode ORL test set uses a laser source with a typical output power of -10dBm . The return loss measurement range is usually between 0 and 50dB or 0 and 60dB. Accuracy in the 0 to 50dB range is normally 0.5dB with a decrease in the 0 to 60dB range of 1dB.

Stabilized Light Source and Optical Power Meter

The stabilized light source and optical power meter work hand in hand. They are typically referred to as an *optical loss test set (OLTS)*. Many different OLTSs are available. Some support only multimode testing, some single-mode, and a few can be used for both. The

multimode OLTS is typically the lowest priced, followed by the single-mode and then the combination multimode and single-mode.

Encircled flux launch conditions for all multimode *insertion loss* measurements has been added to IEC 61280-4-1 Edition 2 and ANSI/TIA-526-14-B. Encircled flux is not required for insertion loss testing performed in accordance with ANSI/TIA-526-14-A. The addition of the encircled flux requirement does not render existing test equipment obsolete, and the publication of ANSI/TIA-526-14-B does not mean that ANSI/TIA-526-14-A will not be called out in an installation or testing contract.

As of this writing, no known multimode stabilized light sources meet the encircled flux requirement without the addition of a modal controller on the transmit jumper. The encircled flux requirement, modal controllers, and mode filters are covered in detail later in this chapter.

This section explains the basic operation of the multimode and single-mode OLTS.

Multimode OLTS

The multimode OLTS is available with either a VCSEL or an LED stabilized light source. This section discusses the differences between applications of these test sets.

Vertical-Cavity Surface-Emitting Laser (VCSEL) OLTS

As discussed in earlier chapters, ANSI/TIA-568-C.3 defines optical fiber transmission performance parameters. This standard includes laser-optimized optical fiber, which was not included in ANSI/TIA/EIA-568-B.3. It would seem that the best source to test laser-optimized optical fiber would be the VCSEL, and several manufacturers do offer VCSEL light sources. However, multimode optical fiber test standards such as TIA-TSB-140 still specify an LED light source. The VCSEL light source should not be used for testing to ANSI/TIA-526-14 revision A or B.

LED OLTS

The multimode LED OLTS consists of an LED-based stabilized light source and an optical power meter. The typical light source consists of an 850nm LED and a 1300nm LED, like the one shown in Figure 14.24. The optical power meter is typically selectable for 850nm, 1300nm, and 1550nm, like the one shown in Figure 14.25.

Two types of LED light sources are available: filtered and unfiltered. A filtered light source does not overfill the optical fiber. An unfiltered light source does overfill the optical fiber. Multimode insertion loss measurements being done in accordance with ANSI/TIA-526-14A should be performed with a light source that meets the launch requirements of ANSI/TIA-455-78-B. This launch requirement can be achieved within the test equipment, which is the case with the filtered light source or with an external mandrel wrap on the transmit test jumper. Most LED light sources in use today are unfiltered and require the mandrel wrap. The mandrel wrap is discussed in detail later in this chapter.



An overfilled light source has a Category 1 Coupled Power Ratio (CPR) value as defined in ANSI/TIA-526-14A, Annex A Tables A.1 and A.2. Table A.1 defines the CPR value range for an 850nm light source, and Table A.2 defines the CPR value range for a 1300nm light source. A mandrel wrap must be used on the transmit test jumper of a Category 1 CPR light source.

FIGURE 14.24 Multimode 850nm and 1300nm light source**FIGURE 14.25** Multimode 850nm, 1300nm, and 1550nm optical power meter

Multimode insertion loss measurements being performed in accordance with ANSI/TIA-526-14-B require a modal controller on the transmit jumper. The modal controller, which replaces the mandrel, is discussed in detail later in this chapter.

The optical output power of an unfiltered LED light source is typically -20dBm for both the 850nm and 1300nm LEDs when measured with a $62.5/125\mu\text{m}$ optical fiber. The optical output power increases roughly 4dB when used with $100/140\mu\text{m}$ optical fiber and decreases roughly 3dB when used with a $50/125\mu\text{m}$ optical fiber.



Remember that ANSI/TIA-568-C.3 recognizes only $50/125\mu\text{m}$ or $62.5/125\mu\text{m}$ multimode optical fiber.

Most multimode light sources are designed to operate between the temperatures of 0° to +40° C or 0° to +50° C. Within these temperature ranges, they should provide a stable optical output power that typically varies less than ± 0.2 dB over an eight-hour period. To achieve this performance, the light source should be allowed to warm up before beginning testing. The manufacturer specifies warm-up times.

The multimode power meter typically uses a single photodiode with a switch to select the proper wavelength. The selector switch compensates for the responsivity of the photodiode at the various wavelengths. The output of the optical power meter is in decibels (dB) referenced to one milliwatt (mW), where 0 dBm is equal to 1 mW of optical power.

Many power meters will support both multimode and single-mode testing. Acceptable optical fiber types may range from 9/125 μ m through 100/140 μ m. The optical input power range is typically +3 dBm to -50 dBm.

As with the light sources, most optical power meters are designed to operate between the temperatures of 0° and +40° C or 0° and +50° C. When operated within this temperature range, absolute accuracy is typically within ± 0.25 dB, relative accuracy within ± 0.15 dB, and repeatability within ± 0.04 dB. The values stated here are typical. Actual values can be obtained from the manufacturer's data sheet.



Real World Scenario

Is My Power Meter Broken?

After years of watching students test various fiber-optic links with an OLTS, I have seen many confused looks when the reading on the power meters drops to a very low value and never stabilizes. As discussed in this chapter, the manufacturer defines the optical input power range for the power meter. This value is typically between +3 dBm and -50 dBm.

The output power of an LED light source is typically -20 dBm with a 62.5/125 μ m optical fiber and -23 dBm with a 50/125 μ m optical fiber. A break in an optical fiber will cause significant attenuation, typically between 20 and 30 dB. If a light source coupled -23 dBm into the fiber-optic link under test and the break added 30 dB of attenuation, the maximum light energy after the break would equal -53 dBm, which is below the power range for the power meter. The actual power would be less because of the losses from the optical fiber and interconnections.

When the light energy into the power meter falls below the lowest value defined by the manufacturer, the power meter typically appears to be hunting. In other words, the numbers slowly increase and decrease; they never seem to stop changing. When this happens to you, remember that the power meter is not broken; however, the optical fiber is probably broken.

Single-Mode OLTS

The single-mode OLTS consists of a laser-based stabilized light source and an optical power meter. The typical light source consists of a 1310nm laser and a 1550nm laser. Some models have a separate output port for each laser, and others combine both wavelength lasers into a single output port. Another popular laser combination is 1550nm and 1625nm.

The optical output power of the laser is typically around -5dBm into a $9/125\mu\text{m}$ single-mode optical fiber. Some single-mode light sources have the ability to attenuate the output of the laser. Most single-mode light sources are designed to operate between the temperatures of 0° to $+40^\circ\text{C}$ or 0° to $+50^\circ\text{C}$. Within these temperature ranges, they should provide a stable optical output power that typically varies no more than $\pm 0.1\text{dB}$ over an eight-hour period. To achieve this performance, the light source should be allowed to warm up before beginning testing. The manufacturer specifies warm-up times.

The single-mode optical power meter is typically designed to handle a range of wavelengths. It may measure common multimode optical fiber wavelengths and single-mode optical fiber wavelengths. A typical power meter will be selectable for 850nm, 1300nm, 1310nm, 1480nm, 1550nm, and 1625nm.

Single-mode optical power meters are typically designed to operate between the temperatures of 0° to $+40^\circ\text{C}$ or 0° to $+50^\circ\text{C}$. When operated within this temperature range, they provide the best accuracy. Absolute accuracy is typically within $\pm 0.25\text{dB}$ when measured at $+25^\circ\text{C}$ with an optical input power of -10dBm , relative accuracy within $\pm 0.15\text{dB}$, and repeatability within $\pm 0.04\text{dB}$. The values stated here are typical. Actual values can be obtained from the manufacturer's data sheet.

Why Is the Loss for This Cable Assembly Different Each Time I Measure It?

One thing I have always enjoyed is the friendly competition between the students during lab. As the students get to know each other and openly communicate, they compete for the lowest-loss cable assembly. Sometimes there is a substantial gap between the loss measurements and the winner is clearly defined. Other times the gap is not measurable with the equipment in the lab and the results vary slightly each time a cable assembly is measured.

In this chapter, the typical values for absolute accuracy, relative accuracy, and repeatability for a power meter were defined. The stability of the light source was also defined. Each of these parameters has a range, and absolute accuracy is typically the largest. What these parameter ranges mean is that insertion loss measurements will vary. There is no such thing as a perfect piece of test equipment.

Patch Cord

ANSI/TIA-568-C.3, section 6 sets the performance specifications for optical fiber patch cords recognized in premises cabling standards. The optical fiber *patch cord* is used at cross-connections to connect optical fiber links. They are also used as equipment or work area cords to connect telecommunications equipment to horizontal or backbone cabling.

A patch cord is a length of optical fiber cable with connectors on both ends. It uses the same connector type and optical fiber type as the optical fiber cabling that it is connected to. The patch cord must meet the cable transmission performance requirements and physical cable specifications of sections 4.2 and 4.3.1 of ANSI/TIA-568-C.3. The patch cord must also meet the connector and adapter requirements of section 5.2 of ANSI/TIA-568-C.3.

Optical fiber patch cords used for either cross-connection or interconnection to equipment shall have a termination configuration defined in section 6.4 of ANSI/TIA-568-C.3. Section 6.4 describes configurations for simplex, duplex, and array patch cords. A simplex patch cord is a single-fiber cable with simplex connection terminations. A duplex patch cord is a two-fiber cable with duplex connectors. An array patch cord is a multifiber cable with array connectors on each end. Array connectors (also known as ferrule connectors) feature a single ferrule that contains multiple optical fibers arranged in a row or rows and columns. This connector is covered in detail in Chapter 9, “Connectors.”

Test Jumper

The terms *patch cord* and *jumper* are often used interchangeably. A test jumper can be a single-fiber cable or a multifiber cable. This section focuses on multimode and single-mode test jumpers as described in ANSI/TIA-568-C.0, Annex E.

In the fiber-optic industry, the *test jumper* has several names. The test jumper connected to the light source is typically called the *transmit jumper*. The test jumper connected to the optical power meter is typically called the *receive jumper*. The U.S. Navy calls the test jumper a *measurement quality jumper (MQJ)*. You may also see the test jumper referred to as a *reference jumper*. Regardless of the name, the test jumper is a critical part of your optical power measurement equipment.

Test jumpers must have a core diameter and numerical aperture nominally equal to the optical fiber being tested. You cannot test a 50/125µm link with a 62.5/125µm test jumper, or vice versa. Per ANSI/TIA-568-C.0, jumpers shall be no less than 1m and no greater than 5m in length. The termination of the test jumper shall be compatible with the cable being tested.

Test jumpers should be cleaned and inspected prior to making measurements. The endface of each connector should be evaluated under a microscope. There should be no scratches, notches, or chips in the endface of the optical fiber. You should also clean and

inspect each connector of the cable under test. Mating a dirty or damaged connector with a test jumper connector can destroy the test jumper connector.

Test jumpers should be tested for insertion loss prior to performing any of the ANSI/TIA-526-14 revision A or B methods. How they are tested depends on which standard or standards are called out by the customer. It may be one of the methods described in ANSI/TIA-526-14 revision A or B or a modified version of one of those methods. The maximum acceptable loss may be defined by the customer; when it is not, 0.4dB is typically a good value. This value is not defined by the ANSI/TIA-568-C.0 standard but serves as a good rule of thumb. Many test jumpers will have a measured loss of less than 0.2dB.

When verifying the quality of a test jumper in accordance with ANSI/TIA-568-C.0, clean and inspect the test jumpers. Verify that your test jumpers have the same optical fiber type as the cable plant you are going to test. While you are cleaning and inspecting the test jumpers, turn on the light source and optical power meter. Set both of them to the test wavelength. Allow sufficient time for the light source and optical power meter to warm up and stabilize. Warm-up information can be found in the manufacturer's data sheet.

After sufficient time has passed for the light source and optical power meter to stabilize, you need to measure the loss of the test jumpers. Connect test jumper 1 as shown in Figure 14.26. When testing multimode optical fiber, test jumper 1 must have a mode filter or modal controller, and when testing single-mode optical fiber a single 30mm loop is required. Record the optical power displayed on the optical power meter. Disconnect test jumper 1 from the optical power meter; then connect test jumper 2 as shown in Figure 14.27 and record the optical power displayed on the optical power meter. The loss for test jumper 2 in that direction is the difference between the recorded values.

FIGURE 14.26 Test jumper 1 optical power measurement

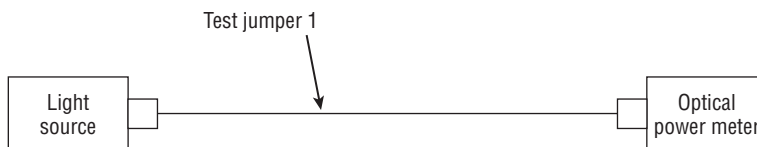
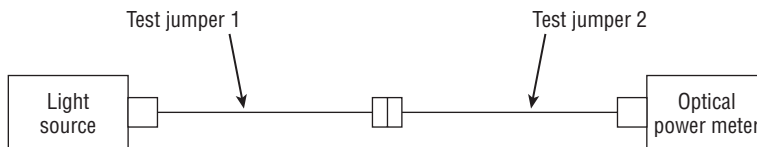


FIGURE 14.27 Test jumpers 1 and 2 optical power measurement



Remove test jumper 2 and interchange the ends. Take the end that was mating with test jumper 1 and connect it to the power meter. Mate the other end with test jumper 1. Record the optical power displayed on the optical power meter. The loss for test jumper 2 in that

direction is the difference between the recorded values. If either value exceeds a specified limit, test jumper 2 cannot be used for testing the cable plant.

Test jumpers should be treated with care, especially multiple fiber measurement quality jumpers (MQJs) like the one shown in Figure 14.28, which can be very expensive. The measured loss of the test jumper will increase gradually as it is used. However, you can typically continue to use the test jumper as long as the loss does not exceed 0.4dB and the endface remains free from defects. This value is not defined by the ANSI/TIA-568-C.0 standard but serves as a good rule of thumb.

FIGURE 14.28 Multiple-fiber MQJ



Photo courtesy of KITCO Fiber Optics

Launch Conditions, Mode Filters, and Encircled Flux

Encircled flux launch conditions for all multimode insertion loss measurements are included in IEC 61280-4-1 Edition 2, ANSI/TIA-568-C.0, and ANSI/TIA-526-14-B. The addition of this requirement arose from the need to improve the accuracy of insertion loss measurements. As the data rates of multimode networks have increased, the maximum allowable channel attenuation has decreased. A channel is the end-to-end transmission path between two pieces of equipment such as a transmitter and receiver.

The maximum allowable channel attenuation for a 100BASE-FX, Ethernet with OM3 optical fiber is 6dB while the maximum allowable attenuation for a 1000BASE-LX Ethernet with the same optical fiber is 2.3dB. Both Ethernets operate at 1300nm, but there is a 3.7dB difference in the maximum allowable channel attenuation between 1000BASE or 1Gbps Ethernet and the 100BASE or 100Mbps Ethernet. When the data rate increases to 10Gbps, the maximum allowable channel attenuation for some Ethernet applications drops to 1.9dB.

Variations in the launch conditions of multimode light sources are well known. Prior to the introduction of encircled flux, the *mandrel wrap* or *mode filter* was required for

light sources that did not meet the launch requirements of ANSI/TIA-455-78-B. It was required because the attenuation in a short link of multimode optical fiber may be higher than calculated when making insertion loss measurements due to the power lost in the high-order modes. The high-order modes are caused by an LED light source that overfills the optical fiber.

Even when light sources met the launch requirements of ANSI/TIA-455-78-B, it was still difficult to get repeatable insertion loss measurements from one piece of test equipment to another. The variations obtained with different pieces of test equipment combined with the reduced channel attenuation led to the development of an encircled flux requirement.

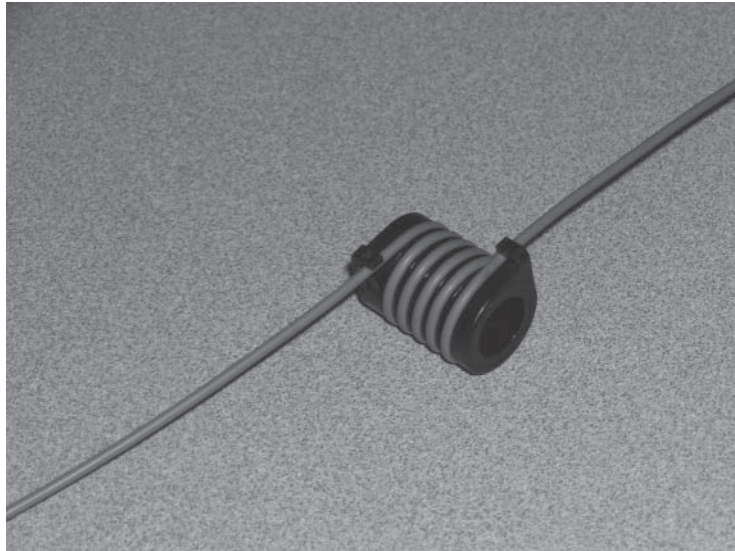
The encircled flux requirement is based on defined lower and upper boundaries of encircled flux values at different distances from the center of the optical fiber core. It corresponds to the ratio between the transmitted power at a given distance from the center of the core and the total power injected into the optical fiber. It is intended to define the light source's near-field output profile and ensure repeatable test results from one piece of test equipment to another.

Just because the encircled flux requirement has been introduced, that does not mean you must throw away your mandrel wraps. The customer typically defines how a link or channel will be tested, and they may call out testing to ANSI/TIA-526-14-A instead of ANSI/TIA-526-14-B. In addition, low-bandwidth systems with a large loss budget will not require encircled flux unless specified by the customer. It is for these reasons that both the encircled flux modal controller and mandrel wrap are discussed in this book.

When you are measuring the insertion loss of a multimode link with an encircled flux requirement, you must use a transmit jumper with a modal controller that meets the requirements of IEC 61280-4-1 Edition 2. Unlike the mandrel wrap, you cannot pick one up for a few dollars and wrap a test jumper around it. A modal controller typically has pigtails, and as with the mandrel wrap, you need a different one for each optical fiber core diameter. However, they are far more expensive than a mandrel wrap. Like the mode filter, the modal controller is a passive optical device; no power is required for the modal controller to operate.

When you're measuring the insertion loss of a multimode link that does not have an encircled flux requirement and using an LED source that overfills the optical fiber, a mandrel wrap must be used on the transmit test jumper. The transmit test jumper should have five nonoverlapping turns around a smooth mandrel. The diameter of the mandrel depends on the cable diameter and optical fiber core diameter, as shown in Table 14.1. Figure 14.29 shows a transmit test jumper with a mandrel wrap attached to an LED light source.

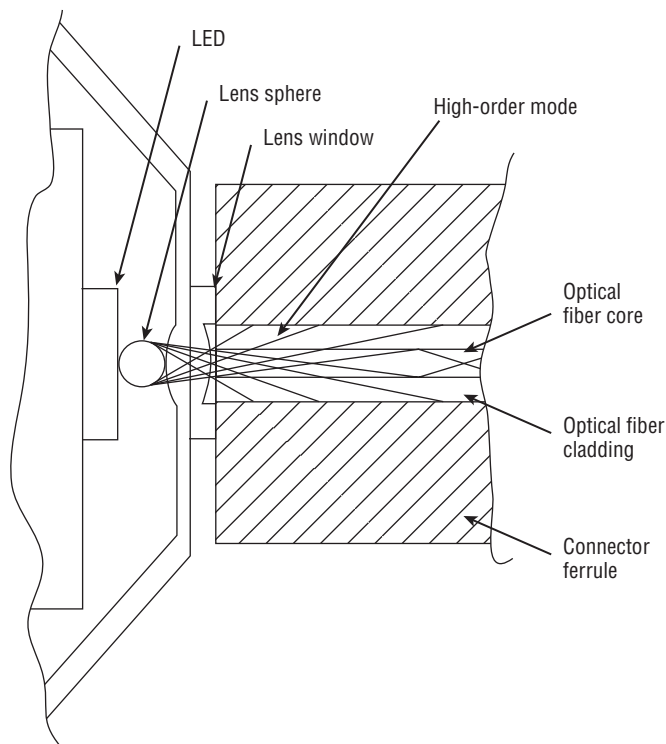
An unfiltered LED light source overfills an optical fiber by launching high- and low-order modes into the core and cladding. The high-order modes experience more attenuation than the low-order modes at interconnections, splices, and bends. This causes higher-than-expected loss in short multimode links. Figure 14.30 shows high-order modes from the LED source entering the core and the cladding of the optical fiber.

FIGURE 14.29 Transmit test jumper with mandrel wrap**TABLE 14.1** Mandrel diameters for multimode optical fiber

Fiber core size (μm)	Mandrel diameter for 900 μm buffered optical fiber (mm)	Mandrel diameter for 2mm jacketed cable	Mandrel diameter for 2.4mm jacketed cable	Mandrel diameter for 3mm jacketed cable
50	25	23	23	22
62.5	20	18	18	17

The mandrel wrap takes advantage of the high loss of high-order modes caused by excessive bending of the optical fiber. The mandrel wrap removes the high-order modes by inserting a series of macrobends in the transmit test jumper. Figure 14.31 shows the high-order modes in the core of the optical fiber refracting into the cladding because of a 90-degree macrobend.

The five small-radius nonoverlapping loops around the mandrel wrap shown in Figure 14.29 significantly attenuate the high-order modes with minimal attenuation of the low-order modes. The mandrel wrap is actually a low-order or low-pass mode filter. It allows the low-order modes to pass with very little attenuation while greatly attenuating the high-order modes. The mandrel wrap should be on the transmit test jumper at every stage of testing when you're making insertion loss measurements.

FIGURE 14.30 LED source overfilling a multimode optical fiber

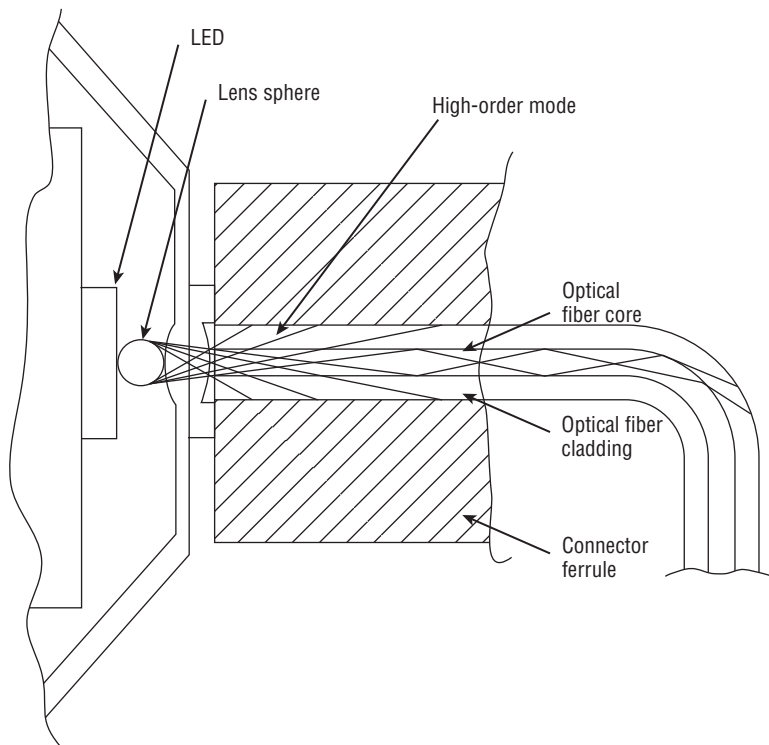
ANSI/TIA-526-14 Optical Loss Measurement Methods

ANSI/TIA-526-14 revisions A and B provide three methods for testing the *cable plant*. The difference between the three methods is how the reference power measurement is taken. Each method uses a different number of test jumpers and is described in detail in this section.

Test methods may be described using either the letter or the number of jumpers. One standard may call out a one test jumper method whereas another may call out method B. A good installer or technician is fluent in both. Therefore, in this section both approaches will be interchanged.

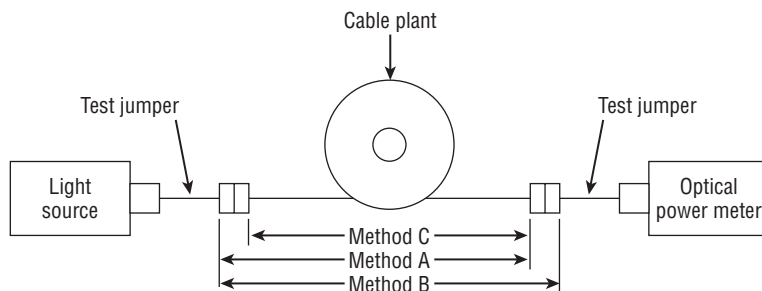
This section of the text does not show a mandrel wrap or modal conditioner on the transmit jumper in the illustrations. This section focuses on how to perform each test method. The test methods are performed exactly the same regardless of whether a mode filter or modal controller is required.

FIGURE 14.31 Macrobend attenuating loosely coupled modes in a multimode optical fiber



The three test methods described in this section each measure the optical loss of the cable plant at different points, as shown in Figure 14.32. The two-test jumper method, or method A, measures the loss of the cable plant plus one connection loss. The one-test jumper method, or method B, measures the loss of the cable plant plus two connection losses. The three-test jumper method, or method C, measures the loss of the cable plant only. Only the one-test jumper method can be used when testing to ANSI/TIA-568-C.0.

FIGURE 14.32 Cable plant measured values for methods A, B, and C

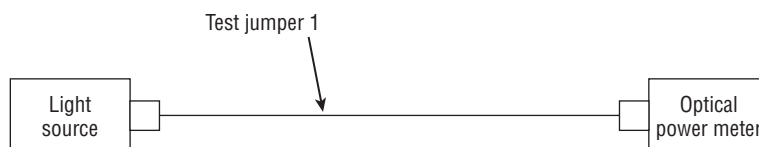


The first step before performing any of the optical power loss measurements is to clean and inspect the test jumpers and connectors at the ends of the cable plant to be tested. Verify that your test jumpers have the same optical fiber type as the cable plant you are going to test.

While you are cleaning and inspecting the test jumpers and cable plant connectors, turn on the light source and optical power meter. Set both of them to the test wavelength. Allow sufficient time for the light source and optical power meter to warm up and stabilize. Warm-up information can be found in the manufacturer's data sheet.

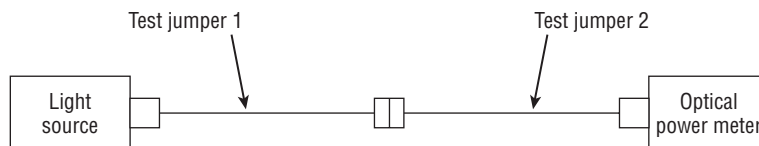
After sufficient time has passed for the light source and optical power meter to stabilize, you need to measure the loss of the test jumpers. Connect test jumper 1 as shown in Figure 14.33. Record the optical power displayed on the optical power meter. This number is the reference power measurement. This number is typically around -20dBm with a $62.5/125\mu\text{m}$ multimode optical fiber and -23dBm with a $50/125\mu\text{m}$ multimode optical fiber. These numbers can vary from OLTS to OLTS.

FIGURE 14.33 Test jumper 1 optical reference power measurement



Disconnect test jumper 1 from the optical power meter. Connect test jumper 2 as shown in Figure 14.34. Record the optical power displayed on the optical power meter. The difference between the reference power measurement and this value is the connection loss for test jumpers 1 and 2. The difference between this value and the reference power measurement should be less than or equal to 0.4dB . If this number is greater than 0.4dB , clean the connectors again and retest. If the loss still exceeds 0.4dB , replace test jumper 2. Repeat this test for test jumper 3 if method C is being used.

FIGURE 14.34 Test jumper 2 optical reference power measurement



Method A, Two-Test Jumper Reference

Method A uses two jumpers to establish the optical power reference. Connect test jumpers 1 and 2 as shown in Figure 14.34. Record the optical power displayed by the optical power meter. This number is the reference power measurement. This number is typically around

–20dBm with a 62.5/125 μ m multimode optical fiber and –23dBm with a 50/125 μ m multimode optical fiber. These numbers can vary from OLTS to OLTS.

After recording the reference power measurement, separate the test jumpers at their point of interconnection, as shown in Figure 14.35. Do this without disturbing their attachment to the light source and optical power meter. Attach the test jumpers to the cable plant as shown in Figure 14.36.

FIGURE 14.35 Test jumpers 1 and 2 separated

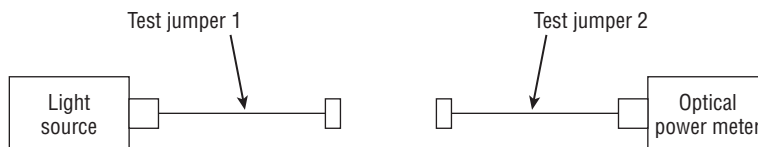
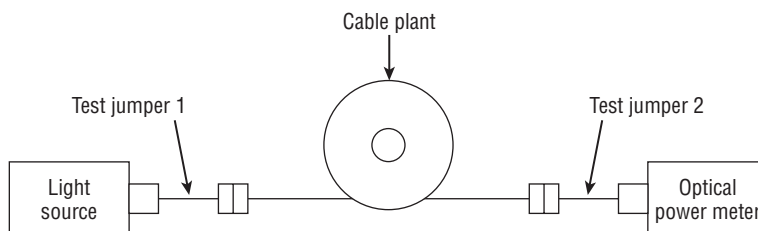


FIGURE 14.36 Test jumpers 1 and 2 connected to the cable plant



With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was –20.4dBm and the test power measurement was –21.6dBm, the optical power loss for the cable plant would be 1.2dB.



Remember the mandrel wrap is always installed on the test jumper connected to the light source when an unfiltered light source is used and a modal controlled is not required.

Method B, One-Test Jumper Reference

Method B uses one jumper to establish the optical power reference. However, two test jumpers are required to perform the test. This method is required by ANSI/TIA-568-C.0. Link attenuation using this method is equal to cable attenuation plus connector insertion loss plus splice insertion loss. The required wavelengths and directions are discussed in

the section “Link Segment and Cabling Subsystem Performance Measurements” later in this chapter.

Connect test jumper 1 as shown earlier in Figure 14.33. Record the optical power displayed by the optical power meter. This number is the reference power measurement. This number is typically around -20dBm with a $62.5/125\mu\text{m}$ multimode optical fiber and -23.5dBm with a $50/125\mu\text{m}$ multimode optical fiber. These numbers can vary from OLTS to OLTS.

After recording the reference power measurement, disconnect the test jumper from the optical power meter. Do not disturb the jumper’s attachment to the light source. Attach test jumper 2 to the optical power meter as shown in Figure 14.35. Attach the test jumpers to the cable plant as shown in Figure 14.36.

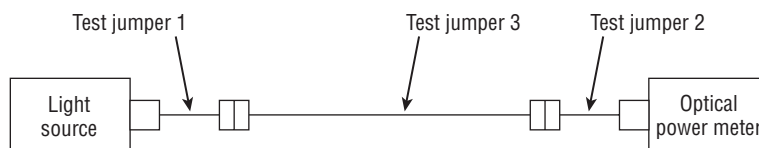
With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.7dBm and the test power measurement was -23.6dBm , the optical power loss for the cable plant would be 2.9dB .

Method C, Three-Test Jumper Reference

Method C uses three jumpers to establish the optical power reference. Connect test jumpers 1, 2, and 3 as shown in Figure 14.37. Record the optical power displayed by the optical power meter. This number is the reference power measurement. This number is typically around -20dBm with a $62.5/125\mu\text{m}$ multimode optical fiber and -23.5dBm with a $50/125\mu\text{m}$ multimode optical fiber. These numbers can vary from OLTS to OLTS.

FIGURE 14.37 Method C reference power measurement

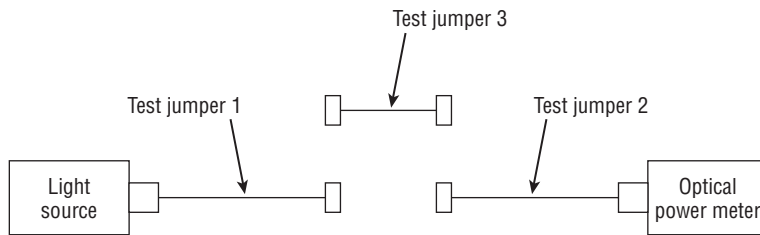


After recording the reference power measurement, separate the test jumpers at their point of interconnection, as shown in Figure 14.38. Do this without disturbing their attachment to the light source and optical power meter. Attach the test jumpers to the cable plant as shown earlier in Figure 14.36.



Test jumper 3 is not used during testing. It is used only to establish reference power.

FIGURE 14.38 Removal of test jumper 3 from method C reference power measurement



With the test jumpers attached, record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the cable plant is the difference between the reference power measurement and the test power measurement.

To obtain the optical power loss for the cable plant, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.2dBm and the test power measurement was -21.6dBm , the optical power loss for the cable plant would be 1.4dB .

Patch Cord Optical Power Loss Measurement

Multimode patch cord optical loss power measurement is performed using the steps described in ANSI/TIA-526-14, method A. The patch cord is substituted for the cable plant. Because patch cords are typically no longer than 5m, the loss for the optical fiber is negligible and testing can be performed at 850nm or 1300nm. The loss measured in this test is the loss for the patch cord connector pair. ANSI/TIA-568-C.3 states that the maximum loss for a connector pair is 0.75dB .

After setting up the test equipment as described in ANSI/TIA-526-14, method A, clean and inspect the connectors at the ends of the patch cords to be tested. Verify that your test jumpers have the same optical fiber type and connectors as the patch cords you are going to test. The transmit jumper should have a mandrel wrap or modal conditioner depending on the revision ANSI/TIA-526-14 being used for testing. Ensure that there are no sharp bends in the test jumpers or patch cords during testing.

Because both patch cord connectors are easily accessible, optical power loss should be measured in both directions. The loss for the patch cord is the average of the two measurements. If the loss for the patch cord exceeds 0.75dB in either direction, the patch cord needs to be repaired or replaced.

Connector Insertion Loss Measurement

Connector insertion loss measurement isolates the loss of a single connector on a cable assembly. It may be referred to as connector loss. Many times cable assemblies are shipped from the manufacturer with the insertion loss for each connector listed on the packaging. The package shown in Figure 14.39 contains a duplex multimode patch cord. In the upper-left corner of the package, a label lists the insertion loss measurements for each connector.

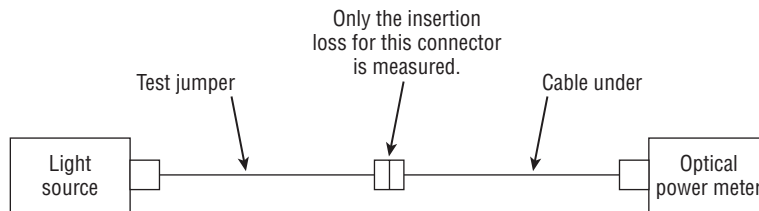
FIGURE 14.39 Duplex multimode fiber patch cable with connector insertion loss measurements on the packaging



Connect test jumper 1 as shown earlier in Figure 14.33. Record the optical power displayed by the optical power meter. This number is the reference power measurement. This number is typically around -20dBm with a $62.5/125\mu\text{m}$ multimode optical fiber and -23.5dBm with a $50/125\mu\text{m}$ multimode optical fiber. These numbers can vary from OLTS to OLTS.

After recording the reference power measurement, disconnect the test jumper from the optical power meter. Do not disturb the jumper's attachment to the light source. Attach the cable to be tested to the optical power meter as shown in Figure 14.40 and record the optical power displayed on the optical power meter. This is your test power measurement. The loss for the connector shown in Figure 14.40 is the difference between the reference power measurement and the test power measurement. The transmit jumper should have a mandrel wrap or modal conditioner depending on the revision of ANSI/TIA-526-14 being used for testing.

FIGURE 14.40 Test jumper and cable under test for connector insertion loss measurement



To obtain the connector insertion loss, subtract the test power measurement from the reference power measurement. If the reference power measurement was -20.7dBm and the test power measurement was -20.9dBm , the connector insertion loss would be 0.2dB .

Link Segment and Cabling Subsystem Performance Measurements

Link segment performance measurements are described in ANSI/TIA-568-C.0. Multimode link attenuation should be measured using the reference methods specified in ANSI/TIA-526-14-B and single-mode link attenuation should be measured using the reference methods specified in ANSI/TIA-526-7. The one-jumper reference method is preferred for both multimode and single-mode; however, other methods described in ANSI/TIA-526-14-B or ANSI/TIA-526-7 may be applied. Ensure that the applied test method is included in testing documentation.

When measuring single-mode link attenuation in accordance with ANSI/TIA-526-7 Method A.1 follow Method B, the one-test jumper reference procedure described earlier in the chapter. To comply with ANSI/TIA-568-C.0, a single turn 30mm in diameter loop must be applied to the transmit test jumper to ensure single-mode operation as described in ANSI/TIA-455-78B.

ANSI/TIA-568-C.0 outlines cabling subsystem performance measurements. There are three cabling subsystems. Cabling Subsystem 1 describes the cabling from the equipment outlet to a connection facility, an intermediate connection facility, or a central connection facility. Cabling Subsystem 2 describes the cabling from a connection facility to either an intermediate connection facility or a central connection facility when an intermediate connection facility is not used. Cabling Subsystem 3 describes the cabling from the intermediate connection facility to the central connection facility.

Cabling Subsystem 1 Testing Requirements These link segments are only required to be tested in one direction at one wavelength. Multimode optical fiber link segments can be tested at either 850nm or 1300nm. Single-mode optical fiber link segments can be tested at either 1310nm or 1550nm.

Cabling Subsystem 2 and 3 Testing Requirements These link segments are required to be tested in at least one direction at both operational wavelengths. Multimode optical fiber link segments must be tested at 850nm and 1300nm. Single-mode optical link segments shall be tested at 1310nm and 1550nm.



A link as defined in ANSI/TIA-568-C.0 is a transmission path between two points not including equipment and cords. A link segment does not include any active or passive devices other than the optical fiber cabling, connectors, and splices if required.

Tier 1 Testing

ANSI/TIA-568-C.0 Annex E provides guidelines for field testing length, loss, and polarity of optical fiber cabling. Section E.2.2 describes Tier 1 testing. Tier 1 testing requires the following:

- Link attenuation measurement
- Fiber length verification
- Polarity verification

Link Attenuation Measurement Link attenuation measurement must be performed with an OLTS. When testing multimode optical fiber, attenuation measurements should be taken in accordance with ANSI/TIA-526-14, Method B, which was covered earlier in this chapter. When testing to ANSI/TIA-526-14-A, ensure the light source meets the launch requirements of ANSI/TIA-455-78B. This can be achieved either within the stabilized light source by the manufacturer or with the edition of a mandrel wrap as described earlier in this chapter. When testing to ANSI/TIA-526-14-B, ensure the transmit jumper has a modal controller.

When testing single-mode optical fiber, attenuation measurements should be taken in accordance with ANSI/TIA-526-7-A, Method A.1. A single 30mm diameter loop is often applied to the source test jumper to ensure single-mode operation.

Fiber Length Verification Fiber length verification may be obtained by using the sequential markings on the cable jacket or sheath as described in Chapter 7, “Fiber-Optic Cables.” It may also be obtained using an OLTS that has length measurement capability.

Polarity Verification Polarity verification is performed to ensure the optical path from the transmit port of one device is routed to the receive port of another device. The polarity may be verified using an OLTS, VFL, or continuity tester.

Documentation of OLTS Testing

As you test the cable plant with the OLTS, you need to properly document the results of the test. The documentation should include at a minimum the following:

- The date of the testing
- Test personnel
- Field test instrument manufacturer
- Field test instrument serial number
- Field test instrument calibration date
- Test jumper type
- Test jumper length
- The number or identification of fiber or cable tested
- The test procedure, method used, wavelengths, directions
- Link loss results

Tier 2 Testing

Tier 2 testing as defined in ANSI/TIA-568-C.0 Annex E is optional. It can be used to supplement Tier 1 testing with the addition of an OTDR trace of the link. However, it is not a replacement for the Tier 1 link attenuation measurement performed with an OLTS. Tier 2 testing with the OTDR can characterize the elements of a link including:

- Fiber segment length
- Attenuation uniformity and attenuation rate
- Interconnection location and insertion loss
- Splice location and insertion loss
- Marcobends

Tier 2 testing documentation is described at the end of the next section.

Optical Time-Domain Reflectometer

So far in this chapter, you have learned about tools and test equipment that can be used to test a fiber-optic link or cable. Of all the tools and test equipment discussed, none provides more information about the fiber-optic cable or link than the *optical time-domain reflectometer* (OTDR). The OTDR can be used to evaluate the loss and reflectance of interconnections and splices. It will measure the attenuation rate of an optical fiber and locate faults.

In general there are two types of OTDRs. One is often referred to as a long-haul OTDR and the other as a high-resolution OTDR. A long-haul OTDR is designed for use with cable lengths typically associated with telecommunication systems. Cable length may range from 50m to many kilometers. A high-resolution OTDR is designed for use with short cable lengths such as those found on an aircraft or ship and may also be used in longer optical fiber length telecommunication applications. The high-resolution OTDR can also be used to evaluate optical sensors.

The use of high-resolution optical reflectometry to characterize and map the optical link is described in AIR6552/2, which as of this writing was under development within SAE International's AS-3 Fiber Optics and Applied Photonics Committee. This standard and its slash sheets are discussed in "Emerging Testing Standards" later in this chapter.

This section of the chapter focuses on basic OTDR theory, setup, and testing that applies to both long-haul and high-resolution OTDRs. The OTDR is a complex piece of test equipment with many variables that must be programmed correctly before testing can be performed. Proper OTDR setup and cable preparation will ensure accurate test results. Always consult the manufacturer's documentation before using any OTDR.

OTDR Theory

The OTDR is a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back. Unlike all the test equipment discussed up to this point, the OTDR provides a graphical representation of what is happening in the fiber-optic link or cable under test. With the OTDR, the fiber-optic link or cable is no longer a black box. The OTDR shows how light passes through every segment of the fiber-optic link.

Light reflecting back in an optical fiber is the result of a reflection or backscatter. Reflections are when the light traveling through the optical fiber encounters changes in the refractive index. These reflections are called Fresnel reflections. Backscatter, or Rayleigh scattering, results from evenly distributed compositional and density variations in the optical fiber. Photons are scattered along the length of the optical fiber. The photons that travel back toward the OTDR, as shown in Figure 14.41, are considered backscatter.

OTDRs come in many different shapes and sizes, as shown in Figure 14.42. The newer models may be almost pocket-sized whereas others require a shoulder strap. The small OTDR is lighter and easier to carry; however, the small screen permits only a

limited viewing area. Some OTDRs may not have a screen and may be controlled with a PC via the USB port.

FIGURE 14.41 Backscattered photons

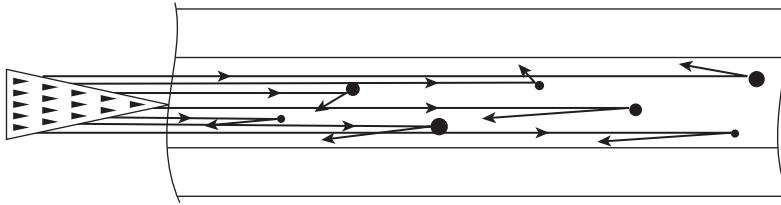


FIGURE 14.42 Large and small OTDRs



The OTDR is typically a battery-powered device. It is battery powered because many places where the OTDR is used have no electrical power available. It is a good idea to bring a charged spare battery with you when you are performing testing.

Many OTDRs can be configured to test both multimode and single-mode optical fibers. A typical OTDR may hold up to three light source modules, like the OTDR shown in Figure 14.43. Modules can be added or removed as testing requirements change. Some OTDRs even contain a VFL like the one shown in Figure 14.44.

A generic OTDR can be broken up into eight basic components: the *directional coupler*, *laser*, *timing circuit*, *single-board computer*, *digital signal processor (DSP)*, *analog-to-digital converter*, *sample-and-hold circuit*, and *avalanche photodiode*. Figure 14.45 is a block diagram of the OTDR showing how light is launched from the laser through the directional coupler into the optical fiber. The directional coupler channels light returned by the optical fiber to the avalanche photodiode.

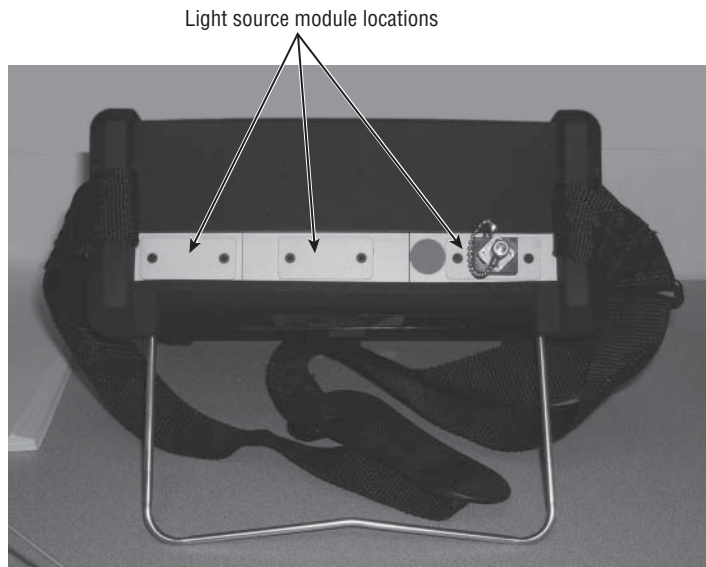
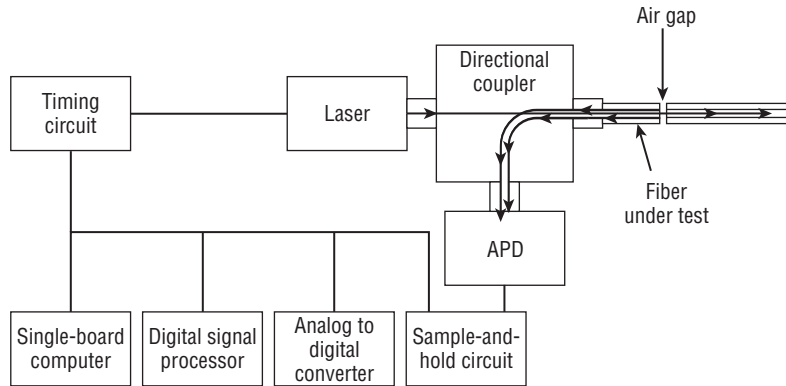
FIGURE 14.43 Light source module locations on an OTDR**FIGURE 14.44** The VFL is built into this OTDR.

FIGURE 14.45 OTDR block diagram

The avalanche photodiode converts the light energy into electrical energy. The electrical energy is sampled at a very high rate by the sample-and-hold circuit. The sample-and-hold circuit maintains the instantaneous voltage level of each sample long enough for the analog-to-digital converter to convert the electrical value to a numerical value. The numerical value from the analog-to-digital converter is processed by the DSP and the result is sent to the single-board computer to be stored in memory and displayed on the screen. This entire process is typically repeated many times during a single test of an optical fiber and coordinated by the timing circuit.

For the OTDR to produce accurate results, the refractive index of the optical fiber under test must be known. The refractive index of an optical fiber is different for each wavelength tested. The operator must enter the correct refractive index into the OTDR for each wavelength. The correct refractive index for a fiber-optic cable can be obtained from the manufacturer.

The OTDR samples light energy from the optical fiber at precise intervals. Each sample taken by the OTDR represents the round-trip time for light energy in the optical fiber. Let's assume that the OTDR is taking 500 million samples per second, or one sample every two nanoseconds. If the refractive index for the optical fiber under test were equal to 1.5, every five samples would represent the distance of 1m, as shown in Figure 14.46.

The following formula can be used to find distance based on time and refractive index. In this formula, the speed of light is rounded up to 3×10^8 m/s:

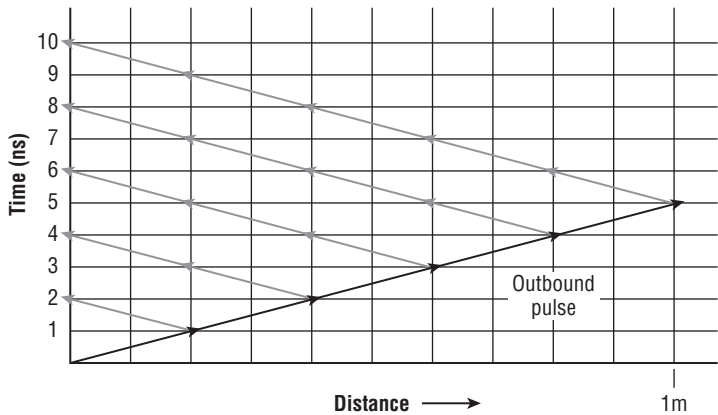
$$\text{Distance} = ((\text{time in ns})/2) \times (\text{speed of light in free space})/\text{refractive index}$$

For our example:

$$\text{Distance} = (10 \text{ ns}/2) \times (3 \times 10^8)/1.5$$

$$\text{Distance} = 1\text{m}$$

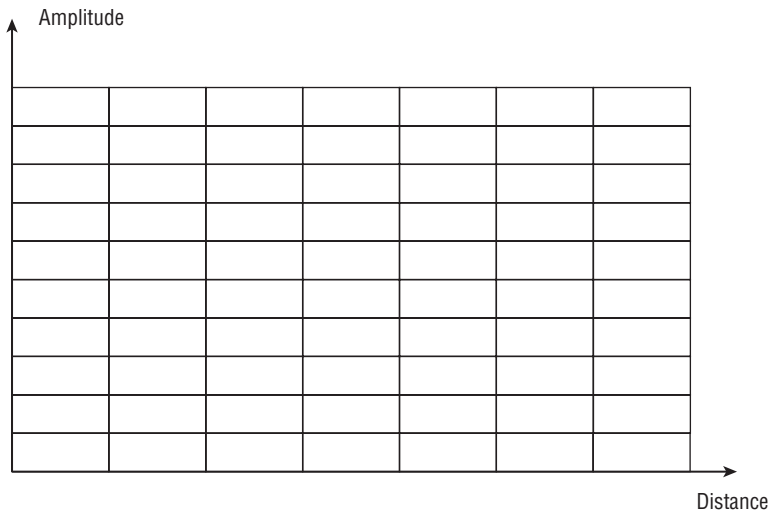
FIGURE 14.46 OTDR sampling at 2ns rate



OTDR Display

The OTDR displays time or distance on the horizontal axis and amplitude on the vertical axis, as shown in Figure 14.47. The horizontal axis can typically be programmed to display distance in feet, meters, miles, or kilometers. The vertical axis is not programmable. The vertical axis displays relative power in decibels.

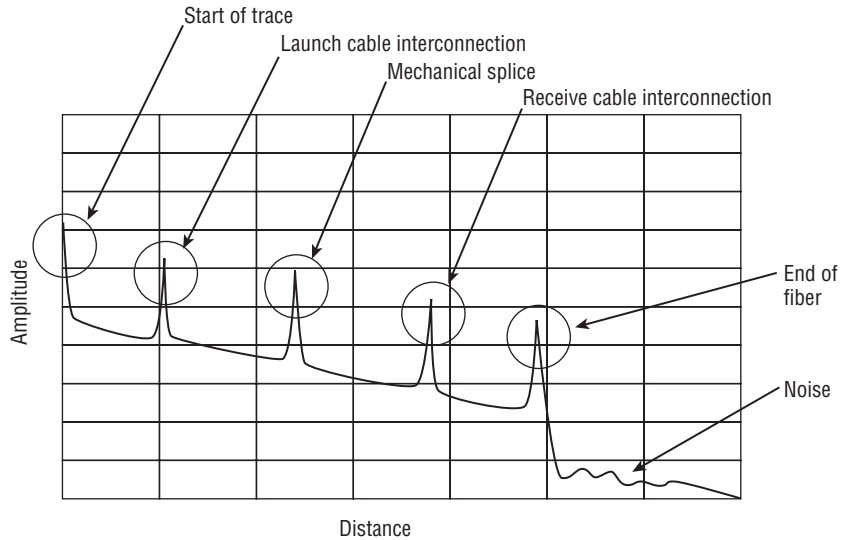
FIGURE 14.47 OTDR display



The OTDR creates a trace like the one shown in Figure 14.48. The trace shows event loss, event reflectance, and optical fiber attenuation rate. The OTDR can horizontally and

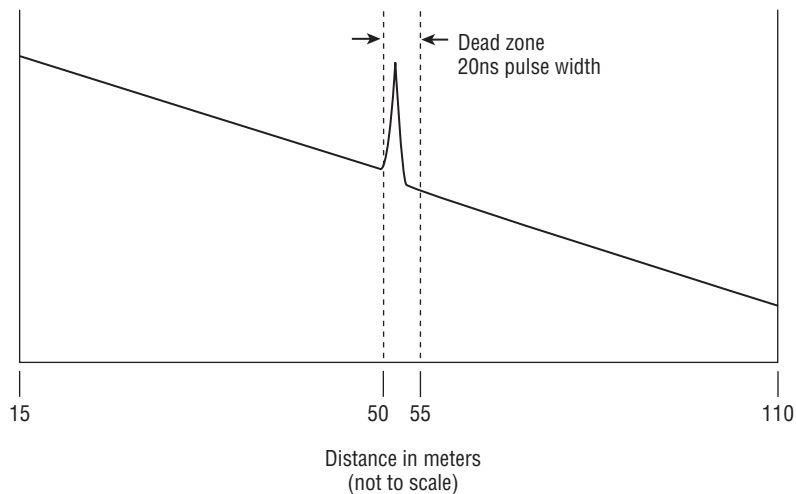
vertically zoom in on any section of the trace. This permits a more detailed inspection of the optical fiber or event.

FIGURE 14.48 Event-filled OTDR trace



The light pulses from the OTDR produce a blind spot or dead zone, like the one shown in Figure 14.49. The dead zone is an area where the avalanche photodiode has been saturated by the reflectance of a mechanical interconnection. The size of the dead zone depends on the length of each light pulse and the amount of light reflected back toward the avalanche photodiode.

FIGURE 14.49 OTDR trace dead zone



An interconnection that has very little reflectance will not saturate the avalanche photodiode. This type of interconnection will produce the smallest dead zone. Dividing the pulse width in nanoseconds by 10 can approximate the length of the light pulse as seen by the OTDR in meters. A pulse width of 20ns will yield an ideal dead zone of 2m whereas a pulse width of 2ns will yield an ideal dead zone of 20cm. Pulse widths less than 10ns are typically available only on high-resolution OTDRs like the one shown in Figure 14.50. The trace shown on the OTDR display in Figure 14.50 contains multiple reflective events over a distance of approximately 17m.

FIGURE 14.50 High-resolution OTDR capable of 2ns pulse widths displaying multiple reflective events over a distance of approximately 17m

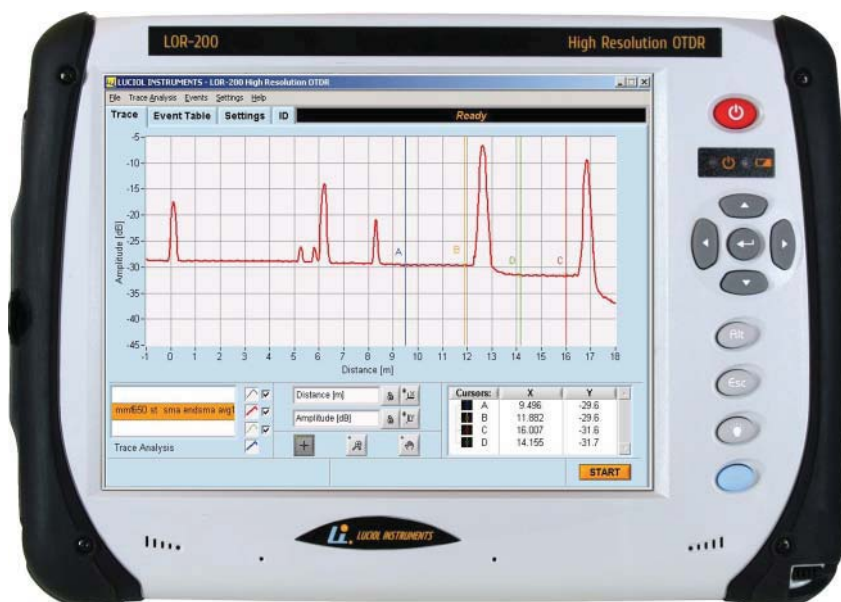


Photo courtesy of Luciol Instruments

The ratio of the dead zone to the pulse width depends on the amount of light energy reflected back toward the OTDR. Poor interconnections or interconnections with an air gap typically saturate the avalanche photodiode and produce a greater-than-ideal dead zone, as much as 10 times greater than ideal.

OTDR Setup

Before testing, the OTDR needs to be set up correctly to provide the most accurate readings. When setting up the OTDR, you must select the correct fiber type, wavelength or wavelengths, range and resolution, pulse width, averages, refractive index, thresholds, and

backscatter coefficient. This process takes only a couple of minutes and ensures the most accurate results possible.

There are many different OTDRs on the market, and it is impossible to describe the setup for each. This section focuses on general setup parameters for both long-haul and high-resolution OTDRs. Some OTDRs may have additional parameters, and some may not include all of the parameters discussed.

Fiber Type

Each light source or light source module in a long-haul or high-resolution OTDR is designed for one or several specific optical-fiber types. A multimode module should not be used to test a single-mode optical fiber, and vice versa. Before heading for the test site, ensure that your OTDR has the correct module for the optical fiber to be tested.

Wavelength

The wavelength that a long-haul or high-resolution OTDR can test with depends on the light source module or modules in your OTDR. Some light source modules contain a single laser whereas others contain two different wavelength lasers. A light source with two lasers allows testing of the optical fiber at two wavelengths without disconnecting the cable under test. This simplifies testing and reduces testing time.

Range and Resolution

The distance range of an unzoomed trace displayed on a long-haul or high-resolution OTDR and the distance between data points is determined by range and resolution. As a general rule of thumb, the OTDR range should be set to 1.5 times the length of the fiber-optic link. If the range is set too short, the results will be unpredictable and the entire link may not be displayed. If the range is set too long, the trace will fill only a small portion of the display area.

Increasing the range automatically increases the distance between the data points. When the distance between the data points is increased, resolution is reduced. When setting range, choose the first range value that exceeds your fiber-optic link length. Selecting a 2km range on a long-haul OTDR for a 1.3km link will yield more accurate results than selecting a 20km range.

Pulse Width

The pulse width determines the size of the dead zone and the maximum length optical fiber that can be tested. A short pulse width produces a small dead zone. However, a short pulse width reduces the length of optical fiber that can be tested.

The pulse width should be selected so that the trace never disappears into the noise floor. If the pulse width is set properly, the trace will stay smooth until the end of the fiber-optic link. If the pulse is set too wide, events may be lost in the dead zone.

Averages

When setting the averages parameter on an OTDR, select the number of averages that produce a smooth trace in the least amount of time. If too few averages are taken, the trace will appear noisy because the noise floor is too high. If too many samples are taken, the trace will be smooth; however, valuable testing time will be wasted.

Refractive Index

The manufacturer should specify the refractive index or the group index of refraction for an optical fiber. The refractive index of similar optical fibers does not vary much from manufacturer to manufacturer. Most values are typically within 1 percent of each other. If the value for the optical fiber you are testing is not known, use the values shown in Table 14.2. Entering a low refractive index will produce measurements that are too long, and entering a high refractive index will produce measurements that are too short.

TABLE 14.2 Refractive index default values

Wavelength	Refractive index
850nm	1.4960
1300nm	1.4870
1310nm	1.4675
1550nm	1.4681

Thresholds

Thresholds can typically be set for end of fiber, event loss, and reflectance. Many OTDRs have a default value preset for each of these. Depending on the testing you are performing, the default values may be too sensitive or not sensitive enough.

Most OTDRs will generate event tables automatically based on the threshold settings. To capture a majority of the events in the event table, set the thresholds on the sensitive side. A good starting place would be to set the values as shown in Table 14.3. If these values are too sensitive, you can always increase them.

Backscatter Coefficient

The OTDR uses the backscatter coefficient to calculate reflectance. As with the refractive index, the optical fiber manufacturer specifies backscatter coefficient. Backscatter coefficient does vary from manufacturer to manufacturer. However, the variation is typically not that great, and the default values shown in Table 14.4 can be used with good results when the manufacturer’s specified value is not known.

TABLE 14.3 Threshold default values

Event	Default value
End of fiber	6.0dB
Event loss	0.05dB
Reflectance	−65dB

TABLE 14.4 Backscatter coefficient default values

Wavelength	Backscatter coefficient
850nm	68.00
1300nm	76.00
1310nm	80.00
1550nm	83.00

Cable Plant Test Setup

Now that the OTDR has been configured with the correct light source modules and the setup parameters have been entered, the OTDR can be connected to the fiber-optic link. There are many different ways that the OTDR can be used to test a single optical-fiber or fiber-optic link. In this chapter, the test setup will include a *launch cable* and *receive cable*. Launch and receive cables may also be referred to as *pulse suppression cables*, *launch and receive fibers*, *dead zone cables*, *test fiber box*, or *access jumpers*.

Launch and receive cables allow the OTDR to measure the insertion loss at the near and far ends of the fiber-optic link. Like test jumpers, launch and receive cables should be constructed of optical fiber similar to the optical fiber under test. Launch and receive cables are available from several manufacturers and come in all shapes and sizes. Some launch and receive cables are designed for rugged environments, like the cable shown in Figure 14.51. Others are designed to be compact, almost pocket-sized, like the ring-type shown in Figure 14.52.

To achieve the compact size, manufacturers coil up coated optical fiber and add a buffer, strength member, and jacket at each end to protect the optical fiber, as shown in Figure 14.53. The open ring-type launch/receive cable shown in Figure 14.53 contains 150m of multimode optical fiber.

FIGURE 14.51 Rugged launch/receive cable

Photo courtesy of OptiConcepts

FIGURE 14.52 Compact ring-type launch/receive cable

You should select the length of the launch and receive cables based on the manufacturer's recommendation or the pulse width you are using to test the optical fiber. Launch and receive cables are typically the same length. A good rule of thumb is to have at least 1m of launch or receive cable for each nanosecond of pulse width, with the minimum length cable being 100m. A 2km link being tested with a 20ns pulse width would require at least a 100m launch and receive cable. A 10km link being tested with a 200ns pulse would require at least a 200m launch and receive cable.

When you are testing generic telecommunications cabling in accordance with ANSI/TIA-568-C.0 Annex E, a launch cable length of 100 meters for multimode optical fiber or 300 meters for single-mode optical fiber is typically acceptable in the absence of manufacturer recommendations.

After the launch and receive cables have been selected, they can be attached to the fiber-optic link, as shown in Figure 14.54. Remember to clean and inspect the connector endfaces on the cable under test and on the launch and receive cables. Dirty or damaged endfaces on the launch and receive cables can damage the endface of the

connectors that you are going to test. A dirty or damaged endface on a launch or receive cable typically results in a high-loss interconnection.

FIGURE 14.53 Open ring-type launch/receive cable

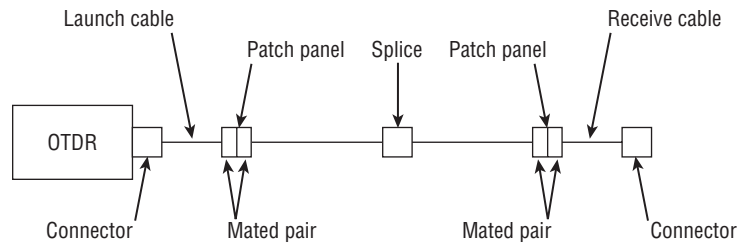


Testing and Trace Analysis

Testing the cable plant or a fiber-optic link with the OTDR should be done in both directions. Multimode optical fiber should be tested at both 850nm and 1300nm. Single-mode optical

fiber should be tested at both 1310nm and 1550nm. Event loss and optical fiber attenuation is typically the average of the bidirectional values.

FIGURE 14.54 OTDR connected to fiber-optic link under test with launch and receive cables



Real World Scenario

If Your Pulse Suppression Cables Don't Match, You May See a Ghost

One day I was watching some students test a reel of multifiber cable with the OTDR. The Army had recently used this cable in the desert. Each fiber in the cable had been terminated on both ends.

After cleaning and visually inspecting the connectors, the pulse suppression cables were connected to both ends of the cable. The OTDR was connected to the launch cable and each optical fiber was tested. Looking at the OTDR trace, the students concluded that each optical fiber had multiple breaks because the OTDR trace had multiple back reflections every 100 meters.

I asked the students to test the continuity of each optical fiber with the continuity tester. Each optical fiber had good continuity. I then asked the students the core diameter of the optical fiber they were testing and the pulse suppression cables.

It turned out that the repetitive pulses or ghosts on the OTDR trace were caused by the core diameter mismatch between the pulse suppression cables and the optical fiber under testing. The students were testing a 50/125 μ m optical fiber with 62.5/125 μ m pulse suppression cables.

There are many standards related to testing both multimode and single-mode optical fiber with an OTDR. This section describes several techniques that will allow you to measure the distances and losses in a typical cable plant or fiber-optic link. We will focus on short links—the type you would typically find in a commercial building.

All measurement techniques discussed in this chapter use the 2-point method. All OTDRs can measure loss using the 2-point method, but not all OTDRs can automatically perform 2-point subtraction. This section assumes that your OTDR can perform 2-point subtraction. If your OTDR can't, you will have to perform the calculations manually.

Some OTDRs are capable of *least-squares averaging (LSA)*. LSA can be used to measure attenuation. This section does not address LSA measurements. If your OTDR can perform LSA measurements, consult the operator's manual for how to perform them.



When testing with the OTDR, always observe the manufacturer's precautions. Never directly view the end of an optical fiber being tested with an OTDR. Viewing the end of an optical fiber being tested with an OTDR directly or with a microscope may cause eye damage.

Baseline Trace

The first step in trace analysis is to generate the baseline trace. You should not press the test button on the OTDR to generate a baseline trace until you ensure that:

- All connectors have been cleaned and inspected and are undamaged.
- Launch and receive cables have optical fiber similar to the optical fiber under test.
- Launch and receive cables are the correct length.
- Launch and receive cables are properly connected to each end of the fiber-optic link under test.
- The correct fiber type, wavelength, range and resolution, pulse width, averages, refractive index, and backscatter coefficient have been entered into the OTDR.

Figure 14.55 is a drawing of a baseline trace as presented on the OTDR screen. This baseline trace contains a launch cable, horizontal segment, and receive cable. The launch and receive cables are 100m in length. The horizontal segment is 85m in length.

Looking at the trace from left to right, you will notice there is a large back reflection at the input to the launch cable. Because a 20ns pulse width was selected, the trace is smooth within 10m. The smooth trace slopes gradually to the back reflection caused by the connector pair where the launch cable and horizontal segment are connected.

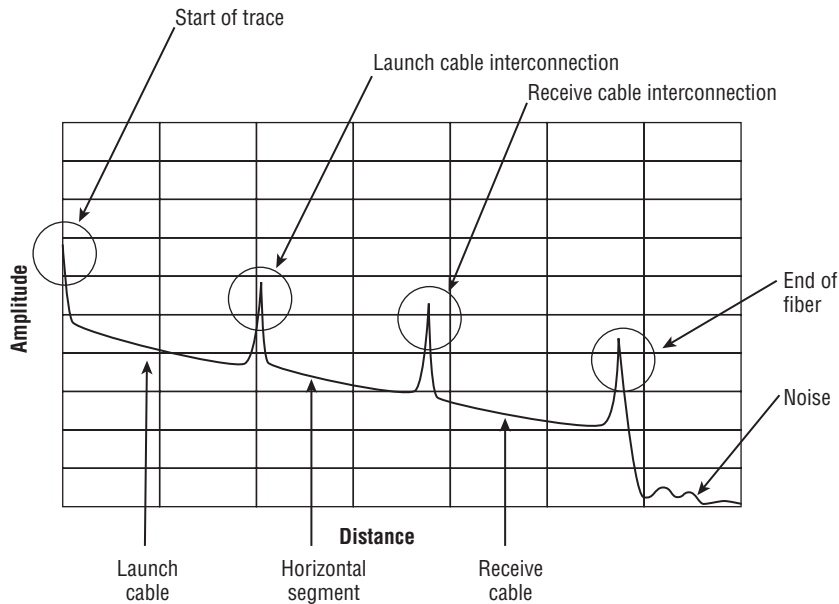
The trace becomes smooth again 10m after the interconnection back reflection. The trace remains smooth up to the back reflection caused by the connector pair where the receive cable and horizontal segment interconnect. The trace again becomes smooth 10m after the interconnection back reflection until a large back reflection is generated by the end of the receive cable. The receive cable back reflection is followed by a large reduction in amplitude, and then the trace disappears into the noise floor.

Measuring the Attenuation of a Partial Length of Optical Fiber

When using the 2-point method, the first thing you should do is measure the attenuation for a partial segment of the cable under test. This should be done for each wavelength that

you are testing the optical fiber with. It has to be done only one time for each cable type being tested. The data should be recorded—this information will be needed to accurately measure interconnection loss.

FIGURE 14.55 Baseline trace of horizontal segment



After taking the baseline trace, position the two cursors on a smooth section of the optical fiber. The longer the section, the more accurate your results will be, because noise will have less impact on the overall measurement. The trace in Figure 14.56 has the A and B cursors 50m apart on the smooth section of the horizontal segment of the trace. The loss for this 50m segment at a wavelength of 850nm is approximately 0.14dB.

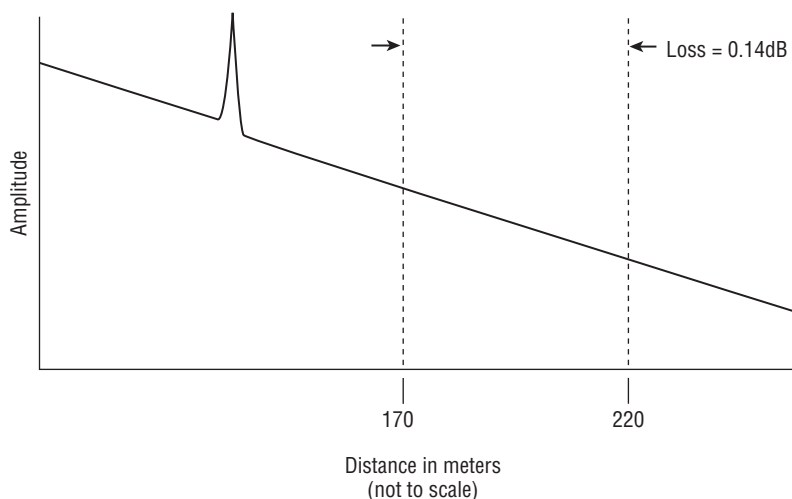


Because of ripple on the smooth trace, you will notice that the A-B loss changes slightly every time a cursor moves. Ripple caused by noise will cause variations from measurement to measurement. This variation may be as much as 0.05dB.

Measuring the Distance to the End of the Optical Fiber

A break in an optical fiber in a fiber-optic cable looks like the end of the optical fiber on the OTDR display because the OTDR cannot see beyond the break. Regardless of the number of optical fiber breaks in a cable, the OTDR can see only the one closest to it. Because a break looks like the end of the optical fiber on the OTDR, the same method is employed to measure the distance to a break in an optical fiber or measure the distance to the end of an optical fiber.

FIGURE 14.56 Measuring the attenuation of a cable segment using the 2-point method



Light from the OTDR exiting the optical fiber under test typically produces a strong back reflection. The strong back reflection is caused by the Fresnel reflection when the light from the optical fiber hits the air. However, a break in an optical fiber is not always exposed to the air. The break may be exposed to index matching gel from a mechanical splice, gel from a loose buffer, or water.

The same method can be used to find the distance to the end of an optical fiber whether or not a Fresnel reflection happens. Figure 14.57 has two OTDR traces. The top trace is the end of an optical fiber with a Fresnel reflection. The bottom trace is the end of an optical fiber without a Fresnel reflection.

To measure the distance to the end of the optical fiber after zooming in on the back reflection, place the A cursor on a smooth section of the trace just in front of the back reflection or the drop in the trace. Move the B cursor toward the A cursor until it is in the leading edge of the back reflection. Keeping moving the B cursor toward the A cursor until the A-B loss is $\pm 0.5\text{dB}$. It should be 0.5dB of loss for the nonreflective trace and 0.5dB of gain for the reflective trace. The distance to the end of the optical fiber is the distance for the B cursor.

Measuring the Length of a Cable Segment

The first step in measuring the length of a cable segment is to horizontally zoom in on the cable segment, as shown in Figure 14.58. Place the cursors in the leading edge of the reflective events for that segment. The cursors should intersect the leading edge of the reflective event at the same vertical height above the smooth part of the trace, as shown in Figure 14.58. The distance between cursors is the length of the cable segment.

FIGURE 14.57 Measuring the distance to the end of an optical fiber using the 2-point method

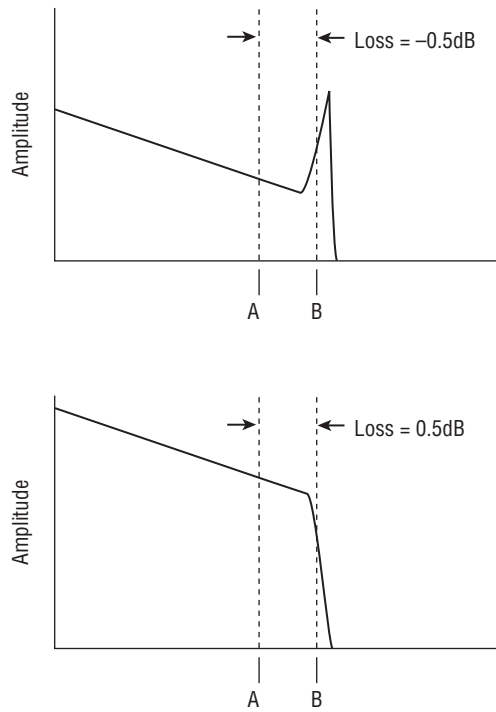
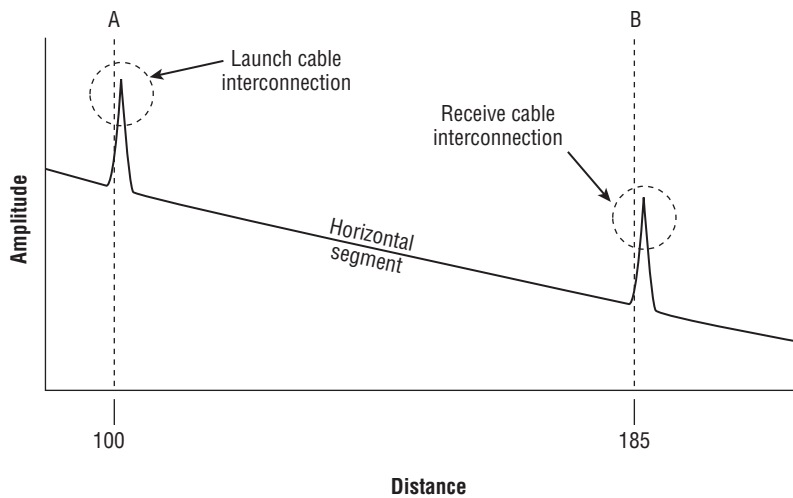


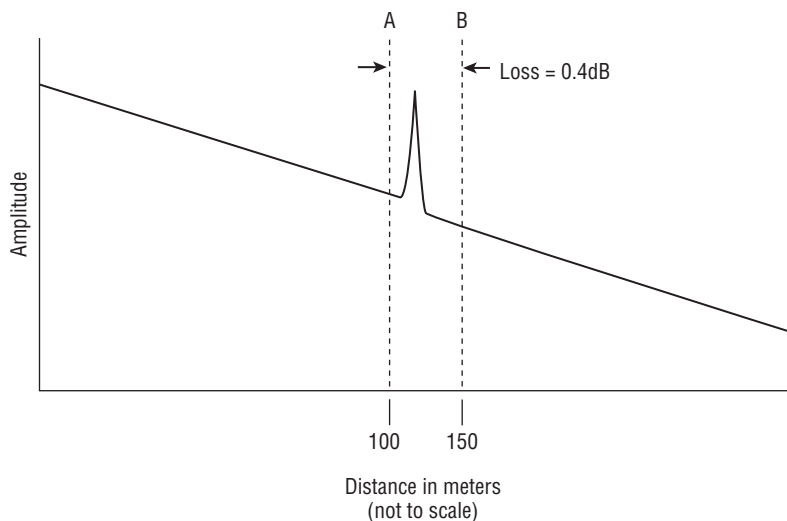
FIGURE 14.58 Measuring the length of a cable segment



Measuring Interconnection Loss

The first step in measuring interconnection loss is to horizontally zoom in on the interconnection. Position the A cursor just in front of the back reflection; then position the B cursor on a smooth area on the trace after the interconnection back reflection. The loss for the interconnection and the optical fiber between the cursors is displayed on the OTDR screen. The loss for the interconnection shown in Figure 14.59 is 0.4dB. The distance between the A and B cursors is 50m.

FIGURE 14.59 Measuring interconnection loss with the OTDR



NOTE

The B cursor should always be in a smooth section of the trace. When a long pulse width is used, the dead zone may be very large. The distance between the A and B cursors may need to be several hundred meters. In that case, you measure the attenuation of a partial length of optical fiber no less than several hundred meters.

To find the loss for only the interconnection, the loss for the optical fiber between cursors A and B needs to be subtracted from the A-B loss displayed by the OTDR. This loss was previously at 0.14dB for 50m. To find the actual interconnection loss, subtract 0.14dB from the 0.4dB A-B loss. The loss for only the interconnection is approximately 0.26dB.

Measuring the Loss of a Fusion Splice or Macrobend

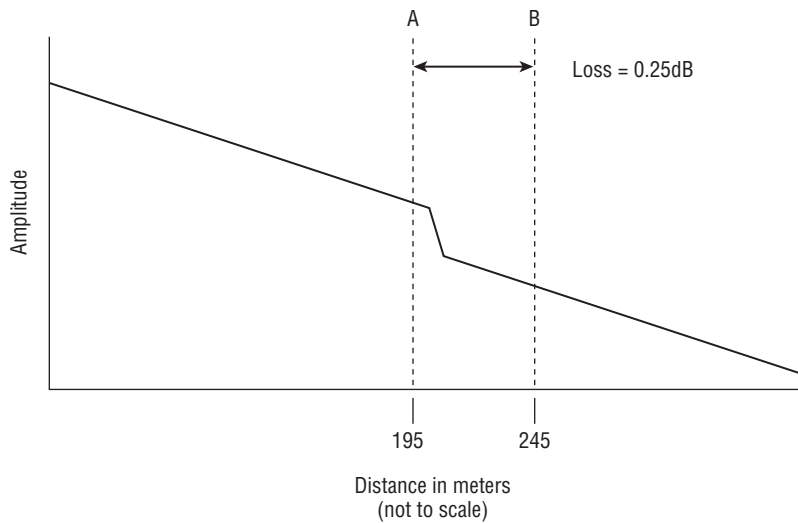
A fusion splice or macrobend does not produce a back reflection. A macrobend will always appear as a loss in the form of a dip in the smooth trace. A fusion splice may appear as a

dip or a bump in the trace. The fusion splice will appear as a dip in the trace when tested in both directions when the optical fibers fusion-spliced together have the same backscatter coefficient.

When optical fibers with different backscatter coefficients are fusion-spliced together, the splice may appear as a loss in one direction and a gain in the form of a bump in the trace when tested in the other direction. This is referred to as a *gainer*. To find the loss for the fusion splice, the splice must be tested in both directions and the results averaged together. The losses or loss and gain should be added together and the sum divided by 2. This is the loss for the fusion splice.

To find the loss of a fusion splice or macrobend, horizontally zoom in on the event. The loss from a fusion splice or macrobend is typically very small and will require vertical zoom in addition to horizontal zoom. Place the A cursor on the smooth part of the trace before the dip in the trace. Place the B cursor on the smooth part of the trace after the dip, as shown in Figure 14.60. The loss for this event is 0.25dB. The loss for the event includes the loss for the fusion splice or macrobend plus the 50m of optical fiber between the cursors. Subtract the loss for the 50m of optical fiber that was previously measured at 0.14dB from the event loss. The loss for this fusion splice or macrobend is 0.11dB.

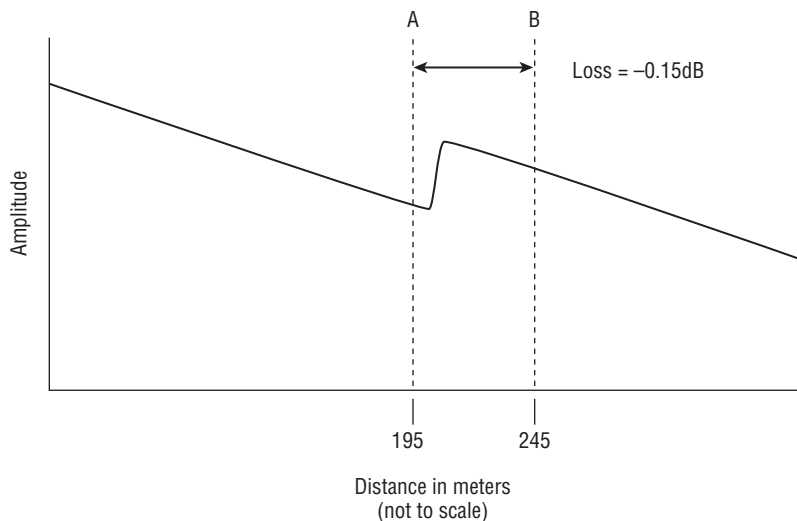
FIGURE 14.60 Measuring the loss of a fusion splice or macrobend



To find the gain of a fusion splice, horizontally zoom in on the event. The gain from a fusion splice is typically very small and will require vertical zoom in addition to horizontal zoom. Place the A cursor on the smooth part of the trace before the bump in the trace. Place the B cursor on the smooth part of the trace after the bump, as shown in Figure 14.61. The gain for this event is 0.15dB. The gain for the event includes the gain for the fusion splice plus the 50m of optical fiber between the cursors. Add the value for the loss for the 50m

of optical fiber that was previously measured at 0.14dB to the event gain. The gain for this fusion splice is 0.29dB.

FIGURE 14.61 Measuring the gain of a fusion splice



Measuring the Loss of a Cable Segment and Interconnections

To find the loss for a cable segment plus the interconnections, the exact length of the segment must be known. This can be done by using the method described to measure the length of a cable segment. The length of the cable segment in this example, as shown in Figure 14.62, is 85m.

The first step is to horizontally zoom in on the cable segment. After zooming in on the cable segment, place the A cursor on a smooth section of the trace just in front of the leftmost cable segment interconnection back reflection. Place the B cursor on the smooth part of the trace after the rightmost cable segment interconnection back reflection. Looking at the A-B distance on the OTDR, move the B cursor to the right until the distance equals the length of the cable segment plus 50m, as shown in Figure 14.62. The A-B loss is 1.5dB.

To find the loss for the cable segment plus the interconnections, subtract the loss for the 50m of optical fiber from the 1.5dB loss for the cable segment and interconnections. The previously measured loss for 50m of optical fiber at 850nm is 0.14dB. The loss for the cable segment and the interconnections is 1.36dB.

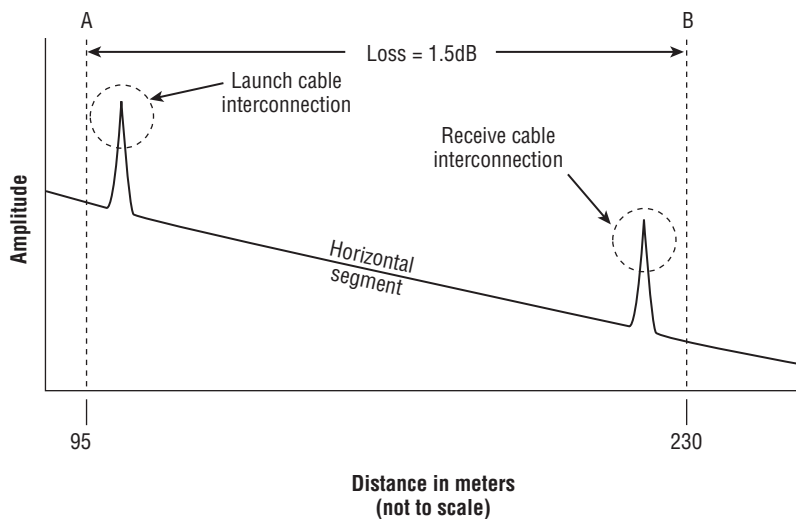
Documentation of OTDR Testing

As you test the cable plant with the OTDR, you need to properly document the results of the test. The documentation should include at a minimum the following:

- The date of the testing
- Test personnel

- The number or identification of fiber or cable tested
- The test procedure number
- OTDR make, model, serial number, and calibration dates
- Pulse width
- Range and resolution
- Number of averages
- Tested wavelength(s)
- Type and length of launch cable

FIGURE 14.62 Measuring the loss of a cable segment and interconnections



Emerging Testing Standards

In April 2013, SAE International's AS-3 Fiber Optics and Applied Photonics Committee began work on a new optical performance monitoring system standard AIR6552 that outlines the sensors and basic architecture required to detect, localize, and isolate impairments as well as assist in failure prediction of the physical layer of a fiber-optic network. This fiber-optic network end-to-end data link evaluation system standard is also known as NEEDLES. Because NEEDLES focuses on the physical layer, it can be applied to any fiber-optic network, including

- Ethernet
- ATM

- RF over fiber
- Passive Optical Networks
 - FTTH
 - FTTB
 - FTTC
 - FTTN

Since NEEDLES is a multisensor architecture, it has a main document AIR6552 and several slash sheets. Each slash sheet addresses a particular type of sensor. As of this writing, the main document and three slash sheets are under development.

AIR6552/1 addresses inline optical power monitoring. Inline optical power monitoring can be used to control optical power, anticipate failures, and detect fluctuations. This document establishes methods to obtain, store, and access data about the health of a fiber-optic network using commercially available inline optical power monitoring sensors.

AIR6552/2 addresses high-resolution optical reflectometry. High-resolution optical reflectometry can characterize and map the optical link. This document establishes methods to obtain, store, and access data about the health of a fiber-optic network using a commercially available high-resolution reflectometer.

AIR6552/3 addresses transceiver health monitoring. Transceiver health monitoring can provide real-time information about the physical, electrical, and optical characteristics of the transceiver. This document establishes methods to obtain, store, and access data about the health of a fiber-optic network using commercially available sensors located in or near the transceiver.

Summary

This chapter took an in-depth look at the key pieces of fiber optic test equipment and their application for testing in accordance with various industry standards. In addition, it described how to operate the equipment, perform various industry standard tests, and document the results.

Exam Essentials

Be able to explain the need for test equipment calibration. Know that calibration should be traceable to the National Institute of Standards and Technology (NIST) calibration standard or some other standard.

Know how to test the continuity of an optical fiber with a continuity tester. Be able to explain how to test the continuity of an optical fiber with the continuity tester.

Know how to locate faults with a VFL. Be able to explain how the VFL is used to identify breaks and macrobends.

Describe the basic operation of a single-mode and multimode OLTS. Be able to describe the difference between a multimode and single-mode light source and an optical power meter. Know that the optical power meter displays optical power in dBm.

Explain the difference between a test jumper and a patch cord. You should know that the terms patch cord and jumper are often used interchangeably. Know that a test jumper can be a single-fiber cable or a multifiber cable no less than 1m and no greater than 5m in length, it must have a core diameter and numerical aperture nominally equal to the optical fiber being tested.

Explain the purpose of a mode filter. Be able to describe the basic operation of a mode filter. Know that a mode filter removes high-order modes from an optical fiber and that the mode filter is always placed on the transmit jumper for multimode insertion loss measurements being performed in accordance with ANSI/TIA-526-14-A.

Explain the purpose of a modal controller. Know that multimode insertion loss measurements being performed in accordance with ANSI/TIA-526-14-B require a modal controller on the transmit jumper instead of a mandrel.

Know how to measure the optical loss in a patch cord using ANSI/TIA-526-14 method A. Be able to describe how to test a patch cord using the two-test jumper reference with an OLTS.

Describe the basic operation of an OTDR. Be able to describe how the OTDR launches pulses of light into an optical fiber and measures the energy returned from the optical fiber over a predetermined period of time.

Review Questions

1. Calibration is not required for a(n) _____.

- A. Continuity tester
- B. Optical power meter
- C. Stabilized light source
- D. OTDR

Hint: It is always required for test equipment that makes absolute measurements.

2. An unfiltered LED light source _____ the optical fiber with high-order modes.

- A. Underfills
- B. Overfills
- C. Fills
- D. Moderately fills

Hint: It causes higher-than-expected loss in short multimode links.

3. Method B uses _____ jumper(s) to establish the optical power reference.

- A. One
- B. Two
- C. Three
- D. Four

Hint: Two test jumpers are required to perform the test.

4. Encircled flux launch conditions were developed to improve the accuracy of _____ insertion loss measurements.

- A. Multimode
- B. Single-mode
- C. Multimode and single-mode
- D. Multifiber

Hint: It has been added to IEC 61280-4-1 Edition 2, ANSI/TIA-568-C.0, and ANSI/TIA-526-14-B.

5. The _____ is a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back.

- A. Continuity tester
- B. OLTS
- C. VFL
- D. OTDR

Hint: It provides a graphical representation of what is happening in the fiber-optic link or cable under test.

Chapter Exercises

Perform Optical Loss Measurement Testing of a Cable Plant

ANSI/TIA-526-14 revisions A and B provide three methods for testing the cable plant. The difference between the three methods is how the reference power measurement is taken. Each method uses a different number of test jumpers. Method A uses two test jumpers to obtain the reference power measurement, method B uses one test jumper, and method C uses three test jumpers.

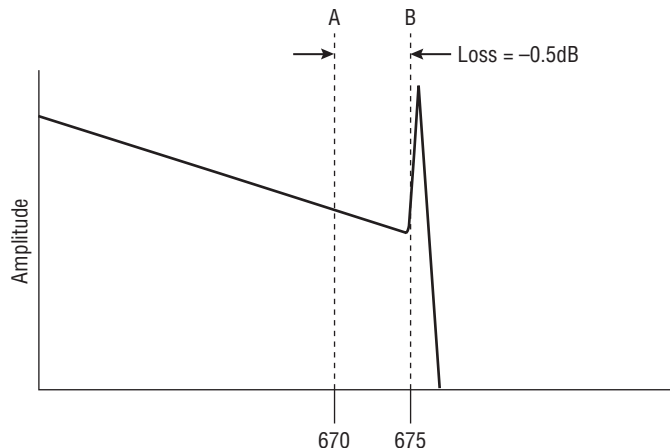
Before testing the cable plant using method B, you obtained your reference power measurement of -23.5dBm . Your test power measurement was -27.6dBm . What was the loss for the cable plant?

To obtain the loss for the cable plant, subtract the test power measurement from the reference power measurement. In this case, the reference power measurement was -23.5dBm and the test power measurement was -27.6dBm , so the loss for the cable plant would be 4.1dB .

Determine the Distance to a Break in the Optical Fiber with an OTDR

A break in an optical fiber in a fiber-optic cable is not the end of the optical fiber. However, a break in an optical fiber looks like the end of the optical fiber on the OTDR display. The same method is employed to measure the distance to a break in an optical fiber and to measure the distance to the end of an optical fiber.

You are troubleshooting fiber-optic cables with the OTDR. One optical fiber appears to be broken. Refer to the following graphic and determine the distance to the break in the optical fiber.

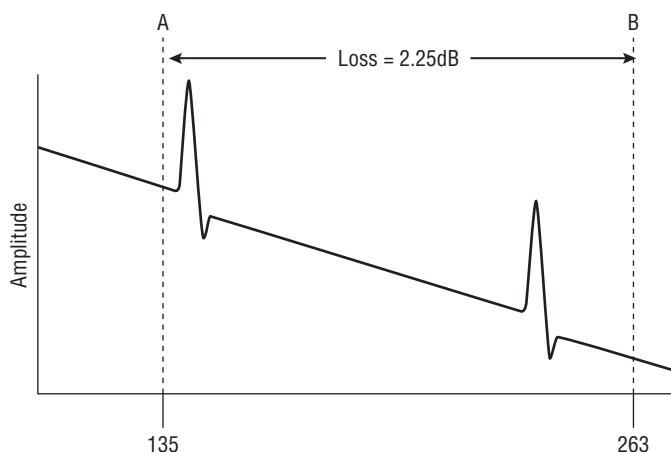


To measure the distance to the break in the optical fiber, zoom in on the back reflection and place the A cursor on a smooth section of the trace just in front of the back reflection. Move the B cursor toward the A cursor until it is in the leading edge of the back reflection. Keeping moving the B cursor toward the A cursor until the A-B loss is -0.5dB . The distance to the break is the distance for the B cursor, which is 675m.

Measure the Loss of a Cable Segment with an OTDR

To find the loss for a cable segment plus the interconnections, the exact length of the segment must be known.

You are testing a new fiber-optic cable installation with the OTDR and the customer has requested the loss information for a cable segment including the interconnections. Refer to the following graphic and determine the loss for the cable segment. The A-B distance equals the length of the cable segment plus 50m.



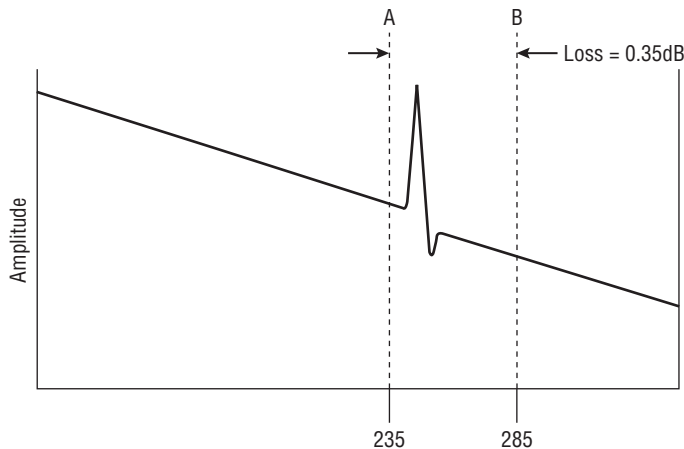
The first step in finding the loss for a cable segment with interconnections is to measure the length of the cable segment. After zooming in on the cable segment, place the A cursor on a smooth section of the trace just in front of the leftmost cable segment interconnection back reflection. Place the B cursor on the smooth part of the trace after the rightmost cable segment interconnection back reflection. Looking at the A-B distance on the OTDR, move the B cursor to the right until the distance equals the length of the cable segment plus 50m. The A-B loss is 2.25dB.

To find the loss for the cable segment plus the interconnections, subtract the loss for the 50m of optical fiber from the 2.25dB loss for the cable segment and interconnections. The previously measured loss for 50m of optical fiber at 850nm is 0.14dB. The loss for the cable segment and the interconnections is 2.11dB.

Measure the Loss of an Interconnection with an OTDR

When measuring the loss for an interconnection with the OTDR, you must remember to subtract the loss for the optical fiber.

You are testing a fiber-optic cable installation with the OTDR and the customer has requested the loss information for each interconnection. Refer to the following graphic and determine the loss for the interconnection.



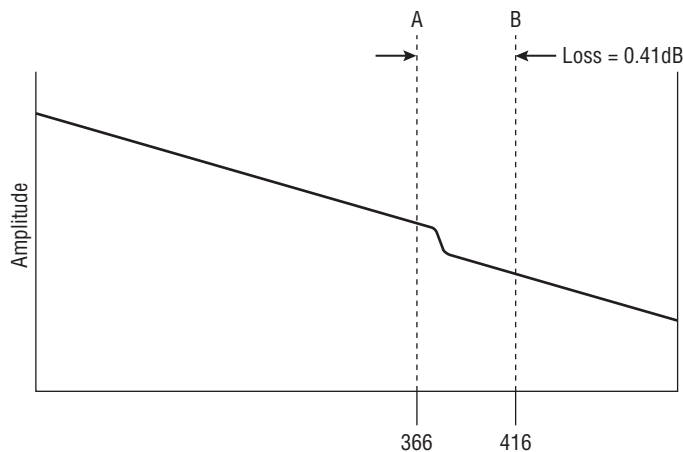
The first step in measuring interconnection loss is to horizontally zoom in on the interconnection. Position the A cursor just in front of the back reflection, then position the B cursor on a smooth area on the trace after the interconnection back reflection. The loss for the interconnection and the optical fiber between the cursors is displayed on the OTDR screen. The loss for the interconnection and 50m of optical fiber shown in the graphic is 0.35dB.

To find the loss for only the interconnection, the loss for the optical fiber between cursors A and B needs to be subtracted from the A-B loss displayed by the OTDR. This loss was previously at 0.14dB for 50m. To find the actual interconnection loss, subtract 0.14dB from the 0.35dB A-B loss. The loss for only the interconnection is approximately 0.21dB.

Measure the Loss of a Fusion Splice or Macrobend with an OTDR

Fusion splices or macrobends do not produce a back reflection. A macrobend will always appear as a loss in the form of a dip in the smooth trace. A fusion splice may appear as a dip or a bump in the trace.

You are testing a fiber-optic cable installation with the OTDR and see a dip in the trace. Since there are no splices in this cable, you determine that this event must be a macrobend. Refer to the following graphic and determine the loss for the macrobend.



To find the loss of a macrobend, horizontally zoom in on the event. The loss from a macrobend may be very small and require vertical zoom in addition to horizontal zoom. Place the A cursor on the smooth part of the trace before the dip in the trace. Place the B cursor on the smooth part of the trace after the dip, as shown in the graphic. The loss for this event is 0.41dB. The loss for the event includes the loss for the macrobend plus the 50m of optical fiber between the cursors. Subtract the loss for the 50m of optical fiber that was previously measured at 0.14dB from the event loss. The loss for this fusion splice or macrobend is 0.27dB.

Appendix

Answers to Review Questions



Chapter 1: History of Fiber-Optics and Broadband Access

1. B. Light travels at approximately 300,000 kilometers per second in air, and it allowed simple messages to be sent quickly over great distances through the use of signal fires.
2. C. The demonstrations performed by Colladon, Babinet, and Tyndall proved that light could be guided through water if the air around it had a lower index of refraction. This demonstrated the principle of total internal reflection.
3. A. Doctors and researchers experimented with bent glass and quartz to create a device that would allow them to see inside the body with minimal intrusion.
4. D. In 1966, Charles K. Kao announced that signal losses in glass fibers could be reduced to 20 decibels per kilometer or less through improved manufacturing processes. This announcement was the beginning of a quest for lower and lower signal loss through optical fiber.
5. D. The first full-scale commercial application of fiber-optic communication systems occurred in 1977, when both AT&T and GTE began using fiber-optic telephone systems for commercial customers.
6. A. Without fiber optics, broadband as we know it today would not exist. No other transmission medium can move the high rates of data over the long distances required to support the global telecommunications system.

Chapter 2: Principles of Fiber-Optic Transmission

1. B. The fiber optic link requires a transmitter to send the signal, a receiver to capture the signal, optical fiber to carry the signal, and connectors to align the fiber end precisely with the light source or photodiode.
2. D. In order for a signal to be transmitted through optical fiber, the transmitter must convert it from electrical energy into light.
3. D. With each 10 decibels of loss, the power remaining is 10 percent of what it was.
4. C. The receiver captures the light and converts it back to an electrical signal.
5. A. To calculate the decibel value for the loss in signal power, use the following equation:
 - $\text{dB} = 10\text{Log}_{10} (P_{\text{out}} \div P_{\text{in}})$
 - $\text{dB} = 10\text{Log}_{10} (12.5 \div 100)$
 - $\text{dB} = 10\text{Log}_{10} 0.125$

- $\text{dB} = -9.03$
 - $\text{Loss} = 9.03\text{dB}$
6. D. To calculate the loss in power from the optical fiber and the output power of the optical fiber, solve first for the loss and then for the output power. The equation would be written as follows:
- $(P_{\text{out}} \div P_{\text{in}}) = \text{antilog}(\text{dB} \div 10)$
 - $(P_{\text{out}} \div 250\mu\text{W}) = \text{antilog}(-4 \div 10)$
 - $(P_{\text{out}} \div 250\mu\text{W}) = \text{antilog}-0.4$
 - $(P_{\text{out}} \div 250\mu\text{W}) = 0.398$
 - $P_{\text{out}} = 0.398 \times 250\mu\text{W}$
 - $P_{\text{out}} = 99.5\mu\text{W}$
7. B. Decibels can be algebraically added and subtracted, $10\text{dB} + 10\text{dB} + 3\text{dB} = 23\text{dB}$.
8. C. To calculate the power in watts, the equation would be written as
- $-8 = 10\text{Log}_{10}(P \div 1\text{mW})$
 - $-0.8 = \text{Log}_{10}(P \div 1\text{mW})$
 - $0.158 = P \div 1\text{mW}$
 - $P = 0.158 \times 1\text{mW}$
 - $P = 158.5\mu\text{W}$
9. D. Each 10-decibel loss divides the percentage of remaining power by another factor of 10; $100\% \div 1000 = 0.1\%$.
10. D. To calculate the power in dBm, the equation would be written as
- $\text{dBm} = 10\text{Log}_{10}(4\text{mW} \div 1\text{mW})$
 - $\text{dBm} = 10\text{Log}_{10} 4$
 - $\text{dBm} = 10 \times 0.6$
 - $\text{dBm} = 6$
 - The power value is 6dBm.

This answer can be checked by locating 4mW in the first column of Table 2.4.

Chapter 3: Basic Principles of Light

1. B. The energy takes two forms: an electrical field and a magnetic field, formed at right angles to each other and at right angles to their path of travel.
2. A. An electromagnetic wave travels through free space or air at approximately 300,000km/s (kilometers per second).

3. C. Light has characteristics of both waves and particles. As a wave, it propagates through a medium and transfers energy without permanently displacing the medium. However, as a massless particle, called a photon, it travels in a wavelike pattern moving at the speed of light.
4. B. Wavelength or frequency can be used to describe electromagnetic energy; however, wavelength is typically used to describe the output of a fiber-optic light source.
5. A. Gamma rays have a shorter wavelength than the other three types of light. Visible light has a longer wavelength than ultraviolet light and a shorter wavelength than infrared light.
6. D. The wavelengths most commonly used for fiber optics are in the infrared range, at windows of 850nm, 1300nm, and 1550nm.
7. B. Refraction, or the change in the direction of light as it changes speed passing from one material into another, is a key component in fiber-optic transmissions.
8. A. The higher the refractive index, the slower the light travels. In other words, the speed of light in a medium is inversely proportional to its index of refraction.
9. C. If the light refracts toward normal, we know that the light has slowed down upon entering medium B, so we know that medium A has a lower n than medium B.
10. B. Total internal reflection takes place when light passing through a medium with a higher n hits the boundary layer of a medium with a lower n at a shallow angle, causing it to reflect from the boundary layer.
11. D. The incident angle required to produce a refracted angle of 90° is called the critical angle.
12. A. We can calculate the amount of refraction using Snell's law, which shows the relationship between incident light and refracted light.
13. B. When light changes speed as it moves from a material with a given index of refraction into a material with a different index of refraction, a Fresnel reflection occurs.
14. C. The greater the difference in the index of refraction between the two materials, the greater the amount of light reflected.
15. C. A Fresnel reflection occurs when light changes speed as it moves from the optical fiber to the air and from the air to the optical fiber.

Chapter 4: Optical Fiber Construction and Theory

1. B. The core is at the center of the optical fiber; the cladding surrounds the core and the coating surrounds the cladding.
2. A. One characteristic of optical fiber that deserves special attention is its tensile strength, or the amount of stress from pulling that the optical fiber can handle before it breaks. Optical fiber has a high tensile strength and resists stretching.

3. D. ANSI/TIA-568-C is applicable to premises cabling components and defines the performance of both multimode and single-mode optical fibers.
4. B. The material that will become the core and cladding of the glass fiber is originally laid down as a soot produced by burning a gas with the materials in it. The soot is then heated and compressed into the high-quality glass fiber.
5. A. Multimode and single-mode depressed clad optical fibers are commonly referred to as bend insensitive.
6. B. There are many potential paths or modes light can follow, depending on the size of the core, wavelength of the light source, and numerical aperture.
7. A. One of the most important characteristics used to distinguish types of optical fiber is the number of potential paths light can take through it.
8. C. Modes are expressed in whole numbers. If the calculation for modes results in a fraction, the result is rounded down to the nearest whole number.
9. D. The refractive index profile graphically represents the relationship between the refractive index of the core and that of the cladding.
10. B. Modal dispersion is caused when light rays follow different paths in the core of the optical fiber.
11. C. Multimode graded-index fiber tackles the problem of modal dispersion by increasing the speed of the high-order mode light rays, allowing them to keep up with the low-order mode rays.
12. D. Single-mode fibers use a small core that doesn't give the light enough space to follow anything but a single path through the fiber.
13. A. The shortest wavelength at which the optical fiber carries only one mode is called the cutoff wavelength.
14. C. Single-mode fibers are typically specified by their mode field diameter rather than their core and cladding size. Mode field diameter is the real estate used by the light within the core and the cladding as it propagates.
15. A. The optical trench surrounds the core of the optical fiber. It has a lower refractive index than the core or the cladding.

Chapter 5: Optical Fiber Characteristics

1. B. In general, dispersion is the spreading of light as it travels away from its source.
2. A. Dispersion limits the optical fiber's bandwidth, or the amount of information it can carry.
3. C. Modal dispersion results from light taking different paths as it passes through the optical fiber.

4. D. Because material dispersion is caused by an overabundance of wavelengths in the optical signal, one solution is to reduce the number of wavelengths by reducing the spectral width of the light source.
5. C. Waveguide dispersion occurs in single-mode fiber as the light passes through not only the core, but also part of the cladding.
6. D. Chromatic dispersion occurs in single-mode optical fiber and results from the combination of effects from material dispersion and waveguide dispersion.
7. A. Dispersion-shifted fiber shifts the “zero-dispersion point,” or the point where chromatic dispersion drops to zero, to the wavelength that allows light to travel through the fiber with the least attenuation.
8. C. Reducing the spectral width of the light source reduces the difference between the slowest and the fastest wavelengths in the optical fiber, thus reducing the material dispersion component of chromatic dispersion.
9. A. Scattering accounts for the greatest percentage of attenuation in an optical fiber—typically between 95 to 97 percent. Absorption accounts for a very small percentage of attenuation in an optical fiber—typically between 3 and 5 percent.
10. B. Light waves travel in different orientations, or polarities. When two different polarities travel through an optical fiber, they can be affected differently by conditions within the fiber, causing one polarity to travel at a different speed than the other.
11. B. ITU does not change letters, only numbers; G.651.1 is a multimode standard while G.652, G.655, and G.657 are single-mode standards.
12. A. The bandwidth decreases as the optical fiber length increases. However, the opposite is true when the length of the optical fiber decreases.
13. D. ISO/IEC 11801 defines the performance of OM1; the maximum attenuation at 850nm is 3.5dB for a length of one kilometer.
14. D. Attenuation in an optical fiber decreases as the infrared wavelength increases.
15. D. The minimum effective bandwidth-length product for a 50/125 μ m laser optimized OM4 multimode optical fiber is 4700MHz · km.
16. C. Many of an optical fiber’s characteristics are determined by the relationship between the core and the cladding. One characteristic, the numerical aperture (NA), expresses the light-gathering ability of the fiber.
17. D. ITU-T G.657 defines the macrobending loss for bend-insensitive single-mode optical fiber. This type of optical fiber is typically employed in passive optical networks (PONs) that run to your home or residence.
18. B. Microbends are small distortions of the boundary layer between the core and cladding caused by crushing or pressure. Microbends are very small and may not be visible when looking at the fiber-optic cable.
19. B. Optical fiber can be broken into two groups: bend sensitive and bend insensitive. There is a considerable difference in the bending attenuation of the two fiber types.
20. A. Plastic optical fibers typically operate in the visible spectrum with attenuation and bandwidth performance defined at 650nm.

Chapter 6: Safety

1. B. The Occupational Safety and Health Administration is responsible for creating and enforcing workplace safety regulations.
2. D. You should inspect your PPE every time you use it to make sure it is not damaged or defective.
3. C. Because infrared lasers are invisible, you cannot tell when they are on, even if you are looking directly into a fiber. You should always make sure the fiber is disconnected from the light source before looking into it with your bare eyes or optical instruments.
4. B. SG1 has a total output power that is less than the accessible emission limit for Class I (400nW), and there is no risk of exceeding the maximum permissible irradiance when viewing the end of a fiber with a microscope, an eye loupe, or the unaided eye.
5. C. These intermediate power lasers have a continuous wave output from 1.0 to 5.0mW. Directly viewing the beam can damage the eye.
6. A. Class 1 levels of laser radiation are safe under normal operating conditions, including the use of optical instruments such as a microscope or eye loupe.
7. B. The Material Safety Data Sheet (MSDS) provides information on specific hazards posed by each chemical and ways to protect yourself through specific handling procedures, protective clothing and equipment, and engineering controls such as ventilation.
8. A. Because solvent molecules can displace oxygen molecules in the blood, they can starve the brain of oxygen and give the appearance of intoxication.
9. B. Use caution when working with the primer portion of these adhesives. With a flash point of -18°C , it is highly flammable.
10. C. If you see someone who is in this situation, do not try to pull them away with your hands. You may be caught up in the circuit as well. Instead, use a nonconducting stick, such as a wooden or fiberglass broom handle, to knock the victim away from the voltage source.

Chapter 7: Fiber-Optic Cables

1. B. The core, cladding, and coating of the optical fiber itself are considered one component of the cable, with the rest being the buffer, strength member, and jacket.
2. A. A loose buffer consists of a buffer layer or tube that has an inner diameter much larger than the diameter of the coated optical fiber.
3. C. Tight-buffered cables have protective buffers attached directly to the optical fiber so that they are protected from damage that would harm bare fibers. This allows the fibers to run outside of a cable for a short distance and be attached to connectors.
4. A. Aramid yarns are flexible, lightweight, strong, and used in bulletproof vests and fire protection gear.

5. C. The jacket is the cable's outer protective layer. It protects the internal components from the outside world.
6. B. Duplex cordage is a convenient way to combine two simplex cords to achieve duplex, or two-way, transmissions without individual cords getting tangled or switched around accidentally.
7. C. Breakout cables consist of two or more simplex cables bundled with a strength member and/or central member covered with an outer sheath.
8. A. Distribution cable consists of multiple tight-buffered fibers bundled in a jacket with a strength member.
9. C. An armored cable protects the optical fibers from gnawing animals and the damage that can occur during direct burial installations.
10. A. When a fiber-optic cable must be suspended between two poles or other structures, a messenger wire is required. The messenger wire can be external to the fiber-optic cable or integrated into the cable. When the messenger wire is integrated into the fiber-optic cable, the cable is typically referred to as a messenger cable.
11. B. Hybrid cable, as applied to fiber optics, combines multimode and single-mode optical fibers in one cable. Hybrid cable should not be confused with composite cable, although the terms have been used interchangeably in the past.
12. C. Composite cable, as defined by the NEC, is designed to carry both optical fiber and current-carrying electrical conductors.
13. D. A fanout kit converts loose-buffered fibers into tight-buffered fibers ready for connectors.
14. B. Conductive cables contain non-current-carrying conductive members, including strength members, armor, or sheath.
15. A. Plenum cables, whether conductive or nonconductive, are suitable for use in ducts, plenums, and other space used for environmental air. These cables will have fire resistance and low smoke-producing characteristics.
16. D. OFNG or OFN are the NEC markings for a nonconductive general-purpose fiber-optic cable.
17. C. Whenever a cable is listed as nonconductive, the only cable that can be substituted for it is another nonconductive cable, such as the OFNP.
18. D. TIA-598-C defines premises cable jacket colors; aqua is the jacket color for laser-optimized multimode optical fiber used for non-military applications.
19. B. The color-coding schemes for premises jackets and optical fibers within a fiber-optic cable described in TIA-598-C are not always used and may be replaced by a number printed every 6" along the entire length of the cable.
20. C. Sequential markings are numbers that appear every 2' or 1m. These markings are useful in determining how much cable is left on a reel, measuring off large runs of cable, or simply determining the length of a piece of cable without pulling out a tape measure.

Chapter 8: Splicing

1. B. Loss from a numerical aperture (NA) mismatch occurs when light exiting the core of the transmitting optical fiber is outside of the cone of acceptance of the receiving fiber.
2. A. Core diameter mismatch occurs when there is a difference in the core diameters of the two optical fibers. A core diameter mismatch loss results when the core diameter of the transmitting optical fiber is greater than the core diameter of the receiving optical fiber.
3. C. Cladding diameter mismatch occurs when the cladding diameters of the transmit and receive optical fibers are not the same. Cladding diameter mismatch loss occurs when the cores of the optical fiber are not aligned because of the cladding diameter mismatch.
4. D. Lateral misalignment occurs when the two optical fibers are offset. Lateral misalignment loss occurs when light from the core of the transmitting optical fiber enters the cladding of the receiving optical fiber, creating a loss.
5. B. Ideally, the core and cladding of an optical fiber are perfectly round and concentric, which means that they share a common geometric center. However, optical fibers are not perfect and there will be concentricity variations.
6. C. A mechanical splice aligns the cleaved optical fibers and holds them in place. Index matching gel inside the mechanical splice reduces or eliminates Fresnel reflections.
7. B. Fusion splicing uses a high-voltage electric arc between two electrodes to heat and melt the fiber end.
8. B. Cleaving is typically a two-step process. The first step in the process is scoring the optical fiber. The next step is to bend the optical fiber.
9. A. Prior to performing the fusion splice, the optical fiber is passed through the center of the protective sleeve and the sleeve is positioned so it will not interfere with the fusion splicing process. After the fusion splice is successfully completed, the optical fiber is removed from the fusion splicer and the protective sleeve is positioned directly over the splice area.
10. B. For inside plant splices, ANSI/TIA-568-C.3 states that optical fiber splices, whether fusion or mechanical, will have a maximum attenuation of 0.3dB.

Chapter 9: Connectors

1. A. Beginning at the working end of the connector, the ferrule holds the fiber in place. The ferrule must hold the fiber exactly centered in its endface for the best possible connection, so its construction is critical.

2. C. Ceramic materials such as aluminum oxide and zirconium oxide are among the best materials for ferrules, offering the best combination of characteristics. They are hard enough to protect the fiber end, and their coefficient of thermal expansion, the measure of how much a material expands and contracts with temperature changes, is about the same as the optical fiber itself.
3. B. When optical fibers are manufactured within specified tolerances, there are still slight variations from one optical fiber to another. The same is true for the ferrule and mating sleeve. Variations in these components can affect the performance of the connection.
4. C. To ensure physical contact (PC) between optical fiber ends, the best endface geometry is a convex curve. This curved, or PC, finish ensures that the highest feature or apex on the endface is the center of the optical fiber end.
5. B. The SC has a standard-sized 2.5mm ferrule and a snap-in connection, which was created as an alternative to connectors that required turning or twisting to keep them in place.
6. A. The LC connector is a small form factor connector developed by Lucent Technologies. This snap-in connector is considered to be a smaller version of the SC connector and is sometimes referred to as a mini-SC.
7. C. The MPO (Multifiber Push On) connector is built on the MT ferrule. It is designed to hold from 4 to 72 fibers.
8. C. Pigtails are fiber ends with connectors factory-attached for future splicing into a system. They require hardware to protect the splice and an investment in mechanical or fusion splicing equipment.
9. D. Oven-cured epoxy is probably the best type you can use, but it is also the most cumbersome. The epoxy itself consists of a resin and a hardener, which must be mixed in the right proportions.
10. B. The epoxy itself consists of a resin and a hardener, which must be mixed in the right proportions. Once it is mixed, it has a limited pot life before it begins to harden, and any epoxy that is unused must be discarded.
11. A. The cure adapter is placed over the ferrule of the connector. It helps transfer heat from the curing oven and protects the optical fiber during the curing process.
12. A. While the adhesive still comes in two parts, each part is applied separately to the fiber and the ferrule. When the two are joined, the epoxy hardens and cures in about 10 seconds.
13. C. The scribe is used for precision work in removing the fiber end once it has been adhered inside the ferrule. To use it, you only need to score or nick one side of the optical fiber lightly. This breaks the surface of the cladding, which provides the fiber's tensile strength.
14. A. An interferometer is a device that uses light to take precision measurements of the fiber-optic connector endface. The measurements are used to generate a three-dimensional (3D) map of the surface and quantitatively define critical parameters.

15. D. The polishing fixture, or puck as it is typically referred to, is used to ensure that the ferrule stays perpendicular to the polishing film during the polishing process.
16. C. Install the strain relief on the cable end, small diameter first. Important: The strain relief boot will not fit over the connector, so it must be placed on the cable first.
17. D. The radius of curvature describes the roundness of the connector endface, and it is measured from the center axis of the connector ferrule. The minimum and maximum values for the radius of curvature as defined in Telcordia GR-326 are 7mm and 22mm.
18. B. IEC 61300-3-35 is a standard that provides methods for quantitatively evaluating the endface quality of a polished fiber-optic connector. This standard breaks the endface into four zones—core, cladding, adhesive or epoxy ring, and contact zone.
19. C. Per ANSI/TIA-568-C.3 the strain relief for a single-mode connector should be blue in color.
20. C. The maximum insertion loss allowed by ANSI/TIA-568-C.3 for a multimode or single-mode mated connector pair is 0.75dB.

Chapter 10: Fiber-Optic Light Sources and Transmitters

1. B. The photons emitted from the junction where the p and n regions meet do not have a fixed relationship; they are out of phase. These out-of-phase light waves are called incoherent light.
2. D. As current flows through the LED, the junction where the p and n regions meet emits random photons. This process is referred to as spontaneous emission.
3. C. Every photon that escapes the optical cavity has the same wavelength, phase relationship, and direction as the first photon. This process of generating light energy is called stimulated emission.
4. A. Every photon that escapes the optical cavity is a duplicate of the first photon to escape. These photons have the same wavelength, phase relationship, and direction as the first photon. This fixed relationship is referred to as coherent light.
5. B. LEDs have wide output patterns and typically use a micro lens within a package to direct as much light as possible into the optical fiber.
6. C. While all of the laser types—Fabry-Pérot, DFB, and VCSEL—are used with multimode optical fiber, only the VCSEL is used for 850nm applications.

7. B. LED spectral widths range from 20nm to 170nm. Laser spectral widths range from 5nm to less than 0.1nm.
8. B. LEDs output less optical power than lasers. Optical output power is the amount of light energy coupled into the core of the fiber. LED optical power output levels vary depending on the core size and NA of the optical fiber.
9. D. Laser light sources may have modulation speeds as great as 10Gbps in a network application. LED modulation speeds in a network application typically do not exceed 100Mbps.
10. B. Currently VCSEL multimode transmitters support only 850nm operation with 50/125 μ m or 62.5/125 μ m optical fiber. The VCSEL can support data rates as high as 10Gbps.
11. D. The light source of choice for parallel optics is the VCSEL. VCSEL technology has matured to where multiple 10Gbps transmitters can be manufactured in the form of an array.
12. B. Achieving a 40Gbps data rate requires four 10Gbps transmitters operating in parallel.
13. D. As defined in IEEE Standard 802.3-2012, Section 6, when multiple transmitters are operating in parallel, multimode OM3 or OM4 optical fiber should be used for the physical layer.
14. A. Under normal operating conditions, Class I lasers do not produce damaging radiation levels and are considered eye safe.
15. B. Never directly view the end of an optical fiber or the endface of a fiber-optic connector without verifying that the optical fiber is dark.

Chapter 11: Fiber-Optic Detectors and Receivers

1. B. In a fiber-optic receiver, it's the job of the photodiode to convert the light energy received from the optical fiber into electrical energy.
2. A. Photons absorbed by a photodiode excite electrons within the photodiode in a process called intrinsic absorption. When stimulated with an outside bias voltage, these electrons produce a current flow through the photodiode.
3. C. With the APD, a small number of photons can trigger an avalanche of electrons. The APD accomplishes this through a process called photomultiplication.
4. C. The dynamic range of the receiver is measured in decibels; it is the difference between the maximum and minimum optical input power that the receiver can accept.
5. B. All receivers generate errors; error generation is typically described by the receiver's bit error rate, or BER.

6. A. The receptacle is the part of the fiber-optic receiver that accepts the connector. It also aligns the ferrule so that the optical fiber within the ferrule is perpendicular with the photodiode in the optical subassembly.
7. D. Optical attenuators are generally used in single-mode long-haul applications to prevent optical overload at the receiver.

Chapter 12: Cable Installation and Hardware

1. A. The minimum installation bend radius is the short-term bend radius, and the minimum operational bend radius is the long-term bend radius. The minimum short-term bend radius is actually the larger of the two because of the tensile stresses that may be placed on the cable during installation.
2. B. A cable's maximum tensile load rating is higher for installation because the loads are not expected to last for the life of the cable.
3. D. A dynamic load is a changing tensile load. Dynamic load is referred to as the installation load.
4. B. The pulling eye is specially designed to attach to the cable's strength member at one end and a pulling line at the other.
5. C. A pullbox is used in long conduit runs or after multiple turns in a conduit to provide an intermediate pulling location for the cable, which relieves the tensile loads that would be imposed if the cable had to be pulled through the entire system at once.
6. A. A splice enclosure is used whenever two fibers have been spliced to protect the splice from exposure and strain.
7. C. A patch panel is an interconnection point for fiber-optic cables. They allow signals to be routed from one cable to another with a patch cord or jumper.
8. D. The weight of the much larger copper cables can cause macrobends if they are piled on top of the optical fiber cable. If there is a risk of this happening, choose an optical fiber cable that will protect the optical fiber inside from the crushing weight of the copper cables.
9. B. Most cable is installed in a conduit by pulling it through with a line attached to a pulling eye, which is attached to the cable's strength member.
10. D. Direct burial is a method of laying a fiber-optic cable directly in the ground. The cable must be protected from damage and from the elements.

11. A. Messenger cable is used in aerial installations. It has a messenger line built into the cable to take the loads imposed on cable that is strung between poles. If a messenger wire is not incorporated into the cable assembly used, the cable will have to be lashed to messenger wire. A messenger wire is a steel or aluminum wire running between the poles to support the fiber-optic cable.
12. C. Blown fiber uses pressurized dry air to blow a fiber through a tube without the need for pulling. It is an alternative to installing cables.
13. C. Article 770.93 places a grounding or isolation requirement on any fiber-optic cable entering a building that contains electrically conductive materials and that may be exposed to or come in contact with electrical light or power conductors. Article 770.100 describes bonding and grounding methods for fiber-optic cables with electrically conductive non-current-carrying metallic members.
14. B. ANSI/TIA-606-B is the administration standard for telecommunications infrastructures. This document contains detailed information on conventions for labeling optical fiber connections within buildings, datacenters, and between buildings that share a telecommunication network.
15. C. Polarity is required to ensure the proper operation of bidirectional fiber optic communication systems that use separate transmit and receive optical fibers. It ensures that there is an end-to-end transmission path between the transmitter and the receiver of a channel.

Chapter 13: Fiber-Optic System Advantages

1. D. Optical fiber offers the highest bandwidth and the least attenuation of any medium currently available.
2. C. Optical fiber is a dielectric or an insulator. Current does not flow through insulators, so EMI has no effect on the operation of an optical fiber.
3. B. Compared to a copper cable with similar transmission frequency performance, a fiber-optic cable will always be smaller and lighter. The difference in size and weight increases considerably as the transmission distance increases.
4. D. Because of total internal reflection, optical fiber does not radiate light energy. Optical fiber is virtually immune to being tapped, which makes it the medium of choice for secure communications.
5. B. Electrical safety is always a concern when working with copper cables. Electrical current flowing through copper cable poses shock, spark, and fire hazards. Optical fiber is a dielectric that cannot carry electrical current and presents no shock, spark, or fire hazard. In addition, it provides electrical isolation between electrical equipment.

Chapter 14: Test Equipment and Link/Cable Testing

1. A. Calibration is always required for test equipment that makes absolute measurements, such as an optical power meter. However, not all test equipment requires calibration. For example, calibration is not required for a continuity tester; however, it is necessary for a stabilized light source.
2. B. An unfiltered LED light source overfills an optical fiber by launching high- and low-order modes into the core and cladding. The high-order modes experience more attenuation than the low-order modes at interconnections, splices, and bends. This causes higher-than-expected loss in short multimode links.
3. A. Method B uses one jumper to establish the optical power reference. However, two test jumpers are required to perform the test.
4. A. Encircled flux launch conditions for all multimode insertion loss measurements has been added to IEC 61280-4-1 Edition 2, ANSI/TIA-568-C.0, and ANSI/TIA-526-14-B. The addition of this requirement arose from the need to improve the accuracy of multimode insertion loss measurements.
5. D. The OTDR is nothing more than a device that launches a pulse or pulses of light into one end of an optical fiber and records the amount of light energy that is reflected back. It provides a graphical representation of what is happening in the fiber-optic link or cable under test.

Glossary



4G (fourth generation) Refers to the fourth generation of standards to support higher mobile telephony and broadband services.

A

absorption The loss of power (signal) in an optical fiber resulting from conversion of optical power (specific wavelengths of light energy into heat). Caused principally by impurities, such as water, ions of copper or chromium (transition metals), and hydroxyl ions, and by exposure to nuclear radiation. Expressed in dB/km (decibels per kilometer). It accounts for a very small percentage of attenuation in an optical fiber—typically between 3 and 5 percent.

acceptance angle With respect to optical-fiber cable, the angle within which light can enter an optical fiber core of a given numerical aperture and still reflect off the boundary layer between the core and the cladding. Often expressed as the half angle of the acceptance cone and measured from the axis. *See also* numerical aperture (NA).

acceptance cone A three dimensional cone that extends outside the optical fiber's core. Light entering the core from outside of the cone will either miss the core or enter at an angle that will allow it to pass through the boundary with the cladding and be lost. The acceptance angle defines the acceptance cone. Light entering the core of the optical fiber at an angle greater than the acceptance angle may not propagate the length of the optical fiber. For light to propagate the length of the optical fiber, it must enter the core at an angle that does not exceed the acceptance angle. It is also referred to as the *cone of acceptance*.

access jumper A cable used to measure insertion loss with an OTDR at the near end or far end of the cable plant. Also referred to as launch fiber, launch cable, pulse suppression cable, dead zone cable, or test fiber box.

acrylate A common type of optical fiber coating material that is easy to remove with hand tools.

active alignment fusion splicer A fusion splicer that can detect and align the cores of the optical fibers prior to splicing.

adapter With respect to optical fiber, a passive device used to join two connectors and fiber cores together to make an optical connection. The adapter is defined by connector type, such as SC, FC, ST, LC, and MT-RJ. Hybrid adapters can be used to join dissimilar connectors together, such as SC to ST. Adapters contain an alignment sleeve that aligns the connector ferrules or plugs and they may be referred to as an alignment sleeve or mating sleeve.

alignment sleeve *See* mating sleeve.

analog-to-digital (A/D) converter A device used to convert analog signals to digital signals for storage or transmission.

angle of incidence With respect to fiber optics, the angle of a ray of light striking a surface or boundary as measured from a line drawn perpendicular to the surface. Also called incident angle.

angle of reflection With respect to fiber optics, the angle formed between the normal and a reflected beam. The angle of reflection equals the angle of incidence.

angle of refraction With respect to fiber optics, the angle formed between the normal (a line drawn perpendicular to the interface) and a refracted beam.

angled physical contact (APC) connector A single-mode optical-fiber connector whose angled endface helps to ensure low mated reflectance and low unmated reflectance. The ferrule is polished at an angle to ensure physical contact with the ferrule of another APC connector.

angular misalignment loss The loss of optical power caused by deviation from optimum alignment of fiber-to-fiber. Loss at a connector due to fiber angles being misaligned because the fiber ends meet at an angle.

apex The highest feature on the endface of the optical fiber end.

apex offset The displacement between the apex of the endface and the center of the optical fiber core.

aramid strength member The generic name for Kevlar® or Twaron® found in fiber-optic cables. A yarn used in fiber-optic cable that provides additional tensile strength, resistance to bending, and support to the fiber bundle. It is not used for data transmission.

array connector A single ferrule connector such as the MPO that contains multiple optical fibers arranged in a row or in rows and columns.

attenuation A general term indicating a decrease in power (loss of signal) from one point to another. In optical fibers, it is measured in decibels per kilometer (dB/km) at a specified wavelength. The loss is measured as a ratio of input power to output power.

attenuator A passive device that intentionally reduces the strength of a signal by inducing loss.

avalanche photodiode (APD) A photodiode designed to use the avalanche multiplication of photocurrent. It multiplies the effect of the photons it absorbs, acting as an amplifier.

B

back reflection *See* return reflection.

backbone A cable connection between telecommunications or wiring closets, floor distribution terminals, entrance facilities, and equipment rooms either within or between buildings. This cable can service voice communications or data communications. In star-

topology data networks, the backbone cable interconnects hubs and similar devices, as opposed to cables running between hub and station. In a bus topology data network, it is the bus cable. Backbone is also called riser cable, vertical cable, or trunk cable.

backbone wiring (or backbone cabling) The physical/electrical interconnections between telecommunications rooms and equipment rooms. *See also* backbone.

backscatter Usually a very small portion of an overall optical signal, backscatter occurs when a portion of scattered light returns to the input end of the fiber; the scattering of light in the direction opposite to its original propagation. Light that propagates back toward the transmitter. Also known as back reflection or backscattering.

backscatter coefficient Also called Rayleigh backscatter coefficient, a value in dB used by an OTDR to calculate reflectance. The optical fiber manufacturer specifies the backscatter coefficient.

bait The mold or form on which silicon dioxide soot is deposited to create the optical fiber preform.

bandwidth Indicates the information carrying capacity of media. For copper cables, bandwidth is defined using signal frequency and specified in hertz (Hz). For multimode optical fiber, bandwidth is defined in MHz · km at a specified wavelength and specified launch.

beam splitter An optical device, such as a partially reflecting mirror, that splits a beam of light into two or more beams. Used in fiber optics for directional couplers.

bend insensitive An optical fiber designed to reduce the amount of attenuation when it is bent.

bend-insensitive multimode optical fiber (BIMMF) Multimode optical fiber designed to reduce the amount of attenuation when it is bent.

bend radius (minimum) The minimum radius optical fiber or fiber-optic cable can be bent without damaging it. It is easy to mistake bend diameter for bend radius; bend diameter is twice the bend radius. If a cable has a diameter of 1" and the bend radius is 10 times the diameter, the bend radius is 10" and the bend diameter is 20". Confusing the two may lead to over bending the cable.

bit A binary digit, the smallest element of information in the binary number system, and thus the smallest piece of data possible in a digital communication system. A 1 or 0 of binary data.

bit error An error in data transmission that occurs when a bit is a different value when it is received than the value it was when it was transmitted.

bit error rate (BER) In digital applications, a measure of data integrity. It is the ratio of bits received in error to bits originally sent or the ratio of incorrectly transmitted bits to total transmitted bits. BERs of one error per billion bits sent are typical.

blown fiber A method for installing optical fiber in which fibers are blown through preinstalled conduit or tube using air pressure to pull the fiber through the conduit.

bonding The National Electrical Code (NEC) defines the term bonding as connecting to establish electrical continuity and conductivity.

breakout cable Multifiber cables composed of simplex interconnect cables where each fiber has additional protection by using additional jackets and strength elements, such as aramid yarn.

breakout kit A collection of components used to add tight buffers, strength members, and jackets to individual fibers from a loose-tube buffer cable. Breakout kits are used to build up the outer diameter of fiber cable when connectors are being installed and designed to allow individual fibers to be terminated with standard connectors.

buffer The protective layer placed around the coating of an optical fiber. There are two buffer types: loose buffer and tight buffer.

buffer tube A tube that has an inner diameter much larger than the diameter of the coated optical fiber.

byte A binary “word” or group of eight bits.

C

c The symbol for the speed of light in a vacuum.

C The symbol for both capacitance and Celsius. In the case of Celsius, it is preceded by the degree symbol (°).

cable (1) Copper: A group of insulated conductors enclosed within a common jacket.
(2) Fiber: One or more optical fibers enclosed within a protective covering and material to provide strength.

cable assembly A cable that has connectors installed on one or both ends.

cable duct A single pathway (typically a conduit, pipe, or tube) that contains cabling.

cable entrance conduits Holes or pipes through the building foundation, walls, floors, or ceilings through which cables enter into the cable entrance facility (CEF).

cable entrance facility (CEF) The location where the various telecommunications cables enter a building. Typically this location has some kind of framing structure (19-inch racks, or plywood sheet) with which the cables and their associated equipment can be organized.

cable management system A system of tools, hold-downs, and apparatus used to precisely place cables and bundles of cables so that the entire cabling system is neat and orderly and that growth can be easily managed.

cable modem A modem that transmits and receives signals, usually through copper coaxial cable. Connects to your CATV connection and provides you with an Internet connection for your computer.

cable plant A combination of cabling components such as cables, jumpers, cords, and connecting hardware that forms an end-to-end transmission path between two points where application-specific equipment is connected.

cable pulling lubricant A lubricant used to reduce the friction against cables as they are pulled through conduits.

cable spool rack A device used to hold a spool of cable and allow easy pulling of the cable off the spool.

cable tray A shallow tray used to support and route cables through building spaces and above network racks.

cabling map A map of how cabling is run through a building network.

campus The buildings and grounds of a complex, such as a university, college, industrial park, or military establishment.

campus backbone Cabling between buildings that share data and telecommunications facilities.

campus distributor (CD) The ISO/IEC 11801 term for the main cross-connect; this is the distributor from which the campus backbone cable emanates.

carbon Carbon is the thinnest coating of all, typically only a fraction of a micron thick. Carbon hermetically seals the glass surface, protecting it from moisture and increasing the optical fiber's fatigue resistance. Carbon cannot be used by itself; it must be combined with another coating such as silicone or polyimide. Unlike the other coatings discussed, carbon coating does not have to be removed before termination.

Category 6A Also called CAT-6A. Recognized in ANSI/TIA-568-C, Category 6A supports transmission at frequencies up to 500MHz over 100-ohm twisted pair. This is the only category cable required to meet alien crosstalk specifications.

central member The center component of a cable, which serves as an antibuckling element to resist temperature-induced stresses. Sometimes serves as a strength element. The central member is composed of steel, fiberglass, or glass-reinforced plastic.

central office (CO) The telephone company building where subscribers' lines are joined to telephone company switching equipment that serves to interconnect those lines. Also known as an exchange center or head end. Some places call this a public exchange.

centralized cabling A cabling topology defined by ANSI/TIA-568-C whereby the cabling is connected from the main cross-connect in an equipment room directly to

telecommunications outlets. This topology avoids the need for workgroup switches on individual building floors.

channel The end-to-end transmission or communications path over which application-specific equipment is connected.

chromatic dispersion The spreading of a particular light pulse because of the varying refraction rates of the different-colored wavelengths. Different wavelengths travel along an optical medium at different speeds. Wavelengths reach the end of the medium at different times, causing the light pulse to spread. This chromatic dispersion is expressed in picoseconds per kilometer per nanometer (of bandwidth). It is the sum of material and waveguide dispersions. *See also* waveguide dispersion, material dispersion.

cladding Name for the material (usually glass, sometimes plastic) that is put around the core of an optical fiber during manufacture. The cladding is not designed to carry light, but it has a slightly lower index of refraction than the core, which causes the transmitted light to travel down the core. The interface between the core and the cladding creates the mode field diameter, wherein the light is actually held reflectively captive within the core.

cladding mode A mode of light that propagates through and is confined to the cladding.

cleaving The process of separating an unbuffered optical fiber by scoring the outside and pulling off the end. Cleaving creates a controlled fracture of the glass for the purpose of obtaining a fiber end that is flat, smooth, and perpendicular to the fiber axis. This is done prior to splicing or terminating the fiber.

closet An enclosed space for housing telecommunications and networking equipment, cable terminations, and cross-connect cabling. It contains the horizontal cross-connect where the backbone cable cross-connects with the horizontal cable. Called a telecommunications room by the TIA/EIA standards; sometimes referred to as a wiring closet.

coating The first true protective layer of an optical fiber. The plastic-like coating surrounds the glass cladding of a fiber and is put on a fiber immediately after the fiber is drawn to protect it from the environment. Do not confuse the coating with the buffer.

coefficient of thermal expansion The measure of a material's change in size in response to temperature variation.

coherent light Light in which photons have a fixed or predictable relationship; that is, all parameters are predictable and correlated at any point in time or space, particularly over an area in a lane perpendicular to the direction of propagation or over time at a particular point in space. Coherent light is typically emitted from lasers.

collimated Light that has been expanded so that the rays are parallel.

concentric Sharing a common geometric center.

conductivity The ability of a material to allow the flow of electrical current; the reciprocal of resistivity. Measured in “mhos” (the word ohm spelled backward).

conductor A material or substance (usually copper wire) that offers low resistance (opposition) to the flow of electrical current.

conduit A rigid or flexible metallic or nonmetallic raceway of circular cross section in which cables are housed for protection and to prevent burning cable from spreading flames or smoke in the event of a fire.

cone of acceptance See *acceptance cone*.

connector A device that attaches to the end of an optical fiber and couples the fiber end mechanically to a piece of equipment or to another optical fiber so that the cores line up accurately and produce the smallest amount of loss.

consolidation point (CP) A location defined by the ANSI/TIA/EIA-568-C standard for interconnection between horizontal cables that extends from building pathways and horizontal cables that extend into work area pathways. Often an entry point into modular furniture for voice and data cables. ISO/IEC 11801 defines this as a transition point (TP).

consumables kit Resupply material for splicing or terminating fiber optics.

continuity An uninterrupted pathway for electrical or optical signals.

continuity tester A tool that projects light from the LED or incandescent lamp into the core of the optical fiber.

cordage Fiber-optic cable designed for use as a patch cord or jumper. Cordage may be either single-fiber (simplex) or double fiber (duplex).

core The central part of a single optical fiber in which the light signal is transmitted. Common core sizes are 8.3 microns, 50 microns, and 62.5 microns. The core is surrounded by a cladding that has a higher refractive index that keeps the light inside the core. The core is typically made of glass or plastic.

core eccentricity A measurement that indicates how far off the center of the core of an optical fiber is from the center of that fiber’s cladding.

coupler With respect to optical fiber, a passive, multiport device that connects three or more fiber ends, dividing the input between several outputs or combining several inputs into one output.

cover plate mounting bracket A bracket within a rack that allows a plate to cover some type of hardware, such as a patch panel.

crimper, crimping, crimp-on A device that is used to install a crimp-on connector. Crimping involves the act of using the crimping tool to install the connector.

critical angle The smallest angle of incidence at which total internal reflection occurs; that is, at which light passing through a material of a higher refractive index will be reflected off the boundary with a material of a lower refractive index. At lower angles, the light is refracted through the cladding and lost. Due to the fact that the angle of reflection equals the angle of incidence, total internal reflection assures that the wave will be propagated down the length of the fiber. The angle is measured from a line perpendicular to the boundary between the two materials, known as normal.

cross-connect A facility enabling the termination of cables as well as their interconnection or cross-connection with other cabling or equipment. Also known as a punch-down or distributor. In a copper-based system, jumper wires or patch cables are used to make connections. In a fiber-optic system, fiber-optic jumper cables are used.

cross-connection A connection scheme between cabling runs, subsystems, and equipment using patch cords or jumpers that attach to connecting hardware at each end.

crosstalk The coupling or transfer of unwanted signals from one pair within a cable to another pair. Crosstalk can be measured at the same (near) end or far end with respect to the signal source. Crosstalk is considered noise or interference and is expressed in decibels.

current The flow of electrons in a conductor.

customer premises The buildings, offices, and other structures under the control of an end user or customer.

cutoff wavelength For a single-mode fiber, the wavelength above which the operation switches from multimode to single-mode propagation. This is the longest wavelength at which a single-mode fiber can transmit two modes. At shorter wavelengths the fiber fails to function as a single-mode fiber.

cycles per second The count of oscillations in a wave. One cycle per second equals a hertz.

D

dark fiber An unused fiber; a fiber carrying no light. Common when extra fiber capacity is installed.

data communication equipment (DCE) With respect to data transmission, the interface that is used by a modem to communicate with a computer.

data packet The smallest unit of data sent over a network. A packet includes a header with addressing information, and the data itself.

data rate The maximum rate (in bits per second or some multiple thereof) at which data is transmitted in a data transmission link. The data rate may or may not be equal to the baud rate.

decibel (dB) A standard unit used to express a relative measurement of signal strength or to express gain or loss.

demarcation point ("dee-marc") A point where the operational control or ownership changes, such as the point of interconnection between telephone company facilities and a user's building or residence.

demodulate To retrieve a signal from a carrier and convert it into a usable form.

dielectric Material that does not conduct electricity, such as nonmetallic materials that are used for cable insulation and jackets. Optical fiber cables are made of dielectric material.

differential group delay (DGD) The total difference in travel time between the two polarization states of light traveling through an optical fiber. This time is usually measured in picoseconds and can differ depending on specific conditions within an optical fiber and the polarization state of the light passing through it.

differential modal dispersion (DMD) The test method used to determine the variation in arrival time of modes in a pulse of light in an optical fiber (typically for multimode fiber).

digital Refers to transmission, processing, and storage of data by representing the data in binary values (two states: on or off). On is represented by the number 1 and off by the number 0. Data transmitted or stored with digital technology is expressed as a string of 0s and 1s. Digital signals are used to communicate information between computers or computer-controlled hardware.

digital signal processor (DSP) A device that manipulates or processes data that has been converted from analog to digital form.

direct burial A method of placing a suitable cable directly in the ground.

directional coupler A device that samples or tests data traveling in one direction only.

dispersion The spreading of light as it travels away from its source.

dispersion shifted fiber A single-mode optical fiber that is sometimes called a zero-dispersion-shifted fiber. It is manufactured so that the zero dispersion point is shifted from 1300nm to 1550nm, where the attenuation is lowest.

distributed feedback (DFB) laser A semiconductor laser specially designed to produce a narrow spectral output for long-distance fiber-optic communications.

distribution subsystem A basic element of a structured distribution system. The distribution subsystem is responsible for terminating equipment and running the cables between equipment and cross-connects. Also called distribution frame subsystem.

distribution cable An optical fiber cable used "behind-the-shelf" of optical fiber patch panels; typically composed of 900 micron tight-buffered optical fibers supported by aramid and/or glass-reinforced plastic (GRP).

distributor An optional connection facility, intermediate connection facility, or central connection facility.

dopants Impurities that are deliberately introduced into the materials used to make optical fibers. Dopants are used to control the refractive index of the material for use in the core or the cladding.

download speed *See* downstream.

downstream A term used to describe the number of bits that travel from the Internet service provider to the person accessing broadband.

draw string A small nylon cord inserted into a conduit when the conduit is installed; it assists with pulling cable through the duct.

drop A single length of cable installed between two points.

drop cable A cable that links a drop terminal to a premises terminal.

duct (1) A single enclosed raceway for wires or cable. (2) An enclosure in which air is moved.

duplex (1) A link or circuit that can carry a signal in two directions for transmitting and receiving data. (2) An optical fiber cable or cord carrying two fibers.

duplex cable A two-fiber cable suitable for duplex transmission. Usually two fiber strands surrounded by a common jacket.

duplex transmission Data transmission in both directions, either one direction at a time (half duplex) or both directions simultaneously (full duplex).

dynamic loads Loads, such as tension or pressure, that change over time, usually within a short period.

dynamic range The difference between the maximum and minimum optical input power that an optical receiver can accept.

E

earth A term for ground (not to mention the planet most of us live on).

edge-emitting LED An LED that produces light through an etched opening in the edge of the LED.

electromagnetic field The combined electric and magnetic field caused by electron motion in conductors.

electromagnetic immunity Protection from the interfering or damaging effects of electromagnetic radiation such as radio waves or microwaves.

electromagnetic interference (EMI) Electrical noise generated in copper conductors when electromagnetic fields induce currents. Copper cables, motors, machinery, and other equipment that uses electricity may generate EMI. Copper-based network cabling and equipment are susceptible to EMI and also emit EMI, which results in degradation of the signal. Fiber-optic cables are desirable in environments that have high EMI because they are not susceptible to the effects of EMI.

encircled flux A term used to describe the ratio between the transmitted power at a given distance from the center of the core of an optical fiber and the total power injected into the optical fiber.

end separation (1) The distance between the ends of two joined fibers. The end separation is important because the degree of separation causes an extrinsic loss, depending on the configuration of the connection. (2) The separation of optical fiber ends, usually taking place in a mechanical splice or between two connectors.

end-to-end loss The optical signal loss experienced between the transmitter and the detector due to fiber quality, splices, connectors, and bends.

energy density For radiation. Expressed in joules per square meter. Sometimes called irradiance.

entrance facility (EF) A room in a building where antenna, public, and private network service cables can enter the building and be consolidated. Should be located as close as possible to the entrance point. Entrance facilities are often used to house electrical protection equipment and connecting hardware for the transition between outdoor and indoor cable. Also called an entrance room.

entrance point The location where telecommunications enter a building through an exterior wall, a concrete floor slab, a rigid metal conduit, or an intermediate metal conduit.

epoxy A glue that cures when mixed with a catalyzing agent.

epoxy-less connector A connector that does not require the use of epoxy to hold the fiber in position.

equilibrium mode distribution (EMD) A term used to describe when light traveling through the core of the optical fiber populates the available modes in an orderly way.

equipment cable Cable or cable assembly used to connect telecommunications equipment to horizontal or backbone cabling systems in the telecommunications room and equipment room. Equipment cables are considered to be outside the scope of cabling standards.

equipment cabling subsystem Part of the cabling structure, typically between the distribution frame and the equipment.

equipment outlet *See* telecommunications outlet.

equipment room (ER) A centralized space for telecommunications equipment that serves the occupants of the building or multiple buildings in a campus environment.

Usually considered distinct from a telecommunications closet because it is considered to serve a building or campus; the telecommunications closet serves only a single floor. The equipment room is also considered distinct because of the nature of complexity of the equipment that is contained in it.

Ethernet A local area network (LAN) architecture developed by Xerox that is defined in the IEEE 802.3 standard. Ethernet nodes access the network using the Carrier Sense Multiple Access/Collision Detect (CSMA/CD) access method.

expanded beam *See* lensed ferrule.

extrinsic attenuation *See* extrinsic loss.

extrinsic factors *See* extrinsic loss.

extrinsic loss When describing a fiber-optic connection, the portion of loss that is not intrinsic to the fiber but is related to imperfect joining, which may be caused by the connector or splice, as opposed to conditions in the optical fiber itself. The conditions causing this type of loss are referred to as extrinsic factors. These factors are defects and imperfections that cause the loss to exceed the theoretical minimum loss due to the fiber itself (which is called intrinsic loss).

F

f *See* frequency (f).

Fabry-Pérot A type of laser used in fiber-optic transmission. The Fabry-Pérot laser typically has a spectral width of about 5nm.

faceplate A plate used in front of a telecommunications outlet.

fanout kit A collection of components used to add tight buffers to optical fibers from a loose-tube buffer cable. A fanout kit typically consists of a furcation unit and measured lengths of tight buffer material. It is used to build up the outer diameter of fiber cable for connectorization.

FC connector A threaded optical fiber connector that was developed by Nippon Telephone and Telegraph in Japan. The FC connector is good for single-mode or multimode fiber and applications requiring low back reflection. The FC is a screw type and is prone to vibration loosening.

Federal Communications Commission (FCC) The federal agency responsible for regulating broadcast and electronic communications in the United States.

ferrule A small alignment tube attached to the end of the fiber and used in connectors. These are made of stainless steel, aluminum, zirconia, or plastic. The ferrule is used to confine and align the stripped end of a fiber so that it can be positioned accurately.

fiber A single, separate optical transmission element characterized by core and cladding. The fiber is the material that guides light or waveguides.

fiber identifier A testing device that displays the direction of travel of light within an optical fiber by introducing a macrobend and analyzing the light that escapes the fiber.

fiber protrusion A term used to describe the distance the optical fiber endface is above the connector endface.

fiber-optic cable Cable containing one or more optical fibers.

fiber optics The optical technology in which communication signals in the form of modulated light beams are transmitted over a glass or plastic fiber transmission medium. Fiber optics offers high bandwidth and protection from electromagnetic interference and radioactivity; it also has small space needs.

fiber undercut A term used to describe the distance the optical fiber endface is below the connector endface.

figure-8 A fiber-optic cable with a strong supporting member incorporated for use in aerial installations. *See also* messenger cable.

fill ratio A term used to describe the percentage of conduit filled by cabling.

fillers Nonconducting components cabled with insulated conductors or optical fibers to impart flexibility, tensile strength, roundness, or a combination of all three.

firestop Material, device, or collection of parts installed in a cable pathway (such as a conduit or riser) at a fire-rated wall or floor to prevent passage of flame, smoke, or gases through the rated barrier.

fish tape (or fish cord) A tool used by electricians to route new wiring through walls and electrical conduit.

floating A floating circuit is one that has no ground connection.

floor distributor (FD) The ISO/IEC 11801 term for horizontal cross-connect. The floor distributor is used to connect between the horizontal cable and other cabling subsystems or equipment.

forward bias A term used to describe when a positive voltage is applied to the p region and a negative voltage to the n region of a semiconductor.

four-wave mixing The creation of new light wavelengths from the interaction of two or more wavelengths being transmitted at the same time within a few nanometers of each other. Four-wave mixing is named for the fact that two wavelengths interacting with each other will produce two new wavelengths, causing distortion in the signals being transmitted. As more wavelengths interact, the number of new wavelengths increases exponentially.

frequency (f) The number of cycles per second at which a waveform alternates—that is, at which corresponding parts of successive waves pass the same point. Frequency is expressed in hertz (Hz); one hertz equals one cycle per second.

Fresnel reflection Reflection of a small amount of light passing from a medium of one refractive index into a medium of another refractive index. In optical fibers, reflection occurs at the air/glass interfaces at entrance and exit ends.

full-duplex A system in which signals may be transmitted in two directions at the same time.

full-duplex transmission Data transmission over a circuit capable of transmitting in both directions simultaneously.

furcation unit A component used in breakout kits and fanout kits for separating individual optical fibers from a cable and securing tight buffers and/or jackets around the fibers.

fusion splicing A splicing method accomplished by the application of localized heat sufficient to fuse or melt the ends of the optical fiber, forming a continuous single strand of fiber. As the glass is heated it becomes softer, and it is possible to use the glass's "liquid" properties to bond glass surfaces permanently.

G

G Green. Used when referring to color-coding of cables.

gain An increase in power.

gainer A fusion splice displayed in an OTDR trace that appears to have gain instead of loss.

gigahertz (GHz) A billion hertz or cycles per second.

graded-index fiber An optical in which the index of refraction of the core is lower toward the outside of the core and progressively increases toward the center of the core, thereby reducing modal dispersion of the signal.

ground The National Electrical Code (NEC) defines the term ground as the earth.

grounding The National Electrical Code (NEC) defines the term grounding as connecting to ground or to a conductive body that extends the ground connection.

H

hackles Irregular defects on a cleaved fiber-optic endface are commonly referred to as hackles.

half-duplex A system in which signals may be sent in two directions, but not at the same time. In a half-duplex system, one end of the link must finish transmitting before the other end may begin.

half-duplex transmission Data transmission over a circuit capable of transmitting in either direction. Transmission can be bidirectional but not simultaneously.

hard-clad silica fiber An optical fiber with a hard plastic cladding surrounding a step-index silica core.

hertz (Hz) A measurement of frequency defined as cycles per second.

heterojunction structure An LED design in which the pn (P-type and N-type semiconductor) junction is formed from similar materials that have different refractive indices. This design is used to guide the light for directional output.

hierarchical star topology A topology standardized in ANSI/TIA-568-C whereby all telecommunications outlets are connected to a centrally located point typically in the main equipment room.

high-order Light rays traveling through the core of an optical fiber with a high number of reflections.

homojunction structure An LED design in which the pn (P-type and N-type semiconductor) junction is formed from a single semiconductor material.

horizontal cabling The cabling between and including the telecommunications outlet and the horizontal cross-connect. Horizontal cabling is considered the permanent portion of a link; may also be called horizontal wiring.

horizontal cross-connect (HC) A cross-connect that connects the cabling of the work area outlets to any other cabling system (like that for LAN equipment, voice equipment).

hybrid adapter A mating or alignment sleeve that mates two different connector types such as an SC and ST.

I

I Symbol used to designate current.

incident angle The angle between the subject light wave and a plane perpendicular to the subject optical surface.

incoherent light Light in which the electric and magnetic fields of photons are completely random in orientation. Incoherent light is typically emitted from lightbulbs and LEDs.

index matching gel A clear gel or fluid used between optical fibers that are likely to have their ends separated by a small amount of air space. The gel matches the refractive index of the optical fiber, reducing light loss due to Fresnel reflection.

index of refraction (1) The ratio of the speed of light in a vacuum to the speed of light in a given transmission medium. This is usually abbreviated *n*. (2) The value given to a medium to indicate the velocity of light passing through it relative to the speed of light in a vacuum.

indoor-outdoor Cable designed for installation in the building interior or exterior to the building.

infrared The infrared spectrum consists of wavelengths that are longer than 700nm but shorter than 1mm. Humans cannot see infrared radiation, but we feel it as heat. The commonly used wavelengths for transmission through optical fibers are in the infrared at wavelengths between 1100nm and 1600nm.

innerduct A separate duct running within a larger duct to carry fiber-optic cables.

insertion loss Optical power loss caused by the insertion of a component into the fiber-optic system.

inside plant (IP) (1) Cables that are the portion of the cable network that is inside buildings, where cable lengths are usually shorter than 100 meters. This is the opposite of outside-the-plant (OP or OSP) cables. (2) A telecommunications infrastructure designed for installation within a structure.

inspection microscope A microscope designed to evaluate the endface of a fiber-optic connector.

installation load The short-term tensile load that may be placed on a fiber-optic cable.

interbuilding backbone A telecommunications cable that is part of the campus subsystem that connects one building to another.

interconnect A circuit administration point, other than a cross-connect or an information outlet, that provides capability for routing and rerouting circuits. It does not use patch cords or jumper wires and is typically a jack-and-plug device that is used in smaller distribution arrangements or connects circuits in large cables to those in smaller cables.

interconnect cabinet Cabinets containing connector panels, patch panels, connectors, and patch cords to interface from inside the plant to outside the plant. The interconnect cabinet is used as an access point for testing and rearranging routes and connections.

interconnection A connection scheme that provides direct access to the cabling infrastructure and the capability to make cabling system changes using patch cords.

interlock armor A helical armor applied around a cable where the metal tape sides interlock giving a flexible, yet mechanically robust cable; specifically adds to the compression resistance in optical fiber cables and acts as an EMI/RFI shield in twisted-pair copper cables. The interlock armor can be made of either aluminum (more common) or steel.

intermediate cross-connect (ICC) A cross-connect between first-level and second-level backbone cabling. This secondary cross-connect in the backbone cabling is used to mechanically terminate and administer backbone cabling between the main cross-connect and horizontal cross-connect (station cables).

intermediate distribution frame (IDF) A metal rack (or frame) designed to hold the cables that connect interbuilding and intrabuilding cabling. The IDF is typically located in an equipment room or telecommunications room. Typically, a permanent connection exists between the IDF and the MDF.

Internet service provider (ISP) A service provider or organization that offers Internet access to a user.

interoffice facility (IOF) A communication channel of copper, fiber, or wireless media between two central offices. Often, IOF refers to telephone channels that can transport voice and/or data.

intrabeam viewing conditions Intrabeam viewing conditions exist when the eye is exposed to the direct or specularly reflected laser beam.

intrabuilding backbone Telecommunications cables that are part of the building subsystem that connect one telecommunications closet to another or a telecommunications room to the equipment room.

intrinsic absorption A term that describes a process in which photons absorbed by the photodiode excite electrons within the photodiode.

intrinsic attenuation Attenuation caused by the scattering or absorption of light in an optical fiber.

intrinsic factors When describing an optical fiber connection, factors contributing to attenuation that are determined by the condition of the optical fiber itself. *See also* extrinsic loss.

isolated ground A separate ground conductor that is insulated from the equipment or building ground.

J

jacket The outer protective covering of a cable, usually made of some type of plastic or polymer.

jumper In fiber optics, a simplex cable assembly that is typically made from simplex cordage.

K

Kevlar A strong, synthetic material developed and trademarked by DuPont; the preferred strength element in cable. Also used as a material in body armor and parts for military equipment. Also known by the generic name aramid; *see also* aramid strength member.

keying A mechanical feature of a connector system that guarantees correct orientation of a connection. The key prevents connection to a jack or an optical fiber adapter that was intended for another purpose.

kHz Kilohertz; 1,000 hertz.

KPSI KiloPSI. A unit of tensile strength expressed in thousands of pounds per square inch.

L

ladder rack A rack used to hold cabling that looks like a ladder.

laser Acronym for light amplification by stimulated emission of radiation. The laser produces a coherent source of light with a narrow beam and a narrow spectral bandwidth (about 2nm). Lasers in fiber optics are usually solid-state semiconductor types. Lasers are used to provide the high-powered, tightly controlled light wavelengths necessary for high-speed, long-distance optical fiber transmissions.

laser chirp When the laser output wavelength changes as the electron density in the semiconductor material changes.

laser diode (LD) A semiconductor that emits laser light when a specific amount of current is applied. Laser diodes are typically used in higher speed applications up to 10Gbps.

lateral misalignment When the core of the transmitting optical fiber is laterally offset from the core of the receiving optical fiber.

launch cable A cable used to measure insertion loss with an OTDR at the near end of the cable plant. Also referred to as launch fiber, pulse suppression cable, dead zone cable, test fiber box, or access jumper.

launch fiber An optical fiber used to measure insertion loss with an OTDR at the near end of the cable plant. Also referred to as launch cable, pulse suppression cable, dead zone cable, test fiber box, or access jumper.

lbf Abbreviation for pounds force.

LC A small form factor connector developed by Lucent Technologies. This snap-in Lucent Connector (LC) is considered to be a smaller version of the SC connector and is sometimes referred to as a mini-SC. The small form factor connector has a 1.25mm ferrule diameter, half the diameter of an SC connector ferrule.

lensed ferrule A ferrule with a lens on the endface, commonly known as an *expanded beam*.

light The electromagnetic radiation visible to the human eye between 400nm and 700nm. The term is also applied to electromagnetic radiation with properties similar to visible light; this includes the invisible near-infrared radiation in most fiber-optic communication systems.

light amplification by stimulated emission of radiation See laser.

light-emitting diode (LED) A semiconductor light source used in a transmitter to convert an electrical signal into light energy to be carried through the optical fiber.

limiting amplifier An amplifier that amplifies the transimpedance amplifier output voltage pulses and provides a binary decision. It determines whether the electrical pulses received represent a binary 1 or a binary 0.

link A transmission pathway between two points using some kind of generic cable. The pathway includes a means to couple the signal to the cable and a way to receive it at the other end in a useful way.

listed Equipment included on a list published by an organization, acceptable to the authority having jurisdiction, that maintains periodic inspection of production of listed equipment, and whose listing states either that the equipment or material meets appropriate standards, or that it has been tested and found suitable for use in a specified manner. In the United States, electrical and data communications equipment is typically listed with Underwriters Laboratories (UL).

local area network (LAN) A network connecting multiple nodes within a defined area, usually within a building. The linking can be wireless or done by cable that carries optical fiber or copper.

local convergence point (LCP) An access point in a passive optical network where the feeder cables are broken out into multiple distribution cables.

loose buffer *See* loose-tube buffer.

loose tube *See* loose-tube buffer.

loose-tube buffer Optical fiber that is carried loosely in a buffer many times the diameter of the fiber. Loose-buffered fiber is typically terminated with a breakout kit or a fanout kit and connected to a patch panel. Also known as *loose buffer* or *loose tube*.

Loose tube gel-filled (LTGF) cable A fiber-optic cable where the loose buffer tubes are filled with a gel that displaces or blocks water and prevents it from penetrating or getting into the cable.

loose tube ribbon cable A cable where fiber ribbons are stacked on top of one another inside a loose-buffered tube.

loss *See* attenuation.

loss budget *See* power budget.

low-order A light ray that experiences a low number of reflections as it travels through the core of an optical fiber.

M

macrobend A major bend in an optical fiber with a radius small enough to change the angle of incidence and allow light to pass through the interface between the core and the cladding rather than reflect off it. Macrobends can cause signal attenuation or loss by allowing light to leave the optical fiber core.

macrobending loss Optical power loss due to bends in the fiber.

main cross-connect A cross-connect for first-level backbone cables, entrance cables, and equipment cables. The main cross-connect is at the top level of the premises cabling tree.

mandrel wrap A device that removes the high-order modes by inserting a series of macro-bends in the transmit test jumper.

material dispersion The spreading of a light pulse caused by different wavelengths of light traveling at different velocities in the optical fiber.

mating sleeve A sleeve used to align two optical fiber ferrules; also known as an alignment sleeve.

maximum tensile rating A manufacturer's specified limit on the amount of tension, or pulling force, that may be applied to a fiber-optic cable.

measurement quality jumper (MQJ) A term used by the U.S. Navy to describe a test jumper.

mechanical splice A splice in which optical fibers are joined mechanically (e.g., glued, crimped, or otherwise held in place).

media *See* medium.

medium Any material or space through which electromagnetic radiation can travel.

messenger cable A cable with a strong supporting member attached to it for use in aerial installations. *See also* figure-8.

metropolitan area network (MAN) An interconnected group of local area networks (LANs) that encompasses an entire city or metropolitan area.

microbend Small distortions of the boundary layer between the core and cladding caused by crushing or pressure. Microbends are very small and may not be visible when looking at the fiber-optic cable.

micrometer Also referred to as a micron; one millionth of a meter, often abbreviated with the symbol μ .

minimum bend radius A fiber-optic cable manufacturer's specified limit on the amount of bending that the cable can withstand without damage.

modal dispersion A type of dispersion or spreading that arises from differences in the amount of time that different modes take to travel through multimode fibers. Modal dispersion potentially can cause parts of a signal to arrive in a different order from the one in which they were transmitted.

mode One of the most important characteristics used to distinguish types of optical fiber is the number of potential paths light can take through it. In fiber optics, the terms "mode" and "path" are often used interchangeably. For example, ANSI/TIA-568-C.3 defines mode as "a path of light in an optical fiber." ARINC Report 805 defines mode as "a term used to describe a light path through a fiber, as in multimode or single-mode."

mode field diameter (MFD) The actual diameter of the light beam traveling through the core and part of the cladding in a single-mode fiber-optic cable. Since the mode field diameter is usually greater than the core diameter, the mode field diameter replaces the core diameter as a practical parameter.

mode filter *See* mandrel wrap.

modulate To convert data into a signal that can be transmitted by a carrier.

modulated A carrier such as light that is varied to transmit information.

MT-RJ connector A duplex fiber-optic connector that looks similar to the RJ-45-type connector.

multimode Transmission of multiple modes of light. *See also* mode.

multimode depressed clad optical fiber *See* bend insensitive.

multimode fiber Optical fiber whose core diameter is large enough to allow light to take more than one possible path through it.

multimode graded-index fiber An optical fiber with a core that consists of many concentric layers with refractive indexes that decrease with the distance from the center. Viewed on end, these layers resemble the rings of a tree.

multimode step-index fiber Also known simply as step-index fiber; multimode step-index fiber has a core with a single refractive index and a cladding with a slightly lower refractive index.

N

National Electrical Code (NEC) The National Electrical Code (NEC) is published by the National Fire Protection Association (NFPA) and contains provisions that are considered necessary for safety. NEC Article 90 states that the purpose of the code is the practical safeguarding of persons and property from hazards arising from the use of electricity. It also states that the NEC is not intended as a design specification or an instruction manual for untrained persons. Local and state governments typically adopt it, making compliance mandatory.

necking A term used to describe a fusion splice where the diameter of the fused optical fiber is smaller near the electrodes than it was prior to being fused.

noise In a cable or circuit, any extraneous signal (electromagnetic energy) that interferes with the desired signal normally present in or passing through the system.

non-current-carrying conductive members Conductive members of a cable such as metallic strength members, metallic vapor barriers, metallic armor, or a metallic sheath whose primary function is not to carry electrical current.

non-zero-dispersion-shifted fiber A type of single-mode optical fiber designed to reduce the effects of chromatic dispersion while minimizing four-wave mixing.

normal A path drawn perpendicular to the *interface*, or a boundary layer between two media, that is used to determine the *angle of incidence* of light reaching the interface.

normal angle The angle that is perpendicular to a surface.

numerical aperture (NA) The light-gathering ability of a fiber, defining the maximum angle to the fiber axis at which light will be accepted and propagated through the fiber. The numerical aperture is determined by the refractive indexes of the core and cladding. The numerical aperture is also used to determine the fiber's acceptance angle.

O

Occupational Safety and Health Administration (OSHA) An organization that is part of the United States Department of Labor whose mission is to assure safe and healthful working conditions for working men and women by setting and enforcing standards and by providing training, outreach, education, and assistance.

operational load *See* static load.

optical amplifier Increases the power of an optical signal without converting any of the signals from optical to electrical energy and then back to optical so that the amplification processes the optical signal wholly within optical amplification equipment. The two most common optical amplifiers are semiconductor laser amplifiers and those made from doped fiber, such as the EDFA (erbium doped fiber amplifier), which amplifies with a laser pump diode and a section of erbium doped fiber.

optical attenuator Reduces the intensity of light waves, usually so that the power is within the capacity of the detector. There are three basic forms of attenuators: fixed optical attenuators, stepwise variable optical attenuators, and continuous variable optical attenuators. Attenuation is normally achieved either by a doped fiber or an offset or core misalignment. *See also* attenuator.

optical loss test set An optical power meter and a light source calibrated for use together to detect and measure loss of signal on an optical cable.

optical power meter A meter designed to measure absolute optical power.

optical return loss (ORL) A ratio expressed in decibels that is used to define the amount of light energy that is reflected back to the transmit end of the fiber-optic cable.

optical splitter A device used to split fiber-optic signals.

optical subassembly The portion of a fiber-optic receiver that guides light from the optical fiber to the photodiode.

optical time-domain reflectometer (OTDR) A device used to test a fiber-optic link, including the optical fiber and connectors, by launching an optical signal (light pulse) through the link and measuring the amount of energy that is reflected back.

optical trench A term used to describe a layer of glass that surrounds the core of the optical fiber. It has a lower refractive index than the core or the cladding. The low refractive index acts as a barrier and prevents light from escaping the core of the optical fiber when it is bent, reducing the amount of attenuation.

outside plant A telecommunications infrastructure designed for installation outside of any structure.

overfilled launch The process of an LED transmitter overfilling a multimode optical fiber.

P

passive alignment fusion splicer A fusion splice where precision-machined V-grooves are used to align the cladding of the optical fibers prior to splicing. This type of fusion splicer may also be referred to as a passive v-groove fusion splicer.

passive optical network An optical network that does not use electrically powered equipment or components excluding the transmitter to get the signal from one place to another.

passive v-groove fusion splicer *See* passive alignment fusion splicer.

patch cable Any flexible piece of cable that connects one network device to the main cable run or to a patch panel that in turn connects to the main cable run; also called *patch cord*. Used for interconnecting circuits on a patch panel or cross-connect. Patch cables are short distance, usually have connectors preinstalled on both ends, are used to connect equipment, and are generally between 3 and 6 meters long.

patch panel A connecting hardware that typically provides a means to connect horizontal or backbone cables to an arrangement of fixed connectors that may be accessed using patch cords or equipment cords to form cross-connections or interconnections. Patch panels may connect either copper or optical fiber cables.

pathway A facility (e.g., conduit, cable tray, raceway, ducting, or plenum) for the placement and protection of telecommunications cables.

PC finish A convex curve connector endface geometry where the highest feature or apex on the endface is the center of the optical fiber end.

pedestal A device, usually mounted on the floor or ground, which is used to house voice/data jacks or power outlets at the point of use. Also commonly referred to as a monument, tombstone, above-floor fitting, or doghouse.

pedestal enclosure *See* pedestal.

Personal Protective Equipment (PPE) Anything that you would wear to protect yourself from materials or situations. It can include protective gloves and eyewear for cutting and grinding operations, respirators for working with chemicals that put out harmful vapors, and specialized goggles for working with lasers.

photodiode A semiconductor diode that when stimulated by light energy, generates electrical current flow through a process called intrinsic absorption.

photon A basic unit of light; the smallest quantum particulate component of light.

physical contact (PC) Description for a connector that places an optical fiber endface in direct physical contact with the optical fiber endface of another connector. The endface geometry is typically a convex curve.

picosecond (PS) One trillionth of a second.

pigtail (1) A short length of fiber with a permanently attached device, usually a connector, on one end. (2) A fiber-optic cable assembly consisting of a connector or hardware device (such as a light source package installed by a manufacturer) on one end and an unterminated fiber at the other end.

PIN photodiode A photodiode that works like a PN photodiode; however, it is manufactured to offer better performance.

plastic-clad silica (PCS) fiber A step-index multimode fiber that has a silica core and is surrounded by a lower index plastic cladding.

plastic fiber Optical fiber having a plastic core and plastic cladding rather than using glass.

plenum The air-handling space between the walls, under structural floors, and above drop ceilings when used to circulate and otherwise handle air in a building. Plenum-grade cable can be run through these spaces if local building codes permit it.

plenum cable Cable whose flammability and smoke characteristics allow it to be routed in plenum spaces without being enclosed in a conduit.

plug The ferrule of a fiber-optic connector is commonly referred to as plug.

PN diode A basic photodiode.

polarity In an electrical circuit, it identifies which side is positive and which is negative. In a fiber-optic network, it is the positioning of connectors and adapters to ensure that there is an end-to-end transmission path between the transmitter and the receiver of a channel.

polarization maintaining fiber Optical fiber that maintains the polarization of light that enters it.

polarization mode The orientation of an electrical or magnetic wave along the path of propagation.

polarization mode dispersion Spreading of the light wave caused by imperfections in a single-mode optical fiber that slow down a polarization mode of the signal. When one polarization mode lags behind another, the signal spreads out and can become distorted.

polyimide A polymer that is used to coat optical fiber for harsh environment applications.

pot life The amount of time after mixing where epoxy can be used before it begins to harden and must be discarded.

power budget A power budget as defined in IEEE Standard 802.3 is the minimum optical power available to overcome the sum of attenuation plus power penalties of the optical path between the transmitter and receiver. These losses are calculated as the difference between the minimum transmitter launch power and the minimum receive power.

preform A short, thick glass rod that forms the basis for an optical fiber during the manufacturing process. The preform is created first, and then melted and drawn under constant tension to form the long, thin optical fiber.

preload A connector with built-in adhesive that must be preheated before the fiber can be installed.

premises A telecommunications term for the space occupied by a customer or an authorized/joint user in a building on continuous or contiguous property that is not separated by a public road or highway.

primary coating The specialized coating applied to the surface of the fiber cladding during manufacture.

puck In fiber optics, a metal disc that holds a connector in the proper position against an abrasive for polishing the connector endface.

pull load *See* tensile load.

pull strength The pulling force that can be applied to a cable without damaging a cable or affecting the specified characteristics of the cable. Also called pull tension.

pull string A string that is tied to a cable and is used to pull cables through conduits or over racks. Similar to fish tape.

pulse dispersion The dispersion of pulses as they travel along an optical fiber.

pulse spreading The dispersion of an optical signal with time as it propagates through an optical fiber.

pulse suppression cable A cable used to measure insertion loss with an OTDR at the near or far end of the cable plant. Also referred to as launch fiber, launch cable, dead zone cable, test fiber box, or access jumper.

Q

quantizer IC An integrated circuit (IC) that measures received optical energy and interprets each voltage pulse as a binary 1 or 0.

R

raceway Any channel or structure used in a building to support and guide electrical and optical fiber wires or cables. Raceways may be metallic or nonmetallic and may totally or partially enclose the wiring (e.g., conduit, cable trough, cellular floor, electrical metallic tubing, sleeves, slots, under-floor raceways, surface raceways, lighting fixture raceways, wireways, busways, auxiliary gutters, and ventilated flexible cableways). *See also* pathway.

rack A frame-like structure where patch panels, switches, and other network equipment are installed. The typical dimension is 19 inches.

radius of curvature A term used to describe the roundness of the connector endface, measured from the center axis of the connector ferrule.

ray A beam of light in a single direction. Usually a representation of light traveling in a particular direction through a particular medium.

Rayleigh scattering The redirection of light caused by atomic structures and particles along the light's path. Rayleigh scattering is responsible for some attenuation in optical fiber, because the scattered light is typically absorbed when it passes into the cladding.

receive cable A cable used to measure insertion loss with an OTDR at the far end of the cable plant. Also referred to as receive fiber, pulse suppression cable, dead zone cable, test fiber box, or access jumper.

receive fiber An optical fiber used to measure insertion loss with an OTDR at the far end of the cable plant. Also referred to as receive cable, pulse suppression cable, dead zone cable, test fiber box, or access jumper.

receive jumper The test jumper connected to the optical power meter.

receiver In fiber-optic systems, an electronic component that converts light energy to electrical energy.

receptacle The part of a fiber-optic receiver that accepts a connector and aligns the ferrule for proper optical transmission.

reference jumper *See* test jumper.

reflectance A percentage that represents the amount of light that is reflected back along the path of transmission from the coupling region, the connector, or a terminated fiber.

refraction The bending of a beam of light as it enters a medium of different density. Refraction occurs as the velocity of the light changes between materials of two different refractive indexes.

refractive index profile A graphical description of the relationship between the refractive indexes of the core and the cladding in an optical fiber.

resistor A passive component in an electrical circuit that restricts the flow of current.

responsivity (1) The ratio of a detector's output to input, usually measured in units of amperes per watt (or microamperes per microwatt). (2) The measure of how well a photodiode converts a wavelength or range of wavelengths of optical energy into electrical current.

restricted mode launch (RML) A launch that limits a laser's launch condition so that fewer modes are used.

return loss As defined in ANSI/TIA-568-C.3, the ratio of the power of the outgoing optical signal to the power of the reflected signal. This ratio is expressed in decibels (dB).

return reflection Light energy that is reflected from the end of a fiber through Fresnel reflection.

reversed biased A term used to describe when the *n* region of the semiconductor is connected to a positive electrical potential and the *p* region is connected to a negative electrical potential.

ripcord A length of string built into optical fiber cables that is pulled to split the outer jacket of the cable without using a blade.

riser (1) A designation for a type of cable run between floors' Fire-code rating for indoor cable that is certified to pass through the vertical shaft from floor to floor. (2) A space for indoor cables that allow cables to pass between floors, normally a vertical shaft or space.

riser cable A type of cable used in vertical building shafts, such as telecommunications and utility shafts. Riser cable typically has more mechanical strength than general use cable and has an intermediate fire protection rating.

S

sample-and-hold circuit A circuit that samples an analog signal such as a voltage level and then holds the voltage level long enough for the analog-to-digital (A/D) converter to change the level to a numerical value.

SC connector An optical-fiber connector with a 2.5mm ferrule diameter and a push-pull latching mechanism.

scattering (1) A property of glass that causes light to deflect from the fiber and contributes to losses. (2) The redirection of light caused by atomic structures and particles along the light's path. *See also* Rayleigh scattering.

score To lightly nick the optical fiber with a blade.

scribe A tool used to score an optical fiber.

secondary coating The acrylate coating layer applied over the primary coating that provides a hard outer surface.

sequential markings Markings on the outside of a fiber-optic cable to aid in measuring the length of the cable.

serial When applied to digital data transmission, it means that the binary bits are sent one after another in the order they were generated.

sheath An outer protective layer of a fiber-optic or copper cable that includes the cable jacket, strength members, and shielding.

Silicone A General Electric trademark for a material made from silicon and oxygen. Can be in thermosetting elastomer or liquid form. The thermosetting elastomer form is noted for high heat resistance.

simplex (1) A link that can only carry a signal in one direction. (2) A fiber-optic cable or cord carrying a single fiber. Simplex cordage is mainly used for patch cords and temporary installations.

simplex cable A term sometimes used to describe a single-fiber cable.

simplex transmission Data transmission over a circuit capable of transmitting in only one direction.

single-board computer A circuit board containing the components needed for a computer that performs a prescribed task. A single-board computer is often the basis of a larger piece of equipment that contains it.

single-mode Transmission of one mode of light. *See also* mode.

single-mode depressed clad optical fiber *See* bend insensitive.

single-mode step-index fiber *See* step-index single-mode fiber.

sintering A process in optical fiber manufacturing in which the soot created by heating silicon dioxide is compressed into glass to make the fiber preform.

slash sheet Within SAE International, a family of documents where the main document refers to the overall/general aspect and the slash sheet refers to specific items.

solar cell A device used to convert light energy into electrical current.

spectral width A measure of the extent of a spectrum. The range of wavelengths within a light source. For a source, the width of wavelengths contained in the output at one half of the wavelength of peak power. Typical spectral widths are between 20nm and 170nm for an LED and between 1nm and 5nm for a laser diode.

speed of light (c) In a vacuum, light travels 299,800,000 meters per second. This is used as a reference for calculating the index of refraction.

splice (1) A permanent joint between two optical fibers. (2) Fusing or mechanical means for joining two fiber ends.

splice enclosure A cabinet used to organize and protect splices.

splice tray A container used to organize and protect spliced fibers.

splitting ratio The ratio of power emerging from two output ports of a coupler.

spontaneous emission The emission of random photons (incoherent light) at the junction of the p and n regions in a light-emitting diode when current flows through it. Occurs when a semiconductor accumulates spurious electrons. Spontaneous emission interferes with coherent transmission.

ST connector A fiber-optic connector with a bayonet housing; it was developed by AT&T and has ferrule diameter of 2.5mm.

stabilized light source An LED or laser diode that emits light with a controlled and constant spectral width, central wavelength, and peak power with respect to time and temperature.

standards Mutually agreed-upon principles of protocol or procedure. Standards are set by committees working under various trade and international organizations.

static load Load such as tension or pressure that remains constant over time, such as the weight of a fiber-optic cable in a vertical run.

step-index fiber An optical fiber that has a core with a single refractive index and a cladding with another, slightly lower refractive index.

step-index single-mode fiber A single-mode optical fiber that has a core with a single refractive index and a cladding with another, slightly lower refractive index.

stimulated emission The process in which a photon interacting with an electron triggers the emission of a second photon with the same phase and direction as the first. Stimulated emission is the basis of a laser.

strength member The part of a fiber-optic cable composed of aramid yarn, steel strands, or fiberglass filaments that increases the tensile strength of the cable.

submarine cable A cable designed to be laid underwater.

surface-emitting LED An LED in which incoherent light is emitted at all points along the *pn* junction.

T

telecommunications outlet A fixed connecting device where the horizontal cable terminates that provides the interface to the work area cabling. Typically found on the floor or in the wall. Sometimes referred to as a telecommunications outlet/connector or a wall plate.

telecommunications closet *See* telecommunications room.

telecommunications room An enclosed space for housing telecommunications equipment, cable terminations, and cross-connect cabling used to serve work areas located on the same floor. The telecommunications room is the typical location of the horizontal cross-connect and is distinct from an equipment room because it is considered to be a floor-serving (as opposed to building- or campus-serving) facility.

tensile load The amount of pulling force placed on a cable.

tensile strength Resistance to pulling or stretching forces.

terminate To install a connector or terminus at the end of an optical fiber.

termini A term used to describe multiple fiber-optic contacts.

terminus A single fiber-optic contact.

test fiber box A box that contains optical fiber and is used as a launch or receive cable when testing the cable plant with an OTDR.

test jumper A single- or multi-fiber cable used for connections between an optical fiber and test equipment.

tight buffer A type of optical fiber cable construction where a 900µm in diameter buffer is applied directly to the 250µm in diameter outer coating layer of an optical fiber.

tight-buffered *See* tight buffer.

timing circuit In an optical time-domain reflectometer, the circuit used to coordinate and regulate the testing process.

total attenuation The loss of light energy due to the combined effects of scattering and absorption.

total internal reflection The reflection of light in a medium of a given refractive index off of the interface with a material of a lower refractive index at an angle at or above the critical angle. Total internal reflection occurs at the core/cladding interface within an optical fiber cable.

tractor A machine used to pull a preform into an optical fiber using constant tension.

transceiver A device in which the fiber-optic transmitter and receiver are combined into a single package.

transimpedance amplifier In a fiber-optic receiver, a device that receives electrical current from the photodiode and amplifies it before sending it to the *quantizer IC*.

transmit jumper The test jumper connected to the light source.

transmitter A device that converts an electrical signal into light energy to be carried through the optical fiber. The signal can be generated by many sources, such as a computer, a voice over a telephone, or data from an industrial sensor.

U

UL Listed If a product carries this mark, it means UL found that representative samples of this product met UL's safety requirements.

UL Recognized If a product carries this mark, it means UL found it acceptable for use in a complete UL Listed product.

underground cable Cable that is designed to be placed beneath the surface of the ground in ducts or conduit. Underground cable is not necessarily intended for direct burial in the ground.

upload speed See upstream.

upstream A term used to describes the number of bits being sent to the Internet service provider; often referred to as *upload speed*.

V

vertical-cavity surface-emitting laser (VCSEL) A type of laser that emits coherent energy along an axis perpendicular to the *pn* junction.

visible light Electromagnetic radiation visible to the human eye at wavelengths between 400nm and 700nm.

visual fault locator (VFL) A testing device consisting of a red laser that fills the fiber core with light, allowing a technician to find problems such as breaks and macrobends by observing the light through the buffer, and sometimes the jacket, of the cable.

W

wall plate (or wall jack) A wall plate is a flat plastic or metal that usually mounts in or on a wall. Wall plates include one or more jacks.

watt A unit of power.

waveguide A structure that guides electromagnetic waves along their length. The core fiber in an optical fiber cable is an optical waveguide.

waveguide dispersion That part of the chromatic dispersion (spreading) that occurs in a single-mode fiber as some of the light passes through the cladding and travels at a higher velocity than the signal in the core, due to the cladding's lower refractive index. For the most part, this deals with the fiber as a waveguide structure. Waveguide dispersion is one component of chromatic dispersion.

wavelength The distance between corresponding points on two consecutive waves. In fiber optics, wavelength (λ) is typically used to describe the light source.

wavelength division multiplexing (WDM) A method of carrying multiple channels through an optical fiber at the same time (multiplexing) whereby signals within a small spectral range are transmitted at different wavelengths through the same optical fiber.

Wi-Fi The technology that allows an electronic device to exchange data or connect to the Internet wirelessly using radio waves. Also known as a wireless LAN.

Z

zero-dispersion point In an optical fiber of a given refractive index, the narrow range of wavelengths within which all wavelengths travel at approximately the same speed. The zero-dispersion point is useful in reducing chromatic dispersion in single-mode fiber.

zero-dispersion-shifted-fiber *See* dispersion shifted fiber.

zero-order When the light rays entering the core of an optical fiber travel in a straight line through the fiber.

zipcord Duplex fiber-optic interconnect cable consisting of two tight-buffered fibers with outer jackets bonded together, resembling electrical wiring used in lamps and small appliances.

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